

Planning, Designing, and Constructing Fixed Offshore Platforms—Load and Resistance Factor Design

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Foreword

This document contains engineering design principles and good practices that have evolved during the development of offshore oil resources. Good practice is based on good engineering; therefore, this recommended practice consists essentially of good engineering recommendations. In no case is any specific recommendation included that could not be accomplished by presently available techniques and equipment. Consideration is given in all cases to the safety of personnel, compliance with existing regulations, and antipollution of water bodies.

Offshore technology continues to evolve. In those areas where the committee felt that adequate data were available, specific and detailed recommendations are given. In other areas, general statements are used to indicate that consideration should be given to those particular points. Designers are encouraged to utilize all research advances available to them. As offshore knowledge continues to grow, this document will be revised. It is hoped that the general statements contained herein will gradually be replaced by detailed recommendations.

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Attention Users of This Publication: This edition supersedes the 1st Edition dated July 1993, as well as Errata dated October 1993 and Supplement 1 dated February 1997. Portions of this publication have been changed from the previous edition. In some cases the changes are significant, while in other cases the changes reflect minor editorial adjustments in order to conform to current API styling. Because the document was significantly reformatted from the previous edition, no attempt was made to indicate the locations of changes with bar notations.

Contents

1	Scope	1
2	Normative References.....	1
3	Terms, Definitions, Symbols, and Abbreviated Terms	2
3.1	Terms and Definitions	2
3.2	Symbols	9
3.3	Abbreviated Terms	14
4	Types of Fixed Steel Offshore Structure.....	16
4.1	General	16
4.2	Jackets.....	16
4.3	Towers	17
4.4	Jack-ups	17
5	Planning.....	18
5.1	General	18
5.2	Foundations and Active Geological Processes	18
5.3	Design for Inspection and Maintenance	18
5.4	Design Situations and Criteria	18
5.5	Regulations	19
6	Overall Considerations	19
6.1	Service and Operational Considerations	19
6.2	Safety Considerations.....	20
6.3	Environmental Considerations	21
6.4	Exposure Levels	24
6.5	Assessment of Existing Structures	26
6.6	Structure Reuse	26
7	General Design Requirements	26
7.1	General	26
7.2	Incorporating Limit States	27
7.3	Determining Design Situations	27
7.4	Structural Modeling and Analysis.....	28
7.5	Design for Pre-service and Removal Situations	28
7.6	Design for the In-place Situation	28
7.7	Determination of Resistances	28
7.8	Strength and Stability Checks.....	29
7.9	Robustness	30
7.10	Reserve Strength	30
7.11	Indirect Loads/Actions	31
7.12	Structural Reliability Analysis.....	31
8	Loads/Actions for Pre-service Situations	31
8.1	General	31
8.2	General Requirements	32
8.3	Loads/Actions Associated with Lifting	34
8.4	Loads/Actions Associated with Fabrication	37
8.5	Loads/Actions Associated with Loadout	37
8.6	Loads/Actions Associated with Transportation	38
8.7	Loads/Actions Associated with Installation	39
8.8	Loads/Actions Associated with Removal	40
9	Loads/Actions for In-place Situations	40
9.1	General	40

9.2	Permanent Loads/Actions (G) and Variable Loads/Actions (Q).....	41
9.3	Extreme Environmental Load/Action due to Wind, Waves, and Current.....	42
9.4	Extreme Quasi-static Load/Action due to Wind, Waves, and Current (E_e).....	43
9.5	Extreme Quasi-static Load/Action Caused by Waves Only (E_{we}) or by Waves and Currents (E_{wce}).....	46
9.6	Loads/Actions Caused by Current.....	54
9.7	Loads/Actions Caused by Wind.....	54
9.8	Equivalent Quasi-static Load/Action Representing Dynamic Response Caused by Extreme Wave Conditions	56
9.9	Factored Loads/Actions.....	58
9.10	Design Situations	59
9.11	Local Hydrodynamic Loads/Actions	61
10	Accidental Situations	62
10.1	General	62
10.2	Vessel Collisions	65
10.3	Dropped Objects.....	66
10.4	Fires and Explosions	67
10.5	Abnormal Environmental Load/Actions (Robustness).....	67
11	Seismic Design Considerations.....	67
11.1	General	67
11.2	Seismic Design Procedure	68
11.3	Seismic Reserve Capacity Factor	68
11.4	Recommendations for Ductile Design	69
11.5	ELE Requirements.....	70
11.6	ALE Requirements	71
11.7	Topsides Appurtenances and Equipment	73
12	Structural Modeling and Analysis.....	74
12.1	Purpose of Analysis	74
12.2	Analysis Principles.....	75
12.3	Modeling	75
12.4	Analysis Requirements.....	83
12.5	Types of Analysis	90
12.6	Nonlinear Analysis	91
13	Strength of Tubular Members	94
13.1	General	94
13.2	Tubular Members Subjected to Tension, Compression, Bending, Shear, or Hydrostatic Pressure	96
13.3	Tubular Members Subjected to Combined Forces without Hydrostatic Pressure	104
13.4	Tubular Members Subjected to Combined Forces with Hydrostatic Pressure.....	106
13.5	Effective Lengths and Moment Reduction Factors.....	108
13.6	Conical Transitions	109
13.7	Dented Tubular Members	112
13.8	Corroded Tubular Members	120
13.9	Grouted Tubular Members.....	120
14	Strength of Tubular Joints.....	125
14.1	General	125
14.2	Design Considerations	126
14.3	Simple Tubular Joints	132
14.4	Overlapping Joints	138
14.5	Grouted Tubular Joints.....	138
14.6	Ring Stiffened Tubular Joints	139

14.7	Other Joint Types	140
14.8	Damaged Joints	140
14.9	Noncircular Joints	140
14.10	Cast Joints	140
15	Strength and Fatigue Resistance of Other Structural Components	140
15.1	Grouted Concentric-tubular Connections	140
15.2	Mechanical Connections	144
15.3	Clamps for Strengthening and Repair.....	149
16	Fatigue	153
16.1	General	153
16.2	General Requirements	154
16.3	Description of the Long-term Wave Environment.....	157
16.4	Performing the Global Stress Analyses	159
16.5	Characterization of the Stress Range Data Governing Fatigue.....	162
16.6	Long-term Local Stress Range History	163
16.7	Determining Long-term Stress Range Distribution by Spectral Analysis	165
16.8	Determining Long-term Stress Range Distribution by Deterministic Analysis	169
16.9	Determining Long-term Stress Range Distribution by Approximate Methods	169
16.10	Geometrical Stress Ranges.....	170
16.11	Fatigue Resistance of the Material	172
16.12	Fatigue Assessment.....	174
16.13	Other Causes of Fatigue Damage Than Wave Load/Action	176
16.14	Further Design Considerations.....	177
16.15	Fracture Mechanics Methods	178
16.16	Fatigue Performance Improvement Techniques	179
17	Foundation Design	179
17.1	General	179
17.2	Pile Foundations.....	180
17.3	General Requirements for Pile Design	181
17.4	Pile Capacity for Axial Compression.....	183
17.5	Pile Capacity for Axial Pullout.....	184
17.6	Axial Pile Performance.....	184
17.7	Soil Reaction for Piles under Axial Compression.....	185
17.8	Soil Reaction for Piles under Lateral Loads/Actions.....	185
17.9	Pile Group Behavior	186
17.10	Pile Wall Thickness	186
17.11	Length of Pile Sections.....	190
17.12	Shallow Foundations	190
18	Corrosion Control.....	192
18.1	General	192
18.2	Corrosion Zones and Environmental Parameters Affecting Corrosivity.....	192
18.3	Forms of Corrosion, Associated Corrosion Rates, and Corrosion Damage.....	193
18.4	Design of Corrosion Control	194
18.5	Fabrication and Installation of Corrosion Control	199
18.6	In-service Inspection, Monitoring, and Maintenance of Corrosion Control	199
19	Materials	201
19.1	General	201
19.2	Design Philosophy	203
19.3	Strength Groups	204
19.4	Toughness Classes	204
19.5	Applicable Steels	206
19.6	Cement Grout for Pile/Pin-to-sleeve Connections and Grouted Repairs	206

20	Welding, Fabrication, and Weld Inspection	207
20.1	General	207
20.2	Welding.....	208
20.3	Inspection.....	215
20.4	Fabrication	215
21	Quality Control, Quality Assurance, and Documentation	218
21.1	General	218
21.2	Quality Management System.....	218
21.3	Quality Control Plan	219
21.4	Inspection of Installation Aids and Appurtenances.....	220
21.5	Inspection of Loadout, Sea-fastening, and Transportation	221
21.6	Installation Inspection.....	222
21.7	Documentation.....	223
21.8	Drawings and Specifications.....	224
22	Loadout, Transportation, and Installation	225
22.1	General	225
22.2	Loadout and Transportation.....	226
22.3	Transfer of Structure from Transport Vessel into Water.....	228
22.4	Placement on Seafloor and Assembly of the Structure	229
22.5	Pile Installation	230
22.6	Installation of Conductors	236
22.7	Topsides Installation.....	237
22.8	Grounding of Installation Welding Equipment	237
23	Structure Reuse	238
23.1	General	238
23.2	Fatigue Considerations for Reused Structures.....	238
23.3	Steel in Reused Structures.....	239
23.4	Inspection of Structures to Be Reused	239
23.5	Removal and Reinstallation.....	240
23.6	In-service Inspection and Structural Integrity Management.....	241
Annex A (informative)	Commentary—Additional Information and Guidance.....	242
Annex B (informative)	CTOD Testing Procedures	459
Annex C (informative)	Material Category Approach.....	464
Annex D (informative)	Welding and Weld Inspection Requirements—Material Category Approach	468
Annex E (normative)	Fabrication Tolerances	471
Bibliography		484
Figures		
6.3-1	Vertical Framing Configurations Not Meeting Robustness Requirements	23
6.3-2	Vertical Framing Configurations Meeting Robustness Requirements	23
9.5-1	Procedure for Calculating the Quasi-static Load/Action Caused by Wave plus Current ...	46
9.5-2	Factor for Reduction of Hydrodynamic Coefficients for Conductors in Array due to Shielding as Function of Conductor Spacing/Diameter Ratio.....	51
9.5-3	Doppler Shift due to Steady Current	52
9.5-4	Regions of Applicability of Stream Function, Stokes V, and Linear Wave Theory	53
11.6-1	Seismic Load/Action–Deformation Curve	72
12.3-1	Definition of Offset	77
12.3-2	Definition of Member Rigid Links	78
13.6-1	Example Conical Transition	110
13.7-1	Sign Convention for Dented Tubular.....	115

14.1-1	Terminology and Geometrical Parameters for Simple Tubular Joints	126
14.2-1	Basic Planar Joint Types	128
14.2-2	Examples of Joint Classification	129
14.2-3	In-plane Joint Detailing	131
14.2-4	Out-of-plane Joint Detailing	132
14.3-1	Examples of Chord Length L_c Calculation.....	137
15.1-1	Grouted Pile-to-structure Connection with Shear Keys.....	142
15.1-2	Recommended Shear Key Details	143
16.7-1	Typical Long-term Cumulative Distribution of GSR Values	168
19.1-1	Flowchart for Material Selection	202
20.1-1	Flowchart for Welding, Inspection, Fabrication, and QA/QC Documentation	208
20.2-1	Vickers Hardness Test Locations.....	214
20.4-1	Positioning of Longitudinal and Circumferential Welds in Tubular Joints	216
A.9.5-1	Definition of Surface Roughness Height and Thickness	261
A.9.5-2	Dependence of Steady-flow Drag Coefficient on Relative Surface Roughness	263
A.9.5-3	Wake Amplification Factor for Drag Coefficients as Function of K/C_{ds}	265
A.9.5-4	Wake Amplification Factor for Drag Coefficients as Function of K	266
A.9.5-5	Inertia Coefficient as Function of K	267
A.9.5-6	Inertia Coefficient as Function of K/C_{ds}	268
A.9.5-7	Current Blockage Flow—Current Vectors Computed from Doppler Measurements	270
A.9.5-8	Shielding Factor for Loads/Actions due to Waves on Conductor Arrays as Function of Conductor Spacing	272
A.9.5-9	Comparison of Linear and Nonlinear Stretching of Current Profiles	275
A.9.8-1	Definition of D_e	278
A.13.2-1	Comparison of Test Data with Representative Column Buckling Strength Equations for Fabricated Cylinders Subjected to Axial Compression	305
A.13.2-2	Comparison of Test Data and the Representative Local Buckling Strength Equations for Cylinders Subjected to Axial Compression.....	307
A.13.2-3	Comparison of Test Data with Representative Bending Strength Equations for Fabricated Cylinders	309
A.13.2-4	Comparison of Test Data with Representative Hoop Buckling Strength Equations for Fabricated Cylinders Subjected to Hydrostatic Pressure.....	311
A.13.3-1	Comparison of Measured and Predicted Strengths of Short Tubulars Subjected to Combined Compression (Local Buckling) and Bending.....	313
A.13.3-2	Comparison of Measured and Predicted Strengths of Long Tubulars Subjected to Combined Compression (Column Buckling) and Bending.....	314
A.13.4-1	Marine Buoyancy Analysis Method vs. Rational Buoyancy Analysis Method	315
A.13.4-2	Comparison of Test Data with Interaction Equation for Cylinders under Combined Axial Tension and Hydrostatic Pressure (F_{hc} Determined from Tests).....	317
A.13.4-3	Comparison of Interaction Equations for Various Stress Conditions for Cylinders under Combined Axial Compressive Load and Hydrostatic Pressure	318
A.13.4-4	Comparison of Test Data with Elastic Interaction Curve for Cylinders under Combined Axial Compressive Load and Hydrostatic Pressure.....	319
A.13.4-5	Comparison of Test Data on Fabricated Cylinders with Elastic Interaction Curve for Cylinders under Combined Axial Load and Hydrostatic Pressure	319
A.13.4-6	Comparison of Test Data with Interaction Equations for Cylinders under Combined Axial Compressive Load and Hydrostatic Pressure (Combined Elastic and Yield Type Failures) ...	320
A.13.5-1	Effective Length Factors for Symmetric X-braces as Function of Force Distribution and Rotational End Stiffnesses for Equal Lengths (Torsional Stiffness of Crossing Member Included).....	321
A.13.5-2	Effective Length Factors for Nonsymmetric X-braces as Function of Force Distribution and Rotational End Stiffnesses (Longer Segment = 0.6 Times the Brace Length)	322

A.13.5-3	Effective Length Factors for Nonsymmetric X-braces as Function of Force Distribution and Rotational End Stiffnesses (Longer Segment = 0.7 Times the Brace Length)	323
A.13.5-4	Alignment Chart for Determining the Effective Length of Unbraced Columns	324
A.13.7-1	Comparison of Measured and Predicted Strengths for Dented Tubulars Subjected to Axial Compression.....	326
A.13.7-2	Comparison of Normalized Test Data of Dented Tubulars with Compression Strength Knock-down Ractor for $\lambda_d < 0.252$	327
A.13.7-3	Comparison of Measured and Predicted Strengths for Dented Tubulars Subjected to Bending (Dent in Compression)	328
A.13.7-4	Comparison of Measured and Predicted Strengths for Dented Tubulars Subjected to Combined Compression and Bending (Dent in Tension, Compression, or on Neutral Axis)	329
A.13.9-1	Comparison of Measured and Predicted Strengths Using the Representative Axial Compressive Strength Equations for Fully Grouted Damaged and Undamaged Tubular Members.....	331
A.13.9-2	Illustration of Relevance of Parameters K_2 and K_3	332
A.14.2-1	Effect of f_y/f_t for K-joints on Joint Strength under Balanced Axial Forces	334
A.14.2-2	Comparison of Fitted Curve to Experimental Data for Force-deformation Behavior of a Simple Tubular Joint.....	335
A.14.2-3	Force Patterns Causing Chord Ovalization Greater Than Considered in Strength Equations	338
A.15.1-1	Measured Bond Strength vs. Cube Compressive Strength	346
A.15.1-2	Measured Bond Strength vs. Cube Compressive Strength Times the Height-to-Spacing Ratio.....	347
A.15.2-1	Linearization of Axial Stress Components	351
A.15.2-2	Definition of Pipe and Notch Stresses	356
A.15.2-3	Transformation of Elastic Stress via Strain Energy	358
A.15.2-4	Relation between Notch and Pipe Stresses	359
A.15.2-5	Example of Calculated $S-N$ Curves for Various SCFs and Constant Pipe Mean Stress for HY-80 Material Compared with the Circumferential Weld between Connector and Pipe ...	362
A.15.3-1	General Design Procedure for Stressed Clamps	366
A.15.3-2	Typical Application of a Clamp on an End-to-end Connection	368
A.15.3-3	Typical Application of a Clamp for the Addition of a Member.....	372
A.15.3-4	Typical Application of a Clamp on a Tubular Joint.....	374
A.16.1-1	Fatigue Illustration: Two Plates of Unequal Thickness Connected by a Full Penetration Butt Weld.....	376
A.16.3-1	Example of Two-dimensional Wave Scatter Diagram.....	380
A.16.3-2	Example of Wave Scatter Diagram	381
A.16.3-3	Typical Long-term Cumulative Distribution of Individual Wave Heights	383
A.16.5-1	Determination of Geometrical Stress at Point A around Tubular Connection	388
A.16.7-1	Flowchart of Spectral Analysis Procedure	389
A.16.7-2	Typical Transfer Functions of Total Applied Loads/Actions due to Waves.....	391
A.16.7-3	Typical Fatigue Damage Scatter Diagram	393
A.16.8-1	Flowchart of Deterministic Analysis Procedure	395
A.16.9-1	Weibull Distributions.....	398
A.16.9-2	Combination of Weibull Distributions	399
A.16.10-1	Distances for Extrapolation to Weld Toes	402
A.16.10-2	Joint Classification and Geometrical Parameters for SCFs	403
A.20.2-1	Welder Test Piece for Single-sided Welds.....	447
A.20.2-2	Welded Tubular Joints—Shield Metal Arc (Stick) Welding	449
B.1	CTOD Specimen Preparation	461
B.2	Notch Locations for CTOD Specimens	462
B.3	CTOD Specimen Sectioning and Examination.....	463
E.1	Global Horizontal Tolerances.....	472

E.2	Global Vertical Tolerances	473
E.3	Tolerance on Straightness of Tubular Members.....	474
E.4	Tolerances on Straightness of Tubular Members Assembled into Structure	475
E.5	Tolerances on Thickness Change Locations	476
E.6	Joint Mismatch Tolerances	477
E.7	Leg Alignment Tolerances at Plan Bracing Level (Structures with More Than Four Legs)....	478
E.8	Tolerances on Alignment of Brace Stubs at Tubular Joints.....	479
E.9	Tolerances for Tubular Joints.....	480
E.10	Ring Stiffener Cross-section Tolerances.....	481
E.11	Ring Stiffener Straightness Tolerance.....	482
E.12	Conductor Guide Alignment Tolerances	483

Tables

6.3-1	Gulf of Mexico Design Criteria and Robustness Analysis	22
6.4-1	Determination of Exposure Level	26
8.2-1	Partial Load/Action Factors for Calculating Internal Forces	33
8.3-1	Dynamic Amplification Factor for a Single Onshore Crane.....	34
9.5-1	Typical Values of Hydrodynamic Coefficients	49
9.5-2	Approximate Current Blockage Factors for Typical Gulf of Mexico Jacket-type Structures	50
9.7-1	Shape Coefficients, C_S , for Perpendicular Wind Approach Angles	55
9.10-1	Partial Load/Action Factors for In-place Situations and Exposure Level L-1.....	61
11.2-1	Applicability of Ductile Design Recommendations	68
11.3-1	Representative Values of Seismic Reserve Capacity Factor, C_r	68
12.4-1	Applicability of Analysis Types for Design Situations	84
13.1-1	Arrangement of Requirements for Members.....	95
13.5-1	Effective Length and Moment Reduction Factors for Member Strength Checking	109
14.3-1	Values for Q_u	134
14.3-2	Values for the Coefficients C_1 , C_2 , and C_3	136
14.5-1	Q_u for Grouted Joints.....	139
15.1-1	Connection Design Limitations	143
16.11-1	Basic Representative $S-N$ Curves for Air and Seawater	173
16.12-1	Fatigue Damage Design Factors, γ_{FD}	175
16.16-1	Achievable Improvement Factors on Fatigue Performance for Weld Improvement Techniques.....	179
17.10-1	Minimum Pile Wall Thickness	189
19.3-1	Steel Group SMYS Requirements.....	204
19.4-1	Minimum Toughness Requirements for Structural Steels.....	205
21.2-1	Quality System Requirements	219
21.7-1	Documentation Requirements	223
A.7.7-1	Statistical Factor, k_n	248
A.7.7-2	Minimum Coefficients of Variation	250
A.7.11-1	Inclusion of Indirect Loads/Actions in Permanent and Variable Loads/Actions	254
A.13.2-1	Comparison of Geometry Requirements in Different Codes for Stiffeners with Flanges When Supported Along Two Edges	312
A.13.4-1	Code Check Equation Differences between Marine Buoyancy Analysis Method and Rational Buoyancy Analysis Method	315
A.16.10-1	Classification Table for a Brace End	405
A.16.10-2	Equations for SCFs in T/Y-joints	410
A.16.10-3	Equations for SCFs in X-joints.....	411
A.16.10-4	Equations for SCFs in Gap/Overlap K-joints.....	413
A.16.10-5	Equations for SCFs in KT-joints	414
A.16.10-6	Joint Classification Tabulation	418

A.16.10-7	Type 1: Material Free from Welding.....	419
A.16.10-8	Type 2: Continuous Welds Essentially Parallel to the Direction of Applied Stress	420
A.16.10-9	Type 3: Transverse Groove Welds in Plates and Tubulars (i.e. Essentially Perpendicular to the Direction of Applied Stress).....	421
A.16.10-10	Type 4: Welded Attachments on the Surface or Edge of a Stressed Member	422
A.16.10-11	Type 5: Load-carrying Fillet and T-butt Welds	423
A.16.10-12	Type 6: Details in Welded Girders and Tubulars	425
A.19.2-1	Recommended Lowest Anticipated Service Temperatures.....	440
A.20.2-1	Root Gap Requirements for Tubular Joints—Shield Metal Arc Welding.....	448
A.20.2-2	Minimum Weld Thicknesses for Tubular Joints—Shield Metal Arc Welding.....	448
C.1	Material Category: Material Selection for the Structure	465
C.2	Correlation of Steel Group and Toughness Class for Steel Plates to US Specifications	466
C.3	Correlation of Steel Group and Toughness Class for Structural Steel Shapes to US Specifications.....	467
C.4	Correlation of Steel Group and Toughness Class for Structural Steel Pipe to US Specifications	467
D.1	Material Category: Minimum Weld Metal Toughness Requirements	468
D.2	Material Category: Minimum Heat Affected Zone Toughness Requirements.....	469
D.3	Material Category Approach: Minimum Extent of Weld Inspection	470

Introduction

The API offshore structures standards constitute a common basis covering those aspects that address design requirements and assessments of all offshore structures used by the petroleum and natural gas industries worldwide. Through their application, the intention is to achieve reliability levels appropriate for manned and unmanned offshore structures, whatever the type of structure and the nature of the materials used.

It is important to recognize that structural integrity is an overall concept comprising models for describing loads/actions, structural analyses, design rules, safety elements, workmanship, quality control procedures, and national requirements, all of which are mutually dependent. The modification of one aspect of design in isolation can disturb the balance of reliability inherent in the overall concept or structural system. The implications involved in modifications, therefore, need to be considered in relation to the overall reliability of all offshore structural systems.

The offshore structures standards are intended to provide a wide latitude in the choice of structural configurations, materials, and techniques without hindering innovation. Sound engineering judgment is therefore necessary in the use of these standards.

Annex A provides background to and guidance on the—much but not all—use of this document and needs to be read in conjunction with the main body of this document. The section numbering in Annex A is the same as in the normative text to facilitate cross-referencing.

Materials, welding, and weld inspection requirements can be based either on a “material category” or on a “design class” approach, as discussed in Sections 19 and 20. If the material category approach is used, the corresponding provisions of Annexes C and D are normative; if the design class approach is used, the corresponding provisions of ISO 19902 are normative.

Annex E gives requirements on fabrication tolerances. To meet certain needs of industry for linking software to specific elements in this document, a special numbering system has been permitted for figures, tables, equations, and bibliographic references.

Planning, Designing, and Constructing Fixed Offshore Platforms—Load and Resistance Factor Design

1 Scope

This document specifies requirements and provides recommendations applicable to the following types of fixed steel offshore structures for the petroleum and natural gas industries:

- caissons, free-standing and braced;
- jackets;
- monotowers;
- towers.

In addition, it is applicable to compliant bottom founded structures, steel gravity structures, jack-ups, other bottom founded structures, and other structures related to offshore structures (such as underwater oil storage tanks, bridges, and connecting structures), to the extent to which its requirements are relevant.

This document contains requirements for planning and engineering of the following tasks:

- design, fabrication, transportation, and installation of new structures as well as their future removal;
- in-service inspection and integrity management of both new and existing structures;
- assessment of existing structures;
- evaluation of structures for reuse at different locations.

Requirements for topsides structures can be found in API 2TOP, for marine operations in API 2MOP, and for the site-specific assessment of jack-ups in ISO 19905-1. Additional requirements for the design of fixed steel offshore structures in arctic environments can be found in API 2N.

2 Normative References

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document applies (including any addenda/errata).

API Recommended Practice 2EQ, *Seismic Design Procedures and Criteria for Offshore Structures*

API Recommended Practice 2FB, *Recommended Practice for Design of Offshore Facilities Against Fire and Blast Loading*

API Recommended Practice 2GEO, *Geotechnical and Foundation Design Considerations*

API Recommended Practice 2GEO, *Geotechnical and Foundation Design Considerations*, First Edition

API Recommended Practice 2MET, *Derivation of Metocean Design and Operating Conditions*

API Recommended Practice 2MOP, *Marine Operations*

API Recommended Practice 2MOP, *Marine Operations*, First Edition

API Recommended Practice 2SIM, *Structural Integrity Management of Fixed Offshore Structures*

ISO 10414-1¹, *Petroleum and natural gas industries—Field testing of drilling fluids—Part 1: Water-based fluids*

ISO 12135, *Metallic materials—Unified method of test for the determination of quasistatic fracture toughness*

ISO 19900, *Petroleum and natural gas industries—General requirements for offshore structures*

3 Terms, Definitions, Symbols, and Abbreviated Terms

3.1 Terms and Definitions

For the purposes of this document, the terms and definitions given in API 2EQ, API 2GEO, API 2MET, API 2SIM, and ISO 19900 and the following apply.

3.1.1

abnormal value

Value of a parameter of abnormal severity used in accidental limit state checks in which a structure should not suffer complete loss of integrity.

NOTE 1 Abnormal design situations are used to provide robustness against events with a probability of exceedance of typically between 10^{-3} and 10^{-4} per annum by avoiding, for example, gross overloading.

NOTE 2 Abnormal values and events have probabilities of exceedance of the order of 10^{-3} to 10^{-4} per annum; in the limit state checks, some or all of the partial factors are set to 1.0.

3.1.2

accidental design situation

Design situation involving exceptional conditions of the structure or its exposure.

EXAMPLE Impact, fire, explosion, local failure, loss of intended differential pressure (e.g. buoyancy).

3.1.3

action

The word “action” is equivalent to the words “load/imposed displacement/accelerations” in various API documents used as references.

3.1.4

after damage design situation

Design situation for which the condition of the structure reflects damage due to an accidental design situation and for which the environmental conditions are specially defined.

3.1.5

analysis type

Method including governing equations for deriving load/action effects.

EXAMPLE Static analysis, transient dynamic analysis, nonlinear analysis.

¹ International Organization for Standardization, Chemin de Blandonnet 8, CP 401, 1214 Vernier, Geneva, Switzerland, www.iso.org.

3.1.6**basic variable**

One of a specified set of variables representing physical quantities that characterize loads/actions, environmental influences, geometrical quantities, or material properties including soil properties.

3.1.7**boundary conditions**

Loads/actions and constraints on a [section of a] structural component [or a group of structural components] by other structural components or by the environment surrounding it.

NOTE Boundary conditions can be used to generate reaction forces at locations of restraint.

3.1.8**braced caisson**

Monotower where the lower part of the monocolumn is supported laterally by one or more inclined braces between the column and one or more foundation piles.

3.1.9**bucket foundation**

Foundation consisting of a cylindrical shell open on one end and installed by suction.

3.1.10**characteristic value**

Value assigned to a basic variable associated with a prescribed probability of not being violated by unfavorable values during some reference period.

NOTE The characteristic value is the main representative value; in some design situations, a variable can have two characteristic values, an upper and a lower value.

3.1.11**compliant bottom founded structure**

Structure that is supported at its base by foundation piles or by another nonsuperficial foundation system and that is sufficiently flexible that applied lateral dynamic loads/actions are substantially balanced by inertial reactions.

NOTE Although this document is applicable to fixed steel offshore structures and is not intended to form a complete standard for compliant bottom founded structures, some of its requirements and guidance can be applied to compliant bottom founded structures; parts of its requirements can also apply to other bottom founded structures (e.g. some forms of minimal structure), as considered appropriate and as agreed upon by parties concerned on a case-by-case basis.

3.1.12**consequence category**

Classification system for identifying the environmental, economic, and indirect personnel safety consequences of failure of a platform.

NOTE See 6.4.3.

3.1.13**critical component**

Structural component, failure of which would cause significant strength loss to or failure of a significant part of or the whole structure.

3.1.14**deformation capacity**

Ability of a structure or structural component to deform without significant loss of resistance, e.g. ductility, or the extent to which it can do so.

3.1.15**design service life**

Assumed period for which a structure is to be used for its intended purpose with anticipated maintenance, but without substantial repair being necessary.

3.1.16**design value**

Value derived from the representative value for use in the design verification procedure.

3.1.17**dynamic amplification factor****DAF**

Ratio of a dynamic load/action effect to the corresponding static load/action effect.

NOTE An appropriately selected dynamic amplification factor can be applied to static loads/actions to simulate the effects of dynamic loads/actions.

3.1.18**explosion**

Rapid chemical reaction of gas or dust in air.

NOTE An explosion results in increased temperatures and pressure impulses; a gas explosion on an offshore platform is usually a deflagration in which flame speeds remain subsonic.

3.1.19**exposure level**

Classification system used to define the requirements for a structure based on consideration of life-safety and of environmental and economic consequences of failure.

3.1.20**extreme value**

Value of a parameter used in ultimate limit state checks, in which a structure's global behavior is intended to stay in the elastic range.

3.1.21**fit-for-service**

Meeting the intent of a standard although not meeting specific provisions of that standard in local areas, such that failure in these areas will not cause unacceptable risk to life-safety or the environment.

3.1.22**fixed structure**

Structure that is bottom founded and transfers all loads/actions on it to the sea floor.

3.1.23**free-standing caisson**

Monotower where the structure consists, over its full height, of a single vertical column that continues into the seabed as the foundation pile.

3.1.24**global analysis**

Determination of a consistent set of internal forces and moments, or stresses, in a structure that are in equilibrium with a defined set of loads/actions on the entire structure.

NOTE When a global analysis is of a transient situation (e.g. seismic), the inertial response is part of the equilibrium.

3.1.25**hazard**

Potential for human injury, damage to the environment, damage to property, or a combination of these.

NOTE Several of the usual hazards to a platform (e.g. extreme storms) are treated as design situations for a structure; in this document, hazards are errors and abnormal and accidental situations.

3.1.26**jacket**

Fixed structure with leg piles and axial force transfer from the structure and topsides into the piles at the top of the structure.

NOTE See 4.2.

3.1.27**jack-up**

Mobile offshore unit that can be relocated and is bottom founded in its operating mode.

NOTE See 4.4.

3.1.28**life-safety category**

Classification system for identifying the applicable level of life-safety for a platform.

NOTE See 6.4.2.

3.1.29**load arrangement**

Identification of the position, magnitude, and direction of a free load/action.

3.1.30**load case**

Compatible load arrangements, sets of deformations and imperfections considered simultaneously with permanent loads/actions and fixed variable loads/actions for a particular design or verification.

3.1.31**local analysis**

Determination of a consistent set of internal forces and moments, or stresses, in a cross-section of a structural component, or in a subset of structural components forming part of the structural system, that are in equilibrium with the boundary conditions.

3.1.32**maintenance**

Set of activities performed during the working life of the structure in order to enable it to fulfil the requirements for reliability.

NOTE Activities to restore the structure after an abnormal, accidental, or seismic event are outside the scope of maintenance.

3.1.33**monotower**

Fixed structure in which the whole structure, or at least the upper part of the structure, consists of a single vertical column (tubular or framed) that carries the topsides.

NOTE Where only the upper part of the structure is a single vertical column, the lower part of the structure consists of tubular members or frames that connect the vertical column to the foundation piles or to another nonsuperficial foundation system that supports the monotower at its base, such as bucket foundations.

3.1.34**nominal geometrical properties**

Properties of a structural component derived from its representative geometrical dimensions, remote from the location under consideration and with any corrosion allowance removed.

NOTE 1 The nominal cross-section of a member at a joint is the cross-section beyond any joint can or brace stub forming part of the joint.

NOTE 2 Nominal geometrical properties are used in a global analysis to calculate the global behavior of the structure.

NOTE 3 The nominal thickness of a component excludes any thickening of the component due to weldments.

3.1.35**nominal stress**

Stress calculated in a sectional area, including the stress-raising effects of the macro-geometrical shape of the component of which the section forms a part, but disregarding the local stress-raising effects from the section shape and any weldment or other fixing detail.

NOTE Overall elastic behavior is assumed when calculating nominal stresses.

3.1.36**nominal value**

Value assigned to a basic variable determined on a nonstatistical basis, typically from acquired experience or physical conditions.

3.1.37**owner**

Representative of the companies that own a development.

NOTE The owner will normally be the operator on behalf of co-licensees.

3.1.38**primary structure**

All main structural components that provide the structure's main strength and stiffness.

3.1.39**quasi-static analysis**

Static analysis of a structure subjected to loads/actions that vary slowly in relation to the structure's fundamental natural period such that the influence of structural accelerations can be either safely neglected or is approximated by using an equivalent quasi-static load/action.

3.1.40**redundancy**

Ability of a structure to find alternative load paths following failure of one or more noncritical components, thus limiting the consequences of such failures.

3.1.41**regulator**

Authority established by a national governmental administration to oversee the activities of the offshore oil and natural gas industries within its jurisdiction, with respect to the overall safety to life and protection of the environment.

NOTE 1 The term "regulator" can encompass more than one agency in any particular territorial waters.

NOTE 2 The regulator can appoint other agencies, such as marine classification societies, to act on its behalf, and in such cases, "regulator" as it is used in this document includes such agencies.

NOTE 3 In this document, the term "regulator" does not include any agency responsible for approvals to extract hydrocarbons, unless such agency also has responsibility for safety and environmental protection.

3.1.42**repair**

Activities performed to preserve or to restore the function of a structure that fall outside the definition of maintenance.

3.1.43**representative value**

Value assigned to a basic variable for verification of a limit state.

3.1.44**representative yield strength**

Stress at which a material exhibits a specified deviation from proportionality of stress and strain.

NOTE An offset of 0.2 % is used for many metals, including the steels covered by this document.

3.1.45**reserve strength ratio****RSR**

Measure of the capacity of a structural system to withstand overload.

NOTE For fixed steel offshore structures, the concept of a reserve strength ratio is usually applied to environmental loads/actions (see 7.10), but is not limited to these loads/actions.

3.1.46**risk-reducing measures**

Measures undertaken to reduce either the probability or the consequences of an accident, including the preparation of contingency plans.

3.1.47**robustness**

Ability of a structure to withstand events with a reasonable likelihood of occurring without being damaged to an extent disproportionate to the cause.

3.1.48**secondary structure**

Structural components that, when removed, do not significantly alter the overall strength and stiffness of the structure.

3.1.49**steel gravity structure**

Fixed structure that is held in place against environmental loads/actions solely by the weight of the structure and any contained ballast, together with foundation resistance resulting from its weight and lateral resistance from any skirts.

NOTE Although this document is not intended to form a complete standard for gravity structures, some of the requirements and guidance could be applied to gravity structures.

3.1.50**strength**

Mechanical property of a material indicating its ability to resist loads/actions, usually given in units of stress.

3.1.51**stress concentration factor****SCF**

Factor relating a nominal stress to the local stress at a detail.

NOTE Stress concentration factors are very important for and especially used in determining fatigue damage (see 16.10.2).

3.1.52**structural analysis**

Process or algorithm for determining load/action effects from a given set of loads/actions.

NOTE Structural analyses are performed at three levels (global analysis of an entire structure, local analysis of a structural component, local analysis of a section of a structural component) using different structural models.

3.1.53**structural component**

Physically distinguishable part of a structure.

NOTE The main structural components of a fixed steel structure are tubular members (see Section 13), tubular joints (see Section 14), mechanical connectors and clamps (see Section 15), foundation piles and bucket foundations (see Section 17).

3.1.54**structural system**

Load-bearing components of a structure and the way in which these components function together.

3.1.55**topsides**

Structures and equipment placed on a support structure (fixed or floating) to provide some or all of a platform's functions.

NOTE 1 For a ship-shaped floating support structure, neither the deck nor a superstructure integral with the hull is part of the topsides.

NOTE 2 For a jack-up, the hull is not part of the topsides.

NOTE 3 A separate fabricated deck or module support frame is part of the topsides.

3.1.56**tower**

Fixed structure that is supported by foundation arrangements at the base of the structure.

NOTE See 4.3.

3.1.57**utilization****member utilization****joint utilization**

Maximum value of the ratio of the generalized representation of the design stress (force) in a structural component to the generalized representation of the design resistance in stress (force) units of the component.

NOTE 1 The utilization is the maximum absolute value of the ratio for all conditions being considered.

NOTE 2 Only utilizations smaller than or equal to 1.0 satisfy the design criteria for a particular limit state.

NOTE 3 The design stress (force) is the stress (force) due to factored loads/actions: $F_d = F_r \gamma_f$; the design resistance is the representative resistance divided by the partial resistance factor: $R_d = R_r / \gamma_R$.

NOTE 4 For members and joints subjected to a single force, the utilization, U , is equal to: $F_d / R_d = F_r \gamma_f \gamma_R / R_r$.

NOTE 5 For members and joints subjected to combined forces, the internal stress (force) pattern and the design resistance combine into an interaction equation; if the interaction equation governing the design check is, or can be, reduced to an inequality of the form $I \leq 1.0$, then the utilization is equal to I .

3.2 Symbols

The following is a summary of the main symbols that are used throughout this document. Many other symbols are locally defined where they are used. Local use includes main symbols with one or more subscripts when a more specific use and associated definition of the symbol is intended.

A	accidental load/action
A	area
C	coefficient in general
C_d	hydrodynamic drag coefficient
C_m	hydrodynamic inertia coefficient
C_m	moment reduction factor
C_r	seismic reserve capacity factor
C_s	shape coefficient
C_s	slamming coefficient
c_u	undrained shear strength of the soil, in stress units
D	diameter of a circular cylindrical element
D	diameter of a tubular joint chord
D	Palmgren-Miner fatigue damage ratio
D	equivalent quasi-static load/action representing dynamic effects
D_r	relative density of sand
d	water depth
d	diameter of tubular joint brace
E	Young's modulus of elasticity, which may be taken as 205,000 MPa for structural steel, in lieu of more detailed information
E	quasi-static environmental load/action
E_e	extreme quasi-static environmental load/action due to wind, waves, and current
E_o	environmental load/action due to owner-defined operating wind, wave, and current parameters
F	load/action
F_d	design value of load/action

F_r	representative value of load/action
f	unit skin friction of a pile, in stress units
f	representative strength in general, in stress units
f_b	representative bending strength, in stress units
f_c	representative axial compressive strength, in stress units
f_{cu}	unconfined cube strength of grout
f_e	Euler buckling strength, in stress units
f_h	representative hoop buckling strength, in stress units
f_t	representative axial tensile strength, in stress units
f_v	representative shear strength, in stress units
f_y	representative yield strength, in stress units
G	permanent load/action
g	acceleration due to gravity
g	gap
H	height of an individual wave
H_s	significant wave height
I	moment of inertia
K	factor (locally defined with subscript for specific meaning)
K	effective length factor
K	dimensionless coefficient of lateral earth pressure
K	Keulegan-Carpenter number
k	factor (locally defined with subscript for specific meaning)
k	wave number
k_{DAF}	dynamic amplification factor
L	length
M	moment (bending or torsion) due to factored loads/actions

M_d	design value of joint bending moment strength, in moment units
M_{uj}	representative value of joint bending moment strength, in moment units
m	inverse slope of the S - N curve
m	mass
N	number of cycles to failure in fatigue of a specified constant amplitude stress range, S
n	number of occurrences of a specified stress range, S , during a specified time
P	percentage
P	member axial force due to factored loads/actions
P_d	design value of joint axial strength, in force units
P_{uj}	representative value of joint axial strength, in force units
P_f	probability of failure
$P-\Delta$	second-order bending effect due to axial compressive force, P , and lateral deflection, Δ
$P(x)$	cumulative probability of variable x
p	factored hydrostatic pressure
p_a	atmospheric pressure
p_0'	effective overburden stress of the soil
$p(x)$	probability density function of variable x
p - y	lateral soil resistance versus local pile displacement
Q	variable load/action
Q_d	design axial pile capacity
Q_f	representative value of the total shaft friction resistance of a pile, in force units
Q_f	chord force factor
Q_g	gap factor for a joint
Q_p	representative value of the end bearing capacity of a pile, in force units
Q_r	representative value of the axial pile capacity
Q_u	chord strength factor

Q - z	pile end bearing resistance versus pile tip displacement
Q_{β}	geometrical factor for a joint
q	unit end bearing of a pile, in stress units
R	radius of a circular cross-section
R_d	design value of a component resistance (from ISO 19900)
Re	Reynolds number
R_{RS}	reserve strength ratio (RSR)
r	radius of gyration
S	internal force (load/action effect)
S	constant amplitude stress range for fatigue assessments
Sr	Strouhal number
S_r	Stress range
T	quasi-static transient load/action
T	wall thickness of the chord member
T	period in general
T_{app}	apparent wave period
T_z	average zero-crossing wave period
t	time
t	wall thickness in general
t	wall thickness of the brace member
t - z	soil pile shear transfer versus local pile displacement
U	utilization
V	beam shear due to factored loads/actions
V	displaced volume
V	coefficient of variation
Z_e	elastic section modulus
Z_p	plastic section modulus
α	slope angle of a cone section

β	ratio of brace outside diameter to chord outside diameter
γ	ratio of chord outside radius to chord wall thickness
γ'	submerged unit weight of soil
γ_f	partial load/action factor of which the value reflects the uncertainty or randomness of the load/action
γ_R	partial resistance factor of which the value reflects the uncertainty or variability of the component resistance including those of material properties (from ISO 19900)
θ	angle in general
θ_m	mean wave direction
λ	wave length
λ	column slenderness parameter
ν	Poisson's ratio, which may be taken as 0.3 for structural steel, in lieu of more detailed information
ρ_a	mass density of air, which may be taken as 1.22 kg/m ³ , in lieu of more detailed information
ρ_s	mass density of steel, which may be taken as 7850 kg/m ³ for structural steel, in lieu of more detailed information
ρ_w	mass density of seawater, which may be taken as 1025 kg/m ³ , in lieu of more detailed information
σ	stress in general
τ	shear stress in general
τ	ratio of brace wall thickness to chord wall thickness
ν	kinematic viscosity of water
ϕ'	angle of internal friction in sand
Ω	ratio of frequency of excitation to natural frequency
ω	circular frequency
ω_e	frequency of excitation
ω_n	natural frequency in general
ω_1	first natural frequency
ξ	factor (locally defined with subscript for specific meaning)
ξ	fraction of critical damping
$\emptyset$	diameter

3.3 Abbreviated Terms

ALE	abnormal level earthquake
ALS	accidental limit states
ASD	allowable stress design
COV	coefficient of variation
CTOD	crack tip opening displacement
CVN	Charpy V-notch
DAF	dynamic amplification factor
DC	design class
ELE	extreme level earthquake
ERW	electric resistance welded
FEA	finite element analysis
FLS	fatigue limit states
GS	geometrical stress
GSR	geometrical stress range
HAT	highest astronomical tide
HAZ	heat affected zone
HISC	hydrogen-induced stress cracking
IC	impressed current
IPB	in-plane bending
JIP	joint industry project
LAST	lowest anticipated service temperature
LAT	lowest astronomical tide
LBZ	local brittle zone
MC	material category
MIC	microbiologically influenced corrosion
MPI	magnetic particle inspection
MPM	most probable maximum

NDT	nondestructive testing
OPB	out-of-plane bending
PLS	progressive collapse limit states
POD	probability of detection
PQR	procedure qualification record
PWHT	post-weld heat treatment
QA	quality assurance
QC	quality control
QMS	quality management system
Q&T	quenched and tempered
RAO	response amplitude operator
RMS	root-mean-square
RSR	reserve strength ratio
RT	radiographic testing
SAW	submerged arc welding
SCF	stress concentration factor
SDOF	single degree of freedom
SIM	structural integrity management (system)
SLS	serviceability limit states
SMYS	specified minimum yield strength
SRA	structural reliability analysis
SRC	seismic risk category
SRD	soil resistance to driving
SSSV	subsurface safety valve
TMCP	thermo-mechanically controlled process
ULS	ultimate limit states
UT	ultrasonic testing
VIV	vortex-induced vibration
WPS	welding procedure specification
WPQ	welding procedure qualification

4 Types of Fixed Steel Offshore Structure

4.1 General

A fixed steel offshore structure can take many forms, including the following:

- a) braced caisson (3.1.8);
- b) free-standing caisson (3.1.23);
- c) jacket (3.1.26, see 4.2);
- d) monotower (3.1.33);
- e) tower (3.1.56, see 4.3);
- f) steel gravity structure (3.1.49);
- g) jack-up (3.1.27, see 4.4).

This document specifies requirements for braced caissons, free-standing caissons, jackets, monotowers, and towers. Some of the requirements may also be used, as appropriate, for other forms of steel bottom founded offshore structures. The design of other forms of fixed steel bottom founded offshore structures shall achieve a level of reliability consistent with that intended by this document.

There have been historical differences in the usage and understanding of the terms “jacket” and “tower,” particularly between the United States and Europe. The difference in such understanding of the terms has no significant impact on the application of this document as long as the differences in structural behavior are considered in the analyses of the different structures.

There have been historical differences, which continue to exist, in the use and understanding of the term “caisson.” In the offshore oil and gas industry, this term, together with its variants “braced caisson” or “free-standing caisson” (see, respectively, 3.1.8 and 3.1.23), has traditionally been used to refer to a particular type of minimum fixed structure, where the main component is a relatively large diameter tubular member, with or without additional lateral support, intended to satisfy various functional requirements such as supporting one or more wells, or supporting small decks and associated facilities. Conversely, in the geotechnical arena the term “caisson” has traditionally been used to refer to “foundation caisson,” i.e. foundation component/system consisting of shorter and more rigid chamber of larger diameter or larger lateral dimensions than the caisson structures described above.

Section 17 covers the geotechnical design of long slender piles (i.e. satisfying the condition $L/D \geq 10$, where L is the length embedded in the soil, and D is the outside diameter) for fixed steel structures. The provisions of that section apply also to braced or free-standing caisson structures with $L/D \geq 10$. However, the content of Section 17 does not apply to the design of short and rigid, large diameter foundations with $L/D < 10$. Guidance on the geotechnical design of this type of foundations is within the scope of API 2GEO.

4.2 Jackets

Jackets and towers are superficially similar structures, the essential differences between them being the arrangement of the foundation piles and, associated with this, the location of axial force transfer from the structure and topsides into the piles. The location and method of transferring the axial forces result in significant differences in the behavior of jackets and towers.

A jacket is a welded tubular space frame with three or more vertical (or near vertical) tubular chords (the legs) and a bracing system between the legs. The jacket provides support for the foundation piles, conductors, risers, and other appurtenances.

A jacket foundation includes leg piles that are inserted through the legs and are connected to the legs either at the top, by welding or mechanical means, or along the length of the legs, by grouting. Additional piles, called “skirt piles,” can be inserted through and connected to sleeves at the base of the structure. Leg and skirt piles jointly anchor the structure and transfer both vertical and horizontal loads/actions to the seabed.

Where the foundation piles are connected to the legs only at the top of the jacket, the axial forces (including moments caused by shear forces) are transferred to the piles at the connection and the jacket “hangs” from the piles.

Where the foundation piles are connected to the legs by grouting along the full length of the leg, the jacket behavior is similar to that of a tower, with the legs and piles acting together as composite components.

4.3 Towers

A tower type structure is supported by foundation arrangements at the base of the structure.

The tower is again made of a welded tubular space frame with three or more vertical (or near vertical) tubular chords, called “legs,” with a bracing system between the legs. The tower provides support for the topsides, conductors, risers, and other appurtenances.

A tower foundation usually includes cluster piles that are inserted through and connected to sleeves around the (corner) legs at the base of the structure. Additional piles, called “skirt piles,” can be inserted through and connected to sleeves at the base and along the perimeter of the structure. As an alternative to piles, a tower can be supported by another nonsuperficial foundation system that supports it at its base, such as bucket foundations. Cluster and skirt piles, or another nonsuperficial foundation system, anchor the structure and transfer both vertical and horizontal loads/actions to the seabed.

The global behavior of a tower is that of a vertical cantilever, with all loads/actions being transferred to the foundation system at the base of the tower.

4.4 Jack-ups

A jack-up comprises a floating hull and one or more legs, which can move up and down relative to the hull. A jack-up reaches its operational mode by lowering the leg(s) to the sea floor and then raising the hull to the required elevation. The majority of jack-ups have three or more legs, each of which can be moved independently and which are supported on the sea floor by spudcans or mats.

The majority of jack-ups are built for short-term operations at different locations around the world. As metocean and foundation conditions vary between locations, such jack-ups are assessed for each particular location (see ISO 19905-1 [4]). A few jack-ups are purpose built for production operations at a single location, albeit that there can be an intent for eventual reuse at further locations.

Two options are available for a jack-up within the framework of the ISO 19900 series; these are as follows:

- a) design, assessment, and operation to the requirements of this document, and
- b) site-specific assessment to the requirements of ISO 19905-1 [4], in which case either the periodic inspection requirements of the IACS member or the alternative requirements for long-term deployment in ISO 19905-1 [4] shall be satisfied.

NOTE ISO 19905-1 [4] applies only to structures holding valid certification as self-elevating units from an IACS member, or equivalent.

Option a) above shall be used for a jack-up purpose built for production at a particular location and where option b) does not apply. Option b) should be used for a jack-up built to operate in a variety of locations. In other cases, when an existing jack-up is to be used for an extended period at a single location for production (rather than for exploration, appraisal, or other short-term application), the owners and regulators shall apply either the reuse requirements (see Section 23) of this document or the requirements of ISO 19905-1 [4].

When option a) is selected, some aspects of the requirements of this document do not apply, including the determination of environmental loads/actions and load/action effects due to dynamic response, structural behavior, and foundation behavior. The alternative procedures given in ISO 19905-1 [4] may be used in such cases; however, the procedures shall be modified as necessary to give an equivalent level of structural integrity to that intended by this document. The foundation strength, the foundation preload, overturning, leg chord strength, and the strength of the holding system shall be checked in accordance with the requirements of ISO 19905-1 [4].

A limitation on the use of this document for jack-ups is the strength and ductility of high-strength jack-up steels (with yield strengths of 700 N/mm² or more) for major structural components—as compared with yield strengths of less than 500 N/mm² for other conventional fixed steel offshore structures. The designer or assessor of such jack-up units shall take note of the limitations of this document for steels with yield strengths higher than 500 N/mm² and shall obtain sufficient data to justify the design for such cases.

5 Planning

5.1 General

Adequate planning should be completed before actual design is started in order to obtain a safe, workable, and economical offshore structure. The initial planning should include the determination of all limit states, design situations, and design criteria upon which the design of the structure is to be based, following the general requirements and conditions specified in ISO 19900.

5.2 Foundations and Active Geological Processes

The capacity and suitability of the foundations and the susceptibility to active geological processes shall be assessed in selecting the type of structure and the location. Requirements and guidance are given in Section 17 and in API 2GEO.

5.3 Design for Inspection and Maintenance

ISO 19900 notes that “Structural integrity, serviceability throughout the intended service life, and durability are not simply functions of the design calculations but are also dependent on the quality control exercised in construction, the supervision on site, and the manner in which the structure is used and maintained.” Therefore, during the planning stage, a philosophy for inspection and maintenance should be developed. The design of the structure as a whole, as well as the structural details, should be consistent with this philosophy. In preparing the philosophy, a realistic assessment should be made of the ability to actually achieve the intended quality of the results of inspection and maintenance efforts. This assessment should be taken into account when progressing the design and fine-tuning design situations and criteria. Relevant requirements related to inspection and maintenance requirements are given in API 2SIM.

5.4 Design Situations and Criteria

Design situations as used herein include all the service and operational requirements resulting from the intended use of the structure and the environmental conditions that can affect the design of the structure in accordance with ISO 19900. In the absence of site-specific information on the environmental conditions, API 2MET contains indicative values for the extreme conditions in certain areas. These values may be used for conceptual design

studies. The owner shall review the validity of these values and, if necessary, use site-specific data for the final design of a structure (see also 6.3.2 and 7.3). Criteria that are to be met by the design are directly related to the specific formulation of the design situations. Therefore, design situations and design criteria shall not be separated from one another. They are jointly specified in Sections 8 to 17.

5.5 Regulations

The requirements of national regulations and standards where a structure will be used can be different from those given in this document. It shall be ensured that the requirements of safety, reliability, and durability implicit in the requirements of this document are met. This applies to all phases of planning, design, fabrication, transportation, installation, service, and removal.

6 Overall Considerations

6.1 Service and Operational Considerations

6.1.1 General Considerations

The principal service and operational requirements to be considered to establish the design basis for fixed steel offshore structures are summarized in ISO 19900. Operational requirements include the following:

- service requirements, i.e. function of the platform;
- manning;
- conductors and risers;
- equipment and material layouts, including rig access to conductor slots;
- personnel and material transfer;
- motions and vibrations;
- any special requirements;
- location and orientation.

In addition, requirements for decommissioning and removal should be considered.

6.1.2 Water Depth

The water depth, surge height, and tide should be determined as accurately as possible, so that elevations can be established for boat landings, fenders, decks, and corrosion protection; see API 2MET for information on these parameters. The potential subsidence of the sea floor and settlement of the structure shall be evaluated when determining the design water depth.

6.1.3 Structural Configuration

6.1.3.1 General

Requirements for the structural configuration are given in ISO 19900; some further requirements are given in 6.1.3.2 and 6.1.3.3.

6.1.3.2 Deck Elevation

Large forces result when waves strike a platform's deck and equipment. To avoid this, the bottom of the lowest deck shall be located at an elevation that will clear the calculated crest of the design wave with adequate allowance for safety to allow the passage of wave crests of abnormal values (see A.6.1.3.2). For new platforms in the Gulf of Mexico, the elevation for the underside of the deck shall not be lower than the 1000-year return period maximum crest elevation provided in API 2MET or site-specific data developed in accordance with the requirements of API 2MET.

An air gap, the distance between the maximum crest elevation used for deck clearance and the bottom of steel on the lower deck, shall be provided for any known or predicted long-term seafloor subsidence, both regional and that due to hydrocarbon extraction. An additional air gap should be allowed to account for structures that experience significant structural rotation or "set down."

In general, no platform components, piping, or equipment should be located below the lower deck. However, when it is unavoidable to position such items as minor subcellars, sumps, drains, or production piping below bottom of steel on the lower deck, provisions should be made for the wave forces developed on these items. These wave forces may be calculated using the crest pressure of the design wave applied against the projected area. These forces may be considered on a "local" basis in the design of the item. These provisions do not apply to vertical members such as deck legs, conductors, risers, etc., which normally penetrate the air gap. The design process accounts for forces on those vertical members.

Whenever this requirement is not satisfied, the structure shall be designed to resist any wave and current loads/actions on the deck(s), including the loads/actions on facilities or equipment supported by the deck(s).

6.1.3.3 Equipment and Material Layouts

Layouts and centers of gravity associated with permanent and variable loads/actions, as defined in 9.2, are required in the development of the design. Heavy concentrated loads/actions on the structure should be located such that proper framing to support these loads/actions can be planned. When possible, consideration should be given to maintenance requirements and future operations.

If portable equipment or materials are to be placed on a lower deck, then adequately sized hatches shall be provided and conveniently located in the upper decks.

6.1.4 Access and Auxiliary Systems

The number and location of stairways and access boat landings on the structure should, as a minimum, be governed by safety considerations. Consideration should be given to providing access to the water level for sea escape and to facilitate self-rescue in man-overboard situations. Furthermore, operating requirements as well as installation, maintenance, and inspection requirements should be considered in determining the number and location of access platforms, walkways, and stairways.

6.2 Safety Considerations

6.2.1 General

The safety of life, environment, and property depends upon the ability of the structure to withstand the loads/actions for which it was designed and to survive the environmental conditions that can occur. Good practice dictates the use of certain structural additions, equipment, and operating procedures such that injuries to personnel and the risks of accidental events are minimized.

6.2.2 Accidental Events

Personnel safety and possible damage to or loss of the structure require that attention be given to accidental events as summarized in ISO 19900. With regard to methods of protecting against fires and explosions, the selection of the system depends upon the function of the platform. Procedures should conform to any applicable national or regional requirements and shall be in accordance with API 2FB. Additional guidance is given in API 2TOP.

6.3 Environmental Considerations

6.3.1 General

Information on the consideration of environmental conditions and their use is provided in ISO 19900. Further requirements and information are contained in API 2MET, as well as in this document that provides specific guidance for the US Gulf of Mexico area.

6.3.2 Selecting Design Metocean Parameters and Load/Action Factors

The reliability of a structure depends on the combination of design loads/actions and design resistances. The design loads/actions are the product of selected representative loads/actions and associated partial load/action factors. Representative environmental loads/actions are governed by the selection of appropriate metocean (meteorological and oceanographic) parameters for design. As a guide, the recurrence interval for metocean design parameters should be several times the design service life of the structure.

Unless noted otherwise, design in accordance with this document shall be based on extreme values with probabilities of exceedance of the order of 10^{-2} per annum, e.g. a return period of 100 years. Where the data are available and sufficient, the 100-year return period may apply to responses (load/action effects) of the structure instead of to metocean design parameters. Further requirements and guidance are provided in Section 9 and in API 2MET.

A robustness level analysis (see 10.5) is required for new platforms based on risk considering life-safety and consequence of failure, as noted in 6.4. Requirements depend on the exposure category (see Table 6.4-1) and platform configuration. Guidelines are also provided in Table 6.3-1 for selecting the abnormal/robustness level hydrodynamic force criteria (see 10.1.5) to be used in the required abnormal/robustness ultimate strength analysis. Information on ultimate strength analysis is provided in API 2SIM.

Table 6.3-1—Gulf of Mexico Design Criteria and Robustness Analysis

Exposure Category	Design Criteria	Robustness Level Ultimate Strength Analysis
L-1 ^a	Use the 100-year full population and associated conditions from API 2MET or site-specific data developed in accordance with the requirements of API 2MET	Use the 1000-year full population wave and associated conditions from API 2MET or site-specific data developed in accordance with the requirements of API 2MET
L-2	Use the 50-year full population and associated conditions from API 2MET or site-specific data developed in accordance with the requirements of API 2MET	Not required if L-2 exposure category platform has a robust configuration For nonrobust configurations, use the 500-year full population wave and associated conditions from API 2MET or site-specific data developed in accordance with the requirements of API 2MET
L-3	Use the 25-year full population and associated conditions from API 2MET or site-specific data developed in accordance with the requirements of API 2MET	Not required
^a Manned nonevacuated platforms are presently not applicable to the US Gulf of Mexico waters where platforms are normally evacuated ahead of hurricane events. The metocean design criteria in 6.3 have not been verified as adequate for manned nonevacuated in the US Gulf of Mexico. However, the winter storm, sudden hurricane, and earthquake criteria for the US Gulf of Mexico have been verified as adequate for the manned nonevacuated situation occurring during those events when platforms in the US Gulf of Mexico waters are not normally evacuated.		

An L-2 exposure category platform has a robust configuration if it has all of the following characteristics.

- a) The structure has four or more legs.
- b) The lower deck bottom of beam elevation is above the 1000-year return “max crest elevation” provided in API 2MET, or site-specific data are developed in accordance with the requirements of API 2MET.
- c) The piles are founded in competent soils that are not susceptible to mudslide or other type of seafloor instability.
- d) The nominal sections of any ungrouted legs have a maximum D/t ratio of 50 at the nominal sections between the joint cans. Alternately, piles are grouted to the jacket leg for the full length of the leg.
- e) The vertical framing transmitting shear forces between horizontal frames consists of X-braces, or single (leg-to-leg) diagonals, arranged such that shear between horizontal frames is carried by braces in both tension and compression. K-bracing cannot be used. See Figures 6.3-1 and 6.3-2.
- f) Horizontal members are provided between all adjacent legs at horizontal framing levels in vertical frames and these horizontal members have sufficient strength in compression to support the redistribution of loads/actions resulting from any buckling of adjacent diagonal braces. See Figure 6.3-2.
- g) The slenderness ratio $(K \cdot L/r)$ of primary diagonal bracing in vertical frames is limited to no more than 80 and $(F_y D)/(Et) \leq 0.069$.
- h) Joints for primary structural members are sized for either the tensile yield load or the compressive buckling load of the members framing into the joint, as appropriate for the ultimate behavior of the structure. This can be accomplished by increasing the 50 % minimum cord capacity requirement of 14.2.3 to 100 % for in-place design conditions.
- i) All pile-jacket shim connections are complete 360-degree welded connections with smooth curved crown shims designed to reduce stress concentrations that affect fatigue life and are designed to carry the ultimate capacity of the pile.

Any robustness level analysis shall follow the ultimate strength analysis procedures provided in API 2SIM.

Extreme metocean parameters for the Gulf of Mexico are provided in API 2MET. Specific annual conditions are also provided in API 2MET with information on how to combine different wave, wind, and currents.

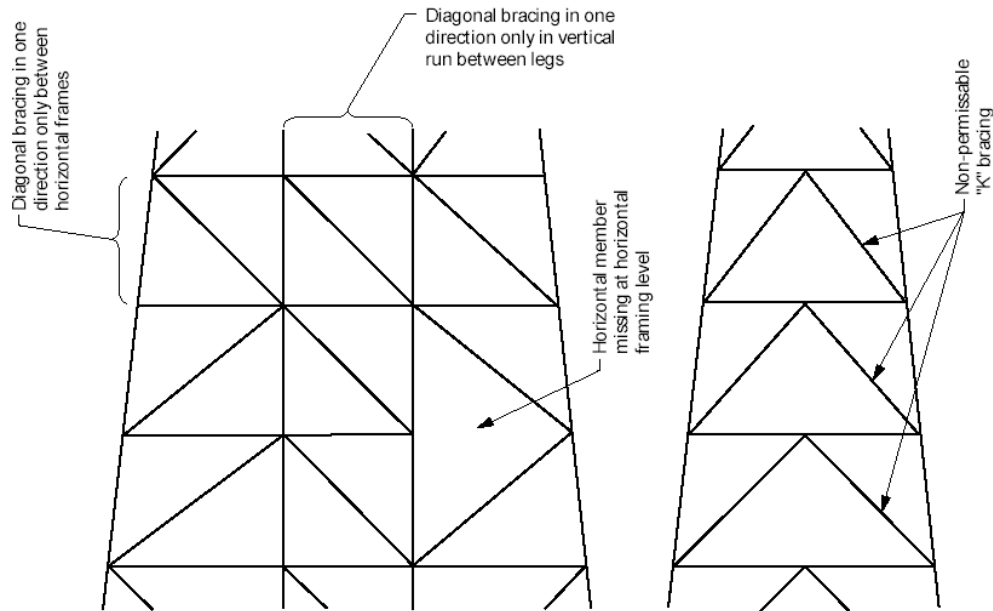


Figure 6.3-1—Vertical Framing Configurations Not Meeting Robustness Requirements

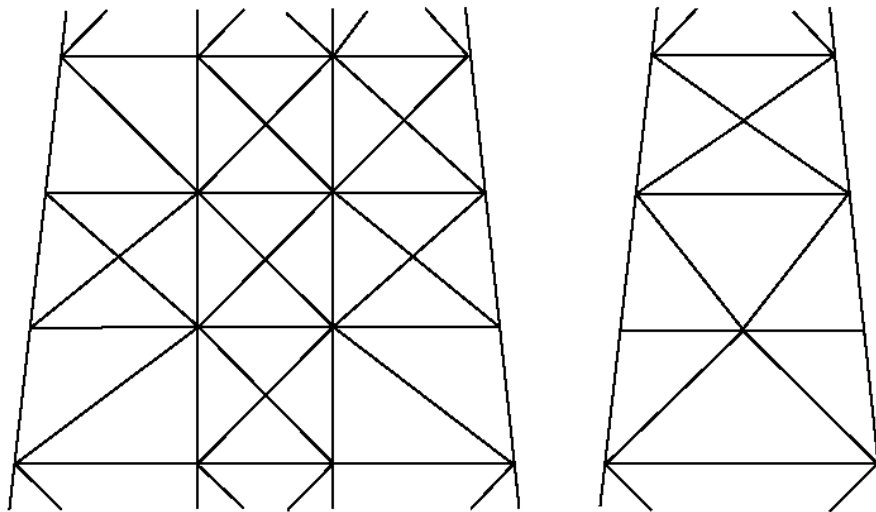


Figure 6.3-2—Vertical Framing Configurations Meeting Robustness Requirements

In addition to specifying extreme and abnormal metocean parameters, the owner shall specify the limiting environmental conditions for performing any particular operation inducing specific loads/actions, or for any particular situation of the platform.

6.4 Exposure Levels

6.4.1 General

Structures can be categorized by various levels of exposure to determine criteria that are appropriate for the intended service of the structure. This applies to the design of new structures and to the assessment of existing structures. The levels are determined by consideration of life-safety and of environmental and economic consequences.

The life-safety category addresses personnel on the platform and the likelihood of successful evacuation before a design environmental event occurs.

The consequence category considers the potential risk to life of personnel brought in to react to any incident, the potential risk of environmental damage, and the potential risk of economic losses.

6.4.2 Life-safety Categories

The category for life-safety should be selected from the following options with specific requirements.

a) S1—Manned Nonevacuated

The manned nonevacuated category refers to a platform that is continuously (or nearly continuously) occupied by persons accommodated and living thereon, and from which personnel evacuation prior to the design environmental event is either not intended or impractical.

A platform shall be categorized as S1 manned nonevacuated unless the particular requirements for S2 or S3 apply throughout the design service life of the platform.

b) S2—Manned Evacuated

The manned evacuated category refers to a platform that is normally manned except during a forecast design environmental event. For categorization purposes, a platform shall not be categorized as a manned evacuated platform unless:

- 1) reliable forecast of a design environmental event is technically and operationally feasible, and the weather between any such forecast and the occurrence of the design environmental event is not likely to inhibit an evacuation;
- 2) prior to a design environmental event, evacuation is planned;
- 3) sufficient time and resources exist to safely evacuate all personnel from the platform, and all other platforms are likely to require evacuation for the same storm.

c) S3—Unmanned

The unmanned category refers to a platform that is only manned for occasional inspection, maintenance, and modification visits. For categorization purposes, a platform shall not be categorized as unmanned unless:

- 1) visits to the platform are undertaken for specific planned inspection, maintenance, or modification operations on the platform itself;
- 2) visits are not expected to last more than 24 hr during seasons when severe weather can be expected to occur;
- 3) the evacuation criteria 1) to 3) for S2 manned evacuated platforms are also met.

A platform in this category may also be described as “not normally manned.”

It is recognized that life-safety category definitions include a degree of judgment. The owner of the structure shall determine the applicable category prior to the design of a new structure or the assessment of an existing structure and shall obtain the agreement of the regulator where applicable.

6.4.3 Consequence Categories

Factors that should be considered in determining the consequence category include:

- life-safety of personnel on or near to the platform brought in to react to any consequence of failure, but not personnel who are part of the normal complement of the platform;
- damage to the environment; and
- anticipated losses to the owner, to other installation owners, to industry, and/or to other third parties as well as to society in general.

The consequence category shall be selected from the following options with specific requirements.

a) C1—High Consequence Category

The high consequence category refers to platforms with high production rates or large processing capability and/or those platforms that have the potential for well flow of either oil or sour gas in the event of platform failure. In addition, it includes platforms where the shut-in of the oil or sour gas production is not planned, or not practical, prior to the occurrence of the design event (such as areas with high seismic activity). Platforms that support trunk oil transport lines and/or storage facilities for intermittent oil shipment are also considered to be in the high consequence category.

A platform shall be categorized as C1, high consequence, unless the particular requirements for C2 or C3 apply throughout the design service life of the platform.

b) C2—Medium Consequence Category

The medium consequence category refers to platforms where production can be shut-in during the design event. For categorization purposes, a platform shall not be categorized as medium consequence unless:

- 1) all wells that can flow on their own in the event of platform failure contain fully functional subsurface safety valves (SSSVs), manufactured and tested in accordance with applicable specifications;
- 2) oil storage is limited to process inventory and “surge” tanks for pipeline transfer;
- 3) pipelines are limited in their ability to release hydrocarbons, either by virtue of inventory and pressure regime, or by check valves or seabed safety valves.

c) C3—Low Consequence Category

The low consequence category refers to minimal platforms where production can be shut-in during the design event. These platforms may support production departing from the platform and low volume in-field pipelines. For categorization purposes, a platform shall not be categorized as low consequence unless:

- 1) all wells that can flow on their own in the event of platform failure contain fully functional SSSVs, manufactured and tested in accordance with applicable specifications;
- 2) oil storage is limited to process inventory;
- 3) pipelines are limited in their ability to release hydrocarbons, either by virtue of inventory and pressure regime, or by check valves or seabed safety valves.

It is recognized that consequence category definitions include a degree of judgment. The owner of the structure shall determine the applicable category prior to the design of a new structure or the assessment of an existing structure and shall obtain the agreement of the regulator where applicable.

6.4.4 Determination of Exposure Level

The three categories for both life-safety and consequence can, in principle, be combined into nine exposure levels. However, the level to be used for structure categorization is the more restrictive level for either life-safety or consequence.

This results in three exposure levels as illustrated in Table 6.4-1.

Table 6.4-1—Determination of Exposure Level

Life-safety Category	Consequence Category		
	C1 High Consequence	C2 Medium Consequence	C3 Low Consequence
S1 Manned nonevacuated	L-1	L-1	L-1
S2 Manned evacuated	L-1	L-2	L-2
S3 Unmanned	L-1	L-2	L-3

The exposure level applicable to a structure shall be determined by the owner prior to the design of a new structure or the assessment of an existing structure, and be agreed upon by the regulator where applicable.

Platform categorization may be revised over the design service life of the structure as a result of changes in factors affecting life-safety or consequence category.

6.5 Assessment of Existing Structures

Any assessment of existing structures to confirm that they comply with this document or are fit-for-service shall be performed in accordance with the assessment requirements of API 2SIM.

6.6 Structure Reuse

Structures may be removed and relocated for use at a new location. When this is considered, the structure should be assessed in accordance with the assessment requirements of Section 23, for the use (including exposure level) and conditions that are applicable at the new location. Any repairs or modifications that are necessary shall be in accordance with this document.

7 General Design Requirements

7.1 General

The general principles on which structural design requirements are based are documented in ISO 19900. ISO 19900 requires that structural design be performed with reference to a specified set of limit states. For each limit state, design situations shall be determined and an appropriate calculation model shall be established. ISO 19900 describes the following:

- the variables that occur in a calculation model, comprising geometrical parameters, loads/actions, and properties of materials and soils;
- the analyses to be performed;
- the design format using partial load/action and partial resistance factors.

This section outlines the overall requirements for the following:

- incorporating limit states (see 7.2);
- determining design situations (see 7.3);
- structural modeling and analysis (see 7.4);
- design of the structure (see 7.5 to 7.11).

These requirements are a minimum.

7.2 Incorporating Limit States

The limit state categories are as follows.

- a) Ultimate limit states (ULS): The design loads/actions to be used in the various ULS are specified in Sections 8, 9, and 11. The design resistances and the application of the ULS are specified in Sections 13, 14, 15, and 17.
- b) Serviceability limit states (SLS): The design shall comply with the serviceability requirements of caissons, conductors, risers, and topsides interfaces, as defined by the owner.

NOTE See also commentary in A.17.3.4 c) on estimating caisson or pile tilt for SLS checks.

- c) Fatigue limit states (FLS): The various FLS are addressed in Section 16, covering methods, loads/actions, and resistances. Additionally, FLS requirements for grouted connections, mechanical connections, and clamps are given in Section 15.
- d) Accidental limit states (ALS): ALS are addressed in Section 10, in respect of loads/actions and modifications to both partial load/action factors and partial resistance factors.

7.3 Determining Design Situations

Determination of situations for which structures are to be designed is the responsibility of the owner in accordance with the requirements of a regulator, where one exists. Aspects to be considered in determining design situations should include the following:

- a) service requirements for the intended function of the structure;
- b) expected service life for each function;
- c) method and duration of construction activities;
- d) expected method of removal of the structure and, where applicable, any intended relocation;
- e) hazards (accidental and abnormal events) to which the structure can be exposed during its service life;
- f) potential consequences of partial or complete structural failure;
- g) the nature and severity of environmental conditions (meteorological, oceanographic, and active geological processes) to be expected during its construction and service life.

Usually, three sets of environmental conditions should be considered: one associated with operating situations (normal conditions), one associated with extreme conditions, and one associated with abnormal conditions.

7.4 Structural Modeling and Analysis

Internal forces in components shall be determined by an analysis method that is suitable for statically indeterminate structures and that accounts for bending as well as for axial load/action effects. In general, a linear elastic model of the structure is sufficient; however, the nonlinear behavior of structure-foundation interaction shall be addressed. For piled foundations, the nonlinear behavior of the axial and lateral pile-soil support shall be explicitly modeled to ensure load-deflection compatibility between the structure and pile-soil system. For a pile analysis, the effects of geometrical and material nonlinearities shall also be assessed within the structure-pile-soil system. In no case shall additional or adjusted partial load/action factors be used as a substitute for a rational analysis to determine internal forces. Other aspects of structural modeling and analysis are discussed in Sections 8 to 17 and API 2SIM.

7.5 Design for Pre-service and Removal Situations

The structure should be designed to resist loads/actions occurring during fabrication, loadout, transportation, installation, and removal, as specified in Section 8. For structures that are intended to be relocated to new sites, loads/actions resulting from removal, on-loading, transportation, upgrading, and re-installation shall also be considered in the design, in addition to the above construction loads/actions.

Methods for fabrication, assembly, loadout, transportation, installation, and removal of the structure should be considered during design in order to identify potential difficulties and limitations for available equipment. Design situations for partially fabricated components of the structure and for the assembled structure as a whole should be established. Fabrication and assembly tolerances assumed in design should be compatible with those specified in the construction specifications and achievable during construction.

7.6 Design for the In-place Situation

The structure should be designed to resist permanent loads/actions, variable loads/actions, environmental loads/actions, repetitive loads/actions, and accidental loads/actions occurring during its service life, as well as rational combinations thereof, to obtain the most onerous conditions for all structural components. The representative values of these loads/actions are given in Sections 9 to 11. Clarification on the selection of representative values and return periods is provided in A.7.6.

7.7 Determination of Resistances

7.7.1 General

The resistances of components in a fixed steel offshore structure shall, wherever possible, be derived from the formulae given in Sections 13 to 15. These formulae provide representative resistances to which the partial resistance factors are applied.

Where the resistance of a component is not adequately described in Sections 13 to 15, the resistances shall be determined from a rational study of the behavior of the component. Both physical testing and computer simulation may be used to derive representative resistances, but in both cases the results shall be compared to any other available test data.

7.7.2 Physical Testing to Derive Resistances

Physical tests may be used to derive resistances for specific components. Care shall be taken to account for the differences between the test arrangements and the arrangement of the component within the structure. Guidance on these differences is given in A.7.7.2.

Comparisons shall be made with other available test data to give an indication of the variability of the component behavior.

7.7.3 Resistances Derived from Computer Simulations Validated by Physical Testing

Computer simulations can provide an indication of resistance and can be particularly valuable for investigating the effects of changes of various parameters. Computer simulations shall be validated against physical test data wherever possible. If test data are not available, the requirements of 7.7.5 apply.

7.7.4 Resistances Derived from Computer Simulations Validated Against Design Formulae

As an alternative to validation against physical test data, computer simulations may be validated against the formulae given in Sections 13 to 15. It should be noted that Sections 13 to 15 give representative resistances and therefore take account of structural behavior variability.

7.7.5 Resistances Derived from Unvalidated Computer Simulations

Unvalidated computer simulations may only be used when no physical test data or design formulae are available. The similarity of available data and the component being simulated is a matter of judgment, and the analyst shall take account of statistical variability of material and component behavior, and limitations and errors that can be introduced in computer simulations.

7.8 Strength and Stability Checks

7.8.1 Load/Action and Resistance Factors

A partial load/action factor shall be applied to each of the external loads/actions in the combinations given in Sections 8 to 11. The combination of factored representative loads/actions results in design values of internal forces. These internal forces are designated by the symbol S in Sections 8 to 11.

A partial resistance factor is applied to the strength of each member, joint, and foundation component to determine its design resistance.

7.8.2 Strength and Stability Equations

Each component shall be proportioned to resist the internal forces, S . The appropriate strength and stability equations are given in Sections 13, 14, 15, and 17. These equations comprise the following:

- the formulae for the representative strength of the component;
- the partial resistance factors;
- the internal force, S , from the design loads/actions specified in Sections 8 to 11.

7.8.3 Unfactored Loads/Actions

Where loads/actions are time-varying and nonsynchronous, the most onerous values of load/action effects cannot generally be determined by simply adding factored individual loads/actions. In some construction and installation situations, the load/action effects are best computed from unfactored loads/actions and by investigating the sensitivity of the load/action effects with realistic variations of the governing parameters. The design values of the internal forces, S , may then be obtained by factoring the load/action effects by the partial load/action factors given in Section 8.

7.9 Robustness

A structure shall incorporate robustness as noted in 6.3 and through assessment of the effects of all hazards and their probabilities of occurrence, to ensure that consequent damage is not disproportionate to the cause. Damage from an event with a reasonable likelihood of occurrence shall not lead to complete loss of integrity of the structure. In such cases, the structural integrity in the damaged state shall be sufficient to allow a process system close down and a safe evacuation (see Section 10).

Robustness is achieved by either of the following:

- a) designing the structure such that the failure of any single load-bearing component carrying its normal design loads/actions will not cause the collapse of the structure or any significant part thereof, or
- b) ensuring (by design or by protective measures) that no critical component exposed to hazard can be made ineffective, or
- c) a combination of a) and b), above.

7.10 Reserve Strength

7.10.1 New Structures

All new structures complying with this document are intended to have a similar margin of system reserve strength for the same exposure level (L-level). The system reserve strength gives the structure a margin to withstand load/action effects that exceed the design load/action effects and is evaluated as the reserve strength ratio (RSR), R_{RS} , expressed in Equation (7.10-1).

$$R_{RS} = F_{\text{collapse}}/F_{100} \quad (7.10-1)$$

where

F_{collapse} is the unfactored global environmental load/action that, when co-existing unfactored permanent and variable loads/actions are added, causes collapse of the structure;

F_{100} is the unfactored 100-year global environmental load/action calculated in accordance with Section 9.

This intent may be achieved in either of two ways, as described in 9.10. The partial load/action factors for environmental loads/actions shall be determined based on the guidelines given in Section 9.

When using this document for structures that provide less inherent system reserve strength (smaller RSR) for the environmental loads/actions than a typical space frame type structure, the partial load/action factor for environmental loads/actions shall be increased to yield the same RSR as for a typical space frame type structure.

7.10.2 Existing Structures

During its service life, the reserve strength of a structure can vary (increase or decrease) due to a variety of causes. API 2SIM provides requirements for the assessment of existing structures, including the use of system strength approaches.

7.11 Indirect Loads/Actions

Consideration should be given to internal forces and stresses induced by indirect loads/actions, including imposed or constrained deformations, temperature changes, relaxation, and prestressing.

7.12 Structural Reliability Analysis

A description of structural reliability analysis (SRA) can be found in A.7.12.

8 Loads/Actions for Pre-service Situations

8.1 General

Users of this document should be aware of API 2MOP. Where the provisions in API 2MOP overlap with this document, API 2MOP may be used in preference unless noted otherwise.

8.1.1 Coverage

This section presents requirements for pre-service situations. The requirements apply both to the structure and to any temporary components such as buoyancy aids, mooring supports, or sea-fastening attached to the structure for pre-service situations. The primary objective is to ensure that a structure begins its service life at an offshore location with its required strength and structural integrity intact. The section covers the following topics:

- a) the type of loads/actions to be included;
- b) procedures for determining representative values for the loads/actions;
- c) the partial load/action factors to be applied;
- d) methods for determining the internal forces due to the effect of factored loads/actions or combinations of factored loads/actions.

8.1.2 Situations

Pre-service design situations are transient in nature and include:

- a) fabrication (including assembly during fabrication);
- b) loadout (including float-out);
- c) transportation;
- d) installation (including launching and uprighting).

Further requirements concerning pre-service situations are given in Sections 19 to 22.

The requirements for b), c), and d) are primarily addressed in API 2MOP; however, some aspects are also addressed within this document.

Loads/actions on (parts of) the structure and internal forces in components of the structure should be calculated for each phase of pre-service in accordance with the principles given in 7.3 to 7.5.

Structural modeling and analysis requirements for the pre-service situations are given in 12.4.3, and operational requirements for loadout, transportation, and installation are given in Section 22.

8.2 General Requirements

8.2.1 Design Situations

Design situations during pre-service situations are generally governed by permanent and variable loads/actions. These loads/actions are predominantly static in nature, in particular due to weights of structural components, appurtenances, and equipment. When dynamic effects can be important, these should be considered. When appropriate, environmental and accidental loads/actions should also be given due attention, following the requirements of Sections 9 and 10.

8.2.2 Weight Control

The weight and center of gravity of each structure or part of a structure should be evaluated during design and during all pre-service situations. This applies in particular prior to any lifting being performed. Where applicable, allowances for weight variations based on weight reports, the status of design and/or fabrication, known outstanding items and/or operational modifications to the structures should be applied.

During pre-service situations the behavior of the structure and the magnitude and distribution of internal forces in the structure can be very sensitive to the position of the center of gravity. In such cases, due allowances shall be made for uncertainties and potential changes in the position of the center of gravity.

NOTE Guidance on weight control is given in ISO 19901-5 [5].

8.2.3 Dynamic Effects

If operations during pre-service situations involve significant motion of large masses, the dynamic loads/actions involved should be considered.

The load/action effects should be adjusted to arrive at adequate representative internal forces for the design of the components affected. These representative internal forces can be derived from an appropriate analysis to determine the static and dynamic effects due to both the external loads/actions and the motion, or they can be based on relevant previous experience.

For dynamic effects during lifting, see the requirements given in 8.3.2.

8.2.4 Internal Forces

8.2.4.1 Internal Forces due to Factored Loads/Actions

Each member, joint, and other relevant component shall be checked for strength using the internal force (load/action effect) (S) resulting from the design load/action (F_d).

The design load/action, F_d , is due to the following:

- factored permanent loads/actions (G);
- factored variable loads/actions (Q);
- factored quasi-static transient loads/actions (T).

Equation (8.2-1) defines F_d for situations not involving lift (additional factors are applied for lift conditions; see 8.3.6):

$$F_d = \gamma_{f,GT} G_T + \gamma_{f,QT} Q_T + \gamma_{f,T} T \quad (8.2-1)$$

where

G_T is the load/action imposed either by the weight of the structure in air during the transient situation being considered, including any permanent equipment or other objects and any piles or conductors installed on the structure;

Q_T is the load/action imposed by the weight of the temporary equipment or other objects, including any rigging installed or carried by the structure, during the transient situation being considered;

T represents the loads/actions from the transient situation being considered, including:

- 1) when appropriate, environmental loads/actions;
- 2) when appropriate, a suitable representation of dynamic effects (see A.8.1 and 8.2.4.2);
- 3) for lifting, the effects of fabrication tolerances and variances in sling length as detailed in 8.3.3 and for a multi-crane lift as detailed in 8.3.4;

$\gamma_{f,GT}$, $\gamma_{f,QT}$ and $\gamma_{f,T}$ are the partial load/action factors.

The three design situations in Table 8.2-1 shall all be assessed.

Table 8.2-1—Partial Load/Action Factors for Calculating Internal Forces

Situation	Partial Load/Action Factor		
	$\gamma_{f,GT}$	$\gamma_{f,QT}$	$\gamma_{f,T}$
1	1.3	1.3	1.0
2	1.1	1.1	1.35
3	0.9	0.9	1.35
NOTE 1 Situation 1 governs for components in which permanent and variable load/action effects are dominant.			
NOTE 2 Situation 2 governs for components in which transient load/action effects are dominant and in which the permanent and variable loads/actions increase the magnitudes of the internal forces.			
NOTE 3 Situation 3 governs for components in which transient load/action effects are dominant, but in which the permanent and variable loads/actions decrease the magnitudes of the internal forces.			

Additional requirements and factors for lifting situations are given in 8.3.

8.2.4.2 Internal Forces due to Unfactored Loads/Actions

In some situations, it is not possible or appropriate to apply load/action factors to each of the individual actions (see A.8.1 and 7.8.3). An example of such a situation is lifting in which substantial dynamic or nonlinear effects can occur. In such cases, each member, joint, and other relevant component shall be checked for strength using the internal force (load/action effect) (S) determined from the following equations:

$$F_{un} = G_T + Q_T + T$$

$$S = \gamma_{f,Sun} S_{un} \quad (8.2-2)$$

where

F_{un} is the total load/action due to the unfactored loads/actions G_T , Q_T , and T defined in 8.2.4.1;

S_{un} is the internal force resulting from F_{un} ;

$\gamma_{f,Sun}$ is the partial factor to be applied to S_{un} , usually 1.3.

Additional requirements and factors for lifting situations are given in 8.3.

If both this method and the one according to 8.2.4.1 can be applied, that giving the more onerous result shall be used.

8.3 Loads/Actions Associated with Lifting

8.3.1 General

Lifting imposes static and dynamic loads/actions on the structure during pre-service situations. The magnitude of these loads/actions shall be based on the weight of the (part of the) structure to be lifted and shall be determined considering static and dynamic equilibrium during the lift. The internal forces in the structure and in the lifting slings and rigging equipment (e.g. spreader bars) should be derived from a rational analysis of the lifting arrangement. General guidance on lifting is given in A.8.3.1.

Dynamic amplification (see 8.3.2), as well as the effects of fabrication and sling length tolerances (see 8.3.3), and weight and center of gravity location inaccuracies (see 8.2.2), should be included in such rational analysis.

Additional factors and considerations apply to members adjacent to lifting attachments (see 8.3.5), and to the lifting attachments themselves (see 8.3.6 and 8.3.7).

Alternative lift criteria can be appropriate, for example, when the characteristics of, and interactions between, the structure and crane are fully understood. Alternative criteria may be used, subject to demonstration of a margin of safety, consistent with the requirements of this document, and to the agreement of the owners of the structure, the contractors, and any appropriate regulator or warranty surveyor.

8.3.2 Dynamic Effects

Dynamic effects of onshore crane(s) taking up load(s)/action(s) and movements of either the crane or the lifted structure shall be accounted for by the use of a dynamic amplification factor (DAF), k_{DAF} , as specified in Table 8.3-1 (extracted from Table 15 in API 2MOP, First Edition). For offshore lifts, reference is made to 8.7.1.

Table 8.3-1—Dynamic Amplification Factor for a Single Onshore Crane

Mass of Lifted Object * tonnes	Gross Weight W kN	k_{DAF} in Air	
		Moving	Static
≤ 100	$W \leq 1000$	1.15	1.00
From 100 to 1000	$1000 < W \leq 10,000$	1.10	1.00
From 1000 to 2500	$10,000 < W \leq 25,000$	1.05	1.00
Greater than 2500	$25,000 < W$	1.05	1.00
* This column is included to facilitate comparison with weight reporting.			

NOTE The lifting DAFs are in accordance with certification society rules such as Lloyds ^[58] and are addressed in API 2MOP.

8.3.3 Effect of Tolerances

Both fabrication misalignments in the structure and sling length variability affect the distribution of forces in the lifting system. However, the effects of fabrication misalignments may be ignored provided they fall within the tolerances specified in Annex E (see 12.3.3.1); 12.3.3.1 also provides for modeling requirements in the cases where fabrication misalignments exceed those specified in Annex E.

When sling length variability does not exceed the greater of 0.25 % of the nominal sling length or 40 mm (sling length tolerance), the effect of sling length variability may be ignored in the analysis of statically determinate lifts, i.e. lifts involving three or fewer slings, although such analyses do not normally account for changes in, e.g. centers of gravity.

However, the effect of sling length variability shall be accounted for when executing statically indeterminate lifts. The requirements and partial load/action factors specified in 8.2.4, 8.3.1, and 8.3.2 apply to situations where the variability in sling length does not exceed the sling length tolerance. Sling forces or sling lengths shall be varied in a stiffness analysis of the structure and lifting system so as to determine the redistribution of forces in both slings and structural members. A.8.3.3 provides specific guidance on how this may be conducted. The resulting sling forces shall be increased by a factor of not less than 1.25 (1.15 for floating spreader beams) to determine the required safe working load of the sling in 8.3.8; the local effects on the structure shall be checked using the local factor given in 8.3.5.

Where sling length variability exceeds the sling length tolerance, the guidance provided in A.8.3.3 may be followed while the resulting sling forces shall be increased by a factor of not less than 1.40. A lower factor may be adopted if both sling stiffnesses and sling lengths have been measured and the analysis performed using these measured values: in no case shall a factor less than 1.25 be used.

8.3.4 Multi-crane Lift

For an onshore lift using multiple cranes, an additional rigging factor should be considered in lieu of assessment of the effects of differential soil settlement under the cranes. Unless special and reliable investigations are performed to substantiate lower values, a minimum value of such a rigging factor of $\gamma_{f,dl} = 1.10$ shall be applied. For single crane lifts, $\gamma_{f,dl} = 1.00$. (In API 2MOP, this factor is referred to as “dual lift rigging factor.”)

Where a multi-crane lift is used, the most onerous orientations of the structure shall be assessed and the design of lifting attachments shall allow for any necessary rotation of the lifting arrangements.

NOTE See API 2MOP.

8.3.5 Local Factor

Due to local inaccuracies that often occur in the analysis of internal force distributions around the lifting points, a further factor ($\gamma_{f,lf}$), for both statically determinate and statically indeterminate lifts, shall be applied to account for the importance of the structural components involved. This local factor includes the effects of tolerances (see 8.3.3) and shall be applied in addition to the DAF (k_{DAF}) in accordance with 8.3.2 and the rigging factor ($\gamma_{f,dl}$) in accordance with 8.3.4.

- For lifting attachments (padeyes, trunnions, padears), spreader beams, and internal members (including both end connections) framing into the joint where the lifting attachment is located and transmitting lift forces, the local factor $\gamma_{f,lf}$ shall be 1.15.
- For other structural members, $\gamma_{f,lf} = 1.00$.

8.3.6 Member and Joint Strengths

The DAF (k_{DAF}), the multi-crane lift factor ($\gamma_{f,dl}$), and the local factor ($\gamma_{f,lf}$) shall be applied to either the factored loads/actions to determine the design load/action (F_d) as in Equation (8.2-1), giving Equation (8.3-1), or to unfactored loads/actions to give the internal forces (S) as in Equation (8.2-2), giving Equation (8.3-2):

$$F_d = k_{DAF} \gamma_{f,dl} \gamma_{f,lf} (\gamma_{f,GT} G_T + \gamma_{f,QT} Q_T + \gamma_{f,T} T) \quad (8.3-1)$$

$$S = k_{DAF} \gamma_{f,dl} \gamma_{f,lf} \gamma_{f,Sun} S_{un} \quad (8.3-2)$$

where

$\gamma_{f,GT}$, G_T , $\gamma_{f,QT}$, Q_T , $\gamma_{f,T}$, T , $\gamma_{f,Sun}$, S_{un} are as defined in 8.2.4 and incorporate the effects of sling length variability as discussed in 8.3.3;

k_{DAF} is as specified in 8.3.2;

$\gamma_{f,dl}$ is as specified in 8.3.4;

$\gamma_{f,lf}$ is as specified in 8.3.5.

8.3.7 Lifting Attachments

Lifting attachments can be of various forms, including the following.

- Padeyes— where a shackle pin passes through a hole in a padeye plate attached to or built into the structure, the sling being connected to the shackle. The padeye plate is normally reinforced to increase strength and avoid undue slack between the shackle and the padeye, thereby reducing eccentricity between the padeye plate and the sling.
- Trunnions— where the sling, or an eye of a sling, passes round a short tubular that transfers the forces into the structure, and that allows rotation of the sling around the axis of the trunnion. Trunnions can be arranged to carry either end of a sling on either side of a central plate.
- Padears— which are similar to trunnions, but in which rotation of the sling is not intended.

Padeye plates should be oriented in such a direction that the possibility for out-of-plane loading of the padeye plate and shackle is minimized.

To account for any side loading on lifting attachments, lifting attachments and the connections to the supporting structural members shall be designed for a lateral force of 5 % of the sling force, in addition to the calculated horizontal and vertical components of the sling force (including DAF, rigging factor, local factor, and partial load/action factors) for the equilibrium lifting condition. This lateral force acts simultaneously with the static sling force and shall be applied perpendicular to the lifting attachment at the center of the pinhole or tubular. However, where sling is out-of-plane with the padeye, the static sling force shall be applied at the bow of the shackle. Where a spreader bar is directly connected to the padeyes, a lateral force of 8 % shall be used.

Where two slings are connected to one padeye, or where a sling is doubled over a trunnion, the padeye or trunnion should be designed for a 45:55 % split of the lift point force between the two slings.

NOTE See API 2MOP.

8.3.8 Slings, Shackles, and Fittings

The calculated sling force is the maximum force on each sling, as calculated in 8.3.1 to 8.3.4, by taking into account all loads/actions, the equilibrium position of the lift, any dynamic amplification, the effect of tolerances, and multi-crane lift considerations. For normal offshore conditions, slings should have a resistance factor of 4.0

applied to the manufacturer's rated minimum breaking strength or a resistance factor of 1.0 applied to the sling safe working load compared to the worst calculated sling force. The total resistance factor includes all necessary partial resistance factors. The total resistance factor should be increased when unusually severe conditions are anticipated. Conversely, the total resistance factor may be reduced to a minimum of 3.0 for carefully controlled conditions.

Where two slings are connected to one padeye, or where a sling is doubled over a trunnion, the slings should be assumed to carry the lift point force in a 45:55 % split of the lift point force between the two slings.

Shackles and fittings should be such that the manufacturer's rated working load is greater than or equal to the calculated sling force, provided the manufacturer's specifications include a minimum resistance factor of 3.0 on the minimum breaking strength.

NOTE See API 2MOP.

8.4 Loads/Actions Associated with Fabrication

In addition to the associated permanent and variable loads/actions, the effects of wind-induced vortex shedding vibrations on long slender members during fabrication should be considered.

8.5 Loads/Actions Associated with Loadout

8.5.1 Direct Lift

Loads/actions on a structure that is lifted onto the transportation barge shall be evaluated in accordance with 8.3. If the lifting arrangement is the same as that used to offload the structure from the transportation barge at sea, it will normally suffice to check the latter load case(s) only (see 8.7.1).

NOTE See API 2MOP.

8.5.2 Horizontal Movement onto Barge

Structures skidded onto transportation barges are subject to loads/actions resulting from horizontal and/or vertical movement of the barge due to tidal fluctuations, nearby marine traffic, environmental conditions, and/or changes in draught, as well as to loads/actions derived from the location, slope, and/or settlement of supports onshore and the flexibility of the barge, at all stages of the skidding operation.

The effect of loads/actions on the structure during loadout should be investigated, taking account of the tolerances on the levels of support on land and on the barge. The effect of pulling/pushing, friction forces, and twisting of the structure due to differential horizontal jacking or winching during loadout should be considered.

In cases of loadout with trailers, the effect of wind and hydraulic pressure variation plus the distributed loads/actions imposed by hydraulic trailers and the effect of dynamic loads/actions due to the braking of the trailers should be considered. This can require consideration of the methods of controlling the elevation and alignment of trailers. Effects of inertial loads during trailer movement shall be addressed for unusually tall and slender structures.

Because movement is normally slow, impact does not need to be considered.

NOTE See API 2MOP.

8.5.3 Self-floating Structures

Self-floating structures are generally not transported on a barge but are towed while floating to the offshore location.

Self-floating structures skidded directly into the water at the fabrication yard should be analyzed to determine the loads/actions on the structures as they move down the slipways and into the floating position. Consideration should be given to local environmental conditions and dynamically induced forces.

This subsection does not apply to self-floating structures built in a dry dock and floated by flooding the dock.

8.6 Loads/Actions Associated with Transportation

8.6.1 General

Loads/actions on a structure or structural part during transportation should be considered in the design of the structure, whether it is transported on barges or is self-floating. Such loads/actions result from the way in which the structure is supported, either by barge or by buoyancy, and from the response of the towing arrangement (structure, tug boats, tow lines, and, where applicable, barge and/or additional buoyancy) to environmental conditions encountered en route to the site.

8.6.2 Environmental Conditions

For the selection of environmental conditions to be used in determining the motions of the towing arrangement and the resulting loads/actions on the structure, the sea-fastenings, and the barge system, the following should be considered:

- a) previous experience along the tow route;
- b) exposure time and reliability of predicted weather windows;
- c) accessibility of safe havens;
- d) seasonal weather systems;
- e) appropriateness of the recurrence interval used in determining design wind, wave, and current conditions in relation to the characteristics of the tow.

8.6.3 Determination of Loads/Actions

The motions of the towing arrangement due to the environmental conditions determined from 8.6.2 should be analyzed using suitable methods. Relevant requirements and guidance are provided in API 2MOP. Sea states with peak periods corresponding to the natural periods of the loaded barge, as well as peak periods corresponding to wave lengths that are critical in relation to the dimensions of the barge, shall be assessed, as these can represent governing conditions. In combining responses, the analysis shall correctly account for relative phasing between motion components (heave, pitch, roll, etc.).

Head, beam, and quartering winds and seas should be considered to determine maximum responses due to the environmental loads/actions on the overall system. In cases of large barge-transported structures, the stiffnesses of both the structure and the barge shall be included in the structural analysis.

Loads/actions on the structure, sea-fastenings, and the barge system should be derived from appropriate combinations of permanent, environmental, and inertial loads/actions resulting from the state of motion, taking due account of the position, orientation, and possible differential deformations of parts of the overall system.

8.6.4 Other Considerations

Large structures overhanging the barge are usually subjected to partial submersion during tow. Submerged members should be investigated for loads/actions due to slamming and buoyancy. These loads/actions should be included in the local and, where relevant, overall design of the structure and the sea-fastenings. Large buoyant overhanging members can also affect motions, and this effect should also be considered. The effects of vortex-induced vibrations (VIVs) due to wind on long slender members also should be investigated.

The effects of repetitive loads/actions during tow can become significant to the fatigue life of certain members, joints, or other components, especially for long transoceanic tows. These effects should be investigated using methods and acceptance criteria agreed upon by the owner and the design or transportation contractor.

8.7 Loads/Actions Associated with Installation

8.7.1 Lifted Structures

Loads/actions on a structure that is lifted off the transportation barge at sea shall be evaluated in accordance with Section 18 of API 2MOP, First Edition.

8.7.2 Launched Structures

Barge-launched structures should be analyzed to determine the loads/actions on the structure throughout the launch. Evaluation shall be given to hydrostatic pressure (see 8.7.4), wind and current loads/actions, and the development of dynamically induced loads/actions resulting from the launch.

Horizontal loads/actions required to initiate movement of the structure should also be evaluated. Expected loads/actions on both the structure and the barge during launching should be considered.

8.7.3 Crane-assisted Uprighting of Structures

Where lifting equipment is used to assist in the uprighting of a structure as part of the installation, the structure should be analyzed for the permanent and variable loads/actions together with the dynamically induced loads/actions during uprighting. The requirements of 8.3 apply to this situation.

NOTE See API 2MOP.

8.7.4 Submergence Pressures

Submerged nonflooded, or partially flooded, components shall be designed to resist pressure-induced stresses during launching and uprighting, in addition to other internal forces acting on the components at the same time.

8.7.5 Member Flooding

Consideration should be given to the effects of unexpected flooding of members or failure of members to flood when intended, either by allowing for such occurrences or by making adequate provisions to prevent such occurrences. Any one failure of the flooding control should be considered, including common-mode failures such as failure of a valve at a manifold.

8.7.6 Loads/Actions on the Foundation during Installation

8.7.6.1 General

The environmental conditions used for installation shall be consistent with the limiting environmental conditions in the installation procedure. A more severe environmental condition should also be checked, corresponding to the last stage of installation, when the structure is upright and placed on the sea floor but not yet permanently anchored.

For structures that are to be piled, the loads/actions to be expected from contact with the sea floor during installation of the structure should be assessed to give adequate assurance that the structure will remain at the planned elevation, location, and attitude until piles can be installed.

The design shall ensure that the following are applied:

- a) the footings or mudmats have adequate capacity against sliding and bearing failure, and that pin-piles, if any, have adequate strength to avoid being damaged;
- b) the footings, mudmats, or other bearing components and structural members supporting these, have adequate strength to avoid being damaged;
- c) the safety margins against overturning of the structure are adequate, with the recommendation that the structure be checked in a piled condition but without the permanent load/action of the topsides if placement of the topsides does not follow shortly after structure installation.

8.7.6.2 Determination of Loads/Actions

Both vertical and horizontal loads/actions should be determined, taking into account changes in configuration and exposure, construction equipment, and additional ballast that can be required for stability during storms.

8.7.6.3 Structure On-bottom Weight

The soil stresses under foundation components at the base of the structure (mudmats, footings, or other bearing components) prior to the installation of the permanent pile foundation can be critical. The distribution of the structure's weight on the sea floor should be considered for each combination of pile sections that are planned to be supported from the structure. Bearing capacity analyses shall be performed in accordance with API 2GEO.

For soils that increase in strength with depth, particularly soft clays and loose sands, the bearing capacity analysis should account for shape effects and for the presence of any holes in the mudmats. This is because any reduction in mudmat area can result in a potential shallow failure surface and hence a reduced bearing capacity. Also, if there are loose sands at the sea floor, vibrations produced by pile driving operations can contribute to additional settlement.

The increase in soil stresses resulting from waves of the maximum height anticipated during the installation period should be considered. The bearing capacity analysis should take account of the combined effect of vertical and horizontal load/action effects (forces and moments) on foundation components at the base of the structure. The more heavily loaded components can experience a reduction of soil stiffness which can allow forces to be transferred to other components. In the analysis, consideration may be given to suction developing under components that are subjected to uplift, provided that the components have been designed with an adequate skirt length and that measures have been taken to prevent ingress of seawater into the skirt compartments, e.g. by the provision of valves.

The structure and foundation components at its base should be checked for the applicable loads/actions specified in this section. The environmental load/action (E_e) in Equation (9.10-1) should be replaced by the load/action determined from waves of the maximum height anticipated during the installation period.

8.8 Loads/Actions Associated with Removal

The owner may specify particular removal requirements to be considered during design.

9 Loads/Actions for In-place Situations

9.1 General

This section presents loads/actions and combinations of loads/actions and their effects that shall be assessed for the in-place situation. It covers the following topics:

- the type of loads/actions to be included;
- procedures for determining representative values for the loads/actions;
- the partial load/action factors to be applied;
- methods for determining the internal forces due to the effect of factored loads/actions or combinations of factored loads/actions.

Design situations to be considered for the structure in-place can be distinguished between persistent situations and accidental situations. In this section, only the persistent situations are addressed; accidental situations are addressed in Section 10.

9.2 Permanent Loads/Actions (G) and Variable Loads/Actions (Q)

9.2.1 Permanent Load/Action 1, G_1

G_1 is the load/action imposed on the structure by the self weight of the structure with associated equipment and other objects. G_1 includes the following:

- weight of the structure in air, including, where appropriate, the weight of piles, grout, and solid ballast;
- weight of equipment and other objects permanently mounted on the structure that do not change with the mode of operation;
- hydrostatic loads/actions acting on the structure below the waterline, including internal and external pressure, and resulting buoyancy;
- the weight of water enclosed in the structure, whether permanently installed or temporary ballast; see 9.2.5 for unintentional flooding.

The representative value of G_1 is the value computed from nominal dimensions and mean values of densities.

9.2.2 Permanent Load/Action 2, G_2

G_2 is the load/action imposed on the structure by the self weight of equipment and other objects that remain constant for long periods of time, but which can change from one mode of operation to another or during a mode of operation. G_2 includes the following:

- weight of drilling and production equipment that can be added to or removed from the structure;
- weight of living quarters, heliport, and other life-support equipment, diving equipment, and utilities equipment, which can be added to or removed from the structure.

The representative value of G_2 is the estimated lift weight of the object plus any field installed appurtenances.

9.2.3 Variable Load/Action 1, Q_1

Q_1 is the load/action imposed on the structure by the weight of consumable supplies and fluids in pipes, tanks and stores, the weight of transportable vessels and containers used for delivering supplies, and the weight of personnel and their personal effects. Where appropriate, the weight of marine fouling and ice should be included in Q_1 .

The weight of scaffolding or other temporary access systems used during operations and maintenance of the platform should also be included in Q_1 . The representative value of Q_1 is computed from the nominal weight of the heaviest material and the largest personnel capacity under the mode of operation considered.

9.2.4 Variable Load/Action 2, Q_2

Q_2 is the short duration load/action imposed on the structure from operations, such as lifting of drill string, lifting by cranes, machine operations, vessel mooring, and helicopters. The additional weight of liquids used for testing of vessels and pipes is also included in Q_2 .

The representative value of Q_2 is computed from the rated maximum capacity of the equipment involved and includes dynamic and impact effects.

9.2.5 Unintentional Flooding

All members that are intended to be internally dry and that are located under the water line shall be checked for strength for the dry as well as for the flooded state.

All flooded members shall also be checked for strength for the dry as well as the flooded state, unless positive means of ensuring full flooding is provided.

In addition, when computing the aggregate weight of the submerged portion of the structure, 10 % of the volume of unflooded members shall be assumed to be flooded if this creates a more severe load case. The weight is to be included as part of G_1 .

9.2.6 Position and Range of Permanent and Variable Loads/Actions

Variations in weights and locations of movable equipment should be considered in order to determine the maximum internal force, S (load/action effect), in each component. Maximum and minimum values of G_2 , Q_1 , and Q_2 for each mode of operation should be considered, in order to ensure that the most onerous condition for each structural component is determined.

9.2.7 Carry Down Factors

Where permanent or variable loads/actions can be moved, or their magnitudes can vary, it is conservative to apply the maximum values over the whole area of their influence. In such cases, a reduction in the resulting loads/actions may be used if operating practices provide adequate safeguards to prevent load/action effects from exceeding the reduced values.

9.2.8 Representation of Loads/Actions from Topsides

Distributed loads/actions over an area may be used to represent permanent and variable loads/actions on platform decks. Partial load/action factors for distributed loads/actions shall be based on the proportion of permanent and variable loads/actions represented by the distributed load/action.

Concentrated loads/actions on the deck, for example from separators, may be included within the distributed loads/actions on the topsides for the design of the structure. If a combined analysis model of the structure and topsides is used, concentrated loads/actions should be represented as such with the appropriate partial load/action factor based on the proportion of permanent and variable loads/actions represented by the concentrated load/action.

9.2.9 Weight Control

During design for in-place situations:

- a) the weight of the structure or part of the structure shall be evaluated using a rational weight-estimating procedure, and a factor for weight growth based on a weight report should be applied;
- b) the center of gravity of the structure or part of the structure shall be evaluated using a rational procedure; allowances shall be made for uncertainties and potential changes in the center of gravity position.

NOTE Guidance on weight control is given in ISO 19901-5 [5].

9.3 Extreme Environmental Load/Action due to Wind, Waves, and Current

9.3.1 General

Wind, wave, and current conditions for calculating extreme environmental loads/actions shall be selected in accordance with API 2MET.

Environmental loads/actions, notably those caused by waves, vary with time. Structural responses (load/action effects) will therefore also vary with time and will be subject to some degree of dynamic amplification. However, in many cases, the dynamic amplification of the global structural responses is negligibly small. The environmental loads/actions can then be treated as quasi-static loads/actions (see 9.4). Where dynamic effects need to be taken into account but remain limited in magnitude, the dynamic amplification of the global structural responses can be adequately approximated by the addition of an equivalent quasi-static load/action (see 9.8). If the dynamic effects exceed the limitations given in 9.8.1, a dynamic analysis is required. Dynamic analysis shall be performed for compliant towers.

9.3.2 Notation

The total quasi-static environmental load/action is denoted by E . Loads/actions due to one of the environmental conditions separately are indicated by E with a subscript i for wind (E_i), w for waves (E_w), and c for current (E_c). Where loads/actions due to waves and current are considered together, both subscripts w and c are attached (E_{wc}). An additional equivalent quasi-static load/action at the sea floor, taking into account dynamic effects, is denoted by D . The extreme value of any one of these loads/actions is indicated by an additional subscript e , e.g. E_e , E_{ie} , E_{wce} , D_e .

9.4 Extreme Quasi-static Load/Action due to Wind, Waves, and Current (E_e)

9.4.1 Procedure for Determining E_e

Subsection 9.4 is directly applicable only to structures that have negligible dynamic amplification; the methods of this subsection cover the quasi-static part of the load/action only. Dynamic amplification shall be included if significant. Methods of treating dynamic effects are described in 9.8. E_e is the extreme quasi-static direct load/action, with a 100-year return period, that is exerted on the structure by the combined effect of wind, waves, and currents in an extreme storm. E_e may therefore include the joint probability of the occurrence of extreme winds, waves, and currents, both in magnitude and in direction.

One of three methods is normally used for defining an environment that generates the extreme direct load/action E_e and generally also the extreme load/action effect, caused by the combined extreme wind, wave, and current conditions:

- a) 100-year return period wave height with associated wave period, wind speed, and current velocities;
- b) 100-year return period wave height and period combined with the 100-year return period wind speed and the 100-year return period current velocity, all determined by extrapolation of the individual parameters considered independently;
- c) any reasonable combination of wave height and period, wind speed, and current velocity that results in:
 - 1) the global extreme environmental load/action on the structure with a return period of 100 years, or
 - 2) a relevant load/action effect (global response) of the structure (e.g. base shear or overturning moment) with a return period of 100 years.

These three methods are discussed in API 2MET, which also describes the selection of appropriate parameters for determining the design conditions.

For structures for which the environmental load/action is dominated by waves, E_e can be estimated by method a) above. For other structures, other combinations of wind, wave, and current can be more appropriate. In some

cases, consideration of a 100-year extreme current velocity with an associated wave height, wave period(s), and wind can provide the best estimate of E_e .

In locations where joint probability information for the environmental conditions is not available, a conservative estimate of E_e can be determined through the sum of the environmental loads/actions caused by the independent extreme values of wind velocities, wave heights and periods, and current velocities, assuming that they act simultaneously and in the same direction [see method b) above]. Use of this method results in Equation (9.4-1):

$$E_e = E_{ie} + E_{wce} \quad (9.4-1)$$

The most general approach for correctly estimating E_e due to combinations of wind, waves, and current is via the calculation of the long-term statistics of global environmental loads/actions (such as applied base shear or overturning moment), or load/action effects for a generic structure at the intended location [see method c) above]. The statistical distributions thus obtained represent response-based global environmental loads/actions for the type of structure and location involved. They contain the most appropriate information on loads/actions for in-place design situations. Based on these long-term statistical distribution(s), a particular combination of wind, wave, and current parameters can be identified that is most likely to generate the 100-year extreme global environmental load(s)/action(s), in conjunction with a corresponding partial load/action factor, $\gamma_{f,E}$, that provides adequate protection against failure under environmental loads/actions in an extreme storm.

An acceptable procedure by which E_e may be calculated is given in 9.5 to 9.7.

The provisions of API 2MET or site-specific data developed in accordance with the requirements of API 2MET shall be used in determining specific values of metocean criteria for the analysis of quasi-static wave loads/actions.

9.4.2 Direction of Extreme Wind, Waves, and Current

In some locations, wave height will vary by direction due to either characteristic storm tracks or topographic features limiting the fetch. Similarly, tidal currents, general circulation currents, or wind-driven currents can also have predominant directions. In such situations, different extreme wind, waves, and/or current magnitudes for different approach directions may be used in design, provided that reliable data are available to derive these. However, the owner shall ensure that the overall safety level of the structure is not compromised by the use of such lower directional environmental conditions.

API 2MET provides wave directions and factors to be applied to the omnidirectional wave heights to be used in the determination of in-place environmental forces. When these directional factors are used, the environmental forces shall be calculated for all directions that are likely to control the design of any structural member or pile. As a minimum, this shall include environmental forces in both directions parallel and perpendicular to each jacket face as well as all diagonal directions, if applicable. These directions shall be determined from the geometry of the base of the jacket.

A minimum of eight directions shall be used for symmetrical, rectangular, and square platforms and a minimum of 12 directions shall be used for tripod jackets. For unsymmetrical platforms or structures with skirt piles, the calculation of the environmental forces from additional directions may also be required. If one of these directions is not the principal direction, then the omnidirectional wave from the principal direction shall also be assessed. The maximum force should be calculated with the crest of the wave at several locations as the crest of the wave passes through the platform.

In locations where reliable directional data are not available, the directions of the extreme wind, waves, and current shall be assumed to coincide when determining E_e .

9.4.3 Extreme Global Loads/Actions

Total base shear and overturning moment are calculated by a vector summation of the following:

- local hydrodynamic drag and inertia loads/actions due to waves and currents (see 9.5.2) integrated over the whole structure;
- dynamic amplification of wave and current loads/actions (see 9.8);
- loads/actions on the structure and the topsides caused by wind (see 9.7).

The above presupposes that there is sufficient air gap as required in 6.1.3.2 and that the extreme wave crest does not impinge on the topsides. If this condition is not satisfied, loads/actions associated with the wave crest impinging on the topsides shall be included in a).

Wave slamming and slapping loads/actions due to water surface impact on structural components may be neglected, as they are only of local significance. Loads/actions due to hydrodynamic lift may be neglected for space frame type structures because they are not correlated from member to member. Axial Froude-Krylov loads/actions may also be neglected. The wave crest should be positioned relative to the structure and the water depth should be selected such that the total base shear and/or the overturning moment have their maximum values. The following should be kept in mind that:

- a) maximum base shear does not necessarily occur at the same wave position as maximum overturning moment;
- b) maximum base shear often occurs at minimum possible water depth (low tide with minimum storm surge);
- c) maximum global load/action can be induced by the trough of a wave in a direction opposite to the direction in which the waves propagate, especially when an opposing current is present and the wave height is moderate.

Shallow water effects and limits on wave models shall be investigated where appropriate. Particularly in shallow water, the range of water depths between the maximum and minimum depth in the 100-year storm shall be investigated to determine the maximum global load/action.

9.4.4 Extreme Local Loads/Actions and Load/Action Effects

The following should be considered in the local design of components:

- a) maximum local loads/actions on, as well as maximum internal forces in, a component can occur for wave positions other than those causing the maximum global load/action, and maximum local forces in all components will not occur for the same wave position;
- b) some components can see their maximum internal force due to global load/action in one direction, whereas others see their maxima due to the global load/action in another direction;
- c) slamming loads/actions on horizontal components and slapping loads/actions on inclined components can occur when the water surface impacts on components at right angles to the component's surface, resulting in local loads/actions perpendicular to the component.

9.4.5 Vortex-induced Vibrations

All slender members should be investigated for the possibility of in-line and/or cross-flow vibrations due to vortex shedding resulting from the flow of water or air past the member, as appropriate.

Where relevant, the impact that cross-flow vibrations can have on the hydrodynamic coefficients given in 9.5.2.3 should be considered.

9.5 Extreme Quasi-static Load/Action Caused by Waves Only (E_{we}) or by Waves and Currents (E_{wce})

9.5.1 Procedure for Determining E_{we} and E_{wce}

The procedure used to determine the extreme, deterministic, quasi-static global load/action exerted on the structure caused by waves alone or by waves and current is applicable only to a fixed structure that satisfies the following conditions:

- negligible distortion of the incident wave by the structure;
- negligible dynamic structural response.

The sequence of steps described below is shown graphically in Figure 9.5-1. The procedure, for a given wave direction, begins with the specification of the design wave height and period, storm water depth, and current profile. If the load/action of waves only is to be calculated, the current velocity is put to zero with no change to the procedure.

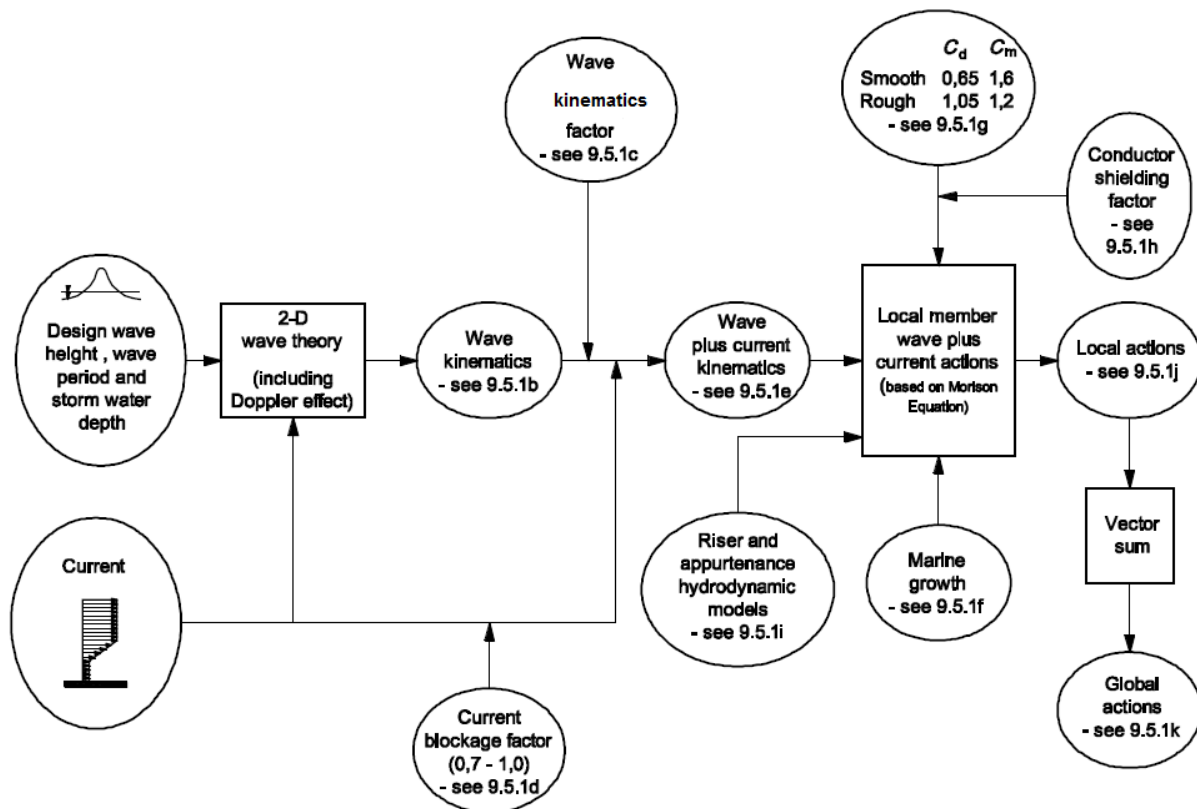


Figure 9.5-1—Procedure for Calculating the Quasi-static Load/Action Caused by Wave plus Current

The procedure for estimating the extreme quasi-static load/action caused by waves and current follows steps a) to k).

- a) Determine the apparent wave period, taking into account the Doppler effect of the current on the wave length and period, in accordance with 9.5.2.6 and A.9.5.2.6.
- b) Determine the two-dimensional wave kinematics for the specified wave height, storm water depth, and apparent period, using an appropriate wave theory (see 9.5.2.7 and A.9.5.2.7).
- c) The horizontal components of the wave-induced water particle velocities and accelerations may be reduced by the wave kinematics factor, which accounts primarily for wave directional spreading (see 9.5.2.8 and A.9.5.2.8).
- d) Determine the effective local current profile by multiplying the specified current profile by the current blockage factor (see 9.5.2.4 and A.9.5.2.4).
- e) Determine the locally incident fluid velocities and accelerations for use in Morison equation (see 9.5.2.1) by vectorially combining the effective local current profile from step d) after stretching it to the instantaneous water surface elevation (see 9.5.2.9) with the wave kinematics from step c).
- f) Increase the member diameters to account for marine growth (see 9.5.2.2).
- g) Determine drag and inertia coefficients for use in Morison equation in accordance with the relevant wave and current parameters, as well as member shape, size, orientation, and roughness (including marine growth). Typical values of these coefficients are given in 9.5.2.3.
- h) Drag and inertia coefficients for the conductor array may be reduced by the conductor shielding factor in 9.5.2.5.
- i) Determine hydrodynamic models for loads/actions on risers and appurtenances in accordance with 9.5.3.
- j) Calculate local loads/actions caused by waves and current for all structural components, conductors, risers, and appurtenances using Morison equation, taking the requirements of steps e) to i) into account.
- k) Calculate the extreme global load/action caused by waves and current as the vector sum of all the local actions from step j).

9.5.2 Models for Hydrodynamic Loads/Actions

9.5.2.1 Morison Equation

The computation of the load/action on a cylindrical object (a member) caused by waves, currents, or a combination of waves and currents depends on the ratio of the wave length to the member diameter. When this ratio is large (> 5), the member does not significantly modify the incident wave. The load/action can then be computed as the sum of a hydrodynamic drag load/action and a hydrodynamic inertia load/action, as given in Equation (9.5-1):

$$F = F_d + F_i = C_d \times \frac{1}{2} \rho_w \times U \times |U| \times A + C_m \times \rho_w \times V \times \frac{\partial U}{\partial t} \quad (9.5-1)$$

where

- | | |
|-------|---|
| F | is the local load/action vector per unit length acting normal to the axis of the member; |
| F_d | is the vector for the drag load/action per unit length acting normal to the axis of the member in the plane of the member axis and U ; |
| F_i | is the vector for the inertia load/action per unit length acting normal to the axis of the member in the plane of the member axis and $\frac{\partial U}{\partial t}$; |
| C_d | is the hydrodynamic drag coefficient; |

ρ_w	is the mass density of water;
A	is the effective dimension of the cross-sectional area normal to the member axis per unit length ($= D$ for circular cylinders);
V	is the displaced volume of the member per unit length ($= \pi D^2/4$ for circular cylinders);
D	is the effective diameter of a member (a circular cylinder), including marine growth;
U	is the component of the local water particle velocity vector (due to waves and/or current) normal to the axis of the member;
$ U $	is the modulus (the absolute value) of U ;
C_m	is the hydrodynamic inertia coefficient;
$\frac{\partial U}{\partial t}$	is the component of the local water particle acceleration vector normal to the axis of the member.

The Morison equation, as stated here, ignores the contribution from the Froude-Krylov load/action associated with the convective acceleration component. This is not always appropriate for inertia dominated structures, small wave heights, steep waves, or very large diameter components. The equation also excludes loads/actions due to hydrodynamic lift, slamming or slapping loads/actions, and axial Froude-Krylov loads/actions.

When the size of a structural object or component is sufficiently large compared to the wave length, the incident waves are scattered or diffracted. This “diffraction regime” is usually considered to occur when the component width exceeds a fifth of the incident wave length. In such cases, diffraction theory, which computes the pressures acting on the structure due to both the incident wave and the scattered wave, should be used instead of the Morison equation to determine the local load/action caused by waves. If a structure has some components in the Morison regime and others in the diffraction regime, or a component is adjacent to another (component of a) structure that is in the diffraction regime, the effect of wave diffraction on the wave kinematics to be used for the Morison component should be considered.

NOTE Depending on its diameter, a free-standing caisson structure can be in the diffraction regime, particularly for the lower sea states associated with fatigue conditions.

9.5.2.2 Marine Growth

The expected effect of marine growth on the hydrodynamic loads/actions on the structure during its design service life should be taken into account.

The cross-sectional dimensions of structural components, conductors, risers, and appurtenances should be increased to account for marine growth thickness. Components with circular cross-sections should further be classified as either “smooth” or “rough,” depending on the amount and size of marine growth expected to have accumulated on them at the time of the loading event.

Structural elements can be considered hydrodynamically smooth if they are located either above the highest astronomical tide (HAT) or sufficiently deep below the lowest astronomical tide (LAT). In both these zones, marine growth accumulation is usually small enough to ignore the effect of roughness. However, caution should be taken because it takes very little roughness to make the rough value of the C_d in 9.5.2.3 realistic. Site-specific data shall be used to reliably establish the extent of the hydrodynamically rough zones. For US waters, the marine growth data provided in API 2MET shall be used if site-specific data are not available. Otherwise, the structural components shall be considered rough down to the sea floor.

9.5.2.3 Drag and Inertia Coefficients

For typical design situations, global hydrodynamic load/action on a structure can be calculated using Morison equation, with the values of the hydrodynamic coefficients for unshielded circular cylinders given in Table 9.5-1.

Table 9.5-1—Typical Values of Hydrodynamic Coefficients

Surface of Component	C_d	C_m
Smooth	0.65	1.6
Rough	1.05	1.2

The values in Table 9.5-1 are appropriate for situations of a steady current with negligible waves and for large waves with $U_{mo}T_{app}/D > 30$, where:

U_{mo} is the maximum horizontal water particle velocity at storm still water level under the wave crest from the two-dimensional wave kinematics theory;

T_{app} is the apparent wave period;

D is the diameter of the structure's legs at storm still water level.

Additional information and guidance on hydrodynamic coefficients in a variety of circumstances is given in A.9.5.2.3.

For situations where waves are dominant but $U_{mo}T_{app}/D < 30$, C_m and C_d for nearly vertical members are modified by “wake encounter” and shall be specially determined.

NOTE Such situations can arise for a large diameter free-standing caisson structure in extreme seas or for ordinary structural components in the lower sea states that are considered in fatigue analyses.

9.5.2.4 Current Blockage

The in-line current speed in the vicinity of a structure is reduced from the specified free stream value by blockage. The presence of the structure causes the incident flow to diverge; some of the flow goes around the structure rather than through it, and the in-line current speed within the structure is reduced.

Since global hydrodynamic load/action on the structure is determined by summing local hydrodynamic load/action contributions from Morison equation, the appropriate local current velocity should be used. Table 9.5-2 gives typical current blockage factors for Gulf of Mexico jacket platforms. Data sets obtained from measurements of the current speed in the vicinity of a structure will generally already include the effect of current blockage in the data. Therefore, current blockage factors should not be applied to such (extrapolated) data sets.

Table 9.5-2—Approximate Current Blockage Factors for Typical Gulf of Mexico Jacket-type Structures

Number of Legs	Heading	Factor
3	All	0.90
4	End-on	0.80
	Diagonal	0.85
	Broadside	0.80
6	End-on	0.75
	Diagonal	0.85
	Broadside	0.80
8	End-on	0.70
	Diagonal	0.85
	Broadside	0.80

For free-standing or braced caissons, the current blockage factor should be 1.0.

9.5.2.5 Conductor Shielding Factor

Depending upon the configuration of the structure and the number of well conductors, the hydrodynamic load/action on the conductors can be a significant portion of the total hydrodynamic load/action on the structure. If the conductors are closely spaced, the global hydrodynamic load/action on them can be reduced as a result of hydrodynamic shielding. To account for this, a shielding factor, k_s , may be applied to the drag and inertia coefficients for the conductor array. The shielding factor can be estimated from Figure 9.5-2, in which S is the center-to-center spacing of the conductors in the wave direction and D is the diameter of the conductors, including marine growth. This shielding factor is appropriate for either:

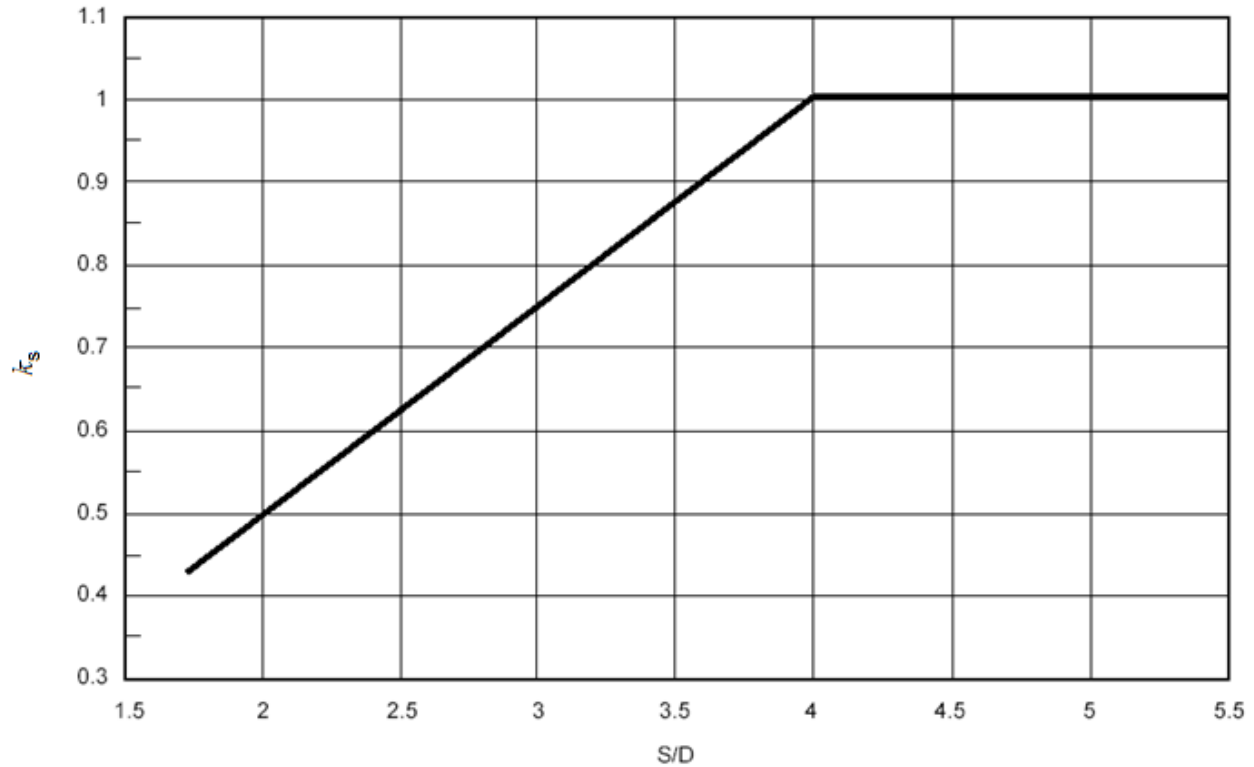
- a) steady current with negligible waves, or
- b) extreme waves, with $U_{mo}T_{app}/S > 5\pi$, where U_{mo} and T_{app} are defined in 9.5.2.3.

For less severe waves, with $U_{mo}T_{app}/S < 5\pi$, as in fatigue analyses, shielding can be less (see A.9.5.2.5).

When the angle between the wave or current direction and the center plane of a row (or column) of conductors is less than 22.5° , S is taken as the center-to-center spacing between the conductors in the row (or column). For angles between 22.5° and 67.5° , S can be approximated as the average of the center-to-center row spacing and the center-to-center column spacing. For a single row of conductors, shielding should not be included if the angle between the center plane of the row and the wave or current direction is greater than 22.5° .

The shielding phenomenon in this subsection does not adequately address the situation of small diameter members adjacent to large diameter members, e.g. an external conductor at a monotower. The interaction between the small and the large diameter member in such situations should be carefully considered (see also 9.11).

A possible increase in the added mass (and inertia loads/actions) for closely spaced conductors shall be considered.

**Key**

- S conductor spacing
- D conductor diameter
- k_s shielding factor

Figure 9.5-2—Factor for Reduction of Hydrodynamic Coefficients for Conductors in Array due to Shielding as Function of Conductor Spacing/Diameter Ratio

9.5.2.6 Apparent Wave Period

A current in the wave direction tends to stretch the wavelength, while an opposing current shortens it. For the simple case of a wave propagating on a uniform in-line current, the apparent wave period seen by an observer moving with the current can be estimated from Figure 9.5-3, in which T is the actual wave period (as seen by a stationary observer). V_i is the current component in the wave direction, d is storm water depth (including storm surge and tide), and g is the acceleration of gravity. This figure provides estimates for $d/gT^2 > 0.01$. For smaller values of d/gT^2 , the equation $(T_{app}/T) = 1 + V_i/\sqrt{gd}$ can be used. Although strictly applicable only to a current that is uniform over the full water depth, Figure 9.5-3 provides acceptable estimates of T_{app} for “slab” current profiles that are uniform over the top 50 m (165 ft) or more of the water column. For other current profiles, T_{app} (see A.9.5.2.6) is generally determined from the iterative solution of a system of simultaneous nonlinear equations. The current used to determine T_{app} should be the free-stream current (not reduced by structure blockage).

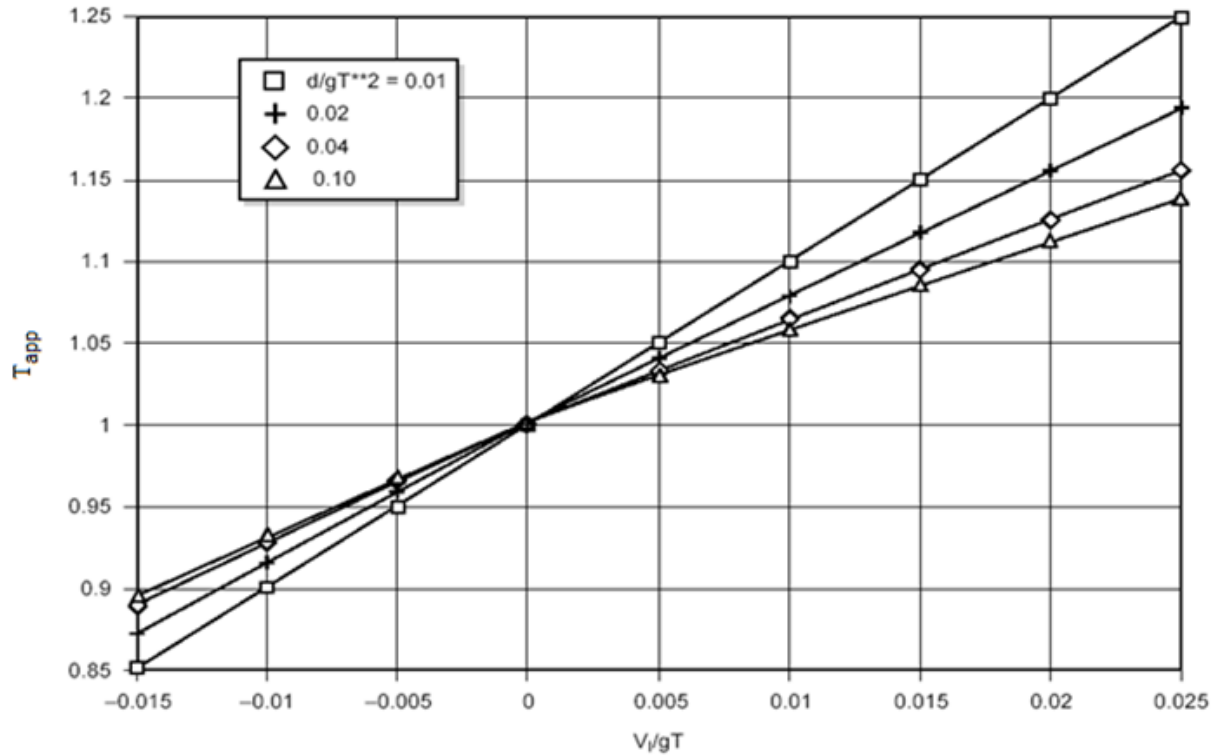


Figure 9.5-3—Doppler Shift due to Steady Current

9.5.2.7 Two-dimensional Wave Kinematics

For the apparent wave period T_{app} , specified wave height H , and storm water depth d , two-dimensional regular wave kinematics can be calculated using the appropriate order of Stream Function wave theory. In many cases, Stokes V wave theory produces acceptable accuracy. Figure 9.5-4 shows the regions of applicability of Stokes V and various orders of Stream Function solutions in the H/gT_{app}^2 , d/gT_{app}^2 plane. Other wave theories, such as Extended Velocity Potential and Chappelear, may be used if an appropriate order of solution is selected.

9.5.2.8 Wave Kinematics Factor

The two-dimensional regular wave kinematics from Stream Function or Stokes V wave theory do not account for wave directional spreading or irregularity in wave profile shape. These “real world” wave characteristics can be approximately modeled in deterministic wave analyses by multiplying the horizontal velocities and accelerations from the two-dimensional regular wave solution by a wave kinematics factor. Wave kinematics measurements support a factor in the range 0.85 to 0.95 for tropical storms and 0.95 to 1.00 for extratropical storms. Particular values within these ranges that shall be used for calculating guideline wave forces are specified for the Gulf of Mexico and for other US waters in API 2MET. See A.9.5.2.8 for additional guidance for calculating the wave kinematics factor for particular sea states whose directional spreading characteristics are known from measurements or hindcasts.

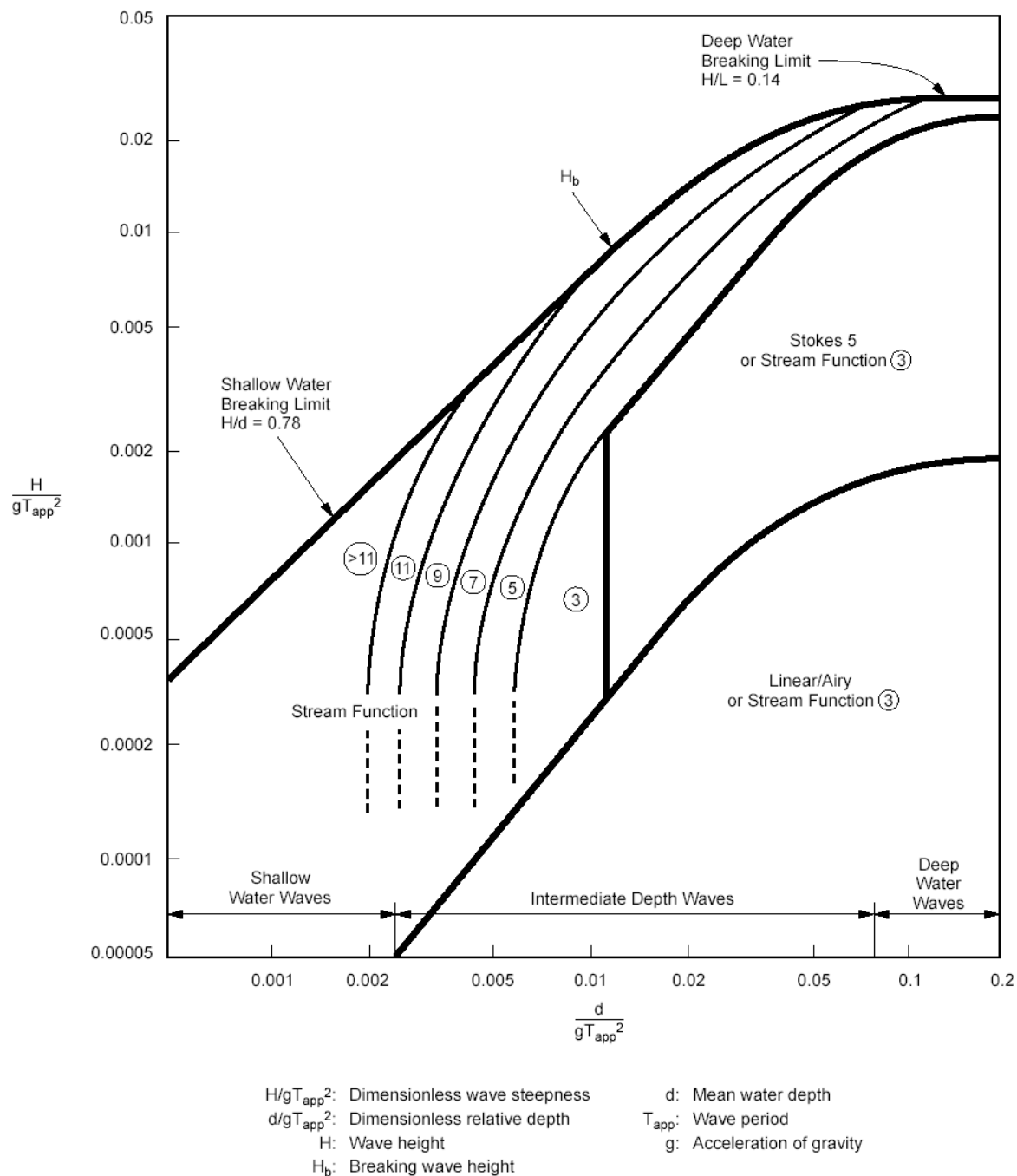


Figure 9.5-4—Regions of Applicability of Stream Function, Stokes V, and Linear Wave Theory
 (from Atkins, 1990; Modified by API Task Group on Wave Force Commentary)

9.5.2.9 Combined Wave/Current Kinematics

Wave kinematics, adjusted for directional spreading and irregularity, should be combined vectorially with the current profile, adjusted for blockage. Since the current profile is specified only to storm mean water level in the design criteria, various methods to stretch (or compress) it to the local wave surface are used. As discussed in A.9.5.2.9, “nonlinear stretching” is the preferred method. For slab current profiles such as those specified for US waters in API 2MET, simple vertical extension of the current profile from storm mean water level to the wave surface is a good approximation to nonlinear stretching. For other current profiles, linear stretching is an acceptable approximation. In linear stretching, the current at a point with elevation z , above which the wave surface elevation is η (where z and η are both positive above storm mean water level and negative below), is computed from the specified current profile at elevation z' . The elevations z and z' are linearly related, as follows:

$$(z' + d) = (z + d) d/(d + \eta) \quad (9.5-2)$$

where

d is the storm water depth.

9.5.3 Hydrodynamic Models for Appurtenances

Appurtenances, such as boat landings, fenders or bumpers, walkways, stairways, grout lines, and anodes, should be considered for inclusion in the hydrodynamic model of the structure. Depending upon the type and number of appurtenances these can significantly increase the global hydrodynamic load/action on the structure. Furthermore, hydrodynamic loads/actions on some appurtenances can be important for local member design. Appurtenances are generally modeled by means of nonstructural members that contribute equivalent hydrodynamic load/action. For some appurtenances, such as boat landings, the hydrodynamic load/action is highly dependent on wave direction due to shielding effects.

9.6 Loads/Actions Caused by Current

The current velocity and current profile as a function of depth shall be determined in accordance with API 2MET.

Where current acts alone, the local load/action due to current can be calculated in accordance with the procedure in 9.5 (as far as relevant) by putting $\partial U/\partial t = 0$. Where current acts together with waves, the current velocity should be stretched and vectorially added to the wave particle velocity before the total local load/action due to waves and current is calculated, as described in 9.5.

Where a global load/action caused by current only is to be determined, the extreme quasi-static global load/action, E_{ce} , shall be calculated as the vector sum of the above local current loads/actions.

9.7 Loads/Actions Caused by Wind

9.7.1 General

The wind criteria for design should be determined by proper analysis of wind data. As with wave loads/actions, wind loads/actions are dynamic in nature, but some structures will respond to them in a nearly static fashion. For conventional fixed steel templates in relatively shallow water, winds are a minor contributor to global loads/actions (typically less than 10 %) except for platforms in very shallow water that have a large deck or superstructure. Sustained wind speeds should be used to compute global platform wind loads/actions, and gust speeds should be used for the design of individual structural elements. The values of wind criteria for different return intervals provided in API 2MET shall be used.

In deeper water and for compliant designs, wind loads/actions can be significant and should be studied in detail. A dynamic analysis of the platform is indicated when the wind field contains energy at frequencies near the natural frequencies of the platform. Such analyses may require knowledge of the wind turbulence intensity, spectra, and spatial coherence (see API 2MET).

9.7.2 Determining Loads/Actions Caused by Wind

For all angles of wind approach to the structure, wind loads/actions on vertical cylindrical objects may be assumed to act in the direction of the wind. Loads/actions on cylindrical objects that are not in a vertical attitude shall be calculated using appropriate formulae that take into account the direction of the wind in relation to the attitude of the object, in the same way as drag loads/actions on submerged cylinders due to wave and current are calculated. Loads/actions on walls of buildings and other flat surfaces that are not perpendicular to the direction of the wind shall also be calculated, using appropriate formulae that account for the skewness between the direction of the wind and the plane of the surface. Where appropriate, loads/actions caused by wind shall be calculated with due account for increased exposure area and surface roughness due to ice. Local wind effects, such as pressure concentrations and internal pressures, should be considered by the designer where applicable.

The wind load/action on an object may generally be calculated using Equation (9.7-1). However, in accordance with the above observations, due attention shall be given to which velocity (in magnitude and direction) and which area is to be used. Similarly, the direction of the resulting load/action vector should be carefully considered. The basic relationship between the wind velocity and the wind load/action on an object is expressed by Equation (9.7-1):

$$F = \frac{1}{2} \rho_a U_w^2 C_s A \quad (9.7-1)$$

where

F is the wind load/action on the object;

ρ_a is the mass density of air (at standard temperature and pressure);

U_w is the wind speed;

C_s is the shape coefficient;

A is the area of the object.

In the absence of data indicating otherwise, the shape coefficients given in Table 9.7-1 are recommended for perpendicular wind approach angles with respect to each projected area.

Table 9.7-1—Shape Coefficients, C_s , for Perpendicular Wind Approach Angles

Component		Shape Coefficients C_s
Flat walls of buildings		1.50
Overall projected area of structure		1.00
Beams		1.50
Cylinders	Smooth, $Re > 5 \times 10^5$	0.65
	Smooth, $Re \leq 5 \times 10^5$	1.20
	Rough, all Re	1.05
	Covered with ice, all Re	1.20
Re is Reynolds number.		

Wind loads/actions on downstream components can be reduced due to shielding by upstream components.

Shielding and shape coefficients for individual components, modules, and lattice structures are described in ENV 1991-2-4.

The extreme quasi-static global load/action, E_{ie} , caused by wind shall be calculated as the vector sum of the above wind loads/actions on all objects.

9.7.3 Wind Loads/Actions Determined from Models

When wind loads/actions are important for structural design, wind pressures and resulting local loads/actions may be determined from wind tunnel tests on a representative model, or from a computational model representing the structure, considering the range and variation of wind velocities. Computational models shall be validated against wind tunnel tests or full-scale measurements of similar structures.

9.8 Equivalent Quasi-static Load/Action Representing Dynamic Response Caused by Extreme Wave Conditions

9.8.1 General

Loads/actions caused by waves vary with time. Any structure subjected to these loads/actions will therefore experience dynamic response to a greater or lesser degree. However, in many cases the dynamic response will remain rather limited in magnitude. In such cases, the global effects of dynamic response can be adequately approximated by the introduction of an equivalent quasi-static load/action, D , representing the overall dynamic load/action effect. The corresponding equivalent quasi-static load/action in design environmental conditions (D_e) is added to the total extreme quasi-static load/action due to wind, waves, and current (E_e).

In general, all (quasi-)static and dynamic loads/actions are vectors. However, for the purposes of this document and for the conditions under which an equivalent quasi-static load/action representing dynamic response may be used, only scalar values are required. The notations D_e , E_e , and E_{wce} , as used in this document, refer to the magnitudes of the corresponding vectors at the sea floor. Once determined, the equivalent extreme quasi-static load/action, D_e , is next distributed over the height of the structure in a set of horizontal loads/actions, d_k , at elevations $k = 1, 2, \dots, K$. The loads/actions d_k represent the inertial loads/actions caused by a global acceleration of the masses lumped at each elevation k . The loads/actions d_k are then added to the corresponding loads/actions due to E_e at the same elevation.

For the applicability of the approximate procedure described in 9.8, it is required that:

- a) the dynamic and the quasi-static responses be approximately in phase, and
- b) the magnitude of D_e be considerably smaller than the magnitude of E_{wce} .

These requirements are considered to be satisfied if the following two conditions are met:

- a) the dynamic response is stiffness controlled, which may be assumed if the fundamental natural period of the structure is less than one-fifth of the peak period of the wave spectrum of the design sea state, and
- b) the magnitude of D_e is less than one-half of the magnitude of E_{wce} .

In all other cases, a more detailed dynamic analysis shall be performed. Requirements for such an analysis are covered in Section 12.

9.8.2 Equivalent Quasi-static Load/Action (D_e) Representing the Dynamic Response

The magnitude of the equivalent quasi-static load/action, D_e , represents the total global inertial load/action that the structure experiences at the time when the total global dynamic response is a maximum. Under the conditions specified in 9.8.1, the return period of the extreme global dynamic response to the environmental load/action is approximately the same as that of the extreme quasi-static response to E_{wce} alone.

The equivalent quasi-static load/action, D_e , is established through a global dynamic analysis in waves. D_e can be determined from the results of the dynamic analysis in one of two ways:

- directly, by subtracting the extreme quasi-static base shear (which is equal to the extreme applied horizontal wave load/action on the structure) from the extreme dynamic base shear obtained from the analysis; or
- indirectly, by multiplying E_{wce} by a factor ($k_{DAF} - 1$), where the k_{DAF} is a DAF on the quasi-static base shear that is derived from the global dynamic analysis as the ratio of the extreme dynamic base shear to the extreme quasi-static base shear.

D_e subsequently needs to be distributed along the height of the structure to model the individual contributions of the total global inertial load/action that act at each mass point (see A.9.8.2).

9.8.3 Global Dynamic Analysis in Waves

9.8.3.1 General

The dynamic analysis is performed at a global level, for which purpose a suitably simplified structural model can be used as appropriate. Subsection 9.8.3 describes the global dynamic analysis for the purpose of determining D_e in 9.8.2.

9.8.3.2 Dynamic Analysis Methods

The time-varying loads/actions caused by waves shall realistically model the sea surface and the frequency content of wave load/action in extreme wave conditions. This necessitates the use of random waves; the procedure described in 9.5 is only appropriate for quasi-static analysis. When there are significant nonlinearities in modeling the environment, the structure, the fluid-structure interaction, or the structure-foundation interaction, time-history methods are preferred. When linearization of hydrodynamic drag loads/actions, inundation effects, and structure-foundation interaction can be justified, frequency domain methods generally provide a more transparent and computationally more efficient means of using random waves and may be used for the global dynamic analysis.

9.8.3.3 Design Sea State

The random waves shall correspond with one or more wave spectra that are plausible representations of sea state(s) that produce the 100-year design load/action effect or, alternatively, the 100-year design wave condition defined in 9.4 (see also 6.3.2). Consideration should be given to including currents in the global dynamic analysis as currents associated with the design sea state can affect dynamic response through the drag load/action in Morison equation [see Equation (9.5-1)]. Wind fluctuations may generally be neglected for conventional fixed steel offshore structures; i.e. loads/actions caused by wind can be considered to be static loads/actions.

9.8.3.4 Hydrodynamic Load/Action on a Member

Morison equation may be used to compute loads/actions on members of rigid structures. Its relative velocity form shall not be used for fixed structures. Hydrodynamic loads/actions in phase with the structure's acceleration are taken into account by the hydrodynamic added mass. Hydrodynamic damping may be taken into account by an equivalent viscous damping as described in 9.8.3.6.

9.8.3.5 Mass

The dynamic model of a fixed structure should include appropriately modeled masses of the following:

- the topsides,
- the structural steel in the structure,
- the appurtenances and conductors,
- the mass of marine growth expected to accumulate on the structure,
- the mass of water within submerged members, and
- the hydrodynamic added mass of submerged members, taking into account increased outside member diameters due to marine growth.

9.8.3.6 Damping

In lieu of an explicit determination of various damping contributions, an equivalent viscous damping model may be used. The equivalent viscous damping coefficient is sensitive to many effects, including structure type and layout, the severity of excitation and response, and the relationship of the natural period of the structure to the peak period of the design wave spectrum. In the absence of substantiating information for damping values for a specific structure, a damping coefficient of 2 % to 3 % of critical may be used for the global dynamic analyses in extreme wave conditions. For less severe sea states used for fatigue analyses, damping values less than 2 % are more appropriate.

9.8.3.7 Stiffness

A linear elastic stiffness model of the structure that includes the interaction between the structure and the foundation is normally adequate for the global dynamic analysis in waves.

9.9 Factored Loads/Actions

9.9.1 General

In accordance with 7.6 and 7.8.1, each representative load/action shall be multiplied by a partial load/action factor, γ_f . The design load(s)/action(s), F_d , for a particular design situation comprise one or more combinations of factored loads/actions.

This section relates to loads/actions and partial load/action factors for strength design satisfying the ULS during extreme conditions for the in-place situation (see 7.2 and 7.6). The corresponding design loads/actions are specified in 9.9.2, 9.9.3, and 9.10. Design loads/actions for operating situations are also specified in 9.10.

Loads/actions and partial load/action factors for strength design satisfying ALS are given in Section 10.

Loads/actions and partial load/action factors for FLS are given in Section 16; those for FLS of grouted and mechanical connections and clamps are given in Section 15.

9.9.2 Factored Permanent and Variable Loads/Actions

The design load/action for permanent and variable loads/actions is given by Equation (9.9-1):

$$F_d = \gamma_{f,G1} G_1 + \gamma_{f,G2} G_2 + \gamma_{f,Q1} Q_1 + \gamma_{f,Q2} Q_2 \quad (9.9-1)$$

where G_1 , G_2 , Q_1 , and Q_2 are defined in 9.2.1 to 9.2.4, taking account of the requirements of 9.2.5 to 9.2.9.

Values for the partial load/action factors, γ_f , for various design situations are given in 9.10.

9.9.3 Factored Extreme Environmental Load/Action

The design load/action for extreme environmental load/action is given by Equation (9.9-2):

$$F_d = \gamma_{f,E} (E_e + \gamma_{f,D} D_e) \quad (9.9-2)$$

where

E_e is the extreme quasi-static load/action due to wind, waves, and current, defined in 9.4, taking account of the requirements of 9.5 to 9.7;

D_e is the equivalent quasi-static load/action representing the dynamic load/action effect defined in 9.8.1;

$\gamma_{f,E}$ is the partial load/action factor for E_e ;

$\gamma_{f,D}$ is an additional partial load/action factor for D_e .

The value of $\gamma_{f,D}$ accounts for the uncertainty of D_e to represent the global dynamic response of the structure, as well as the variability of the influence of global dynamic response on individual components of the structure.

Values for the partial load/action factors $\gamma_{f,E}$ and $\gamma_{f,D}$ are provided in 9.10. (See A.9.9 for a further discussion on the partial load/action factors.)

9.10 Design Situations

9.10.1 General Considerations on the Ultimate Limit State

The basic requirement for satisfying the ULS under design environmental conditions in the in-place situation for structures that respond predominantly quasi-statically in accordance with the conditions of 9.9.1 is that the structure possesses an adequate strength to meet a minimum safety level commensurate with its exposure level (see 7.10).

This requirement can in principle be met in several ways, two of which are listed below and both of which are within the scope of this document:

- by implicitly following a partial factor design format, which is the usual design and assessment procedure and which is described in 9.10.3; or
- by explicitly demonstrating that a structure has a certain minimum RSR, as discussed in 9.10.2.

NOTE The RSR approach is not applicable to the exposure category levels for the Gulf of Mexico criteria contained herein.

For structures where dynamic response exceeds the limitations of 9.8.1 and a proper dynamic analysis is performed, the requirements for the ULS shall be explicitly assessed.

9.10.2 Demonstrating Sufficient RSR under Environmental Loads/Actions

A structure's ultimate system strength may be determined by means of a suitable nonlinear pushover analysis, from which the RSR can be calculated. If the RSR meets an appropriate criterion for the following, the design satisfies the ULS for the situation being considered:

- a) the relevant exposure level L-1, L-2, or L-3 (see 6.4), and
- b) the geographical location concerned.

This approach represents a significant departure from conventional design practice and can be hampered by several difficulties. These difficulties include the need for special and advanced knowledge and experience of the designers, the limited relevant experience that is available with this approach, the fact that the calculated ultimate strength depends on the method and the details by which the assessment is carried out, and the lack of common

agreement on the setting of appropriate RSR criteria. Consequently, the approach can only be adopted if it is approved by the owner and the regulator, where one exists; such approval shall include the methodology to be used and the RSR value to be achieved. For a discussion of the elements involved, see A.9.9.

Notwithstanding this, when these difficulties can be overcome and the owner, the designer, and the regulator, where one exists, can reach agreement on all the aspects involved, the approach is fully in agreement with the general requirements of ISO 19900 and the requirements for fixed steel offshore structures in this document.

9.10.3 Partial Factor Design Format

9.10.3.1 General

For a quasi-statically responding structure, adequate proof of satisfying the ULS should be deemed to be provided by the partial design format described in this subsection. Each member, joint, and foundation component should be checked for strength using the internal force (load/action effect) (S) due to the combinations of factored:

- permanent loads/actions,
- variable loads/actions,
- extreme quasi-static environmental loads/actions, and
- where relevant, equivalent quasi-static loads/actions representing dynamic response caused by extreme wave conditions.

The combinations and factors are specified in 9.10.3.2.

9.10.3.2 Design Loads/Actions for In-place Situations

The general equation for determining the design action (F_d) for in-place situations is given in Equation (9.10-1), and the appropriate partial load/action factors for each design situation are given in Table 9.10-1:

$$F_d = \gamma_{f,G1} G_1 + \gamma_{f,G2} G_2 + \gamma_{f,Q1} Q_1 + \gamma_{f,Q2} Q_2 + \gamma_{f,E0} (E_o + \gamma_{f,D} D_o) + \gamma_{f,Ee} (E_e + \gamma_{f,D} D_e) \quad (9.10-1)$$

where

G_1, G_2	are the permanent loads/actions defined in 9.2;
Q_1, Q_2	are the variable loads/actions defined in 9.2;
E_o	is the environmental load/action due to the owner-defined operating wind, wave, and current parameters;
D_o	is the equivalent quasi-static load/action representing dynamic response in accordance with 9.8, but caused by the wave condition that corresponds with that for E_o ;
E_e	is the extreme quasi-static load/action due to wind, waves, and current as defined in 9.4 and taking account of the requirements of 9.5 to 9.7;
D_e	is the equivalent quasi-static load/action representing dynamic response defined in 9.8.1;
$\gamma_{f,G1}, \gamma_{f,G2}, \gamma_{f,Q1}, \gamma_{f,Q2}$	are the partial load/action factors for the various permanent and variable loads/actions discussed in 9.9 and for which values for different design situations are given in Table 9.10-1 (see A.9.10.3.2.1);
$\gamma_{f,E0}, \gamma_{f,Ee}$	are partial load/action factors applied to the total quasi-static environmental load/action plus equivalent quasi-static load/action representing dynamic response for operating and extreme environmental conditions, respectively, and for which values for different design situations are given in Table 9.10-1;
$\gamma_{f,E}, \gamma_{f,D}$	are the partial load/action factors for the environmental loads/actions discussed in 9.9 with $\gamma_{f,E} = 1.35$ and $\gamma_{f,D} = 1.25$.

Table 9.10-1—Partial Load/Action Factors for In-place Situations and Exposure Level L-1

Design Situation	Partial Load/Action Factors ^a					
	$\gamma_{f,G1}$	$\gamma_{f,G2}$	$\gamma_{f,Q1}$	$\gamma_{f,Q2}$	$\gamma_{f,E0}$	$\gamma_{f,Ee}$
Permanent and variable loads/actions only	1.3	1.3	1.5	1.5	0.0	0.0
Operating situation with corresponding wind, wave, and/or current conditions ^b	1.3	1.3	1.5	1.5	$0.9 \gamma_{f,E}$	0.0
Extreme conditions when the load/action effects due to permanent and variable loads/actions are additive ^c	1.1	1.1	1.1	0.0	0.0	$\gamma_{f,E}$
Extreme conditions when the load/action effects due to permanent and variable loads/actions oppose ^d	0.9	0.9	0.8	0.0	0.0	$\gamma_{f,E}$
^a A value of 0 for a partial load/action factor means that the load/action is not applicable to the design situation. ^b For this, check that G_2 , Q_1 , and Q_2 are the maximum values for each mode of operation. ^c For this, check that G_1 , G_2 , and Q_1 include those parts of each mode of operation that can reasonably be present during extreme conditions. ^d For this, check that G_2 and Q_1 exclude any parts associated with the mode of operation considered that cannot be ensured of being present during extreme conditions.						

9.11 Local Hydrodynamic Loads/Actions

Internal forces are due to both global loads/actions transferred from the rest of the structure and local hydrodynamic loads/actions on a member. In some cases, VIVs add to the internal forces.

The calculation of the internal forces for the design of members and adjacent joints should include these local loads/actions with the partial load/action factor for environmental loads/actions.

Transferred loads/actions are due to the global environmental loads/actions and the dynamic response of the entire structure. For stiffness controlled dynamic response, the latter will not be too severe and is taken into account as described in 9.8.

Local loads/actions include not only the drag and inertia loads/actions modeled by Morison equation [see Equation (9.5-1)] but also loads/actions due to hydrodynamic lift, axial Froude-Krylov loads/actions, distributed weight, and buoyancy effects. Horizontal members near storm still water level can experience vertical slamming loads/actions as a wave passes. Loads/actions due to hydrodynamic lift, slamming, and slapping can dynamically excite individual members, thereby increasing internal forces. For smaller members adjacent to larger members, such as appurtenances supported by legs, the calculation of the kinematics used in Morison equation should take account of the flow modification caused by the larger member.

The fraction of total internal forces due to locally generated loads/actions is generally greater for members higher in the structure. Therefore, local loads/actions due to hydrodynamic lift, slamming, and slapping should be considered in designing those members and adjacent joints. The maximum local internal forces can also occur at a different position of the wave crest relative to the structure centerline than those that cause the greatest global environmental load/action on the structure. For example, some members of conductor guide frames can experience their greatest internal forces due to vertical hydrodynamic loads/actions, which generally peak when the wave crest is far from the structure centerline. Also, maximum internal forces in some members can occur for wind, wave, and current directions other than those that cause maximum global environmental loads/actions.

10 Accidental Situations

This section describes how hazards are treated and designed for in the context of this document.

10.1 General

10.1.1 Hazards

Offshore structures are exposed to various hazards of greatly different types and having vastly different probabilities of occurrence. Hazards can be categorized in several ways, one of which is based on probability of occurrence. Using this criterion, hazards may be grouped into the following three main groups:

- group 1: hazards with a probability of occurring or being exceeded of the order of 10^{-2} per annum (return periods of the order of 100 years);
- group 2: hazards with a 10 to 100 times lower probability of occurring or being exceeded, i.e. probabilities of the order of 10^{-3} to 10^{-4} per annum (return periods of the order of 1000 to 10,000 years);
- group 3: hazards with a probability of occurring or being exceeded markedly lower than 10^{-4} per annum (return periods well in excess of 10,000 years).

NOTE See Table 6.3-1 for L-2 and L-3 structures.

These probabilities (return periods) should be seen as an indication of the order of magnitude rather than as precise numbers, since accurate databases for such low probabilities of occurrence rarely exist.

The main hazards that are faced by an offshore structure include the following:

- vessel (ship) collisions,
- dropped objects,
- fires and explosions, and
- abnormal environmental loads/actions, including abnormal seismic loads/actions.

Discussions of hazards and how these should be dealt with are often hampered by the use of various terminologies and concepts. In this document, a hazard (see 3.1.25) is defined quite generally as the potential for human injury, damage to the environment, damage to property, or a combination of these.

10.1.2 Designing for Hazards

Offshore structures and their structural components shall be designed to satisfy particular limit states. Each limit state is verified by defining a number of design situations and requiring that the associated load/action effects shall meet given design criteria. Design situations are classified into three categories:

- a) persistent situations, with a duration similar to the design service life of the structure;
- b) transient situations, with a much shorter duration and varying levels of intensity;
- c) accidental situations, which are of short duration and low probability of occurrence.

Designing for hazards of group 1 from exposure to design environmental conditions (wind, waves, current, as well as earthquakes) is normally treated by the regular design process and incorporated in the verification of relevant limit states for persistent and transient design situations. Design wind, wave, and current conditions and the corresponding extreme metocean parameters are described in API 2MET, whereas the associated design loads/actions (load/action effects) are given in Section 9. The extreme level earthquake (ELE) is described in API 2EQ, and the associated design loads/actions (load/action effects) are given in Section 11. Verification of the ULS of structural components is described in Sections 13, 14, 15, and 17. Verification of FLS for the accumulation of load/action effects resulting from environmental conditions is described mainly in Section 16 and partly in Section 15.

Other hazards belonging to group 1 and not treated by the regular design process, as well as hazards belonging to group 2, are specially addressed by a requirement that the structure satisfy particular ALS. ALS are verified through specified accidental situations (see 10.1.3 to 10.1.6).

Hazards falling into group 3 are sometimes referred to as residual accidents and may normally be neglected for design.

When a hazardous event occurs, the structure shall not be damaged disproportionately to the original cause. The intention of the ALS is to ensure that the structure can tolerate specified accidental situations and, if damage occurs, that it subsequently maintains structural integrity for a sufficient period under specified environmental conditions to enable evacuation to take place. This requirement is sometimes called the “progressive collapse limit state” (PLS).

Irrespective of a need for analysis and evaluation, due consideration should always be given to providing robustness as specified in 7.9, both in the design of the structure and in the layout and arrangement of facilities and equipment on or in the topsides.

10.1.3 Accidental Situations

Accidental situations used in this document, relate to two types of hazards as follows.

- a) Hazards associated with specially identified accidental events: These hazards belong to group 1 or group 2 and are not included in the regular design process.
- b) Hazards associated with abnormal environmental loads/actions: Abnormal environmental loads/actions can occur due to the possible exposure to very rare and abnormally severe environmental conditions. The return period of the loads/actions (load/action effects) exceeds the return period of the loads/actions (load/action effects) due to design environmental conditions. The corresponding hazards fall into group 2.

The two types of hazards are different by nature. In principle, accidental events can in some cases be avoided by taking appropriate measures to eliminate the source of the event or by bypassing and overcoming its structural effects. In contrast with this, the possible occurrence of abnormal loads/actions cannot be influenced by taking such measures.

An accidental situation normally comprises the occurrence of an identified accidental event or of abnormal environmental loads/actions, in combination with expected concurrent operating conditions and associated permanent and variable loads/actions.

In the occurrence of an accidental event or abnormal environmental loads/actions, the structure can sustain damage. Such consequential damage creates a new situation, which is characterized by the after damage design situation. Subsequent to the hazardous event having occurred, the after damage design situation, which considers the structure's further behavior, shall also be addressed. The structure shall consequently be designed following a two-stage procedure:

- a) design the structure for the accidental situation involving the hazard considered;
- b) after the hazardous event has occurred, check the after damage design situation in relation to specified environmental loads/actions.

The second step is only necessary if the resistance of the structure is reduced by structural damage caused by the hazard (see also 10.1.6).

Identified accidental events are evaluated using accidental design situations (see 3.1.2 and 10.1.4).

Abnormal environmental loads/actions are evaluated using abnormal design situations (see 3.1.1 and 10.1.5).

Consequential damage is evaluated using after damage design situations (see 3.1.4 and 10.1.6).

10.1.4 Identified Accidental Events

Typical examples of accidental events are as follows:

- collision from vessels (see 10.2);
- impact from dropped objects (see 10.3);
- fire or explosion (see 10.4).

Designers can choose between avoiding a hazard (e.g. by taking special preventive measures such as operational restrictions), minimizing the (structural) consequences of the considered hazard (e.g. by implementing protective measures) or designing for the hazard.

If the design option is chosen, one or more accidental design situations shall be defined (see 10.1.3), and design requirements shall be established taking account of the operational conditions and the type, function, and location of the structure. Structural components shall meet the strength requirements of Sections 13, 14, 15, and 17. However, when checking ALS for accidental events, all partial load/action and resistance factors may be set to 1.0.

As accidental events are on the whole statistically independent, only one accidental event need be considered for a particular accidental design situation. If certain accidental events are not statistically independent, a corresponding accidental design situation or situations should be specially considered.

10.1.5 Abnormal Environmental Loads/Actions

The appropriate return period for the abnormal design situation for abnormal loads/actions due to wind, wave, and current is dependent on the exposure level of the platform, the environmental climate (see API 2MET), the acceptability of overall risk to a platform and its inhabitants, and the way in which that overall risk is apportioned to individual hazards facing the platform and its inhabitants. Provided a sufficient deck elevation (see 6.1.3.2) is available to avoid wave impingement on the topsides, the abnormal environmental design situation can be reasonably accounted for.

In lieu of other information (e.g. Table 6.3-1), this abnormal design situation may be based on a return period of 1000 years for an exposure level L-1 platform; see 10.5 and API 2MET for a discussion of such conditions. This return period is an indication of the order of magnitude rather than a precise number, since accurate databases for such small probabilities of exceedance rarely exist.

The abnormal design situation for abnormal earthquake loads/actions may be based on the abnormal level earthquake (ALE) according to API 2EQ. See Section 11 for further seismic design considerations.

Structural components shall meet the strength requirements of Sections 13, 14, 15, and 17. However, as for accidental events, when checking ALS for abnormal environmental loads/actions, all partial load/action and resistance factors may be set to 1.0.

10.1.6 Damaged Structures

10.1.6.1 Requirements for Damage Tolerance

For new structures, or those being assessed for compliance with this document, the occurrence of damage following an accidental event or abnormal environmental loads/actions can only be based on analysis. If a linear structural analysis of a corresponding design situation indicates no damage, then no further checks of the ALS for that situation are required. However, if damage is indicated, a reliable prediction of the nature and the extent of the damage normally requires the application of nonlinear structural analysis methods.

The requirements of a platform having sustained damage are, to an extent, dependent on the environmental climate in the geographical area, regulatory requirements, and the owner's preference. The possible requirements vary between the following:

- a) the platform shall remain intact for a period sufficient for all personnel to be evacuated (if manned at the time of the hazardous occurrence), for all process plant to be made safe, and for all risk of pollution to be removed;
- b) the platform shall remain intact and without further damage for such time as it takes for repairs to be effected and for the platform to be made fit-for-service;
- c) the platform shall remain fit-for-service throughout.

In lieu of specific requirements, the after damage design situation shall be analyzed using environmental conditions corresponding to loads/actions (load/action effects) of a return period commensurate with twice a conservative assessment of the time required to effect suitable repairs (including inspection, design, fabrication, and installation), by which the structure's strength would be restored to the design strength; the minimum return period shall be 1 year. The specified partial load/action factors for the in-place situation (see Section 9) shall be used. The strength of damaged components shall either be assessed following a rational approach, e.g. using the procedures of 7.7, or be neglected. Equations for the strength of certain dented members are given in Section 13. Otherwise, the partial resistance factors and strength requirements for the in-place situation given in Sections 13, 14, 15, and 17 shall be applied.

10.1.6.2 Assessment of Structures Following Damage

For existing structures that have been damaged during service following an accidental event or abnormal environmental loads/actions, the nature and extent of the actual structural damage shall be established by inspection in accordance with API 2SIM. The modeling of the structure for the corresponding after damage design situation shall reflect the assessment of the actual damage as accurately as possible. Verification of the after damage design situation and the associated design criteria shall be in accordance with API 2SIM.

Appropriate analysis combined with the information on the actual damage gathered by inspection shall determine any requirements for a shut-in and/or evacuation of the platform, as well as the need for any immediate (possibly temporary) repairs, while awaiting a decision and plan for the implementation of definitive repairs or for abandonment.

10.2 Vessel Collisions

10.2.1 General

Vessel impact shall be addressed for structures with exposure levels L-1 and L-2.

The effect of a vessel impact shall be evaluated if the probability of collision is not negligible. In such an evaluation, the nature of all vessel operations in the platform vicinity shall be taken into account. Such vessel operations can include vessels servicing the platform, fishing vessels working in the area, and passing vessels.

Potential vessel impact on the structure's waterline members, risers, and external wells shall be evaluated. Barge bumpers, boat landings, and other external fendering may be used as protection.

Depending on the risk of collision and the consequences for the structural integrity of the structure, an analysis of vessel impact conditions can be required. Irrespective of whether an analysis is required, robustness in relation to vessel collisions should be incorporated into the design by indirect means, such as the following:

- avoiding weak elements in the structure (particularly at joints);
- selecting materials with sufficient toughness;
- ensuring that critical components are not placed in vulnerable locations.

10.2.2 Collision Events

In a rigorous impact analysis, if required, accidental design situations shall be established representing bow, stern, and beam-on impacts on all exposed components. Evaluation of accidental design situations shall be in accordance with 10.1.

The collision events shall represent both a fairly frequent condition, during which the structure should only suffer insignificant damage, and a rare event where the emphasis is on avoiding a complete loss of integrity of the structure.

The following two energy levels shall be assessed:

- a) low-energy level—representing the most frequent condition, based on the type of vessel that would routinely approach alongside the platform (e.g. a supply boat) and that would have a velocity representing normal maneuvering of the vessel approaching, leaving, or standing alongside the platform;
- b) high-energy level—representing a rare condition, based on the type of vessel that would operate in the platform vicinity, drifting out of control in the worst sea state in which it would be allowed to operate close to the platform.

The general requirements given in 10.1 apply to the accidental design situations for both energy levels. Level a) above represents an SLS to which the owner can set their own requirements based on practical and economical considerations, whereas level b) represents an ULS in which the structure is damaged but progressive collapse shall not occur.

In both cases, the analysis shall account for the vessel's mass, its added mass, orientation, and velocity. Effective operational restrictions on vessel approach sectors can limit the exposure to impacts in some areas of the structure.

The vertical height of the impact zone shall be established, based on the dimensions and geometry of the structure and the vessel, and shall account for tidal ranges, operational sea state restrictions, vessel draught, and motions of the vessel.

10.2.3 Collision Process

The energy absorbing mechanisms during the collision should be evaluated. Typically, local member denting, elastic and plastic deflection of the impacted member, global elastic and plastic response of the whole structure, and denting of the ship are the main mechanisms.

In a rigorous impact analysis, the collision loads/actions should be evaluated based on a dynamic time simulation. The duration of the simulation should be sufficient to cover all relevant phases of the collision and the energy dissipation process.

10.3 Dropped Objects

When evaluating the impact risk from dropped objects, the nature of all crane operations in the platform vicinity shall be taken into account. If the probability of impact is not negligible, relevant accidental design situations shall be defined and evaluated in accordance with 10.1. Depending on the consequences for the structural integrity of the structure, the need for a rigorous impact analysis shall be determined.

Irrespective of whether a rigorous analysis is required, robustness in relation to dropped objects should be incorporated into the design by indirect means, such as the following:

- avoiding weak elements in the structure (particularly at joints);
- selecting materials with sufficient toughness;
- ensuring that critical components are not placed in vulnerable locations.

10.4 Fires and Explosions

Hydrocarbon pool fires on the sea surface and jet fires from ruptured risers can cause heating of structural components and hence degradation of their properties. Sources of hydrocarbons can include conductor or riser fracture, or spillage from the topsides following a process vessel rupture, whereas ignition sources can include radiation from oil burners and flares. API 2FB contains requirements and recommendations for fires and explosions. If the probability of exposure of the structure to fires is not negligible, relevant accidental design situations shall be defined and evaluated following the requirements of 10.1. Consequential damage resulting from such accidental design situations should be provided for in the structure design as required by 10.1.

Generally, conventional steel-framed structures with relatively small topsides that do not have enclosed compartments containing flammable fluids or ignition sources do not require design against explosion. However, if explosion studies of the topsides indicate significant or unusual loads/actions on the structure or specific support requirements to ensure that the topsides integrity is maintained, then such loads/actions or support requirements shall be provided for in the structure design in accordance with 10.1.

NOTE API 2TOP^[20] contains specific requirements for topsides structures for fires and explosions.

10.5 Abnormal Environmental Load/Actions (Robustness)

Robustness shall be provided by evaluating situations for abnormal environmental loads/actions due to wind, wave, current, and, where applicable, earthquake.

Abnormal environmental loads/actions due to wind, wave, and current, with a return period in accordance with Table 6.3-1, shall be determined and used to check structural integrity with all partial load/action and resistance factors set to 1.0. Where applicable, abnormal environmental loads/actions and load/action effects due to ALE shall also be determined; see Section 11 for further seismic design considerations.

11 Seismic Design Considerations

11.1 General

The procedures in API 2EQ for design against seismic events shall be followed. This section complements API 2EQ by presenting the following:

- a) the design loads/actions, combinations of loads/actions, and load/action effects resulting from ground motions;
- b) the requirements for a fixed steel offshore structure subjected to seismic loads/actions.

As outlined in API 2EQ, a two-level seismic design procedure shall be followed. The structure shall be designed to the ULS for strength and stiffness when subjected to an ELE, from which it should sustain little or no damage. The structure is next checked when subjected to an ALE to ensure that it meets reserve strength and energy dissipation requirements. The structure may sustain considerable damage from an ALE, but structural failures causing loss of life and/or major environmental damage shall not be expected to occur.

11.2 Seismic Design Procedure

API 2EQ gives alternative procedures for determining seismic loads/actions and alternative methods of evaluation of seismic activity. The selection of the procedure and the method of evaluation of activity depend on a structure's seismic risk category (SRC). The SRC according to API 2EQ depends on the platform's exposure level and the seismic zone in which it stands. The API 2EQ requirements are the following:

- determine seismic loads/actions using either the simplified seismic load/action procedure or the detailed seismic load/action procedure, as specified in API 2EQ;
- evaluate seismic activity and the associated response spectra for the design of a structure against excitation of its base by ground motions using either ISO maps, regional maps, or a site-specific seismic hazard analysis, as specified in API 2EQ;
- demonstrate the ALE performance of the structure, which can require a nonlinear analysis.

Specific recommendations for the ductile design of fixed steel offshore structures, dependent on the structure's SRC as given in Table 11.2-1, are presented in 11.4.

Table 11.2-1—Applicability of Ductile Design Recommendations

SRC	Recommendations for Ductile Design
1	Not Applicable
2	Optional
3	Recommended
4	Recommended

11.3 Seismic Reserve Capacity Factor

Both simplified and detailed seismic load/action procedures require an estimate of the seismic reserve capacity factor, C_r . This factor represents a structure's ability to sustain ground motions due to earthquakes beyond the level of ELE. It is defined as the ratio of spectral acceleration that causes structural collapse or catastrophic system failure to the ELE spectral acceleration. For fixed steel offshore structures, the representative value of C_r may be estimated from the general characteristics of a structure's design in accordance with Table 11.3-1.

Table 11.3-1—Representative Values of Seismic Reserve Capacity Factor, C_r

Characteristics of Structure Design	C_r
The recommendations for ductile design according to 11.4 are followed and a nonlinear static pushover analysis according to 11.6.3 and API 2EQ is performed to verify the global performance of the structure under ALE conditions.	Variable up to 2.80, as demonstrated by analysis
The recommendations for ductile design according to 11.4 are followed, but a nonlinear static pushover analysis to verify ALE performance is not performed.	2.00
The structure has a minimum of three legs and a bracing pattern consisting of leg-to-leg diagonals with horizontals, or X-braces without horizontals. The slenderness ratio ($K \cdot L/r$) of primary bracing in vertical frames is limited to no more than 80 [with the corresponding column slenderness parameter λ not exceeding $80/\pi \times \sqrt{(f_{yc}/E)}$; see Section 13] and $f_y D/E \cdot t \leq 0.069$. For X-bracing in vertical frames, the same restrictions apply, where the length L to be used depends on the loading pattern of the X-bracing. A nonlinear analysis to verify ALE performance is not performed.	1.40
If none of the above characterizations apply.	1.10

Where the values of C_r according to Table 11.3-1 are not adopted, previous experience may be used to assume a value. In such cases, both of the following conditions shall apply.

- a) If the simplified seismic load/action procedure is followed, the assumed value of C_r shall not exceed 2.8 for L-1 platforms, 2.4 for L-2 platforms, or 2.0 for L-3 platforms.
- b) A nonlinear time-history analysis in accordance with 11.6.4 shall be performed to ensure survival in the ALE event. As an alternative, a static pushover analysis in accordance with 11.6.3 and API 2EQ may be performed to confirm that C_r is equal to or higher than that assumed.

11.4 Recommendations for Ductile Design

In seismically active areas, platform response to rare, intense earthquake motions can involve inelastic loads/actions, and structural damage can occur. The provisions of this section are intended to ensure that structure-foundation systems planned for such areas remain stable in the event of a rare, intense earthquake at the site. This can be achieved by providing sufficient system redundancy such that load redistribution and inelastic deformation will occur before collapse and by minimizing abrupt changes in stiffness in the vertical configuration of the structure. Adequate ductility can be demonstrated by adhering to the design practices outlined below or by nonlinear analysis, where applicable.

The number of legs is very important for the reserve strength and energy dissipation capacity of a structure (sometimes also termed its ductility) and hence for the values of the seismic reserve capacity factor C_r given in 11.3. A structure with a minimum of three legs may be assumed to have a C_r value of 1.40 if the associated requirements given in 11.3 are met. A structure may be assumed to have a C_r value greater than or equal to 2.0 only if the structure-foundation system meets all of the requirements listed below.

- a) The structure shall have eight or more legs supported by piles.
- b) The piles shall be founded in competent soils that are not susceptible to liquefaction during the ELE and the ALE events.
- c) The legs of the structure, including any enclosed piles, shall meet the requirements of 11.5.1 using twice the design environmental load/action E during the ELE event.
- d) The vertical framing transmitting shear forces between horizontal frames shall consist of X-braces, or single (leg-to-leg) diagonals, and shall be arranged so that the shear between horizontal frames is carried by braces in both tension and compression (see Figure 6.3-2). K-bracing should not be used.
- e) Horizontal members shall be provided between all adjacent legs at horizontal framing levels in vertical frames (see Figure 6.3-1) and these horizontal members shall have sufficient strength in compression to support the redistribution of loads/actions resulting from any buckling of adjacent diagonal braces.
- f) The slenderness ratio ($K \cdot L/r$) of primary bracing in vertical frames shall be limited to no more than 80 [with the corresponding column slenderness parameter λ not exceeding $80/\pi \times \sqrt{(f_{yc}/E)}$; see Section 13] and $f_y D/E \cdot t \leq 0.069$.
- g) All nontubular members at connections in vertical frames shall have greater local buckling strength than global buckling strength, can develop fully plastic behavior (i.e. are compact sections), and shall meet the requirements of 11.5.1 using twice the design environmental load/action E during the ELE event.
- h) All joints for primary structural members in the structure shall be sized to meet the minimum strength requirements given in 14.2.3. In lieu of this requirement, joint strengths may be verified by time-history analyses simulating the ALE event following the requirements of 11.6.4.

11.5 ELE Requirements

11.5.1 Partial Load/Action Factors

Each member, joint, and foundation component shall be checked for strength using the internal force (load/action effect), S , resulting from the design load/action, F_d , calculated using Equations (11.5-1) and (11.5-2):

$$F_d = 1.1 G_1 + 1.1 G_2 + 1.1 Q_1 + 0.9 E \quad (11.5-1)$$

where E is the inertia load/action induced by the ELE ground-motion and determined using dynamic analysis procedures such as response spectrum analysis or time-history analysis. G_1 , G_2 , and Q_1 are defined in 9.2.1 to 9.2.3 and shall include loads/actions that are likely to be present during an earthquake.

When contributions to the internal forces due to weight oppose the inertia loads/actions due to the earthquake, the partial load/action factors for permanent and variable loads/actions shall be reduced such that:

$$F_d = 0.9 G_1 + 0.9 G_2 + 0.8 Q_1 + 0.9 E \quad (11.5-2)$$

where G_1 , G_2 , and Q_1 shall include only loads/actions that are reasonably certain to be present during an earthquake.

11.5.2 ELE Structural and Foundation Modeling

The mass used in the dynamic analysis shall consist of the mass of the structure associated with the following:

- a) the permanent loads/actions G_1 and G_2 ,
- b) 75 % of the variable loads/actions Q_1 ,
- c) the mass of entrapped water, and
- d) the added mass.

The added mass may be estimated as the mass of the displaced water for motion transverse to the longitudinal axis of individual structural members and of appurtenances. For motions along the longitudinal axis of the structural members and appurtenances, the added mass may be neglected.

The structural model shall include the three-dimensional distribution of the stiffness and mass of the structure. Joints in the model may be treated as rigid. Asymmetry in the distribution of the stiffness and mass of the structure can lead to significant torsion and should be considered in design.

In computing the dynamic characteristics of braced, pile-supported fixed steel offshore structures, a modal damping ratio of up to 5 % of critical may be used in the dynamic analysis of the ELE event. Additional damping, including hydrodynamic or soil-induced damping, shall be substantiated by special studies.

Pile-soil performance and pile design requirements should be determined on the basis of studies that consider the design loads/actions, installation procedures, cyclic and strain rate effects on soil properties, and characteristics of soils as appropriate for the axial or lateral capacity algorithm being used. When an equivalent linear foundation model is used, the stiffness of the pile foundation should be compatible with the expected level of nonlinearity in the axial and lateral foundation response. See API 2GEO for guidance.

The foundation stiffness can have a large effect on the natural period(s) of the structure; hence, upper and lower bound foundation stiffness values shall be used to evaluate the sensitivity to a range of possible natural periods.

For the design of piles for the ELE event, a partial resistance factor of 1.25 shall be used to determine the axial pile capacity (see 17.3.4) and a partial resistance factor for the p - y curves of 1.0 shall be used to determine the lateral pile performance (see 17.8).

11.6 ALE Requirements

11.6.1 General

The structure-foundation system shall be analyzed to demonstrate an ability to withstand the rare, intense ALE earthquake. This analysis shall establish that:

- a) the structure does not globally collapse during the earthquake, and
- b) the structural integrity of the topsides is maintained.

The characteristics of the rare, intense ALE event shall be developed in accordance with API 2EQ. The stability of the structure-foundation system during the ALE event shall be demonstrated by analytical procedures that are rational and reasonably representative of the expected response of the structural and soil components to intense ground shaking. Performance requirements of the structure-foundation system shall be in accordance with API 2EQ.

The designer should develop a thorough insight into the performance of the structure and its foundation during the ALE event. The expected nonlinear effects, including material yielding, buckling of structural components, and pile failures, shall be adequately modeled and captured. The time-history method of analysis is recommended; however, a static pushover analysis may be used. Whereas an analysis for the ELE event focuses on internal forces and stresses, the focus of an analysis for the ALE event is on strains and displacements.

11.6.2 ALE Structural and Foundation Modeling

Analysis of the ALE event should be based on the most realistic estimate of values of parameters such as steel yield strength, member slenderness, member strength, and soil strength used for determining the axial capacity and the lateral performance of piles. The partial resistance factors for axial capacity and lateral pile performance under ALE conditions shall be 1.0. Models of the structural and soil elements should include, as appropriate, the representative degradation of strength and stiffness under abnormal load/action reversals, the interaction of axial forces and bending moments in structural members, hydrostatic pressures, local inertial effects caused by members vibrating out of plane, and the P - Δ effect of earthquake loads/actions.

In an analysis of the ALE event, the most realistic estimate of the yield strength of the material and of the nonlinear behavior of the joints (including any design strength bias) should be used. Local joint deformations greater than 5 % of the chord diameter shall be specifically investigated; otherwise, the joint should be assumed to fracture at this deformation level.

NOTE 1 See API 2GEO for general guidance on foundation modeling.

NOTE 2 The foundation strength can have a large effect on the energy dissipation mechanisms within the structure-foundation system, and sensitivity to higher than expected soil stiffness should be investigated.

11.6.3 Nonlinear Static Pushover Analysis

The objective of a static pushover analysis is to verify that the seismic reserve capacity factor, C_r , of the structure as designed is greater than that initially estimated for design. The loads/actions used in a static pushover analysis should represent the pattern of ALE seismic loads/actions on the structure and foundation. Load/action patterns in a pushover analysis may be constructed to match the shear and moment distributions determined from an ALE

response spectrum analysis along the height of the structure. Pushover analyses should be performed in several directions, as follows, in order to identify the structure's weakest direction:

- with the pattern of seismic loads/actions aligned with the longitudinal (end-on) axis of the structure;
- with the pattern of seismic loads/actions aligned with the transverse (broadside) axis of the structure;
- with the pattern of seismic loads/actions aligned with one or more diagonal axes of the structure.

Diagonal direction(s) can be the weakest direction(s), especially with regard to foundation performance.

Yielding of structural members or piles shall not occur at global load/action levels lower than or equal to the global ELE load/action, F_{ELE} (Figure 11.6-1). The seismic reserve capacity factor, C_r , shall be the smallest value computed among all pushover analyses (i.e. weakest direction). C_r may be estimated using Equations (11.6-1) and (11.6-2) from the global seismic load/action–deformation curve obtained in a static pushover analysis, e.g. from global shear vs. deck displacement (see Figure 11.6-1):

$$C_r = C_{sr} C_{dr} \quad (11.6-1)$$

where C_{sr} is a factor corresponding to the strengthening regime of the load/action–deformation curve, estimated as follows:

$$C_{sr} = \Delta_u / \Delta_{ELE} \quad (11.6-2)$$

where Δ_{ELE} is the deformation caused by the global ELE load/action F_{ELE} and Δ_u is the deformation corresponding to F_u , the ultimate load/action where the slope of the load/action–deformation curve becomes negative (see Figure 11.6-1).

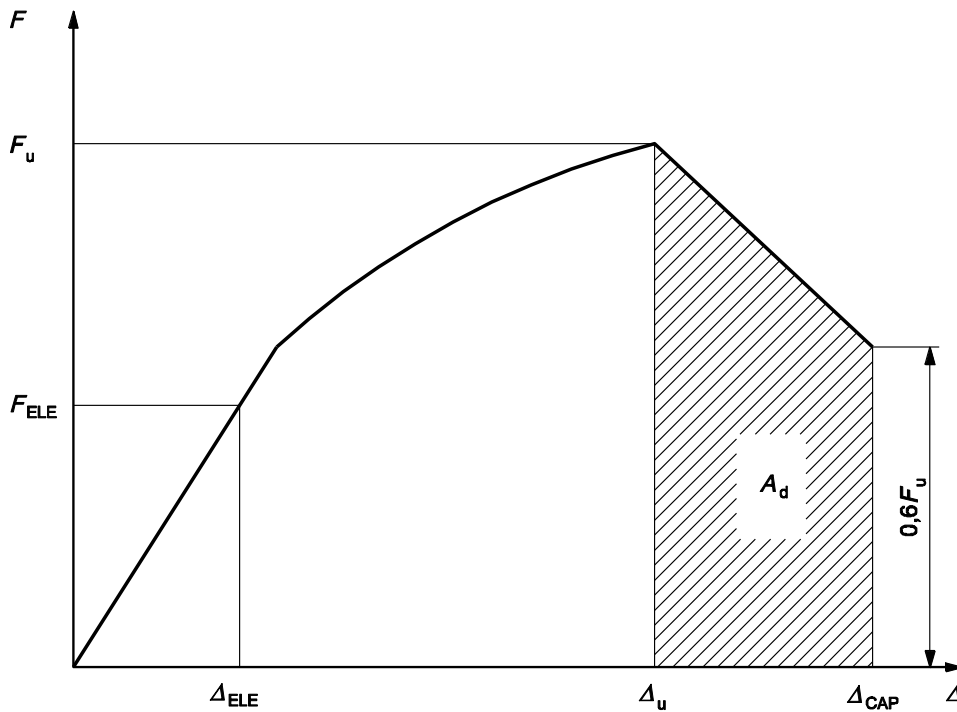


Figure 11.6-1—Seismic Load/Action–Deformation Curve

C_{dr} is a factor corresponding to the degrading regime of the load/action–deformation curve. It is a measure of the energy dissipation capacity of the structure beyond the ultimate seismic load/action and the corresponding deformation. C_{dr} is estimated using Equation (11.6-3):

$$C_{dr} = \sqrt{1 + \frac{A_d}{F_u \Delta_u}} \quad (11.6-3)$$

A_d is the area under the load/action–deformation curve starting from Δ_u and ending with Δ_{CAP} . For the purpose of a nonlinear static pushover analysis, the deformation capacity shall be assumed to be the deformation where the global load/action falls to 60 % of F_u .

The deformation that should be used is a displacement of the deck (location of largest mass). As ratios of deformations and areas are used, the exact elevation within the deck is unlikely to have any effect. It is expected that the nonlinearity will be confined to the structure below the deck and that the seismic load/action–deformation curves for different deck elevations will be similar.

The above determination of C_r presupposes that the primary sources of degradation of the global resistance have been properly modeled in the static pushover analysis to represent the structure's weakest direction, e.g. soil degradation, buckling of compression members, local buckling of members due to rotations at the member end (reducing the plastic moment capacity) and pushover direction.

Alternatively, Δ_u can be defined as the deformation where the slope of the load/action–deformation curve is reduced to 5 % of the initial elastic slope, with C_{dr} assumed to be equal to 1.0.

NOTE Dynamic time-history techniques may also be used to effectively replicate static pushover analyses; see, for example, Reference [24].

11.6.4 Time-history Analysis

The objective of a nonlinear time-history analysis is to demonstrate that the structure can be expected to sustain the ALE seismic event without collapse and without major structural failure of the topsides. It shall be demonstrated that the structure-foundation system is expected to remain stable under the deformations induced by the ALE ground motions. The structure-foundation system shall be considered unstable when the deformations and degradation are severe enough to cause collapse under the influence of applicable permanent and variable loads/actions.

The response of the structure-foundation shall be determined by a minimum of four sets of ground motion records characterizing the likely intensity, frequency content, and duration of the ALE event. If seven or more time-history records are used, global structural survival shall be demonstrated in half or more of the time-history analyses. If fewer than seven time-history analyses are used, global survival shall be demonstrated in at least four time-history analyses.

11.7 Topsides Appurtenances and Equipment

Topsides design accelerations shall include the effects of the global dynamic response of the structure and, if appropriate, of the local dynamic response of the topsides and appurtenance itself.

Topsides equipment, appurtenances, and piping that are both small (less than 5 % of the topsides weight) and stiff (local natural period less than half of the three-dimensional global 5th mode) may be designed quasi-statically to resist peak deck accelerations from modal or time-history analyses of the overall structure.

It is recommended that general deck or topsides response spectra for design of major topsides equipment be obtained from time-history analyses of the complete structure. General deck or topsides spectra should account for the variability in seismic responses of different locations on the deck to properly envelope horizontal, vertical,

and torsional motions of the deck. For a tall or seismically sensitive topsides subsystem (major equipment or module, e.g. drilling system, living quarters, flare), more specific response spectra can be developed, based on the motion of some support location point. Flexibility between that chosen support location and the subsystem should be considered in order to properly capture the dynamic characteristics of the subsystem in an uncoupled analysis. A coupled analysis may be used, but care shall be exercised when coupling exists between the subsystem and the global modes. When coupling occurs, the sensitivity to modeling assumptions shall be evaluated.

The topsides response spectra should be the median values from at least four time-history analyses. Direct spectra-to-spectra generation techniques may also be used; however, such methods should be calibrated against the time-history method. The topsides and other equipment specific response spectra shall be broadened to account for uncertainties in structure frequencies and soil-structure interaction.

Seismic loads/actions on major topsides equipment, piping, and appurtenances should be derived by dynamic analysis using either:

- a) an uncoupled analysis with deck-level floor response spectra as input, or
- b) a coupled analysis that properly includes a simple dynamic model of the relevant part of the topsides or the appurtenance in the global structural model (beware of modeling uncertainty and coupled interaction when including major equipment in dynamic structural models).

Equipment, piping, and other topsides appurtenances shall be designed and supported such that induced ELE loads/actions can be resisted. Induced ELE displacements and deformations of the topsides shall be limited or designed against to avoid damage to the equipment, piping, appurtenances, and supporting structures.

Drilling and well servicing structures shall be designed for ELE loads/actions, using the response spectrum or time-history methods of analysis. It is important that these movable structures and their associated setback and pipe-rack tubulars are tied down or restrained at all times, except when the structures are being moved.

Safety critical systems and structures on or in the topsides shall be designed such that they are functional during and after the ALE event. Hazardous systems shall be designed such that they do not fail catastrophically (e.g. rupture) during the ALE event. In lieu of performing an ALE analysis of deck-supported structures, topsides equipment, and equipment tie-downs, they shall be designed with an increased partial load/action factor on E of 1.15 rather than 0.9 in Equations (11.5-1) and (11.5-2). This partial load/action factor increase is only applicable if the ratio of ALE to ELE ground motions is less than 2.0. In areas where the ALE to ELE ratio exceeds 2.0, an additional increase in partial factor should be considered.

The use of walk-down techniques that are described in API 2TOP [20] is recommended for ensuring that all necessary equipment is adequately supported for seismic conditions.

12 Structural Modeling and Analysis

12.1 Purpose of Analysis

Structural analysis is the process of determining the load/action effects in a structure, or part thereof, in response to a given set of loads/actions. Load/action effects required for the design of fixed steel offshore structures typically include the following:

- internal section forces, which shall not exceed the strength of the section;
- displacements and vibrations, which shall be within acceptable limits for operation of the structure;
- support reactions, from which the required foundation capacity can be determined.

This section presents the requirements for structural analyses necessary for the design of fixed steel offshore structures for the following design situations:

- a) pre-service situations including fabrication, loadout, transportation, and installation (for which the design loads/actions are given in Section 8);
- b) in-place situations, including:
 - 1) permanent, variable, and environmental loads/actions (for which the design loads/actions are given in Section 9), but excluding fatigue, which is dealt with in full in Section 16;
 - 2) accidental situations (see Section 10); and
 - 3) seismic events (see Section 11);
- c) removal situations (see Section 8);
- d) structure reuse (see Section 23).

Structural analysis shall be based on loads/actions and partial load/action factors specified in Sections 8 to 11. Load/action effects calculated by structural analysis shall be used as input to the design or assessment process for components as described in Sections 13 to 17.

12.2 Analysis Principles

12.2.1 Extent of Analysis

Structural analyses shall be performed to provide load/action effects suitable for checking the structure for all relevant design situations.

The number, types, and extent of analyses to be performed should cover all stages of the lifetime of the structure, i.e. pre-service or in-place. However, if it can be demonstrated that particular stages in the design service life of a component of the structure do not govern its design, such stages need not be analyzed for that component.

Requirements for typical analyses to be performed for a fixed steel structure and its components are given in 12.4. Complex or unusual structures can require other forms of analysis to determine all significant load/action effects in the structure.

Loads/actions shall be applied simultaneously, if they can coexist, and shall include partial load/action factors, in accordance with Sections 8 to 11, for each design situation being checked.

12.2.2 Calculation Methods

Various calculation methods may be used for the determination of load/action effects in response to a given set of loads/actions. These include, but are not limited to, hand calculations and computer methods, such as spreadsheets and finite element analyses (FEAs).

Nonlinear FEA is appropriate for the analysis of foundation components and may also be used to estimate ultimate strengths of structural components, and hence RSRs.

12.3 Modeling

12.3.1 General

This subsection provides guidance on the modeling of a structure, its structural components, their interactions, and support conditions for the purpose of structural analysis.

Structural analyses can normally be limited to the structure being considered. However, where adjacent structures interact with the structure, the interaction should be included in the analysis, to the extent required to predict the load/action effects accurately.

Due consideration should be given to the interaction between primary structure, secondary structure, and the topsides structure (see 12.3.5).

Consideration should further be given to any imperfections due to fabrication and to accidental damage.

12.3.2 Level of Accuracy

Material properties and geometrical tolerances are defined in Section 19 and Annex E. Where a particular component does not conform to these tolerances, the effect of the nonconformities on structural behavior shall be evaluated. The effect of tolerances shall also be incorporated into the analysis where the structural design is particularly sensitive to their magnitude. The need for nonlinear analysis to account for nonconformities shall be evaluated.

Where FEA is performed, consideration should be given to the type and accuracy of the element formulations, to ensure that sufficient elements are used to predict the structural behavior, particularly in areas where the load/action effects change rapidly.

12.3.3 Geometrical Definition for Framed Structures

12.3.3.1 General

Dimensions used in structural analysis calculations shall represent the structure as accurately as necessary to produce reliable values of the load/action effects.

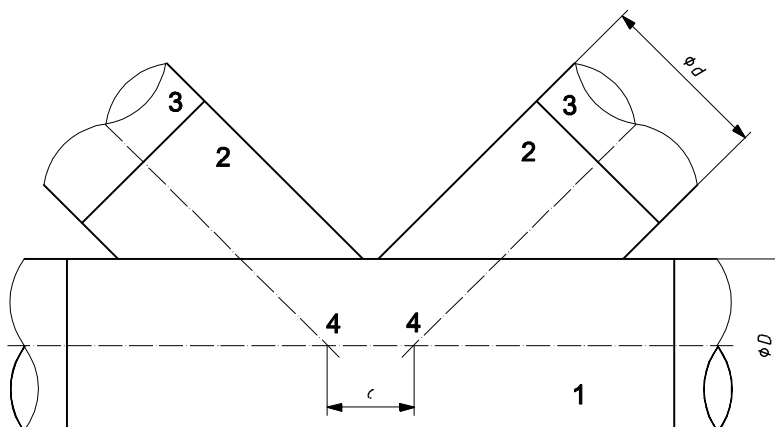
A space frame model shall be created to describe the three-dimensional geometry of the structure. The model is usually based on nominal sizes and dimensions. Where as-built dimensions differ from nominal sizes by more than the permissible tolerances set out in Annex E, the effect of this dimensional mismatch shall be assessed and incorporated into the analysis if appropriate.

The dimensions used in structural analysis calculations shall produce reliable, conservative values of the structural resistance and load/action effects. Stiffness and strength assessment shall normally be based on nominal sizes and dimensions. Mass, weight, and hydrodynamic load/action calculations shall account for corrosion allowances (if any) and for the presence of marine growth.

A space frame model consists of members that are connected at nodes. The continuous member at a connection point (a node) is called the “chord”; members that end or are interrupted at the chord are called “braces.” The assembly of chord and braces is a joint.

The nodes of the space frame model shall be the intersection points (working points) of the centerlines of the braces with the centerline of the chord (see Figure 12.3-1). The offset is the distance between brace working points. Joints with zero offset are called “concentric.”

For a global analysis, offsets smaller than $D/4$ (see Figure 12.3-1) need not be included in the space frame model of the structure. However, for a local analysis, the axial forces in the brace members of K- and X-joints (see 14.2.4) cause secondary moments in the chord that can be important and should generally be assessed (see A.12.3.3.1).

**Key**

- 1 chord can
- 2 stub
- 3 brace
- 4 working points

- D can diameter
- c offset
- d brace diameter

Figure 12.3-1—Definition of Offset**12.3.3.2 Member Modeling**

The structural model of a framework should primarily consist of beam elements representing the axial, bending, shear, and torsional stiffnesses of the structural members. In some cases, special modeling arrangements are necessary for pile clusters and large diameter members provided for storage or flotation or they may become unintended major load paths.

The structural members of the framework should be modeled using one or more beam elements for each span between the nodes of the model of the framework. The number of beam elements depends on the element formulation, loads/actions, and potential for local dynamic response. The element properties should be varied along the spans to account for variations in the component cross-section where appropriate. The element properties should also be adjusted to account for any corrosion allowance.

The structure should be modeled in detail to include the primary and secondary structures, conductors, and appurtenances to ensure that load/action effects are accurately predicted. If this is not possible, the necessary detail of the model should be prioritized as follows, in the order given:

- a) primary structure;
- b) secondary structure—if representation of secondary framework is simplified, a more rigorous analysis of the secondary framework should be undertaken for its local design with local analysis boundary conditions applied;
- c) components (provided for temporary conditions such as launch framing, mudmats, etc.)—these should be included in the model for appropriate analyses unless:
 - 1) they are removed prior to the design situation under consideration, or
 - 2) the load/action effects on these components (and the effect of their stiffness on the rest of the structure) are not significant;
- d) conductors—these should be included in the model if they contribute significantly to the overall stiffness or strength of the structural system, including the foundation; otherwise, they may be treated as appurtenances (see 12.3.6, 12.3.7.4, and 12.3.7.5).
- e) appurtenances.

Care should be taken not to exclude secondary structure or appurtenances that can be subjected to excessive load/action effects induced by deformation of the primary structure.

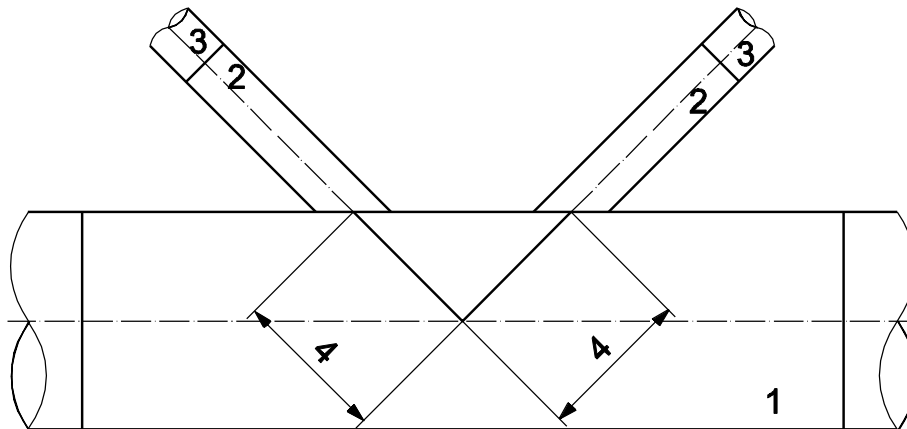
Interaction with adjacent structures shall be considered and appropriate loads/actions applied to represent any relative movement. Particular consideration shall be given to loads/actions that do not result in overcoming friction at bearings, and loads/actions resulting from pipe-work movements.

When the structural contribution of any component is neglected, the self weight, buoyancy, and hydrodynamic loads/actions on the component shall still be included in the model.

12.3.3.3 Joint Modeling

For typical structures, depending on the diameter of the chord, the length between the physical end of the brace stub and the centerline of the chord can be significant and can affect the calculation of member end forces and stresses, weights, masses, hydrodynamic and hydrostatic loads/actions. In such cases, it is customary to model the length of braces between the outer surface of the chord and its centerline as rigid connections (rigid links); joint flexibility of brace and chord connections is thus neglected.

For joints between small diameter braces and large diameter chords, rigid links may be introduced between the longitudinal axis of the chord and the intersection of the chord can with the brace stub (see Figure 12.3-2). Where rigid links are included, care shall be taken that they only contribute stiffness to the structural model; their stiffness shall be at least one order of magnitude greater than that of the brace and the brace stub, if present. Rigid links shall not contribute to the mass and shall not attract direct hydrodynamic or aerodynamic loads/actions. Some computer analysis systems allow the introduction of member end offsets to achieve these requirements. Care should be taken when using elements with very high stiffnesses to represent rigid links, as computational problems can occur.



Key

- 1 chord can
- 2 brace stub
- 3 brace member
- 4 rigid link

Figure 12.3-2—Definition of Member Rigid Links

Joint flexibility is an elastic effect in which bending moments are redistributed. Where several braces connect with the chord at the same joint, the flexibility effects are dominated by the rotation of the brace ends as a result of chord wall rotation. This chord wall rotation results from the forces and moments in all the braces at the node and can increase or decrease any particular member end moments. In practice, full joint modeling is rarely performed; where joint flexibility is modeled, it shall be modeled throughout the structure.

12.3.4 Modeling of Material Properties

Material properties (e.g. modulus of elasticity, Poisson's ratio, yield strength, ultimate tensile strength) used in the structural analyses of a new design shall reflect the materials specified for construction.

For existing structures, material properties may be based on statistical analysis of the material certificate data. Where no documentation can be found for assessing existing structures, coupon tests may be used to indicate which grades of steel were used to fabricate the structure. For additional information, see 8.5.4.4.2 of API 2SIM, First Edition.

The value of yield strength found from a coupon test shall be compared with the specified minimum yield strength (SMYS) for the component in question. Although the term "specified minimum yield strength" is adopted in this document, alternative terminology for "yield strength" is used in different regions of the world and in different standards from the same region in the world. Other commonly used terms include "yield point" and "nominal yield strength." The term "characteristic yield strength" has also been used when it is in fact just the nominal value.

Material properties used for nonlinear analysis (e.g. true stress-strain curve) shall be based on published or test data for the appropriate steel specification and grades, including, if appropriate, strain hardening and plasticity effects (see also 12.6.7 and API 2SIM). Mean yield strength may be used instead of nominal yield strength.

When assessing the resistance of a structure to elevated temperatures arising from fires, account shall be taken of the dependency of material properties, including yield strength, f_y , and Young's modulus, E , on temperature.

12.3.5 Topsides Structure Modeling

Where the support structure and the topsides structure interact significantly, a combined model of the support structure and topsides structure shall be used. This applies equally to the design of new structures and the assessment of existing structures. The modeling of the topsides structure shall represent the interaction of the two structures and the sequence of installation. The primary topsides structure shall be included in the structural model of the whole platform. The model shall include any deck members and deck plating and stiffening that affect or contribute to the distribution of forces into the support structure. Differential deflections shall be taken into account when these can have a significant effect on the performance of the structures, particularly for serviceability conditions. The sequence of stressing introduced into rigid jointed structures arising from installation should be considered.

If separate models for the support structure and the topsides structure can be justified, the stiffness of the topsides structure and its interface with the support structure shall be modeled in sufficient detail to ensure that its self weight and applied loads/actions are appropriately distributed to the structure.

NOTE Detailed requirements for the topsides structure are given in API 2TOP [20].

12.3.6 Appurtenances

The support and movement of appurtenances shall be modeled with appropriate constraint conditions so that the loads/actions applied to them, including those due to self weight, are correctly transferred to the main structure. An example of an arrangement that often occurs is that unrestrained differential axial displacement and rotations between an appurtenance and its guide(s) are possible but that the guide(s) constrain the appurtenance to have the same lateral movements. The effect of any eccentricity in the support arrangement on the appurtenance and the structure shall be checked and included in the analysis if significant.

12.3.7 Soil-structure Interaction

12.3.7.1 General

The stiffness of a piled foundation generally displays nonlinear characteristics. The appropriate soil properties shall be determined in accordance with API 2GEO and Section 17.

The foundation shall be modeled and analyzed using nonlinear soil p - y , t - z , and Q - z curves generated from site-specific data, when available. Particular care should be taken in modeling relatively thin layers near the sea floor, where p - y soil properties change rapidly. The soil layers shall be modeled accurately to at least the depth at which the piles are expected to terminate. If the pile tip rests on weak soil layers, the possibility of punch-through shall be investigated.

For an analysis to determine the behavior of the structure, an equivalent simple linear elastic model may be used to model a piled foundation. When such a simple model is used, the forces and displacements at the pile heads (at the sea floor) shall be checked for compatibility with the predicted nonlinear foundation behavior.

Pile penetrations should be conservatively assessed based on upper and lower bound considerations of soil properties and pile driving hammer properties. Either upper or lower bounds can be more critical for different situations and different components of the overall structural system.

The effects of global sea floor scour, of local scour in granular soils, and of the partial loss of soil-pile contact in cohesive soils shall be taken into account.

12.3.7.2 Pile Groups

When modeling the individual piles in a pile group (or cluster), the nonlinear soil p - y , t - z , and Q - z curves shall be adjusted to account for pile group effects. Alternatively, an interaction model to account for the interaction between individual piles shall be included in the analysis. See also 17.9.

For pile groups, each pile-sleeve connection may be represented by one or more coaxial tubular member(s). Any bracing, shear plates, or similar connections between a pile-sleeve and a leg shall be simulated by an appropriate combination of bar, beam, and plate elements that models the method of transfer of forces between the leg and pile-sleeve connection.

12.3.7.3 Pile Connectivity

The constrained movement of ungrouted piles within legs shall be modeled. Each pile shall be allowed to move independently of its leg, both along its axis and around all rotational axes, but the leg and piles shall be constrained to have the same lateral displacements (normal to the longitudinal axis of the leg/pile). Such constraints should be applied at locations where contact is expected to occur, such as at shims or centralizer locations or at plan bracing levels. Where grout is used in the leg/pile annulus, the effects of this connectivity shall be included.

12.3.7.4 Conductor Modeling

Conductors can contribute significantly to the foundation stiffness and lateral strength of a structure. In such cases, the conductors should be modeled and analyzed as structural components with their own foundation stiffness.

The lateral behavior of the conductors below the sea floor may be modeled in the same way as piles, either as equivalent members or individually, employing nonlinear soil p - y curves generated from site-specific data when available. For vertical, self-supporting conductors that are constrained only in the horizontal direction, modeling the conductors' axial behavior using the soil t - z and Q - z characteristics may be omitted. Group effects of conductors shall be assessed.

When equivalent members are used, the displacements at the sea floor shall be checked for compatibility with the predicted nonlinear foundation behavior.

It should be noted that the installation of conductors can cause significant disturbance to the integrity of the soil and the contribution from the conductors to the foundation stiffness can be much lower than anticipated. Where the foundation stiffness associated with the conductors is included in the analysis, consideration should be given to the effectiveness of the conductor foundations and to situations where not all of the conductors are installed. In the latter cases, although the hydrodynamic loads/actions on the structure are lower, the load/action effects on the main structure piles can be higher. An upper bound value of the load/action effects on the main structure piles should be determined by releasing the horizontal and bending (but not torsion) restraints at the base of each conductor. An upper bound value of the load/action effects on any conductor bracings close to the sea floor should be determined by modeling an upper bound foundation stiffness.

12.3.7.5 Conductor Connectivity

The constrained movement of conductors within their guide frames shall be modeled. Each conductor shall be allowed to move independently of its guide, both along its axis and around all rotational axes, but the conductor and its guide shall be constrained to have the same lateral displacements (normal to the longitudinal axis of the conductor) as the guide. The contact load/action, including friction, between curved or slanting conductors and their guide frames requires special consideration—in some cases, it may be necessary to account for the effects of gaps between the conductor and guide frame.

12.3.8 Other Support Conditions

The load/action of water pressure on a structure (or on a supporting barge) while floating (e.g. during transportation or installation) shall be evaluated from suitable hydrostatic and hydrodynamic analyses for wave and current, as appropriate, and shall be applied to appropriate external surfaces of the structure (and/or the supporting barge). See also 12.4.3.2.

Where balanced loads/actions are applied to a structure or component, the minimum number of artificial restraints shall be applied for certain types of analysis (e.g. lift) to prevent rigid body movement. Artificial restraints shall be applied as follows:

- a) as translational restraints that do not inhibit relative deflections within the structure or the component;
- b) at points where no direct loads/actions are applied, although distributed self weight may be applied at a restraint;
- c) to extreme locations in the structure or the component;
- d) to relatively stiff points in the structure or the component.

Where components of the structure are not fully restrained, such as conductors within guides (see 12.3.7.5) and bearing surfaces between topsides structure and bridge structure, consideration should be given in the analysis for movement at such interaction points.

12.3.9 Local Analysis Structural Models

Local models may be created to study the behavior of a structural component in greater detail. The boundaries of the local model shall be sufficiently remote from the component being studied that the method of applying the boundary conditions does not affect the component's behavior. The loads/actions applied to the local model shall be the same as those applied to that part of the global model represented by the local model; the application of the loads/actions to the local model may be more detailed than in the global model.

The boundary conditions applied to the local model shall be one of the following:

- a) forces extracted from an analysis of the overall structural system and applied as direct loads/actions at the boundaries of the local model together with the minimum number of restraints to prevent rigid body movement; or
- b) displacements extracted from an analysis of the overall structural system and applied as imposed deformations at the boundaries of the local model; or
- c) a set of related forces, displacements, and stiffnesses determined by substructuring the overall structural system in which one or more substructured parts represent local models with connections to the remainder of the structural system at the intended boundaries of the local model.

In the absence of forces, displacements, and stiffnesses at the boundaries of the local model derived from the overall structural system, appropriate conditions shall be applied to the boundary of the component to simulate the behavior of the surrounding structure. Where there is uncertainty over the boundary conditions, a range of possible conditions should be considered.

12.3.10 Loads/Actions

In a structural analysis, the applied loads/actions shall include partial load/action factors appropriate to the load/action being assessed and the limit state checks to be performed; it is common practice to subdivide loads/actions into categories to allow separate application of partial load/action factors and to provide flexibility in the analysis.

Loads/actions shall be determined by recognized methods, taking into account the variation of actions in time and space, relevant environmental and soil conditions, and the limit states being addressed (see Sections 8 to 11).

Permanent and variable loads/actions shall be based on the most likely representative values for the situation being analyzed. Consideration should be given to both representative maximum and minimum values. In a number of cases representative minimum values can govern the design of some components.

Hydrostatic pressures shall be based on the likely range of fluid surface elevations and densities. Hydrostatic pressures on structures while afloat, such as during transportation, installation, and removal, shall include the effects of tilt of the structure due to wind, wave load/action, or damage.

The load/action effects due to wave, current, and wind loads/actions shall include the influence of such loads/actions on structure motions when the structure is afloat or on a transportation barge. In cases where dynamic load/action can be of importance, such influence should be considered in determining load/action effects. Quasi-static or dynamic analyses shall be used in accordance with 12.5.

Less well-defined loads/actions, such as variations in topsides weight and center of gravity and actions during loadout, should be represented by ranges of representative values, the structure being checked for the most onerous values.

Loads/actions on a local model shall be included in that model (see 12.3.9).

12.3.11 Mass Simulation

For dynamic analysis and for motion prediction during transportation and installation, a suitable representation of the mass of a structure is required. Generally, masses should be determined as a representative combination of mass values for the situation considered. Such a mass simulation shall include relevant contributions from, at least, the following:

- a) all structural components and appurtenances;
- b) the operating mass of all intended equipment;
- c) the estimated mass of temporary items, such as slings, storage, lay-down, etc.;
- d) masses of any fluids contained within the structure or topsides, including vessels, piping contents, oil, and bulk storage, etc.;
- e) snow and ice accumulation on the structure, if significant;
- f) drill cuttings or other deposits on the structure, if significant;
- g) the mass of marine growth;
- h) the added mass representing hydrodynamic reaction to structure motions (allowing for any marine growth thickness);
- i) the internal mass (flooded members) moving with the structure.

The magnitude and distribution of mass shall be as accurate as necessary for the determination of all significant rigid body motions or modes of vibration for the structural analysis being performed. Particular attention shall be paid to the location of individual centers of mass and their moment of inertia, in particular for heavy equipment and storage tanks in the topsides structure. Sufficient dynamic analyses shall be carried out to determine the sensitivity of load/action effects to anticipated variations in the magnitude and distribution of mass.

Calculation of the hydrodynamic reactions to structure motions (represented in the form of added mass and damping) shall be based on the best available published information or suitable hydrodynamic analysis. In lieu of such analysis, the added mass may conservatively be taken as equal to the full mass of water displaced by the structure.

12.3.12 Damping

Damping arises from a number of sources, including material damping, soil damping, hydrodynamic damping, and frictional damping between moving parts. For fixed steel offshore structures, damping is relatively small. In the absence of substantiating values obtained from measurements of existing structures, conservative (lower bound) damping values shall be used.

12.4 Analysis Requirements

12.4.1 General

All structural analyses performed shall simulate, with sufficient accuracy, the load/action effects in the structural system or the structural component for the limit state being considered. This shall be achieved by appropriate selection of the analysis type and method of solution with due regard to the nature of the loads/actions and the nature of the structural behavior. More than one analysis type can be required to simulate the behavior of the structural system during a given phase of its life.

Sufficient global analyses of the structural system shall be performed to allow subsequent assessment of the structural components. As a minimum, the structural system shall be assessed for ULS conditions. Local analysis shall be performed to assess any detail that is complex in form and/or appears from the global analysis to experience large load/action effects. Such local analysis may be based on nonlinear methods.

Table 12.4-1 summarizes the applicability of analysis types for different design situations.

Table 12.4-1—Applicability of Analysis Types for Design Situations

Situation	Analysis Type		
	Static/Quasi-static Linear Elastic	Dynamic Linear	Nonlinear
1. Pre-service and Removal Situations			
1.1 Fabrication Fatigue during fabrication	Appropriate	Appropriate (needed for wind-induced vibrations for long slender braces only; see Section 16)	
1.2 Loadout	Appropriate		
1.3 Transportation Fatigue during transportation	Appropriate	Appropriate (needed for long sea tows only; see Section 16)	
1.4 Installation	Appropriate	Appropriate for fatigue analysis of piles during pile driving	
1.5 Removal	Appropriate		Appropriate if needed to demonstrate that no failures will occur during removal
2. In-place Situations			
2.1 Normal operating and extreme environmental situations Fatigue during the in-place situation	Appropriate for structures that are not unduly dynamically sensitive (see Section 9) Appropriate for structures that are not unduly dynamically sensitive (see Section 16)	Appropriate for structures that are appreciably dynamically sensitive Appropriate for structures that are appreciably dynamically sensitive (see Section 16)	Appropriate for foundation components
2.2 Accidental situations	Appropriate but can be unduly conservative	Appropriate but can be conservative	Appropriate
2.3 Seismic events	Appropriate for extreme level earthquake (ELE) event (see Section 11)	Appropriate (see Section 11)	Appropriate for abnormal level earthquake (ALE) event (see Section 11)
NOTE 1 Quasi-static linear analysis includes inertia loads/actions to represent rigid body structural accelerations. NOTE 2 Linear static or quasi-static analysis as defined here include simple nonlinearities, e.g. from gaps opening and closing. NOTE 3 Nonlinear local analysis for component buckling shall be included (e.g. using the provisions of Section 13). NOTE 4 Nonlinear global analysis may be used to demonstrate that a structure is safe when a linear analysis predicts failure. The assumptions used in the nonlinear model shall be demonstrated to be pessimistic. NOTE 5 All the analyses in the table shall be performed using partial load/action and resistance factors as specified in Sections 9 to 11 and 13 to 15, respectively. In cases where these partial factors are deemed inappropriate, the analyses may be performed within a structural reliability framework. When reliability methods are used, the input data and methods used shall be demonstrated to be satisfactory.			

12.4.2 Fabrication

Fixed steel offshore structures, particularly large or slender structures, or those with particularly slender structural components, should be reviewed to determine whether analysis for fabrication is required.

Where such analysis is undertaken, assessment shall be given to the sequence and to the completeness of fabrication (e.g. whether welding has been completed at particular joints) in determining load/action effects. Specific evaluation shall also be given to the stability and strength of structural components during fabrication and to the need for any temporary supports or strengthening. Adequate support for equipment subjected to temporary loads/actions, such as for crane footings, shall be demonstrated.

Fabrication phase analyses may generally be performed using static analysis. However, dynamic analysis shall be considered for the determination of load/action effects in tall, slender structures and long thin appurtenances in view of vibrations induced by wind turbulence.

12.4.3 Other Pre-service and Removal Situations

12.4.3.1 General

Loadout, transportation, and installation (including launch, floating, upending, and/or lifting) are situations in which the structure is in motion and mass inertial forces due to accelerations need to be considered. In most cases, in particular for loadout and installation, static linear elastic analysis is adequate, but care should be taken in the definitions of the loads/actions and the application of partial load/action factors to ensure that only feasible and stable conditions are analyzed. In general, the analysis should be undertaken for unfactored loads/actions, with load/action effects factored after the internal forces are calculated (see 8.2.4.2).

Requirements for the calculation of loads/actions and partial load/action factors for pre-service and removal situations are given in Section 8.

12.4.3.2 Loadout

Loadout refers to the series of operations needed to transfer the structure from its assembly position in the fabrication yard to the transportation or launch barge. During loadout, the structure is generally supported unevenly, with changing support conditions. Limitations on the acceptable support conditions should be identified during the design phase and should not be exceeded during loadout. Because loadout occurs slowly, inertial loads/actions may be neglected, and linear static analysis is sufficient. However, the nonlinearities from lift-off of supports should be taken into account.

12.4.3.3 Transportation

Fixed steel offshore structures and topsides are normally transported to location by barge. During transportation, the structures are subjected to loads/actions generated by heave, sway, surge, pitch, and roll accelerations. Parts of the structure overhanging the barge are also susceptible to loads/actions due to buoyancy, wave load/action, and wave slam if the structure penetrates the water. Structural components can also be subjected to excitations associated with wind-induced vortex shedding. These load/action effects should neither exceed the strength of structural components nor cause excessive fatigue damage.

The structural integrity of a steel structure should be assessed for the transportation phase (sea tow) of the structure, whether it is self-floating, barge-supported, or barge-assisted. Analysis of situations during sea tow may normally be based on quasi-static analysis, representing the motion of the steel structure due to the most onerous combination of heave, sway, surge, pitch, and roll accelerations, as predicted by hydrodynamic analysis. For a long sea tow, where the duration of the tow precludes assurance of calm weather for the whole length of the tow, fatigue damage should be assessed (see Section 16).

The model of the structure (see 12.3) shall include all sea-fastenings. If the bending and torsional stiffness of the barge is significantly greater than that of the structure, the barge may be assumed to be rigid and appropriate boundary conditions applied where the sea-fastenings are connected to the deck of the barge. A quasi-static analysis shall be carried out for design combinations of self weight, roll, pitch, and heave, with roll and pitch accelerations applied about the appropriate roll and pitch centers for the combined barge and structure, with the barge and structure inclined at the maximum angle of roll and/or pitch.

If barge flexibility is important, the model shall include the structure and the barge, as well as a representation of the hydrostatic and hydrodynamic loads/actions due to the surrounding water. The barge should preferably be modeled using plate/shell elements, having both in-plane (membrane) and out-of-plane bending stiffness. The total loads/actions on the barge and the structure, with the water pressure distribution derived from a motion analysis, should be in quasi-static equilibrium.

Sea-fastening and grillage supporting the structure on the barge shall be checked for strength using design situations, including appropriate combinations of the following:

- self weight, and/or
- roll and pitch angles, and/or
- roll, pitch, and heave accelerations.

The modeling should recognize the sequence of loadout of the structure onto the barge and the attachment of the sea-fastenings. In particular, as sea-fastenings are installed after the structure loadout on the transport barge is completed, it is common practice to assume that they do not resist any of the structure dead weight in the still water condition. The validity of this assumption should be verified in consideration of the relative flexibility of the barge and the structure.

Structural components, sea-fastenings, and grillage shall be checked for strength in accordance with Sections 13 to 15. The fatigue life of structural components and sea-fastenings shall be assessed, when relevant, in accordance with Section 16. Structural components that are subjected to wave slam shall be assessed for strength and fatigue due to the associated loads/actions.

12.4.3.4 Installation

Structures are either launched or lifted into the water. When launched, a time-history analysis should be performed to:

- a) determine the launch trajectory to ensure sufficient clearance from the sea floor;
- b) determine the stability of the structure after separating from the barge;
- c) check that the structure is not overstressed as it:
 - 1) rotates and enters the water,
 - 2) upends, with or without crane assistance,
 - 3) is ballasted so that it rests on the sea floor; and
- d) check that the barge, including the rocker arm, is not overstressed at any stage of the launch.

The strength of the structure shall be checked using representative loads/actions generated at various stages of the launch and upending process. The loads/actions shall include hydrostatic and hydrodynamic loads/actions on the structure, loads/actions from self weight at the appropriate orientation relative to the structure and the loads/actions due to the instantaneous translational and rotational accelerations of the structure. Sufficient boundary conditions shall be applied to prevent rigid body movement of the structure and to provide a datum for deflections. The structure shall be analyzed for each set of representative loads/actions.

Transients are important for lift installations, but a full dynamic analysis is not normally carried out. Instead static analysis is performed employing representative loads/actions to cover motions of the structure and crane vessel, loads/actions due to “snatching” as the structure is lifted off the barge and due to impact between the structure and the sea floor, etc. As with the launch, sufficient boundary conditions shall be applied to prevent rigid body movement and to provide a datum for deflections. Except at the lifting points, the support reactions should be negligible. When performing single-point lift analysis, it is often convenient to determine the center of gravity and orient the model so that the center of gravity is vertically below the lifting point. The effects of variances in the lengths of the slings should be taken into account.

Installation of the topsides by lifting does not normally impose loads/actions on the support structure that are greater than, or in a different direction to, those induced by the topsides in in-place situations. Accordingly, analysis of the loads/actions on the support structure due to the installation of the topsides may be omitted.

For float-over topsides installation, the design of the structure legs and other components can be substantially affected by both the vertical and lateral loads/actions induced by the topsides/barge system. This design situation shall be appropriately analyzed.

For each installation analysis, structural components shall be checked for strength in accordance with Sections 13 to 15. In particular, structural components and appurtenances attached to the structure can be subjected to significant slamming load/action upon entering the water. Evaluation shall be given to the stresses induced by slamming.

The structure should also be assessed for on-bottom stability, when the structure is resting on the sea floor with the aid of mudmats prior to piling being completed; the cases considered should include that of the structure with and without pile(s) stabbed.

12.4.3.5 Removal

Analysis of the structure for removal shall represent the structure in all appropriate configurations, and shall include the suction and pull out resistances of pile stubs (i.e. any length of pile attached to the structure following pile cutting) prior to separation from the sea floor. Weights of accumulated debris and marine growth left in place at the time of removal shall also be assessed, together with the weights of the piles and pile grout to be lifted during removal.

The effect of any structural damage, including degradation of the materials during the life of the structure, shall be evaluated. The determination of the material degradation shall take into account any results from underwater surveys and inspections.

12.4.4 In-place Situations

12.4.4.1 General

The design of a fixed steel structure for in-place situations can generally be based on a static or quasi-static analysis, using a linear elastic model of the structure coupled with a nonlinear foundation model, unless there is a likelihood of significant dynamic or nonlinear response to a given type of load/action. In such cases, linear dynamic or nonlinear static or dynamic analysis approaches, respectively, are normally required, as described in 12.5.

It is generally acceptable to base static structural analysis on the principles of deterministic analysis, predicting load/action effects to specific events.

The structure shall be analyzed for operating and extreme environmental loads/actions. Section 9 defines the partial load/action factors for permanent and variable loads/actions for in-place situations.

Selection of environmental parameters for the analysis of in-place situation(s) shall be in accordance with API 2MET. The analysis shall ensure that sufficient variations and combinations of the parameters are analyzed to capture the most onerous load/action effects throughout the structure.

If soils data are available, a pile-soil interaction analysis shall be carried out using nonlinear soil p - y , t - z , and Q - z curves (see 12.3.7). When, during preliminary analysis, soils data are not yet available, a simplified model of the foundation may be used. Such a simplified model may also be used if design or assessment of the foundation and the lower portion of the structure is not the purpose of the analysis.

The foundation capacity shall be verified in accordance with Section 17 and the strength of structural components in accordance with Sections 13 to 15.

12.4.4.2 Extreme Environmental Conditions

To determine the governing load/action effects for each structural component, the structure shall be analyzed for environmental loads/actions due to the most onerous combinations of wind, current, and wave conditions, as specified in Section 9. To achieve this, a sufficient number of appropriate combinations of the following environmental parameters should be considered:

- a) water depth, including effects of tides and storm surge;
- b) wind speed, omnidirectional or in relation to wind direction;
- c) current speed, omnidirectional or in relation to current direction;
- d) wave height and period, omnidirectional or in relation to wave direction;
- e) wave position (phase angle) relative to the structure.

In regions with seasonal ice cover, loads/actions due to ice also shall be evaluated.

12.4.4.3 Accidental Situations

Analyses for accidental situations shall be performed in accordance with Section 10.

The analysis of collisions (such as ship, helicopter, or iceberg collisions) may be performed by considering the energy absorption arising from the combined effect of local and global deformation. The energy absorbed shall be compared with, and equated to, the impact (kinetic) energy due to collision, and the results shall be documented.

The analyses should allow for the following:

- a) the energy absorbed in local deformation of the structures at the point of impact, e.g. denting of the colliding ship and denting of a structural component or a group of structural components, and
- b) the energy absorbed in elastic and plastic deformation of the structural system, e.g. using quasi-static nonlinear analysis of the structural system under the estimated impact.

The resistance of the impact area may be studied using local models. These will generally involve nonlinear geometrical and material analyses, since the local structure may be allowed to yield and deform substantially under the accidental loads/actions. Appropriate boundary conditions should be applied far enough away from the damaged region for inaccuracies to be minimized.

Global analysis of the structure under accidental situations, including those due to dropped objects, fire, and explosion, shall normally be required to ensure that a progressive collapse is not initiated. The analysis shall include the weakening effect of damage to the structural system in the affected area. Linear elastic redundancy analysis may be used to demonstrate that removal of a damaged member will not initiate progressive collapse. Nonlinear analysis may be used to simulate the redistribution of load/action effects as section resistances are exceeded. The global analysis may be based on a simplified representation of the structure that is sufficient to simulate progressive collapse.

The natural frequencies of the structural system shall be estimated and compared with the frequency content of the accidental loads/actions. Analysis of the overall structural system shall include any significant dynamic response under these loads/actions.

12.4.4.4 Seismic Events

Analyses for seismic events shall be performed in accordance with Section 11 for the ELE and ALE. The structure shall sustain little or no damage due to ELE events for which linear analysis is appropriate. During ALE events, the structure may sustain considerable local damage (but not a total collapse), provided structural failures do not cause loss of life and/or environmental damage. Nonlinear analysis can be required to demonstrate conformance with these safety requirements.

12.4.4.5 Fatigue Analysis

Requirements for, and guidance on, performing a fatigue analysis are provided in Section 16.

12.4.4.6 Analysis for Reserve Strength

Reserve strength analyses can be used to determine the adequacy of unconventionally framed structures and for the assessment of existing structures (see 7.10). They are intended to identify the collapse strength of the space frame structure and foundation under the influence of abnormal environmental loads/actions. The uncertainties associated with foundation capacity are significantly greater than those associated with the ultimate strength of space frame structures (see A.12.5.4). In performing reserve strength analyses, it is therefore important to make this distinction and to evaluate both structural and foundation failure modes. Owing to this, the following strategy is recommended:

- a) structural or foundation failure should be identified using an analysis based on the mean (or best estimates) of structural steel properties and soil properties;
- b) where foundation failure occurs before structural failure, structural failure should be determined by assuming a foundation capacity based on upper bound estimates of soil properties;
- c) the upper bound approach, b) above, provides an assessment of the steel structure strength.

In each case, the foundation should be included in the model in order to evaluate potential combined structure and foundation collapse mechanisms.

For dynamically responding structures (see 12.5.2), dynamic effects can influence collapse due to environmental loads/actions.

Reserve strength evaluation is used to estimate the most likely collapse strength of a structure with partial resistance factors set to 1.0. Since partial resistance factors are omitted, an ultimate strength evaluation shall be interpreted and used with care.

12.5 Types of Analysis

12.5.1 Natural Frequency Analysis

A natural frequency analysis shall be carried out to determine whether dynamic behavior is significant. A reasonably accurate structural model, including both stiffness and mass, shall be developed and analyzed so that the natural frequencies of the structural system can be compared to the frequencies of any excitations.

Dynamic behavior is likely to be significant if any natural frequency, particularly the fundamental frequency, is similar to the frequency of an excitation. Analyses should be carried out to determine any sensitivity to structural stiffness, mass distribution, and foundation stiffness.

12.5.2 Dynamically Responding Structures

Structures for which dynamic behavior is significant are generally referred to as dynamically responding structures. Redundant, multi-legged fixed structures (e.g. jackets, towers, etc.), with fundamental natural periods or having one or more components with natural periods greater than 2.5 s to 3 s usually respond dynamically to wave load/action during sea tow or in-place situations. For other types of structures, such as monotowers and caissons, dynamic behavior can be significant even with natural periods of 1 s or less.

Other loads/actions to which the structural system can be subjected, such as wind turbulence, seismic events, impact, and explosion, can also cause dynamic effects of significant magnitude at other natural periods.

12.5.3 Static and Quasi-static Linear Analysis

Static analysis is appropriate when dynamic effects are minimal and can be assumed to be covered by the partial load/action and resistance factors.

Quasi-static analysis is appropriate when dynamic effects can be assumed to be approximately uniform throughout the structural systems and so small that one static analysis or a series of static analyses, with a small correction for dynamic effects, is adequate. The correction for dynamic effects may be determined using one or the other of the following methods:

- a) by performing one static analysis in which the loads/actions are enhanced by a set of equivalent quasi-static inertial loads/actions representing the dynamic response;
- b) by performing a series of static analyses over an appropriate range of frequencies of excitation, where, at each frequency, the corresponding loads/actions are applied and the calculated load/action effects are multiplied by the DAF of a single degree of freedom (SDOF) system at that frequency.

Guidance on the application of the chosen method is given in A.12.5.3.

12.5.4 Static Ultimate Strength Analysis

The collapse strength of a space frame structure is usually expressed as the ratio of the global environmental load/action at collapse, measured in terms of base shear or overturning moment, to the global environmental load/action used for design (see 7.10). The collapse strength of the structural system may therefore be defined by the maximum value of the global environmental load/action sustained by the structural system during the analysis; guidance for the analysis is provided in A.12.5.4.

The following analysis procedure is suggested. Unfactored permanent and variable loads/actions are applied followed by the environmental loads/actions due to wind, wave, and current. The environmental load/action is gradually increased until structural collapse occurs.

For a static strength analysis, the following checks shall be performed:

- a) assess the deformation patterns of the structural system to identify whether a clear failure mechanism has developed;
- b) check the plastic strains in all members that have yielded in tension against the ductility limits set by fracture considerations (see 12.6.6);
- c) check the internal axial forces acting on the pile heads against pile capacities and performances;
- d) check the joint forces against the joint strength.

The structural behavior should also be studied to determine how forces are redistributed as members, joints, and piles fail.

12.5.5 Dynamic Linear Analysis

When dynamic response is considered significant, the structural system shall be analyzed for dynamic behavior. An accurate structural model including both stiffness and mass shall be used. Lumped and consistent mass modeling shall be employed as appropriate.

The type of analysis is governed by the form of applied loads/actions:

- steady state analysis in response to harmonic loads/actions, as required for spectral analysis;
- transient analysis in response to arbitrary time-history loads/actions, as can be required for accidental situations and nonlinear loads/actions due to waves or earthquakes.

For both types of analysis, the behavior of the structure and the foundation are assumed to be linear elastic.

12.5.6 Dynamic Ultimate Strength Analysis

Dynamic nonlinear structural analysis may be performed in one or the other of the following ways.

- Full transient dynamic nonlinear analysis, in which the environmental load/action is simulated in time.
- Quasi-static analysis, in which static nonlinear analysis procedures are carried out using the environmental loads/actions enhanced by a set of equivalent quasi-static inertial loads/actions representing the dynamic response. The set of equivalent quasi-static inertial loads/actions may be determined in a manner analogous to the procedures in 9.8 and A.9.8. A static pushover analysis can then be performed using the procedure suggested in 12.5.4.

NOTE The guidance given in A.12.5.4 is equally applicable to dynamic ultimate strength analysis.

12.6 Nonlinear Analysis

12.6.1 General

When, for example, the structure is subjected to abnormal environmental loads/actions due to wind, wave, and current or an earthquake, or to accidental loads/actions from ship impact, fire or explosion, and when a linear analysis predicts:

- displacements of a magnitude that are likely to cause second-order $P-\Delta$ effects,
- joint failure,
- member buckling, and/or
- stresses that exceed the yield strength of the material,

then nonlinear analysis may be performed to demonstrate that overall structural integrity is not impaired.

Nonlinear analysis shall include all relevant loads/actions to represent the maximum loads/actions to which the structure or component will be subjected. The loads/actions shall be applied to simulate the load/action history, e.g. the loads/actions due to permanent and variable loads/actions shall be applied first, to be followed by the loads/actions due to wave and current. The modeling of load/action history is necessary because nonlinear behavior is dependent on the sequence of the force distribution through the structure. The magnitude of the loads/actions should be the representative values, i.e. without partial load/action factors. Similarly, no partial resistance factors should be applied to the resistances. Any suspected sensitivity of behavior to material properties, e.g. Young's modulus and yield strength, shall be investigated by additional nonlinear analyses with modified material properties (see 12.6.7).

The behavior of the structural system to the applied loads/actions can be static or dynamic (see 12.5.2). Depending on this, a (quasi-)static or transient dynamic nonlinear analysis shall be performed in accordance with 12.5.4 or 12.5.6.

As a nonlinear analysis is undertaken without partial factors, an appropriate margin of safety—dependent on the situation being considered—should be included in the interpretation of the results.

NOTE Reference DNV-SINTEF-BOMEL gives guidance on nonlinear analysis methods including representation of nonlinear member and joint behavior.

12.6.2 Geometry Modeling

In addition to the requirements of 12.3, the following apply.

When creating models for a nonlinear analysis, due consideration should be given to the types of elements and size of mesh required to predict nonlinear structural behavior. For example, sufficient elements shall be employed along the length of a member to simulate elasto-plastic buckling involving yielding at the ends and middle of the member; insufficient elements will overestimate the strength of the member. In addition, sufficient elements shall be employed to simulate all possible modes of failure—especially beams that may fail due to lateral torsional buckling. Thus, a model for nonlinear analysis will usually be more complex than the model required for extreme environmental and fatigue analysis.

12.6.3 Component Strength

The collapse of a space frame structure usually results from progressive failure of its components, in particular its primary members and/or joints. Correct modeling of component behavior is therefore essential for predicting nonlinear behavior. This may be achieved using a finite element formulation or phenomenological models.

The redistribution of internal forces following a component failure, and the prediction of collapse behavior, therefore depends on the accuracy of the models used to describe the full path of member and joint behavior, from their initial stress-free state to failure and including their post-failure response.

12.6.4 Models for Member Strength

The ultimate strengths of undamaged tubular members may be established using the formulae given in Section 13, with all partial resistance factors set to 1.0. The models for describing the ultimate strength and the subsequent post-collapse behavior of members in a space frame structure shall be able to describe the following failure modes:

- a) tensile and compressive material yielding of a member's cross-section, which may be achieved by adopting a plastic hinge formulation or by monitoring the stresses and strains in the member;
- b) buckling of a member and the post-buckling redistribution of internal forces that can involve local buckling; for open section members this includes both Euler and lateral torsional buckling as well as taking the interaction between yielding and buckling into account;
- c) local buckling.

The models should allow for the effects of initial geometrical imperfections and hydrostatic pressure. Consideration should be given to any locked-in forces due to fabrication.

12.6.5 Models for Joint Strength

The ultimate strengths of undamaged tubular joints may be established using the formulae given in Section 14, with all partial resistance factors set to 1.0. If both the ultimate strengths for axial forces and moments, as provided by these joint strength formulae, exceed the axial yield force and plastic moment strengths of the brace cross-section, the brace will fail before the joint. If this is the case, joint failure does not need to be included in the structural model. If it is not the case, joints which can participate in the failure mechanism shall be identified and their behavior included in the nonlinear structural model.

12.6.6 Ductility Limits

When joints are stronger than the incoming brace members, the model for member tensile failure shall include a ductility criterion that will disconnect (sever) a member end when the plastic strains have reached the tensile fracture strain. In lieu of more advanced analysis, the member end shall be conservatively assumed to disconnect when the tensile strains in the extreme fibers of a steel member exceed 5 %.

12.6.7 Yield Strength of Structural Steel

If the mean yield strength of the steel for a part can reliably be determined, this mean yield strength may be used for the part under consideration instead of the SMYS of the material. The SMYS is the representative value of the yield strength and should be used in all other cases.

The mean yield strength can be determined from either generic studies of the specific grades or from statistical analysis of coupon tests from particular plates used in the specific structure. However, coupon tests from the particular material used in the structure may be used only if they belong to a single population (see A.12.6.7). Benefit may be taken of strain rate effects; however, the strain rates used for nominally static tests shall be taken into account.

Increased strength due to strain hardening should be considered carefully as often there are uncertainties in the post-yield stress-strain behavior, such as the following:

- the effect is important in reducing strains in plastic hinges;
- for compression members, strain hardening occurs only in the post-collapse range and reduces the amount of internal force redistribution;
- for tensile members, ductility criteria limit strains to values that have to be exceeded for strain hardening to become significant;
- for joints, strain hardening will, to some extent, be included in the ultimate strength formulae.

12.6.8 Models for Foundation Strength

The foundation shall be included in the overall model and shall be analyzed as a fully integrated part of the structural system using nonlinear soil p - y , t - z , and Q - z curves. For detailed design, these curves shall be generated from site-specific data. The effect of cyclic loads/actions on the soil curves shall be included. The model shall include the nonlinear geometrical and material behavior of the piles.

Simplified models for foundation failure may be used when modeling collapse mechanisms that involve part of the framework and when structural failure occurs before foundation failure.

For nonlinear ultimate strength analysis of the structural system, it can be more appropriate to use mean soil properties as opposed to representative properties. Representative soil properties shall be used to determine a lower bound of the system collapse strength if system collapse is governed by the foundation.

12.6.9 Investigating Nonlinear Behavior

Having performed a nonlinear analysis of a structural system or one of its components, the magnitude of the stresses, particularly the equivalent von Mises stresses, shall be examined. The plastic strains shall also be examined to ensure that the ductility limit (fracture strain) has not been exceeded (see 12.6.6), and to confirm areas of yield. For analyses involving high temperatures, the material properties are significantly less than those at a normal ambient temperature of 15 °C.

The deformation of members provides information on their behavior, e.g. members failing due to elastic buckling will have large deformations but no areas of yield, whereas members yielding in tension will be straight and have areas of yield that can extend along the whole length of the member.

Any excessive rotations at the ends of members shall be examined to determine whether they are due to yielding of the member (plastic hinge) or to joint failure. The response of failed joints shall be checked to ensure that the forces in the chord and braces are consistent with respect to the defined behavior (see 12.6.4 and 12.6.5).

The structural deflections shall also be checked to ensure that serviceability requirements are satisfied with regard to appurtenances such as risers and caissons, failure of which can necessitate platform shutdown or impair the platform's safety systems.

The behavior of the structure in a sequence of high waves shall be assessed. If repeated plastic strains occur then a low cycle fatigue analysis shall be carried out.

The above checks shall be carried out for various stages of the nonlinear analysis in order to gain an understanding of the history of the structure's behavior, in particular the sequence of component failures and internal force redistribution.

13 Strength of Tubular Members

13.1 General

The requirements given in this section apply to unstiffened and ring stiffened cylindrical tubulars having a thickness $t \geq 6$ mm, a diameter to thickness ratio $D/t \leq 120$, and material meeting the general requirements of Section 19. In addition, yield strengths shall be less than 500 MPa and the ratio of yield strength as used to ultimate tensile strength shall not exceed 0.90.

For structural shapes, other than circular tubulars, the requirements and recommendations given in API 2TOP are applicable. API 2TOP provides for specific guidance on detailing and dimensioning to be taken from an onshore building standard and describes a methodology in which a correspondence factor is introduced to address any differences in the derivation of the load/action and resistance factors of different standards.

The requirements for the different components and types and combinations of loads/actions are contained in different subsections, as detailed in Table 13.1-1. Where no reference is included for a particular component and loads/actions, there exist insufficient test data to enable comprehensive design equations to be prepared, and these circumstances shall be assessed on a case-by-case basis.

Table 13.1-1—Arrangement of Requirements for Members

Subsection	Component	Loads/Actions				
		Tension	Compression	Bending	Shear	Hydrostatic
13.2.2	Tubular	X				
13.2.3	Tubular		X			
13.2.4	Tubular			X		
13.2.5	Tubular				X	
13.2.6	Tubular					X
13.3.2	Tubular	X		X		
13.3.3	Tubular		X	X		
13.4.2	Tubular	X		X		X
13.4.3	Tubular		X	X		X
13.6.3	Cone	X	X	X		
13.6.4	Cone	X	X	X		X
13.7.2.2	Dented tubular	X				
13.7.2.3	Dented tubular		X			
13.7.2.4	Dented tubular			X		
13.7.2.5	Dented tubular				X	
13.7.3.1	Dented tubular	X		X		
13.7.3.2	Dented tubular		X	X		
13.8	Corroded tubular	X	X	X	X	X
13.9.2.2	Grouted tubular	X				
13.9.2.3	Grouted tubular		X			
13.9.2.4	Grouted tubular			X		
13.9.3.1	Grouted tubular	X		X		
13.9.3.2	Grouted tubular		X	X		

For tubulars subjected to hydrostatic pressure, it can be necessary to ensure that the circumferential value of yield strength is consistent with the value adopted in design.

The requirements in this section assume that the tubular is constructed in accordance with the fabrication tolerances given in Section 21. The requirements allow structural design to proceed on the basis that stresses due to the forces from the capped-end loads/actions of hydrostatic pressure are either included in or excluded from the analysis.

In Section 13, y and z are the axes of a tubular cross-section used to define in-plane and out-of-plane behavior respectively. “In-plane” is the plane common to the longitudinal axis of the brace member under consideration and the longitudinal axes of the chord member providing restraint, whereas “out-of-plane” is perpendicular to this plane.

In Equations (13.2-1) to (13.9-39), the stresses are always the absolute values of the stresses as computed, and hence—whether tensile or compressive—are always positive.

13.2 Tubular Members Subjected to Tension, Compression, Bending, Shear, or Hydrostatic Pressure

13.2.1 General

Tubular members subjected independently to axial tension, axial compression, bending, shear, or hydrostatic pressure shall be designed to satisfy the strength and stability requirements specified in 13.2.2 to 13.2.6.

13.2.2 Axial Tension

Tubular members subjected to axial tensile forces shall be designed to satisfy the following condition:

$$\sigma_t \leq \frac{f_t}{\gamma_{R,t}} \quad (13.2-1)$$

where

σ_t is the axial tensile stress due to forces from factored loads/actions;

f_t is the representative axial tensile strength, $f_t = f_y$;

f_y is the representative yield strength, in stress units;

$\gamma_{R,t}$ is the partial resistance factor for axial tensile strength, $\gamma_{R,t} = 1.05$.

The utilization of a member, U_m , under axial tension shall be calculated from Equation (13.2-2):

$$U_m = \frac{\sigma_t}{f_t / \gamma_{R,t}} \quad (13.2-2)$$

13.2.3 Axial Compression

13.2.3.1 General

Tubular members subjected to axial compressive forces shall be designed to satisfy the following condition:

$$\sigma_c \leq \frac{f_c}{\gamma_{R,c}} \quad (13.2-3)$$

where

σ_c is the axial compressive stress due to forces from factored loads/actions;

f_c is the representative axial compressive strength, in stress units (see 13.2.3.2);

$\gamma_{R,c}$ is the partial resistance factor for axial compressive strength, $\gamma_{R,c} = 1.18$.

The utilization of a member, U_m , under axial compression shall be calculated from Equation (13.2-4):

$$U_m = \frac{\sigma_c}{f_c / \gamma_{R,c}} \quad (13.2-4)$$

13.2.3.2 Column Buckling

In the absence of hydrostatic pressure, the representative axial compressive strength in 13.2.3.1 for tubular members shall be the smaller of the in-plane and the out-of-plane buckling strengths determined from the following equations:

$$f_c = (1.0 - 0.278\lambda^2) f_{yc} \quad (\text{for } \lambda \leq 1.34) \quad (13.2-5)$$

$$f_c = \frac{0.9}{\lambda^2} f_{yc} \quad (\text{for } \lambda > 1.34) \quad (13.2-6)$$

$$\lambda = \sqrt{\frac{f_{yc}}{f_e}} = \frac{K L}{\pi r} \sqrt{\frac{f_{yc}}{E}} \quad (13.2-7)$$

where

f_c is the representative axial compressive strength, in stress units;

f_{yc} is the representative local buckling strength, in stress units (see 13.2.3.3);

λ is the column slenderness parameter;

f_e is the smaller of the Euler buckling strengths in the y- and z-directions, in stress units (see 13.3.3);

E is Young's modulus of elasticity;

K is the effective length factor in the y- or z-direction selected so that $K L$ is the larger of the values in the y- and z-directions (see 13.5);

L is the unbraced length in y- or z-direction;

r is the radius of gyration, $r = \sqrt{I/A}$;

I is the moment of inertia of the cross-section;

A is the cross-sectional area.

13.2.3.3 Local Buckling

The representative local buckling strength, f_{yc} , in 13.2.3.2 shall be determined from:

$$f_{yc} = f_y \quad (\text{for } \frac{f_y}{f_{xe}} \leq 0.170) \quad (13.2-8)$$

$$f_{yc} = \left(1.047 - 0.274 \frac{f_y}{f_{xe}} \right) f_y \quad (\text{for } 0.170 < \frac{f_y}{f_{xe}}) \quad (13.2-9)$$

and

$$f_{xe} = 2 C_x E t / D \quad (13.2-10)$$

where

f_y is the representative yield strength, in stress units;

f_{xe} is the representative elastic local buckling strength, in stress units;

C_x is the elastic critical buckling coefficient (see below);

E is Young's modulus of elasticity;

D is the outside diameter of the member;

t is the wall thickness of the member.

The theoretical value of C_x for an ideal tubular is 0.6. However, a reduced value of $C_x = 0.3$ should be used in Equation (13.2-10) to account for the effect of initial geometrical imperfections within the tolerance limits given in Section 21. A reduced value of $C_x = 0.3$ is implicit in the value of f_{xe} used in Equations (13.2-8) and (13.2-9).

13.2.4 Bending

Tubular members subjected to bending moments shall be designed to satisfy the following condition:

$$\sigma_b = \frac{M}{Z_e} \leq \frac{f_b}{\gamma_{R,b}} \quad (13.2-11)$$

where

σ_b is the bending stress due to forces from factored loads/actions; when $M > M_y$, σ_b is to be considered as an equivalent elastic bending stress, $\sigma_b = M/Z_e$;

f_b is the representative bending strength, in stress units [see Equations (13.2-13) to (13.2-15)];

$\gamma_{R,b}$ is the partial resistance factor for bending strength, $\gamma_{R,b} = 1.05$;

M is the bending moment due to factored loads/actions;

M_y is the elastic yield moment;

Z_e is the elastic section modulus, $Z_e = \frac{\pi}{64} \left(D^4 - (D - 2t)^4 \right) / \left(\frac{D}{2} \right)$.

The utilization of a member, U_m , under bending moments shall be calculated from Equation (13.2-12):

$$U_m = \frac{\sigma_b}{f_b / \gamma_{R,b}} = \frac{M / Z_e}{f_b / \gamma_{R,b}} \quad (13.2-12)$$

The representative bending strength for tubular members shall be determined from:

$$f_b = \left(\frac{Z_p}{Z_e} \right) f_y \quad \left(\text{for } \frac{f_y D}{E t} \leq 0.0517 \right) \quad (13.2-13)$$

$$f_b = \left[1.13 - 2.58 \left(\frac{f_y D}{E t} \right) \right] \left(\frac{Z_p}{Z_e} \right) f_y \quad \left(\text{for } 0.0517 < \frac{f_y D}{E t} \leq 0.1034 \right) \quad (13.2-14)$$

$$f_b = \left[0.94 - 0.76 \left(\frac{f_y D}{E t} \right) \right] \left(\frac{Z_p}{Z_e} \right) f_y \quad \left(\text{for } 0.1034 < \frac{f_y D}{E t} \leq 120 \frac{f_y}{E} \right) \quad (13.2-15)$$

where, additionally,

f_y is the representative yield strength, in stress units;

D is the outside diameter of the member;

t is the wall thickness of the member;

Z_p is the plastic section modulus, $Z_p = \frac{1}{6} [D^3 - (D - 2t)^3]$.

13.2.5 Shear

13.2.5.1 Beam Shear

Tubular members subjected to beam shear forces shall be designed to satisfy the following condition:

$$\tau_b = \frac{2V}{A} \leq \frac{f_v}{\gamma_{R,v}} \quad (13.2-16)$$

where

τ_b is the maximum beam shear stress due to forces from factored loads/actions;

f_v is the representative shear strength, in stress units, $f_v = f_y / \sqrt{3}$;

$\gamma_{R,v}$ is the partial resistance factor for shear strength, $\gamma_{R,v} = 1.05$;

V is the beam shear due to factored loads/actions, in force units;

A is the cross-sectional area.

The utilization of a member, U_m , under beam shear shall be calculated from Equation (13.2-17):

$$U_m = \frac{\tau_b}{f_v / \gamma_{R,v}} = \frac{2V / A}{f_v / \gamma_{R,v}} \quad (13.2-17)$$

13.2.5.2 Torsional Shear

Tubular members subjected to torsional shear forces shall be designed to satisfy the following condition:

$$\tau_t = \frac{M_{v,t}D}{2I_p} \leq \frac{f_v}{\gamma_{R,v}} \quad (13.2-18)$$

where, in addition to the definitions in 13.2.5.1,

τ_t is the torsional shear stress due to forces from factored loads/actions;

$M_{v,t}$ is the torsional moment due to factored loads/actions;

I_p is the polar moment of inertia, $I_p = \frac{\pi}{32} [D^4 - (D - 2t)^4]$.

The partial resistance factor, $\gamma_{R,v}$, for shear, is the same for both torsional shear and beam shear (see 13.2.5.1).

The utilization of a member, U_m , under torsional shear shall be calculated from Equation (13.2-19):

$$U_m = \frac{\tau_t}{f_v / \gamma_{R,v}} = \frac{M_{v,t}D / 2I_p}{f_v / \gamma_{R,v}} \quad (13.2-19)$$

13.2.5.3 Combined Beam Shear and Torsional Shear

For tubular members subjected to combined beam shear forces and torsional shear forces, the representative shear strength, f_v , shall be reduced to $f_{v,t}$ to account for the presence of torsion using:

$$f_{v,t} = \left(1 - \frac{\tau_t}{f_v / \gamma_{R,v}} \right) f_v \quad (13.2-20)$$

The utilization of a member, U_m , under beam shear shall then be calculated from Equation (13.2-21):

$$U_m = \frac{\tau_b}{f_{v,t} / \gamma_{R,v}} = \frac{2V / A}{f_{v,t} / \gamma_{R,v}} \quad (13.2-21)$$

The utilization of the member, U_m , under torsional shear shall still be calculated from Equation (13.2-19).

13.2.6 Hydrostatic Pressure

13.2.6.1 Calculation of Hydrostatic Pressure

The effective depth at the location being checked shall be calculated taking into account the depth of the member below still water level and the effect of passing waves. The factored hydrostatic pressure (p) shall be calculated from Equation (13.2-22):

$$p = \gamma_{f,G1} \rho_w g H_z \quad (13.2-22)$$

where

$\gamma_{f,G1}$ is the partial load/action factor for permanent loads/actions 1 (see Table 9.10-1);

ρ_w is the density of the seawater that may be taken as 1025 kg/m³;

g is the acceleration due to gravity (m/s²);

H_z is the effective hydrostatic head (m);

$$H_z = -z + \frac{H_w}{2} \frac{\cosh[k(d+z)]}{\cosh(kd)} \quad (13.2-23)$$

where

z is the depth of the member relative to still water level (measured positive upwards);

d is the still water depth to the sea floor;

H is the wave height;

k is the wave number, $k = 2\pi/\lambda$;

where

λ is the wave length.

For installation conditions, z shall be the maximum depth of submergence during launch, or the maximum differential head during the upending and installation sequence plus an amount to allow for deviations from the planned sequence, and $\gamma_{f,G1}$ in Equation (13.2-22) shall be replaced by $\gamma_{f,T}$ (see Section 8).

13.2.6.2 Hoop Buckling

Tubular members subjected to external pressure shall be designed to satisfy the following condition:

$$\sigma_h = \frac{p D}{2 t} \leq \frac{f_h}{\gamma_{R,h}} \quad (13.2-24)$$

where

σ_h is the hoop stress due to the forces from factored hydrostatic pressure;

p is the factored hydrostatic pressure (see 13.2.6.1);

D is the outside diameter of the member;

t is the wall thickness of the member;

f_h is the representative hoop buckling strength, in stress units [see Equations (13.2-25) to (13.2-27)];

$\gamma_{R,h}$ is the partial resistance factor for hoop buckling strength, $\gamma_{R,h} = 1.25$.

For tubular members satisfying the out-of-roundness tolerances given in Annex E, f_h shall be determined from:

$$f_h = f_y \quad (\text{for } f_{he} > 2.44 f_y) \quad (13.2-25)$$

$$f_h = 0.7 \left(f_{he} / f_y \right)^{0.4} f_y \leq f_y \quad (\text{for } 0.55 f_y < f_{he} \leq 2.44 f_y) \quad (13.2-26)$$

$$f_h = f_{he} \quad (\text{for } f_{he} \leq 0.55 f_y) \quad (13.2-27)$$

where

f_y is the representative yield strength, in stress units;

f_{he} is the representative elastic critical hoop buckling strength, in stress units;

$$f_{he} = 2C_h E t / D \quad (13.2-28)$$

where the elastic critical hoop buckling coefficient C_h is:

$$C_h = 0.44 t/D \quad (\text{for } \mu \geq 1.6 D/t) \quad (13.2-29)$$

$$C_h = 0.44 t/D + 0.21 (D/t)^3 / \mu^4 \quad (\text{for } 0.825 D/t \leq \mu < 1.6 D/t) \quad (13.2-30)$$

$$C_h = 0.737 / (\mu - 0.579) \quad (\text{for } 1.5 \leq \mu < 0.825 D/t) \quad (13.2-31)$$

$$C_h = 0.80 \quad (\text{for } \mu < 1.5) \quad (13.2-32)$$

where μ is a geometrical parameter and

$$\mu = \frac{L_r}{D} \sqrt{\frac{2D}{t}}$$

where L_r is the length of tubular between stiffening rings, diaphragms, or end connections.

For tubular members exceeding the out-of-roundness tolerances, see A.13.2.6.2.

The utilization of a member, U_m , under external pressure shall be calculated from Equation (13.2-33):

$$U_m = \frac{\sigma_h}{f_h / \gamma_{R,h}} = \frac{p D / 2 t}{f_h / \gamma_{R,h}} \quad (13.2-33)$$

13.2.6.3 Ring Stiffener Design

For $\mu \geq 1.6 D/t$, the elastic critical hoop buckling stress is approximately equal to that of a long unstiffened tubular. Hence, to be effective, stiffening rings, if required, should be spaced such that:

$$L_r < 1.6 \sqrt{\frac{D^3}{2t}} \quad (13.2-34)$$

The circumferential stiffening ring size may be calculated from Equations (13.2-35) or (13.2-36) as appropriate, provided, if the yield strength of the ring stiffener is less than that of the member, that this smaller value of yield strength is used instead of f_y in Equations (13.2-25) to (13.2-27).

$$I_c = f_{he} \frac{t L_r D^2}{8 E} \quad (\text{for internal rings}) \quad (13.2-35)$$

$$I_c = f_{he} \frac{t L_r D_r^2}{8 E} \quad (\text{for external rings}) \quad (13.2-36)$$

where, in addition to the definitions given in 13.2.6.2,

I_c is the required moment of inertia for the composite ring section;

L_r is the ring spacing;

D is the outside diameter of the member;

D_r is the diameter of the centroid of the composite ring section;

E is Young's modulus of elasticity.

The composite ring section may be assumed to include an effective width of the member wall of $1.1\sqrt{Dt}$.

Where out-of-roundness is in excess of that permitted by Annex E, larger stiffeners can be required. In such cases the bending due to excess out-of-roundness shall be specifically investigated.

Local buckling of ring stiffeners with flanges may be excluded as a possible failure mode, provided that the following requirements are fulfilled:

$$\frac{h}{t_w} \leq 1.1 \sqrt{\frac{E}{f_{y,r}}} \quad (13.2-37)$$

and

$$\frac{b}{t_f} \leq 0.6 \sqrt{\frac{E}{f_{y,r}}} \quad (13.2-38)$$

where, in addition,

h is the web height;

t_w is the web thickness;

b is half the flange width of T-stiffeners or the full flange width for angle stiffeners;

t_f is the flange thickness;

$f_{y,r}$ is the representative yield strength of the ring stiffeners, in stress units.

Local buckling of ring stiffeners without flanges may be excluded as a possible failure mode, provided that,

$$\frac{h}{t_w} \leq 0.6 \sqrt{\frac{E}{f_{y,r}}} \quad (13.2-39)$$

For flat bar stiffeners, the minimum dimensions should be 10 mm x 76 mm for internal rings and 13 mm x 102 mm for external rings.

13.3 Tubular Members Subjected to Combined Forces without Hydrostatic Pressure

13.3.1 General

The following gives requirements for members subjected to combined forces, which give rise to global and local interactions between axial forces and bending moments, without hydrostatic pressure. Generally, the secondary moments from factored global loads/actions and the associated bending stresses ($P-\Delta$ effects) do not need to be considered. However, when the axial member force is substantial, or when the component on which the axial force acts is very flexible, the secondary moments due to $P-\Delta$ effects from factored global loads/actions should be taken into account.

13.3.2 Axial Tension and Bending

Tubular members subjected to combined axial tension and bending forces shall be designed to satisfy the following condition at all cross-sections along their length:

$$\frac{\gamma_{R,t} \sigma_t}{f_t} + \frac{\gamma_{R,b} \sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_b} \leq 1.0 \quad (13.3-1)$$

where, in addition to the definitions in 13.2.2 and 13.2.4,

$\sigma_{b,y}$ is the bending stress about the member y-axis (in-plane) due to forces from factored loads/actions;

$\sigma_{b,z}$ is the bending stress about the member z-axis (out-of-plane) due to forces from factored loads/actions.

The utilization of a member, U_m , under combined axial tension and bending shall be calculated from Equation (13.3-2):

$$U_m = \frac{\gamma_{R,t} \sigma_t}{f_t} + \frac{\gamma_{R,b} \sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_b} \quad (13.3-2)$$

13.3.3 Axial Compression and Bending

Tubular members subjected to combined axial compression and bending forces shall be designed to satisfy the following conditions at all cross-sections along their length:

$$\frac{\gamma_{R,c} \sigma_c}{f_c} + \frac{\gamma_{R,b}}{f_b} \left[\left(\frac{C_{m,y} \sigma_{b,y}}{1 - \sigma_c / f_{e,y}} \right)^2 + \left(\frac{C_{m,z} \sigma_{b,z}}{1 - \sigma_c / f_{e,z}} \right)^2 \right]^{0.5} \leq 1.0 \quad (13.3-3)$$

and

$$\frac{\gamma_{R,c} \sigma_c}{f_{yc}} + \frac{\gamma_{R,b} \sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_b} \leq 1.0 \quad (13.3-4)$$

where, in addition to the definitions given in 13.2.3, 13.2.4, and 13.3.2,

$C_{m,y}$, $C_{m,z}$ are the moment reduction factors corresponding to the member y- and z-axes, respectively (see 13.5);

$f_{e,y}$, $f_{e,z}$ are the Euler buckling strengths corresponding to the member y- and z-axes, respectively, in stress units;

$$f_{e,y} = \frac{\pi^2 E}{(K_y L_y / r)^2} \quad (13.3-5)$$

$$f_{e,z} = \frac{\pi^2 E}{(K_z L_z / r)^2} \quad (13.3-6)$$

where

K_y , K_z are the effective length factors for the y- and z-directions, respectively (see 13.5);

L_y , L_z are the unbraced lengths in the y- and z-directions, respectively.

The utilization of a member, U_m , under axial compression and bending shall be the larger value calculated from Equations (13.3-7) and (13.3-8):

$$U_m = \frac{\gamma_{R,c} \sigma_c}{f_c} + \frac{\gamma_{R,b}}{f_b} \left[\left(\frac{C_{m,y} \sigma_{b,y}}{1 - \sigma_c / f_{e,y}} \right)^2 + \left(\frac{C_{m,z} \sigma_{b,z}}{1 - \sigma_c / f_{e,z}} \right)^2 \right]^{0.5} \quad (13.3-7)$$

$$U_m = \frac{\gamma_{R,c} \sigma_c}{f_{yc}} + \frac{\gamma_{R,b} \sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_b} \quad (13.3-8)$$

13.3.4 Piles

Overall column buckling is normally not a problem in the design of pile segments below the sea floor because the surrounding soils inhibit overall column buckling. However, whenever laterally loaded piles are subjected to significant axial loads/actions, the secondary moments ($P-\Delta$ effects) should be considered in stress computations. An effective method of analysis is to model the pile as a beam-column on an elastic foundation. When such an analysis is used, the pile segment should be designed to satisfy Equation (13.3-4), except that $\sigma_{b,y}$ and $\sigma_{b,z}$ in this formula should include the stresses from the secondary moments ($P-\Delta$ effects) computed from factored loads/actions.

13.4 Tubular Members Subjected to Combined Forces with Hydrostatic Pressure

13.4.1 General

A tubular member below the water line is subjected to hydrostatic pressure unless it has been flooded. Flooding is normally used only for a structure's legs in order to assist in upending and placement and for pile installation. Even where members are flooded in the in-place condition, they can be subjected to hydrostatic pressure during launch and installation. The effects of hydrostatic pressure shall be taken into account when conducting member checks, including the axial components of such pressure (i.e. capped-end loads/actions). When conducting an analysis of the axial components of hydrostatic pressure, such load/action effects can be taken directly into account during the analysis or can be included subsequently. The formulations presented in 13.4 allow either approach for accounting for the axial effects of hydrostatic pressure to be used.

When checking tubular members subjected to hydrostatic pressure, four checks are required:

- check for hoop buckling under hydrostatic pressure alone [Equation (13.2-24)];
- check for tensile yielding when the combination of load/action effects, including those due to capped-end forces, results in tension in the member (see 13.4.2);
- check for compression yielding and local buckling when the combination of load/action effects, including those due to capped-end forces, results in compression in the member (see 13.4.3);
- check for column buckling when the load/action effects, excluding those due to capped-end forces, result in compression in the member (see 13.4.3).

For analyses using factored loads/actions that include capped-end loads/actions (i.e. only frame loads/actions in analyses; namely, marine buoyancy analysis method):

$\sigma_{t,c}$ is the axial tensile stress due to forces from factored loads/actions;

$\sigma_{c,c}$ is the axial compressive stress due to forces from factored loads/actions.

For analyses using factored loads/actions that do not include the capped-end loads/actions:

$$\sigma_{t,c} = \sigma_t - \sigma_q \quad (\text{if } \sigma_t \geq \sigma_q) \quad (13.4-1)$$

$$\sigma_{c,c} = \sigma_q - \sigma_t \quad (\text{if } \sigma_t < \sigma_q) \quad (13.4-2)$$

$$\sigma_{c,c} = \sigma_c + \sigma_q \quad (13.4-3)$$

where

σ_t is the axial tensile stress due to forces from factored loads/actions without capped-end loads/actions;

σ_c is the axial compressive stress due to forces from factored loads/actions without capped-end loads/actions;

σ_q is the compressive axial stress due to the capped-end hydrostatic loads/actions calculated using the value of pressure from Equation (13.2-22).

NOTE 1 In some circumstances, the use of factored loads/actions leads to conditions in which $\sigma_{t,c}$ is tensile, whereas under unfactored loads/actions $\sigma_{t,c}$ is compressive. These cases usually occur for relatively low values of $\sigma_{t,c}$, and the error is not considered to be significant.

The capped-end stresses (σ_q) may be approximated as half the hoop stress due to forces from factored hydrostatic pressure, i.e.:

$$\sigma_q = 0.5 \sigma_h \quad (13.4-4)$$

NOTE 2 In accordance with 13.1, σ_q always has a positive value.

In reality, the magnitude of these stresses depends on the restraint on the member provided by the rest of the structure and its value can be more or less than $0.5 \sigma_h$. The approximation $0.5 \sigma_h$ may be replaced by a stress computed from a more rigorous analysis, using factored loads/actions.

When an analysis uses factored loads/actions that include capped-end loads/actions, σ_c for net compression cases may be approximated by:

$$\sigma_c = \sigma_q - \sigma_{t,c} \quad (\text{if } \sigma_{t,c} < \sigma_q) \quad (13.4-5)$$

$$\sigma_c = \sigma_{c,c} - \sigma_q \quad (\text{if } \sigma_{c,c} > \sigma_q) \quad (13.4-6)$$

13.4.2 Axial Tension, Bending, and Hydrostatic Pressure

When member longitudinal tensile stresses, bending stresses, and hoop compressive stresses (collapse) occur simultaneously, the following interaction equation should be satisfied:

$$A^2 + B^2 + 2\nu|A|B \leq 1.0 \quad (13.4-7)$$

where

$$A = \frac{\gamma_{R,t}\sigma_{t,c} + \gamma_{R,b}\sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_y}$$

the term “A” should reflect the maximum tensile stress combination;

$$B = \frac{\gamma_{R,h}\sigma_h}{f_h};$$

ν = Poisson's ratio = 0.3.

The utilization of a member, U_m , under axial tension, bending, and hydrostatic pressure shall be calculated from Equation (13.4-8):

$$U_m = A^2 + B^2 + 2\nu|A|B \quad (13.4-8)$$

13.4.3 Axial Compression, Bending, and Hydrostatic Pressure

When member longitudinal compressive stresses, bending stresses, and hoop compressive stresses occur simultaneously, the following equations should be satisfied:

$$\frac{\gamma_{R,c}\sigma_{c,c}}{f_{yc}} + \frac{\gamma_{R,b}\sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_y} \leq 1.0 \quad (13.4-9)$$

Equation (13.4-9) should reflect the maximum compressive stress combination. Equation (13.3-3) also shall be satisfied, with σ_c in the axial term.

If the maximum combined compressive stress, $\sigma_x = \sigma_b + \sigma_{c,c}$, exceeds the limit given below, then Equation (13.4-11) shall also be satisfied:

$$\sigma_x > 0.5 \frac{f_{he}}{\gamma_{R,h}}$$

$$\frac{\sigma_x - 0.5 f_{he} / \gamma_{R,h}}{f_{xe} / \gamma_{R,c} - 0.5 f_{he} / \gamma_{R,h}} + \left(\frac{\gamma_{R,h}\sigma_h}{f_{he}} \right)^2 \leq 1.0 \quad (13.4-10)$$

where

f_{he} is the representative elastic critical hoop buckling strength as defined in 13.2.6.2;

f_{xe} is the representative elastic local buckling strength as defined in 13.2.3.3.

If $\sigma_b > \sigma_{c,c}$, both Equation (13.4-8) and Equation (13.4-9) shall be satisfied.

The utilization of a member, U_m , under axial compression, bending, and hydrostatic pressure shall be the largest value calculated from Equations (13.4-11) and (13.4-12):

$$U_m = \frac{\gamma_{R,c}\sigma_{c,c}}{f_{yc}} + \frac{\gamma_{R,b}\sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_y} \quad \text{when Equation (13.4-9) applies} \quad (13.4-11)$$

$$U_m = \frac{\sigma_x - 0.5 f_{he} / \gamma_{R,h}}{f_{xe} / \gamma_{R,c} - 0.5 f_{he} / \gamma_{R,h}} + \left(\frac{\gamma_{R,h}\sigma_h}{f_{he}} \right)^2 \quad \text{when Equation (13.4-10) applies} \quad (13.4-12)$$

13.5 Effective Lengths and Moment Reduction Factors

The effective lengths and moment reduction factors may be determined using a rational analysis that includes joint flexibility and side-sway. In lieu of such a rational analysis, values of effective length factors (K) and moment reduction factors (C_m) may be taken from Table 13.5-1. These factors include the following:

- do not apply to cantilever members, and
- assume both member ends are rotationally restrained in both planes of bending (see A.13.5).

NOTE Examples of the use of rational analysis can be found in the relevant references cited in Annex A.

Lengths to which the effective length factors K are applied are normally measured from centerline to centerline of the end joints. However, for members framing into legs, the following modified lengths may be used, provided that no interaction between the buckling of members and legs affects the utilization of the legs:

- face-of-leg to face-of-leg for main diagonal braces;
- face-of-leg to centerline of end joint for K-braces.

Lower K factors than those according to Table 13.5-1 may be used provided they are supported by more rigorous analysis.

Table 13.5-1—Effective Length and Moment Reduction Factors for Member Strength Checking

Structural Component	K	C_m^a
Topsides legs		
Braced	1.0	1)
Portal (unbraced)	K^b	1)
Structure legs and piling		
Grouted composite section	1.0	3)
Ungouted legs	1.0	3)
Ungouted piling between shim points	1.0	2)
Structure brace members		
Primary diagonals and horizontals	0.7	2) or 3)
K-braces ^c	0.7	2) or 3)
X-braces		
Longer segment length ^c	0.8	2) or 3)
Full length ^d	0.7	2) or 3)
Secondary horizontals	0.7	2) or 3)
^a C_m values for the three cases defined in this table are as follows: 1) 0.85; 2) for members with no transverse loading, other than self weight, $C_m = 0.6 - 0.4 \times M_1/M_2$ where M_1/M_2 is the ratio of smaller to larger moments at the ends of the unbraced portion of the member in the plane of bending under consideration; M_1/M_2 is positive when the member is bent in reverse curvature, negative when bent in single curvature; C_m shall not be larger than 0.85; 3) for members with transverse loading, other than self weight, $C_m = 1.0 - 0.4 \times (\sigma_c/f_e), \text{ or } 0.85, \text{ whichever is less,}$ and $f_e = f_{ey}$ or f_{ez} as appropriate. ^b See effective length alignment chart in A.13.5. This may be modified to account for conditions different from those assumed in the development of the chart. ^c For either in-plane or out-of-plane effective lengths, at least one pair of members framing into a K- or X-joint shall be in tension, if the joint is not braced out-of-plane. ^d When all members are in compression and the joint is not braced out-of-plane.		

13.6 Conical Transitions

13.6.1 General

The recommendations in this section may be applied to a concentric cone frustum between two cylindrical tubular sections. In addition, the rules may be applied to conical transitions at brace ends, with the cone-cylinder junction ring rules applicable only to the brace end of the transition.

NOTE See Reference [25].

13.6.2 Cone Section Properties

The cone section properties should be chosen to satisfy the axial and bending stresses at each end of the cone. The nominal axial and bending stresses at any section in a cone transition are given approximately by $(\sigma_a + \sigma_b)/\cos \alpha$,

where α equals one-half the projected apex angle of the cone (see Figure 13.6-1) and σ_a and σ_b are the nominal axial and bending stresses computed using the section properties of an equivalent cylinder with diameter and thickness equal to the cone diameter and thickness at the section.

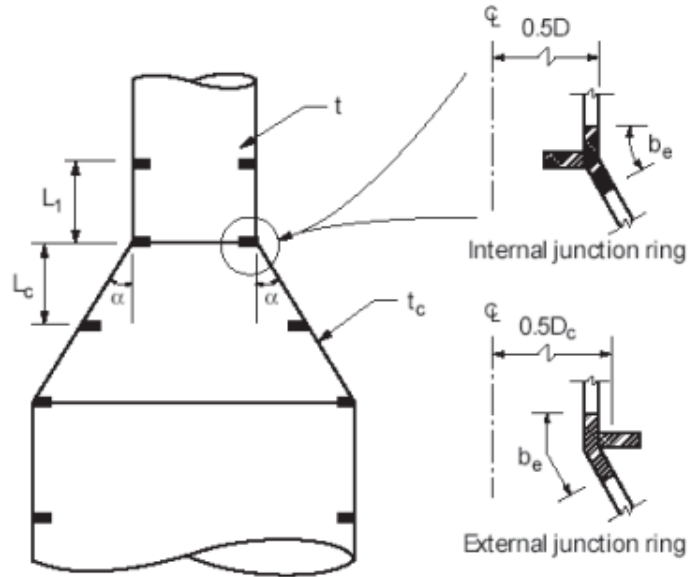


Figure 13.6-1—Example Conical Transition

13.6.3 Local Buckling

For local buckling under axial compression and bending, conical transitions with an apex angle less than 60° may be considered as equivalent cylinders with diameter equal to $D/\cos \alpha$, where D is the cone diameter at the point under consideration. This diameter is used in Equations (13.2-8) and (13.2-9) to determine f_{yc} . For cones of constant thickness, using the diameter at the small end of the cone would be conservative.

13.6.4 Unstiffened Cone-cylinder Junctions

13.6.4.1 General

Cone-cylinder junctions are subject to unbalanced radial forces due to longitudinal axial and bending loads and to localized bending stresses caused by the angle change. The longitudinal and hoop stresses at the junction may be evaluated as described in 13.6.4.2 and 13.6.4.3.

13.6.4.2 Longitudinal Stress

In lieu of a detailed analysis, the localized bending stress due to unfactored global loads at an unstiffened cone-cylinder junction may be estimated, based on results presented in Reference [25] from:

$$\sigma_{b,u}' = \frac{0.6t\sqrt{D(t+t_c)}}{t_e^2} (\sigma_{a,u} + \sigma_{b,u}) \tan \alpha \quad (13.6-1)$$

where

- D is the cylinder diameter at junction, in meters (inches);
- t is the cylinder thickness, in meters (inches);
- t_c is the cone thickness, in meters (inches);
- t_e is the t for stress in cylinder section or t_c for stress in cone section, in meters (inches);
- $\sigma_{a,u}$ is the acting axial stress due to unfactored global loads in cylinder section at junction, in megapascals (kilopounds per square inch);
- $\sigma_{b,u}$ is the acting resultant bending stress due to unfactored global loads in cylinder section at junction, in megapascals (kilopounds per square inch);
- α is one-half the apex angle of the cone, in degrees.

For strength requirements, the total stress ($\sigma_{a,u} + \sigma_{b,u} + \sigma_{b,u}'$) should be limited to the minimum tensile strength of the cone and cylinder material. For fatigue considerations, the cone-cylinder junction should satisfy the requirements of Section 16 with a stress concentration factor (SCF) equal to $1 + \sigma_{b,u}'/(\sigma_{a,u} + \sigma_{b,u})$, where $\sigma_{b,u}'$ is given by Equation (13.6-1). For equal cylinder and cone wall thicknesses, the SCF is equal to $1 + 0.6 \sqrt{2D/t} \tan \alpha$.

13.6.4.3 Hoop Stresses

The hoop stresses due to factored global loads caused by the unbalanced radial line load may be estimated from:

$$\sigma_{h,f}' = 0.45 \sqrt{\frac{D}{t}} (\sigma_{a,f} + \sigma_{b,f}) \tan \alpha \quad (13.6-2)$$

where

- $\sigma_{a,f}$ is the acting axial stress due to factored global loads in cylinder section at junction, in megapascals (kilopounds per square inch);
- $\sigma_{b,f}$ is the acting resultant bending stress due to factored global loads in cylinder section at junction, in megapascals (kilopounds per square inch).

For hoop tension, $\sigma_{h,f}'$ should be limited to $f_y/\gamma_{R,t}$. For hoop compression, $\sigma_{h,f}'$ should be limited to $f_h/\gamma_{R,h}$, where f_h is computed using Equations (13.2-25) to (13.2-27) with $f_{he} = 0.4 Et/D$. This suggested value of f_{he} is based on results presented in Reference [26].

13.6.5 Cone-cylinder Junction Rings

If stiffening rings are required, the section properties should be chosen to satisfy both the following requirements:

$$A_c = \frac{tD}{f_y} (\sigma_{a,f} + \sigma_{b,f}) \tan \alpha \quad (13.6-3)$$

$$I_c = \frac{tDD_c^2}{8E} (\sigma_{a,f} + \sigma_{b,f}) \tan \alpha \quad (13.6-4)$$

where

D is the cylinder diameter at junction, in meters (inches);

D_c is the diameter to centroid of composite ring section, in meters (inches);

A_c is the cross-sectional area of composite ring section, in square meters (square inches);

I_c is the moment of inertia of composite ring section, in meters to the fourth power (inches to the fourth power).

In computing A_c and I_c , the effective width of shell wall acting as a flange for the composite ring section may be computed from:

$$b_e = 0.55 \left(\sqrt{Dt} + \sqrt{Dt_c} \right) \quad (13.6-5)$$

NOTE 1 For flat bar stiffeners, the minimum dimensions should be 10 mm × 76 mm (³/₈ in. × 3 in.) for internal rings and 13 mm × 102 mm (¹/₂ in. × 4 in.) for external rings.

NOTE 2 For internal rings, D should be used instead of D_c in Equation (13.6-4).

13.7 Dented Tubular Members

13.7.1 General

Neither the effects of external hydrostatic pressure nor the presence of cracks have been considered in the derivation of the equations of 13.7. Such effects should be considered on a case-by-case basis. Fatigue problems can arise at dent locations and shall also be considered.

The limits of calibration for the formulae in this subsection are a dent depth, h , of $h \leq 0.3 D$ and $h \leq 10 t$. The effects of dent depths in excess of either of these values shall be examined by other means.

13.7.2 Dented Tubular Members Subjected to Tension, Compression, Bending, or Shear

13.7.2.1 General

Dented tubular members subjected independently to axial tension, axial compression, bending or shear shall be assessed using the strength and stability requirements specified in 13.7.2.2 to 13.7.2.5.

13.7.2.2 Axial Tension

Dented tubular members subjected to axial tensile forces shall satisfy the following condition:

$$\sigma_{t,d} \leq \frac{f_y}{\gamma_{R,t,d}} \quad (13.7-1)$$

where

$\sigma_{t,d}$ is the axial tensile stress due to forces from factored loads/actions on the undamaged cross-section,
 $\sigma_{t,d} = T/A$;

T is the member axial tensile force;

A is the cross-sectional area of the undamaged section;

f_y is the representative yield strength, in stress units;

$\gamma_{R,t,d}$ is the partial resistance factor for axial tensile strength for dented members, $\gamma_{R,t,d} = 1.05$.

The utilization of a dented member, $U_{m,d}$, under axial tension shall be calculated from Equation (13.7-2):

$$U_{m,d} = \frac{\sigma_{t,d}}{f_y / \gamma_{R,t,d}} \quad (13.7-2)$$

13.7.2.3 Axial Compression

13.7.2.3.1 General

Dented tubular members subjected to axial compressive forces shall satisfy the following condition:

$$\sigma_{c,d} \leq \frac{f_{c,d}}{\gamma_{R,c,d}} \quad (13.7-3)$$

where

$\sigma_{c,d}$ is the axial compressive stress due to forces from factored loads/actions on the undamaged cross-section, $\sigma_{c,d} = P/A$;

P is the member axial compressive force;

$f_{c,d}$ is the representative axial compressive strength of dented members, in stress units (see 13.7.2.3.2);

$\gamma_{R,c,d}$ is the partial resistance factor for axial compressive strength of dented members, $\gamma_{R,c,d} = 1.18$.

The utilization of a dented member, $U_{m,d}$, under axial compression shall be calculated from Equation (13.7-4):

$$U_{m,d} = \frac{\sigma_{c,d}}{f_{c,d} / \gamma_{R,c,d}} \quad (13.7-4)$$

13.7.2.3.2 Column Buckling

The representative axial compressive strength for the dented members considered in 13.7.2.3.1 shall be determined from the following equations:

$$f_{c,d} = f_{c,d,o} \quad (\text{for } \Delta y / L \leq 0.001) \quad (13.7-5)$$

$$\frac{f_{c,d}}{f_{c,d,o}} + \frac{f_{c,d} A_d (\Delta y - 0.001L)}{\left(1 - \frac{f_{c,d}}{\xi_c f_{e,d}}\right) \xi_m f_b Z_e} = 1.0 \quad (\text{for } \Delta y / L > 0.001) \quad (13.7-6)$$

where

Δy is the maximum out-of-straightness of the dented member;

L is the unbraced member length, in the plane of buckling which coincides with the plane of Δy ;

$f_{c,d}$ is the representative axial compressive strength of dented members, in stress units;

$f_{c,d,o}$ is the representative axial compressive strength of dented members when $\Delta y / L \leq 0.001$, in stress units;

$$f_{c,d,o} = (1.0 - 0.278 \lambda_d^2) \xi_c f_{yc} \quad (\text{for } \lambda_d \leq 1.34) \quad (13.7-7)$$

$$f_{c,d,o} = \frac{0.9}{\lambda_d^2} \xi_c f_{yc} \quad (\text{for } \lambda_d > 1.34) \quad (13.7-8)$$

f_{yc} is the representative local buckling strength as defined in 13.2.3.3, in stress units;

f_b is the representative bending strength as defined in 13.2.4, in stress units;

$f_{e,d}$ is the Euler buckling strength of the dented member, in stress units;

$$f_{e,d} = \frac{\pi^2 E}{(K_d L / r_d)^2} \quad (13.7-9)$$

Z_e is the elastic section modulus of the undamaged member as defined in 13.2.4;

λ_d is the slenderness parameter of the dented member;

$$\lambda_d = \sqrt{\frac{f_{yc}}{f_{e,d}}} = \frac{K_d L}{\pi r_d} \sqrt{\frac{f_{yc}}{E}} \quad (13.7-10)$$

K_d is the effective length factor of the dented member, which may be assumed to be the same as that for the undamaged member as defined in 13.2.3.2;

r_d is the radius of gyration of the dented member, which shall be calculated as:

$$r_d = \sqrt{\frac{I_d}{A_d}} \quad (13.7-11)$$

I_d is the effective moment of inertia of the dented cross-section;

$$I_d = \xi_m I_o \quad (13.7-12)$$

I_o is the moment of inertia of the undamaged member as defined in 13.2.3.2;

A_d is the effective cross-sectional area of the dented section;

$$A_d = \xi_c A \quad (13.7-13)$$

where

A is the cross-sectional area of the undamaged section as defined in 13.2.3.2;

ξ_c and ξ_m are coefficients defined by:

$$\xi_c = e^{-0.08 h/t} \quad \left(\text{for } \frac{h}{t} \leq 10.0\right) \quad (13.7-14)$$

$$\xi_m = e^{-0.06 h/t} \quad \left(\text{for } \frac{h}{t} \leq 10.0\right) \quad (13.7-15)$$

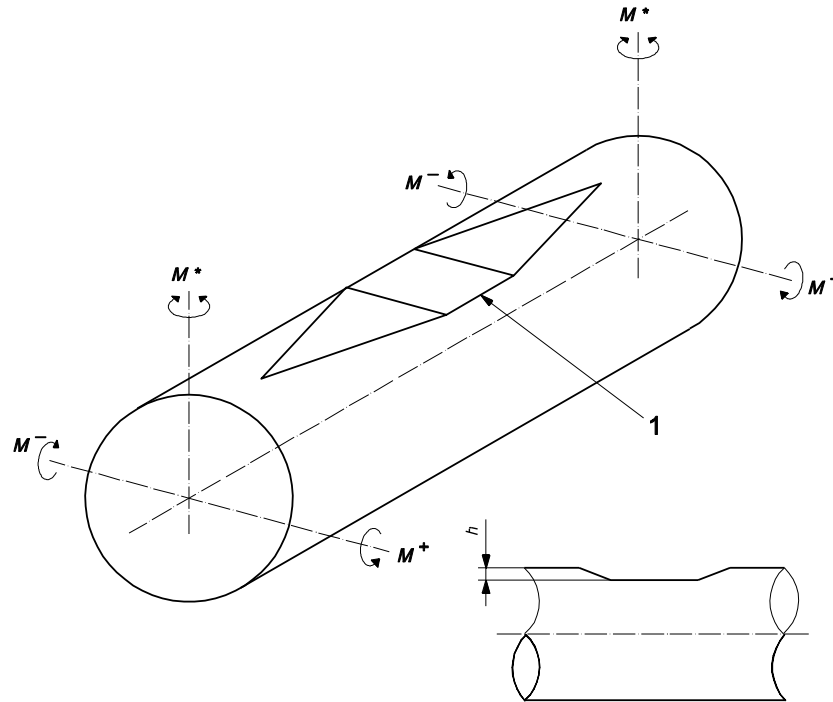
h is the maximum depth of the dent (see Figure 13.7-1);

t is the thickness of the member.

13.7.2.4 Bending

13.7.2.4.1 General

Depending upon the plane of bending, dented cylindrical members subjected to bending can experience tensile, compressive, or zero (neutral) stress conditions existing in the central region of the dent (see Figure 13.7-1). Accordingly, they shall satisfy the requirements of 13.7.2.4.2 to 13.7.2.4.4.



Key

- 1 dented section of the member
- M^+ positive moment—dent in tension
- M^- negative moment—dent in compression
- M^* neutral moment—zero stress in center of dent
- h maximum depth of the dent

Figure 13.7-1—Sign Convention for Dented Tubular

13.7.2.4.2 Positive Bending

When the dented region is subjected to a positive bending moment M^+ :

$$\sigma_{b,d}^+ = \frac{M^+}{Z_e} \leq \frac{f_b}{\gamma_{R,b,d}} \quad (13.7-16)$$

where

f_b is the representative bending strength defined in 13.2.4, in stress units;

$\sigma_{b,d}^+$ is the positive bending stress due to forces from factored loads/actions with respect to the undamaged cross-section, $\sigma_{b,d}^+ = M^+/Z_e$;

$\gamma_{R,b,d}$ is the partial resistance factor for bending strength for dented members, $\gamma_{R,b,d} = 1.05$.

The utilization of a dented member, $U_{m,d}$, under positive bending shall be calculated from Equation (13.7-17):

$$U_{m,d} = \frac{\sigma_{b,d}^+}{f_b / \gamma_{R,b,d}} = \frac{M^+ / Z_e}{f_b / \gamma_{R,b,d}} \quad (13.7-17)$$

13.7.2.4.3 Negative Bending

When the dented region is subjected to a negative bending moment M^- :

$$\sigma_{b,d}^- = \frac{M^-}{Z_e} \leq \xi_m \frac{f_b}{\gamma_{R,b,d}} \quad (13.7-18)$$

where $\sigma_{b,d}^-$ is the negative bending stress due to forces from factored loads/actions with respect to the undamaged cross-section, $\sigma_{b,d}^- = M^-/Z_e$.

The utilization of a dented member, $U_{m,d}$, under negative bending shall be calculated from Equation (13.7-19):

$$U_{m,d} = \frac{\sigma_{b,d}^-}{\xi_m f_b / \gamma_{R,b,d}} = \frac{M^- / Z_e}{\xi_m f_b / \gamma_{R,b,d}} \quad (13.7-19)$$

where ξ_m is as defined in 13.7.2.3.2.

13.7.2.4.4 Neutral Bending

When the dented region is subjected to a neutral bending moment M^* :

$$\sigma_{b,d}^* = \frac{M^*}{Z_e} \leq \frac{f_b}{\gamma_{R,b,d}} \quad (13.7-20)$$

where $\sigma_{b,d}^*$ is the neutral bending stress due to forces from factored loads/actions with respect to the undamaged cross-section, $\sigma_{b,d}^* = M^*/Z_e$.

The utilization of a dented member, $U_{m,d}$, under neutral bending shall be calculated from Equation (13.7-21):

$$U_{m,d} = \frac{\sigma_{b,d}^*}{f_b / \gamma_{R,b,d}} = \frac{M^* / Z_e}{f_b / \gamma_{R,b,d}} \quad (13.7-21)$$

13.7.2.5 Shear

Dented tubular members subjected to beam shear forces shall satisfy the condition specified in 13.2.5.1 in which the maximum shear stress due to forces from factored loads/actions should be approximated by the following equations:

$$\tau_{b,d} = \frac{2V}{A} \quad (\text{for } h \leq 0.25D) \quad (13.7-22)$$

$$\tau_{b,d} = \frac{2V/A}{(1.5 - 2h/D)} \quad (\text{for } 0.25D < h \leq 0.3D) \quad (13.7-23)$$

where V and A are as defined in 13.2.5.1.

13.7.3 Dented Tubular Members Subjected to Combined Forces

13.7.3.1 Axial Tension and Bending

13.7.3.1.1 General

Dented tubular members subjected to combined axial tension and bending forces shall satisfy the following conditions at all cross-sections along their length.

13.7.3.1.2 Axial Tension, Positive Bending, and Neutral Bending

Dented tubular members shall satisfy Equation (13.7-24):

$$\frac{\gamma_{R,t,d} \sigma_{t,d}}{f_y} + \frac{\gamma_{R,b,d} \sqrt{(\sigma_{b,d}^+)^2 + (\sigma_{b,d}^*)^2}}{f_b} \leq 1.0 \quad (13.7-24)$$

The utilization of a dented member, $U_{m,d}$, under axial tension, positive bending, and neutral bending shall be calculated using Equation (13.7-25):

$$U_{m,d} = \frac{\gamma_{R,t,d} \sigma_{t,d}}{f_y} + \frac{\gamma_{R,b,d} \sqrt{(\sigma_{b,d}^+)^2 + (\sigma_{b,d}^*)^2}}{f_b} \quad (13.7-25)$$

13.7.3.1.3 Axial Tension, Negative Bending, and Neutral Bending

Dented tubular members shall satisfy Equation (13.7-26):

$$\frac{\gamma_{R,t,d} \sigma_{t,d}}{f_y} + \sqrt{\left(\frac{\gamma_{R,b,d} \sigma_{b,d}^-}{\xi_m f_b} \right)^\alpha + \left(\frac{\gamma_{R,b,d} \sigma_{b,d}^*}{f_b} \right)^2} \leq 1.0 \quad (13.7-26)$$

where

$$\alpha = 2 - 3 \frac{h}{D} \quad (13.7-27)$$

and ξ_m is as defined in 13.7.2.3.2.

The utilization of a dented member, $U_{m,d}$, under axial tension, negative bending, and neutral bending shall be calculated from Equation (13.7-28):

$$U_{m,d} = \frac{\gamma_{R,t,d} \sigma_{t,d}}{f_y} + \sqrt{\left(\frac{\gamma_{R,b,d} \sigma_{b,d}^-}{\xi_m f_b} \right)^\alpha + \left(\frac{\gamma_{R,b,d} \sigma_{b,d}^*}{f_b} \right)^2} \quad (13.7-28)$$

13.7.3.2 Axial Compression and Bending

13.7.3.2.1 General

Dented tubular members subjected to combined axial compressive and bending forces shall satisfy the following conditions at all sections along their length.

13.7.3.2.2 Axial Compression, Positive Bending, and Neutral Bending

Dented tubular members shall satisfy Equations (13.7-29) and (13.7-30):

$$\frac{\gamma_{R,c,d} \sigma_{c,d}}{f_{c,d}} + \frac{\gamma_{R,b,d}}{f_b} \left[\left(\frac{\sigma_{b,d}^+}{(1 - \sigma_{c,d}/f_e)} \right)^2 + \left(\frac{\sigma_{b,d}^*}{(1 - \sigma_{c,d}/f_e)} \right)^2 \right]^{0.5} \leq 1.0 \quad (13.7-29)$$

$$\frac{\gamma_{R,c,d} \sigma_{c,d}}{f_{yc,d}} + \frac{\gamma_{R,b,d} \sqrt{(\sigma_{b,d}^+)^2 + (\sigma_{b,d}^*)^2}}{f_b} \leq 1.0 \quad (13.7-30)$$

where

$f_{yc,d}$ is the representative local buckling strength of the dented member, in stress units;

$$f_{yc,d} = \xi_c f_{yc} \quad (13.7-31)$$

ξ_c is as defined in 13.7.2.3.2;

f_{yc} is the representative local buckling strength of the undamaged member as defined in 13.2.3.3;

f_e is the smaller of the Euler buckling strengths of the undamaged member in the positive and neutral bending directions, in stress units, as defined in 13.3.3.

The utilization of a dented member, $U_{m,d}$, under axial compression, positive bending, and neutral bending shall be the larger value calculated from Equations (13.7-32) and (13.7-33):

$$U_{m,d} = \frac{\gamma_{R,c,d} \sigma_{c,d}}{f_{c,d}} + \frac{\gamma_{R,b,d}}{f_b} \left[\left(\frac{\sigma_{b,d}^+}{(1 - \sigma_{c,d}/f_e)} \right)^2 + \left(\frac{\sigma_{b,d}^*}{(1 - \sigma_{c,d}/f_e)} \right)^2 \right]^{0.5} \quad (13.7-32)$$

$$U_{m,d} = \frac{\gamma_{R,c,d} \sigma_{c,d}}{f_{yc,d}} + \frac{\gamma_{R,b,d} \sqrt{(\sigma_{b,d}^+)^2 + (\sigma_{b,d}^*)^2}}{f_b} \quad (13.7-33)$$

13.7.3.2.3 Axial Compression, Negative Bending, and Neutral Bending

Dented tubular members shall satisfy Equations (13.7-34) and (13.7-35):

$$\frac{\gamma_{R,c,d} \sigma_{c,d}}{f_{c,d}} + \left\{ \left[\frac{\gamma_{R,b,d} \sigma_{b,d}^-}{\left(1 - \frac{\sigma_{c,d}}{\xi_c f_{e,d}}\right) \xi_m f_b} \right]^\alpha + \left[\frac{\gamma_{R,b,d} \sigma_{b,d}^*}{\left(1 - \frac{\sigma_{c,d}}{f_e}\right) f_b} \right]^2 \right\}^{0.5} \leq 1.0 \quad (13.7-34)$$

$$\frac{\gamma_{R,c,d} \sigma_{c,d}}{f_{yc,d}} + \left[\left(\frac{\gamma_{R,b,d} \sigma_{b,d}^-}{\xi_m f_b} \right)^\alpha + \left(\frac{\gamma_{R,b,d} \sigma_{b,d}^*}{f_b} \right)^2 \right]^{0.5} \leq 1.0 \quad (13.7-35)$$

where

α is as defined in 13.7.3.1.3;

$f_{e,d}$ is as defined in 13.7.2.3.2;

f_e is the Euler buckling strength of the undamaged member in the neutral bending direction as defined in 13.3.3.

The utilization of a dented member, $U_{m,d}$, under axial compression, negative bending, and neutral bending shall be the larger value calculated from Equations (13.7-36) and (13.7-37):

$$U_{m,d} = \frac{\gamma_{R,c,d} \sigma_{c,d}}{f_{c,d}} + \left\{ \left[\frac{\gamma_{R,b,d} \sigma_{b,d}^-}{\left(1 - \frac{\sigma_{c,d}}{\xi_c f_{e,d}}\right) \xi_m f_b} \right]^\alpha + \left[\frac{\gamma_{R,b,d} \sigma_{b,d}^*}{\left(1 - \frac{\sigma_{c,d}}{f_e}\right) f_b} \right]^2 \right\}^{0.5} \quad (13.7-36)$$

$$U_{m,d} = \frac{\gamma_{R,c,d} \sigma_{c,d}}{f_{yc,d}} + \left[\left(\frac{\gamma_{R,b,d} \sigma_{b,d}^-}{\xi_m f_b} \right)^\alpha + \left(\frac{\gamma_{R,b,d} \sigma_{b,d}^*}{f_b} \right)^2 \right]^{0.5} \quad (13.7-37)$$

13.8 Corroded Tubular Members

The presence of local or overall corrosion should be taken into account in determining the residual strength of a corroded tubular member. At this time, there is insufficient information on the strength of corroded tubular members for the inclusion of requirements in this document. However, a methodology for estimating the effect of corrosion on member strength is discussed in A.13.8.

13.9 Grouted Tubular Members

13.9.1 General

Subsection 13.9 applies to both fully grouted undamaged members and fully grouted dented tubular members, with the dent depth h limited to either $h \leq 0.3 D$ or $h \leq 10 t$. For grouted tubular members, neither the effect of external hydrostatic pressure nor the presence of cracks has been taken into account in the derivation of the equations given in this subsection. Such effects should be considered on a case-by-case basis.

Requirements for partially grouted tubular members are excluded, owing to the limited availability of relevant test data (see A.13.9.1 for additional information).

The strength model recognizes that grout filling inhibits the development of local buckling of undamaged tubulars and the growth of dents in damaged tubulars. Therefore, it is assumed that completely grouted tubular members can undergo column buckling, but not local buckling. However, in cases where local buckling can be important (e.g. large D/t), the representative steel yield strength, f_y , may be replaced by the representative local buckling strength, f_{yc} (see 13.2.3.3).

13.9.2 Grouted Tubular Members Subjected to Tension, Compression, or Bending

13.9.2.1 General

Grouted tubular members subjected independently to axial tension, axial compression, or bending shall be assessed using the strength and stability requirements specified in 13.9.2.2 to 13.9.2.4.

13.9.2.2 Axial Tension

Fully grouted tubular members subjected to axial tensile forces shall normally satisfy the following condition:

$$\sigma_{t,g} \leq \frac{f_t}{\gamma_{R,t,g}} \quad (13.9-1)$$

where

$\sigma_{t,g}$ is the axial tensile stress in the steel tubular member due to forces from factored loads/actions, neglecting the grout;

f_t is the representative axial tensile strength of the steel, in stress units, $f_t = f_y$;

f_y is the representative yield strength of the steel, in stress units;

$\gamma_{R,t,g}$ is the partial resistance factor for axial tensile strength of the grouted member, $\gamma_{R,t,g} = 1.05$.

Where it can be demonstrated that complete grouting of the tubular has been achieved, f_t may be taken as $1.12 f_y$.

The utilization of a grouted tubular member, $U_{m,g}$, under axial tension shall be calculated from Equation (13.9-2):

$$U_{m,g} = \frac{\sigma_{t,g}}{f_t / \gamma_{R,t,g}} \quad (13.9-2)$$

13.9.2.3 Axial Compression

Fully grouted tubular members subjected to axial compressive forces shall satisfy the following condition:

$$\sigma_{c,g} \leq \frac{f_{c,g}}{\gamma_{R,c,g}} \quad (13.9-3)$$

where

$\sigma_{c,g}$ is the axial compressive stress in the fully grouted member due to forces from factored loads/actions acting on the transformed area, $\sigma_{c,g} = P/A_{tr}$;

$f_{c,g}$ is the representative axial compressive strength of the grouted member, in stress units, as defined in Equations (13.9-8) and (13.9-9);

$\gamma_{R,c,g}$ is the partial resistance factor for axial compressive strength of the grouted member, $\gamma_{R,c,g} = 1.18$;

P is the axial compressive force in the grouted member due to factored loads/actions;

A_{tr} is the transformed area of the fully grouted member;

$$A_{tr} = A_s + \frac{A_g}{m} \quad (13.9-4)$$

A_s is the cross-sectional area of the steel;

$$A_s = (D - t)t \left[\pi - (\alpha_g - \sin \alpha_g) \right] \quad (13.9-5)$$

$$\alpha_g = \cos^{-1} [1 - 2h/(D - t)] \quad (13.9-6)$$

A_g is the cross-sectional area of the grout;

$$A_g = (D - 2t)^2 (\pi - \alpha_g + 1/2 \sin 2\alpha_g) / 4 \quad (13.9-7)$$

m is the ratio of elastic moduli of steel and grout, $m = E_s/E_g$ ($m = 18$, in lieu of actual data);

E_s is Young's modulus of elasticity for steel;

E_g is the modulus of elasticity of grout;

h is the maximum dent depth, if present;

D is the outer diameter of the steel tubular member;

t is the thickness of the steel.

The representative axial compressive strength of fully grouted tubular members is due to column buckling and shall be determined from the following equations:

$$f_{c,g} = (1.0 - 0.278\lambda_g^2) f_{ug} \quad (\text{for } \lambda_g \leq 1.34) \quad (13.9-8)$$

$$f_{c,g} = \frac{0.9}{\lambda_g^2} f_{ug} \quad (\text{for } \lambda_g > 1.34) \quad (13.9-9)$$

$$\lambda_g = \sqrt{\frac{f_{ug}}{f_{e,g}}} \quad (13.9-10)$$

where, in addition to the definitions given above,

λ_g is the column slenderness parameter of the grouted member;

f_{ug} is the axial squash strength of the grouted member, in stress units;

$$f_{ug} = (A_s f_y + 0.67 A_g f_{cu}) / A_{tr} \quad (13.9-11)$$

f_y is the representative yield strength of the steel, in stress units;

f_{cu} is the representative unconfined cube strength of grout, in stress units;

$f_{e,g}$ is the Euler buckling strength of the fully grouted member, in stress units;

$$f_{e,g} = \frac{\pi^2 (E_s I_s + 0.8 E_g I_g)}{A_{tr} (KL)^2} \quad (13.9-12)$$

K is the effective length factor (see 13.5);

L is the longer of the unbraced lengths in the y- and z-directions;

I_s is the effective moment of inertia of the steel cross-section;

$$I_s = \left[(D-t)^3 t \left(\pi - \alpha_g - \frac{1}{2} \sin 2\alpha_g + 2 \sin \alpha_g \cos^2 \alpha_g \right) / 8 \right] - A_s e_s^2 \quad (13.9-13)$$

I_g is the effective moment of inertia of the grout cross-section;

$$I_g = \left[(D-2t)^4 \left(\pi - \alpha_g + \frac{1}{4} \sin 4\alpha_g \right) / 64 \right] - A_g e_g^2 \quad (13.9-14)$$

$$e_s = \frac{1}{2} (D-t)^2 t \sin \alpha_g (1 - \cos \alpha_g) / A_s \quad (13.9-15)$$

$$e_g = (D-2t)^3 \sin^3 \alpha_g / (12 A_g) \quad (13.9-16)$$

The utilization of a grouted member, $U_{m,g}$, under axial compression shall be calculated from Equation (13.9-17):

$$U_{m,g} = \frac{\sigma_{c,g}}{f_{c,g} / \gamma_{R,c,g}} \quad (13.9-17)$$

13.9.2.4 Bending

Fully grouted tubular members subjected to bending forces shall satisfy the following condition:

$$\sigma_{b,g} = \frac{M}{Z_e} \leq \frac{f_{b,g}}{\gamma_{R,b,g}} \quad (13.9-18)$$

where, in addition to the definitions given in 13.2.4, 13.9.2.2, and 13.9.2.3,

$\sigma_{b,g}$ is the bending stress due to forces from factored loads/actions and, when $\sigma_{b,g} > f_{b,g}$ is to be considered as an equivalent elastic bending stress, $\sigma_{b,g} = M/Z_e$;

M is the bending moment in the grouted member due to factored loads/actions;

$f_{b,g}$ is the representative bending strength of the grouted member, in stress units, defined as:

$$f_{b,g} = \frac{Z_p}{Z_e} f_y \delta (1 + 0.01k) \quad (13.9-19)$$

$$\delta = 1 - 0.5 \frac{h}{D} - 1.6 \left(\frac{h}{D} \right)^2 \quad (13.9-20)$$

$$k = 5.5 \delta \left(\rho \frac{D}{t} \right)^{0.66} \quad (13.9-21)$$

$$\rho = \frac{0.6 f_{cu}}{f_y} \quad (13.9-22)$$

Z_e, Z_p are calculated as in 13.2.4 for undamaged and ungrouted members;

$\gamma_{R,b,g}$ is the partial resistance factor for the bending strength of grouted members, $\gamma_{R,b,g} = 1.05$.

The utilization of a grouted tubular member, $U_{m,g}$, under bending shall be calculated from Equation (13.9-23):

$$U_{m,g} = \frac{\sigma_{b,g}}{f_{b,g} / \gamma_{R,b,g}} = \frac{M / Z_e}{f_{b,g} / \gamma_{R,b,g}} \quad (13.9-23)$$

13.9.3 Grouted Tubular Members Subjected to Combined Forces

13.9.3.1 Axial Tension and Bending

No requirements for fully grouted tubular members subjected to axial tension and bending forces are presented because of the limited availability of relevant test data. However, several approximate methods that may be used to estimate the interaction are discussed in A.13.9.3.1.

13.9.3.2 Axial Compression and Bending

Fully grouted tubular members subjected to combined axial compressive and bending forces shall satisfy the following conditions at all cross-sections along their length for undamaged tubulars, and along the dented section for damaged tubulars:

$$\frac{\gamma_{R,c,g} \sigma_{c,g}}{f_{c,g}} + \frac{\gamma_{R,b,g} T_1 \sigma_{b,g}}{f_{b,g}} + \frac{T_2 (\gamma_{R,b,g} \sigma_{b,g})^2}{f_{b,g}^2} \leq 1.0 \quad \left(\text{for } \frac{\gamma_{R,c,g} \sigma_{c,g}}{f_{c,g}} \geq \frac{K_2}{K_1} \right) \quad (13.9-24)$$

$$\frac{\gamma_{R,b,g} \sigma_{b,g}}{f_{b,g}} \leq 1.0 \quad \left(\text{for } \frac{\gamma_{R,c,g} \sigma_{c,g}}{f_{c,g}} < \frac{K_2}{K_1} \right) \quad (13.9-25)$$

where, in addition to the definitions given in 13.9.2.3 and 13.9.2.4,

$$T_1 = 1 - \frac{K_2}{K_1} - T_2 \quad (13.9-26)$$

$$T_2 = 4 \frac{K_3}{K_1} \quad (13.9-27)$$

$$K_1 = 1.0 - 0.278 \lambda_g^2 \quad \left(\text{for } \lambda_g \leq 1.34 \right) \quad (13.9-28)$$

$$K_1 = 0.9 / \lambda_g^2 \quad \left(\text{for } \lambda_g > 1.34 \right) \quad (13.9-29)$$

$$K_2 = \frac{K_{20} [115 - 30(2\beta - 1)(1.8 - \theta) - 100\lambda_g]}{50(2.1 - \beta)} \quad \left(\text{for } 0 \leq K_2 \leq K_{20} \right) \quad (13.9-30)$$

$$K_3 = K_{30} + \frac{\lambda_g [(0.5\beta + 0.4)(\theta^2 - 0.5) + 0.15]}{1 + \lambda_g^3} \quad (13.9-31)$$

$$K_{20} = (0.9\theta^2 + 0.2) \leq 0.75 \quad (13.9-32)$$

$$K_{30} = (0.04 - \theta/15) \geq 0 \quad (13.9-33)$$

$$\theta = \frac{0.67 A_g \left(f_{cu} + \frac{C_1 f_y t}{D} \right)}{f_{ug} A_{tr}} \quad (13.9-34)$$

β is the ratio of the smaller to the larger end moment, with $\beta = 1$ if no end moments apply;

$$C_1 = 4\phi\varepsilon \left(1 + \phi + \phi^2 \right)^{-0.5} \quad (13.9-35)$$

$$\phi = 0.02 \left(25 - \frac{K L}{D} \right) \geq 0 \quad (13.9-36)$$

$$\varepsilon = 0.25 \left(25 - \frac{K L}{D} \right) \geq 0 \quad (13.9-37)$$

The utilization of a grouted tubular member, $U_{m,g}$, under axial compression and bending shall be the larger value calculated from Equations (13.9-38) and (13.9-39):

$$U_{m,g} = \frac{\gamma_{R,c,g} \sigma_{c,g}}{f_{c,g}} + \frac{\gamma_{R,b,g} T_1 \sigma_{b,g}}{f_{b,g}} + \frac{T_2 \left(\gamma_{R,b,g} \sigma_{b,g} \right)^2}{f_{b,g}^2} \quad (13.9-38)$$

$$U_{m,g} = \frac{\gamma_{R,b,g} \sigma_{b,g}}{f_{b,g}} \quad (13.9-39)$$

14 Strength of Tubular Joints

14.1 General

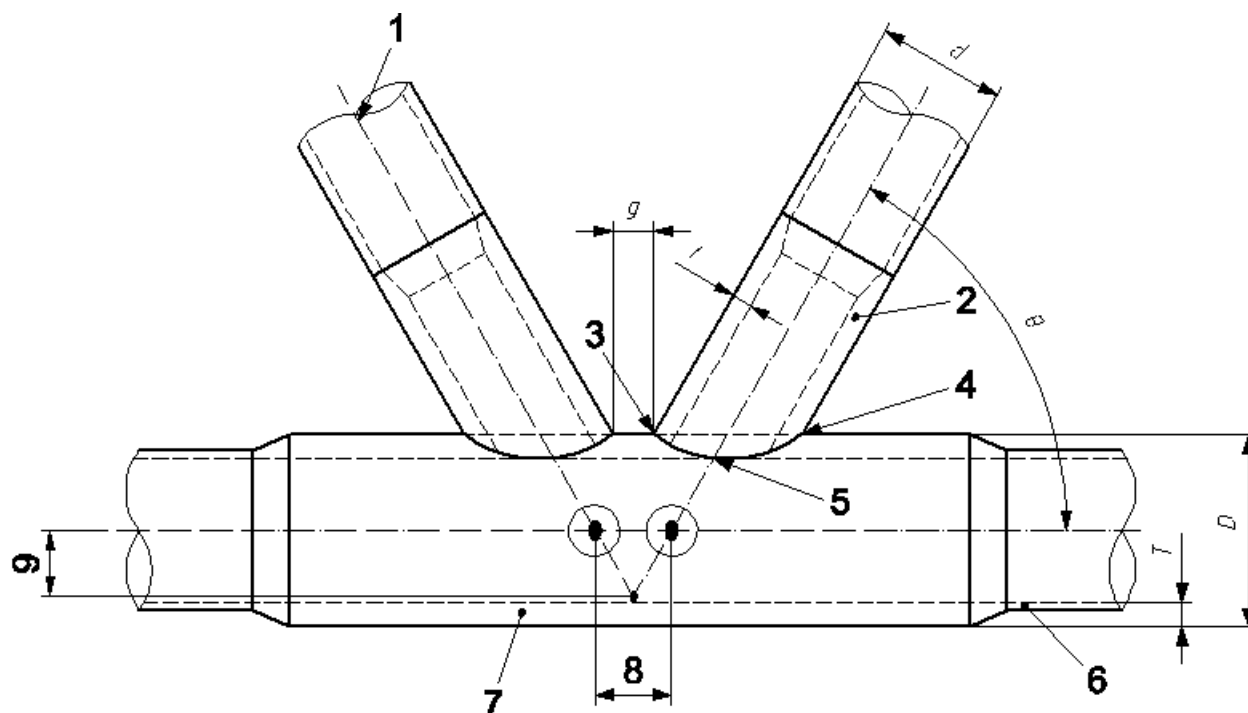
The requirements given in this section apply to the static design of tubular joints formed by the connection of two or more members. Members shall satisfy the requirements of 13.1 together with the additional requirements of 14.2.

Generic requirements for nontubular joints are also given. Joint types are classified in 14.2.5. See A.14.1 for additional information regarding the joint strength equations contained herein.

In lieu of the requirements in this document, reasonable alternative methods may be used for the design of joints. Test data and analytical techniques may be used as a basis for design, provided that the strength of such joints can be determined reliably. Such analytical or numerical techniques should always be calibrated and benchmarked to suitable test data.

The requirements have been derived from a consideration of the representative strength (as opposed to the mean strength) of tubular joints. Representative strength is comparable to lower bound strength. The background to, and a discussion of, the requirements are presented in A.14.

The nomenclature for simple joints is given in Figure 14.1-1.



Key

1	brace	θ	included angle between chord and brace axes
2	stub (where present)	g	gap between braces, negative for overlapped stubs
3	crown toe	t	brace wall thickness at intersection
4	crown heel	T	chord wall thickness at intersection
5	saddle	d	brace outside diameter
6	chord	D	chord outside diameter
7	can		
8	offset	$\beta = \frac{d}{D}$	$\gamma = \frac{D}{2T}$
9	eccentricity		$\tau = \frac{t}{T}$

Figure 14.1-1—Terminology and Geometrical Parameters for Simple Tubular Joints

14.2 Design Considerations

14.2.1 Materials

The provisions of this section apply to joints made of steels having representative yield strengths not exceeding 500 MPa. However, the usable value of yield strength as a fraction of ultimate tensile strength shall not exceed 0.80 (see 14.3.2).

There is sometimes uncertainty in the material properties in structures that are being assessed (see API 2SIM) or reused (see Section 23). In these instances, testing of material removed from the actual structure can be required. If the through-thickness toughness of joint can steel cannot be determined, inspection for possible cracks or lamellar tearing shall be performed.

Recommendations for grout materials for use in grouted joints are given in 19.6.

14.2.2 Design Forces and Joint Flexibility

Joints shall be designed and assessed using internal forces resulting from factored loads/actions in accordance with Sections 8 to 11.

The reduction in secondary (deflection induced) bending moments due to joint flexibility or due to inelastic relaxation may be considered. For ultimate strength analysis of the structure, information on the force-deformation characteristics of joints may be used. These characteristics are dependent on the joint type, configuration, geometry, material properties, load case under consideration and, in certain instances, hydrostatic pressure effects; see A.14.2.2 for a further discussion on joint flexibility.

14.2.3 Minimum Joint Strength

Chord cans, in addition to developing the strength required by design loads/actions, shall have a minimum axial capacity of at least 50 % of the effective strength of each incoming brace for each design situation (in-place, loadout, lifting, launch, accidental, etc.).

For earthquake loads/actions, the chord can capacity shall be at least 100 % of the brace effective strength of each incoming brace for the in-place design situation.

The effective strength of the brace is defined as the representative yield strength for braces where the axial component is tensile, Equation (13.2-1), or the compression buckling strength where the axial component is compressive, Equation (13.2-5) or Equation (13.2-6). The effective strength calculations shall be based on the nominal material and geometrical properties of the brace, not those of the brace stub (should one exist).

For the purposes of this requirement, the chord can capacity shall be determined using the representative strength given in Equation (14.3-1). The strength factor (Q_u) and the chord force factor (Q_f) shall be consistent with the design situation being considered.

14.2.4 Weld Strength

Welds in fabricated joints shall be designed to develop a strength greater than or equal to both the yield strength of the nominal brace cross-section (ignoring any brace stubs) and the full strength of the joint. Further guidelines for welds for circular tubular joints are given in Section 20.

Joints often involve welds from several brace connections in close proximity. The high restraint of joints can cause large strain concentrations and a potential for cracking or lamellar tearing. Hence, the chord material (and brace/stub material, if overlapping is present) shall have adequate through-thickness toughness (see Section 19).

14.2.5 Joint Classification

There are three basic planar joint types, Y-, K-, and X-joints (also called cross-joints), as shown in Figure 14.2-1 and as described below.

- a) A brace and a chord constitute a Y-joint if the component of the brace axial force perpendicular to the chord is reacted by beam shear in the chord. The brace axial force parallel to the chord is balanced by axial force in the chord.
- b) A pair of braces in the same plane and on the same side of a chord constitutes a K-joint if the components of the brace axial forces perpendicular to the chord balance each other to within 10 %. The components of the axial brace forces parallel to the chord are balanced by an axial force in the chord.
- c) A pair of braces in the same plane and on opposite sides of a chord or a brace in combination with other structural components on opposite sides of the chord (e.g. padeyes, launch rails) constitutes an X-joint if the axial force is transferred through the chord to the opposite side: there is no overall reaction in the chord.

Fixed steel offshore structures are normally space frames containing mixtures of the above joint types either in one plane or in several planes (multiplanar joints). For the purposes of classification, all members in planes within $\pm 15^\circ$ of each other may be considered as being in one plane. Joint classification is based solely on consideration of the axial forces in the braces.

Variants of the above joint types are T-joints that are Y-joints where the angle between the brace and the chord is approximately 90° and DT-joints, or Double T-joints, which look like X-joints with angles of approximately 90° but behave as two T-joints because the brace axial forces are transferred to the chord rather than to the opposite brace.

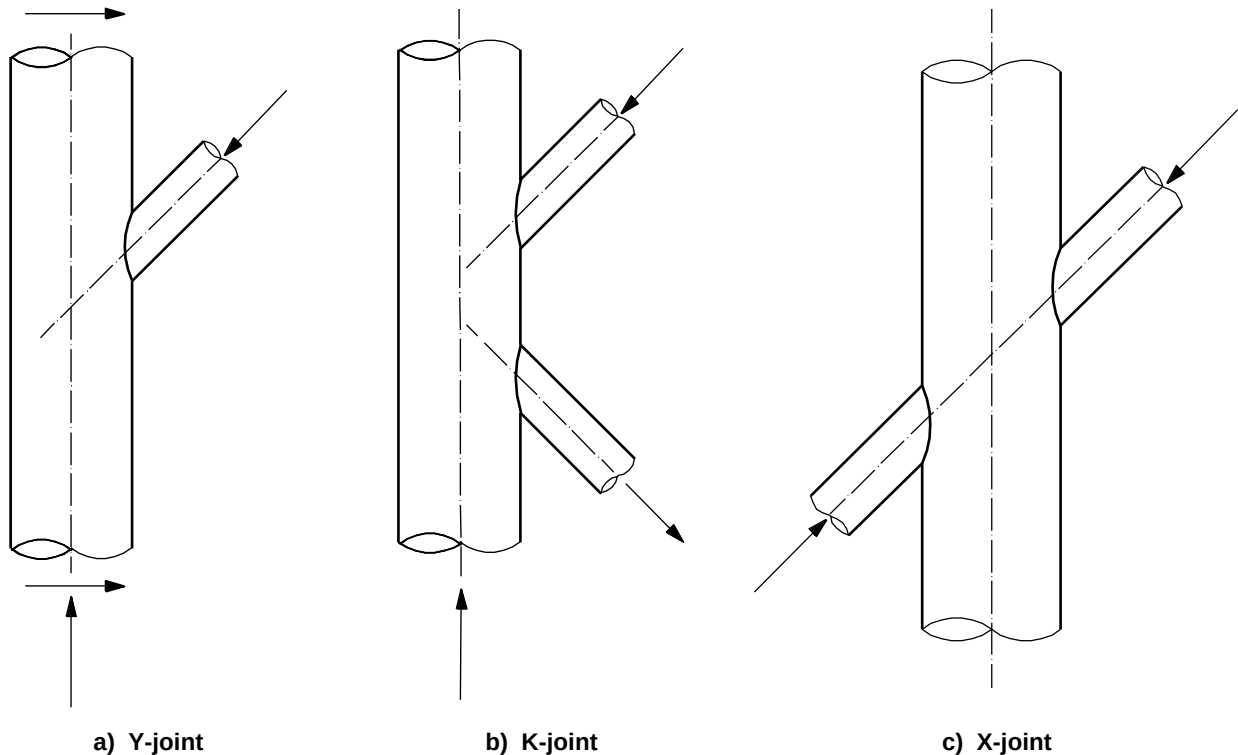
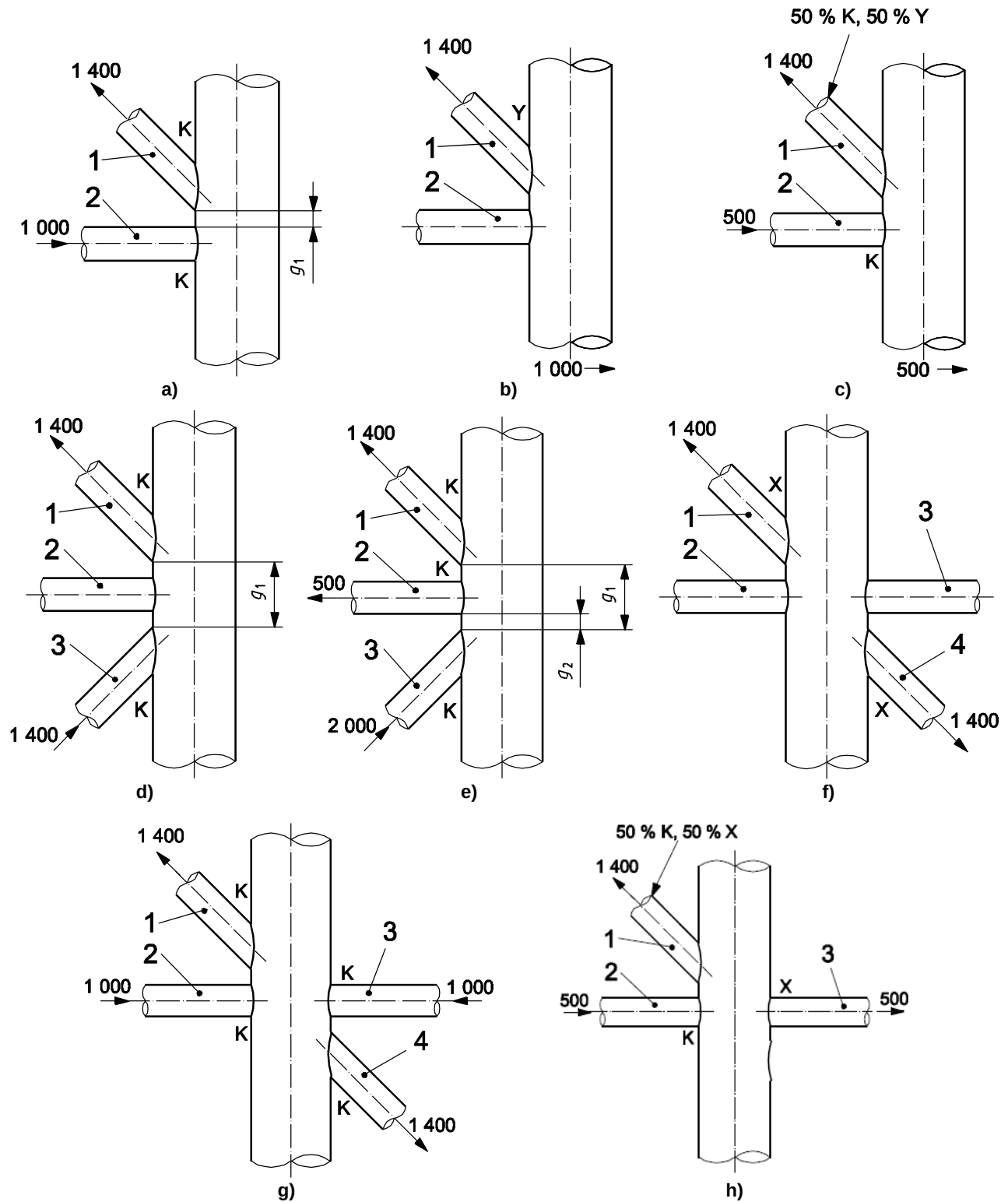


Figure 14.2-1—Basic Planar Joint Types

The design strength of most joints can be determined using the parametric formulae given in 14.3 for the three basic planar joint types. The practical use of the basic joint formulae shall reflect, as closely as possible, the force pattern assumed in deriving the formulae by classifying each combination of brace(s) and chord according to the flow of the axial force in the brace(s).

Classification shall apply to the combination of an individual brace with a chord (Y-joint) or combination of a pair of braces with a chord (K- or X-joints), rather than to the whole joint on the basis of the axial force pattern for each load case. Thus, for example, an individual brace can form part of a K-joint and part of a Y-joint and shall be classified accordingly as a proportion of each relevant type, e.g. 50 % K-joint and 50 % Y-joint. This classification is relevant to both fatigue and strength considerations. Classification shall proceed on the basis of assessing whether the joint is, firstly, a K-joint and, secondly, a Y-joint or an X-joint. Figure 14.2-2 shows some simple examples of the brace joint classification scheme.



Key

g_1 gap 1

g_2 gap 2

Figure 14.2-2—Examples of Joint Classification

Figure 14.2-2 h) is a good example of the axial force flow and classification hierarchy that should be adopted in the classification of braces in joints. The braces 1 and 2 on the left hand side of the chord act as a K-joint accounting for 50 % of the axial force in the diagonal brace. The other 50 % of the axial force in brace 1 forms an X-joint with brace 3. Replacement of brace axial forces by a combination of tension and compression forces to give the same net force is not permitted. For the example shown in Figure 14.2-2 h), replacing the axial force in brace 2 by a compression force of 1000 and a tension force of 500 is not permitted, as this will result in an inappropriate X-joint classification for this horizontal brace and a full K-joint classification for brace 1.

Careful consideration should be given to determining the correct gap between braces in a K-joint. In Figure 14.2-2 a), the appropriate gap is between adjacent braces. However, if an intermediate brace exists, as in Figure 14.2-2 d), the appropriate gap is between the outer braces acting as the K-joint. In this case, since the gap is often large, the K-joint strength can revert to that of a Y-joint. Figure 14.2-2 e) is instructive in that the appropriate gap for brace 2 is g_2 , whereas for brace 1 it is g_1 . Although brace 3 is classified wholly as a K-joint (with brace 2 for 500 normal to the chord and with brace 1 for the remainder of the normal component of brace 3), the strength is determined by weighting the strength with gaps of g_1 and g_2 by the proportions of the axial force balancing from braces 1 and 2.

There are some instances where the above joint classification methodology is difficult to apply or is not applicable. Two of the more common cases in the latter category are associated with loads/actions on a launch frame and with in situ loads/actions on skirt pile-sleeves. Some guidance for such instances is given in A.14.2.5.

Once the joint classification is established, the joint strength can be determined using the procedures in 14.3.

14.2.6 Detailing Practice

Joint detailing is an essential element of joint design. For unreinforced joints, the recommended detailing nomenclature and dimensioning are shown in Figures 14.2-3 and 14.2-4. Where an increased wall thickness or higher yield or toughness properties is required for the chord, this material should extend beyond the outside edge of incoming braces by the greater of a minimum of one quarter of the chord diameter, or 300 mm. The strength of Y- and X-joints is a function of the can length (see 14.3.5) and short can lengths can lead to a reduction in joint strength. Increasing the can lengths beyond the minimum values given here should be considered to avoid the need for downgrading strength.

In an X-joint, the larger diameter member shall continue through the joint, and the other should frame onto the through member. Where members of equal diameter meet at an X-joint, it is more efficient to make the through member that which carries the greater forces.

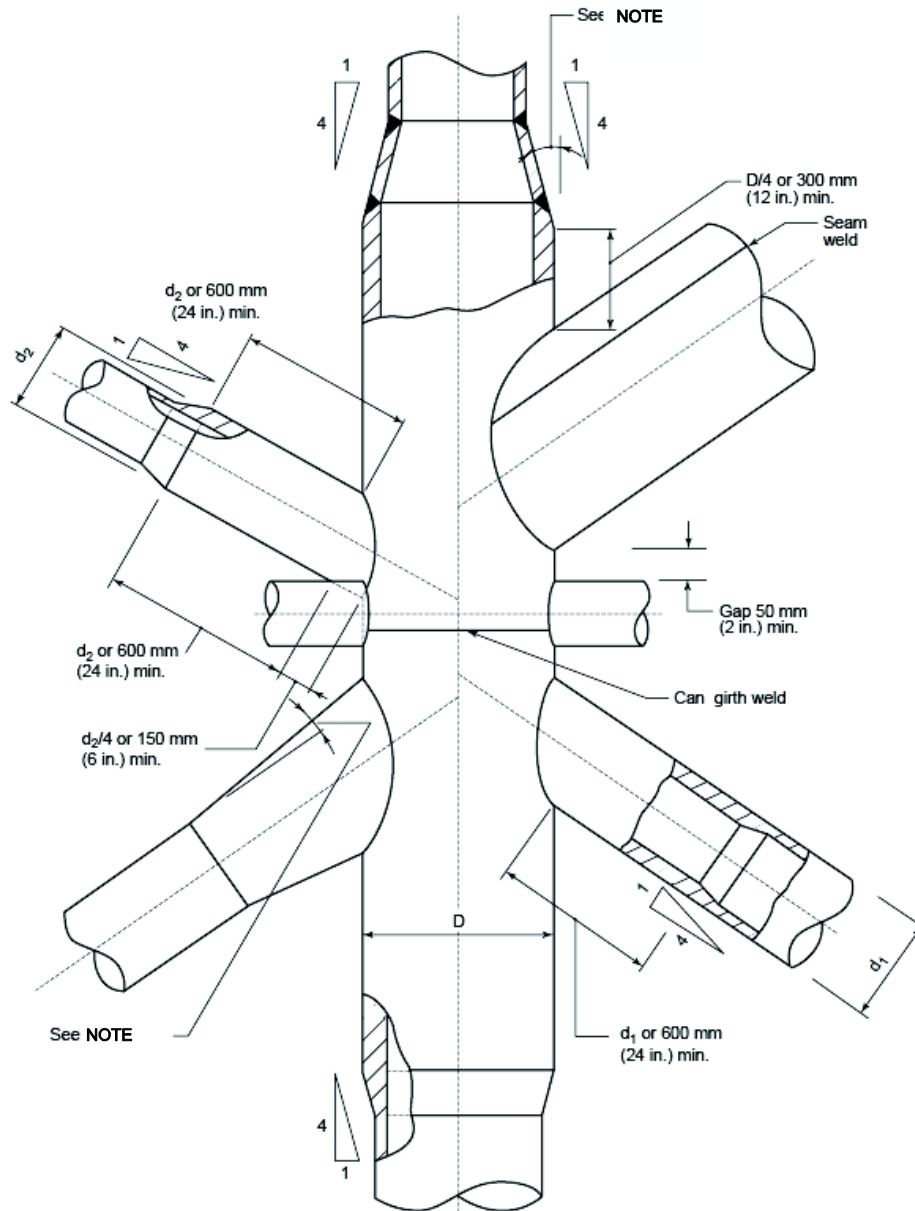
When two or more braces intersect at a joint, the order in which each member frames into the joint should be determined by wall thickness and/or diameter. The member with the thickest wall should be the continuous or through member, and the sequence for framing the remaining members shall be based on the order of decreasing wall thickness. If two or more members have the same wall thickness, the larger diameter member shall be the continuous or through member. The through brace can require an end stub to ensure that its thickness is at least equal to that of the overlapping brace. If two or more members have the same diameter and wall thickness, either member may be the through member unless the designer has designated a through member. Sections of brace welds, which will be covered by other brace connections, shall be inspected by nondestructive testing (NDT) prior to cover up.

When a brace requires increased wall thickness, higher yield strength, and/or more stringent toughness properties, the required change material should extend from its connection with the chord or beyond its connection with any overlapping brace(s) by the greater of the brace diameter or 600 mm.

Neither the chord can nor the brace stub minimum dimensions given above include the length over which any thickness taper occurs; any difference in thickness between chord can and chord member or between brace stub and brace member shall be tapered at 1:4 or lower (see Figure 14.2-3). For joints where fatigue considerations are important, tapering on the inside can have both an undesirable influence on crack origin and make early detection of cracks more difficult; for such joints, tapering should be on the outside (i.e. matching internal diameters).

The nominal gap (i.e. excluding weld toes) between adjacent braces, whether in-plane or out-of-plane, should not be less than 50 mm. Overlapping of welds of nonoverlapping braces at the weld toes shall be avoided. When braces overlap, the overlap should be at least $d/4$ (where d is the diameter of the through brace) or 150 mm, whichever is greater. This dimension is measured along the axis of the through member (see Figure 14.2-3).

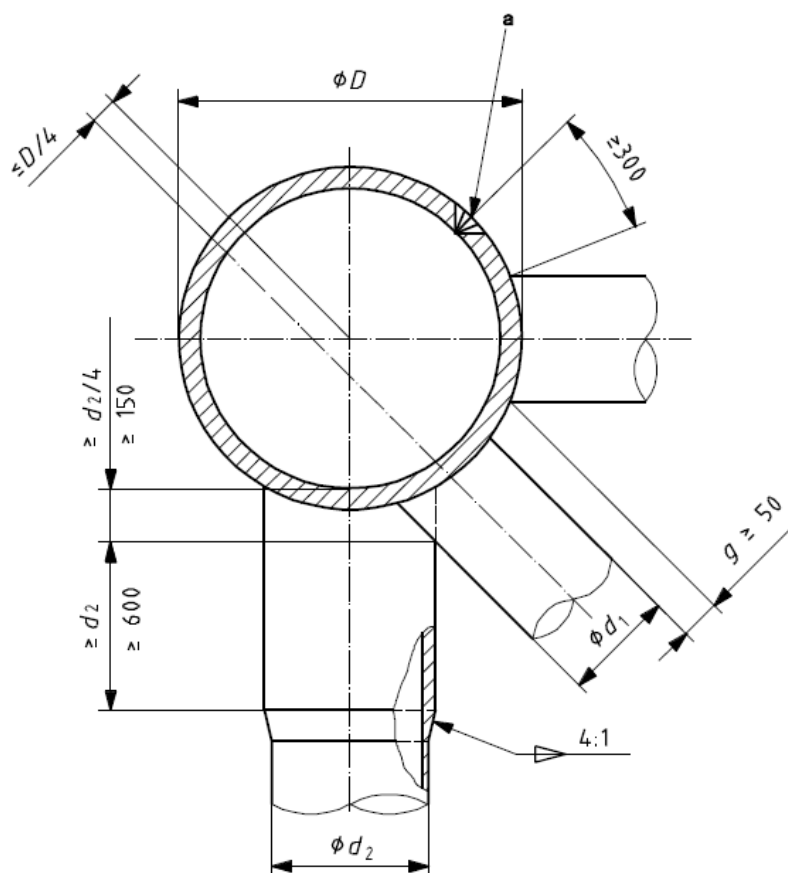
Longitudinal seam welds and circumferential welds should be located to minimize or eliminate their impact on joint performance. The longitudinal seam weld of a brace should be located near the crown heel of the joint (see Figure 14.2-3). The longitudinal seam weld of the chord should be separated from incoming braces by at least 300 mm (see Figure 14.2-4). Where a chord requires a circumferential weld to achieve the desired can length, the weld should be positioned at a lightly loaded brace intersection, between saddle and crown locations (see Figure 14.2-3).



NOTE Slope of conical transition 1 to 4 for $D/t < 30$; flatter slope necessary for higher D/t ratios.

Figure 14.2-3—In-plane Joint Detailing

Dimensions in millimeters



^a Longitudinal seam weld of chord can.

Figure 14.2-4—Out-of-plane Joint Detailing

14.3 Simple Tubular Joints

14.3.1 General

Simple tubular joints are joints having no gussets, diaphragms, grout, or stiffeners. Simple Y- and X-joints have no overlap of principal braces, but simple K-joints may have overlaps up to $0.6 D$.

The validity ranges for the parameters that are used in the formulae given in the remainder of 14.3 are as follows:

- $0.2 \leq \beta \leq 1.0$;
- $10 \leq \gamma \leq 50$;
- $30^\circ \leq \theta \leq 90^\circ$;
- $\tau \leq 1.0$;
- $f_y \leq 500 \text{ MPa}$.

For K-joints, the following validity range also applies: $0.6 < g/D$. Annex A discusses approaches that can be adopted for joints that fall outside the above ranges.

14.3.2 Basic Joint Strength

The strength of a joint varies not only with its material properties and geometry but also with the pattern of forces on each brace. Consequently, values of strength can vary between load cases.

The strength of simple tubular joints should be calculated for each brace, for each individual force component of tension, compression, in-plane bending (IPB) and out-of-plane bending (OPB), and for each load case consisting of a combination of forces.

Representative strengths for simple tubular joints are given in Equations (14.3-1) and (14.3-2):

$$P_{uj} = \frac{f_y T^2}{\sin \theta} Q_u Q_f \quad (14.3-1)$$

$$M_{uj} = \frac{f_y T^2 d}{\sin \theta} Q_u Q_f \quad (14.3-2)$$

where

P_{uj} is the representative joint axial strength, in force units;

M_{uj} is the representative joint bending moment strength, in moment units;

f_y is the representative yield strength of the chord (the lesser of SMYS or 0.8 of the tensile strength), in stress units;

T is the chord wall thickness at the intersection with the brace;

d is the brace outside diameter;

θ is the included angle between brace and chord;

Q_u is a strength factor (see 14.3.3);

Q_f is a chord force factor (see 14.3.4).

For braces with a mixed classification, P_{uj} and M_{uj} should be calculated by weighting the contributions from Y-, K-, and X-joint behavior by the proportions of that behavior in the joint. This means that P_{uj} and M_{uj} can be different for each load case considered (see A.14.3.2).

For X- and Y-joints with chord cans, P_{uj} shall not exceed the strength limits defined in 14.3.5.

The design values of the strength of simple tubular joints are as follows:

$$P_d = \frac{P_{uj}}{\gamma_{R,j}} \quad (14.3-3)$$

$$M_d = \frac{M_{uj}}{\gamma_{R,j}} \quad (14.3-4)$$

where

P_d is the design value of the joint axial strength, in force units;

M_d is the design value of the joint bending moment strength, in moment units;

$\gamma_{R,j}$ is the partial resistance factor for tubular joints, $\gamma_{R,j} = 1.00$.

14.3.3 Strength Factor, Q_u

In general, the strength factor, Q_u , varies with joint classification and type of brace force, as given in Table 14.3-1. For IPB and OPB, Q_u is independent of joint classification.

Table 14.3-1—Values for Q_u

Joint Classification	Brace Force			
	Axial Tension	Axial Compression	In-plane Bending	Out-of-plane Bending
K	$(16 + 1.2\gamma)\beta^{1.2}Q_g$ but $\leq 40\beta^{1.2}Q_g$		$(5 + 0.7\gamma)\beta^{1.2}$	$2.5 + (4.5 + 0.2\gamma)\beta^{2.6}$
T/Y	30β	$2.8 + (20 + 0.8\gamma)\beta^{1.6}$ but $\leq 2.8 + 36\beta^{1.6}$		
X	$6.4\gamma^{0.6}\beta^2$	$[2.8 + (12 + 0.1\gamma)\beta]Q_\beta$		

In Table 14.3-1, Q_β is a geometrical factor defined by:

$$Q_\beta = \frac{0.3}{\beta(1 - 0.833\beta)} \quad (\text{for } \beta > 0.6) \quad (14.3-5)$$

$$Q_\beta = 1.0 \quad (\text{for } \beta \leq 0.6) \quad (14.3-6)$$

and Q_g is a gap factor defined by:

$$Q_g = 1 + 0.2 [1 - 2.8g/D]^3 \text{ but } \geq 1.0 \quad \text{for } g/D \geq 0.05 \quad (14.3-7)$$

$$Q_g = 0.13 + 0.65\phi\gamma^{+0.5} \quad \text{for } g/D \leq -0.05 \quad (14.3-8)$$

For $-2.0 < g/T < +2.0$, the gap factor, Q_g , may be found by linear interpolation between the results of Equations (14.3-7) and (14.3-8) for the limiting values of $g/T = -2.0$ and $g/T = +2.0$,

where

$$\phi = t \times f_{y,b} / (T \times f_y)$$

and, in addition to the definitions given in 14.3.2,

$f_{y,b}$ is the representative yield strength of the brace at the intersection with the chord;

t is the brace wall thickness at the intersection with the chord.

Note the following applications:

- The overlap should preferably not be less than $0.25\beta D$.
- The Q_u term for tension loading is based on limiting the capacity to first crack. The Q_u associated with full ultimate capacity of tension loaded Y- and X-joints is given in A.14.
- The X-joint, axial tension, Q_u term for $\beta > 0.9$ applies to coaxial braces (i.e. $e/D \leq 0.2$ where e is the eccentricity of the two braces). If the braces are not coaxial ($e/D > 0.2$) then 23β should be used over the full range of β .

14.3.4 Chord Force Factor, Q_f

The chord force factor, Q_f , is a factor that accounts for the presence of forces from factored loads/actions in the chord:

$$Q_f = \left[1 + C_1 \left(\frac{P_c}{P_y} \right) - C_2 \left(\frac{M_{ipb}}{M_p} \right) - C_3 A^2 \right] \quad (14.3-9)$$

where

$$A = \left[\left(\frac{P_c}{P_y} \right)^2 + \left(\frac{M_c}{M_p} \right)^2 \right]^{0.5} \quad (14.3-10)$$

where

P_c is the axial force in the chord from factored loads/actions;

M_c is the bending moment resultant in the chord from factored loads/actions, $M_c = (M_{ipb}^2 + M_{opb}^2)^{0.5}$;

P_y is the representative axial strength due to yielding of the chord can ignoring buckling, in force units;

$$P_y = A_c f_y$$

f_y is the representative yield strength of the chord can, in stress units;

A_c is the cross-sectional area of the chord or chord can at the brace intersection;

M_p is the representative plastic moment strength of the chord can;

$\gamma_{R,q}$ is the partial resistance factor for yield strength, $\gamma_{R,q} = 1.05$;

ipb refers to in-plane bending;

opb refers to out-of-plane bending;

C_1, C_2, C_3 are the coefficients given in Table 14.3-2.

Table 14.3-2—Values for the Coefficients C_1 , C_2 , and C_3

Joint Type	C_1	C_2	C_3
K-joints under brace axial loading	0.2	0.2	0.3
T/Y-joints under brace axial loading	0.3	0	0.8
X-joints under brace axial tension ^a			
$\beta \leq 0.9$	0.2	0	0.5
$\beta = 1.0$	0.2	0	0.2
X-joints under brace axial compression ^a			
$\beta \leq 0.9$	0.2	0	0.5
$\beta = 1.0$	-0.2	0	0.2
All joints subject to brace bending moments	0.2	0	0.4
^a Linearly interpolated values between $\beta = 0.9$ and $\beta = 1.0$ for X-joints under brace axial loading.			

The average of the chord forces and bending moments on either side of the brace intersection should be used in Equations (14.3-9) and (14.3-10). Chord axial force is positive in tension and chord resultant bending moment is positive when it produces compression at the brace footprint.

14.3.5 Effect of Chord Can Length on Joint Strength

For simple Y- and X-joints with a chord can, the joint representative axial strength shall be calculated using Equation (14.3-11):

$$P_{uj} = \left[r + (1-r)(T_n/T_c)^2 \right] P_{uj,c} \quad (14.3-11)$$

where

P_{uj} is the representative joint axial strength, in force units;

$P_{uj,c}$ is the value of P_{uj} from Equation (14.3-1), based on chord can geometrical and material properties, including Q_f calculated from chord can properties and dimensions;

r = $L_c/(2.5 D)$ for joints with $\beta \leq 0.9$;
 = $(4 \beta - 3) L_c/(1.5 D)$ for joints with $\beta > 0.9$;
 ≤ 1.0

L_c is the effective total chord length (see Figure 14.3-1);

T_n is the lesser of the nominal chord's thicknesses on either side of the chord can (see Figure 14.3-1);

T_c is the chord can thickness (see Figure 14.3-1).

Figure 14.3-1 gives examples for the calculation of L_c .

As an alternative to Equation (14.3-11), joint axial strength, P_{uj} , may be evaluated using an approximate closed ring analysis. Such an analysis should include plastic behavior with appropriate safety factors, and an effective chord length up to $1.25 D$ on either side of the line of load/action of the brace force at the chord face, taking into account any thickness changes within this distance. Where multiple braces are in the same plane and dominantly loaded in the same sense (tension or compression), the relevant perpendicular force is derived as the summation of relevant components over all the braces.

NOTE There is no reduction in the strength of K-joints due to length effects.

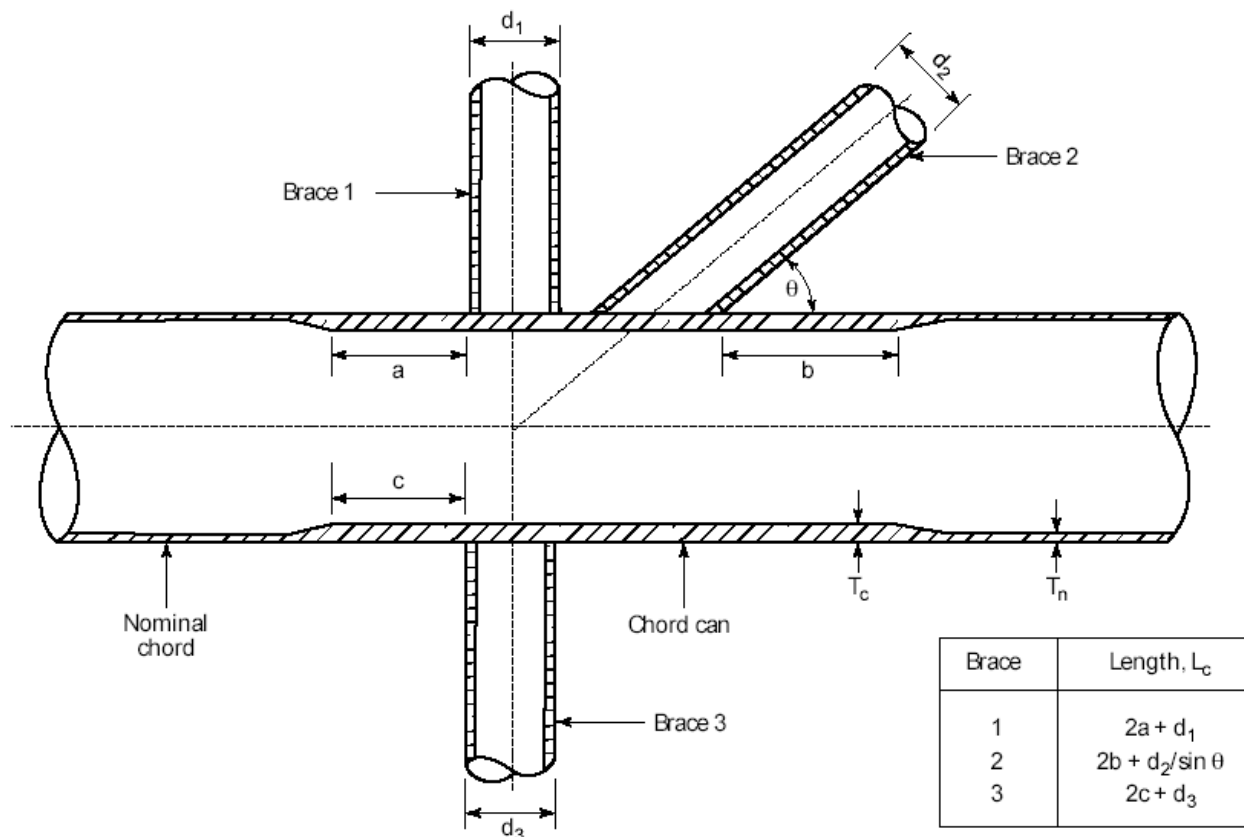


Figure 14.3-1—Examples of Chord Length L_c Calculation

14.3.6 Strength Check

Each brace in a joint that is subjected either to an axial force or a bending moment alone, or to an axial force combined with bending moments, shall satisfy the following:

$$U_j = \left| \frac{P_B}{P_d} \right| + \left(\frac{M_B}{M_d} \right)_{ipb}^2 + \left| \frac{M_B}{M_d} \right|_{opb} \leq 1.0 \quad (14.3-12)$$

where

U_j is the joint utilization;

P_B is the axial force in the brace from factored loads/actions;

M_B is the bending moment in the brace from factored loads/actions;

P_d is the design value of the joint axial strength (see 14.3.2);

M_d is the design value of the joint bending moment strength (see 14.3.2);

ipb represents in-plane bending;

opb represents out-of-plane bending.

14.4 Overlapping Joints

The strength of joints with in-plane overlaps involving two or more braces shall be determined using the requirements for simple joints defined in 14.3, with the following exceptions and additions.

- a) Shearing of the brace parallel to the chord face is a potential failure mode and shall be checked.
- b) Subsection 14.3.5 shall not apply to overlapping joints.
- c) If axial forces in the overlapping and through braces have the same sign (both compression or both tension), when determining the strength of the through brace, P_B shall be calculated as the combined axial force equal to the force in the through brace plus a portion of the overlapping brace force(s). The portion of the overlapping brace force is calculated as the ratio of the cross-sectional area of the overlapping brace that bears onto the through brace to the full area of the overlapping brace multiplied by the force in the brace.
- d) When checking the strength of the through brace, the combined moments on the overlapping and through braces, taking account of the sign of the moments, shall be used to compute M_B .
- e) The overlap onto the through brace shall be checked by using the through brace as the chord in the equations in 14.3. The through brace strength shall also be checked for combined axial force and bending moment in the overlapping brace in accordance with 14.3.6 using the value of Q_f calculated for the through brace. Should this lead to $\beta > 1$, β shall be taken as 1.0.
- f) Where nominal thicknesses of the overlapping and through braces differ by more than 10 %, the thicker brace shall be treated as the through brace.

Joints having out-of-plane overlap can be checked on the same general basis as in-plane overlapping joints, except that the axial strength should normally revert to that for Y-joints.

14.5 Grouted Tubular Joints

Two varieties of grouted joints are commonly used in practice. The first is a fully grouted chord; the second has a double-skin, whereby grout is placed in the annulus between an outer chord and an internal member. In both cases, the grout is unreinforced and no benefit to joint strength shall be derived from shear keys (if present).

The strength of grouted joints that are otherwise of simple configuration may be determined using the simple joint requirements defined in 14.3, with the modifications and limitations below.

- a) For fully grouted and double-skin joints, the Q_u values in Table 14.3-1 may be replaced with the values pertinent to grouted joints given in Table 14.5-1. Classification and joint can derating may be disregarded. The adopted Q_u values should not be less than those for simple joints.

Table 14.5-1— Q_u for Grouted Joints

Brace Load	Q_u
Axial tension	$2.5\beta\gamma K_a$ where $K_a = \frac{1}{2} \left(1 + \frac{1}{\sin \theta} \right)$
Bending	$1.5\beta\gamma$
NOTE No term is provided for axial compression since most grouted joints cannot fail under compression. Compression capacity is limited by that of the brace.	

The minimum capacity requirements of 14.3 should still be observed.

- b) For double-skin grouted joints, an effective thickness, T_e , should be determined from Equation (14.5-1):

$$T_e = (T^2 + T_p^2)^{0.5} \quad (14.5-1)$$

where

T is the wall thickness of the chord;

T_p is the wall thickness of the inner member.

The effective thickness, T_e , should be used in place of T in the simple joint equations, including the determination of the γ parameter, but taking into account b) and c) hereafter.

- c) The Q_f value calculated from Equations (14.3-9) and (14.3-10) depends on joint type and brace loading through the coefficients in Table 14.3-2, but not on thickness.
- d) Axial and moment design strengths calculated for the joint shall exceed the brace design strengths, for noncompact members inclusive of the local buckling strength at the brace end.

14.6 Ring Stiffened Tubular Joints

Primary joints along launch frames are often strengthened by ring stiffening. Ring stiffening is also used in some structures to address fatigue requirements or to avoid very thick chord cans. A.14.6 outlines the salient features of several common approaches available to design ring-stiffened joints.

14.7 Other Joint Types

Joints not covered by 14.3 to 14.6 may be designed on the basis of appropriate experimental, numerical, or in-service evidence. For circular joints with doubler or collar plates, material strength approaches may be employed, although extreme care is needed in identifying all components that are expected to participate in resisting incoming brace forces and in establishing the acting force envelopes prior to conducting strength checks. Often material strength checks are complemented with calibrated FEA to establish the magnitude and location of acting stresses.

14.8 Damaged Joints

Joints in existing structures can become damaged as a result of fatigue, corrosion, or overload (environmental or accidental). In such cases, the reduced joint strength shall be estimated from simple models, e.g. based on the use of reduced area or section modulus, from more extensive numerical analysis using FEA models or experimental evidence.

14.9 Noncircular Joints

Joints with noncircular chord or brace sections are typically used in topsides structure (see API 2TOP [20]). Common types include I-beam, column, and plate girder sections and rectangular or square hollow sections. For some configurations, detailed land-based design practice is available (see A.14.9). For arrangements for which little or no practice is available, the requirements are similar to those according to 14.7.

14.10 Cast Joints

Cast joints can be of any geometry and are usually of variable wall thickness; they are of particular benefit when there are many similar joints or where fatigue is causing difficulties. The design of a cast joint requires calibrated FEA. A conservative design approach for strength is to limit stresses in the joint due to forces from factored loads/actions to below the design yield strength of the material, $f_y/\gamma_{R,q}$ (see 14.3.4). The design of cast joints is often carried out in conjunction with the manufacturer.

15 Strength and Fatigue Resistance of Other Structural Components

15.1 Grouted Concentric-tubular Connections

15.1.1 General

Platform loads/actions may be transferred to steel piles/pins by grouting the annulus between the jacket leg (or sleeve) and the pile/pin. The load is transferred to the pile/pin from the structure across the grout. Experimental work indicates that the mechanism of load transfer is a combination of bond and confinement friction between the grout and the steel surfaces and the bearing of the grout against mechanical aids such as shear keys.

Centralizers should be used to maintain a uniform annulus or space between the pile/pin and the surrounding structure. A minimum annulus width of 38 mm (1 1/2 in.) should be provided where grout is the only means of load transfer. Adequate clearance between pile and sleeve should be provided, taking into account the shear keys' outstand dimension, h . Packers should be used as necessary to confine the grout. Proper means for the introduction of grout into the annulus should be provided so that the possibility of dilution of the grout or formation of voids in the grout will be minimized. The use of wipers or other means of minimizing mud intrusion into the spaces to be occupied by piles should be considered at sites having soft mud bottoms.

15.1.2 Factors Affecting the Connection Strength

Many factors affect the strength of a grouted connection. These include, but are not limited to, the unconfined compressive strength of the grout; size and spacing of the shear keys; type of admixture; method of placing grout; condition of the steel surfaces; presence of surface materials that would prevent bonding of grout to steel; and the amount of disturbance from platform movement while the grout is setting. For high D/t ratios, the hoop flexibility of the sleeve and the pile is also known to be a factor.

15.1.3 Computation of Applied Axial Force

In computing the axial force applied to a grouted pile/pin-to-structure connection, due account should be taken of the distribution of overall structural loads/actions among various piles in a group or cluster. The design load/action for the connection should be the highest computed load/action determined from factored loads/actions with due consideration given to the range of axial pile and in situ soil stiffness.

15.1.4 Computation of Allowable Axial Force

15.1.4.1 General

In the absence of reliable comprehensive data that would support the use of other values of connection strength, the allowable axial load transfer should be taken as the smaller value (pile or sleeve) of the force calculated by a multiplication of the contact area between the grout and steel surfaces and the allowable axial load transfer stress f_{ba} , where f_{ba} is computed by the appropriate value in 15.1.4.2 or 15.1.4.3 for the grout/steel interface. This allowable axial force should be greater than or equal to the applied axial force determined from factored loads/actions computed according to 15.1.3.

15.1.4.2 Plain Pipe Connections

The value of the allowable axial load transfer stress, f_{ba} , should be taken as 0.248 MPa (36 psi).

15.1.4.3 Shear Key Connections

Where shear keys are used at the interface between steel and grout, the value of the allowable axial load transfer stress, f_{ba} , should be taken as:

$$f_{ba} = 0.248 \text{ MPa (36 psi)} + 0.90 f_{cu} \times \frac{h}{s} \quad (15.1-1)$$

where

f_{cu} is the unconfined grout compressive strength, in megapascals (pounds per square inch), as defined in 19.6.2;

h is the shear key outstand dimension, in millimeters (inches) (see Figure 15.1-1 and Figure 15.1-2);

s is the shear key spacing, in millimeters (inches) (see Figure 15.1-1 and Figure 15.1-2).

The maximum axial load transfer stress calculated for loading combinations given in 9.10.3 should satisfy the following condition:

$$\sigma_{b,\max} \leq \frac{f_{ba}}{\gamma_{ba}} \quad (15.1-2)$$

where

$\sigma_{b,max}$ is the maximum load transfer stress from factored loads;

γ_{ba} is the partial resistance factor for axial load transfer, $\gamma_{ba} = 1.1$.

Shear keys designed according to Equation (15.1-1) should be detailed in accordance with the following requirements.

- Shear keys may be circular hoops at spacing “s” or a continuous helix with a pitch of “s.” See 15.1.4.4 for limitations.
- Shear keys should be one of the types indicated in Figure 15.1-2.
- For driven piles, shear keys on the pile should be applied to sufficient length to ensure that, after driving, the length of the pile in contact with the grout has the required number of shear keys.
- Each shear key cross section and weld should be designed to transmit that part of the connection capacity that is attributable to the shear key for operating loads/actions in 9.10.3. The shear key and weld should be designed at basic allowable steel and weld stresses to transmit an average force equal to the shear key bearing area multiplied by $1.7f_{cu}$, except for a distance of 2 pile diameters from the top and the bottom end of the connections where $2.5f_{cu}$ should be used.

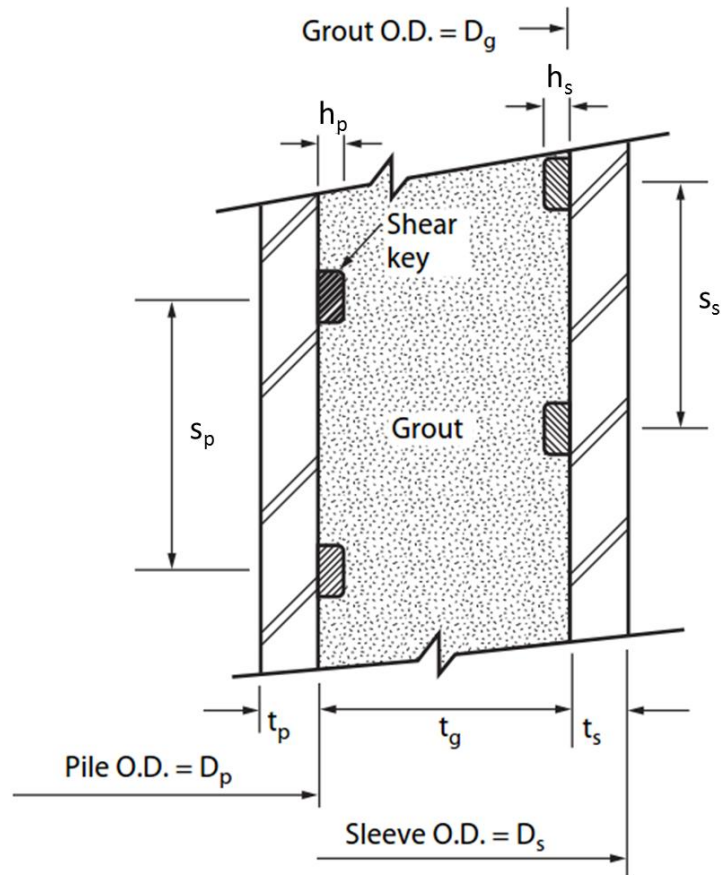


Figure 15.1-1—Grouted Pile-to-structure Connection with Shear Keys

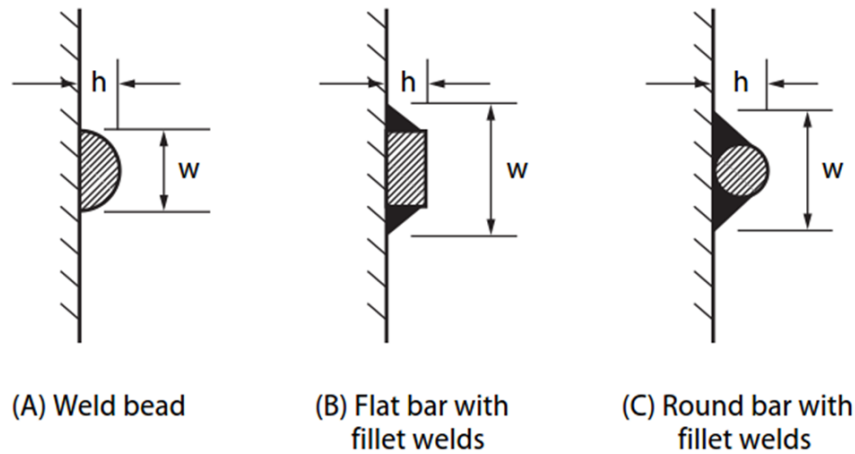


Figure 15.1-2—Recommended Shear Key Details

15.1.4.4 Limitations

The limitation $17 \text{ MPa (2500 psi)} \leq f_{cu} \leq 110 \text{ MPa (16,000 psi)}$ should be observed when designing a connection in accordance with 15.1.4.2 or 15.1.4.3.

The limitations in Table 15.1-1 should be observed when designing a connection according to 15.1.4.3 (see Figure 15.1-1 and Figure 15.1-2).

Table 15.1-1—Connection Design Limitations

Attribute	Limitation
Sleeve geometry	$\frac{D_s}{t_s} \leq 80$
Pile geometry	$\frac{D_p}{t_p} \leq 40$
Grout annulus geometry	$7 \leq \frac{D_g}{t_g} \leq 45$
Shear key spacing ratio ^a	$2.5 \leq \frac{D_p}{s} \leq 8$
Shear key ratio	$h/s \leq 0.10$
Shear key shape factor	$1.5 \leq \frac{w}{h} \leq 3.0$
Product of f_{cu} and h/s	$\leq 5.5 \text{ MPa (800 psi)}$
^a For helical shear keys only.	

15.1.4.5 Other Design Methods

Other methods, which are based on testing and verification, may be used for calculating the allowable load transfer stress f_{ba} . One such method is described in A.15.1.4.5.

15.1.4.6 Loadings Other Than Axial Load

Grouted pile-to-sleeve connections will be subjected to loading conditions other than axial load/action, such as transverse shear and bending moment or torque. The effect of such loadings, if significant, should be considered in the design of connections by appropriate analytical or testing procedures.

15.1.5 Fatigue Assessment

Grouted connections that comply with the static strength requirements of 15.1.4 and that are subjected primarily to cyclically varying forces due to wave load/action do not require a detailed fatigue assessment.

In other cases, the fatigue strength may be evaluated with the provisions in A.15.1.5.

15.2 Mechanical Connections

15.2.1 Types of Mechanical Connectors

Mechanical connectors may be used to join structural components of offshore structures. These components include, but are not limited to, (foundation) piles, conductors, production and export risers, and tendons. Although there are a variety of connector designs, the following four general types of connectors are most commonly used for joining coaxial tubular members:

- a) threaded or grooved connectors of the pin-and-box type assembled by torque or radial interference;
- b) flanged connectors assembled by threaded fasteners;
- c) dogged connectors that use radial wedges between the pin and box parts;
- d) swaged connectors.

Requirements for conventional mechanical connectors a) to c) above are given in 15.2.2 to 15.2.7; specific requirements for threaded connections only are given in 15.2.8; and requirements for swaged connections are given in 15.2.9.

Connectors that rely on impinging grip or friction grip by means of collets or sliding wedges to affect load transfer are generally not amenable to the analysis procedures discussed in A.15.2. However, the structural and validation requirements still apply, and testing is recommended for validating performance.

15.2.2 Design Requirements

15.2.2.1 General

The design of mechanical connectors for structural applications shall comply with the following structural and functional requirements. Structural requirements relate to the static strength and fatigue performance, whereas functional requirements include assembly, disassembly, installation, and sealing.

15.2.2.2 Static Strength Requirements

Connectors should be as strong as, or stronger than, the weakest of the connected tubular members. In all cases, connectors shall be designed to the strength criteria in 15.2.5. If the connector has a strength lower than the representative yield strength of the attached full tubular cross-section, the effect of relevant extreme forces imparted by waves and current, as well as forces due to deformation of the connector, forces due to possible restraints, and accidental loads/actions shall all be assessed. The effects of deformations and sudden disengagement, such as thread jump-out, shall also be considered in detail.

15.2.2.3 Fatigue Performance Requirements

Connectors should have a fatigue endurance in excess of full penetration, circumferential welds between tubular members of the same size. Failing this, the fatigue life of the connector shall exceed the planned life for the tubular component by a fatigue damage design factor (see 16.12.2), taking into account the following:

- a) inspectability and inspection program (see API 2SIM);
- b) likelihood and consequence of failure;
- c) uncertainty in forces due to loads/actions and analysis methods;
- d) uncertainty in the fatigue life calculation method;
- e) remaining life, if through-thickness cracking does occur.

For structurally critical applications, where in-service inspection of the connector is unreliable or not possible, the calculated fatigue life of the connector shall be a minimum of 10 times the planned life of the connected tubular members.

15.2.2.4 Functional Requirements

Connectors shall assemble and disassemble (when applicable) in a reliable manner consistent with the assumptions made in the analysis of the model used to demonstrate compliance with both strength and fatigue criteria. In cases where the connector is to contain internal or external pressure, seals shall be provided such that they remain sound after installation and for the design life of the connection. All necessary functions of the connector shall remain effective upon application of the design and installation forces.

15.2.3 Loads/Actions and Forces on the Connector

Loads/actions can be applied directly to a connector or indirectly through the tubular members being connected. The loads/actions can include any combination of pressures (internal, external, or both), forces and moments in the connected tubular members, deformations and thermal gradients. A simplified approach for calculating an equivalent axial force on the tubular members when they are simultaneously subjected to axial forces and bending moments is given in A.15.2.3.

15.2.4 Resistance of the Connector

The strength of the connector can be determined by a calibrated analysis (see A.15.2.4) or prototype testing to failure (see 7.7). The strength shall be determined as the maximum applied typical force that can be sustained by the fully made up connector in combination with all other relevant deformations and loads/actions applied repeatedly and in any order, without giving rise to gross deformation, disengagement, or, if applicable, loss of sealing. The representative resistance of the connector, R_c , shall be derived in accordance with 7.7.

15.2.5 Strength Criteria

Connectors shall be designed to sustain the forces due to factored loads/actions and deformations arising from external loads/actions applied to the tubular members, thermal gradients, internal and external pressures, and assembly, without exceeding the representative resistance of the connector as defined in 15.2.4 and reduced by the partial resistance factor γ_R . The potential effect of temperature on material strength shall be taken into account. One approach to conducting strength checks for connectors based on linear FEA is given in A.15.2.5.

15.2.6 Fatigue Criteria

To satisfy the requirements given in 15.2.2.3, the fatigue life of the connector should be determined. For an approximate fatigue life calculation, either of two methods may be used: the stress life approach (the $S-N$ curve method) or the strain life approach (the initiation life method). Either approach requires local elastic stresses at the fatigue critical location(s) in the connector induced by the forces in the connected tubular members and material-specific fatigue data (see A.15.2.6).

15.2.7 Stress Analysis Validation

15.2.7.1 General

Although FEA programs are used to analyze connectors, some degree of experimental verification of the results obtained is highly desirable for confirming the validity of boundary conditions, simplifications in geometry, and preload simulation. Hence, direct measurements of actual strains/stresses using experimental techniques, e.g. strain gauges or photo elasticity, shall be included in a validation program.

15.2.7.2 Strength Validation

Experimental validation of the static strength of connectors shall either:

- a) demonstrate that the connector is stronger than the weakest of the connected tubular members, or
- b) establish the actual strength of the connector.

The advantage of b) is that it gives a measure of the actual reserve strength of the connector. In planning strength tests, specimen size and force-moment-pressure ($P-M-p$) interaction should be considered in light of the requirements imposed by the particular application.

15.2.7.3 Fatigue Validation

Because of the many uncertainties involved in fatigue calculation methods and the nature of the phenomenon itself, prototype tests shall be conducted to confirm the fatigue performance of the connector, unless equivalent lifetime service records are available for the connector in question. Uncertainties arise from effects that are difficult, if not impossible, to account for in the analysis procedures, but which can have a significant impact on fatigue life. Examples are connector size, surface finish, metallurgical and environmental effects. The extent and conditions of the fatigue testing should be commensurate with the level of conservatism built into the design process, the refinement of the analytical tools, the criticality of the connector application, and the degree of reliability sought through the validation exercise.

15.2.7.4 Functionality Validation

Analytical modeling of variables, such as surface contact, surface roughness and friction, is difficult and unreliable. These variables affect the functional performance of connectors in terms of assembly and sealing. Thus, tests

shall be conducted to demonstrate that connectors meet the functionality requirements. The tests shall use full-scale prototypes under conditions equivalent to, or more stringent than, those expected during operation.

15.2.8 Threaded Fasteners

15.2.8.1 General

In some structural components, such as flanges and clamps, threaded fasteners are the primary means to transfer forces across the assembled mechanical connector. Threaded fasteners should normally be pretensioned, in order to provide resistance to disengagement and improve fatigue performance. Stresses in the threaded fastener induced by preload alone should be treated as secondary (see A.15.2.5.1).

15.2.8.2 Threaded Fastener Materials and Manufacturing

Threaded fastener materials should be selected taking account of the application of the connector. Factors to be considered should include the following:

- a) maximum stress level to which the fastener is subjected;
- b) material yield strength, ultimate strength, and ductility;
- c) corrosion resistance and galvanic corrosion;
- d) durability at service temperature in the offshore environment;
- e) selection of suitable coatings to provide the resistance to corrosion;
- f) embrittlement;
- g) loss of stress in the fastener due to relaxation and creep.

The yield strength of carbon steel threaded fasteners should be less than 725 MPa to avoid potential embrittlement due to hydrogen absorption from exposure to cathodic protection potentials.

Manufacture of threaded fastenings shall take account of the intended service (see A.15.2.8.2).

15.2.8.3 Threaded Fastener Installation

To ensure that the threaded fasteners achieve their design tension, a suitable installation technique is required. A comprehensive installation methodology shall be derived and documented. Simultaneous use of hydraulic jacks or tensioners, one for each threaded fastener, is recommended. This technique avoids uneven preload, which can result from sequential stressing techniques. For clamps where torquing devices are used, the accuracy of the relation between applied torque and tension shall be allowed for during installation (see A.15.2.8.3).

When using tensioners, the installation stress shall allow for transfer losses due to immediate elastic strains in the fastener thread and structural components that occur when the preload transfers from the tensioner to the threaded fastener. The installation stress shall also allow for longer-term losses caused by creep in the threaded fastener and clamp components and for shrinkage of grout annuli or liner where used.

Spherical washers can relieve bending of the threaded fastener due to initial nut/bearing and plate misalignment.

Methods to prevent loosening of the fasteners should be considered.

15.2.8.4 Threaded Fastener Inspection

Threaded fasteners should be periodically inspected during operation to ensure that they remain sound, protected from corrosion, and properly tensioned. The inspection of the threaded fasteners should include measurements with respect to cathodic protection and tension (see A.15.2.8.4 and API 2SIM).

The inspection of threaded fasteners may take into account any redundancy provided by the number of threaded fasteners.

15.2.8.5 Threaded Fastener Strength Criteria

Threaded fasteners shall be designed to sustain the combined effect of assembly preload and the factored loads/actions applied to the connected tubular members.

15.2.8.6 Threaded Fastener Fatigue Criteria

The fatigue performance of threaded fasteners is directly linked to the stress ranges induced in them by the variation of the applied loads/actions. Such stress ranges shall be calculated by accepted engineering methods, taking into account the effect of preload, and the relative stiffness of the fasteners and the structural component parts they fasten. Palmgren-Miner rule, in conjunction with stress histories and $S-N$ curves, based on tests of actual fasteners, may be used to calculate fatigue damage.

15.2.9 Swaged Connections

15.2.9.1 Strength of Swaged Connections

A swaged connection between concentric tubular members is made by radially expanding, or swaging, the inner tubular into circumferential grooves machined on the internal surface of the outer tubular. Upon swaging, a permanent radial preload and a longitudinal mechanical interlock develop between the inner and the outer tubular walls. The resulting connection is referred to as a swaged connection.

Swaged connections shall be designed to sustain the static forces in the tubular members with no significant relative axial slippage between the inner member and the outer member, and no permanent deformation upon loading the connection within its rated strength. The static strength of swaged connections may be determined by calibrated analytical means or by representative tests with regard to size, material, and forming process. In doing so, due consideration should be given to the variables of the connection configuration. These variables include, but should not be limited to, the following:

- a) the number, depth, and width of the grooves;
- b) the spacing between adjacent grooves;
- c) the radii of the re-entrant and protruding corners inside and between the grooves;
- d) the pressure used in the swaging process.

The hoop stiffness of both tubular members shall be selected so that sufficient residual radial pressure remains at the contact points between the two parts, in order to develop adequate radial preload upon swaging.

15.2.9.2 Fatigue Performance of Swaged Connections

The fatigue performance of swaged connections shall be demonstrated by tests representative of in-service loads/actions, in which the definition of fatigue failure shall include cracking anywhere within the connection and unacceptable progressive slippage between the two tubular members.

15.2.9.3 Material for Swaged Connections

The material properties of the tubular members shall be selected to ensure that no cracks develop during the expansion of the tubular wall. In corrosive environments, such as offshore, the potential deleterious effects on the performance of the connection arising from general corrosion, stress corrosion cracking, corrosion fatigue, and crevice corrosion should be considered.

For swaged connections of foundation piles, the fracture performance of the material should remain within acceptable limits for the intended purpose following pile driving and swaging. Particular attention in this regard should be given to the weld material present in the pile and the sleeve.

15.2.9.4 Installation of Swaged Connections

Swaged connections may be produced by mechanical or hydraulic means applied to the inside of the inner tubular member. In either case, the tool and the installation procedure shall ensure that the inner tubular member is swaged onto all of the grooves in the outer tubular member and that the intended amount of swaging of the tubular is reliably attained. One way to accomplish this is by conducting onshore trials and calibrating the tools.

15.3 Clamps for Strengthening and Repair

15.3.1 General

Subsection 15.3 applies to the design of clamps to strengthen or repair tubular members or joints, or to connect tubular members.

The common features among these clamps are that they are deployed in two or more pieces, are fastened by bolts, and typically surround a structural component, such as a joint or a member; the surrounded component is further denominated as the substrate component. However, a key distinction can be made on the basis of the function and effect of the bolts, resulting in two generic types of clamps: split sleeve clamps and prestressed clamps.

15.3.2 Split Sleeve Clamps

In split sleeves the initial tension applied to the bolts is merely intended to provide continuity of the sleeve parts. When sufficiently stiff and properly preloaded, the longitudinal joint can provide hoop stiffness continuity in the split sleeve. Otherwise, the bolts simply maintain the pieces together so as to produce an encasement around the substrate component. In both cases, the annulus or space created between the outer clamp pieces and the substrate component is filled with grout, but no prestress develops between the clamp and the substrate component.

15.3.3 Prestressed Clamps

In prestressed clamps, the pretensioned bolts induce a continuous radial pressure on the substrate component, allowing friction on the contact surfaces, which then allows shear forces to be transferred between the parts of the clamp. The bolts can also provide a direct load path between the clamps parts. Three types of prestressed clamps are used, as follows.

- a) Prestressed grouted clamps—have the cavity left between the substrate members and the clamp filled with grout. After the grout is cured, prestress is applied normal to the grout surface to enhance the friction capacity strength. This prestress is normally achieved by stressing external bolts.
- b) Prestressed mechanical clamps—are similar to the prestressed grouted clamps, except that the grout is omitted and the clamp is in direct metal-to-metal contact with the substrate structure.
- c) Prestressed lined clamps—are the same as prestressed mechanical clamps, except that an elastomeric material is placed between the clamp and the substrate structure to accommodate local irregularities better.

Caution—Test data and in-service performance indicate that elastomeric lined clamps can be unreliable (see 15.3.6.3). Care should be taken in their application.

15.3.4 Forces on Clamps

15.3.4.1 Mechanism of Force Transfer

The internal forces in the substrate components and in the attached members generally translate into the following forces on the clamp:

- shear forces and bending moments that tend to separate the clamp pieces;
- axial force and torque along the clamp axis that tend to produce relative slippage between the clamp and the substrate members.

Prestressed clamps transfer forces by radial bearing contact and friction at the clamp to member interface. Therefore, the long-term bolt forces shall be of such magnitude as to ensure that the contact pressure induced by the bolts is not overcome by the separation load/action due to the forces applied to the clamp. At the same time, the contact pressure shall develop sufficient friction to resist slippage. Similar to grouted connections, the interaction of the bending moments and shear forces with the axial force, all acting at the free ends of the clamp, tends to enhance the slip strength of the clamp. Therefore, it is conservative to assume that the separation and slip forces do not interact.

15.3.4.2 Member Forces

In determining the forces in the members of the structure, the effects of both the modified stiffness and the modified environmental loads/actions, due to the presence of the clamp, shall be taken into account.

15.3.4.3 Bolt Forces

To evaluate the bolt forces necessary to resist the separation and slip loads/actions on the clamp, the contributions from all loads/actions on the clamped members (the substrate members) and the members attached to the clamp body shall be taken into account. The forces induced in the bolts may be determined by accepted analytical or numerical engineering methods or by testing. Particular attention shall be paid to the forces in the bolts at the ends of the clamp. Approximate methods for calculating the forces induced in the bolts by the separation forces and by the acting slip force at the clamp interface are presented in A.15.3.4.3.

15.3.5 Clamp Design

15.3.5.1 General Approach

Clamps shall be designed to fully sustain and transfer all of the forces and moments acting in the following:

- a) members attached to the clamp;
- b) substrate members on which the clamp is installed.

However, in cases where adequate static or fatigue strength of the intact or damaged substrate structural component can be demonstrated, the clamp may be designed to transfer only the portion of the forces and moments attracted by the clamp.

The design process involves selecting the size, number, distribution, and preload of the bolts, in conjunction with a clamp configuration such that the clamp shall sustain and transfer the forces in the clamped and attached members without opening, slipping, or crushing the clamped members. If time-varying fatigue forces are acting on the clamp, the fatigue performance of both the clamp and the bolts shall also be evaluated.

15.3.5.2 Check of the Clamped Member

Because the bolt forces are directly resisted by hoop forces in the clamped tubular members, the substrate members shall be checked to avoid crushing at the free end of the clamp. For mechanical and lined clamps, crimping (local buckling of the part of the substrate member not contained within the clamp) shall also be checked, if appropriate. In prestressed grouted clamps, advantage may be taken of the strength of the grout annulus to carry compression forces when calculating the radial crushing pressure on the clamped member, provided that the grout annulus is circumferentially continuous. Crushing of the clamped member can be alleviated by lengthening the clamp or filling the member with grout.

15.3.5.3 Static Design of Bolts

The long-term bolt tension shall be determined as the largest of the bolt forces required to:

- a) maintain a positive contact between the clamp and the substrate member under the load/action of the separating forces from factored loads/actions, and
- b) resist slippage under the load/action of the separating forces from factored loads/actions.

Bolts shall be sized to sustain the long-term tension in accordance with the requirements given in 15.2.8.5 for threaded fasteners.

The contribution from individual bolts to the overall strength of the clamp should be considered at the design stage to ensure that failure of any individual bolt or loss of tension in one or more bolts does not lead to overall failure of the clamp. The degree of bolt redundancy should be linked to the extent of inspection (see 15.2.8.4 for threaded fasteners).

15.3.5.4 Fatigue Design of Bolts

The evaluation of the variable bolt stresses caused by the time-varying forces shall be conducted by FEA or by any other demonstrably conservative engineering method (see A.15.3.5.4).

Fatigue lives may be predicted using the cumulative Palmgren-Miner damage ratio with an appropriate $S-N$ curve. In selecting the $S-N$ curve, account shall be taken of the type of bolts, the diameter, and the exposure conditions (see A.15.2.8.6 for threaded fasteners).

15.3.5.5 Interface Transfer Strength of Prestressed Clamps

The ability of prestressed clamps to transfer force at the clamp-substrate interface depends on numerous variables. These include the bolt size and the level of preload, the number and distribution of the bolts, the length and stiffness of the clamp, the size and surface finish of the clamped member, and, if applicable, the grout strength.

When calculating the slip strength of a prestressed clamp, the loss of bolt tension from transfer losses upon tensioning and long-term effects on the grout and steel shall be allowed for in the derivation of the long-term bolt forces (see 15.2.8 for threaded fasteners).

In prestressed clamps comprising a continuous ring of grout, a proportion of the bolt preload is resisted by hoop compression in the grout, leading to a reduction of the effective prestress at the grout/member interface. This loss of effective prestress should be considered in the slip strength calculations.

NOTE Additional information can be referenced in ISO 19902 ^[1].

15.3.5.6 Interface Transfer Strength of Split Sleeve Clamps

The interface transfer strength of split sleeve clamps may be taken as that given for grouted connections (see 15.1.4.2), provided that the hoop stiffness continuity of the longitudinal seams can be demonstrated by showing that no separation will occur on any part of the contact faces along the splits under any combination of forces and allowing for all losses of bolt stress, including time dependent effects. Where hoop stiffness continuity cannot be

demonstrated, the strength of the split sleeve clamp is reduced, whereas the degree of reduction depends on the contribution of the sleeve size to hoop stiffness, the configuration of the seam flanges, and the bolting arrangement. Again, tests are the only reliable approach to quantifying the strength of this type of clamp (see 7.7).

15.3.6 General Requirements for Bolted Clamps

15.3.6.1 Mechanical Clamps

Reliance on metal-to-metal contact to develop the interface transfer strength in mechanical clamps requires that due account shall be taken of the following:

- a) fabrication and installation tolerances, and
- b) the finish of the contacting surfaces (see A.15.3.6.1).

15.3.6.2 Grouted Clamps

Clamps using grout as a filler medium provide greater flexibility for installation tolerances than mechanical clamps. The following additional points should be considered when using grouted clamps:

- a) the completeness of grout filling between the clamp and the substrate member, which can be affected by the position of grout inlets and outlets, the effectiveness of seals, and grout bleed;
- b) the interface transfer strength, calculated in accordance with Sections 15.1.4.2 and 15.1.4.3, provided the clamp body can be regarded as continuous;
- c) the shrinkage and creep of grout, effectively reducing the contact pressure between the clamp and the clamped member;
- d) the relative movement between the clamp and the substrate member during grout curing and prior to bolt force application;
- e) the requirements of 19.6 in respect of grout materials.

15.3.6.3 Lined Clamps

Elastomeric lined clamps should not be used where stiffness is important (see A.15.3.6.3). Relative stiffness is particularly important for partial strength repairs, where the original damaged structure can be significantly stiffer than the repair or where the repair relies to any extent on the stiffness contribution from the existing structure through the load path being repaired.

Elastomeric lined clamps should not be used for primary structural members unless specific data on strength and long-term effects are available.

15.3.6.4 Corrosion Protection

Adequate corrosion protection should be provided when attaching clamps and additional strengthening of members to an existing structure. Underwater corrosion protection is preferably achieved by providing the clamp with sacrificial anodes or, alternatively, by ensuring electrical continuity between the clamp and the structure, if appropriate. In either case, continuity between the clamp body and the bolts shall be provided to ensure protection of the bolts. The use of coated bolts, whether these are cathodically protected or not, is recommended.

It is also important to consider the interaction between any repairs and the existing structure with regard to levels of cathodic protection, and to any possible damaging effects on the repair components, especially the bolts.

15.3.7 Bolting Considerations

The requirements of 15.2.8 on threaded fasteners apply.

16 Fatigue

16.1 General

16.1.1 Applicability

Section 16 presents general considerations and methods for verifying that fixed steel offshore structures satisfy the FLS at all locations in the structure during their entire life, from fabrication via the in-place situation to the removal of the structure.

16.1.2 The Fatigue Process

Fatigue refers to the cumulative damage done by repeated application of time-varying stresses at a specific location in the structure. These time-varying stresses are caused by variable loads/actions, especially, but not exclusively, due to wave load/action. For general reference, a brief overview of the fatigue process is given in A.16.1.2.

16.1.3 Fatigue Assessment by Analysis Using $S-N$ Data

Normally, during design a fatigue assessment is performed by analysis using $S-N$ data where S is the constant amplitude stress range and N is the predicted number of cycles to failure under constant amplitude stress range, S . The following major aspects for determining fatigue damage due to wave load/action are addressed in Section 16:

- a) general requirements for a fatigue assessment (see 16.2);
- b) a description of the long-term wave environment during the design service life (see 16.3);
- c) a description of particular aspects of the stress analyses needed for a fatigue assessment (see 16.4);
- d) the characterization of the stress range data governing fatigue (see 16.5);
- e) the long-term local stress range history (see 16.6 to 16.9);
- f) the geometrical stress range (GSR) concept, including SCFs (see 16.10);
- g) the fatigue resistance of the material (see 16.11);
- h) the fatigue damage assessment during a period of exposure (see 16.12).

Points e) to h) are specific to the particular location in the structure being considered and shall be repeated for all locations that are fatigue sensitive.

The various aspects a) to h) are integral parts of a fatigue assessment and cannot be seen as independent elements. Despite the fact that they generally involve distinctly different technologies, which are often developed by different specialists, they shall be treated in an interrelated manner. This is essential in view of the large influence they can jointly have on the results obtained. Lack of appreciation of these interrelationships presents an inherent danger to any fatigue assessment. However, the various steps in the assessment are usually performed separately and in sequence; therefore, it is most convenient to also discuss them separately.

Section 16 provides the fundamental requirements that shall be met to achieve reliable results. It leaves flexibility where possible or necessary to accommodate differences in the details of calculation procedures that are embodied in established practices or in computer programs that are used in the industry.

The design assessment is concluded with some observations on fatigue damage due to causes other than wave load/action (see 16.13) and some practical design considerations (see 16.14).

16.1.4 Fatigue Assessment by Analysis Using Fracture Mechanics Methods

In some special cases, fatigue lives can be determined using fracture mechanics methods (see 16.15). Fracture mechanics methods can also be used for special purposes associated with, but not directly aimed at, designing components of a new structure, and for determining the remaining fatigue life of components in which a crack has developed.

16.1.5 Fatigue Assessment by Other Methods

Normally, a fatigue assessment requires detailed analyses using rational methods. However, when it can be demonstrated that either the FLS will not be reached in the design service life or, alternatively, when relevant prior experience exists to reliably indicate that this will be the case, a detailed fatigue analysis as in accordance with 16.1.3 or 16.1.4 may be replaced by a simpler assessment(s).

16.1.6 Fatigue Assessment of Existing Components

Fatigue assessments of components in existing structures may be based on fracture mechanics methods or on a careful evaluation of inspection results, in lieu of a detailed fatigue analysis using the long-term stress range history and $S-N$ curves. Additionally, fatigue lives of existing components can be improved by a number of weld improvement methods (see 16.16).

16.2 General Requirements

16.2.1 Applicability

The requirements given in this document apply specifically to fixed steel structures of welded tubular space frame configuration. However, the principles described are much more widely applicable to steel structures in general. This includes structure types such as free-standing or braced caissons, compliant bottom founded structures, jack-ups, and topside structures. Assessing the degree of applicability, or the need for specialization of the requirements for a particular type of structure, is the responsibility of the user of this document.

When used for components of nontubular configuration, or for framed structures containing such components [e.g. components constructed of welded, rolled or fabricated beams, or of welded (un)stiffened plates], care should be exercised to adapt the procedure for selecting the most appropriate stress parameter that governs the fatigue process in these cases.

All components of a structure are potentially sensitive to fatigue damage accumulation. In particular, every welded connection, every connection joined by other means than welding, every attachment, every structural discontinuity, and every place where some form of stress concentration is present is a potential location of fatigue cracking, and individual assessment should be considered. It is therefore often useful to first perform lower bound screening assessments to determine those locations in the structure that are most susceptible to fatigue damage accumulation and should be subsequently assessed in detail on an individual basis.

The requirements in this document are applicable to the fatigue design of new structures as well as the fatigue assessment of existing structures. However, they only relate to fatigue evaluations of “uncracked” locations; therefore, in the case of existing structures, the proviso is that there be no crack already present. The reason for

this is that the local stress situation at a crack is not captured by the stress calculations described herein, whereas the fatigue crack growth process is neither adequately modeled by the $S-N$ curves describing the fatigue resistance of the material or the structural detail. An assessment of the acceptability of a pre-existing crack requires a fatigue crack growth evaluation using fracture mechanics methods (see 16.15).

In this context “uncracked” refers to the absence of macroscopic cracks of a size that would be detectable by suitable inspection techniques. Microscopic defects of a size that cannot be detected by suitable inspection techniques are always present in welded connections, and these are not considered as cracks in the present sense. Successful application of the requirements given in this document to the fatigue design of new structures implies that the structure at the end of its design service life will not have cracks of critical size.

Requirements for fatigue design may be relaxed when they are complemented with requirements for in-service inspection to verify actual fatigue performance (see 16.14.8 and A.16.14.8).

16.2.2 Fatigue Crack Initiation and Crack Propagation

The small crack-like defects present in welds, or the material in the heat affected zones (HAZs) adjacent to welds, will act as initiators for crack growth under the influence of variable stresses. Therefore, the procedures described in this document include the number of stress cycles required for the crack initiation and the crack propagation phases in welded construction.

16.2.3 Sources of Variable Stresses Causing Fatigue

The predominant source of variable stresses causing fatigue in an offshore structure is wave load/action in the in-place situation during the design service life. However, other sources of variable stresses causing fatigue damage should also be considered (see 16.13). The observations and requirements given in 16.3 to 16.12 largely refer to the in-place situation, but with due alteration of details they are much more widely applicable.

16.2.4 Service Life and Fatigue Life

Design service life is defined in ISO 19900 as the assumed period during which a structure is to be used for its intended purpose with anticipated maintenance, but without substantial repair being necessary. For a fixed offshore structure, this is the period during which it is in-place at its offshore location. The design service life is preceded by a pre-service period, which generally comprises a period of onshore fabrication in the yard, a period of ocean transportation, and a period of (offshore) installation.

The structure's design service life shall be specified by the owner—where appropriate, in accordance with the requirements of a regulator where one exists. For the purpose of a fatigue assessment, the design service life shall not be taken as less than 10 years. The fatigue life at any location in the structure, calculated in accordance with 16.12, shall not be less than the design service life.

Fatigue damage incurred prior to the design service life, and fatigue damage due to causes other than wave load/action in the in-place situation, shall be taken into account (see 16.12 and 16.13).

16.2.5 Characterization of the Stress Range Data Governing Fatigue

Due to the highly localized nature of fatigue damage, it is very important to select appropriate stress range parameter(s) for describing the local stress range history as discussed in 16.5 (see also A.16.1.2).

16.2.6 The Long-term Stress Range History

The stress variations occurring during a particular period at a specified location in the structure shall be specified in the form of a long-term stress range history. For example, the period considered can be the design service life for the in-place situation, or the duration of yard fabrication or the duration of an ocean tow for the pre-service situation.

The long-term stress range history can be given as a function of time or as a statistical distribution. Specification as a function of time is usually impractical and, more importantly, for nearly all cases of practical interest, the sequence of events producing the stress range time-history during the period considered is, a priori, unknown. A selected time-history would hence merely be an arbitrary choice from an infinite number of equally probable possibilities. Experience further indicates that the impact of the sequence of events on the assessment of fatigue damage averages out with time. When considered over a long period, as is normal for a fatigue assessment, the effect is hence very small if noticeable at all. Therefore, the long-term stress range history is normally specified in the form of a statistical distribution.

The long-term stress range history can be determined by means of the following:

- a series of spectral analyses if a spectral fatigue analysis method is used, as described in 16.7;
- a series of deterministic analyses if a deterministic fatigue analysis is used, as described in 16.8;
- other approximate procedures that are usually based on previous experience for the type of structure and the geographical area concerned (see 16.9).

Guidance on the choice between a spectral and a deterministic fatigue analysis is given in 16.6.

The use of other approximate procedures can be justified only for screening analyses during initial design phases or if previous experience indicates that fatigue is not an important design issue. Even in these cases, great care combined with sound knowledge shall be exercised.

16.2.7 Partial Load/Action and Resistance Factors

In determining stress variations for a fatigue analysis, the partial load/action factors shall be taken as 1.0. The partial resistance factor on the fatigue resistance shall also be taken as 1.0. Safety against fatigue failure is provided by using fatigue resistance in the form of a design $S-N$ curve, which is the mean minus two standard deviations of $\log N$. Overall safety against failures associated with fatigue damage accumulation is further provided by an additional fatigue damage design factor $\gamma_{FD} \geq 1.0$. Local experience can be taken into account by a local experience factor k_{LE} , which can be larger or smaller than 1.0 (see 16.12.3).

16.2.8 Fatigue Resistance

The fatigue resistance of the material in or around a weld to variable stresses is given in the form of empirical $S-N$ curves, where S is a constant amplitude stress range and N the number of cycles to failure. Many $S-N$ curves have been determined and can be found in the literature. Each curve is strictly associated with a particular definition of S and N , as well as a particular structural detail; they shall not be used with variables that are defined in a different manner or for different structural details (see also A.16.1.2). The $S-N$ curves that shall be used are presented in 16.11.

16.2.9 Fatigue Damage Calculation

The long-term stress range history and the fatigue resistance both relate to a particular location in the structure. These two items are combined via the Palmgren-Miner rule to determine the fatigue damage at the location during the period under consideration (see also A.16.1.2). Application of the Palmgren-Miner rule is discussed in 16.12.

16.2.10 Weld Improvement Techniques

Weld improvement techniques should not be used for the design of a new structure in lieu of reducing fatigue damage by means of appropriate modifications to the design, as long as such modifications are still practically possible.

Appropriate design modifications can include:

- reducing the variable loads/actions causing stress variations, if possible;
- changing the geometry or the dimensions of the component(s) involved to reduce the magnitudes of the stress variations (the load/action effects) occurring;
- changing the construction details of the component(s) concerned.

Weld improvement techniques (see 16.16) may be considered as a last resort measure that can be justified only when the project is too far advanced to still enable design changes to be introduced in a timely and practical manner.

16.3 Description of the Long-term Wave Environment

16.3.1 General

In a description of the wave environment, short-term and long-term conditions are distinguished.

Short-term conditions refer to a stationary situation in which the sea remains the same in a statistical sense. Such stationary situations are limited in duration and are normally assumed to last for a standard period of 3 hr. A stationary situation (short-term condition) of the sea is referred to as a sea state. A short-term condition of the sea can be realistically described by a theoretical model in the form of a wave spectrum, the parameters of which can be chosen to reflect the actual wave conditions at a particular time and location.

Long-term conditions refer by definition to situations in which the wave environment is no longer stationary but changes. Long-term wave conditions hence always consist of a succession of sea states. As there are no theoretical models to describe the long-term wave climate, long-term conditions at a particular location are always empirical.

Long-term conditions typically refer to the duration of a season, a year or the design service life of a structure. For a fatigue assessment of the in-place situation, the long-term conditions during the design service life or the long-term conditions during a typical year can be used. In the latter case, it is assumed that the conditions in a typical year repeat themselves every year during the design service life. For a fatigue assessment of pre-service situations, the long-term conditions relate to the duration of yard fabrication, the duration of the transportation from the yard to the offshore location, or the duration of the offshore installation.

16.3.2 Wave Scatter Diagram

The most suitable representation of the long-term wave environment at a location is the wave scatter diagram. Ideally, a wave scatter diagram specifies the probability density function, $p(H_S, T_R, \theta_m)$, of the joint occurrence of the three main parameters defining a sea state, being the significant wave height H_S , a representative period, T_R , and the mean wave direction, θ_m . However, joint statistics of these three parameters are rare and the best that is usually available is a two-dimensional wave scatter diagram, $p(H_S, T_R)$.

In some cases, the available data permit the determination of directional wave scatter diagrams for eight or preferably 12 equally spaced directions. Directional wave scatter diagrams are discrete subsets, $p(H_S, T_R)$, of the three-dimensional wave scatter diagram, $p(H_S, T_R, \theta_m)$, for specific wave directions.

Wave scatter diagrams usually represent the long-term wave environment during a (typical) year and should be based on several years of site-specific data in order to ensure that they adequately represent the wave environment at the location of the structure. The data can be determined by measurements, by hindcast modeling, or by a combination of the two.

16.3.3 Mean Wave Directions

Where no information on wave directions is available, an omnidirectional wave scatter diagram can be used that does not distinguish the mean direction of individual sea states. It is then assumed that the three-dimensional wave scatter diagram, $p(H_S, T_R, \theta_m)$, can be transformed into an omnidirectional two-dimensional wave scatter diagram, $p(H_S, T_R)$, by a separation of variables:

$$p(H_S, T_R, \theta_m) = p(\theta_m) p(H_S, T_R) \quad (16.3-1)$$

where $p(\theta_m)$ is the probability of occurrence of each mean wave direction irrespective of the significant wave height H_S and the representative period T_R .

Mean wave directions are difficult to measure and information on them is not often available. In such cases, the statistics of wind directions can be used as an approximation to the probability of occurrence of mean wave directions. It is then assumed that the distribution of the mean wave direction $p(\theta_m)$ coincides with the distribution of the mean wind direction $p(\theta_w)$.

Where applicable, care shall be taken to account for special geographic features, such as restrictions in fetch from certain directions or special sea floor topography, which can impose limitations on wave generation for specific wind directions.

16.3.4 Wave Frequency Spectra

Each sea state in the wave scatter diagram is described by a two-parameter wave spectrum. For fatigue analysis purposes, a wave spectrum shape shall be chosen that is representative of the average distribution of wave energy over wave frequency for a large number of sea states. For fully developed sea states, the two-parameter Pierson-Moskowitz spectral formulation is normally most appropriate. For sea states that are not fully developed, the two-parameter JONSWAP wave spectral formulation can be preferred.

The wave period used in the spectral formulation shall be consistent with the representative period, T_R , for the data in the wave scatter diagram.

NOTE See API 2MET.

16.3.5 Wave Directional Spreading Function

The water surface elevation in a real sea is short-crested (see A.16.3.1), i.e. the frequency components are spread over a range of directions around the mean wave direction. To accommodate the effect of wave spreading on structural responses, a directional spreading function may be used where appropriate.

NOTE See API 2MET.

16.3.6 Periodic Waves

In all fatigue analysis methods, the basic stress variations in the structure are calculated using individual periodic waves. The particular way in which the periodic waves are used in the various fatigue analysis methods is described in 16.7 to 16.9.

16.3.7 Long-term Distribution of Individual Wave Heights

In a deterministic fatigue analysis, the long-term wave environment is specified by the long-term distribution of individual wave heights, which can be derived from the wave scatter diagram (see A.16.3.7).

NOTE See API 2MET.

16.3.8 Current

Currents are normally not taken into account in calculating applied wave loads/actions for a fatigue analysis (see A.16.3.8). However, fatigue damage due to VIV caused by currents should be considered (see 16.13).

16.3.9 Wind

For a fixed offshore structure, variable loads/actions due to wind are small in comparison with variable loads/actions due to waves and do not need to be considered for the in-place fatigue analysis.

There is no direct wave load/action on the topsides of a fixed offshore structure, and therefore variable loads/actions due to wind are normally the most important source of fatigue stresses in components of the topsides structure in the in-place situation, particularly for long slender structures such as flare structures.

Wind-induced fatigue should be considered for other situations than the in-place situation (see 16.13).

16.3.10 Water Depth

In a fatigue analysis of the in-place situation, the average water depth during the design service life of the structure shall be used; the potential effect of subsidence should be considered (see A.16.3.10).

16.3.11 Marine Growth

For new structures, the design marine growth thickness shall be used.

16.4 Performing the Global Stress Analyses

16.4.1 General

General guidance on structural modeling and analysis is given in Section 12. The guidance in 16.4 relates specifically to stress analyses for a fatigue assessment.

All methods of fatigue assessment by analysis are based on the results of a large series of global stress analyses in individual periodic waves (see 16.3.6). The calculated stress variations are the input data for all subsequent steps in the fatigue assessment and should therefore be fully reliable. The computer models used for the stress analyses should be suitably detailed to ensure that the nominal stress variations at a particular location in the structure are adequately captured and are compatible with the purpose and the chosen method of fatigue analysis. This applies to both the load/action model and to the structural model. Neither model used for the strength analysis of the structure is normally detailed enough for stress analyses for fatigue assessment purposes and both require suitable additions and refinements.

For the load/action model, only time-varying direct loads/actions are required; permanent loads/actions do not need to be considered.

The structural model should include all relevant characteristics of the structure and its foundation, in order to correctly reflect the magnitude and distribution of its stiffness. Where dynamic structural response is involved, the magnitude and distribution of the structure's mass should also be correctly represented. Care should further be taken to model the geometry of the structure in sufficient detail to capture any asymmetry in the structure's stiffness, in its mass, or in the applied loads/actions in order to ensure that torsional responses are also properly taken into account. Where required, damping can normally be modeled by viscous damping using a damping ratio (a nondimensional damping coefficient).

For space frame structures the time-varying stresses in an individual structural component are due to overall frame load/action and to direct wave loads/action on and around the component considered. Direct wave load/action usually becomes more important the higher the component's elevation in the structure. Therefore, all relevant members in the (near) vertical frames and in the plan framing levels should normally be included and modeled in a realistic manner. Recognizing the potential influence of asymmetry, the lack of symmetry in bracing patterns, with respect to the overall configuration of the structure in combination with the lack of symmetry in

applied loads/actions for seemingly similar wave directions, should not be overlooked. The transfer of applied loads/actions to foundation piles, any foundation constraint provided by conductors, where present, and associated details in the modeling of the plan framing at the sea floor, should also be included in the model, in order to ensure that the distribution of support reactions through the structure is correctly represented for different levels of the applied loads/actions.

In many respects, modeling particulars for the different fatigue assessment methods are the same. Those aspects that are common are described here. Aspects that are specific to one of the methods are described in 16.7 to 16.9.

16.4.2 Loads/Actions Caused by Waves

The loads/actions on the structure for the stress analyses are the direct loads/actions caused by periodic waves. The basis of the procedure for calculating these loads/actions is described in 9.5, with the following specializations and modifications:

- a) current and the current blockage factor shall not be used (see 16.3.8);
- b) instead of the storm water depth, the water depth specified in 16.3.10 shall be used;
- c) in the context of the procedure used in 9.5, the wave directional spreading factor shall be 1.0;
- d) care should be taken that the influence of marine growth on the stresses calculated for fatigue analyses is correctly taken into account, as marine growth affects both the drag and the inertia term in Morison equation (see below);
- e) for fatigue analyses, the values of the drag and inertia coefficients shall not be taken from 9.5.2.3 (see below);
- f) the conductor shielding factor shall be 1.0.

For spectral and deterministic fatigue analysis methods, different periodic wave theories can be applicable (see 16.7 and 16.8).

All fatigue assessment methods are based on stress ranges. Each stress range in an individual periodic wave is due to a full wave cycle. The resulting stress cycle at a particular location consists of contributions from an axial and a bending stress cycle, which are in general not synchronous. To determine the appropriate resulting stress range in the time domain, a full cycle of each periodic wave shall be stepped past the structure, requiring a substantial number of separate wave positions with the associated calculations (see A.16.4.2). When the whole formulation for the structural analysis is adequately linearized, the stress calculations may alternatively be performed in the frequency domain. In this latter case, two calculations for each periodic wave suffice, one for the real (in-phase) and one for the imaginary (out-of-phase) components of the applied wave loads/actions.

For all fatigue assessment methods, the loads/actions due to waves should be calculated for a sufficient number of wave directions to adequately capture the influence of wave direction on local stress variations resulting both from overall frame load/action due to global hydrodynamic loads/actions and from direct wave load/action on individual components. For typical configurations of steel structures a minimum of eight to 12 wave directions covering the full 360° of the compass are required for a comprehensive representation of the stresses at a particular location. The selected wave directions should normally coincide with the mean wave directions for which statistical information in the wave scatter diagram is available (see 16.3.2).

The model for hydrodynamic loads/actions should include a three-dimensional representation of all components of the structure that are exposed to wave load/action. Whereas for a global strength analysis horizontal loads/actions are usually the most important, for a fatigue analysis careful attention should be paid to the accurate modeling of loads/actions on components in all directions. This notably includes vertical applied loads/actions on submerged plan frames.

Appurtenances subjected to wave load/action should be included and properly modeled in their correct location and with their appropriate connection details, so as to realistically represent the transfer of any loads/actions on them to the structure. Appurtenances that should be modeled in this manner include, but are not limited to, conductors, caissons, risers, J-tubes, boat landings, fenders, launch rails, pile guides, pile sleeve assemblies, etc. Miscellaneous attachments such as anodes also attract hydrodynamic loads/actions, which should be included as well. However, wave loads/actions on anodes are representative of a more distributed type of load/action and may be modeled with an increase in the drag and/or inertia coefficients in the Morison equation.

The influence of marine growth should be included by taking into account the influence of an increase in the diameter on the drag term as well as on the inertia term in Morison equation. The drag coefficient shall additionally reflect the influence of roughness. All this is normally achieved by using an effective hydrodynamic diameter in conjunction with appropriately adjusted drag and inertia coefficients.

The analyses should be run for a wide range of wave heights and periods, the majority of these being much lower and shorter than the design wave height and period for the strength analysis. The inertia term in Morison equation is hence much more important in fatigue analyses than for the strength analysis in an extreme storm condition, up to the point that the inertia term can become dominant over the drag term. Therefore, the cross-sectional area (or the equivalent volume of components with shapes that are not tubular) and the associated choice of inertia coefficients require careful attention.

As short wave periods are associated with short wave lengths, it is generally necessary to calculate the applied loads/actions at a larger than usual number of discrete points along the length of components for extreme wave analyses, in order to adequately define the distribution of wave loads/actions over individual components as well as the total wave load/action on the structure as a whole. Only in this manner can the influence of distributed loads/actions on member end moments be properly taken into account.

In fatigue analyses, an inertia coefficient of $C_m = 2.0$ shall be used for tubular members (see A.16.4.2). The drag coefficient C_d may be chosen in accordance with A.9.5.2.3. The coefficients that should be used for nontubular members and nontubular appurtenances are subject to special consideration.

16.4.3 Quasi-static Analyses

As the loads/actions on the structure are by definition time-varying, the stress responses are also time-varying. Structures for which dynamic behavior is significant are referred to as “dynamically responding structures” (see 12.5.2). When the influence of structural accelerations can be neglected, stresses may be calculated using a quasi-static linear structural analysis (see 12.5.3). The acceptability of neglecting dynamic effects should be carefully reviewed; some guidance is provided in 12.5.2 and A.16.4.3.

16.4.4 Dynamic Analyses

16.4.4.1 General

In global structural analyses to determine member nominal stresses, the steady state dynamic response to each of the periodic waves stepped past the structure should be included by performing a dynamic linear analysis (see 12.5.5). The influence of dynamic response on the long-term local stress range history should be determined using spectral analysis and a frequency domain dynamic solution.

The position of the platform natural frequencies relative to the peaks and troughs of the transfer function of the applied wave load/action can have a profound effect on the level of dynamic response. Since the natural frequencies can vary significantly, depending upon design assumptions and operational deck mass, the theoretical natural frequencies should be reviewed and their position adjusted if necessary (see A.16.4.4.1).

16.4.4.2 Mass

The mass model includes all structural elements, conductors, appurtenances, miscellaneous attachments, grout, marine growth, entrapped water, and added mass. A lumped mass model is sufficient to obtain global structural response. However, it does not always adequately predict local dynamic response. Where necessary, local responses should be examined using a consistent mass model.

The model of the topsides should represent the total topsides mass, the center of mass, and the rotational inertia of the deck topsides mass in order to correctly account for torsional response.

16.4.4.3 Stiffness

The structural model should include an appropriate three-dimensional distribution of structure stiffness. The stiffness of the deck structure should be modeled in sufficient detail to adequately represent its effect on the behavior of the structure due to fatigue excitation.

Foundation springs should be included for all foundation components that carry significant reactions (e.g. main piles or pile groups, skirt piles, and conductors); the relative stiffness of different components should be taken into account.

The foundation stiffness should be modeled by an equivalent linear representation, such that the foundation deflections and rotations in a sea state representing wave conditions that contribute significantly to fatigue damage are correctly reflected. The center of fatigue damage sea state, calculated as described in A.16.7.2.3, is appropriate for this purpose.

The foundation stiffness can have a large effect on the natural period(s) of the structure. When assessing the range of natural period values that can occur, upper and lower bound foundation stiffness values should be considered.

16.4.4.4 Damping

Energy dissipation due to damping has a profound effect on structural response calculations at and in the immediate vicinity of the resonant frequency or frequencies. Outside the resonance region, the influence of damping is negligible.

Offshore structures are very lightly damped. The physical sources of damping are difficult to determine and to quantify with any certainty. For the purpose of a fatigue analysis damping should be modeled by a viscous damping coefficient that accounts for all sources of damping including structural, foundation and hydrodynamic effects. For typical pile founded tubular space frame structures, a total damping value of 1 % to 2 % of critical is appropriate. For other structural configurations and for topsides structures, damping values are typically of similar magnitude but should be selected with care.

16.5 Characterization of the Stress Range Data Governing Fatigue

The general conditions of a fatigue assessment are discussed in 16.1.2 and A.16.1.2. The requirements in 16.5 to 16.12 are specifically concerned with the fatigue assessment of welded connections of tubular and nontubular components in a fixed steel offshore structure caused by variable loads/actions due to waves during the structure's design service life. However, the concepts discussed are more widely applicable to fatigue assessment in general.

Residual stresses in or around welds can be assumed to have magnitudes equal to the stress at which the material yields in tension. Stress variations in or around welds can hence be assumed to always range downwards from the yield strength in tension. Consequently, other than for plain steel, the mean stress and the

stress ratio, R (the ratio of the minimum to the maximum stress during a cycle), of the applied stresses are inappropriate parameters to assess fatigue damage potential in welded components. For applied stresses that are less than or equal to half the yield strength, the local R value will always be greater than or equal to zero. Therefore, the stress range is universally accepted as the sole stress parameter that governs fatigue of welded components.

The most important factors influencing fatigue damage are the stress range at a location, the number of applied cycles of a particular stress range magnitude, and the fatigue resistance of the material.

A global analysis model is normally used to determine nominal stress ranges in the vicinity of the connection. Requirements for performing global analyses are given in Section 12 and in 16.4. These analyses shall provide the full cycle of relevant stress components (e.g. axial stresses and bending stresses), which are generally not synchronous; for harmonic stresses, this manifests itself in phase differences. The nonsynchronous nature (phase differences) between stress components should be duly taken into account.

For beam elements, nominal stresses are calculated at member ends. The relevant stresses are the stresses that are parallel to the longitudinal axis of the beam, i.e. axial stresses, IPB stresses, and OPB stresses. The stress variations of these three nominal stress components shall be kept separate and shall not be combined. Shear stresses and torsional stresses may be neglected. For elements other than beam elements, engineering principles should be applied in determining which stress component(s) are to be considered.

For tubular connections, GSRs are determined at specific locations of the connection by multiplying the nominal stresses first by SCFs and then combining the stress components as described in A.16.5. The SCFs are described in 16.10.2.

For nontubular connections, GSRs are determined at specific locations of the connection by assigning a joint classification to the connection (or to a particular construction detail of the connection) and specifying which stress range is to be used as the GSR (see 16.10.3).

The GSR is subsequently used to determine the long-term local stress range history, which is ultimately combined with the corresponding $S-N$ curve (see 16.11) in performing the fatigue assessment (see 16.12).

Both test data and analytical techniques may be used to determine an SCF for the specific location at the tubular joint and the type of stress considered, in relation to the corresponding $S-N$ curve, provided that it can be demonstrated that the magnitude of the SCF can be estimated reliably with the selected method. Care should be taken when using results obtained from limited test programs with models that are not entirely adequate for the purpose. Similar care should be exercised when using results from analytical investigations performed with techniques that are not entirely adequate for modeling the local situation.

16.6 Long-term Local Stress Range History

16.6.1 General

The long-term local stress range history is a statistical distribution that combines information on the magnitudes and numbers of cycles of the stress ranges at a specific point during a given period of time. It thus provides the number of occurrences, n_i , of local stress ranges, S_i , where S_i is the GSR (see 16.5). All stress ranges that can potentially contribute to fatigue damage should be included in the distribution.

For the in-place situation, the stress ranges are caused by wave load/action. The long-term distribution is then determined by summing the weighted short-term distributions for all individual sea states and wave approach directions occurring during the design service life. The weight factor for the contribution of each short-term distribution is the product of the probability of occurrence of each sea state and the probability of occurrence of its wave approach direction.

The GSR stress data from which the short-term distribution of S_i for each individual sea state is compiled can be obtained from:

- a probabilistic determination using spectral analysis methods, or
- a deterministic determination using individual periodic waves.

The two methods require different computational effort and provide different levels of accuracy. Their main fundamental difference lies in the manner in which the long-term wave climate is represented, with corresponding differences in the subsequent steps (see 16.6.2 and 16.6.3).

16.6.2 Probabilistic Determination Using Spectral Analysis Methods

In the spectral analysis method, a sea state (the short-term wave environment) is represented by a two-parameter wave frequency spectrum (see 16.3.4). This spectral description provides the most comprehensive representation of the features of waves in a real sea. The description retains the random nature as well as the frequency content of a real sea. As a result of this, it is able to realistically model the effect of wave frequency on applied wave loads/actions and structural response. Thus, the spectral procedure:

- enables a probabilistic determination of the short-term distribution of S_i for each sea state and at each location of interest in the structure, and
- is the only method that can be applied to both (quasi-)statically responding and dynamically responding structures.

A disadvantage of the spectral method is that the global analyses by which the nominal stresses are determined require linearization of the hydrodynamic drag load/action, of wave inundation effects around the still water level and of foundation behavior. Appropriate treatment of these aspects should be carefully considered in all applications. Application of the spectral method is detailed in 16.7.

The spectral method is the preferred method for the final fatigue assessment of all structures, unless it can be reliably demonstrated that another method can be justified.

16.6.3 Deterministic Determination Using Individual Periodic Waves

Instead of a probabilistic description using wave spectra, the ocean environment is sometimes described by a series of deterministic individual waves. These are periodic (regular) waves with a particular height, period, and direction, and an associated number of occurrences. Such periodic waves are merely an abstraction of reality for analysis purposes and do not attempt to produce a realistic representation of the features of waves in a real sea. Consequently, they reflect neither the random nature nor the frequency content of real waves. Hence, the deterministic procedure:

- produces a deterministic short-term distribution of S_i for each sea state and at each location of interest in the structure, and
- is not suitable for application to dynamically responding structures.

Guidance on whether the dynamic behavior of a structure needs to be taken into account or may be neglected is given in 16.4.3.

The deterministic method does not require linearization of the global analyses by which the nominal stresses are determined. Application of the deterministic method is detailed in 16.8.

Where a deterministic method can be justified, it may be used for screening evaluations during the initial design phases and for the final fatigue assessment of structures that are not critically fatigue sensitive.

16.6.4 Approximate Determination Using Simplified Methods

Experience from many analyses as well as some measurements from existing structures has shown that in many cases the long-term distribution of local stress ranges can be represented reasonably well by a Weibull distribution or combination of two Weibull distributions.

In extratropical areas (outside the areas where tropical cyclones can occur), the whole range of sea states—from very light wave conditions to severe storms—belongs essentially to one and the same weather population. Consequently, the short-term distribution of GSRs for each sea state and the long-term distribution of GSRs, being the weighted summation of all the short-term distributions, are also of this type. Generally, the distribution is well approximated by a Weibull distribution.

In tropical areas, both wave conditions due to normal atmospheric conditions (including winter storms and monsoon conditions) and wave conditions due to tropical cyclones (depending on the area, cyclones are called “hurricanes” or “typhoons”) belong to two different populations. This results in two different types of short-term distributions of GSRs for sea states belonging to the two populations. In these circumstances, the long-term distribution of local GSRs can be approximated by the combination of two Weibull distributions.

The applicability of these simplified methods is limited (see 16.9 for further information).

16.7 Determining Long-term Stress Range Distribution by Spectral Analysis

16.7.1 General

Spectral analysis is a practical method that is best able to represent the random nature of the wave environment. Accordingly, the spectral method is the most comprehensive and most reliable assessment method for fatigue due to wave load/action. As it is based on superimposition of many individual frequency components, it is, strictly speaking, only applicable to linear systems. However, by suitable linearization of nonlinear elements in the problem formulation (principally the drag term in Morison equation and wave inundation effects around the still water level), this formal constraint can be adequately overcome.

The spectral method is performed in the frequency domain and explicitly accounts for the influence of the frequency of excitation (the wave frequency) on loads/actions as well as on structural response.

The two crucial steps in the procedure are the determination of the frequency response functions (also referred to as transfer functions) for all GSRs and the subsequent calculation of the short-term response statistics in each sea state. These steps are described in 16.7.2 and 16.7.3, respectively, together with the information provided in A.16.7.

The final step in the procedure is to determine the long-term response distribution by accumulating the short-term response statistics for all sea states in the scatter diagram (see 16.7.4).

16.7.2 Stress Transfer Functions

16.7.2.1 General

When a linear system is subjected to harmonic excitation at a particular frequency (the input), the response of the system (the output) is also harmonic with the same frequency and a phase shift between input and output. The transfer function is defined as the ratio of the amplitude of the output to the amplitude of the input. This ratio is also known as the response amplitude operator (RAO). The complete operator between the input and the output, comprising both the RAO and the phase shift, is referred to as the frequency response function. Both the RAO and the phase shift are frequency dependent. The frequency response function is thus a complex valued function of frequency. The transfer function (or RAO) is the modulus of the frequency response function and hence a real

function of frequency. Both the frequency response function and the transfer function are system properties and do not depend on the magnitude of the excitation. The distinction in terminology is, however, not always adhered to and the terms “frequency response function” and “transfer function” are often considered as synonyms. This practice will be followed in this document as well: the transfer function can hence consist of the RAO and phase shift or of the RAO alone, depending on circumstances.

During the calculations, phase shifts are very important and shall be strictly maintained in order to obtain the correct response values. In the global analyses for a fatigue assessment of the in-place situation, the single input is the water surface elevation (the wave) and the multiple outputs are the nominal stresses at all locations of interest in the structure. All calculated nominal stresses are functions of wave frequency as well as wave approach direction.

The stress transfer functions may be determined by performing global stress analyses directly in the frequency domain for the real (in-phase) and imaginary (out-of-phase) parts of the applied wave load/action separately. If this method is chosen, the global analyses shall be performed using linear wave theory and the drag term in Morison equation shall be used in a linearized form. The calculated stresses are then linearly dependent on the wave amplitude (height) and nonlinear wave height influences are not included. Guidance on the selection of wave frequencies is given in 16.7.2.2.

Alternatively, the stress transfer functions may be determined by performing global stress analyses in the time domain by stepping a full wave cycle past the structure. If this method is chosen, the global analyses may be performed using various wave theories. The use of linear (Airy) wave theory is in strict accordance with the linearity of spectral analysis. A sinusoidal wave input to the structural analysis produces sinusoidal stress outputs. However, for the determination of transfer functions in engineering applications, the linearity demand can be somewhat relaxed. Use of nonlinear waves (e.g. stretched Airy waves, Stokes V waves) and a nonlinear drag term can be allowed. The wave input to the structural analysis is then no longer sinusoidal, and/or the model for the hydrodynamic loads/actions is no longer fully linear, as a result of which the stress outputs are no longer purely sinusoidal either.

Using this relaxation, it is possible to approximate the influence of nonlinear wave loads/actions on the calculated stresses by an appropriate selection of the wave height input and suitable linearization of the results. Guidance on the selection of wave heights to achieve this is given in 16.7.2.3. The selection of wave frequencies is the same as for the calculations in the frequency domain and is described in 16.7.2.2.

Once the transfer functions of the GSRs (see 16.5) have been determined, the RAO is the only variable that is needed in the further calculations.

16.7.2.2 Selection of Wave Frequencies

The frequencies of the regular waves used in the determination of the transfer functions shall be selected so that the transfer functions adequately define response features due to applied wave excitation as well as structural behavior over the entire frequency range of interest. The selection shall hence be based on the characteristics of the environment as well as on the characteristics of the structure. Guidance on the selection is given in A.16.7.2.2.

The outcome of the further numerical calculations depends critically on the definition of the transfer functions. Therefore, their adequacy should be checked. This can be done by plotting the transfer function of total applied loads/actions on the structure (statically and dynamically) that are due to waves for each wave direction to ensure good definition of peaks and valleys and to verify the reliability of interpolation routines, if these are used. Figure A.16.7-2 shows an example of the shape of transfer functions of total applied loads/actions and depicts typical features of the frequency grid selection.

16.7.2.3 Selection of Wave Heights

Where applicable, the heights of the regular waves used in the determination of the transfer functions should be selected in a suitable manner, so that an appropriate level of nonlinear (drag) load/action due to waves is included in the linearized calculations. This is normally achieved by means of choosing a constant wave steepness, which provides a simple relationship between the wave height and the wave frequency. Typical wave steepness values are in the range of 1:15 to 1:25. The value used can, for example, be determined by calibration for the structure under consideration in the applicable environment. For this calibration, the sea state at the center of the fatigue damage scatter diagram may be used, which is representative of the conditions that contribute most to fatigue damage (see A.16.7.2.3).

NOTE Wave steepness is the ratio of wave height to wave length; the wave length is related to the wave frequency by the appropriate wave theory that is used in the analysis.

16.7.3 Short-term Stress Range Statistics

Once the GSR transfer functions for one wave direction and all locations in the structure have been determined, short-term GSR statistics shall be calculated for each sea state using standard statistical procedures for Gaussian processes. The short-term statistics give the number of times that the GSR exceeds a certain value during one occurrence of the sea state and the wave approach direction considered. This shall be repeated for all sea states in the wave scatter diagram and for all wave approach directions.

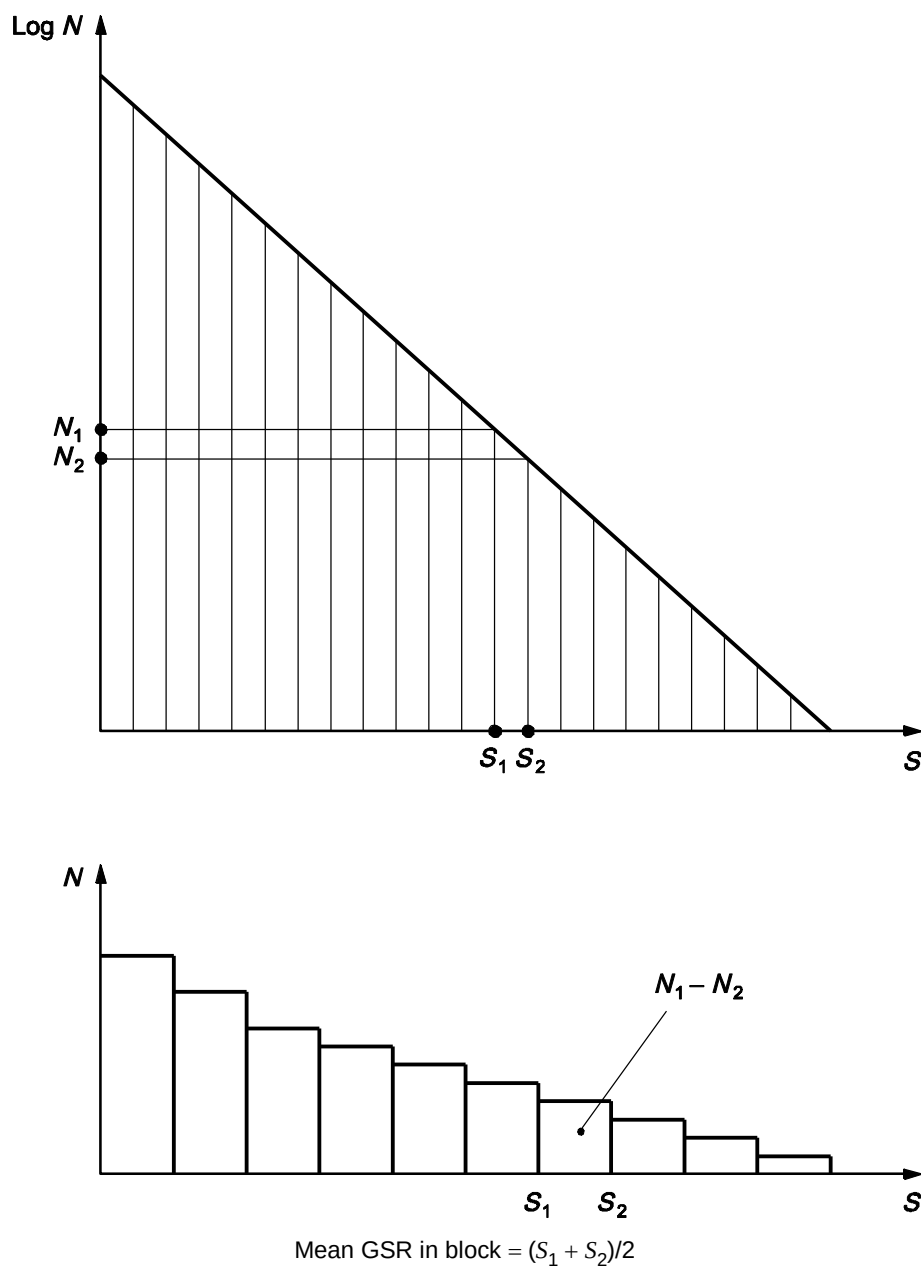
16.7.4 Long-term Stress Range Statistics

The long-term GSR statistics for each location in the structure shall be calculated by accumulating the corresponding short-term GSR statistics over all sea states in the wave scatter diagram and all wave approach directions. Prior to summation, the short-term GSR statistics for an individual sea state and direction are weighted by the product of the probability of occurrence of the sea state and the probability of occurrence of the wave approach direction.

From the long-term cumulative distribution of GSR values, the numbers of occurrences in discrete GSR blocks are determined, which are used in the fatigue damage calculation as described in 16.12. An example of a long-term cumulative distribution of GSR values and its subdivision into discrete GSR blocks is shown in Figure 16.7-1. The distribution is usually found to be close to an exponential distribution and plots approximately as a straight line on log-linear scales.

A legitimate alternative procedure that is sometimes followed is to calculate the fatigue damage in accordance with the sea state directly from the short-term GSR statistics. If this is done, these partial fatigue damages are summed after weighting them by the product of the probability of occurrence of the sea state and the probability of occurrence of the wave approach direction.

In some computer programs, the fatigue damage in accordance with the sea state is calculated using a closed-form solution based on analytical expressions for the short-term GSR statistics and the applicable $S-N$ curve. Provided the closed-form procedure can handle bilinear $S-N$ curves, where applicable, this is another legitimate procedure.

**Key**

- S stress range at location, $S = C \times S_{\text{nom}} = C \times (\sigma_{\text{max}} - \sigma_{\text{min}})$, linear scale where C is the SCF and σ_{max} and σ_{min} are the maximum and minimum nominal stresses
- N number of cycles, logarithmic scale
- $N_1 - N_2$ number of cycles in GSR block being considered
- N_1 number of cycles by which stress range exceeds S_1
- N_2 number of cycles by which stress range exceeds S_2
- S_1 lower value of stress range in GSR block being considered
- S_2 upper value of stress range in GSR block being considered

Figure 16.7-1—Typical Long-term Cumulative Distribution of GSR Values

16.8 Determining Long-term Stress Range Distribution by Deterministic Analysis

16.8.1 General

The deterministic fatigue analysis procedure uses an artificial representation of the true random nature of the wave environment. The method has similarities with conventional design wave analyses in that environmental loads/actions and structural responses are calculated directly by periodic (regular) waves. The method can offer some computational saving compared with a spectral analysis, but it still requires a substantial computational effort.

A deterministic fatigue analysis method is not recommended for final checking of structures in harsh fatigue environments. It can find application for screening evaluations during the initial design stages, or for a final fatigue assessment of structures for which dynamic effects can be neglected and that are not critically fatigue sensitive. It is included in this document only in order to cover such general applications. Guidance on dynamically responding structures is given in 16.4.4 and 16.6.3.

However, by exception, members in the splash zone between the highest submerged plan level and the lowest plan level above still water are often best analyzed by means of a deterministic fatigue analysis (see also 16.14.6).

The deterministic fatigue analysis method uses the long-term distribution of individual wave heights to represent the long-term wave environment (see 16.3.7). Stresses are calculated in a series of analyses, using deterministic waves with wave heights that are selected from this distribution (see 16.8.2) and wave periods that are appropriate for the chosen wave height classes (see 16.8.3).

The influence of nonlinear effects in the applied wave loads/actions for the selected regular waves, as well as nonlinear aspects of structural behavior, can be included in the analysis.

16.8.2 Wave Height Selection

The periodic (regular) waves used for the deterministic analysis are selected to represent the long-term distribution of individual wave heights. This distribution is discretized into a number of wave height blocks. Typically, a minimum of six to 10 blocks with uniform increments in wave height are selected. Assuming that omnidirectional wave height statistics are used, then the same set of wave heights will apply for all wave directions.

16.8.3 Wave Period Selection

The wave periods for the selected regular wave heights should be determined, based on the expected joint distribution of wave heights and periods for the location. Considerations similar to the selection of wave frequencies for the spectral method apply (see 16.7.2.2).

16.8.4 Long-term Stress Range Distribution

For each location in the structure, the long-term distribution of GSR values is calculated by accumulating the GSR values from all regular wave analyses. Assuming that omnidirectional wave height statistics are used, the total number of GSR occurrences for each wave height block is the sum of the number of occurrences in the wave height block, weighted with the probability of occurrence of the wave approach direction.

16.9 Determining Long-term Stress Range Distribution by Approximate Methods

As noted in 16.6.4, the long-term distribution of local GSRs can in many cases be represented by a Weibull distribution. The Weibull distribution is a very general class of statistical distributions by which natural phenomena (e.g. wave heights, wind speeds) and engineering applications (e.g. reliability problems) can be suitably described.

See A.16.9 for a discussion of the Weibull distribution, the determination of its parameters, and their application to fatigue assessments.

Methods using estimated Weibull distributions are intended for initial screening of structures, or for providing supposedly conservative fatigue checks of structures in benign fatigue environments. The methods are not suitable for final checking of structures in harsh fatigue environments or for structures that respond dynamically (see also 16.2.7).

The methods can also be used to establish approximate allowable GSR values, which may be used as a guide for initial joint design. The expected adequacy of the structure during the design process in satisfying final fatigue requirements is then assessed by comparing an allowable design value of the local GSR with the GSR calculated by stepping the associated fatigue design wave through the structure (see A.16.9). The final design should be checked by performing a spectral or a deterministic fatigue analysis, as appropriate.

Due to the increase in computing power and the availability of computer programs for performing detailed fatigue analyses in a relatively straightforward and rapid manner the relevance of, and interest in, approximate procedures for determining the long-term stress range distribution has distinctly diminished.

16.10 Geometrical Stress Ranges

16.10.1 General

The stress parameter that shall be used to evaluate fatigue damage at any weld is the GSR (see 16.5).

16.10.2 Stress Concentration Factors for Tubular Joints

16.10.2.1 General Requirements for the Determination of the Stress Concentration Factor

The welds at tubular joints are among the most fatigue-sensitive areas in offshore structures, due to the high local stress concentrations. Fatigue damage at locations around the connection weld shall be estimated by evaluating the GSR using the associated SCFs at each location of interest. Specific information and guidance is provided in A.16.10.2.1.

The nominal brace stress should be based on the section properties of the brace end under consideration, taking due account of the brace stub or a flared member end, if present. The evaluation of the SCF should be based on the same section dimensions. Nominal variable stress in the chord can also influence the GSR and should be considered (see A.16.10.2.1.5).

The SCF shall include all stress-raising effects associated with the geometry and force pattern of the joint, except the local (microscopic) weld notch effect, which is included in the $S-N$ curve.

SCFs may be derived from FEA, model tests, or empirical equations based on such methods. In general, the SCFs depend on the type of variable forces exerted by the brace on the joint, the type of joint, and details of the joint geometry; see the classification of tubular joints into types in Section 14 and in A.16.10.2.1.3. Brace axial forces, brace IPB, and brace OPB moments cause different patterns of variable stresses around the joint.

For welded tubular joints subjected to all three types of variable forces and moments, a minimum SCF of 1.5 should be used. For ring-stiffened joints, the minimum SCF for the brace side of the weld and brace axial forces or brace OPB moments should be taken as 2.0.

16.10.2.2 Unstiffened Tubular Joints

General guidance and discussion of SCFs for unstiffened welded tubular joints is given in A.16.10.2.2. The SCFs may be evaluated using the Efthymiou equations (see A.16.10.2.2.2). For this purpose, the tubular joints are

typically classified into T/Y-, X-, K-, and KT-joints, depending on the joint configuration, the brace under consideration, and the brace force pattern, as described in A.16.10.2.1.

As a generalization of the classification approach, the influence function algorithm discussed in A.16.10.2.2.4 may be used to evaluate the GSRs. This algorithm can handle generalized forces and moments on the braces. Moreover, the influence function algorithm can handle multiplanar joints for the important case of axial forces.

For a discussion on tubular joints welded from one side, see A.16.10.2.2.5.

Thickness transitions between butt welded tubular members are also subject to stress concentrations. This can apply to all joint-can-to-chord and brace-stub-to-brace connections. It is recommended that thickness transitions be designed to be internally flush (see A.16.10.2.2.6). The transition should normally have a slope of 1:4. Similarly, diameter transitions achieved through the use of a truncated cone are subject to stress concentrations (see 13.6).

16.10.2.3 Internally Ring-stiffened Tubular Joints

The SCF concept also applies to internally ring-stiffened joints, including the stresses in the stiffeners and the stiffener to chord weld. Ring-stiffened joints can have stress peaks at the brace to ring intersection points. Special consideration should be given to these locations. SCFs for internally ring-stiffened joints can be determined by applying the Lloyds reduction factors to the SCFs for the equivalent unstiffened joint (see A.16.10.2.3). Ring stiffeners without flanges on the internal rings should be used with care because high stress concentrations can occur at the inner edge of the ring.

16.10.2.4 Grouted Tubular Joints

Grouting tends to reduce the SCF of the joint since the grout reduces the chord deformations. In general, the larger the ungrouted SCF, the greater the reduction in SCF for grouted joints. Hence, the reductions are typically higher for X- and T-joints than for Y- and K-joints. Approaches for calculating SCFs for grouted joints are discussed in A.16.10.2.4.

16.10.2.5 Cast Joints

For cast joints, the SCF is defined as the maximum principal stress at any point on the surface of the casting (including the inside) divided by the nominal brace stress outside the casting. The SCFs for castings are not extrapolated values, but are based on directly measured or calculated values at any given point. Consideration should be given to the brace to casting circumferential butt weld, which can be the most critical location for fatigue.

16.10.3 Geometrical Stress Ranges for Other Fatigue-sensitive Locations

This subsection covers all types of fatigue-sensitive locations other than tubular joints, which are considered in 16.10.2. These locations include, but are not limited to, material away from the weld, plate to tubular connections, and weld attachments. Each location at which a fatigue crack can potentially develop should be classified according to the appropriate type of constructional detail (see A.16.10.3).

For nontubular connections, the GSR is obtained by assigning a joint classification to the plain metal, to the connection (or to a constructional detail of a connection) and specifying which stress range in the vicinity of the detail is considered the most appropriate for the local geometry of the class and the type of variable forces involved. This stress range shall be used as the GSR. Any additional stress-raising effect due to the gross geometry of the joint (e.g. stress concentrations resulting from holes, shear lag, or local through-wall bending) should be separately included where appropriate.

A procedure for the fatigue assessment of bolts is given in Section 15.

16.11 Fatigue Resistance of the Material

16.11.1 Basic $S-N$ Curves

The basic representative $S-N$ curve is of the form:

$$\log_{10} N = \log_{10} k_1 - m \log_{10} S \quad (16.11-1)$$

where

N is the predicted number of cycles to failure under constant amplitude stress range, S ;

k_1 is a constant ($k_1 = N$ for $S = 1$);

m is the inverse slope of the $S-N$ curve;

S is the constant amplitude stress range, which is the GSR according to 16.10.

Representative $S-N$ curves for tubular joints (TJ), cast joints (CJ), and other joints (OJ) are given in Table 16.11-1. These $S-N$ curves are based on steels with a yield strength less than 500 MPa; see 16.11.2 for steels with higher yield strengths.

The representative $S-N$ curves presented in Table 16.11-1 are the mean minus two standard deviations in $\log_{10} N$ from the database of test results for each class. The curves are applicable for joints in both air and seawater with adequate corrosion protection. Information concerning $S-N$ curves for joints without adequate protection is given in A.16.11.1.

For welded tubular joints exposed to constant amplitude or random variations of stress due to environmental or operational loads/actions, the TJ curve should be used. The brace to chord tubular intersection for ring-stiffened joints should also be designed using the TJ curve. Fabrication of welded joints should be in accordance with Section 20.

The set of OJ $S-N$ curves should be used for joints classified in 16.10.3, including plated joints. For ring-stiffened joints, the ring stiffener to chord connection as well as the ring inner edge should also be designed using the OJ curve.

Fatigue crack growth from the weld root into the section under the weld is possible and should be considered. This type of cracking is referred to in A.16.10.3, with examples in Table A.16.10-8.

For cast joints, see 16.11.3.

Table 16.11-1—Basic Representative $S-N$ Curves for Air and Seawater
(with adequate corrosion protection)

Curve	Air		Seawater with Adequate Corrosion Protection	
	$\log_{10}k_1$ (for S in units of MPa)	m	$\log_{10}k_1$ (for S in units of MPa)	m
Tubular joints (TJ)	12.48 16.13	3.0 for $N \leq 10^7$ 5.0 for $N > 10^7$	12.18 16.13	3.0 for $N \leq 1.8 \times 10^6$ 5.0 for $N > 1.8 \times 10^6$
Cast joints (CJ)	15.17	4.0	See A.16.10.2.5 and A.16.11.3	
Other joints (OJ)				
B	15.01 17.01	4.0 for $N \leq 10^7$ 5.0 for $N > 10^7$	14.61 17.01	4.0 for $N \leq 10^5$ 5.0 for $N > 10^5$
C	13.63 16.47	3.5 for $N \leq 10^7$ 5.0 for $N > 10^7$	13.23 16.47	3.5 for $N \leq 4.68 \times 10^5$ 5.0 for $N > 4.68 \times 10^5$
D	12.18 15.63	3.0 for $N \leq 10^7$ 5.0 for $N > 10^7$	11.78 15.63	3.0 for $N \leq 10^6$ 5.0 for $N > 10^6$
E	12.02 15.37	3.0 for $N \leq 10^7$ 5.0 for $N > 10^7$	11.62 15.37	3.0 for $N \leq 10^6$ 5.0 for $N > 10^6$
F	11.80 15.00	3.0 for $N \leq 10^7$ 5.0 for $N > 10^7$	11.40 15.00	3.0 for $N \leq 10^6$ 5.0 for $N > 10^6$
F ₂	11.63 14.71	3.0 for $N \leq 10^7$ 5.0 for $N > 10^7$	11.23 14.71	3.0 for $N \leq 10^6$ 5.0 for $N > 10^6$
G	11.40 14.33	3.0 for $N \leq 10^7$ 5.0 for $N > 10^7$	11.00 14.33	3.0 for $N \leq 10^6$ 5.0 for $N > 10^6$
W ₁	10.97 13.62	3.0 for $N \leq 10^7$ 5.0 for $N > 10^7$	10.57 13.62	3.0 for $N \leq 10^6$ 5.0 for $N > 10^6$
See A.16.10.3 and Tables A.16.10-6 to A.16.10-11 for application of the relevant OJ curve from B to W ₁ to a particular constructional detail.				

16.11.2 High-strength Steels

There is generally insufficient data on the fatigue behavior of high-strength steels to be able to confidently predict the fatigue performance of higher-strength steels (see A.16.11.2 for some information).

16.11.3 Cast Joints

For cast joints, the CJ curve in Table 16.11-1 should be used. The curve for cast joints is only applicable to castings having an adequate fabrication inspection plan.

16.11.4 Thickness Effect

The TJ and OJ curves presented in Table 16.11-1 are based on 16 mm material thickness. For material thickness above 16 mm, the following thickness effect should be applied:

$$S = S_0(16/t)^{0.25} \quad (16.11-2)$$

where

S is the stress range of the S – N curve, when adjusted for thickness effects;

S_0 is the stress range of the S – N curve in Table 16.11-1;

t is the member thickness in mm for which the fatigue life is predicted.

The CJ curve is based on 38 mm material thickness. For material thickness above 38 mm, the following thickness effect shall be applied:

$$S = S_0(38/t)^{0.15} \quad (16.11-3)$$

No thickness effect shall be applied to material thicknesses less than 16 mm for the TJ and OJ curves and less than 38 mm for the CJ curve.

16.12 Fatigue Assessment

16.12.1 Cumulative Damage and Fatigue Life

Fatigue assessment due to variable loads/actions is normally based on the hypothesis of a linear accumulation of fatigue damage under constant amplitude stress ranges, according to the Palmgren-Miner rule:

$$D = k_{LE} \times \gamma_{FD} \times \sum_i \frac{n_i}{N_i} \quad (16.12-1)$$

where

D is a nondimensional number, the Palmgren-Miner sum or damage ratio for a time T ;

k_{LE} is a local experience factor (see 16.12.3);

γ_{FD} is a fatigue damage design factor (see 16.12.2);

n_i is the number of cycles of stress range, S_i , occurring during time period, T ;

N_i is the number of cycles to failure under constant amplitude stress range, S_i , taken from the relevant S – N curve.

In the fatigue assessment procedure, fatigue failure is assumed to occur when $D = 1.0$.

Equation (16.12-1) may be employed for the following:

- variable stress ranges associated with a given sea state,
- multiple sea states and wave directions, and
- multiple sources of cyclic loads/actions.

It may also be used to evaluate multiple sets of in situ conditions, such as for a structure that has been relocated.

Equation (16.12-1) may be used to estimate fatigue life. If all damage is due to the same circumstances (e.g. in the in-place situation), the calculated fatigue life L may be calculated using Equation (16.12-2):

$$L = T/D \quad (16.12-2)$$

where

L is the calculated fatigue life based on the calculated fatigue damage;

T is the time period ($T \geq 1$ year) over which the Palmgren-Miner sum was determined;

D is the calculated fatigue damage during time, T , according to the Palmgren-Miner sum.

If some prior fatigue damage, D_1 , has occurred, Equation (16.12-2) shall be applied to the remaining fatigue damage, D_2 . As failure is assumed to occur when $D_1 + D_2 = 1.0$, the allowable remaining fatigue damage is $D_2 = (1.0 - D_1)$. If D_2 is associated with a time period, T_2 , over which the Palmgren-Miner sum is determined, the remaining fatigue life, L_2 , is given by Equation (16.2-3):

$$L_2 = T_2/(1.0 - D_1) \quad (16.12-3)$$

The prior damage, D_1 , can, for example, be due to damage during fabrication in the construction yard, or damage during the tow to location, or damage due to prior structure use.

16.12.2 Fatigue Damage Design Factors

For in situ conditions, the fatigue damage design factors, γ_{FD} , for fatigue of steel components primarily depend on the failure consequence (i.e. the component's criticality) and in-service inspectability. In lieu of more detailed assessment, a γ_{FD} of 2.0 is recommended for inspectable and nonfailure critical locations. For failure-critical and/or noninspectable locations, γ_{FD} values of up to 10 are recommended (see Table 16.12-1). "Inspectability" implies meeting the appropriate in-service inspection requirements as detailed in API 2SIM. For new designs, the γ_{FD} shall not be taken as less than 1.0.

In lieu of more detailed assessment, the fatigue damage design factors can be taken from Table 16.12-1.

Table 16.12-1—Fatigue Damage Design Factors, γ_{FD}

Failure Critical Component	Inspectable	Not Inspectable
No	2	5
Yes	5	10

The factors given in Table 16.12-1 should be considered to relate to exposure level L-1, but should also be used for exposure levels L-2 and L-3. There is currently insufficient background to establish different factors for lower exposure levels.

16.12.3 Local Experience Factor

As stated in several places, fatigue analyses and assessment procedures are subject to many and considerable uncertainties. Therefore, actual experience with existing structures is of great value and may justifiably be taken into account through the application of an overall local experience factor, k_{LE} , provided this experience is directly relevant to the case being considered, can be substantiated by reliable evidence, and is fully documented (see A.16.12.3).

16.13 Other Causes of Fatigue Damage Than Wave Load/Action

16.13.1 General

Wave load/action in the in-place situation is the predominant source of fatigue damage to offshore structures. However, other sources of fatigue damage should also be given consideration. The most relevant ones are described in this subsection. If different damage sources are not acting concurrently, damage due to each source can in general be evaluated separately and damage ratios can be added.

16.13.2 Vortex-induced Vibrations

Structural members that are susceptible to VIV due to either wind, current, or waves can experience large cyclic stresses causing fatigue damage. The susceptibility of components to VIV should be assessed. For those members for which it cannot be confidently ensured that they will not experience VIV, an evaluation of vibration amplitudes, stresses, and associated fatigue damage shall be undertaken.

16.13.3 Wind-induced Vibrations

During construction in the fabrication yard some members can experience vibrations due to wind. If this occurs, the associated fatigue damage should be assessed.

Flexible topsides structures such as vent or flare structures with comparatively long natural periods can be susceptible to resonant vibrations due to wind gust effects. The resulting stresses can be evaluated using dynamic spectral analysis techniques using a suitable wind spectrum. If these stresses are significant, the resulting fatigue damage should be evaluated.

16.13.4 Transportation

Structures with long, exposed ocean tows from the fabrication yard to the in-place location should be checked for fatigue damage arising during transportation. The fatigue damage during tow in all components of the structure shall be combined with the fatigue damage from other sources using suitable fatigue damage design and/or experience factors to account for uncertainties in its determination. Sea-fastening members should be included in the fatigue damage assessment.

16.13.5 Installation

The main source of cyclic stresses during installation is due to pile driving. If hard driving is expected, the fatigue damage in piles and conductors due to driving should be evaluated and combined with the fatigue damage from other sources. Information on the expected number of hammer blows and the magnitude and number of stress cycles per blow can be determined from the driveability analysis and should be used as the basis for the fatigue calculation. For pile driving analyses, worst case soil conditions should be applied.

During pile driving, large cyclic stresses can further occur due to vibrations of miscellaneous attachments such as anodes and installation piping in the vicinity of the pile, pile sleeves, and hammer. Rigorous analysis of these vibrations is usually not feasible, but simplified assessments using appropriate assumptions on peak

accelerations, dynamic amplification, and number of vibration cycles can provide insight into the severity of the problem. In practice, design measures to avoid the potential fatigue problem are most effective.

16.13.6 Risers

The design of risers normally requires input from both the structure designer and the pipeline designer. Risers will normally be modeled in the structure analysis to assess fatigue damage due to wave load/action on the risers (see also 16.14.2). This damage should be combined with fatigue damage due to other sources that are normally assessed by the pipeline designer, including damage due to pipeline start-up and shut-down forces and slug-induced forces. Variations in the loads/actions on the riser often also act on the riser supports that should be included in the assessment.

16.14 Further Design Considerations

16.14.1 General

In addition to the description of the procedures to determine the fatigue resistance during design (see 16.11 and 16.12), close attention to good detailing for both the main structure and all significant constructional details is of great importance in ensuring that adequate fatigue performance is achieved.

In 16.14, some aspects requiring particular attention during design and analysis with regard to fatigue are discussed.

Furthermore, the approach normally followed in fatigue evaluations is inherently linked to fabrication and inspection methods and to the in-service inspection philosophy. Some guidance in relation to these aspects is given.

16.14.2 Conductors, Caissons, and Risers

To check the fatigue of conductors, caissons, and risers and their supports, it is normally necessary to include a detailed structural model of these components in the overall structural analysis. Where there are several components with a standard arrangement, it can be adequate to model only one typical example of each in detail. Depending on structural connection details, these components can also contribute stiffness to the members of the space frame and thereby influence the stresses, and hence the fatigue damage experienced, by the members in the frame(s) to which they are connected.

16.14.3 Miscellaneous Nonload-carrying Attachments

To check the structure for miscellaneous nonload-carrying attachments, such as brackets or padeyes that are left from construction, all members of the structure should be screened using the F_2 S - N curve in conjunction with a GSR that is equal to the maximum combination of nominal stresses in the member (i.e. all SCFs equal to unity).

16.14.4 Miscellaneous Load-carrying Attachments

It is recommended that connections of miscellaneous attachments such as anodes, ladders, walkways, minor piping, etc., which are subjected to local wave load/action and are not explicitly modeled in the structural analysis, are separately checked for possible fatigue damage. Simplified evaluations based on the principles of deterministic analyses are normally adequate for these applications, using realistic numbers of cycles and load/action magnitudes.

Where the connections of attachments to primary members are found to be sensitive to fatigue, doubler plates are recommended. Doubler plates reduce the stresses in the primary member, and any cracking due to the attachment to the doubler plate is remote from the primary member.

16.14.5 Conical Transitions

Guidance on SCFs for stiffened and unstiffened conical transitions is given in DNV-RP-C203.

16.14.6 Members in the Splash Zone

Fatigue damage for members close to the water surface is subject to additional uncertainty due to the complexities of local loads/actions (see A.16.14.6) and sensitivity of the loads/actions to the actual structure elevation, both initially and because this can change during the design service life. Where uncertainty exists in the water depth or final structure settlement, all members in the splash zone, the loads/actions on which can be significantly affected by this uncertainty, should be designed with an increased fatigue damage design factor or by explicitly assessing the possible range of water depths.

16.14.7 Topsides Structure

Topsides structures supported by conventional steel jacket or tower structures do not normally experience cyclic stresses from wave load/action. However, exceptions can occur. For example, for structures with an unbraced portal arrangement supporting the topsides structure, cyclic wave load/action on the structure can introduce significant bending moments and associated variable stresses into the topsides structure. Another exception concerns conductors or other components that span from the water surface to the deck without intermediate supports.

Where the configuration is such that the deck can be subjected to the effects of cyclic wave load/action, fatigue of the topsides structure should be evaluated. This evaluation shall take appropriate account of the effect of wave frequency on wave loads/actions applied to structural components in the wave zone. This normally requires that a spectral analysis is performed.

For structures that are subject to significant dynamic response, the mass inertia forces of the topsides can cause significant cyclic stresses in the topsides structure. This can, for example, be the case for structures with relatively long natural periods and associated large dynamic excitation. In such cases, fatigue of the topsides structure should be considered. This will normally require that a spectral analysis be performed of the combined structure and topsides structure.

16.14.8 Inspection Strategy

The predicted fatigue performance of structural components provides valuable input for establishing a strategy for underwater inspection, in addition to other considerations that need to be considered when formulating such a strategy. However, designing new structures with a comfortable margin in calculated fatigue lives generally provides for a cost-effective investment as compared with a marginal design requiring more extensive inspection for fatigue.

16.15 Fracture Mechanics Methods

Fracture mechanics methods may be employed to quantify fatigue design lives of welded details or structural components in situations where the normal $S-N$ fatigue assessment procedures are inappropriate. Some typical applications are to assess the fitness-for-service and inspection requirements of a joint with and without known defects, or to assess the structural integrity of castings.

It is important that the fracture mechanics formulation that is used should be shown to predict, with acceptable accuracy, either the fatigue performance of a joint class with a detail similar to that under consideration, or test data for joints that are similar to those requiring assessment.

16.16 Fatigue Performance Improvement Techniques

For welded joints, improvement factors on fatigue performance can be obtained by a number of methods, including controlled burr grinding of the weld toe, hammer peening, or as-welded profile control to produce a smooth concave profile that blends smoothly with the parent metal. Table 16.16-1 shows improvement factors that can be applied, provided adequate control procedures are followed. The grinding improvement factor is not applicable for joints in seawater without adequate cathodic protection.

Table 16.16-1—Achievable Improvement Factors on Fatigue Performance for Weld Improvement Techniques

Weld Improvement Technique	Improvement Factor on S	Improvement Factor on N
Profile (see A.20.2.3.3)	$r^{-0.1 a}$	Varies
Weld toe burr grind	1.25	2
Hammer peening	1.56	4
^a Chord side only.		

For welds with profile control where the weld toe has been profiled, by grinding if required, to merge smoothly with the parent metal, and magnetic particle inspection (MPI) demonstrates the weld toe is free of surface and near-surface defects, the improvement on fatigue performance can be considered as shown in the table, where r is the ratio of brace/chord thickness. This improvement is in addition to the use of hot spot stress at the actual weld toe location, and the reduced size effect exponent. Either the factor on S or on N is used, but not both.

Table 16.16-1 indicates associated improvement factors on fatigue life that can be achieved. Adequate quality control (QC) procedures shall be applied if the appropriate improvement factor is to be attained.

The grinding improvement factor is not applicable to joints in seawater without adequate cathodic protection.

Benefit shall be taken only for one improvement technique, even if more than one technique is employed.

Specific requirements for the various weld improvement techniques are discussed in A.16.16.

17 Foundation Design

17.1 General

17.1.1 Applicability

This section establishes requirements for foundation design. Pile foundations and, more specifically, foundations consisting of steel cylindrical (pipe) piles or structural caissons (see 6.1.1) with slenderness ratio $L/D \geq 10$, where L is the embedded length and D is the outside diameter, are addressed in 17.1 to 17.11. Considerations for the design of shallow foundations are given in 17.12, whereas design requirements and guidance can be found in API 2GEO. A.17 contains discussion and guidance on the requirements of Section 17.

Additional guidance on matters relating to foundations is given in API 2GEO.

17.1.2 Overall Considerations

The foundation shall be designed to carry static, cyclic, and transient loads/actions without excessive deformations or vibrations in the structure. Special attention should be given to the effects of cyclic and transient loads/actions on the structural behavior of piles, as well as on the strength of the supporting soils. The criteria for foundation capacity provided in 17.3, 17.4, and 17.5 are based on static, monotonic loads/actions. Unless 17.3.4 c) is applicable, the foundation capacity refers to axial pile capacity only, as lateral soil behavior is of sole interest for foundation displacements and not for capacity. Furthermore, these criteria do not necessarily apply to so-called “problem” soils, such as carbonate material, volcanic sands, or highly sensitive clays. The possibility of movement of the sea floor against the foundation components should be investigated. The loads/actions caused by such movements, if anticipated, should be considered in the design. The potential for disturbance to foundation soils by installation of well conductors or shallow well drilling should also be assessed (see 22.6). The design of foundations differs from that of the rest of the structure in that the soil behavior is nonlinear as well as dependent on the rate of change of the loads/actions.

The geotechnical investigation for pile-supported structures should provide, as a minimum, the soil engineering property data needed to determine the following parameters: axial capacity of piles in tension and compression, force-deflection characteristics of piles subjected to axial and lateral loads/actions, pile driveability characteristics, and mudmat bearing capacity. Mudmat design requires careful determination of the slope of the sea floor and accurate assessment of the shear strength of soft surface layers.

17.1.3 Exposure Levels

It is not normal practice to apply the concept of exposure levels, as discussed in 6.6, to the foundation design for new structures. Therefore, the partial load/action factors that are applicable to exposure level L-1 shall be used for the design of foundations for structures of all exposure levels. If enhanced or reduced reliability of the foundations is justified, a proper risk assessment should be undertaken and the resistance factor varied accordingly. For existing structures, the use of reduced partial load/action factors for the assessment of exposure level L-2 and L-3 structures may be considered.

17.2 Pile Foundations

17.2.1 Types of Pile Foundation

The types of pile foundation used to support offshore structures and considered in this document are the following:

- driven piles,
- drilled and grouted piles,
- belled piles, and
- vibro-driven piles.

17.2.2 Driven Piles

Open-ended piles are normally used in foundations for offshore structures. These piles are usually driven into the seabed with impact hammers, which use steam, diesel fuel, or hydraulic power as the source of energy. Driveability studies are carried out in accordance with the principles given in 22.5.5 in order to define the type of hammer necessary to reach the target design pile penetration.

The design penetration of driven piles shall be determined in accordance with the principles outlined in 17.3 to 17.6 and 17.9, rather than upon any correlation of pile capacity with the number of blows required to drive the pile a certain distance into the seabed.

17.2.3 Drilled and Grouted Piles

Drilled and grouted piles can be used in soils and rocks that will hold an open hole with or without drilling mud. The design penetration of drilled and grouted piles shall be determined in accordance with the principles outlined in 17.3 to 17.6. Force transfer between grout and pile should be designed in accordance with 15.1, provided that the radial stiffness term for the pile sleeve is replaced by a stiffness term for the rock mass.

17.2.4 Belled Piles

Bells may be constructed at the tip of piles so as to give increased bearing and uplift capacity through direct bearing on the soil. The end bearing capacity of belled piles shall be determined in accordance with the principles given for the design of drilled and grouted piles.

17.2.5 Vibro-driven Piles

The capability of hydraulic vibratory driving hammers to install piles has been demonstrated, in particular for the installation of small diameter piles in cohesionless soils. Owing to the lack of data with respect to the effect of the installation method on the pile axial capacity, the use of vibratory hammers for installing piles subjected to significant axial loads/actions is not recommended.

Vibratory hammers may be considered for installing well conductors, or piles which are predominantly subjected to horizontal loads/actions, such as reaction piles for start-up of pipelines or anchor piles. They are also frequently used where extraction and repositioning can be required. Vibratory hammers may further be considered as complementary tools to impact hammers, i.e. for initial driving.

17.3 General Requirements for Pile Design

17.3.1 Foundation Size

When sizing a pile foundation, the following items should be considered: design loads/actions, diameter, penetration, wall thickness, type of tip, number of piles, spacing, geometry, location, pile head fixity, material strength, installation method, and other parameters as appropriate.

17.3.2 Foundation Response

A number of different analysis procedures may be used to determine the requirements of a foundation. As a minimum, the procedure used shall properly simulate the nonlinear behavior of the soil and ensure force-deflection compatibility between the structure and the pile-soil system.

17.3.3 Deflections and Rotations

Deflections and rotations of individual piles and the total foundation system should be checked at all critical locations, which include pile tops, points of contraflexure, sea floor, etc. Deflections and rotations should not exceed serviceability limits that if exceeded would render the structure inadequate for its intended function (see 7.2).

17.3.4 Foundation Capacity

The foundation capacity is the lower of the pile strength and the pile resistance as below.

a) Pile strength: The pile strength shall be verified using the steel tubular strength checking equations given in 13.3 or 13.4 for conditions of combined axial force and bending. Internal pile forces at the location being checked shall be those due to the design loads/actions according to 9.10 using a coupled structure-soil nonlinear foundation model. When lateral restraint normally provided by the soil is inadequate or nonexistent, column buckling effects on the pile shall also be checked in accordance with 17.10.2.

b) Pile axial resistance: The axial pile capacity shall satisfy the following conditions:

$$P_{d,e} \leq Q_d = Q_c / \gamma_{R,Pe} \quad (17.3-1)$$

$$P_{d,p} \leq Q_d = Q_c / \gamma_{R,Pp} \quad (17.3-2)$$

where

- Q_d is the design axial pile capacity, i.e. the design resistance of the pile;
- Q_c is the representative value of the axial pile capacity as determined in 8.1 and 8.2 of API 2GEO, First Edition;
- $P_{d,e}$ is the design axial load/action on the pile, determined from a coupled linear structure and nonlinear foundation model using the design loads/actions for extreme conditions according to 9.10.3;
- $P_{d,p}$ is the design axial load/action on the pile, determined from a coupled linear structure and nonlinear foundation model using the design loads/actions for permanent and variable loads/actions or the design axial load/action for operating situations according to 9.10.3;
- $\gamma_{R,Pe}$ is the pile partial resistance factor for extreme conditions ($\gamma_{R,Pe} = 1.25$);
- $\gamma_{R,Pp}$ is the pile partial resistance factor for permanent and variable loads/actions or operating situations ($\gamma_{R,Pp} = 1.50$).

c) Pile overturning resistance: If the overturning moment capacity of the foundation is provided primarily by the lateral resistance of the soil along pile length, such as with free-standing caissons, instead of by coupling the axial resistance of multiple piles, then the following conditions shall also be satisfied:

$$M_{d,e} \leq O_d = O_c / \gamma_{R,Oe} \quad (17.3-3)$$

$$M_{d,p} \leq O_d = O_c / \gamma_{R,Op} \quad (17.3-4)$$

where

- O_d is the design foundation overturning capacity, i.e. the design overturning resistance of the foundation;
- O_c is the representative value of foundation overturning capacity provided by soil lateral resistance as determined in 8.5 of API 2GEO, First Edition; O_c may be augmented by the contribution of the skin friction;
- $M_{d,e}$ is the design overturning moment/action on the foundation, determined from a coupled linear structure and nonlinear foundation model using the design loads/actions for extreme conditions according to 9.10.3, and accounting for secondary moment/action due to lateral deflection above the sea floor and moment/action resulting from the installation tolerance for out-of-vertical, i.e. for any $P-\Delta$ effects;

- $M_{d,p}$ is the design overturning moment/action on the foundation, determined from a coupled linear structure and nonlinear foundation model using the design loads/actions for permanent and variable loads/actions or the design overturning moment/action for operating situations according to 9.10.3, and accounting for secondary moment/action due to lateral deflection above the sea floor and moment/action resulting from the installation tolerance for out-of-vertical, i.e. for any $P-\Delta$ effects;
- $\gamma_{R,Oe}$ is the overturning partial resistance factor for extreme conditions ($\gamma_{R,Oe} = 1.25$);
- $\gamma_{R,Op}$ is the overturning partial resistance factor for permanent and variable loads/actions or operating situations ($\gamma_{R,Op} = 1.50$).

17.3.5 Scour

Sea floor scour shall be assessed in the design of the foundation, as it affects both lateral and axial pile performance and capacity. Scour depends on very local circumstances and its prediction remains an uncertain art. Sediment transport studies can assist in defining scour design criteria, but local experience is the best guide. The uncertainty regarding design criteria should be taken into account by robust design and/or an operating strategy of monitoring and remediation as needed. Scour design criteria will usually be a combination of local and global scour and overall movement of the sea floor (see also API 2GEO).

17.4 Pile Capacity for Axial Compression

17.4.1 General

Pile capacity for axial compression, as discussed in 17.4.2 to 17.4.5, relates to the axial resistance of a pile when the pile head is subjected to compressive loads/actions along the pile axis. Pile capacity for axial tension is addressed in 17.5.

Further to the introductory discussion in 17.1.2, pile capacities are commonly determined using the simplified calculation model described in 17.4.2; the parameters that are used in this model are determined in accordance with 17.4.3 to 17.4.5. This simplified model has been developed and applied in many years of offshore practice and represents the current industry standard. However, the model does not provide any information about axial pile displacements that are important for serviceability requirements, especially in nonextreme conditions for loads/actions due to permanent, variable, and operating environmental loads/actions that are generally well below the design loads/actions. Axial pile behavior aimed at meeting service requirements is referred to as “axial pile performance” and is discussed in 17.6. Methods for determining pile performance are described in API 2GEO.

The simplified model for pile capacity presented in 17.4.2 is based on a (quasi-)static and monotonic application of the axial loads/actions and has no ability to reflect the complex occurrences that take place in the interaction between pile and soil during field conditions. To enhance understanding of the limitations of the model and to assist in applying engineering judgment to its results, it is useful to gain a better insight in the actual occurrences through the investigation of pile performance (see 17.6).

The relationships between mobilized axial shear transfer between pile and soil and the local pile displacement, and between mobilized end bearing resistance and the pile tip displacement can be determined using 17.7.

17.4.2 Representative Axial Pile Capacity

The representative value of the axial capacity of piles, Q_c , including belled piles, shall be determined using Equation (16) in API 2GEO, First Edition.

The foundation configurations should be based on those that experience has shown can be installed consistently, practically, and economically under similar conditions with the pile size and installation equipment being used. Alternatives for possible remedial load/action in the event design objectives cannot be obtained during installation should also be investigated and defined prior to construction.

NOTE See C.8.1.2 in API 2GEO, First Edition.

17.4.3 Shaft Friction and End Bearing in Cohesive Soils

For pipe piles in cohesive soils, the unit shaft friction, f , in kPa (lb/ft²), at any point along the pile shall be calculated by Equation (17) in API 2GEO, First Edition.

For piles end bearing in cohesive soils, the unit end bearing q , in kPa (lb/ft²), shall be computed by Equation (20) in API 2GEO, First Edition.

Alternative means of determining pile capacity that are based on sound engineering principles and are consistent with industry experience are permissible. A more detailed discussion of alternative prediction methods is included in A.17.4.3.

NOTE See C.8.1.3.1 and C.8.1.3.2 in API 2GEO, First Edition.

17.4.4 Shaft Friction and End Bearing in Cohesionless Soils

See 8.1.4 in API 2GEO, First Edition, for alternative methods for assessing pile capacity in cohesionless soils. Both simple methods and CPT-based methods are addressed.

For pipe piles in cohesionless soils, the unit shaft friction at a given depth, f , may be calculated by Equation (21) in API 2GEO, First Edition.

For piles end bearing in cohesionless soils, the unit end bearing, q , may be computed by Equation (22) in API 2GEO, First Edition.

NOTE See C.8.1.4 in API 2GEO, First Edition.

17.4.5 Shaft Friction and End Bearing of Grouted Piles in Rock

The unit shaft friction of grouted piles in jetted or drilled holes in rock shall not exceed half the uniaxial compressive strength of the rock or grout, but in general should be much less than this value. The reduction depends on pile construction factors (such as roughness on the side of the hole) and on rock mass factors (such as the presence of discontinuities). The sidewall of the hole can develop a layer of slaked mud or clay, which will never gain the strength of the rock. The bond stress between the pile and the grout shall be checked in accordance with 15.1.

NOTE See API 2GEO for additional information.

17.5 Pile Capacity for Axial Pullout

The ultimate pile pullout capacity is less than or equal to, but shall not exceed, Q_f , the total skin friction resistance. In computing the tensile design load/action on the pile, the weight of the pile should be considered; the weight of the soil plug may also be considered, if this can be justified. For cohesive soils, f shall be the same as stated in 17.4.3. For cohesionless soils, f should be computed according to 17.4.4. For rock, f should be the same as stated in 17.4.5.

17.6 Axial Pile Performance

17.6.1 General

The axial capacity of a pile is its axial resistance, whereas pile performance refers to a specified service requirement by the owner [e.g. deflection(s) at the pile head]. Both axial capacity and pile performance are dependent upon many variables (e.g. the types of soils, the pile characteristics, the installation methods, and the characteristics of the applied loads/actions). The influence of these variables should be considered in pile design.

NOTE See C.8.3 in API 2GEO, First Edition.

17.6.2 Static Axial Behavior of Piles

Pile axial deflections shall be within acceptable serviceability limits, and these deflections shall be compatible with the internal forces and movements of the structure. Axial pile behavior is affected by directions, types, rates, and sequence of the applied loads/actions; by the installation technique; by soil type; by axial pile stiffness; and by other parameters (see API 2GEO).

17.6.3 Cyclic Axial Behavior of Piles

Unusual conditions of loads/actions on piles or limitations on design penetrations of piles warrant detailed consideration of cyclic effects.

Cyclic loads/actions (including inertial loads/actions) due to environmental conditions such as storm waves and earthquakes can have two potentially counteractive effects on the static axial capacity. Repetitive loads/actions can cause a temporary or permanent decrease in resistance and/or an accumulation of deformation. Rapidly applied loads/actions can cause an increase in resistance and/or stiffness of the pile. Very slowly applied loads/actions can cause a decrease in resistance and/or stiffness of the pile. The resultant influence of cyclic loads/actions will be a function of the combined effects of the magnitudes, cycles, and rates of change of applied loads/actions; the structural characteristics of the pile; and the types of soils.

The design pile penetration shall be sufficient to develop the design pile capacity, as discussed in 17.3 and 17.4, to resist the design static and cyclic loads/actions. The design pile penetration can be determined by performing analyses of the pile-soil system subjected to static and cyclic loads/actions. Analytical and numerical methods to perform such analyses are described in API 2GEO. The pile-soil, resistance-displacement, t - z and Q - z , characterizations are discussed in 17.7.

17.6.4 Overall Axial Behavior of Piles

When any of the effects discussed in 17.6.3 are explicitly considered in a pile analysis, the design static and cyclic loads/actions should be imposed on the pile top and the displacements of the pile determined. At the completion of the design loads/actions, the maximum pile resistance and displacement should be determined. Pile deformations shall meet structure serviceability requirements. The total pile resistance after the design loads/actions shall meet the requirements of 17.3.4.

17.7 Soil Reaction for Piles under Axial Compression

17.7.1 General

The pile foundation shall be designed to resist the static and cyclic axial loads/actions. The axial resistance of the soil for pile compression is provided by a combination of axial soil-pile adhesion and associated shear transfer along the sides of the pile, and end bearing resistance at the pile tip. The relationship between mobilized soil-pile shear transfer and local pile displacement at any depth is described using a t - z curve. Similarly, the relationship between mobilized end bearing resistance and axial tip displacement is described using a Q - z curve. Information on soil reaction t - z and Q - z curves for axially loaded piles can be found in 8.4 of API 2GEO, First Edition.

NOTE See C.8.4 in API 2GEO, First Edition.

17.7.2 End Bearing Resistance-displacement, Q - z , Curve

The representative end bearing capacity shall be determined as described in 17.4. However, relatively large pile tip displacements are required to mobilize the full end bearing resistance. See API 2GEO for information regarding Q - z curves.

17.8 Soil Reaction for Piles under Lateral Loads/Actions

The pile foundation shall be designed to resist static and cyclic lateral loads/actions. The lateral resistance of the soil near the surface is significant to pile design, and the possible effects of scour on this resistance shall be assessed. In the absence of more definitive criteria, the procedures given in 8.5 of API 2GEO, First Edition, may be used for constructing the relationships between lateral soil resistance and lateral displacement, p - y curves.

NOTE See C.8.5 in API 2GEO, First Edition.

17.9 Pile Group Behavior

17.9.1 General

Consideration should be given to the effects of closely spaced adjacent piles on the resistance-displacement characteristics of the pile group. Generally, for pile spacings less than eight diameters, group effects have to be evaluated.

NOTE See C.8.6 in API 2GEO, First Edition.

17.9.2 Axial Behavior

For piles embedded in clays, the group capacity can be less than the single isolated pile capacity multiplied by the number of piles in the group; conversely, for piles embedded in sands, the group capacity can be higher than the sum of the capacities of the isolated piles. The group settlement in either clay or sand is normally larger than that of a single pile subjected to the average load/action per pile of the pile group.

17.9.3 Lateral Behavior

For piles with the same pile head fixity conditions and which are embedded in either cohesive or cohesionless soils, the pile group normally experiences greater lateral displacements than those undergone by a single pile subjected to the average load/action per pile of the corresponding group. The major factors influencing the group displacements and distribution of loads/actions over the piles are the pile spacing, the ratio of the pile penetration to the pile diameter, the pile flexibility relative to the soil, the dimensions of the group, and the variations in the shear strength and stiffness modulus of the soil with depth.

17.9.4 Pile Group Stiffness and Structure Dynamics

When the dynamic behavior of a structure is sensitive to variations in foundation stiffness, parametric analyses should be performed to bound the vertical and lateral foundation stiffness values to be used in the dynamic structural analyses.

17.9.5 Resistance Factors

The pile group capacity shall comply with the requirements of 17.3.4. Where there is a nonuniform distribution of loads/actions into the piles, the partial load/action factors for individual piles in the group may be less than those specified in Sections 9 to 11, provided that it can be demonstrated that the displacements and corresponding deformations of the piles and associated structural members are acceptable.

17.10 Pile Wall Thickness

17.10.1 General

The wall thickness of the pile may vary along its length and can be controlled at a particular point by any one of several load cases or requirements that are discussed in 17.10.2 to 17.10.9. The designer should note the pile hammers that have been evaluated for use during driving on the installation drawings or specifications.

17.10.2 Pile Stresses

Pile stresses due to the design load(s)/action(s) shall be checked in accordance with Section 13. A rational analysis considering the restraints placed upon the pile by the structure and by the soil should be used to check the pile stresses for the portion of the pile that is not laterally restrained by the soil. General column buckling of the portion of the pile below the sea floor need not be considered, unless the pile is believed to be laterally unsupported because of extremely low soil shear strengths, large computed lateral deflections, or for some other reason.

Pile stresses and, as a result thereof, the pile wall thickness in the vicinity of the sea floor and possibly at other points are normally controlled by the combined axial force and bending moment that result from the design load(s)/action(s) on the structure. The moment curve for the pile can be computed with soil resistance determined in accordance with 17.8, giving consideration to possible soil removal due to scour.

17.10.3 Pile Design Checks

When lateral deflections associated with cyclic loads/actions at or near the sea floor are relatively large (e.g. exceeding y_c as defined in API 2GEO for soft clay), consideration should be given to reducing or neglecting the soil-pile adhesion through this zone.

17.10.4 Check for Load Case due to Weight of Hammer during Hammer Placement

Each pile or well conductor section on which a hammer (or pile top drilling rig, etc.) is placed should be checked for the loads/actions associated with placing the equipment. This load case can be the limiting factor in establishing the maximum length of add-on sections. This is particularly so in cases where piles are driven or drilled under an angle. The load/action effects that should be checked include static bending, axial forces, and lateral forces generated during initial hammer placement.

Experience indicates that reasonable protection from failure of the pipe wall due to the above load case is provided if the static strength is calculated as follows.

- The projecting add-on section is considered as a free-standing fixed-end column with its appropriate effective length factor K (see A.17.10.4).
- Axial forces and bending moments are calculated using the full factored weight of the hammer, pile cap, and leads, acting through the center of gravity of their combined masses with $\gamma_{f,G} = 1.3$ or $\gamma_{f,Q} = 1.5$, depending on how well the weight of each item is known, and the factored weight of the add-on section with $\gamma_{f,G} = 1.3$, taking into account the angle and eccentricities of the centers of mass. Nearly vertical add-ons are considered as inclined cantilevers, having an initial or realistic small out-of-plumb inclination of at least 2 % when determining their design moment.
- The secondary bending moment, also to be determined, is the sum of the $P-\Delta$ moments due to the first-order lateral deflections of the add-on (considered as a fixed end cantilever). The deflection at the top should be associated with the factored weight of the hammer, etc. and the deflection at mid-height of the add-on should be associated with the factored weight of the add-on.
- In accordance with 13.3.3, the beam-column resistance check shall be satisfied in accordance with Equation (13.3-3), but the bending stresses should include the secondary bending stresses due to deflection of the add-on. If the equation is applied to the plane in which bending takes place, the equation can be simplified to:

$$\frac{\gamma_{R,c} \sigma_c}{f_c} + \frac{\gamma_{R,b} \sigma_b}{f_b} \left(1 + \frac{\sum P \times \Delta}{M} \right) \leq 1.0 \quad (17.10-1)$$

where

- | | |
|--|--|
| $\sum P \times \Delta$ | is the sum of the secondary $P-\Delta$ moments discussed in item b) above, where P is the factored weight of the item and Δ is the associated first-order lateral deflection; |
| M | is the primary bending moment at the foot of the add-on (in undeflected state); |
| σ_b | is the primary bending stress due to M at the foot of the add-on; |
| $\gamma_{R,c}$, σ_c , f_c , $\gamma_{R,b}$, and f_b | are as defined in Section 13. |

17.10.5 Stresses during Driving

Consideration should be given to the stresses that occur in the free-standing pile section during driving. Generally, stresses are checked against the conservative criterion that the sum of the stresses due to the impact of the hammer (the dynamic stresses) and the stresses due to axial force and bending moment(s) (the static stresses) should not exceed the SMYS of the steel. Less conservative criteria are permitted, provided that these are supported by sound engineering analyses and empirical evidence. The potential fatigue damage due to driving and its impact on the in-service fatigue life of the piles shall also be considered.

A method of analysis based on wave propagation theory should be used to determine the dynamic stresses (see 22.5.5). In general, it may be assumed that column buckling will not occur as a result of the dynamic portion of the driving stresses. The unfactored dynamic stresses should not exceed 80 % to 90 % of yield, depending on specific circumstances such as the location of the maximum stresses down the length of pile, the number of blows, previous experience with the pile-hammer combination, and the confidence level in the analyses. Separate considerations apply when significant driving stresses are transmitted into the structure and damage to appurtenances should be avoided.

When using hydraulic hammers, the driving energy can exceed the rated energy, and this should be considered in the analyses. Also, the static stresses induced by hydraulic hammers need to be computed with special care due to the possible variations in driving configurations, for example, when driving vertical piles without lateral restraint that are exposed to environmental loads/actions (see also 22.5.14).

17.10.6 Minimum Wall Thickness

The D/t ratio of the entire length of a pile should be small enough to preclude local buckling at stresses up to the yield strength of the pile material. Consideration should be given to the different situations and associated load cases occurring during the installation and the service life of a pile.

Experience indicates that for piles with a high D/t ratio, minor local damage near the pile tip (e.g. out-of-roundness or denting) can propagate during installation (e.g. by driving) to cause more extensive deformation and collapse of the pile. In addition to selecting an appropriate D/t ratio, the pile bottom section shall be checked for all load cases occurring during handling (e.g. pick-up, upending, stabbing, etc.) to ensure that local damage does not occur.

For piles that are to be installed by driving where sustained hard driving [820 blows/m (250 blows/ft) with the largest size hammer to be used] is anticipated, the minimum piling wall thickness used shall not be less than the values calculated from Equation (17.10-1) or Equation (17.10-2).

In SI units:

$$t = 6.35 + \frac{D}{100} \quad (17.10-1)$$

In USC units:

$$t = 0.25 + \frac{D}{100} \quad (17.10-2)$$

where

t is the wall thickness, in millimeters (inches);

D is the diameter, in millimeters (inches).

Minimum wall thickness for normally used pile sizes should be as listed in Table 17.10-1.

The preceding requirement for a lesser D/t ratio when hard driving is expected may be relaxed when it can be shown by past experience or by detailed analysis that the pile will not be damaged during its installation.

Table 17.10-1—Minimum Pile Wall Thickness

Pile Diameter mm (in.)	Nominal Wall Thickness, t mm (in.)
610 (24)	13 ($1/2$)
762 (30)	14 ($9/16$)
914 (36)	16 ($5/8$)
1067 (42)	17 ($11/16$)
1219 (48)	19 ($3/4$)
1524 (60)	22 ($7/8$)
1829 (72)	25 (1)
2134 (84)	28 (1.125)
2438 (96)	31 (1.25)
2743 (108)	34 (1.375)
3048 (120)	37 (1.50)

17.10.7 Allowance for Underdrive and Overdrive

With piles having thickened sections at the sea floor, consideration should be given to providing an extra length of heavy wall material in the vicinity of the sea floor so that the pile will not be overstressed at this point if the design penetration is not reached. The amount of underdrive allowance provided in the design will depend on the degree of uncertainty regarding the penetration that can be obtained. In some instances, consideration should also be given to providing an overdrive allowance in a similar manner, in the event that an expected bearing stratum is not encountered at the anticipated depth.

The possibility of underdrive and overdrive should also be considered for determining the required length of the piles and for determining the position and extent of shear keys in grouted pile-to-sleeve connections.

17.10.8 Driving Shoe

The purpose of driving shoes is to assist piles to penetrate through hard layers or to reduce driving resistances, thereby allowing greater penetrations to be achieved than would otherwise be the case. Different design considerations apply for each use.

If an internal driving shoe is provided for driving through a hard layer it should be designed to ensure that unacceptably high driving stresses do not occur at and above the transition point between the normal and the thickened section of the pile tip. It should also be checked that the shoe does not reduce the end bearing capacity of the soil plug below the value assumed in the design. External shoes are not normally used, as they tend to reduce the skin friction along the length of pile above them.

For reducing the internal skin friction during driving in cohesive soils, an internal driving shoe at the pile tip can be considered. The effect of an internal driving shoe should be taken into account when evaluating the total representative capacity of the pile.

17.10.9 Driving Head

Any driving head at the top of the pile should be designed in association with the installation contractor to ensure that it is fully compatible with the proposed installation procedures and equipment.

17.11 Length of Pile Sections

In selecting pile section lengths, consideration should be given to the following:

- a) the capability of the lift equipment to raise, lower, and stab the sections;
- b) the capability of the lift equipment to place the pile driving hammer on the sections to be driven;
- c) the possibility of a large amount of downward pile movement (e.g. due to self weight of the pile or immediately following the penetration of the closure of one of the structure's legs, placement of the hammer, or of a new add-on);
- d) stresses developed in the pile section while lifting;
- e) the wall thickness and material properties at field welds;
- f) interference with the planned concurrent driving of neighbouring piles;
- g) the type of soil in which the pile tip is positioned during driving interruptions for field welding to attach additional sections.

In addition, static and dynamic stresses due to the hammer weight and operation should be considered, as discussed in 17.10.4 and 17.10.5.

Each pile section on which driving is required should contain a cut-off allowance to permit the removal of material damaged by the impact of the pile driving hammer. The normal allowance is 0.5 m to 1.5 m (2 ft to 5 ft) per section. Where possible, the cut for the removal of the cut-off allowance should be made at a conveniently accessible elevation.

17.12 Shallow Foundations

17.12.1 General

The shallow foundations considered herein are typically used for the support of a piled offshore structure, e.g. mudmats, or the support of subsea structures. Detailed advice on the design of shallow foundations is contained in API 2GEO, whereas ISO 19903 ^[6] and ISO 19905-1 ^[4] provide special information for gravity base structures and jack-ups, respectively. It should be noted that the design practice in those documents applies material factors to soil properties, rather than a resistance factor to the capacity calculations, as is the case in Section 17. Care should be taken when applying design procedures from one document to formulations in another.

The design of shallow foundations should include, where appropriate to the intended application, consideration of the following:

- a) stability, including failure due to overturning, bearing, or sliding or combinations thereof;
- b) static foundation deformations, including possible damage to components of the structure and its foundation or attached facilities;
- c) dynamic foundation characteristics, including the influence of the foundation on load/action effects in the structure and the performance of the foundation itself under dynamic loads/actions;
- d) hydraulic instability, such as scour or piping load/action due to wave pressures, including the potential for foundation instability and damage to the structure;
- e) installation and removal, including penetration and pull-out of shear skirts or the foundation base itself, and the effects of pressure build-up or draw-down of trapped water underneath the base;
- f) flat-bottomed components or structures, which should be designed with due consideration given to the possibility of nonuniform soil reactions that can cause hard spots acting on the base of the structure, and where under-base grouting, differential ballasting, and/or site preparation may be used to mitigate the effects from such loads.

General recommendations pertaining to these aspects of shallow foundation design are given in API 2GEO. It should be noted that the complexity of the analyses can vary enormously from situation to situation. For example, a shallow circular foundation on a strong uniform soil under vertical loads/actions only may be designed using simple techniques that can be found in most undergraduate soil mechanics text books. On the other hand, a mudmat of irregular shape on soft variable soil, subject to cyclic horizontal and vertical loads/actions, could need a three-dimensional FEA to produce an economical and reliable design.

NOTE See A.7 in API 2GEO, First Edition.

17.12.2 Stability of Shallow Foundations

The equations to be considered in evaluating the stability of shallow foundations are given in API 2GEO. The equations are for static conditions (monotonic loads/actions only). This is normally acceptable since a limiting weather criterion applies during the installation of offshore structures and, for subsea structures, design situations other than static are normally not applicable. However, if cyclic loads/actions are likely to occur, e.g. during prolonged installation periods for structures with insufficient piles installed, these shall be taken into account separately.

The design bearing and sliding capacities of shallow foundation shall be determined from the representative values shown in these equations in API 2GEO with relevant resistant factors.

a) Bearing Failure Mode

$$P_{d,b} \leq Q_{d,b} = Q_d/\gamma_{R,b} \text{ or } Q_d'/\gamma_{R,b} \quad (17.12-1)$$

b) Sliding Failure Mode

$$P_{d,s} \leq H_{d,s} = H_d/\gamma_{R,s} \text{ or } H_d'/\gamma_{R,s} \quad (17.12-2)$$

where

$Q_{d,b}$ is the design bearing capacity of the shallow foundation;

Q_d or Q_d' is the representative value of the bearing capacity as determined in 7.4, 7.5, and 7.6 of API 2GEO, First Edition;

$P_{d,b}$	is the design bearing load/action on the shallow foundation;
$\gamma_{R,b}$	is the bearing partial resistant factor for shallow foundation ($\gamma_{R,b} = 1.50$);
$H_{d,s}$	is the design sliding capacity of the shallow foundation;
H_d or H_d'	is the representative value of sliding capacity as determined in 7.15 of API 2GEO, First Edition;
$P_{d,s}$	is the design sliding load/action on the shallow foundation;
$\gamma_{R,s}$	is the sliding partial resistant factor for shallow foundation ($\gamma_{R,s} = 1.25$).

In many offshore applications the lateral loads/actions and overturning moments as well as vertical loads/actions are highly variable. In assessing capacity of foundations, the uncertainty of all these factored loads/actions should be considered. A consistent method for accomplishing this is construction of an envelope of load/action combinations that constitute failure and comparing these limiting conditions with design capacity. See Reference [11] for a more detailed discussion of this procedure.

18 Corrosion Control

18.1 General

Fixed steel offshore structures require efficient control of corrosion to ensure that their strength is not reduced by progressive corrosion degradation. Corrosion damage can affect structural integrity and the ability to resist operational, environmental, and accidental loads/actions in various ways. One primary objective of corrosion control is to prevent such damage in fatigue-sensitive areas, as corrosion damage can result in stress concentrations for the initiation of fatigue cracks. Corrosion control will also avoid a potential reduction of the strength of structural components subjected to static loads/actions only.

NOTE 1 Guidance for coatings is given in References [28] to [31].

NOTE 2 Applicable cathodic protection codes are given in References [32] to [39].

18.2 Corrosion Zones and Environmental Parameters Affecting Corrosivity

Marine corrosion environments can be divided into the following corrosion zones:

- atmospheric zone;
- splash zone (or intermediate zone);
- submerged zone;
- buried zone.

The splash zone is defined as that area of the structure that is frequently wetted due to waves and tidal variations. However, surfaces that are only wetted during major storms are excluded. The extent of the splash zone is subject to large local variations and can vary approximately between 2 m and 10 m. In some areas, progressive subsidence of the structure should be considered when defining the splash zone.

The atmospheric zone includes freely exposed and semi-sheltered areas above the splash zone. The seawater submerged zone extends below the splash zone and is defined here to include any seawater-flooded internal compartments. The buried zone includes any structural parts buried in the sea floor sediments or covered by disposed solids.

The corrosivity of corrosion zones varies as a function of geographical location. Of the primary environmental factors influencing corrosion, the main factor is temperature, which is also subject to significant seasonal variations. Salinity, sea currents, wave load/action, and cycles of fresh water intrusion are other important parameters.

In the atmospheric zone, the frequency and duration of wetting (“time-of-wetness”) is a main factor affecting corrosion. The corrosive conditions are typically most severe in areas sheltered from direct rainfall and sunlight, but freely exposed to sea spray and condensation that facilitates accumulation of sea salts and moisture, with a resultant high time-of-wetness. A combination of high ambient temperature and time-of-wetness creates the most corrosive conditions.

In the upper submerged zone and the lower part of the splash zone, the corrosion environment is normally modified by marine growth. Depending on the type of growth and the local conditions, the net effect can be either to enhance or retard corrosion attack. Enhancement of corrosion processes by marine growth (e.g. through corrosive metabolites) is commonly referred to as “microbiologically influenced corrosion” (MIC). Marine growth can also interfere with systems for corrosion control, including coatings/linings and cathodic protection. Ice scoring in arctic waters can cause very high corrosion rates by removal of corrosion retarding rust layers, corrosion protective coatings, or marine growth. In deeper waters, variations in oxygen content and sea currents are the dominating parameters, although the corrosivity is relatively low.

In addition to freely exposed surfaces in the submerged zone, MIC can potentially occur in flooded steel members unless special precautions are taken to ensure that water remains sterile (e.g. by addition of a biocide). Internal flooded components may not be subjected to corrosion if there is no possible circulation of seawater (e.g. closed flooded legs, piles, skirt piles, etc.).

In the buried zone, corrosion is predominantly related to MIC. In undisturbed sediments, MIC will only be significant in the uppermost layer. However, disturbance of sediments and discharge of drill-cuttings or other effluents can enhance bacterial activity and hence MIC.

NOTE Other splash zone definitions can apply in regulations or in other design codes.

18.3 Forms of Corrosion, Associated Corrosion Rates, and Corrosion Damage

Corrosion damage to uncoated carbon steel in the atmospheric zone and in the splash zone is associated with oxygen attack and is typically more or less uniform. In typical conditions for the atmospheric zone, the steady state corrosion rate for carbon steel (i.e. as uniform attack) is normally around 0.1 mm/year or lower. In the splash zone and the most corrosive conditions for the atmospheric zone (i.e. high time-of-wetness and high ambient temperature), corrosion rates can amount to 0.3 mm/year, and for internally heated surfaces in the splash zone even much higher (up to an order of magnitude of 3 mm/year).

In the submerged and buried zones, corrosion is mostly governed by MIC causing colonies of corrosion pits. Welds are often preferentially attacked. Corrosion as uniform attack is unlikely to significantly exceed about 0.1 mm/year, but the rate of pitting can be much higher: 1 mm/year and even more under conditions favoring high bacterial activity (e.g. ambient temperature of 20 °C to 40 °C and access to organic material).

In most cases, the static strength of large structural components is not jeopardized by MIC, due to its localized form. However, MIC can initiate fatigue cracking of components subjected to cyclic loads/actions.

Aluminium alloys of the 5000 and 6000 series as used for topsides structural components are highly resistant to marine atmospheres and suffer only superficial staining or micropitting. However, galvanic coupling (i.e. metallic plus electrolytic coupling) to structural steel, stainless steels and copper alloys shall be avoided. Otherwise, severe galvanic corrosion of aluminium can result at the point of contact. Similarly, structural steel can suffer enhanced corrosion if in galvanic contact with stainless steel or with copper base alloys. The propensity for galvanic corrosion is highest in the submerged zone but is normally prevented by cathodic protection.

Very high-strength steels (yield strength in excess of 1200 MPa) and certain high-strength aluminium, nickel, and copper alloys are sensitive to stress corrosion cracking in marine atmospheres.

NOTE The term “stress corrosion cracking” refers to cracking that is caused by an interaction between static tensile stresses in a material and a specific corrosion medium.

18.4 Design of Corrosion Control

18.4.1 General

The following main systems for corrosion control of fixed steel offshore structures apply:

- coatings, linings, and wrappings;
- cathodic protection;
- corrosion-resistant materials;
- corrosion allowance.

18.4.2 Considerations in Design of Corrosion Control

The initial selection and subsequent detailed design of systems for corrosion control of fixed steel offshore structures shall take the following factors into account:

- a) regulatory requirements;
- b) criticality of the overall system and the functional requirements to individual structural components to be protected;
- c) type and severity of corrosion environment(s);
- d) design service life (and likelihood of lifetime extension);
- e) accessibility for inspection and maintenance, including overall maintenance philosophy;
- f) suitability, reliability, and economy of optional techniques for corrosion control.

18.4.3 Coatings, Linings, and Wrappings

Coatings are defined as relatively thin (< 1 mm) organic or metallic layers, single or multiple, that are applied by spraying, brushing, or dipping. Linings and wrappings are defined as thicker (> 1 mm) corrosion protective layers applied with the objective of resisting mechanical wear by wave load/action, facilitating removal of marine fouling, protecting against impacts, etc. Organic materials used for linings and wrappings are normally reinforced (e.g. by glass fibers or flakes). Metallic materials are typically copper based (to incorporate antifouling properties).

Special precautions are required to prevent corrosion under coatings, linings, and wrappings, including under fire protective coatings. Metallic materials shall preferably be seal welded to structural components.

Coating systems in the atmospheric zone include various forms of organic (paint) coatings and certain metallic coatings. Of the latter, zinc layers are applied as hot dipping or thermal spraying. Thermally sprayed aluminium coatings have been used more recently, particularly for more demanding applications.

In the seawater submerged zone, organic coatings or thermally sprayed aluminium may be used to reduce weight or hydrodynamic loads/actions due to drag caused by galvanic anodes and/or to improve the current distribution by cathodic protection. On subsea production systems, paint coatings are primarily used to ease visibility. In the upper submerged zone coatings may also be chosen to improve periodic removal of marine growth.

Coating and lining systems should primarily be selected based on proven experience for a specific application and environment. Comprehensive field testing is required when practical experience is lacking. For the atmospheric and splash zones, maintainability is a major criterion, whereas compatibility with cathodic protection is normally required in the submerged and buried zones. Resistance to damage, e.g. due to the removal of marine fouling, is also required.

The design of all components to be paint coated shall take into account relevant measures to ease both the initial application and later maintenance. This includes a preference for tubular shapes, rounding of sharp edges, and requirements for securing scaffolding. Structural components exposed to sea spray, rain, or intermediate wetting, either externally or internally, shall be designed to prevent accumulation of moisture, e.g. by using continuous welding and making provisions for drainage.

18.4.4 Cathodic Protection

18.4.4.1 Cathodic Protection Systems

Cathodic protection may be effected using either of the following:

- galvanic (sacrificial) anodes, or
- impressed current (IC) from one or more rectifiers.

Cathodic protection is applicable to the submerged and the buried zones. In the splash zone, cathodic protection is generally fully effective for any part of the structure extending below the LAT and marginally effective in the tidal zone.

With adequate design, any form of corrosion damage can be prevented, including galvanic corrosion between dissimilar materials. Cathodic protection is also efficient in preventing corrosion damage in narrow crevices between structural components. It is generally accepted that cathodic protection restores resistance to initiation of fatigue cracks to the same level as in dry air. Laboratory testing has shown that cathodic protection can enhance propagation of fatigue cracks at high stress intensities. However, practical experience indicates that this effect is not critical to normal structural materials at the potential range required below.

Cathodic protection by galvanic anodes has proven reliability for long-term protection, even in the harshest weather conditions. IC systems are less tolerant of any shortcomings of design, installation, and maintenance and require a dependable external current source. IC systems have also been used for upgrading cathodic protection systems based on galvanic anodes. Reliability and costs for installation, operation, and maintenance shall be taken into account when choosing between IC and galvanic anode systems.

Owing to the risk of hydrogen-induced stress cracking (HISC), steels with SMYS in excess of 720 MPa (105 ksi) shall not be used for critical cathodically protected components without due evaluations. This imposes some restrictions on the subsea use of high-strength parts, e.g. bolting. Furthermore, any welding (or other construction method affecting ductility or tensile properties) shall be carried out according to a qualified procedure that limits hardness to HV350. This restricts the use of welded structural steels to a maximum SMYS of approximately 550 MPa (80 ksi).

The cathodic protection system for fixed steel offshore structures shall account for risers, conductors, pull-tubes, and other submerged objects electrically connected to the structure. In some cases, subsea pipelines are provided with electrical insulation to prevent interaction between the structure and pipeline cathodic protection systems.

Current drain to piles, well casings, concrete reinforcing bars, or other components not considered to require cathodic protection shall be assessed.

Internal compartments flooded with seawater shall be provided with sufficient galvanic anodes, suitably arranged, to last the design service life of the structure. IC systems are not normally applicable for this purpose.

The design shall ensure a protection potential within the range -0.80 V to -1.1 V relative to Ag/AgCl/seawater. More negative potentials can be achieved by IC systems, but these can be harmful to coatings and promote HISC in ordinary structural steels.

The current demand, I_c , for cathodic protection of external surfaces in the submerged and buried zones and for wetted surfaces in any internal compartments shall be calculated from Equation (18.4-1):

$$I_c = A_c \times k_c \times i_c \quad (18.4-1)$$

where

I_c is the current demand (A);

A_c is the actual surface area (m^2);

k_c is the coating break-down factor for any coated surfaces, with $k_c = 1$ for bare steel;

i_c is the design current density (A/m^2).

I_c shall be calculated as the average current demand, $I_{c,average}$, to maintain cathodic protection throughout the design service life, t_L , in years, of the system. The initial and final current demands, i.e. $I_{c,initial}$ and $I_{c,final}$, respectively, required to polarize the relevant surfaces to a protection potential of -0.80 V relative to Ag/AgCl/seawater, shall also be calculated.

Design current densities and coating break-down factors for calculations of average and initial/final current demands are given in the applicable cathodic protection design codes (see A.18).

For large structures, it is always convenient, and often necessary, to subdivide the structure into units to be protected. The division may, for example, be based on depth zones. Each internal compartment shall comprise at least one unit for individual design.

The design life, t_L , in years, of cathodic protection systems shall normally be equal to the design service life of the structure itself. If the design is based on replacement of anodes, provisions to facilitate retrofitting shall be addressed during the design.

Possible interaction with other structures, pipelines, and vessels shall be evaluated. The need for temporary corrosion protection prior to the energizing of the IC system shall be assessed.

To ensure consistency, design parameters shall all be collected from the same cathodic protection design code selected (see A.18).

18.4.4.2 Galvanic Anode Systems

Based on the average total current demand, $I_{c,average}$, for each unit (including any current drain), the total net anode mass, m_{total} (kg) required to maintain cathodic protection throughout the design service life t_L (in years) is calculated from Equation (18.4-2):

$$m_{\text{total}} = \frac{I_{\text{c,average}} \times t_L \times 8760}{u \times \varepsilon} \quad (18.4-2)$$

where

u is the anode's utilization factor;

ε is the anode materials electrochemical efficiency (A·h/kg).

Design values for ε and u are, again, given in the applicable cathodic protection design codes (see A.18).

From the required total net anode mass, m_{total} , a tentative selection of anode dimensions and number of anodes can be determined. It shall subsequently be demonstrated that this selection meets the requirements for the initial/final current output, I_a (A), per anode, and the total current capacity, C_a (A·h).

The anode current output, I_a , is calculated from Ohm's law according to:

$$I_a = \frac{E_c - E_a}{R_a} \quad (18.4-3)$$

where

E_c is the design protective potential (V) matching the initial/final design current densities (i.e. normally – 0.80 V relative to Ag/AgCl/seawater);

E_a is the design closed circuit anode potential (V);

R_a is the anode resistance (ohms).

E_c , E_a , and R_a are also given in the applicable cathodic protection design codes (see A.18).

The current capacity c_a of one anode is given by:

$$c_a = m \times u \times \varepsilon \quad (18.4-4)$$

where m is the net mass per anode.

The total current capacity C_a for n anodes thus becomes:

$$C_a = n \times c_a \quad (18.4-5)$$

Anode dimensions and net weight shall be selected to match all requirements for current output (initial/final) and current capacity for a specific number of anodes. Calculations shall be carried out to demonstrate that the following requirements are met:

$$C_a = n \times c_a \geq I_{\text{c,average}} \times t_L \times 8760 \quad (18.4-6)$$

$$n \times I_{\text{a,initial}} \geq I_{\text{c,initial}} \quad (18.4-7)$$

$$n \times I_{\text{a,final}} \geq I_{\text{c,final}} \quad (18.4-8)$$

The final current output shall be calculated using the estimated anode resistance when the anode has been consumed to its utilization factor.

The calculated number of anodes shall be distributed to provide a uniform current distribution, taking into account the current demand of individual surface areas and any current drain.

Design of galvanic anode cores shall take into account forces induced by environmental loads/actions and installation activities (including any pile driving operations). For large stand-off anodes, the use of doubler plates should be considered. Effects of added weight, increased drag loads/actions, and subsea operations should be evaluated.

Anodes shall be located in the submerged zone for the protection of both splash zone, submerged zone, and buried zone. The uppermost part of the submerged zone should be avoided, in order to reduce the effects of tide, waves, and marine fouling. Anodes should not be located close to joints and other areas that are critical to structural integrity. The distribution of anodes should otherwise reflect the current demand of individual areas to be protected.

For sacrificial anode systems, equipping a few anodes with instruments to enable current output monitoring can aid in future maintenance and retrofit decisions.

18.4.4.3 Impressed Current Systems

The design of IC cathodic protection systems shall include extra capacity (i.e. compared to the calculated current demand) to compensate for a more uneven current distribution from the relatively few, high output IC anodes. Furthermore, redundancy shall be included to compensate for some deficiency of individual anodes and rectifiers. The design shall include detailed procedures for maintenance (replacement) of anodes and other subsea equipment.

IC systems shall have a structure-to-seawater monitoring system that is able to show that cathodic protection is maintained within specified limits at both areas closest to and furthest from the anodes.

18.4.5 Corrosion-resistant Materials

The selection of corrosion-resistant materials for structural components shall take into account their anticipated corrosion resistance for the intended application (including resistance to environmentally induced cracking) and compatibility with other materials, mechanical properties, and ability to fabricate.

Precautions shall be taken to prevent galvanic corrosion of less resistant materials. In the submerged and buried zones, galvanic corrosion can be efficiently prevented by cathodic protection. However, in the atmospheric and splash zones, special precautions shall be taken to prevent galvanic corrosion. These can include coating of the component with the highest electrochemical potential or electric insulation.

NOTE The term “environmentally induced cracking,” sometimes also called “environmentally assisted cracking,” includes stress corrosion cracking, corrosion fatigue, and hydrogen-induced cracking associated with hydrogen generated by cathodic protection.

18.4.6 Corrosion Allowance

A corrosion allowance, i.e. extra steel thickness to compensate for the effect of metal loss by corrosion on structural integrity, is sometimes applicable, e.g. for tubular sections in the splash zone. It can be used alone or in combination with a coating. The thickness of any corrosion allowance shall be determined taking expected corrosivity, design service life, and maintenance plans into account.

18.5 Fabrication and Installation of Corrosion Control

18.5.1 General

Fabrication procedures can affect the corrosion resistance of materials, in particular for certain corrosion-resistant materials. All fabrication involving welding or brazing to structural components shall be performed in accordance with Section 20, regulatory requirements, applicable codes/standards, and approved project-specific procedures and drawings.

18.5.2 Coatings and Linings

Recommendations for surface preparation, materials, coating application, inspection, and repairs are given in applicable standards and practices.

QC during surface preparation, coating application, and repairs is most essential to ensure consistent performance of coatings and linings. Follow-up of coating work, from surface preparation to final inspection by a certified coating inspector, is highly recommended.

18.5.3 Cathodic Protection

Manufacturing and QC of sacrificial anodes shall be performed in accordance with a procedure that defines compositional limits of anode and anode core materials, weights and dimensional tolerances, visual inspection and permissible surface defects, marking, and documentation. Short-term electrochemical testing may be specified for the verification of electrochemical performance on a heat basis.

Installation of galvanic anodes by welding or brazing shall be in accordance with 18.5.1. After completing installation, anode locations and sizes shall be verified.

Commissioning of IC cathodic protection systems shall include detailed structure-to-seawater potential measurements for verifying readings from fixed reference electrodes and to confirm that the protective potential range is achieved for all concerned parts of the structure.

18.5.4 Corrosion-resistant Materials

Fabrication of corrosion-resistant materials shall be performed with due consideration of how the applicable techniques (welding, grinding, etc.) affect their corrosion resistance and mechanical properties. Improper fabrication methods can easily cause staining and incipient pitting of stainless steel and aluminium surfaces.

18.6 In-service Inspection, Monitoring, and Maintenance of Corrosion Control

18.6.1 General

In-service inspection of corrosion control is a periodic activity for confirmation of the physical condition and integrity of the corrosion control system(s) and/or the actual components to be protected. Monitoring of corrosion control refers to regular recording of data associated with corrosion control. It is frequently carried out online.

EXAMPLE Recordings of structure-to-seawater potentials from fixed reference electrodes.

Planning of schemes for inspection and monitoring of corrosion control shall take into account:

- a) regulatory requirements;
- b) the criticality of the overall system and of the individual components being protected;
- c) the type and severity of the corrosion environment(s);
- d) potential forms of corrosion damage;
- e) the capability of inspection and monitoring tools, as well as accessibility for inspection;
- f) results and consequences of previous inspections and/or monitoring.

NOTE 1 See API 2SIM for in-service inspection and corrosion control requirements for structural integrity management (SIM).

NOTE 2 In other standards, other definitions of inspection, monitoring, and maintenance of corrosion control can apply; for example, corrosion monitoring is sometimes defined to include corrosion inspection and vice versa.

18.6.2 Coatings and Linings

Inspection of coatings and linings is primarily performed by visual inspection and has as its objective the assessment of the need for maintenance (i.e. repairs). A close visual examination will also disclose any areas where coating degradation has allowed corrosion to develop to a degree requiring repair or replacement of structural components.

18.6.3 Cathodic Protection

Inspection of cathodic protection may include visual examination of anodes and measurements of structure-to-seawater potentials and anode-to-seawater potentials. A protection potential of -0.80 V relative to Ag/AgCl/seawater or more negative is a generally accepted criterion for cathodic protection of carbon steel in seawater. Accurate and detailed potential measurements require a suitable reference electrode positioned by a diver or an underwater vehicle. A general assessment of the protection of the structure or parts thereof can be obtained by lowering a reference electrode from the surface.

Galvanic anode potential measurements shall be performed as part of a detailed structure survey. Average anode and cathode potentials can be used to estimate cathode current demand and anode service life.

Galvanic anode potential measurements shall be performed with minimum disruption of any deposit layers. Accurate estimates of anode consumption can require removal of deposits and should therefore be performed after potential measurements have been completed.

Monitoring of galvanic anode cathodic protection systems is sometimes performed using recordings from fixed reference electrodes.

For galvanic anode systems, an initial survey comprising structure-to-seawater potential measurements (with emphasis on the more critical areas) and preferably anode-to-seawater potential measurements should be performed within 1 year after installation. The frequency of further potential surveys shall be assessed on the basis of results from any such surveys performed earlier and from periodic visual inspection of structural components and anodes.

For IC cathodic protection systems, a comprehensive mapping of the potential distribution shall be performed during commissioning (see 18.5.3). Fixed reference electrodes used to control current output should be calibrated at regular intervals; rectifier voltage and total current outputs should be recorded on a frequent basis, together with readings from individual anodes and fixed reference electrodes.

Safety regulations can prohibit diver activities whilst IC systems are in operation. Divers can suffer electric shocks from faulty electric equipment or from direct contact with anodes. The following shall be followed when divers inspect structures with IC systems installed:

- a) when divers are not involved in or not aware of cathodic protection (such as for marine growth cleaning), it is recommended that the IC system be switched off during diving;
- b) for divers involved in cathodic protection inspection, a safe distance shall be kept between divers and IC anodes that are in operation;
- c) for the performance of close visual inspection of IC anodes, the current supply to anodes shall be switched off.

18.6.4 Corrosion-resistant Materials

Inspection of corrosion control based on corrosion-resistant materials can be integrated with visual inspection of the structural integrity of critical components associated with such materials.

19 Materials

19.1 General

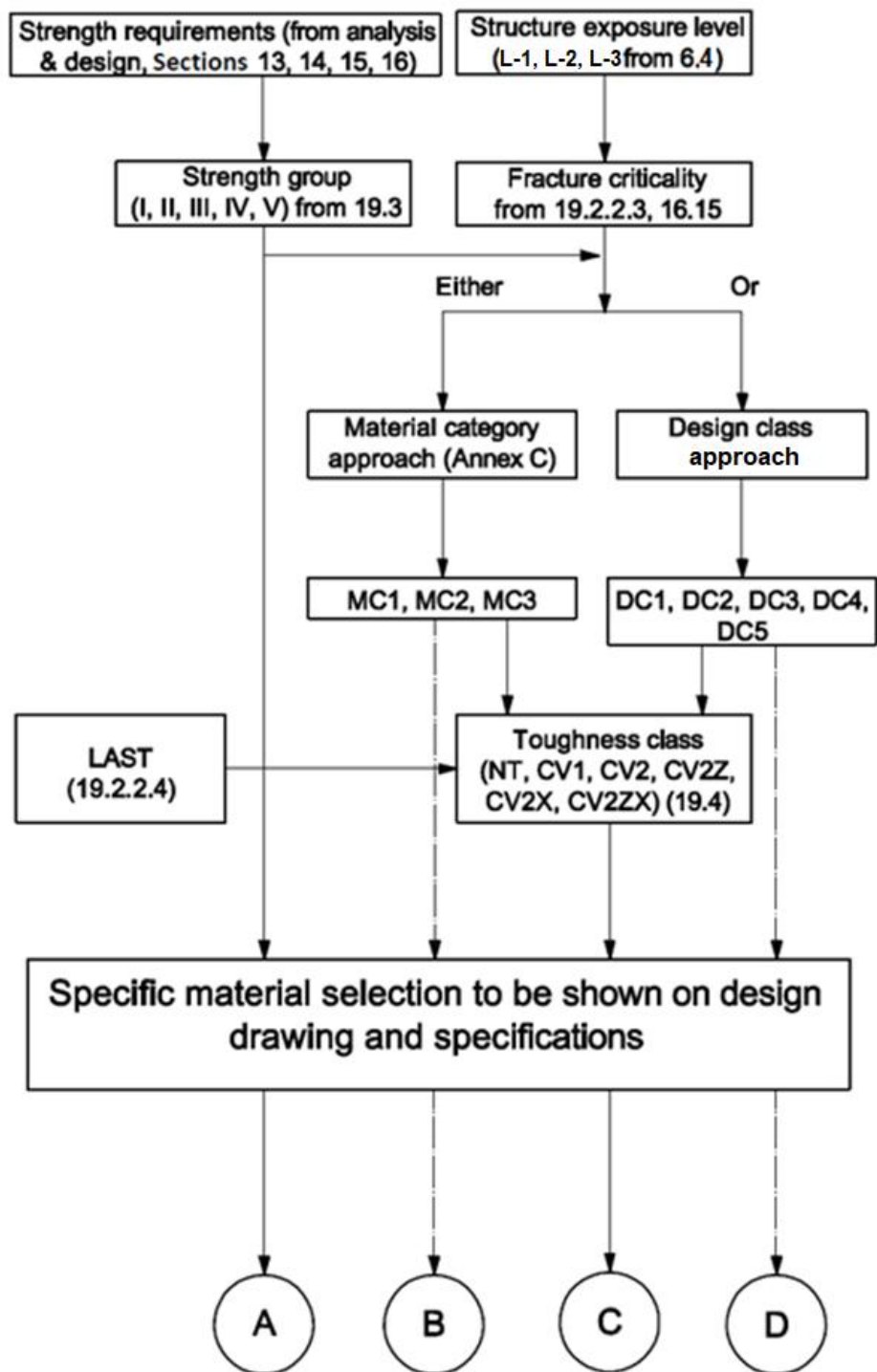
This section presents good practice for the selection of steels that are expected to perform effectively over the design service life of a structure, while allowing practical and economical fabrication and inspection.

Two methods are used for determining the particular steel specifications for a specific structure and the accompanying welding, fabrication, and inspection requirements. These methods, briefly introduced in 19.2.4 and 19.2.5 and described in detail in Annex C and ISO 19902, are generally referred to as:

- the material category (MC) approach that shall be used for all structures in US waters, and
- the design class (DC) approach that is not yet approved for US waters.

A flowchart of the process followed by this section is given in Figure 19.1-1. The MC and DC methods are mutually exclusive. Once the method has been selected, it is not interchangeable at any stage with the other. The objective is to guide the designer in arriving at specific material selections, which shall be shown on design drawings and in project specifications. The strength grades and toughness classes of the selected steels carry over to related welding, inspection, fabrication, and quality requirements in Section 20.

For any specific project application, the owner shall define the mutual responsibilities between design and fabrication.



A, B, C, D continue to Figure 20.1-1, as input to welding, inspection, fabrication, and QA/QC documentation considerations.

Figure 19.1-1—Flowchart for Material Selection

19.2 Design Philosophy

19.2.1 Material Characterization

For the purposes of material selection and utilization in offshore structures, steels are characterized as belonging to a strength group and a toughness class.

Strength groups (see 19.3) are defined by a range of yield strengths, which are determined by tensile testing of specimens in accordance with steel manufacturing specifications. The specified yield strength is a design criterion used to size members in accordance with selected design standards.

Toughness classes (see 19.4) are determined by the ability of steels to achieve a minimum Charpy V-notch (CVN) test energy at a specified minimum temperature.

Toughness becomes more important as the magnitudes of varying loads/actions increase, as the criticality of the structure increases and as service temperatures decrease. The toughness of the steel affects the fracture resistance.

The designer can also consider (at least qualitatively) the extent of plastic deformation needed before fracture, the degree of restraint as influenced by section thickness, as well as residual stress as influenced by weld shrinkage and general section thickness of the parts.

19.2.2 Material Selection Criteria

19.2.2.1 Yield Strength Requirements

The minimum yield strength for a given component geometry and thickness shall satisfy the strength requirements defined in Sections 13, 14, 15, and 16, in conjunction with the loads/actions described in Sections 8, 9, 10, and 11.

19.2.2.2 Structure Exposure Level

The material selection shall take into account the structure's exposure level (L-1, L-2, L-3), established at the outset of the design process in accordance with 6.6.

19.2.2.3 Component Criticality

In the selection of the appropriate material, consideration should be given to the criticality of the component. A component (member or joint) is considered critical if its sole failure is likely to lead to a catastrophic structural collapse (3.1.13). In general, the toughness requirements and inspection of this document ensure that the material selected for a critical component has higher toughness and ductility requirements and more stringent inspection requirements than the material selected for a component with high redundancy, where alternative load paths in the structure lessen the risk of a major failure.

19.2.2.4 Lowest Anticipated Service Temperature

The steel performance characteristics are affected by the lowest anticipated service temperature (LAST). The LAST value to be used in the material selection shall be in accordance with applicable regulatory requirements in the region of application (regional or national standards, regional information in API 2MET, etc.). Suggested LAST values for certain offshore areas are given in A.19.2.2.4.

The LAST value establishes the temperature for CVN toughness, which shall be taken as the LAST minus the temperature margin shown in Table 19.4-1.

19.2.2.5 Other Considerations

Consideration should be given to other factors, such as resistance to fracture in the presence of plastic deformation, residual stress, thickness restraint, cost, availability, weldability, resistance to lamellar tearing, and

demonstrated successful use. Guidance on these topics is generally included in complementary standards or guidelines that need to be used in conjunction with this document.

19.2.3 Selection Process

Once the criteria described in 19.2.2 have been established, the steel selection process follows the logical flowchart shown in Figure 19.1-1.

A key part of the process is the method described in 19.2.4, the MC approach, which has been successfully used within the US offshore industry for many years. The DC approach, referenced in 19.2.5, provides an alternative method. Both methods address the issues described in 19.2.2.

Either approach provides a framework for material selection, welding qualification, and extent of inspection commensurate with the level of reliability implicit in the design process.

19.2.4 Material Category Approach

In the MC approach (see Annex C for details) the steel selection is based on the interrelation of the structure's exposure level, the material yield strength and toughness, and the component consequence ranking. Thickness is considered in connection with component type. The corresponding welding and inspection requirements are given in Annex D.

19.2.5 Design Class Approach

The DC method has evolved from North Sea practices, consistent with the use of BS, EEMUA, NORSOK, and EN standards. See ISO 19902 for informative information regarding the DC approach.

19.3 Strength Groups

Steels are grouped according to SMYS, as presented in Table 19.3-1.

Table 19.3-1—Steel Group SMYS Requirements

Group	SMYS Range
I	220 MPa (32 ksi) to 275 MPa (40 ksi)
II	> 275 MPa (40 ksi) to 395 MPa (57 ksi)
III	> 395 MPa (57 ksi) to 455 MPa (66 ksi)
IV	> 455 MPa (66 ksi) to 495 MPa (72 ksi)
V	> 495 MPa (72 ksi)

NOTE 1 See A.19 for information.

NOTE 2 Group V steels are steels having an SMYS above 500 MPa (72 ksi) together with high levels of impact toughness and weldability. Experience to date is very limited (see Reference [A.19.2-1]).

19.4 Toughness Classes

This subsection describes the toughness classes used in this document, and the types of steel for which they are considered applicable. All testing shall conform to an internationally recognized standard.

Class NT steels (not CVN tested) are considered suitable for application in critical welded components at service temperatures above 0 °C (32 °F).

Class CV1 steels are suitable for use where service temperatures, thickness, cold work, restraint, stress concentration, impact loading, and/or lack of redundancy indicate the need for improved notch toughness. Class CV1 steels shall, as a minimum, exhibit CVN test energy as shown in Table 19.4-1.

Class CV2 steels have a large margin between CVN testing temperature and service temperature, as prescribed in Table 19.4-1. Such steels are suitable for major primary structures or structural components and for critical or nonredundant components, particularly in the presence of factors such as the following:

- high stresses and stress concentrations,
- high residual stresses,
- severe cold work from fabrication,
- low temperatures,
- high calculated fatigue damage, and/or
- impact loading.

In addition to all the other characteristics of class CV2 steel, class CV2Z steels shall have through-thickness (short transverse direction) ductility for resistance to lamellar tearing caused by tensile stress in the direction of thickness. Through-thickness ductility shall be demonstrated by either having a minimum reduction in area of 30 % in a tension test conducted on a specimen cut from the through-thickness direction, or by specifying a sulphur content by weight (P_S) of 0.006 % or less in the product analysis.

Class CV2X steels shall have a formal crack tip opening displacement (CTOD) prequalification in addition to the other characteristics of class CV2 steel.

Class CV2ZX steels shall have a formal CTOD prequalification in addition to the characteristics of class CV2Z steel.

Testing of material shall be performed in accordance with suitable national or internationally recognized standards.

Minimum steel toughness requirements are given in Table 19.4-1. This table is not intended to supersede the designer's judgment for special situations, such as the need to increase the toughness level for any of the reasons described in 19.2.1.

Table 19.4-1—Minimum Toughness Requirements for Structural Steels

Steel Group	SMYS Range	Charpy Toughness	Toughness Classes and Charpy Impact Test Temperature			
			NT	CV1	CV2	CV2Z, CV2X, and CV2ZX
	MPa (ksi)		(CVN testing not required)	Test at LAST	Test at 30 °C (54 °F) below LAST	Test at 30 °C (54 °F) below LAST
I	220–275 (32–40)	20 J (15 ft-lbs)	No test	X	X	Not applicable
II	> 275–395 (> 40–57)	35 J (25 ft-lbs)	No test	X	X	X
III	> 395–455 (> 57–66)	45 J (35 ft-lbs)	Combination not allowed	X	X	X
IV	> 455–495 (> 66–72)	60 J (45 ft-lbs)	Combination not allowed	X	X	X
V	> 495 (> 72)	60 J (45 ft-lbs)	Combination not allowed	Combination not allowed	X	X

X Denotes required tests to minimum Charpy toughness at the specified temperature.

19.5 Applicable Steels

Annex C lists commonly used specifications for steel plates, steel shapes (open sections), and steel tubulars, within the context of the MC approach.

Steels that are not included in the lists may be used, as long as they satisfy the provisions of Sections 19 and 20.

19.6 Cement Grout for Pile/Pin-to-sleeve Connections and Grouted Repairs

19.6.1 Grout Materials

In general, Portland cement grouts should be used with or without inert fillers mixed with either seawater or fresh water. However, there may be special circumstances where the use of seawater is undesirable because of corrosion or other durability effects.

Any water used to mix grout should be free of hydrocarbons and other deleterious matter.

In grout mix composition, the water/cement ratio should not exceed 0.40.

Grout mix design should be submitted for approval by the owner prior to works.

High alumina cement (HAC) grout shall be used only with extreme care, and restricted to below splash zone applications in order to reduce conversion and associated strength loss.

Admixtures may be used to improve the grout slurry properties, provided it can be demonstrated that they do not adversely affect the required grout durability and performance.

Grout constituent materials should be supplied to a recognized national standard and stored in accordance with the manufacturer's recommendations.

19.6.2 Onshore Grout Trial

In the absence of previous production data, the compressive strength of the grout mix shall be confirmed by an onshore trial. An onshore trial shall consist of at least three separate batches of grout. A minimum of six cubes shall be cast from each batch for each age at which strength is to be determined. The compressive strength of 75 mm cubes at 28 days should be used as the basis for strength compliance. However, it is recommended that strength gain with age should be measured at 30 hours, at 3 days and at 7 days, or at other ages as considered appropriate. Cubes of other sizes may be used provided appropriate conversion factors relating the strength to a 75 mm cube are available. Cylinders shall not be used. Test procedure shall be in accordance with ASTM C109.

The mean cube strength calculated from the data should be greater than either:

$$f_{cu} + 1.64\sigma \left(0.86 + \sqrt{2/n} \right) \quad (\text{for } 10 \leq n \leq 100) \quad (19.6-1)$$

or

$$f_{cu} + 1.64\sigma \quad (\text{for } n \geq 100) \quad (19.6-2)$$

where

f_{cu} is the required specified minimum compressive strength;

σ is the standard deviation of the data;

n is the number of cubes.

The grout mixing equipment and mixing times used in the onshore trials, or in previous production grouting from which data are taken, should both be the same as those to be used during offshore grouting to ensure the grout slurry is similarly sheared.

The specific gravity of the grout slurry should be correlated against the mean compressive strength for offshore monitoring.

19.6.3 Offshore Grout Trial

Prior to grouting, an offshore trial shall be performed, using the equipment and materials to be used for the production grouting, to check the 30 hr grout strength compliance. Grouting shall not proceed until compliance has been achieved. Compliance is satisfied if all cube results exceed the specified minimum 30 hr compressive strength, calculated as defined in 19.6.2.

19.6.4 Offshore Quality Control

Specific gravity measurements of the grout slurry prior to pumping shall, as a minimum, be recorded once per batch or once per cubic meter, whichever is the more frequent, and preferably should be measured and recorded continuously. The specific gravity of the in-place grout slurry should also be measured at the most appropriate of the following locations: surface returns, the highest point within the grout volume, or from a suitably sized vertical standpipe. An in-place specific gravity greater than or equal to that determined during the onshore trial shall be achieved. Specific gravity measurements shall be made using a pressurized fluid density balance in accordance with ISO 10414-1 or monitoring devices calibrated to this type of balance.

In situations where subsea measuring devices fail, a percentage over volume may be pumped in lieu of direct measurements, if the following conditions are met:

- a) it can be demonstrated that on similar connections on the structure, where measurements are available, the required specific gravity was achieved with a recorded net level of grout over volume having been pumped;
- b) an additional 20 % in-place volume of grout over and above the net recorded volume in a) above is pumped;
- c) no major grout system losses are evident.

During the grouting operation, samples shall be taken from randomly selected batches. At least four samples, each of three cubes, should be taken for each grout volume to give a representative distribution throughout the in-place grout. The cubes should be cast and tested to a recognized standard and match cured to the in-place grout environmental conditions. At least one cube should be tested from each sample to confirm compliance at 28 days. Compliance is satisfied if all individual cube results, in a set of four, exceed the specified minimum compressive strength, f_{cu} , as defined in 19.6.2.

20 Welding, Fabrication, and Weld Inspection

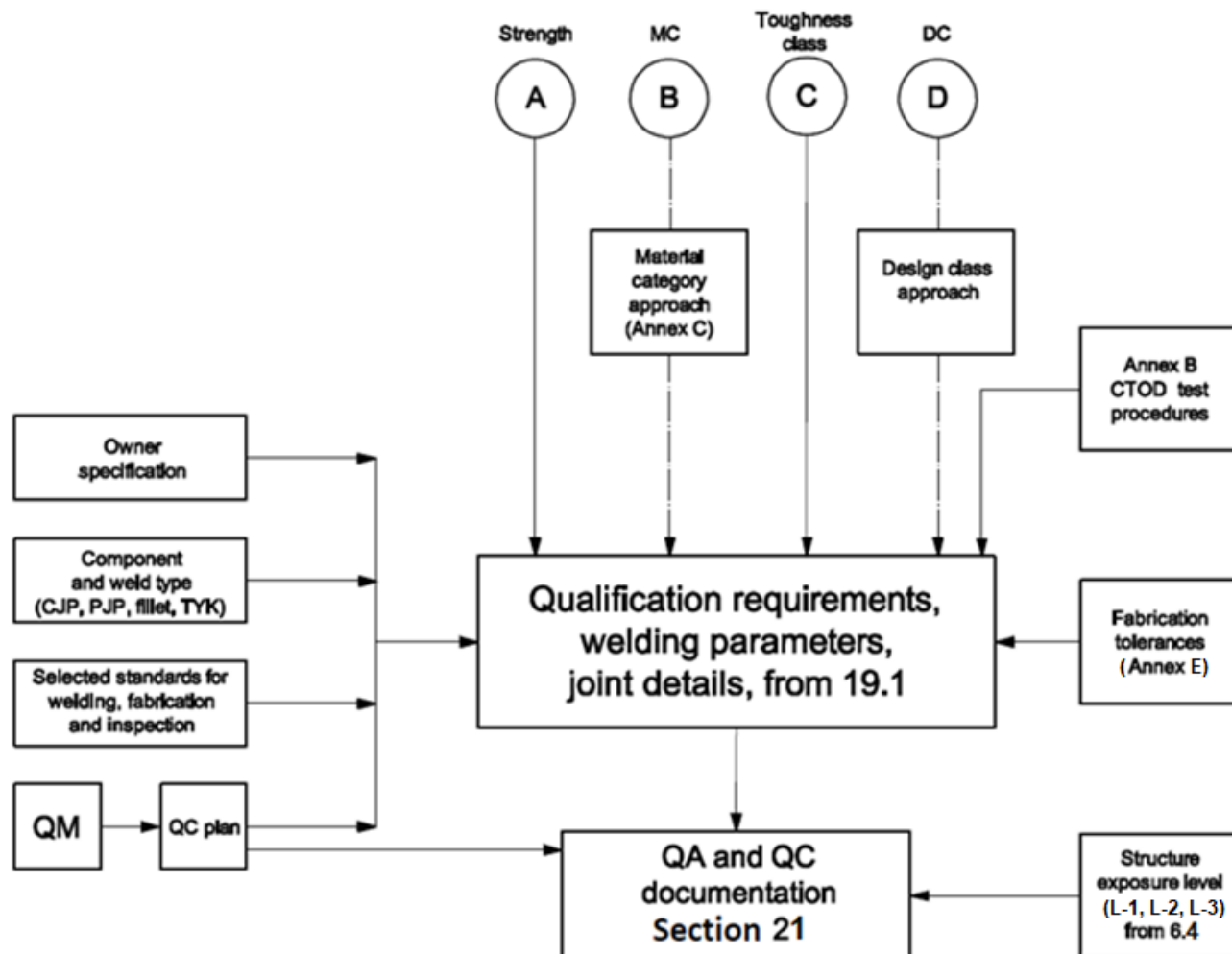
20.1 General

This section is to be used in conjunction with owner specifications, selected international, national, or regulatory standards for welding, fabrication, and inspection. It describes complementary provisions for welding, inspection, fabrication, and quality issues, in which requirements specific to fixed steel offshore structures are adapted to steel strength, toughness class, and criticality of the structure or component.

An overview of the process is given in Figure 20.1-1. Inputs to procurement and construction are derived from engineering design processes described in Section 19 and Annex C and ISO 19902.

The provisions of Section 20 shall be supplemented by Annexes B and E, as well as by either Annex D or ISO 19902, as appropriate.

Guided by the constructor's embedded quality management system (QMS), a project-specific QC plan provides proactive QC inspections and controls (e.g. qualification) prior to and during fabrication (see Section 21).



Key

- CJP complete joint penetration
- PJP partial joint penetration
- TYK T-, Y-, or K-joint
- QM quality management

Figure 20.1-1—Flowchart for Welding, Inspection, Fabrication, and QA/QC Documentation

20.2 Welding

20.2.1 Selected Generic Welding and Fabrication Standards

The welding of fixed offshore structures shall be performed in accordance with selected generic international or national standards that complement requirements given herein to jointly address, at a minimum, the following topics:

- a) applicable welding processes;
- b) approved consumables (electrodes, flux, etc.);
- c) approved base metals;
- d) interchangeability of steels and consumables within groups;
- e) matching, undermatching, and overmatching of base metals and consumables;
- f) minimum/maximum preheat and interpass temperatures;
- g) welding procedure and welding personnel qualification, including:
 - 1) weld metal tensile testing and CVN testing of weld metal and HAZ (for procedures only),
 - 2) range of applicability (positions, thicknesses, etc.),
 - 3) essential variables,
 - 4) number and type of test specimens,
 - 5) test specimen descriptions,
 - 6) test specimen locations,
 - 7) test specimen acceptance criteria, and
 - 8) welding of single-sided T-, Y-, and K-joints;
- h) production weld joint details and tolerances, including:
 - 1) welding of single-sided T-, Y-, and K-joints, and
 - 2) other welds;
- i) post-weld heat treatment (PWHT);
- j) heat straightening limits;
- k) weld size/profile limits;
- l) environmental limitations for welding;
- m) back-gouging/backing/runoff tab criteria;
- n) preparation of base metal (cleaning, bevelling, etc.);
- o) fatigue improvement techniques (if applicable);
- p) NDT procedures;
- q) NDT discontinuity acceptance criteria;
- r) repairs.

The standard or standards used to address a) to r) above can be generic documents applicable to a wide range of steel structures. In the remainder of this section, these standards are referred to as “the selected standard.” The

requirements of this section deal with issues specific to fixed offshore structures and shall be used to supplement the selected standard.

For a particular project application, the owner shall identify suitable standard(s) to satisfy the above requirements. Examples are given in A.20.2.1. In the United States, American Welding Society (AWS) D1.1/D1.1M^[12] is the most commonly used welding standard. This standard is compatible with the MC approach for selecting steels, which is identified as the required approach in 19.1.

20.2.2 Weld Metal and HAZ Properties

20.2.2.1 General

Subsection 20.2.2 describes required properties for weld metal and HAZ for the strength grade and toughness class of the connected base metal.

Strength matching shall be as specified in the selected welding standard (see 20.2.1). Where undermatching is permitted, this shall be as specified in the design. Where materials of two different strengths are joined, the protocol for matching shall be as described in design documents.

20.2.2.2 Material Category Toughness

Annex D defines the minimum toughness requirements for weld metal and HAZ, respectively, for use with the MC approach. Group V steels are not fully described, as these steels require the designer to specify the minimum toughness requirements.

20.2.2.3 Design Class Toughness

See ISO 19902 for requirements.

20.2.2.4 Charpy V-notch Toughness

20.2.2.4.1 Testing

For specimen tests, the selected welding standard's requirements shall be followed.

The test temperatures and minimum energy values in weld metal and HAZ shall be as required to match the performance of the given steels to be joined as tabulated in Annex D or E.

20.2.2.4.2 Additional Essential Variables

For welding procedures that are Charpy-tested, the following changes shall be considered essential variables, in addition to the essential variables listed in the selected welding standard.

- a) An increase in carbon equivalent value, P_{CE} , computed from Equation (20.1-1), greater than 0.03 %, or an increase in modified carbon equivalent parameter for cracks, P_{CM} , computed from Equation (20.1-2), greater than 0.02 %, with respect to the values specified in the procedure qualification test:

$$P_{CE} = P_C + P_{Mn}/6 + (P_{Cr} + P_{Mo} + P_V)/5 + (P_{Ni} + P_{Cu})/15 \quad (20.1-1)$$

$$P_{CM} = P_C + P_{Si}/30 + (P_{Mn} + P_{Cu} + P_{Cr})/20 + P_{Ni}/60 + P_{Mo}/15 + 5 P_B \quad (20.1-2)$$

where

- P_C is the weight percentage of carbon;
 P_{Mn} is the weight percentage of manganese;
 P_{Cr} is the weight percentage of chromium;
 P_{Mo} is the weight percentage of molybdenum;
 P_V is the weight percentage of vanadium;
 P_{Ni} is the weight percentage of nickel;
 P_{Cu} is the weight percentage of copper;
 P_{Si} is the weight percentage of silicon;
 P_B is the weight percentage of boron.

- b) An increase of more than 28 °C (50 °F) over the maximum interpass temperature attained during the procedure qualification test.
- c) A change greater than 25 % in the heat input of the welding passes from the average value calculated in the supporting procedure qualification test. Heat input average for fill passes shall be calculated separate from cap and root passes.
- d) Any change in electrode or electrode/flux brand name or place of manufacture when the CVN test temperature is lower than –18 °C (0 °F).

20.2.2.5 CTOD Toughness

20.2.2.5.1 General

CTOD tests are only required for CV2X and CV2ZX steels. For testing details, see Annex B.

CTOD testing of the HAZ may be omitted if the requirements of 20.2.2.5.2 are met. HAZ CVN test results on the procedure qualification record (PQR) should be greater than or equal to the results obtained by the steel manufacturer during preproduction qualification.

20.2.2.5.2 Preproduction Qualification

CTOD tests of the HAZ may be omitted if the steel is preproduction qualified by the manufacturer in accordance with a recognized standard.

CTOD tests shall be performed for each condition of material supply from each manufacturer. Essential variables to be considered in material supply shall include melting practice, rolling practice, maximum thickness, specific chemical composition range and plate manufacturing process [e.g. normalized, thermo-mechanically controlled process (TMCP), or quenched and tempered (Q&T)].

Repetitions of HAZ CTOD testing may be waived for tests in which the CTOD properties are not expected to be degraded (e.g. where a second procedure is within the range of heat input and preheat/interpass temperatures of those to be qualified by a first).

CTOD data for the HAZ is generally provided by the steel supplier's weldability data. However, specific CTOD testing of the HAZ of the test weld may be specified in the welding procedure qualification (WPQ) requirements. When reliance is placed on the steel supplier's CTOD tests, Charpy test results on the PQR should not be substantially lower than those obtained by the steel supplier in parallel with the CTOD test.

Qualification welding shall be conducted at both the highest heat input/interpass temperature and lowest input/interpass temperature to be allowed for production welding, as governed by the welding procedure specification (WPS).

The test welding positions shall be selected as those representing the highest and lowest heat input of the procedure(s) to be qualified by such tests.

20.2.2.5.3 CTOD Fracture Toughness Requirements

CTOD requirements are a function of design philosophy. The test temperature shall not be greater than the minimum design temperature for a given service.

All tests shall meet the target CTOD acceptance level set by the designer.

Representative CTOD requirements range from 0.10 mm at 4 °C (0.004 in. at 40 °F) to 0.38 mm at –10 °C (0.015 in. at 14 °F). Achieving the higher levels of toughness can require some difficult compromises, set against other desirable attributes of the welding process—for example, the deep penetration and relative freedom from trapped slag of uphill passes.

CTOD testing reports complying with the testing standard shall be included in the PQR documentation. Photographs of at least 1× magnification, clearly showing the fatigue pre-crack and the fracture face of one of the broken halves of all CTOD specimens, shall be included in the PQR documentation.

20.2.2.5.4 Additional Essential Variables

For welding procedures that are CTOD-tested, the following changes shall be considered essential variables, in addition to the essential variables listed in the selected welding standard:

- a) any increase in the maximum interpass temperature over that qualified in the high heat input CTOD procedure qualification test;
- b) an increase in joint thickness greater than that used for the CTOD procedure qualification tests;
- c) a decrease in heat input from the heat input qualified during low heat input CTOD procedure prequalification test for the HAZ;
- d) any increase in the heat input from the average heat input qualified of the fill passes during the high heat input CTOD procedure qualification tests;
- e) any change in electrode or electrode/flux brand name or the place of manufacture;
- f) a reduction in the depth of back-gouging as measured during the CTOD qualification testing;
- g) a reduction in the width of back-gouging as measured during the CTOD qualification testing;
- h) an increase in the layer thickness (as measured at the fusion line) greater than 10 %, versus that measured during CTOD prequalification testing.

20.2.2.5.5 Qualification Range

The CTOD qualification tests qualify production welding thickness up to and including the thickness tested.

20.2.2.5.6 Documentation and Results

CTOD testing reports shall be included in the PQR documentation. Photographs of at least 1× magnification, clearly showing the fatigue pre-crack and the fracture face of one of the broken halves of all CTOD specimens shall also be included.

20.2.2.6 Hardness Testing

A Vickers hardness survey shall be performed on one of the macrosections from each test weld in accordance with a recognized standard, using an applied load of 100 N or less. The sample shall be ground, polished, and etched with a suitable solution to give clear definition of the weld, HAZ, and fusion line microstructures under the microscope.

Hardness test locations shall be located as detailed below and as shown in Figure 20.2-1:

- a) HAZ—starting at 0.25 mm to 0.50 mm from the fusion line with three indentations at 0.50 mm to 0.75 mm intervals;
- b) weld metal—a minimum of three equally spaced indentations for each hardness traverse.

The hardness traverse on fillet welds shall be similar to that for T-connections.

Maximum hardness values shall not exceed 325 HV10 for below water locations and 350 HV10 for above water. If any individual hardness value exceeds the limit specified, two further sections shall be taken, both of which shall meet the specified requirements.

The maximum HAZ hardness limit of 325 HV10 is considered an acceptable safe level for typical carbon and low-alloy structural steels. Weldability tests have shown that critical hardness decreases with decreasing carbon equivalent values (P_{CE} and P_{CM}) (see A.19.3.2 and A.19.3.3). Newer low-carbon structural steels can crack at maximum HAZ hardness of 325 HV10 and less.

Hardness test sample locations for qualification tests performed on pipe shall be at the 3 o'clock or 9 o'clock position for downhill cap and at the 6 o'clock position for uphill cap passes.

A photograph or photo macrograph of 1× to 3× magnification, clearly showing the hardness indentations, HAZ, and the weld zone, shall be included in the test results. For acceptable qualification, the macroetch test specimen, when inspected visually, shall also conform to the requirements for visual examination.

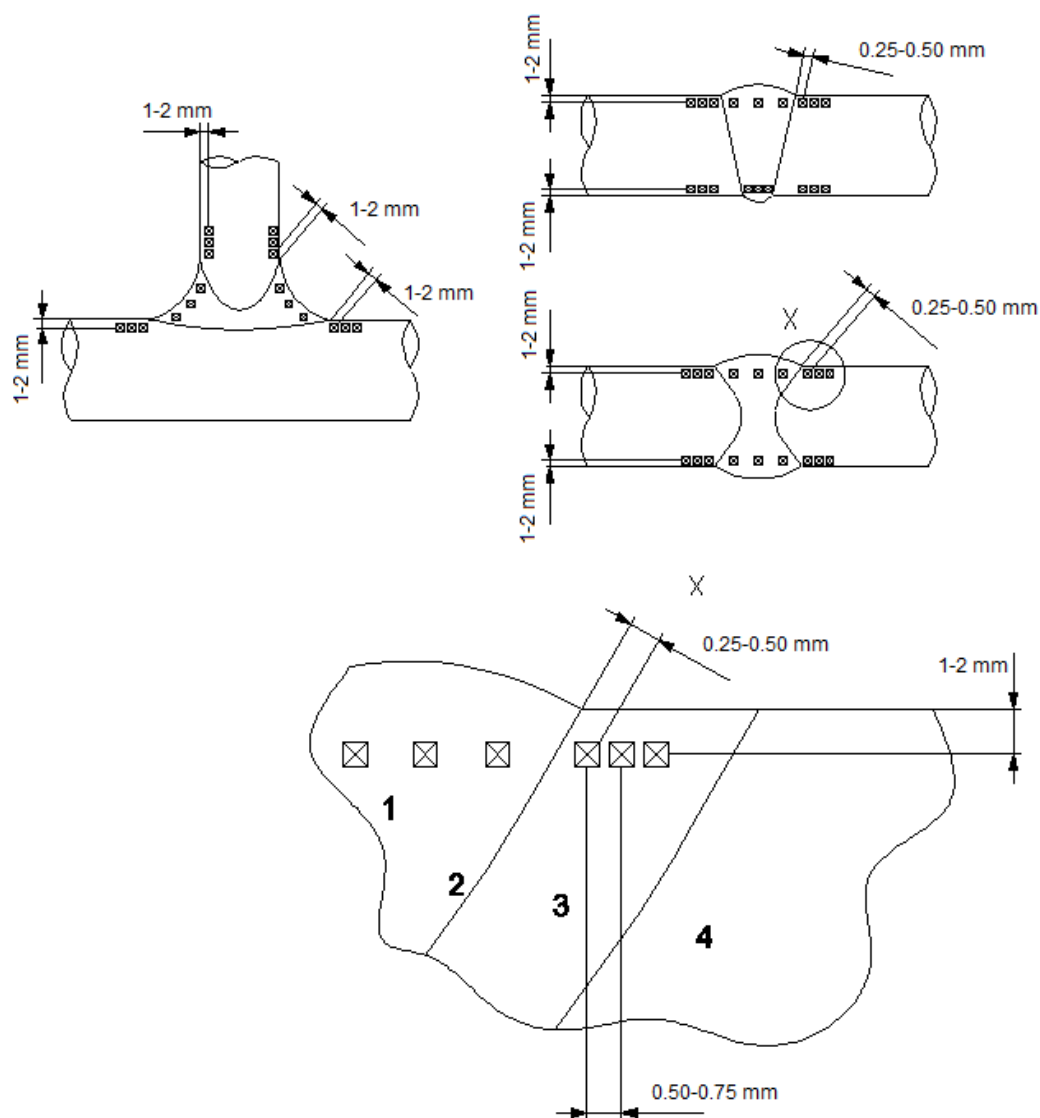
20.2.2.7 Other Mechanical Tests

Other mechanical tests shall be conducted during the WPQ in accordance with the selected standard's requirements. These include, as applicable, the following:

- tension tests,
- bend tests, and
- macroetch sectioning.

20.2.3 Tubular T-, Y-, and K-joints

Welded connections of tubular intersections at T-, Y-, and K-joints are a dominant feature of fixed steel offshore platforms. Such welds are made from the outside only in tubular bent (or point to point) fabrication. Here, achieving the performance attributes of complete joint penetration welds requires specialized consideration of geometry and position variations, weld joint bevel preparation, fit-up tolerances, WPQ, welding personnel qualification, welder qualifications, cap profiling, and fitness-for-service NDT flaw acceptance criteria. The selected standard should provide such guidance for welding, fabrication, and inspection (see A.20.2).

**Key**

- 1 weld metal
- 2 fusion line
- 3 HAZ
- 4 base metal
- X test location

Figure 20.2-1—Vickers Hardness Test Locations

20.3 Inspection

The inspection requirements for the MC and DC methodology are described in Annex D and ISO 19902, respectively.

Additional requirements for QC, quality assurance (QA), and documentation are presented in Section 21.

20.4 Fabrication

20.4.1 General

The following pertains specifically to fixed steel offshore structures and shall take precedence over selected standards for generic industrial applications.

20.4.2 Weld Requirements

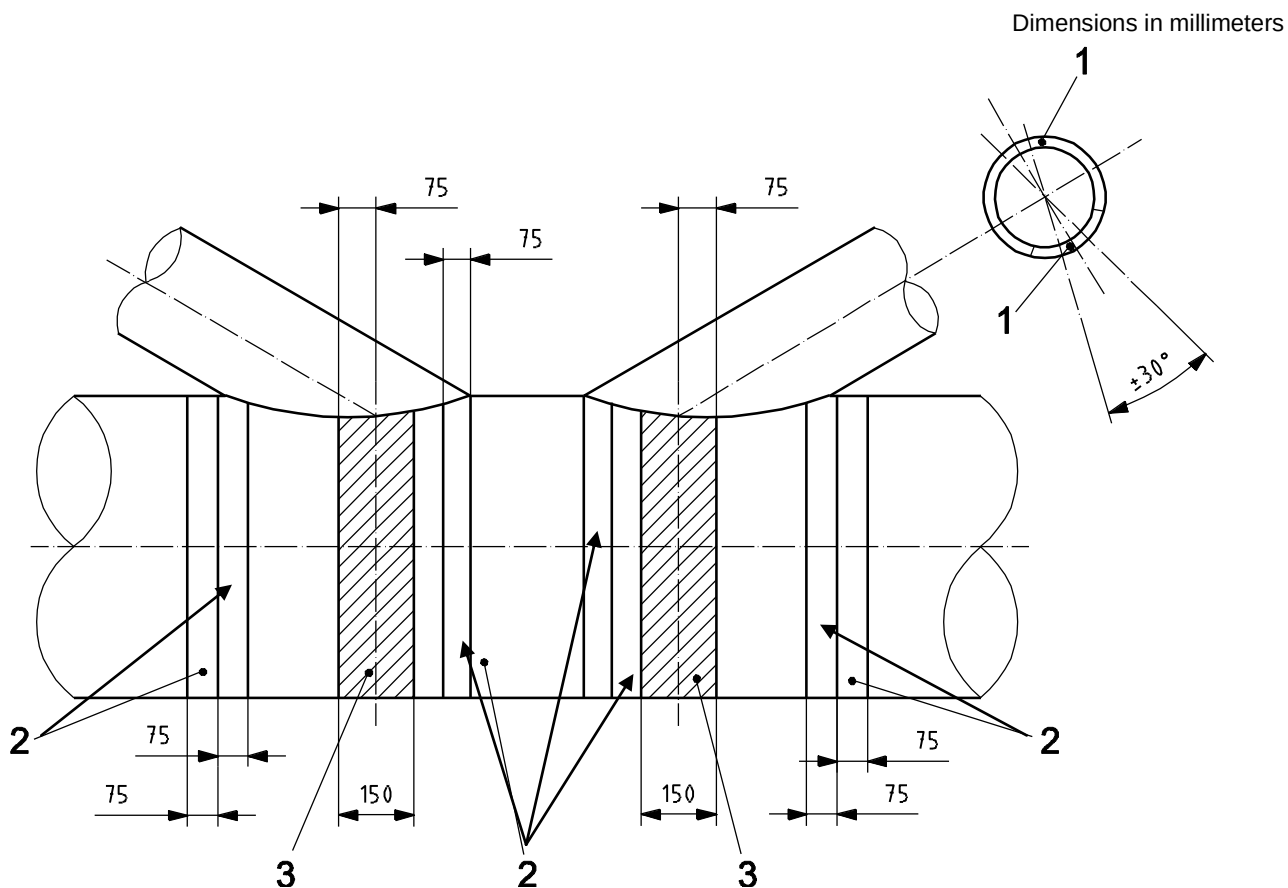
20.4.2.1 Splice Welds

Splices should be fabricated using full penetration welds.

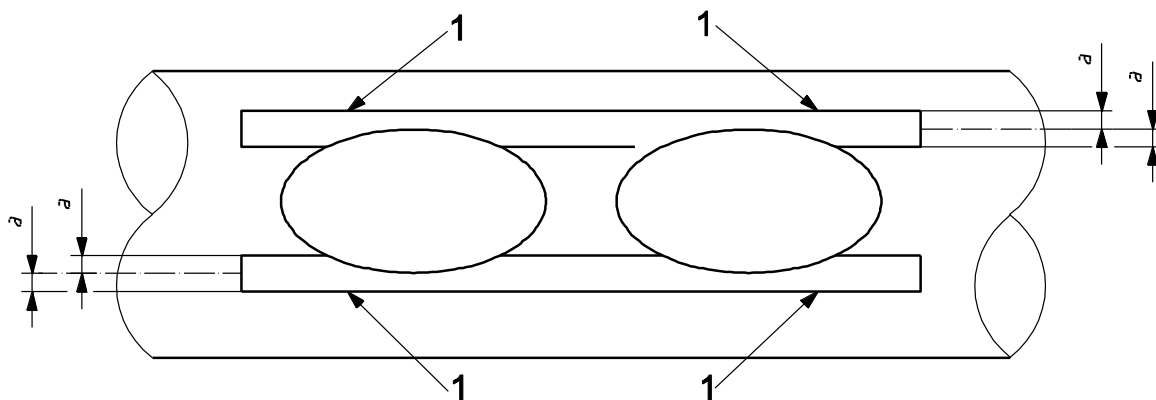
20.4.2.2 Weld Proximity

Weld proximity should satisfy the following requirements.

- a) Circumferential welds are not permitted within cones and node stubs unless specified on design or approved shop drawings.
- b) The minimum spacing between circumferential welds in tubular members should be the smaller of 915 mm (36 in.) or the diameter of the member.
- c) Ring stiffeners in tubular members should be at least 100 mm (4 in.) from circumferential welds. Where this is not possible, the welds should overlap by at least 10 mm ($\frac{3}{8}$ in.).
- d) Rings used to stiffen cone-cylinder junctions shall not be offset and the welds shall be overlapped by at least 10 mm ($\frac{3}{8}$ in.), to avoid coincident location of weld toes.
- e) Longitudinal and circumferential welds in tubular joints should avoid the most critical areas of a joint, as shown in Figure 20.4-1.
- f) The offset between longitudinal seams in adjacent cans in tubular joints should be maximized, taking into account connections at each end of each member, and should be in no case less than 90°, except that the location of a longitudinal seam may be adjusted by no more than 300 mm (12 in.) to avoid the prohibited locations as shown in Figure 20.4-1 b).
- g) In any members to be slotted to receive gusset plates, the slot should be at a distance of 300 mm (12 in.) or twelve times the member wall thickness, whichever is greater, from any circumferential weld. To avoid notches, the slotted member should be drilled or cut and ground smooth at the end of the slot with a diameter of at least 3 mm ($\frac{1}{8}$ in.) greater than the width of the slot. Where the gusset plate passes through the slot, the edge of the gusset plate should approximately conform to the shape of the slot to provide a better fit-up and welding condition.



a) Prohibited Locations for Circumferential Welds on Chords and Longitudinal Seams on Stubs



b) Prohibited Locations for Longitudinal Seams on Chords

Key

- 1 longitudinal seams not permitted in these areas
- 2 circumferential welds not permitted in these areas
- 3 circumferential welds should be avoided in these areas
- ^a Lesser of 150 mm or twice the chord thickness (measured along the chord circumference).

Figure 20.4-1—Positioning of Longitudinal and Circumferential Welds in Tubular Joints

20.4.2.3 Fabrication Aids

When fabrication aids and temporary attachments occur in areas that become part of a completed weldment within 300 mm (12 in.) of an intersecting weld or in areas that require coating, the attachment should be removed by oxy-fuel, cutting a minimum of 3 mm ($\frac{1}{8}$ in.) above the surface of the structural member and grinding the remainder flush with the surface of the member. If such ground areas lie close to or underneath a weld, the surface should be inspected using MPI after grinding and before the permanent weld is executed.

Fabrication aids and temporary attachments that occur in areas that do not require coating and that are located more than 300 mm (12 in.) from an intersecting weld should be seal welded and removed to within 6 mm ($\frac{1}{4}$ in.) of the surface of the member with sharp edges removed by grinding. Oxy-fuel cutting may be used for the bulk removal of a fabrication aid or temporary attachment, but should not be used within 3 mm ($\frac{1}{8}$ in.) of the surface of the underlying component. Fabrication aids and temporary attachments shall not be removed by hammering or oxy-fuel washing.

20.4.3 Forming

Guidelines for forming steel plates into a circular tubular member are given below.

For a diameter-to-thickness (D/t) ratio of less than 20, mechanical tests shall be undertaken to demonstrate that the steel retains its mechanical properties. Post-forming heat treatments, or hot forming, may be used to mitigate the effects of cold forming.

Any materials that require straightening before use to achieve the specified tolerances may be cold straightened if the strain does not exceed 5 %. Similarly, tubular cans for members or joints may be re-rolled after welding to meet the dimensional tolerances if the strain during re-rolling does not exceed 5 %.

Consideration should be given to strain concentration and discontinuity growth in the weld metal or HAZ, if the nominal strain during re-rolling exceeds 4 %, or the product of SMYS times thickness exceeds 3200 MPa · cm (180 ksi · in.).

Re-rolling, hot forming, or post-forming heat treatments, if used, shall be represented in WPQ testing of weld metal and HAZ.

20.4.4 Fabrication Tolerances

The location of each member of the structure should be in accordance with the design drawings within the final fabrication tolerances. The fabrication tolerances are given in Annex E. Tolerances not stated in Annex E shall follow applicable national standards.

Tolerances should be checked at each stage in accordance with the fabrication procedures and the final survey should meet the defined tolerances. Fitters should perform their own checking as the job progresses; their tape measures and straight edges do not need calibration. The final survey should be on complete subassemblies and on the completed structure after PWHT, if applicable.

The personnel responsible for the final survey should be either qualified surveyors or have had at least 5 years experience of similar work. Instruments used shall be in accurate adjustment and have current valid calibration certificates.

Allowance should be made for weld gap tolerances and weld shrinkage in all components, subassembly, and global tolerance calculations.

Where tolerances have to be derived from a formula (i.e. tolerances expressed in terms of the component dimension; e.g. wall thickness), the results shall be taken to the nearest mm (0.04 in.).

20.4.5 Grouted Pile/Pin-to-sleeve Connections

Requirements for the detailing of pile/pin-to-sleeve connections are given in 15.1.3.

Steel surfaces in contact with the in-place grout and on which interface shear is to be developed should be free of loose material, mill scale, grease, and any other material that can reduce the grout to steel friction.

Care should be taken during and after the installation of packers to prevent damage from handling, high temperatures, and splatter from welding and debris. Any debris that could cause damage to the packers during the installation process should be removed. Measures such as dressing of the pile tip to a smooth profile and positioning of centralizers on the sleeve directly above the packers should be taken to minimize potential packer damage during pile stabbing.

21 Quality Control, Quality Assurance, and Documentation

21.1 General

This section provides the QC, QA, and documentation requirements for fabrication, transportation, and installation phases of a typical offshore structure construction project.

QC inspection and testing should be performed to ensure adherence to the plans and specifications containing the detailed instructions necessary to obtain the desired quality and service in the finished product, during all phases of construction, including the fabrication, loadout, sea-fastening, transportation, and installation.

QA/QC, inspection, and documentation requirements should be commensurate with the structure's exposure level (L-1, L-2, or L-3), as described in 6.6, and should be consistent with welding and fabrication provisions, in accordance with recognized international, de facto international, regional, or national standards. Additional requirements and modifications to suit the project application can be specified by the owner. These should be coordinated with the material selection and fracture control philosophy used in design.

The extent of NDT of welds during fabrication should be in compliance with the inspection category. The selection of an inspection category for each weld should be in accordance with Annex D or ISO 19902, as appropriate.

The most effective QC scheme is one that prevents the introduction of defective materials or workmanship into a structure, rather than depending upon QA and documentation that come after the fact.

21.2 Quality Management System

All design, fabrication, and installation shall be performed under a documented QMS, certified to ISO 9000, and following ISO 9001 and ISO 9004, or an equivalent QMS.

The QMS shall, as a minimum and where applicable, address the items listed in Table 21.2-1, in accordance with the structure's exposure level. Items may be covered by reference to the QC plan. The QMS shall relate to the elements of the structure and any additional elements that are critical to the fabrication of the structure, e.g. design and testing of lifting beams and lifting procedures.

Conformance and/or accreditation to a formalized QMS is preferred, e.g. ISO 9000 with ISO 9001 and ISO 9004, API Q1, or similar recognized standard.

The requirements of Table 21.2-1 should be considered even when formalized QMS documentation is not required.

Table 21.2-1—Quality System Requirements

Documentation Description	Exposure Level		
	L-1	L-2	L-3
Quality assurance manual	X	X	—
Organizational chart	X	—	—
Documentation and drawing control procedures	X	X	—
Design change procedures	X	X	—
Design calculations	X	X	—
Fabrication procedures	X	X	—
Lifting procedures	X	X	—
Material tracing control procedures	X	X	—
Dimensional control/survey procedures	X	X	X
Welding consumable storage and handling procedures	X	X	X
Pressure/leak, and equipment operability test procedures	X	—	—
Preheat procedures	X	X	X
NDT procedures	X	X	—
Nonconformance/remedial procedures	X	X	—
Pressure/leak test procedures	X	X	—
Inspection equipment calibration procedures and certificates	X	—	—
Subcontractor quality procedures	X	X	—
Weight reports procedures	X	—	—
Procurement procedures for materials/services	X	X	X
Procedures for handling special processes (e.g. heat treatment)	X	—	—
X Minimum requirements for applicable phases of the overall project.			
— Not required.			

21.3 Quality Control Plan

21.3.1 General

QC is normally performed as appropriate prior to and during fabrication, loadout, transportation, and installation. The purpose of QC is to ensure that materials and workmanship meet the specified requirements. QA is generally performed during and after QC activities, to provide records and documents.

A QC plan shall be developed that makes reference to procedures for all inspection and NDT techniques, including proposed report formats and the names and qualifications of inspection and NDT personnel.

The QC plan shall define the scope of inspection required for fabrication while meeting the requirements of Section 20 and either Annex D or ISO 19902, as appropriate.

The QC plan shall include the elements in the following subsections.

21.3.2 Inspector Qualifications

Inspectors should be qualified to carry out their duties by education, experience, or practical testing according to recognized international, de facto international, regional, or national standards.

Personnel who assist inspectors during any phase of construction of an offshore structure should demonstrate ability and experience, or be qualified to an appropriate standard for the required inspection of a particular project.

21.3.3 Nondestructive Testing Personnel Qualifications

NDT personnel should be certified to an approved program consistent with a recognized standard that requires experience and passing written and practical tests applicable to the project inspection methods and type of construction.

21.3.4 Inspection of Materials

Inspection should verify that all materials being incorporated into any portion of the fabrication are in accordance with the specified requirements.

Steel that does not have supporting material traceability record (MTR) documentation from the steel manufacturer or supplier should not be used.

21.3.5 Inspection of Fabrication

Inspections of the structure should be made upon completion of fabrication to confirm compliance with the dimensional requirements, using the tolerances established by the selected standard and the specific requirements of Section 20. Inspections made during fabrication should be used to determine compliance with in-process requirements, such as joint fit-up, that cannot be confirmed upon general fabrication completion.

21.3.6 Inspection of Welding

Welding inspection and testing should be performed to verify compliance with the requirements specified in Annex D or ISO 19902.

21.4 Inspection of Installation Aids and Appurtenances

Inspections should verify that all installation aids and appurtenances have been installed and tested in accordance with the specified requirements, including any manufacturer's recommendations.

Installation aids include the following:

- a) launch systems;
- b) flooding systems;
- c) grouting systems;
- d) mudmats;
- e) jetting systems;
- f) lugs and guides;
- g) monitoring systems.

Appurtenances include the following:

- a) boat landings;
- b) riser guards;
- c) risers and clamps;
- d) J-tubes;
- e) sump and pump caissons.

During transportation and installation a structure can carry piles and conductors. The inspections should verify that these are installed and tested in accordance with the specified requirements.

The location, size, and orientation should be checked, and weld attachments (including temporary restraints) should be subjected to NDT in accordance with Section 20.

Inspections should include functional tests of all mechanical and electrical equipment and systems, including instrumentation. Cabling and instrumentation should be checked to ensure continuity, and all hydraulic and pneumatic lines should be pressure tested. All nonsteel components (i.e. diaphragms, packers, valve seats, etc.) should be protected from damage from weld spatter, debris, and/or any other construction activities, and hydraulic lines shall be thoroughly flushed and drained before and after testing. The inside of legs, skirt piles, etc. should be inspected to ensure complete removal of debris (e.g. welding rods, miscellaneous pieces of wood, steel, etc.), which could damage nonsteel components during installation.

21.5 Inspection of Loadout, Sea-fastening, and Transportation

Inspection should be performed for all areas related to loadout, sea-fastening, and transportation, to confirm compliance with the specified requirements. Prior to loadout, a final visual inspection of the structure shall be conducted to ensure that all components are in place; all welds have been properly completed and inspected; all temporary transportation/installation aids are included and secure; all hydraulic and pneumatic lines have been properly installed, tested, flushed, and secured; all temporary fabrication aids and debris have been removed; and that all temporary welded attachments have been removed and attachment marks repaired, according to the specified requirements.

The support foundations, including the loadout pathway, the dock, the transport vessel, and the sea floor at the dockside should be inspected, in order to ensure compliance with specified requirements.

Other areas for inspection include the lifting/pulling/pushing components attached to the structure (which require NDT) and those between the structure and lifting equipment (i.e. lifting slings, shackles, spreader beams). For vendor supplied items, documentation is required in addition to the inspections. The capacity and condition of loadout equipment should be confirmed by inspection and documentation.

For skidded loadout, inspection should be performed to confirm that the skidway and/or launch surface is clean and properly lubricated (if required) prior to loadout. The winches, jacks, and pulling cables should be inspected for proper capacity and condition.

If ballast and deballast operations are required to compensate for tidal variations, inspection of the ballast system is required to confirm adequacy and condition of the equipment. The operation should be monitored to ensure compliance with the loadout procedure.

Inspection for sea-fastening of the structure and all deck cargo is required to confirm compliance with the specified requirements. This includes temporary tie-downs and bracing required for transport. Materials, fabrication, and weld inspection requirements shall be in accordance with Sections 19 and 20. Inspection for structure launch items should be conducted where possible prior to sea transport.

Seaworthiness of tugs, towing attachments, and the transport vessel should be confirmed. For preparation of self-floating structures for transport to location, visual inspection should be performed to confirm seaworthiness and that all towing/restraining lines are properly attached.

21.6 Installation Inspection

21.6.1 Structure Launch and Upending

Prior to launch, visual inspection should confirm that all tie-downs and temporary bracings are cut loose and tow lines and loose items are removed from the launch barge or safely secured. Inspection is required to confirm that the structure flooding system is undamaged, flooding valves are closed and that the launching arm system is in the proper mode of operation. For lifted structures, inspection should confirm removal of all restraints and proper attachment of lifting equipment, as well as the undamaged and properly configured operation mode of the flooding system. For self-floating structures, inspection should confirm removal of tow lines, as well as the undamaged and properly configured operation mode of the flooding system.

Inspection should be carried out after the structure is secured in place. If inspection is necessary before then (i.e. inspection for suspected damage to the flooding system), it should be limited to those items required to upend and secure the structure.

21.6.2 Piling and Conductor Installation

All pile and conductor welds performed during fabrication shall be inspected in accordance with the requirements of Section 20 prior to loadout, including lifting devices, lugs, and attachments. During installation, inspection should be conducted to ensure that the correct pile make-up is followed and that the welding of add-on sections (if applicable) is performed in accordance with the specified requirements.

Prior to each use, pile hammers should be inspected for proper hook-up and alignment for operation.

During pile installation, NDT should be performed on the following:

- a) on the welded connections at pile add-ons,
- b) between pile and deck support members, and
- c) between the pile and leg,

as well as elsewhere, to confirm compliance with the specified requirements. NDT inspection shall be performed in accordance with either Annex D or ISO 19902, as appropriate.

21.6.3 Topsides Installation

Prior to lifting, visual inspection should be performed to confirm that tie-downs and other items not considered in the lifting design are removed from the topsides. Proper rigging and connection of all lifting components should be confirmed.

Immediately after lifting, visual inspection should be performed on all scaffolding and other temporary support systems to confirm their adequacy for completion of weld out. Materials, fabrication, and welding requirements shall be in accordance with Sections 19 and 20. Visual inspection should be performed on the structure and deck mating points to confirm proper alignment and fit-up and to ensure that weld preparations are in accordance with the specified requirements. Following weld-out, inspection should be performed on the welded connection in accordance with Section 20 and/or other specified requirements.

These inspections should be performed for each component of a multiple lift, with inspection for alignment during each lift.

21.6.4 Underwater Inspection

In the event the installation requires underwater operations, the inspection should verify, either by direct communications with divers or through the use of a remote monitoring device, that the operation has been conducted in accordance with the specified requirements.

21.7 Documentation

21.7.1 General

During the structure's fabrication, construction, loadout, and installation, inspection-related data are generated. This information, along with any pertinent additional data should be recorded as the job progresses, and should be compiled in a timely manner and in a form suitable to be retained as a permanent record.

The inspection results and other documents listed in Table 21.7-1 should be prepared. All documentation referenced in 21.7 should be retained on file for the life of the structure.

Table 21.7-1—Documentation Requirements

Documentation Description	Exposure Level		
	L-1	L-2	L-3
Material tracing records	X	X	—
Engineering drawings	X	X	X
Shop drawings	X	X	—
Design calculations for construction activities	X	X	—
Dimensional control/survey records	X	—	—
Mill certificates	X	X	—
Welding procedure matrix	X	X	—
Welding procedure specification	X	X	X
Welding procedure (PQR) test results	X	X	X
Site welding instruction sheets (short WPS)	X	X	—
Welder qualifications	X	X	X
NDT results	X	X	—
NDT personnel qualifications	X	X	—
Inspection records	X	X	—
Pressure/leak and equipment operability test reports	X	X	—
Weight reports	X	—	—
Other inspection	X	—	—
Foundation verification report (pile driving records, etc.)	X	X	X
As-built drawings	X	X	—
Weld numbering system	X	X	—
X Minimum requirements for fabrication. — Not required.			

21.7.2 Calculations

The structural integrity of components and adequacy of equipment during all phases of construction should be addressed.

Calculations should include the following:

- a) address the structure, attachments, temporary works, cranes, and rigging;
- b) address structural strength and stability;
- c) address all loads/actions, stresses, deflections, equipment, and rigging;
- d) reference appropriate drawings and specifications.

Calculation documentation should provide the source of calculation methods.

21.7.3 Weight and Center of Gravity Reports

Weight and center of gravity reports should be based on design, shop, or as-built drawings as the job progresses.

NOTE Guidance on weight control is given in ISO 19901-5 [5].

21.7.4 Fabrication Inspection Documentation

Documentation related to fabrication activities should be provided in accordance with Table 21.7-1, as appropriate.

21.7.5 Installation Inspection Documentation

Documentation for materials, testing, and welding inspection performed during the installation phase should be recorded and retained. Pile blow count with depth and final pile penetration should be documented, and a continuous log of events, including climatic conditions (i.e. temperature, wind, barometric pressure, humidity), sea states, operational activities, etc. should be retained.

21.8 Drawings and Specifications

The drawings and specifications prepared through the course of the project should comprise the following:

- a) conceptual drawings (see A.21.8.2);
- b) bid drawings and specifications (see A.21.8.3);
- c) design drawings and specifications (see A.21.8.4);
- d) fabrication drawings and specifications (see A.21.8.5)—the contractor should prepare and furnish fabrication procedures and assembly drawings describing and showing the proposed methods and order of assembly on the structure; these procedures and drawings should include rigging components, rigging configuration, lifting crane capacity and location, temporary aids and attachments to the structure;
- e) shop drawings (see A.21.8.6)—shop drawings should include all shop details, including material types, cuts, copes, joint details, holes, bolts, and piece numbers in accordance with the contract drawings; all welds shall be identified according to Section 21;
- f) installation drawings and specifications (see A.21.8.7);
- g) as-built drawings and specifications (see A.21.8.8)—the contractor should prepare and furnish as-built versions of the design drawings that accurately reflect the structure as built; these will be clear and legible, permanently reproducible drawings with all changes highlighted by clouds around the details concerned; the material type and SMYS should be shown on the drawings for any changes; only those dimensions outside the fabrication tolerances should be marked on the as-built drawings.

22 Loadout, Transportation, and Installation

22.1 General

22.1.1 Planning

The installation of a platform consists of loading out and transporting its various components from the fabrication site to the installation location, positioning the structure on location, and assembling the various components into a stable structure in accordance with the design drawings and specifications.

An installation plan shall be prepared for each platform. This plan should include the methods and procedures developed for the loadout, the sea-fastening, the transportation of all components, the complete installation of the structure, with piles/conductors and topsides, as well as include the equipment used in executing the procedures. The information may be provided in the form of written descriptions, specifications, and/or drawings. Depending upon the complexity of the installation, more detailed instructions can be required for special items such as grouting, diving, welding, and inspection. Any restrictions or limitations to operations due to, for example, environmental conditions, barge stability, or structural strength (e.g. lifting capacity), should be stated. Any contingency plans shall be clearly documented in the installation plan.

The installation plan is normally subdivided into phases, for example, for the structure: loadout, transportation, transfer from the transport barge into the water, placement on the sea floor, pile installation, conductor installation and topsides installation.

Detailed guidance on marine operations and their requirements is provided in API 2MOP; references to Section 8 should also be taken as references to API 2MOP, where applicable.

22.1.2 Records and Documentation

To aid in compiling the records and documentation required by API 2MOP, during loadout, transportation, and installation, all daily reports, logs, test reports, pile driving records, etc. shall be prepared, compiled, and retained. These documents should also record any variation from intended installation procedures and all unusual environmental conditions that occurred during the installation. Any field modifications made shall be noted to record the as-built condition of the structure. At the completion of the work, all documentation shall be collected and retained by the owner (see Section 21), together with other documents related to the construction and inspection of the structure.

22.1.3 Loads/Actions and Required Resistance

The loads/actions associated with each phase of the installation shall be calculated in accordance with Section 8. Structural analysis and design shall be performed in accordance with the appropriate sections of this document to ensure that the structure has sufficient strength and stiffness to resist the internal forces from all relevant combinations of loads/actions.

The structural strength and deformation of any installation equipment (barges, tie-downs, etc.) should also be considered in as much as they can affect the structure.

22.1.4 Temporary Bracing and Rigging

The requirements of this document shall also apply to any installation aids, temporary struts, bracing, or rigging required during any phase of the installation. Any welding of installation aids, temporary struts, or bracing to the structure should be in accordance with Section 20.

22.2 Loadout and Transportation

22.2.1 General

The transfer of all parts of a platform from their fabrication sites to the installation location presents a complex task that requires detailed planning. Considerations vary with the type of platform to be transported, but the following aspects should be considered where appropriate.

A platform normally consists of a structure, piling, topsides, and other miscellaneous items. These parts are usually transported to site as deck cargo on barges or other vessels. In some cases, the structure is designed with sufficient buoyancy to be floated to site without the use of a barge; such buoyancy can include temporary buoyancy removed after installation of the structure.

Additional information on considerations for marine operations can be found in API 2MOP.

22.2.2 Loadout

Loadout shall be performed in accordance with the requirements of the installation plan. The plan should include allowable environmental conditions during loadout operations and design environmental conditions for any mooring system. All items of cargo should be positioned on the barge as shown on loadout plans.

If the structure is loaded out onto a free-floating barge, the barge's ballast system shall be capable of compensating the changes in draught, trim, heel, and tide. An adequate standby ballast system should be provided.

If the structure is loaded out onto a grounded barge, it should be demonstrated by analysis or by previous experience that the barge has sufficient structural strength to distribute the concentrated loads/actions on its deck to the supporting foundation. In addition, the seabed or pad should be smooth, level, and free of any obstructions that could damage the hull.

Loads/actions resulting from the loadout operation are addressed in 8.5.

22.2.3 Cargo and Launch Vessels

22.2.3.1 General

An adequate number of seaworthy cargo vessels shall be provided. The vessels selected shall be of proper size and structural strength to ensure that the stresses in the vessel, cargo, and sea-fastening during loadout and transportation are within acceptable limits. If the structure is to be launched from a barge without the use of a crane vessel, the launch barge shall be suitable for this operation. More detailed and additional cargo and launch barge requirements for 22.2.3 can be found in API 2MOP.

22.2.3.2 Vessel Strength and Stability

The structure and any other platform part to be transported on the same vessel shall be loaded out in such a manner that a balanced and stable condition is ensured at all times. The vessel trim is then adjusted, and ballasted to the appropriate draught for transportation to site. Such operations should be performed at the dockside or in a sheltered area before sea-fastenings are attached and before reaching open water.

Static and dynamic stresses in the vessel hull and framing due to loadout (e.g. from localized loads/actions), transportation, and launching should be in accordance with the requirements of the classification society responsible for classification of the barge.

The stability of the vessel together with its cargo shall be verified to comply with international regulations and with the flag state requirements.

22.2.3.3 Sea-fastenings

Adequate sea-fastenings shall be designed and installed for all cargo components to prevent shifting while in transit. The sea-fastenings should be described and detailed in the installation plan. They shall be attached to the structure, topsides, and other platform components only at locations approved by the structure designer. Additionally, they shall be attached to the barge at locations capable of distributing the loads/actions to the internal framing. The sea-fastenings should be designed to facilitate their easy removal onsite.

Sea-fastenings shall be designed for loads/actions calculated in accordance with 8.6. Where there is substantial operational experience for routine installations, for similar cargoes, in a particular area and at a similar time of year, then sea-fastening arrangements may be based on that previous experience.

In lieu of a specific assessment for the transportation being considered, the sea-fastenings may be designed for environmental conditions determined from the time required to transport the barge to shelter. The probability of exceeding the selected environmental conditions during such time should be in the range of 1 % to 5 %, taking into account the season of the year in which the tow will take place and the length and reliability of the weather forecast.

22.2.4 Towing Vessels

A sufficient number of seagoing tugs shall be provided, with ample power and size to operate safely for each particular route or ocean to be travelled. The size and power requirement of the towing vessel and the design of the towing arrangement should be calculated or determined from past experience.

The selection of towing vessels should consider the length of the tow route, the proximity of sheltered water, and the weather conditions and sea states expected for the season of the year. When more than one towing vessel is required, the total calculated required bollard pull should be increased to take into account the loss of efficiency due to a multiple tow.

A standby or alternative towing line should be provided and rigged for easy access, in the event of tow line failure.

More detailed and additional towing vessel requirements can be found in API 2MOP.

22.2.5 Loads/Actions on the Platform Components

Consideration should be given to the loads/actions applied to the various platform components as they are lifted on and off the vessel, or as they are rolled on, and launched from, the vessel. The loads/actions, inclusive of the loads/actions from transportation, shall be determined in accordance with Section 8. The resistance to such loads/actions shall be determined in accordance with Sections 13 to 16, as appropriate.

22.2.6 Buoyancy and Flooding Systems

22.2.6.1 General

Any platform part intended to float during transport and/or installation shall be checked to ensure adequate buoyancy for flotation, with sufficient reserve buoyancy for the planned operations. All components shall be designed to withstand the loads/actions associated with transport and/or installation, including hydrodynamic pressures during launching, hydrostatic collapse, or those associated with another installation method. Consideration should be given to accidental flooding of compartments and the implication of this for other platform parts.

The flooding system, the buoyancy compartments, and any necessary lifting connections on the structure should be designed to ensure that the structure can be safely uprighted and landed on the sea floor.

22.2.6.2 Flooding Controls

The location and accessibility of all controls for selective flooding and righting, as well as the protection of the controls from environmental and operational hazards, should be carefully considered.

22.2.6.3 Model Tests and Analysis

Model tests and detailed calculations should be considered for any platform part to be towed to site to determine towing and stability characteristics during the towing and upending procedures.

22.3 Transfer of Structure from Transport Vessel into Water

22.3.1 Lifting Operations

For all lifting operations, the strength of the structure and the suitability of lifting equipment, including acceptable margins on capacity and reach, should be determined. The loads/actions shall be derived in accordance with Section 8 and Section 18 of API 2MOP, First Edition, with members and joints checked to ensure that they are adequate for the lift conditions.

Lifting operations shall be performed in accordance with the agreed plans. The operations should not be performed under more severe environmental conditions than those considered in the design phase.

Prior to lifting, the lifted weight should be assessed to ensure that it is within the limits defined by the design and within the capacity of all lifting equipment. If no weighing is carried out, an adequate margin should be applied to cover mill tolerances and growths in piping, equipment weights, etc.

The derivation of design loads/actions for lifting points is addressed in Section 18 of API 2MOP, First Edition. Padeye plates should be oriented in such a direction that the possibility for out-of-plane loads/actions on the padeye plate and shackle are minimized.

The rigging should be designed to carry loads/actions derived in accordance with Section 18 of API 2MOP, First Edition, with the appropriate resistance requirements according to this document to allow the structure to be lifted off the barge and lowered into the water. The slings should be attached above the center of gravity of the structure being lifted to provide good control of the operation.

22.3.2 Launching

22.3.2.1 General

A structure shall not be launched from a vessel if the significant wave height exceeds approved Installation Manual values (see Section A.5 of API 2MOP, First Edition) or if it is expected to exceed approved Installation Manual values before sufficient on-bottom stability is achieved (see 8.7.6).

For those structures that are to be launched, the requirements of Section 17 of API 2MOP, First Edition, shall be met with a launching system considering the items listed in subsections 22.3.2.2 to 22.3.2.5.

22.3.2.2 Launch Vessel

Any launch vessel shall be equipped with launchways, rocker arms, a controlled ballast and dewatering system, and an appropriate power unit (hydraulic ram, winch, etc.), to assist the structure to slide down the ways.

22.3.2.3 Loads/Actions on the Structure

The structure to be launched shall be designed and fabricated to withstand the loads/actions of the launch. This can be done by either strengthening those members that can become overstressed by the launching operation or by designing a special frame into the structure, commonly referred to as a launch frame. A combination of the above two methods may also be used.

22.3.2.4 Flotation

A structure that is to be launched shall be watertight and buoyant. If its upending is to be assisted by a crane vessel, the launched structure should float in a position such that lifting slings from the crane can be safely attached. The attachment points of any pre-installed slings should be exposed and accessible.

22.3.2.5 Crane Vessel Requirements

The crane vessel shall be of sufficient size to:

- a) change the position of the launched structure from its floating position to its upright position, and setdown for placement on seabed, or
- b) hold the launched structure at the site until it can be righted by a controlled flooding system.

22.4 Placement on Seafloor and Assembly of the Structure

22.4.1 General

Placement on the sea floor and assembly of the structure should be in accordance with the installation plan.

22.4.2 Safety of Navigation

Safety and navigational aids should be positioned and maintained as required throughout the installation phase, particularly when the structure is to be left for some time before the installation of the topsides.

22.4.3 Offshore Mooring

During marine operations, the crane and cargo vessels shall be held in position using either dynamic positioning systems or mooring (see API 2MOP).

Where mooring is connected to anchors, the following applies, in accordance with ISO 19901-7 ^[7].

- a) The length of anchor lines should be adequate for the water depth at the site.
- b) Anchor sizes and shapes should be selected such that the anchors will bite and hold in the seabed at the site. The holding capacity should be sufficient to resist the strongest tides, currents and winds that can reasonably be expected to occur at the site.

22.4.4 Positioning of the Structure

22.4.4.1 General

The term “positioning” includes the placement of the structure on the sea floor at the installation location, in a position and with an orientation within the allowable tolerances, and leveling the structure ready for the installation of the foundation piles, in accordance with the installation plan and the requirements of Section 17 of API 2MOP,

First Edition. This often requires upending the platform parts that have been towed to the site or launched from a barge at the site.

If the structure is to be installed over one or more existing wells, the wellheads should be protected from damage through accidental contact with the structure. Advance planning and preparation should be in such detail as to minimize hazards to both the wells and the structure. If the structure is not to be installed over an existing well, and not to be located adjacent to an existing structure, parameters for the accuracy of the positioning should be stated in the installation plan.

When the design of the platform is based on the directional variation of environmental forces, proper orientation of the structure is essential. The required orientation of the platform, as well as the acceptable tolerance for out-of-alignment as discussed in 9.4.2 should be shown on the drawings and stated in the installation plan. Procedures should be included in the installation plan to ensure that the structure can be positioned within the acceptable orientation tolerances.

22.4.4.2 Upending

The upending process is generally accomplished by a combination of crane barge assistance and controlled or selective flooding systems. The upending phase requires advanced planning to predetermine the simultaneous lifting and controlled flooding steps necessary to set the structure on the sea floor. Closure devices, lifting connections, etc. should be provided where necessary. The flooding system shall be designed to withstand the water pressures that will be encountered during the positioning process.

22.4.4.3 Leveling of the Structure

The structure should be leveled within the tolerances specified in the installation plan. Leveling the structure before any of the piles are installed is preferred, but often not practical. It can therefore be necessary to level the structure by jacking or lifting after a minimum number of piles have been driven. Leveling the structure after all the piles have been installed should be avoided if possible. Procedures should minimize bending stresses in the piles. After the structure has been leveled, care should be exercised to maintain its levelness during further pile installation operations.

22.4.4.4 Structure On-bottom Weight

In the event of rough seas, or if the installation equipment leaves the site for other reasons before the structure has been adequately secured with piles, it can be necessary to adjust the effective weight on bottom to minimize the possibility of movement of the structure due to skidding, overturning, or soil failure. See 8.7.6.3 for additional information regarding on-bottom analysis.

22.5 Pile Installation

22.5.1 General

Proper installation of piling, including conductor piles, is vital to the life and permanence of the structure and requires each pile to be driven to or near design penetration, without damage. All field-made structural connections shall be compatible with the design requirements. Pile sections should be marked in a manner to facilitate installing the pile sections in the proper sequence. The closure device on the lower end of the structure's legs and pile sleeves, if required, should be designed to avoid interference with the installation of the piles.

The pile wall thickness shall be adequate to resist axial and lateral loads/actions as well as the stresses during pile driving (see 17.10). The stresses resulting from pile driving can be predicted approximately from computer analyses based on the principles of one-dimensional elastic stress-wave theories, run as part of driveability studies (see 22.5.5).

22.5.2 Stabbing Guides

Add-on pile sections should be provided with guides to facilitate stabbing and alignment. The guide should provide a tight and uniform fit for proper alignment. The guides should be capable of safely supporting the full weight of the add-on pile section prior to welding.

22.5.3 Lifting Methods

If lifting padeyes are used for the handling of the pile sections, the padeyes should be designed (with due regard for the influence of impact) for the stresses developed during the initial pick-up of the section as well as those occurring during the stabbing of the section. If lifting padeyes or weld-on lugs are used to support the initial pile sections from the top of the structure, the entire hanging weight should be considered to be supported by a single padeye or lug. The lifting padeyes or support lugs should be removed by torch cutting 6 mm (0.25 in.) from the pile surface and ground smooth. Care should be exercised to ensure that any remaining protrusion does not prevent driving of the pile or cause damage to elements such as packers. If burned holes are used in lieu of lifting padeyes, they should comply with the applicable requirements of this subsection, and consideration should be given to possible detrimental effects during hard driving.

As an alternative to providing lifting padeyes on the piles, pile handling tools may be used, provided they are of the proper size and capacity for the piles being driven and the operating conditions anticipated. The tools should be inspected prior to each use, to ensure that they are in proper working condition. They should be used in strict accordance with the manufacturer's instructions and/or recommendations. For installations that require the use of pile followers, the followers should be inspected prior to their first use and periodically during the installation, depending on the severity of pile driving.

22.5.4 Field Welds

The add-on pile sections should be carefully aligned and the bevel inspected to ensure that a full penetration weld can be obtained before welding is initiated. If necessary, the bevel should be opened up by grinding or gouging. Welding should be performed in accordance with Section 20. NDT of the field welds, using one or more of the methods referenced in Section 20, should be performed.

22.5.5 Driveability Studies

Computer analyses (commonly known as wave equation analyses) can be used to simulate the hammer-pile-soil system and pile driving behavior, with the objective of defining the range of blow counts necessary to reach the target design pile penetration and assessing the stresses in the pile resulting from pile driving. The predicted range of blow counts to reach a given penetration is governed by the assumed profile of soil resistance to driving (SRD), but will also be affected by the assumed hammer efficiency, or energy transferred to the pile and, to a lesser extent, by the so-called quake and damping parameters in the wave equation model. Therefore, selection of these input parameters should be based on previous pile driving experience and engineering judgment. The energy transferred to the pile is highly dependent on the type of hammer (i.e. diesel, steam, or hydraulic hammer) for a given rated energy and should be based on pile driving experience with reliable measurements from pile instrumentation.

The definition of SRD is the main factor that governs the results of the driveability studies, i.e. the hammer type required to reach the target pile penetration. Several methods for calculating the SRD in different types of soil have been proposed in the literature (see A.22.5.5).

General procedures cannot be applied at all sites, as piling behavior is highly site-dependent. Therefore, back-analysis of previous pile driving experience at the site, or at sites with similar soil conditions, is recommended to calibrate SRD calculation procedures and improve driveability predictions for other structures at the site. For cohesive soils, the SRD calculation should take into account the increase in resistance due to pore pressure dissipation (set-up) during delays, in particular when delays are necessary for welding add-on elements.

To confirm that the hammer performs in accordance with the specifications and with the assumptions made in the driveability predictions, the pile/hammer should be instrumented and monitored during driving. Pile instrumentation is preferable, as hammer monitoring does not provide sufficient information about the driving energy actually transferred into the pile.

Pile driving instrumentation data, based on measurements from strain and acceleration transducers fixed near the top of the pile, may be used for verifying pile hammer efficiency, soil stratification, assessing the actual SRD during driving, as well as giving additional information for estimating the ultimate static pile capacity—particularly, if re-strike test data are available. The actual SRD during driving, as back-calculated from pile instrumentation data, should be compared with the predicted range in soil resistance. Such analyses can be used to improve the reliability of subsequent driveability predictions at the site.

References [40] to [57] provide relevant information on driveability analyses and the parameters used in these analyses.

22.5.6 Obtaining Required Pile Penetration

The adequacy of the structure's foundation depends upon each pile being driven to or near its design penetration. Where applicable, the driving of each pile should be carried to completion with as little interruption as possible to minimize the increased driving resistance that often develops during delays. It is often necessary to work one pile at a time during the driving of the final one or two sections, so as to minimize set-up time. Workable back-up hammers with leads should be available, especially when critical pile set-up is anticipated.

The fact that a pile has met premature refusal does not ensure that it is capable of supporting the design loads/actions. Final blow count alone cannot be considered as assurance of piling adequacy. Continued driving beyond the defined refusal (see 22.5.7) can be justified if it offers a reasonable chance of significantly improving the capacity of the foundation without risk of damaging the pile (including pile fatigue), hammer, or structure.

In some instances, when continued driving is not successful, the penetration and associated capacity of a pile can be improved by the methods described in 22.5.8. Such methods should be approved by the design engineer.

22.5.7 Driven Pile Refusal

Pile refusal requires defining, primarily to:

- a) establish the point at which pile driving with a particular hammer should be stopped and other methods instituted (see 22.5.8), and
- b) prevent damage to the pile and hammer.

The definition of refusal should be consistent with the individual soil characteristics anticipated at the specific location. Refusal should be defined for all hammer sizes to be used and is contingent upon the hammer being operated at the energy and rate recommended by the manufacturer.

In establishing the pile driving refusal criteria, the recommendations of the pile hammer manufacturer should be considered. The exact definition of pile refusal for a particular installation should be defined in the installation specification. Examples of refusal criteria (for use only in the event that no other requirements are included in the installation specification) are given in A.22.5.7.

22.5.8 Pile Refusal Remedial Measures

If a pile refuses before it reaches design penetration, one or more of the following measures can be taken.

- a) Review of hammer performance: A review of all aspects of hammer performance, possibly with the aid of hammer and/or pile head instrumentation, can identify problems that can be solved by improved hammer operation and maintenance, or by the use of a more powerful hammer.
- b) Re-evaluation of design penetration: Reconsideration of loads/actions, displacements and required capacities of individual piles, of other foundation elements and of the foundation as a whole, can identify available reserve capacity. An interpretation of driving records in conjunction with instrumentation [see a) above] can allow design soil parameters or stratification to be revised and the calculated pile capacity to be increased.
- c) Modifications to piling procedures: Modifying procedures, usually the last course of taken, can permit the piles to be driven to the required penetration after all. The following modifications can be considered.
 - 1) Plug removal: The soil plug inside the pile can be removed by jetting and air lifting, or by drilling, to reduce pile driving resistance. Several soil plug removals and redrives can be required to reach target penetration.

If plug removal results in inadequate pile capacity, the removed soil plug should be replaced by a grout or concrete plug. The minimum axial capacity of the plug should be equal to the pile end bearing capacity in a plugged condition. Attention should be paid to the characteristics of shear transfer between plug and pile. In some circumstances plug removal is not effective, particularly in cohesive soils.

- 2) Soil removal below the pile tip: Soil below the pile tip can be removed, either by drilling an undersized hole or by jetting and possibly air lifting. The drilling or jetting equipment is lowered through the pile, which acts as the casing pipe for the operation.

Considering the resulting uncertainties with respect to the pile axial capacity, the soil below the pile tip should not be removed to reduce the soil resistance during driving in uncemented soils.

Under special circumstances, e.g. in the case of an intermediate layer of strong cemented material, undersized drilling can be applied to partially remove the hard layer before pile driving can be resumed. The depth of drilling should be restricted to the thickness of the hard cemented layer.

Undersized drilling should be restricted to relatively thin and not too hard layers. In thick and very hard rock layers preferably under-reaming of the hole to at least the full pile size should be applied to avoid potential risk of pile tip buckling.

Where soil removal below the pile tip has been performed by drilling (undersized or otherwise), the contribution of the relevant zone of soil to the pile capacity should be ignored, unless this zone has been grouted.

Jetting below the pile tip should, in general, be avoided because of the unpredictability of the results.

- 3) Two-stage driven piles: A first-stage or outer pile can be driven to a predetermined depth, after which the soil plug is removed, and a second-stage or inner pile is driven inside the first-stage pile. The annulus between the two piles is grouted to permit shear transfer between the first- and second-stage piles and to develop composite load/action.
 - 4) Drilled and grouted piles and belled piles: See 22.5.10 and 22.5.11, and also 17.2.3 and 17.2.4.

To minimize delays in installation, a pile acceptance procedure should be established. The procedure should outline the measures to be taken on location for adjusting planned pile driving scenarios, in case of, for example, premature pile driving refusal or a significantly lower blow count than anticipated at design target pile penetration.

22.5.9 Selection of Pile Hammer

If piles are to be installed by driving, the influence of the hammers to be used should be evaluated as part of the design process in accordance with 17.10. The type(s) of pile hammer considered for use during driving should be noted by the designer on the installation drawings or specifications.

Any change in the hammers to be used for pile driving shall be assessed to ensure that the consequences of the change are acceptable (including pile drivability, pile capacity, pile and structure strength, and fatigue).

22.5.10 Drilled and Grouted Piles

The hole for drilled and grouted piles may be drilled with or without drilling mud to facilitate maintaining an open hole. Drilling mud can be detrimental to the surface of some soils. If used, consideration should be given to flushing the mud with circulating water upon completion of drilling, provided the hole will remain open.

Reverse circulation should normally be used to maintain sufficient flow for removal of cuttings. Drilling operations should be done carefully, in order to maintain proper hole alignment and minimize the possibility of hole collapse. An insert pile with an upset drill bit on its tip may be used as the drill string, so that it can be left in place after completion of the hole.

Centralizers should be attached to the pile to provide a uniform annulus between the insert pile and the hole. A grouting shoe may be installed near the bottom of the pile to permit grouting of the annulus without grouting inside the pile. If a grouting shoe is used, it can be necessary to tie the pile down to prevent flotation in the grout. The time before grouting the hole should be minimized in soils that may be affected by exposure to seawater. The quality of the grout should be tested at intervals during the grouting of each pile. Means should be provided for determining that the annulus is filled. Holes for closely positioned piles should not be open at the same time unless there is assurance that this will not be detrimental to pile capacity and that grout will not migrate during placement to an adjacent hole.

22.5.11 Belled Piles

Drilling of the bell is carried out through the pile by under-reaming with an expander tool. A pilot hole may be drilled below the bell to act as a sump for unrecoverable cuttings. The bell and the pile are filled with concrete to a height sufficient to develop the necessary transfer of forces between the bell and the pile. Bells are connected to the pile to transfer both full uplift and compressive forces using steel reinforcing such as structural members with adequate shear lugs, deformed reinforcement bars, or prestressed tendons. In general, drilling of bells for belled piles should employ only reverse circulation methods. Drilling mud should be used where necessary to prevent caving and sloughing. The expander or under-reaming tool used should have a positive indicating device to verify that the tool has opened to the full width required. The shape of the bottom surface of the bell should be concave upward (sides higher than the center) to facilitate later filling of the bell with tremie concrete.

To aid in concrete placement, longitudinal bars and spiral steel reinforcements should be well spaced. Reinforcing steel may be bundled or grouped to provide larger openings for the flow of concrete. Special care should be taken to prevent undue congestion at the throat between the pile and the bell, where such congestion can trap laitance. Reinforcing steel cages or structural members should extend far enough into the pile for an adequate transfer of forces to be developed.

Concrete should be placed as for tremie concrete, with the concrete being ejected from the lower end of a pipe at the bottom of the bell, always discharging into fresh concrete. Concrete with aggregates of 10 mm ($3/8$ in.) and less in size may be placed by direct pumping. Because of the long drop down along the pile and the possibility of a vacuum forming with subsequent clogging, an air vent should be provided in the pipe near the top of the pile. To start placement of concrete, the pipe should have a steel plate closure with soft rubber gaskets to exclude water from the pipe. Care should be taken to prevent unbalanced fluid heads and a sudden discharge of concrete. The pile should be filled to a height above the design concrete level equal to 5 % of the total volume of concrete, placed so as to displace all laitance above the design level. Suitable means should be provided to indicate the level of the concrete in the pile.

Concrete placement in the bell and adjoining section of the pile should be as continuous as possible.

22.5.12 Grouting Pile-to-sleeve Connections and Grouted Repairs

The grouting operation should be performed using an approved and proven procedure by experienced grouting operatives. The design of pre-installed grout line attachment details should consider the effects of pile installation and driving from impact and vibration.

The equipment should have sufficient capacity to achieve the grout filling in a single continuous operation. Grouting should not commence unless there are sufficient materials available, including a contingency, to complete the task. Grout slurry stored in a holding tank should be continuously stirred and should not be held for more than 30 min prior to pumping. In case of rapid hardening mixes, the storage duration in a holding tank should be reduced to less than 30 min.

Prior to grouting and after activating any sealing devices, dyed water should be flushed through the complete grouting system to both remove any deleterious matter and to prove its functionality. A pressure test can be appropriate for closed systems. The annulus should then be carefully filled by maintaining a continuous grout flow through the lowest practical point.

Grout returns to allow surface sampling are preferable. If these are provided, grout samples for strength compliance testing may be taken from the returns in addition to the slurry specific gravity measurements. If surface returns are not provided, visual inspection to confirm that grout has completely filled the annulus should be performed immediately after cessation of grout pumping and again after initial grout set, typically 12 hr.

22.5.13 Pile Installation Records

Throughout the driving of main or skirt piles, comprehensive driving and associated data should be recorded and reviewed for conformance with the installation plan. If significant deviations are observed, it can be necessary to take appropriate measures. The recorded data can include, as appropriate, the following:

- a) structure and pile identification, water depth, and reference elevation of readings;
- b) relevant information on pile stabbing;
- c) penetration of the pile under its own weight or under the weight of a new add-on;
- d) penetration of the pile under the weight of the hammer;
- e) where applicable, data on followers used;
- f) blow counts throughout driving, with hammer identification and energy observations;
- g) at relevant penetrations, the cumulative number of blows;
- h) the hammer blow rate (blows/minute) after every few meters of penetration;
- i) if available, pile instrumentation data;
- j) unusual behavior of the hammer or the pile during driving;
- k) date and time of starts and stops in driving, including set-up time;
- l) elapsed time for driving each section;
- m) elevations of soil plug and internal water surface after driving;
- n) actual length of each pile section and cut-offs;
- o) pertinent data of a similar nature covering driving, drilling, grouting, or concreting of grouted or belled piles.

22.5.14 Use of Hydraulic Hammers

Hydraulic hammers tend to be more efficient than steam hammers and the energy transferred to the pile for a given rated energy tends to be greater. They can be used both above and below water for driving battered or vertical piles, through legs or through sleeves and guides, as well as vertical piles through sleeves alone. In calculating pile stresses, full account should be taken of wave, current, and wind loads/actions, both during driving and during hammer stabbing (which can be either above or below water). While for steam hammers the weight of the cage is generally held by a crane, for hydraulic hammers the whole weight of the hammer is borne by the pile.

The energy output is generally varied by the installation contractor to maintain a fairly low blow count. Therefore, blow counts do not give a direct guide to soil stratification and resistance. Since the ram is encased, hammer performance cannot be judged visually. It is therefore important that measurements be made to give a record of the hammer's performance, including ram impact velocity, stroke, pressure of accelerating medium, and blow rate. Reliable instrumentation of some piles should also be considered, to verify the energy transferred to the pile to aid interpretation of soil stratification and to limit pile stresses.

Monitoring of underwater driving requires that easily identified, unambiguous datum points be used, together with robust television cameras or remotely operated vehicles capable of maintaining station. Alternatively, for shallow water sites, it is possible to extend the hammer casing so that blow counts can be monitored above water.

Because no cushion block is used, there is no change in characteristics between ram and anvil as driving progresses and no requirement for cushion changes. However, because of the steel-to-steel contact, particular attention should be paid to the design of the pile head.

In selecting hydraulic hammers for deeper water applications, account should be taken of possible decreases in efficiency due to increased friction between the ram and its surrounding air. Sufficient air should be supplied to the hammer so that water ingress is prevented. Water in the pile should be able to escape freely. It should be noted that hydraulic hammer changes take much longer than steam hammer changes.

22.6 Installation of Conductors

The planning and execution of conductor installation and shallow well drilling should recognize the potential for disturbance to foundation soils and the consequent risk of a reduction in stability of the fixed structure or of adjacent conductors.

During drilling operations, soil disturbances can result from hydraulic fracture, from wash-out, or from encountering shallow gas pockets. Hydraulic fracture occurs where drilling fluid pressure is too high and fluid is lost into the formation, possibly softening the surrounding soil. Wash-out (uncontrolled enlargement of the drilled hole) generally occurs in granular soils and can, in part, be induced by high drilling fluid circulation rates. Wash-out leads to stress relief in the surrounding soils. These incidents can be accompanied by loss of circulation of drilling fluids, by return of these fluids to the seabed other than through the conductor, or by the creation of seabed craters.

If piles are installed within the zone of influence of soil disturbance, reduction in axial or lateral capacity and foundation stiffness can occur. Similarly, the stability of shallow foundations can be reduced and settlements increased.

It should be noted that these detrimental effects can occur whether the drilling takes place either after installation of the structure or before, e.g. for a pre-installed template or for an exploration well.

The proximity of conductor slots to as-installed or future pile locations is critical, and risks are clearly greater for narrow structures with vertical piles.

The following recommendations should be considered for conductor installation and shallow well drilling.

- a) The conductor setting depth should be selected taking due account of hydraulic fracture pressure profiles. The depth should preferably be chosen at a cohesive stratum that is a sufficient distance from the proposed pile tip penetration to minimize the risk of disturbance of foundation soils.
- b) Particular care should be exercised in the installation of conductors that are installed by drilling or drill-drive techniques instead of by driving alone.
- c) In conductor or shallow well drilling operations, fluid pressures should be kept within the calculated hydraulic fracture pressure profile. Flow rates should be controlled to minimize wash-out, particularly in granular soils.
- d) Records of conductor installation and shallow well drilling should be available to the structure's design engineer. The implications for foundation soils of any incidents of excessive loss of circulation, of return of drilling fluids to the seabed other than through the conductor, or of creation of seabed craters should be assessed.

22.7 Topsides Installation

22.7.1 General

The topsides installation will normally consist of lifting items such as deck sections, module support frames, modules, and packages from the transport barges onto the structure, connecting them to the structure and to each other as specified by the design.

22.7.2 Alignment and Tolerances

After the piling has been driven and cut off to grade, the topsides should be set, with proper care being exercised to ensure proper alignment and elevation. The topsides parts shall be aligned within the tolerance specified in the design documents. Unless otherwise specified, the deck elevation shall not vary more than ± 75 mm (3 in.) from the design elevation shown in the drawing. The finished elevation of the deck shall be within 13 mm (0.5 in.) of level.

22.7.3 Securing Topsides

Once the topsides parts have been set, they should be secured to provide the support and fixity as required by the design.

22.7.4 Appurtenances

Once the topsides are installed, all stairways, handrails, and other similar appurtenances should be installed as specified.

22.8 Grounding of Installation Welding Equipment

22.8.1 General

Normal welding procedures use reverse polarity, wherein the welding rod is positive and the ground is negative. An adequate and properly placed ground wire is necessary to prevent stray currents, which, if uncontrolled, can cause severe corrosion damage.

22.8.2 Welding Equipment

The welding machine should be located on, and grounded to, the structure whenever possible. When this is impossible or impractical, and the welding machine is located on the barge (or vessel), both leads from the output of the welding machine should be run to the structure and the ground lead secured to the structure as close as practical to the area of welding. Under no conditions should the hull of the barge (or vessel) be used as a current path. The case or frame of the welding machine should be grounded to the hull to eliminate shock hazards to personnel.

The welding cables should be completely insulated to prevent stray currents. Damaged cables should not be allowed to hang in the water.

Grounding cable lugs should be tightly secured to grounding plates. The contact should be thoroughly cleaned to bare metal. The resistance of the connection should be a maximum of $125\ \mu\Omega$ per connection or the voltage drop across the connection should be a maximum of 62.5 mV for a current of 500 A.

The minimum cross-sectional area of the return ground cable should be $645\ \text{mm}^2/1000\ \text{A}$ per 30 m (100 ft) of cable. One or more cables connected in parallel may be used to meet minimum cross-section requirements.

More than one ground cable of sufficient size should be used to guard against a single return or ground becoming loose.

Connecting several welding machines to a common ground cable that is connected to the structure being welded will control stray currents if adequately sized and properly insulated from the barge (or vessel) containing welding machines.

22.8.3 Monitoring Remote Ground Efficiency

If welding is conducted using generators remote from a structure, grounding efficiency can be monitored by simultaneously measuring the potential of the structure and the barge (or vessel) housing the welding generators. A change in potential reading from either indicates insufficient grounding.

23 Structure Reuse

23.1 General

In general, structures are designed for onshore fabrication, loadout, transportation, and offshore installation, as well as for the in-place situations. By reversing the construction sequence, structures can be removed, back-loaded, transported, modified (if required), and then reinstalled at new sites.

Any or all parts of a platform may be reused, depending on their suitability and condition, and it is not necessary to reuse a structure with the same topsides or vice versa. Components may be renewed or replaced as appropriate; in particular, the majority of piles are likely to require major refurbishment or replacement.

Structures that are reused shall comply with this section as well as with the other design sections of this document. This section contains additional requirements for structures to be reused with respect to fatigue, materials, inspection, removal, and reinstallation.

23.2 Fatigue Considerations for Reused Structures

Fatigue sensitive locations in reused structures, especially tubular joints, should be inspected in accordance with 23.4. The calculation of fatigue damage shall include appropriate allowances for fatigue damage accumulated during the prior in-service period(s) and all transportation phases, in addition to the design service life at the new

location. For structures from, or to be reused in, areas of the world where fatigue can be a governing design criterion, the sum of the previously accumulated fatigue damage, D_1 , and the future fatigue damage, D_2 , for the intended reuse period should not exceed a value of 1.0, including appropriate fatigue damage design factors for both periods (see 16.12).

Remedial measures such as grinding welds, grouting, and reinforcing can improve fatigue performance and can be beneficial when a structure is to be reused.

23.3 Steel in Reused Structures

The type and grade of steel used in primary structure components of reused structures should be determined from the original records. Reused structures with tubular connections in which heavy-wall tubular joints have been fabricated from toughness class CV1 (or better) steel should be inspected in accordance with the requirements of 23.4, including ultrasonic testing (UT) inspection to detect the occurrence of unacceptable defects. Where Section 19 requires the use of toughness class CV2Z or better steel, steel inspection (including UT inspection) should be used to confirm the absence of unacceptable defects.

The chemical composition and mechanical properties of all materials should be verified for consistency with the assumptions made for the design analysis at the new location. Particular attention shall be paid to confirming the identification and properties of any steel of group II or higher.

Mill certificates or other documentation from the original fabrication with adequate material traceability may be used. If material certificates are unavailable, or if there is any doubt about the correlation of certificates with the locations of steel within the structure, specimens should be taken from the structure and tested by a certified laboratory to confirm both chemical and mechanical properties.

23.4 Inspection of Structures to Be Reused

23.4.1 General

When structures are considered for reuse, adequate inspection and testing shall be undertaken to verify suitability for the intended application. Care shall be taken in evaluating the inspection results if the structure is inspected prior to its removal from the original location; in such cases the limitations of in situ inspection shall be taken into account, both in its ability to provide a thorough inspection and the possibility of further damage during removal and transportation.

Inspection programs prepared for evaluation of existing structures being considered for reuse should be sufficiently detailed to establish the condition of the structures as described in API 2SIM. Additionally, inspection should be performed to verify the absence of any damage that can impair the structure's ability to withstand loads/actions imposed during all phases of removal operations from the prior location.

All design assumptions should be verified by inspection, including material composition and properties (see 23.3), connection integrity and extent of any corrosion or other degradation due to prior service.

Assessment of the condition of used structures should begin with a review of existing documentation from the original construction of the structure, together with results of any in-service surveys. Any evidence of damage or repairs, for which adequate investigation and assessment has not previously been undertaken, shall be reviewed and assessed in accordance with API 2SIM. Such damage can occur from environmental overload, ship collisions, and operational activities. Validation of any repair systems and their integrity should be conducted.

23.4.2 Initial Condition Assessment of Structural Members and Connections

The extent, quality, timing, and findings of NDT performed during the original fabrication and during periodic in-service inspections of the structure should be reviewed. Where adequate documentation exists, and the extent of inspection and weld qualities are consistent with current acceptance criteria, inspection may be limited to an investigation of in-service damage due to overload or fatigue.

Where adequate NDT documentation is not available, an initial spot survey of the structure should be made to provide guidance for the assessment and to assist in the formulation of a detailed inspection plan. The spot survey should include a general visual inspection of the whole structure, to detect any gross structural damage (parted connections, missing members, dented or buckled members, corrosion damage, etc.). Structural members and connections having in-service damage should be 100 % inspected, using appropriate NDT techniques.

23.4.3 Extent of Weld Inspection

The extent of weld inspection undertaken on a structure to be reused should be determined from assessments of the utilizations of members and joints in the structure in both prior service and future reuse. The inspection of welds at joints, particularly where they are fatigue-sensitive, should be directed to the higher stressed areas where fatigue cracks are more likely to initiate.

Welds should be sufficiently cleaned for effective NDT. NDT inspection methods and deployment of inspection equipment are described in API 2SIM. Guidance on the extent of inspection for various existing parts of the structure is given in A.23.4.3.

Weld inspection may take place either with the structure still at the previous location, or with the structure removed from the water, as appropriate, but the requirements for the extent of the inspection should be the same.

Inspection of all new components and connections shall be in accordance with Sections 20 and 21. Inspection of existing welds should generally comply with the requirements of API 2SIM.

23.4.4 Corrosion Protection Systems

The integrity of corrosion protection systems should be verified in accordance with Section 18 and API 2SIM. Verification should include assessment of remaining anode materials, anode connections, IC functionality and the condition of any protective coatings (e.g. splash zone coatings and wraps), and possible hidden damage under coatings.

23.4.5 Inspection for Removal of Structures from Prior Location

The inspection required for reuse may be undertaken before or after removal of the structure from the previous location (see 23.4.3). All inspection required for the safe removal of the structure should be determined during the planning stages (see 23.5) and executed prior to the commencement of the removal. As part of this process, the structure and equipment weights should be verified.

23.5 Removal and Reinstallation

Removal and reinstallation shall follow all relevant requirements of Section 22 and API 2MOP.

An offshore construction plan should be prepared for structure removal and reinstallation. This plan should include the methods and procedures for the safe execution of the entire removal and reuse operation, including the following specific activities and considerations:

- a) removal of the topsides, equipment, appurtenances, conductors, piling, and structure;
- b) loading of these items onto the transportation barges, including the crane barge if appropriate;
- c) sea-fastening and transportation of the topsides, equipment, appurtenances, conductors, piling, and structure to the new location or to and from an onshore location used for storage and/or refurbishment;
- d) any necessary modifications or refurbishment of topsides, equipment, appurtenances, conductors, piling, and structure;
- e) reinstallation of the reused or replaced structure, piling, appurtenances, topsides, equipment, and conductors.

Lift weights and centers of gravity should be defined in the procedures and checked during lifting.

Particular emphasis should be placed on the prevention of damage from the removal operations of any structure components intended for reuse.

23.6 In-service Inspection and Structural Integrity Management

A new inspection and SIM strategy shall be developed for the structure at the new location, taking full account of the structure's history, design service life, and the requirements of API 2SIM.

Annex A

(informative)

Commentary—Additional Information and Guidance

NOTE The sections in this Annex provide additional information and guidance on the sections in the body of this document. The same numbering system and heading titles have been used for ease in identifying the subsection in the body of this document to which it relates. Guidance is provided only on the identified clauses.

A.6 Overall Considerations

A.6.1 Service and Operational Considerations

A.6.1.1 General Considerations

If the structure is to be set over an existing well with the wellhead above water, information is needed on the dimensions of the tree, size of conductor pipe, and the elevations of the casing head flange and top of wellhead above mean low water. If the existing well is a temporary subsea completion, plans should be made for locating the well and setting the structure such that the well can later be extended above the surface of the water.

A.6.1.3 Structural Configuration

A.6.1.3.2 Deck Elevation

A safety margin or air gap is required between the crest of the design wave and the lowest point (beam, equipment, or fixing) of the lowest deck of the platform such that abnormal wave crests do not impinge on the deck. This is necessary, since very large loads/actions can occur if a wave hits the deck. If there is insufficient deck elevation, wave impact can determine the reliability of the structure. Where possible, deck height should be chosen so that the frequency of wave impact on the deck is compatible with the target failure rate of the structure. Structural reliability arguments indicate that air gaps selected in accordance with traditional procedures should be increased or reduced, depending on the region in which the platform is located. The air gap further ensures that green water does not interfere with platform operation or safety devices. For the Gulf of Mexico, the lowest point of the lowest deck is required to be above the 1000-year crest based on experience and reliability analysis in lieu of a 1.5 m clearance above the 100-year design wave.

Any determination of the air gap should account for uncertainty in water depth, structure settlement, sea floor subsidence, sea level rise, storm surge and tide, and abnormal wave crest elevation. The deck elevation can be set by either of the following methods:

- a) a rational process, using long-term surface elevation statistics and reliability considerations, going to various levels of complexity; or
- b) experience and judgment, if a rational approach is not possible.

Method a) above can be followed if the metocean database is sufficiently accurate and comprehensive. Account may be taken of the joint probability of tide, surge height, and crest heights to estimate the maximum surface elevation relative to the deck. In this case, a probability of nonexceedance close to the target failure rate of the structure may be used with no additional air gap allowance added.

When method a) above is not appropriate, the deck elevation, h , above the mean sea level can be estimated from Equation (A.6.1-1) or Equation (A.6.1-2).

If storm surge is not expected to occur at the same time as the abnormal wave crest, Equation (A.6.1-1) applies:

$$h = \sqrt{a^2 + s^2 + t^2} + f \quad (\text{A.6.1-1})$$

If storm surge is expected to occur at the same time as the abnormal wave crest, Equation (A.6.1-2) applies:

$$h = \sqrt{(a + s)^2 + t^2} + f \quad (\text{A.6.1-2})$$

where

a is the abnormal wave crest height;

s is the extreme storm surge;

t is the maximum elevation of the tide relative to the mean sea level;

f is the expected sum of subsidence, settlement and sea level rise over the design service life of the structure.

For deep and intermediate water depths a can be approximated to:

$$a > 1.3 a_{100} \quad (\text{A.6.1-3})$$

$$a > a_{100} + 1.5 \text{ m} \quad (\text{A.6.1-4})$$

where a_{100} is the extreme wave crest height with a return period of 100 years.

The estimate for h obtained with this procedure is indicative and suitable for conceptual design studies. The owner should review the deck elevation prior to detailed design.

In general, no platform processing elements, piping, or equipment should be located below the lower deck in the designated air gap. However, when it is unavoidable to position such items as minor subcellars, sumps, drains, or production piping in the air gap, provisions should be made for the loads/actions due to waves developed on these items.

NOTE 1.5 m is the traditional value used for air gap, but analysis of metocean data has shown that it does not always allow sufficient reliability in certain geographical areas.

A.6.3 Environmental Considerations

A.6.3.2 Selecting Design Metocean Parameters and Load/Action Factors

Use of this document should result in safe but not necessarily optimal structures. Platform owners may find justification for designing structures for conditions more or less severe than indicated by these guidelines. Design criteria depends upon the overall loading, strength, and exposure characteristics of the installed platform. These guidelines should not be taken as a condemnation of platforms designed by different practices. Historical experience, loading, and strength characteristics of these structures may be used for such evaluations. The provisions of this section are intended to accommodate such considerations. The actual platform experience and exposure and the amount of detailed oceanographic data available vary widely among the different U.S. waters. The Gulf of Mexico is characterized by a substantial amount of experience, exposure, and data. For other areas, there may be less experience and data.

Worldwide experience of drilling and production structures supports the use of 100-year return metocean parameters. The partial load/action and resistance factors recommended herein are intended for use in designing the structure in accordance with this practice. In the US Gulf of Mexico, it has been the practice to design low-consequence man-evacuated and unmanned structures for less than the 100-year full-population hurricane.

Where the recommended partial load/action factors are not used, the values used should be based on a risk and reliability analysis. This analysis should include the estimated cost of the structure designed to resist loads/actions from environmental conditions using various load/action factors, the probability of damage or loss when subjected to environmental conditions of various recurrence intervals, the financial loss due to damage or loss including lost production, cleanup, structure and well replacement, etc. The reliability approach used to develop the partial load/action and resistance factors is reasonable for such a study.

A.6.4 Exposure Levels

A.6.4.1 General

Guidance in the following subsections relates to life-safety and the consequences of platform loss. Life-safety concerns personnel belonging to the normal complement of personnel on the platform. "Consequences" is a broad notion, which includes factors such as the life-safety of personnel who are not part of the normal complement of the platform (e.g. rescue personnel or those who might be brought in to clean up and make safe a platform after an incident), damage to the environment, and anticipated economic losses (see 6.4.3). The emphasis is normally on the first and second of these factors. Owners may, however, choose to consider economic factors more specifically in setting exposure levels to give a higher level of reliability of a structure.

When considering economic losses, in addition to loss of the platform and associated equipment, and damage to connecting pipelines, the loss of reserves can be considered if the field is subsequently abandoned. Removal costs include the salvage of the collapsed structure, re-entering and plugging damaged wells, and cleanup of the sea floor at the site. If the site is not to be abandoned, restoration costs include replacing the structure and equipment and re-entering the wells. Other costs include repair, re-routing, or reconnecting pipelines to the new structure.

A.6.4.2 Life-safety Categories

When determining the length of time required for evacuation of a category S2 manned evacuated platform, consideration should be given to the:

- a) number of personnel to be evacuated;
- b) distances involved;
- c) capacity and operating limitations of the evacuating equipment;
- d) type and size of docking/landings, refuelling, egress facilities on the platform;
- e) environmental conditions anticipated to occur throughout the evacuation effort.

An occasionally manned platform (e.g. one manned for only short duration activities such as maintenance, construction, workover operations, drilling, and decommissioning) may be classified as S3, unmanned, provided that the activities are scheduled such that all exposure of personnel to any design environmental event is both minimal and as low as reasonably practicable.

A.6.4.3 Consequence Categories

The degree to which negative consequences could result from platform collapse is a judgment that should be based on the importance of the structure to the safety of any personnel associated with the platform on a normal basis or in relation to the failure, the potential damage to the environment, the owner's overall operation, and the level of economic losses that could be sustained as a result of the collapse.

When considering pollution and environmental damage and the cost of mitigation, particular attention should be given to the hydrocarbons contained in the topsides process inventory, possible leakage of damaged wells or pipelines, and the proximity of the platform to the shoreline or to environmentally sensitive areas such as coral reefs, estuaries, and wildlife refuges. The potential amount of liquid hydrocarbons or sour gas released from these sources should be considerably less than the available inventory from each source. The factors affecting the release from each source are discussed below.

- **Topsides inventory:** At the time of a platform collapse, liquid hydrocarbon in the vessels and piping is not likely to be suddenly released. Due to the continuing integrity of most of the vessels, piping, and valves, it is most likely that very little of the inventory will be released. Thus, it is judged that significant liquid hydrocarbon release is a concern only in those cases where the topsides inventory includes large capacity containment vessels.
- **Wells:** The liquid hydrocarbon or sour gas release from wells depends on several variables. The primary variable is the reliability of the SSSVs, which are fail-safe closed or otherwise activated when an abnormal flow situation is sensed. Where regulations require the use and maintenance of SSSVs, it can be judged that uncontrolled flow from wells is not a concern for the platform assessment. Where SSSVs are not used and the wells can freely flow (e.g. under well pressure), the flow from wells can be a significant concern.

Any liquid hydrocarbon or sour gas above the SSSV could be lost over time in a manner similar to a ruptured pipeline; however, the quantity will be small and does not necessarily have a significant impact.

- **Pipelines:** The potential for liquid hydrocarbon or sour gas release from pipelines or risers is a major concern because of the many possible causes of rupture (e.g. platform collapse, soil bottom movement, intolerable unsupported span lengths, and anchor snag). Only platform collapse is addressed in this document. Platform collapse is likely to rupture pipelines or risers near to or within the structure. For the design environmental event where the lines are not flowing, the maximum liquid hydrocarbon or sour gas release is likely to be substantially less than the inventory of the line. The amount of product released will depend on several variables such as the line size, the residual pressure in the line, the gas content of the liquid hydrocarbon, the undulations of the line along its route, and other secondary parameters.

Of significant concern are major oil transport lines that are long and large in diameter and have a large inventory. In-field lines, which are smaller and have less inventory, are not necessarily a concern.

A.7 General Design Requirements

A.7.4 Structural Modeling and Analysis

In general, partial load/action factors are applied to external loads/actions before calculating internal forces. If the structure and support conditions are linear, it does not matter whether the partial load/action factor is applied to the external load/action or to the internal force. However, when nonlinearity exists, applying the partial load/action factors to the external loads/actions ensures compatible deformations and equilibrium of internal forces at a load level closer to an overload condition.

A.7.6 Design for the In-place Situation

Except for fatigue, the design requirements for exposure level L-1 are the same for structures of all design service lives. For example, the return period and load/action factor for wind, wave, and current loads/actions are the same whether the structure has a design service life of 5 years or 30 years. This ensures that all structures have the same reliability in any single year. This is desirable for the safety of personnel on the structure and to control the exposure to potential consequences of failure.

A shorter design service life reduces the probability of encountering an extreme environmental event and so increases the lifetime probability of survival. The owner of exposure levels L-2 and L-3 structures may take advantage of this fact. However, any change in the return period and design load/action factors impacting the risk to life, injury, damage to the environment, or economic losses to parties other than the owner should be determined by a risk analysis. Changes impacting solely on economic losses to the owner are at the discretion of the owner.

A.7.7 Determination of Resistances

A.7.7.1 General

In many cases it is possible to derive the resistance of a complex component from a computer simulation, usually using finite element techniques. Guidance on the use of such techniques, including the selection of appropriate elements, is given in Section 12.

There is a significant possibility of errors being introduced in both physical tests and in computer simulations. Considerations for physical tests are given in A.7.7.2. When using computer simulations, potential errors in deriving a representative resistance should be taken into account by:

- a) calibrating the results of analyses to similar results from tests or design formulae;
- b) using computer simulations where the statistical variation is introduced in the derivation of the representative resistance value based on “known” values of the variance.

Where no test data are sufficiently similar, requirements are given in 7.7.5.

A.7.7.2 Physical Testing to Derive Resistances

Large scale testing is a valid method of determining the behavior and the resistance of structures and of components of structures; however, such tests can be expensive and can also introduce additional unknowns. The factors that should be considered in designing tests include the following:

- a) the expected mode of failure;
- b) the scale of the test pieces and consequent scaling factors required;
- c) the methods of application of the loads (e.g. by the use of hydraulic rams);
- d) the arrangement and stiffness of the support and reaction framing;
- e) the method of controlling the application of the load (e.g. by controlling the applied load or by controlling the applied displacements);
- f) whether the method of load application and control will affect the test conclusions;
- g) other loads/actions on the real structure that are not replicated in the test (e.g. hydrostatic pressure);
- h) the use of different materials to represent the materials intended for the real component;
- i) the methods of monitoring the behavior of the test piece including strain and displacement gauges;
- j) the determination of the resistance of the component from the test results, such as the load/displacement relationships.

The test arrangement should be designed to minimize the influence or take account of these considerations. The experience of the test contractors can be of great benefit in optimizing the test program and in understanding the results of testing.

Where the resistance is derived by physical testing, and where the variability of the behavior is expected to follow a normal distribution, Equation (A.7.7-1) may be used to derive the design resistance R_d :

$$R_d = \frac{\eta_d}{\gamma_R} \times R_{\text{mean}} (1 - k_n \times V_R) \quad (\text{A.7.7-1})$$

where

η_d is a correction factor;

γ_R is the partial resistance factor according to this document;

R_{mean} is the mean of the test results;

k_n is a statistical factor;

V_R is the coefficient of variation.

Factor η_d is intended to account for the differences between the test arrangements and the real component, such as scaling, differences in support and restraint between test piece and real conditions, differences in yield strengths and strain rates, etc. Consequently, the value of η_d is specific to the arrangement of both the real component and the test arrangement and specific guidance cannot be given in this document.

The partial factor approach adopted in this document takes account of the statistical variations in the loads/actions applied to a structure and in the resistances of the components of the structure. In deriving the representative resistances formulated in Sections 13 to 15, large numbers of tests have been undertaken and the mean strengths and coefficients of variation are well understood. Where few tests are available the natural coefficient of variation, V_R , is estimated from the sample using Equation (A.7.7-2):

$$V_R = \frac{1}{R_{\text{mean}}} \sqrt{\frac{1}{n-1} \sum (x_i - R_{\text{mean}})^2} \quad (\text{A.7.7-2})$$

where

R_{mean} is the mean of the test results;

x_i are individual test results;

n is the number of tests.

The factor, k_n , used in Equation (A.7.7-1) is a statistically derived factor for taking account of the low number of test results that could be available from the test program. Where no data from similar testing can be found for deriving an appropriate coefficient of variation, a minimum of three tests should be undertaken. Less conservative values of k_n may be used if comparisons with data from similar tests can be used. Factor k_n is given in Table A.7.7-1 for cases where V_R can be established from similar data (V_R known), and where similar data are not available (V_R unknown). Table A.7.7-1 assumes a normal distribution and is for a 95 % exceedance at the 50 % confidence level.

Table A.7.7-1—Statistical Factor, k_n [A.7.7-1]

n	1	2	3	4	5	6	8	10	20	30	∞
V_R known	2.31	2.01	1.89	1.83	1.80	1.77	1.74	1.72	1.68	1.67	1.64
V_R unknown	—	—	3.37	2.63	2.33	2.18	2.00	1.92	1.76	1.73	1.64

Where similar tests are available, the coefficient of variation may be taken from those tests and used in Equation (A.7.7-1). The degree of similarity of tests is a matter of judgment, but this approach should only be used if no additional behavioral effects, such as a change in mode of failure, are likely. Cases where appropriate data are available and where the coefficient of variation from those data is used in Equation (A.7.7-1) are classed as “ V_R known” in Table A.7.7-1.

It is not always necessary to conduct a series of tests on nominally similar specimens to establish k_n and V_R if the mode of failure is well defined. In such cases, it is sometimes possible to replicate only that part of the specimen that causes the failure. For example, if it is known that failure of a particular bolted connection is always associated with bolt shear, then simpler tests on bolts in shear may be used to establish V_R and k_n . The mean R_{mean} should, however, be established with reference to at least three complete specimens. Care needs to be exercised to ensure that all relevant parts of the complete component that can affect the failure (including stiffness) are included in any simpler tests.

A.7.7.3 Resistances Derived from Computer Simulations Validated by Physical Testing

A methodology for deriving resistances from computer simulations validated by physical testing is presented below. In some cases it will be possible to use data from existing tests rather than requiring new tests. This methodology essentially captures both any bias in the predictions and the natural variability of the structural behavior, and is particularly appropriate when there are a series of components with varying parameters, such that the test series can cover the parameter ranges. A series of tests should be simulated and the ratios of measured resistances (R_m) to predicted resistances (R_p) calculated. The predicted resistances may be based on the measured yield strengths of the test pieces. A representative ratio $(R_m/R_p)_{\text{rep}}$ is calculated using Equation (A.7.7-3):

$$(R_m/R_p)_{\text{rep}} = (R_m/R_p)_{\text{mean}} (1 - k_n V_{m/p}) \quad (\text{A.7.7-3})$$

where

$(R_m/R_p)_{\text{mean}}$ is the mean ratio of measured to predicted strengths;

k_n is a statistical factor based on the number of tests (see A.7.7.2);

$V_{m/p}$ is the coefficient of variation of the ratio R_m/R_p .

The design resistance of the in situ component, R_d , is given by Equation (A.7.7-4):

$$R_d = \frac{1}{\gamma_R} \times (R_m / R_p)_{\text{rep}} \times R_{p,s} \quad (\text{A.7.7-4})$$

where

γ_R is the partial resistance factor from this document;

$(R_m/R_p)_{\text{rep}}$ is the representative ratio from Equation (A.7.7-3);

$R_{p,s}$ is the predicted resistance of the in situ component from the simulation, based on its SMYS.

In order to use Equation (A.7.7-4), the analyst should be satisfied that the variability of the behavior of the in situ component is no greater than the variability of the test specimens; this requires sufficient tests to be undertaken to evaluate the variability to the appropriate level of confidence. Consideration should be given to the effects of any additional parameters such as welding stresses and geometrical imperfections.

The value of $R_{p,s}$ in Equation (A.7.7-4) is the predicted resistance of the component based on computer simulation and taking into account the accuracy of computer simulations of available test configurations. As the analysis introduces additional parameters (e.g. mesh size selection), the same techniques should be used for analyses of both the real component and the available test configurations.

A.7.7.4 Resistances Derived from Computer Simulations Validated Against Design Formulae

When validating computer simulations against design formulae, it should be noted that the formulae from Sections 13 to 15 give representative resistances and therefore take account of structural behavior variability. The design resistance, R_d , is given by Equation (A.7.7-5):

$$R_d = \frac{1}{\gamma_R} \times (R_f/R_p)_{\text{mean}} \times R_{p,s} \quad (\text{A.7.7-5})$$

where

γ_R is the partial resistance factor from this document;

$(R_f/R_p)_{\text{mean}}$ is the mean ratio of design formulae resistances (R_f) to simulation resistances (R_p) for the same values of parameters;

$R_{p,s}$ is the predicted strength of the real component from the simulation.

Both R_p and $R_{p,s}$ should be calculated using the SMYS.

The value of $R_{p,s}$ in Equation (A.7.7-5) is the predicted resistance of the component based on the computer simulation and taking into account the accuracy of computer simulations of arrangements covered by design formulations. As the analysis introduces additional parameters (e.g. mesh size selection) the same techniques should be used for analyses of both the real component and the simulations of the arrangements covered by design formulae. Depending on the accuracy of the simulations, it may be appropriate to adjust modeled material properties or to apply a factor to the results of the analysis in determining $R_{p,s}$.

In order to use Equation (A.7.7-5), the analyst should be satisfied that the arrangement of the component is sufficiently similar to that from which the design formula was derived and that both will have the same mode of failure. An example of an appropriate use of Equation (A.7.7-5) is to extrapolate just beyond the limits of validity of available design formulae.

A.7.7.5 Resistances Derived from Unvalidated Computer Simulations

It is preferable that computer simulations be validated against true structural performance; 7.7.3 directly achieves this, while 7.7.4 achieves validation through design formulae that in themselves are derived from test data. It is, however, recognized that there are occasions in which physical testing is unrealistic, for example, due to scale and complexity, and hence this guidance is given for deriving resistances from computer simulations for such cases. Where no physical testing or design formulae validation is available, R_d may be derived from Equation (A.7.7-6):

$$R_d = \frac{\eta_a}{\gamma_R} \times R_{p,rep} \quad (\text{A.7.7-6})$$

where

η_a is a modeling conversion factor;

γ_R is the partial resistance factor from this document;

$R_{p,rep}$ is the representative resistance derived from the analysis;

$$R_{p,rep} = R_{p,s}(1 - k_1 V_R) \quad (\text{A.7.7-7})$$

where

$R_{p,s}$ is the resistance derived from the simulation;

k_1 is a statistical factor equal to 2.31;

V_R is a coefficient of variation to reflect the variability of structural behavior.

Since the value of k_1 is quite onerous, multiple analyses may be performed to allow the use of k_n in place of k_1 , with $R_{p,s}$ becoming the mean resistance from the analyses. Where this is done, and to reduce some of the systematic errors being introduced, the multiple analyses should be undertaken using different analysis techniques and different analysts.

A similar approach to that in A.7.7.2 is used for the determination of the design value from computer simulations, with a modeling factor, η_a , provided to take account of differences between the simulation and the real conditions. Modeling factor η_a will approach a value of 1.0 as the real conditions are more closely simulated; it should be determined by the analyst and is not intended to be used as an indiscriminate reduction factor to enhance safety.

Each of the analysis techniques should be validated by comparison to available test data, and the value of $R_{p,s}$ used in Equation (A.7.7-7) should be the mean of the resistances from the various analyses. The appropriate value of V_R depends on the expected mode of failure. The analyst should consider any available data on testing and select the value of V_R accordingly; however, V_R should not be less than the values given in Table A.7.7-2.

Table A.7.7-2—Minimum Coefficients of Variation

Type of Failure Expected	Minimum Coefficient of Variation, V_R
Excessive yielding or gross deformation	0.05
Local buckling	0.11
Overall buckling	0.17

It is customary to use the SMYS as the yield strength when determining $R_{p,s}$. The use of SMYS together with the use of V_R from Table A.7.7-2 can, however, lead to over-conservatism. Where the analyst has sufficient understanding of the statistical variation of the material yield strength, $f_{y,n}$, the value of yield strength, f_y , used in the analysis may be modified. In the absence of data on the material yield strength, the analysis may be based on a yield strength from Equation (A.7.7-8):

$$f_{y,a} = \frac{f_{y,n}}{(1 - 1.64 \times 0.05)} = 1.09 f_{y,n} \quad (\text{A.7.7-8})$$

where

$f_{y,a}$ is the yield strength used in the analysis when determining $R_{p,s}$;

$f_{y,n}$ is the characteristic yield strength for a normal distribution that is not truncated by removing rejected samples with yield strengths below SMYS ($f_{y,n}$ is normally slightly larger than SMYS).

The values of 1.64 and 0.05 are valid for a large sample size and a typical coefficient of variation for structural steel yield strength.

It should be noted that steel for offshore structures is usually purchased by stipulating the SMYS, and that, although steel with yield strengths falling below the SMYS will be rejected, it cannot be assumed that the yield strength distribution of purchased steel is normal. For this reason, $f_{y,a}$ should not be taken as greater than the mean of the SMYS of the steel being used and the SMYS of the next grade of steel available.

The value of $k_1 = 2.31$ for use in Equation (A.7.7-7) is the value appropriate for a single physical test from Table A.7.7-1. As the solution of a particular computer simulation will always produce the same result, there is a requirement to use different analysis techniques and different analysts if the opportunity to use a lower value of k_n is desired. The variations between the analyses should include as many of the following as possible:

- a) different analysts, with little collaboration in establishing their simulations;
- b) different analysis programs;
- c) different formulations for the behavior of the finite elements;
- d) different mesh sizes and arrangements.

Where computer simulations are used to extend or explore the range of a parameter, which is difficult to control in physical tests (e.g. imperfections in columns), then it is possible to use the results to establish V_R if reasonable correspondence to the existing results is obtained with relevant simulations.

A.7.8 Strength and Stability Checks

These requirements refer to the partial factor design format. In general, different components (tubular members, joints, piles, etc.) have different partial resistance factors (also known as strength reduction factors or ϕ factors), whereas different loads/actions (self weight, variable loads/actions due to movable equipment, environmental loads/actions, etc.) have different partial load/action factors (also known as γ factors). The partial load/action factors can further vary depending on which other loads/actions are combined with them.

A.7.9 Robustness

The robustness concept is closely related to accidental loads/actions, consequences of human error, and failure of equipment. These situations are denoted “hazardous circumstances” or “hazards.” Robustness is also important in the event of serious but unidentified fatigue damage.

Robustness is achieved by considering ALS that represent the structural effects of hazards. Ideally all such hazards should be identified and quantified by means of rational analyses. However, in many cases it is possible,

based on experience and engineering judgment, to identify and reasonably quantify the most important ALS. They will often be those from ship impact, dropped objects, fires, and explosions.

ISO 19900 requires the following approach:

- a) careful planning of all phases of development and operation,
- b) avoidance of the structural effects of the hazards either by eliminating the source or by bypassing and overcoming them,
- c) minimizing the consequences, or
- d) designing for hazards.

When the hazard cannot reliably be avoided, the designer has a choice between minimizing the consequences (i.e. the consequences of losing a structural component due to a hazard), or designing for the hazard (i.e. making the component strong enough to resist the hazard). In the first case, the structure should be designed such that all structural components that can be exposed to hazards are noncritical. In the second case, critical components that can be exposed to hazards are made strong enough to resist the hazards considered.

It should be noted that robustness requirements do not imply that structures can survive removal of any structural component. If there is no hazard, then there is no requirement in relation to robustness. Also, only one hazard at the time should be considered.

A.7.10 Reserve Strength

A.7.10.1 New Structures

The global strength of structural systems able to safely withstand environmental overload situations is greater than that inferred from component-based design. This is due to different sources of reserve strength in the design such as:

- a) conservatism in code design equations;
- b) the difference between actual and representative material yield strengths (SMYS);
- c) additional strength required by load cases other than in-place environmental loads/actions;
- d) a designer's choice to oversize members in order to comply with standard sizes, reduce the number of different sizes used, facilitate fabrication, or speed up the design process;
- e) reserve strength offered by system behavior.

A nonlinear pushover analysis may be used to estimate system reserve strength in terms of the RSR and thus quantify the structure's ability to resist environmental overload.

The partial load/action factors for environmental loads/actions specified in Section 9 are derived from a database of space frame structures, which assumes a certain minimum degree of reserve strength due to various sources such as listed above.

The determination of system reserve strength is not required for design of new space frame structures. For structures with less inherent reserve system strength than that of a typical space frame type structure, an adequate safety level can be achieved by determining the structure's RSR and adjusting $\gamma_{f,E}$ so as to yield the required system reserve strength.

In lieu of more refined methods, this can be done following ISO 19900 to determine the partial load/action factor, $\gamma_{f,Em}$, for the extreme environmental load/action:

$$\gamma_{f,Em} = \frac{R_{RS,Typ}}{R_{RS,M}} \gamma_{f,E} \quad (A.7.10-1)$$

where

$R_{RS,Typ}$ is the minimum acceptable value of RSR for a conventionally framed structure for the relevant exposure level;

$R_{RS,M}$ is the RSR for the structural arrangement being considered, found either from a pushover analysis or from lower bound estimates from component resistances;

$\gamma_{f,E}$ is the partial load/action factor for extreme environmental loads/actions (see 9.9 and A.9.9).

$R_{RS,Typ}$ should be close to 1.85 for an L-1 structure. This value was calculated in References [A.7.10-1] and [A.7.10-2] from an analysis of North Sea type structures. As the value of $R_{RS,M}$ depends on the component sizes, which are themselves dependent on $\gamma_{f,Em}$, an iterative process can be required to determine both $\gamma_{f,Em}$ and $R_{RS,M}$. Lower values of RSR can be appropriate for L-2 and L-3 structures.

A.7.10.2 Existing Structures

Assessment of existing structures may be based on different levels of refinement. It is recommended that simple techniques be exhausted prior to the use of technically complex analyses with additional information provided in API 2SIM.

A.7.11 Indirect Loads/Actions

In general, the ultimate strengths of ductile structural systems are not sensitive to indirect actions, because the behavior of ductile components beyond full plastic loading is not affected by the initial level of internal stress. Nevertheless, indirect loads/actions can be particularly important in the following instances:

- a) where a component loses resistance during buckling (e.g. a thin curved shell or a torsionally buckling nominally straight stiffener);
- b) where the indirect stresses move the component's response from the linear region to a near buckling geometrically nonlinear region (as a guide, this effect can be important if the indirect load/action effect is greater than about 20 % of the elastic critical buckling resistance);
- c) where the indirect stresses result in significant triaxial tensile stresses, such as the result of weld shrinkage in a complex component (a triaxial stress state can reduce the fracture toughness of the affected steel).

The effects of indirect loads/actions should therefore be checked on a case-by-case basis. Where a) above applies, the indirect loads/actions should be included in the ULS check; where b) or c) applies, the indirect loads/actions need not be included in the ULS check, but the enhanced stress ranges should be included in fatigue calculations.

Where indirect loads/actions are considered in detail, they should be included with permanent and variable loads/actions as detailed in Table A.7.11-1 (see 9.2).

Table A.7.11-1—Inclusion of Indirect Loads/Actions in Permanent and Variable Loads/Actions

Indirect Load/Action	Included in
Prestressing	Permanent load/action, G_1
Creep/relaxation	
Settlement	
Permanent temperature loads (air, seawater)	Permanent load/action, G_2
Production oriented temperature loads	Variable load/action, Q_1

A.7.12 Structural Reliability Analysis

SRA methods are used to assess the effects of uncertainties in the loads/actions, resistances, and modeling of (parts of) a structure and its performance. Uncertainties exist due to the natural variation, the physical uncertainty or the randomness in the basic variables (aleatoric or Type I uncertainties), and due to those factors that are a function of a lack of complete understanding or knowledge (epistemic or Type II uncertainties).

SRA is not normally undertaken as part of a new design in accordance with this document, but may be used during the initial design process to provide comparative data. SRA may be used in the (re)calibration of partial load/action and resistance factors for special or unusual circumstances, in decision analyses to support inspection and monitoring programs, and in some cases in the structural assessment of existing structures. The design procedures in this document should be used for new designs.

The objective of SRA is to determine the probability of an event (usually a failure event) occurring during a specified reference period. The failure probability during the structure's design service life (or for some other specified reference period) can be of use in economic risk models; annual failure probabilities are generally evaluated when considering life-safety issues.

Where SRA methods are used, it is generally necessary to compare evaluated failure probabilities with appropriate target values. Target reliabilities should allow for the consequences and nature of failure, and where possible, should be calibrated against well established cases that are known to have adequate safety.

A.8 Loads/Actions for Pre-service Situations

A.8.1 General

Loads/actions and corresponding safety checks for pre-service and removal situations are often ignored in standards because their possible consequences are viewed as owner risks and do not affect the general public. For offshore structures, failures during this phase are primarily economic and do not normally involve operating personnel or environmental hazards. The situation changes, however, if structures are partially damaged during pre-service or removal situations and this fact escapes detection. Subsequent capacity to perform functions or resist environmental hazards can then be reduced.

During lifting operations and during the installation stage, the structure is in a state of dynamic equilibrium such that arbitrary use of partial load/action factors can lead to incorrect and even unsafe results. Unlike other situations, the representative values to be used for lifting and installation analyses, including stiffness parameters and external loads/actions, should be best estimates rather than conservative values. Safety considerations should in these cases be taken into account by performing appropriate sensitivity analyses, instead of applying partial load/action and resistance factors.

Partial load/action factors should not be confused with dynamic or impact factors. The latter are the increase in the effect of loads/actions due to inertia. Partial load/action factors, on the other hand, account for the uncertainty in estimating the effect of individual static and/or dynamic loads/actions.

A.8.2 General Requirements

A.8.2.2 Weight Control

Cases where the behavior of the structure and the magnitude and distribution of the forces in the structure can be very sensitive to the position of the center of gravity include:

- floating situations,
- uprighting,
- lifts with a very eccentric location of the center of gravity, and
- cases where the arrangement of equipment items leads to high uncertainty in the center of gravity location.

In normal cases, where the effect of a shift in the center of gravity is mainly local, this effect can be assumed to be included in the local factor of 8.3.5.

A.8.3 Loads/Actions Associated with Lifting

A.8.3.1 General

Lifting loads depend on the nature of the object being lifted (size, weight, stiffness, etc.), the particulars of the lifting equipment (crane stiffness, hook speed, sling arrangement, etc.), and on the conditions and procedures under which the lift is made (e.g. onshore, sheltered inshore, or exposed offshore). The general guidance on lifting loads provided in this document assumes typical lifts in reasonably controlled conditions.

Critical offshore lifts, such as those involving heavy loads approaching the capacity of the lifting equipment or those using dual crane (two cranes on one vessel) or tandem lift (two crane vessels) operations, should be subject to special investigation. This should investigate the dynamic lift forces in cranes, rigging, lifting points, and in the lifted structure, as well as the loads/actions due to impact during lift-off and setting down of the load, and should account for the limiting weather conditions in which the operation may proceed. Frequency domain and/or time domain techniques may be used to analyze the relative motions of the lift vessel, the transportation barge, and the structure to determine the magnitude of these dynamically induced forces.

Specific lift criteria used for final design should be determined in conjunction with the installation contractor.

For the lifting of an object, the following considerations should be taken into account:

- a) weight and weight growth;
- b) dynamic amplification due to lift-off from a cargo barge and due to motions of the crane vessel;
- c) effects of a shift in the center of gravity;
- d) tilt, which, for a single crane lift with four slings, is caused by the fact that the center of gravity is not located exactly under the crane hook, and for a dual crane lift, by the fact that the center of gravity position is not exactly below the line through both crane hooks and by possible uneven hoisting speeds;
- e) yaw, an effect present only in dual crane lift operations, caused by yawing of the lifted object;
- f) sling force distribution, which depends on sling stiffness, the stiffness of the lifted object, sling length, and the hook assembly.

Loads/actions due to lifting on lifting attachments and on other components of the structure should include both vertical and horizontal components, the latter occurring when lift slings are other than vertical. Vertical loads/actions on the lifted object should include the effect of buoyancy, as well as loads/actions imposed by the lifting equipment.

When suspended, the structure will occupy a position such that its center of gravity is below the centroid of all upward loads/actions on it and such that the structure is in static equilibrium. The attitude of the lift in this state of static equilibrium should be used to determine internal forces in the structure and in the slings. The movement of the lift as it is picked up and set down should be taken into account in determining critical combinations of vertical and horizontal loads/actions at all points, including those to which lifting slings are attached. For lifts where either the crane or the structure to be lifted is on a floating vessel, the selection of the nominal lifting forces should consider the influence of vessel motions.

The lift should be designed such that all structural steel members are proportioned for factored resistances as specified in 7.7. In addition, all critical structural connections and primary members should be designed to have adequate reserve strength to ensure structural integrity during lifting.

A.8.3.3 Effect of Tolerances

The effect of tolerances in a lift analysis of a standard four-point lift may be taken into account by one of the following methods:

- a) an analysis with one pair of opposite slings assumed to carry 75 % and the other pair 25 % of the hook force, and vice versa;
- b) an analysis with modified sling lengths, e.g. two diagonally opposite slings with increased length, each by an amount corresponding to the total tolerance, for each diagonal in turn.

The analyses described above are very detailed, and yet reality is more complex. For instance, during design, the sling lengths, final location of the center of gravity, hook geometry and effect of hook assembly hinges, and the stiffness of the slings are not exactly known. The minimum sling force increase given in 8.3.3 compensates for these unknowns.

A.8.3.5 Local Factor

The objective of the local factors is to account for the sensitivity of framing members to tolerances in the fabrication of lifting attachments and for tolerances in the load transfer from the structure through the lifting attachments to the slings. As the whole weight of the structure normally passes through four lifting attachments, which are essentially pinned connections, the effect of misalignment can be significant. The effect of misalignment, however, reduces quickly away from the lifting attachment, thereby allowing $\gamma_{f,lf}$ to be reduced for members not framing directly into lifting attachments.

A.8.5 Loads/Actions Associated with Loadout

A.8.5.3 Self-floating Structures

Self-floating structures can be launched from the fabrication yard to float with their own buoyancy for tow to the installation site. The last portion of such a structure leaving the launchways can be subjected to localized forces as the first portion of the structure to enter the water gains buoyancy and causes the structure to rotate from the slope of the launchways. Loads/Actions should be evaluated for the full travel of the structure down the launchways.

A.8.6 Loads/Actions Associated with Transportation

A.8.6.2 Environmental Conditions

The environmental conditions used in determining the motions of the towing arrangement should be established by the owner (in consultation with the marine warranty surveyor, if appropriate), taking account of the expected tow route and season. For long ocean tows where the structure and barge are unmanned, the extreme environmental conditions are typically selected to have a probability of exceedance during the tow duration in the range of 1 % to 10 %. The specific value will depend on an evaluation of acceptable risks and consequences. For short duration tows, the environmental conditions should generally have a return period of not less than 1 year for the season in which the tow takes place.

A.8.7 Loads/Actions Associated with Installation

A.8.7.2 Launched Structures

Barge-launched structures are usually launched at or near the intended destination. The structure is generally moved along slipways, which terminate in rocker arms, on the deck of the barge; these rocker arms distribute reactions into strong points of the structure. A barge can have both primary and secondary rocker arms. As the center of gravity of the structure moves over the edge of the barge, the structure starts to rotate, causing the rocker arms at the end of the slipways to rotate while the structure continues to slide from the rocker arms. Reactions supporting the structure on the slipways should be evaluated for the full travel of the structure. Deflection of the rocker beam and the effect on internal forces throughout the structure should also be considered. In general, the most severe forces will occur as rotation starts.

A.8.7.6 Loads/Actions on the Foundation during Installation

A.8.7.6.1 General

The wind, wave, and current conditions used to determine the environmental loads/actions should be established for the installation season, and are typically selected to have a probability of exceedance during the exposure period in the range of 1 % to 5 %. The specific value will depend on an evaluation of acceptable risks and consequences. The exposure period used should take account of the time of the year, the installation sequence and the structure configuration for the various phases, considering relevant operations, such as buoyancy tank removal, pile stabbing, driving and securing, conductor installation, and deck placement. For short exposure durations, the seasonal environmental conditions should generally have a return period of not less than 1 year, unless restrictions are imposed in installation procedures to limit the conditions in which operations may proceed.

A.8.8 Loads/Actions Associated with Removal

Examples of loads/actions associated with removal are those due to explosive cutting, the sudden transfer of pile weight to structure and mudmats, pile and mudmat suction forces, lifting forces, reduced buoyancy and increased weight (e.g. from marine growth or grout) compared with those at installation, and concentrated loads/actions during landing the structure on a barge.

A.9 Loads/Actions for In-place Situations

A.9.1 General

The partial load/action factors provided in Section 9 are intended to cover variations in the intensity of direct loads/actions from the specified representative values and as far as appropriate the uncertainties in predicting internal forces.

The values have been primarily calibrated to conditions experienced in the Gulf of Mexico, accounting for the analysis procedures normally followed in applying API 2A-LRFD [A.9.1-1].

In no instance should partial load/action factors be used as a substitute for rational analysis, e.g. to account for dynamics.

Partial load/action factors may be obtained from this document, or from national application documents, or where sufficient information is available by an appropriate reliability analysis. Such a reliability analysis includes:

- a) uncertainty in the intensity of direct loads/actions,
- b) probability of simultaneous occurrence of independent loads/actions,
- c) accuracy of the prediction of load/action effects (e.g. internal force), and
- d) interpretation of relevant field experience, verification, and control.

Acceptable reliability should be based on failure consequences and compatibility with minimum safety levels intended to be achieved by this document (see 7.10).

A.9.2 Permanent Loads/Actions (G) and Variable Loads/Actions (Q)

A.9.2.2 Permanent Load/Action 2, G_2

This category has been separated from traditional variable loads/actions because there is little variability in its magnitude when the equipment or package is known.

A.9.2.5 Unintentional Flooding

As noted in 9.2.1, buoyancy (and hydrostatic pressure) is considered to be a permanent load/action. This is consistent with the relatively high certainty of the displaced volume of a component.

One uncertainty in the load/action due to buoyancy is whether the component remains internally dry. The occurrence of unintentional flooding is not well documented, so rather than developing an ad hoc partial load/action factor, several additional checks are called for. The check of unflooded members as flooded can be done for all members at once, rather than each singly. For intentionally flooded members, the need to ensure positive flooding can be critical to prevent hydrostatic collapse, particularly during rapid submergence that can occur during installation.

The requirement for flooding some fraction of all unflooded members can be modeled by a set of equivalent loads/actions on joints.

A.9.4 Extreme Quasi-static Load/Action due to Wind, Waves, and Current (E_e)

A.9.4.5 Procedure for Determining E_e

The return period in years means the inverse of the annual probability of exceedance of a parameter (e.g. a wave height or wind speed). In this context, a 100-year storm, for example, has no meaning; the quantity of interest should be qualified in terms of metocean parameters.

The three methods noted in 9.4.1 are all used to determine the extreme direct load/action, E_e ; further discussion of these methods is given in API 2MET. Method a) in 9.4.1 (100-year return period wave with other associated parameters estimated from correlations) has been used for Gulf of Mexico designs, while method b) (100-year return period wave with 100-year return period wind and 100-year return period current) has been used in the

North Sea and many other areas, whereas method c) (100-year return period load/action or load/action effect) is a more recent development, suitable when a database of joint occurrences of wind, waves, and current is available.

Additional considerations should be given to obtaining the extreme direct load/action, E_e , for locations where there are strong currents that are not driven by local storms. Such currents can be driven by tides or by deep water currents, such as the Loop Current in the Gulf of Mexico or the Gulf Stream. In this case, method 9.4.1 a) can be acceptable if the storm generated conditions are the predominant contributors to the extreme global environmental load/action (load/action effect) and if the appropriate “associated” value of tidal and circulation current can be determined. However, method c) is conceptually more straightforward and preferable. Method b) is the simplest method and ensures an adequate design environmental load/action (load/action effect) that can be very conservative compared to the true 100-year return period global environmental load/action (load/action effect).

For some areas, substantial databases are becoming available with which it is possible to establish statistics of joint occurrence of wind, wave, and current magnitudes and directions. When such a database is available, this should be used to develop environmental conditions based on method 9.4.1 c), which provides the 100-year return period extreme global environmental load/action on the structure. The corresponding partial load/action factors and RSR to be used in conjunction with the 100-year return period global environmental load/action (load/action effect) should be determined using SRA principles, in order to ensure that an appropriate safety level is achieved. This approach provides more consistent reliability (safety) for different geographic areas than has been achieved by the practice of using separate (marginal) statistics of winds, currents, and waves.

A.9.5 Extreme Quasi-static Load/Action Caused by Waves Only (E_{we}) or by Waves and Currents (E_{wce})

A.9.5.1 Procedure for Determining E_{we} and E_{wce}

The procedure depicted in Figure 9.5-1 can be used with any combination of wave and current, not just the 100-year wave and associated current. However, the wave-current combination used as input to the procedure should account for wave-current joint probability insofar as possible. The procedure is intended to account for the principal hydrodynamic phenomena as realistically as is practical within the framework of the calculation of a deterministic quasi-static global environmental load/action, and has been validated using measured hydrodynamic loads/actions on full-scale structures in simultaneously occurring waves and currents [A.9.5-1]. Therefore, use of wave-current combinations that do not account for wave-current joint probability will result in hydrodynamic loads/actions that are biased high.

When appropriate, the procedure is repeated for various relevant and foreseeable combinations of extreme wave height, wave period, current and wind speeds.

A.9.5.2 Models for Hydrodynamic Loads/Actions

A.9.5.2.1 Morison Equation

The use of the local acceleration rather than the total (local plus convective) acceleration in the inertia term of Morison equation is the subject of ongoing debate. There have been several publications on this topic (see References [A.9.5-2], [A.9.5-3], [A.9.5-4], and [A.9.5-5]). These publications all conclude that the total acceleration should be used. However, it should be noted that these publications all unrealistically assume that the flow does not separate from the cylinder. Realistically, except for very small amplitudes of oscillation (Keulegan-Carpenter number, $K < 3$; see A.9.5.2.3.4), the flow separates on the downstream side of the cylinder, creating a wake of reduced velocity. The local change in velocity across the cylinder due to the convective acceleration in the undisturbed far-field flow is generally much less than the change in velocity due to local flow separation, as implied in Reference [A.9.5-6]. The convective acceleration can also be nearly in phase with the locally incident flow velocity, which leads the undisturbed far-field velocity in oscillatory flow because of “wake encounter” [A.9.5-7]. Therefore, it can be argued that the convective acceleration should be neglected, either because it is small

relative to local velocity gradients due to flow separation or because it is already implicitly included in drag coefficients derived from measurements of local loads/actions in separated flow. As a practical matter, the convective acceleration only exceeds 15 % of the local acceleration in steep waves, for which inertia load/action is generally much smaller than drag load/action [A.9.5-5].

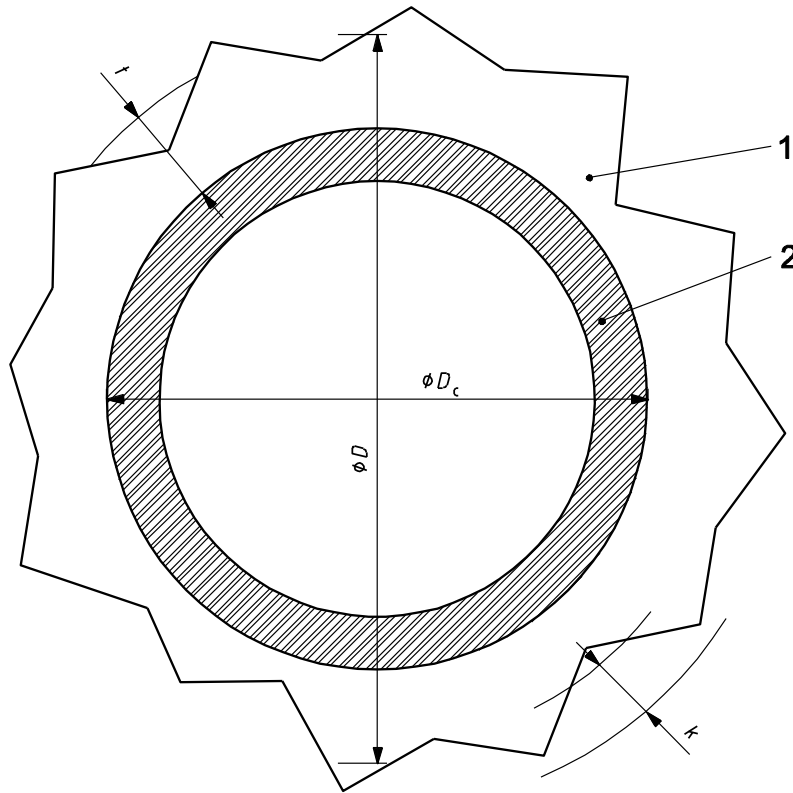
Only the components of velocity and acceleration normal to the member axis are used in computing drag and inertia loads/actions, based on the “flow-independence” or “cross-flow” principle. This principle has been verified in steady subcritical flow in Reference [A.9.5-8] and in steady post-critical flow in Reference [A.9.5-9]. The data given in Reference [A.9.5-10], as re-interpreted by Reference [A.9.5-11], have shown the flow-independence principle to be acceptable also for inertia loads/actions in one-dimensional oscillatory flow. Therefore, it is reasonable to assume that the flow-independence principle is valid in general for both steady and multidimensional oscillatory flows, with the exception of flows near the unstable, critical Reynolds number regime.

A.9.5.2.2 Marine Growth

It is necessary to account for the foreseen influence of marine growth on hydrodynamic load/action that the structure can experience during its design service life. This influence arises from an increased drag coefficient due to roughness, increased diameter, and addition to density. It will vary over depth and with time throughout the design service life of the structure.

All components of the structure (members, conductors, risers, appurtenances, etc.) are increased in cross-sectional area by marine growth (see Figure A.9.5-1). The effective component diameter (or cross-sectional width for noncircular members or prisms) is $D = D_c + 2t$, where D_c is the “clean” outer diameter and t is the average marine growth thickness that would be obtained by circumferential measurements with a 25 mm to 100 mm wide tape. An additional parameter that affects the drag coefficient of elements with circular cross-sections is the relative roughness, $e = k/D$, where k is the average peak-to-valley height of “hard” growth organisms. Marine growth thickness and roughness are illustrated in Figure A.9.5-1 for a circular cylinder. Marine organisms generally colonize a structure soon after installation. They grow rapidly at the beginning, but growth tapers off after a few years. In some areas, the initial colonization is by one species (e.g. mussels), which is subsequently removed by a predatory species (e.g. starfish), and ultimately replaced by another (e.g. sea anemones). Marine growth has been measured on structures in many areas but should be estimated for other areas.

The thickness of marine growth depends on location. Experience in one area of the world cannot necessarily be applied in another. Where possible, site-specific studies should be conducted to establish marine growth thickness and its dependence on depth below water.



Key

- 1 hard growth
- 2 member
- t average marine growth thickness
- k average marine growth peak to valley height
- D effective component diameter, $D = D_c + 2 t$
- D_c diameter of clean member
- e relative roughness, $e = k/D$

Figure A.9.5-1—Definition of Surface Roughness Height and Thickness

A.9.5.2.3 Drag and Inertia Coefficients

A.9.5.2.3.1 General

In the ocean environment, hydrodynamic load/action predicted by Morison equation is only an engineering approximation. Morison equation can match measured drag and inertia loads/actions reasonably well in any particular half wave cycle with the constants, C_d and C_m , but the best-fit values of C_d and C_m vary from one half wave cycle to another. Most of the variation in C_d and C_m can be taken into account by expressing C_d and C_m as functions of:

- relative surface roughness $e = k/D$
- Reynolds number $Re = U_m D/\nu$

— Keulegan-Carpenter number $K = U_m T/D$

— current/wave velocity ratio $r = U_c/U_{mo}$

where

k is the average roughness height;

D is the effective diameter (including marine growth);

U_m is the maximum speed (including current) normal to the cylinder axis in a wave cycle;

ν is the kinematic viscosity of water;

T is the wave period;

U_c is the current speed measured in-line with the waves;

U_{mo} is the maximum wave-induced orbital velocity.

A.9.5.2.3.2 Surface Roughness

The dependence of the steady-flow drag coefficient at post-critical Reynolds numbers (C_{ds}) on relative surface roughness is shown in Figure A.9.5-2 for “hard” roughness elements. All the data in this figure have been adjusted, where necessary, to account for wind tunnel blockage and the drag coefficient has been referenced to the effective diameter, D , including the roughness elements.

Natural marine growth on structures will generally have a relative surface roughness, $e > 10^{-3}$. Thus, in the absence of better information on the expected value of surface roughness and its variation with depth for a particular site, it is reasonable to assume C_{ds} in the range 1.00 to 1.10 for all members below high tide level. It is still necessary to estimate the thickness of marine growth that will ultimately accumulate in order to estimate D . For elements above high tide level, a reasonable estimate of surface roughness is $k = 0.05$ mm, which will give C_{ds} in the range of 0.6 to 0.7 for typical diameters.

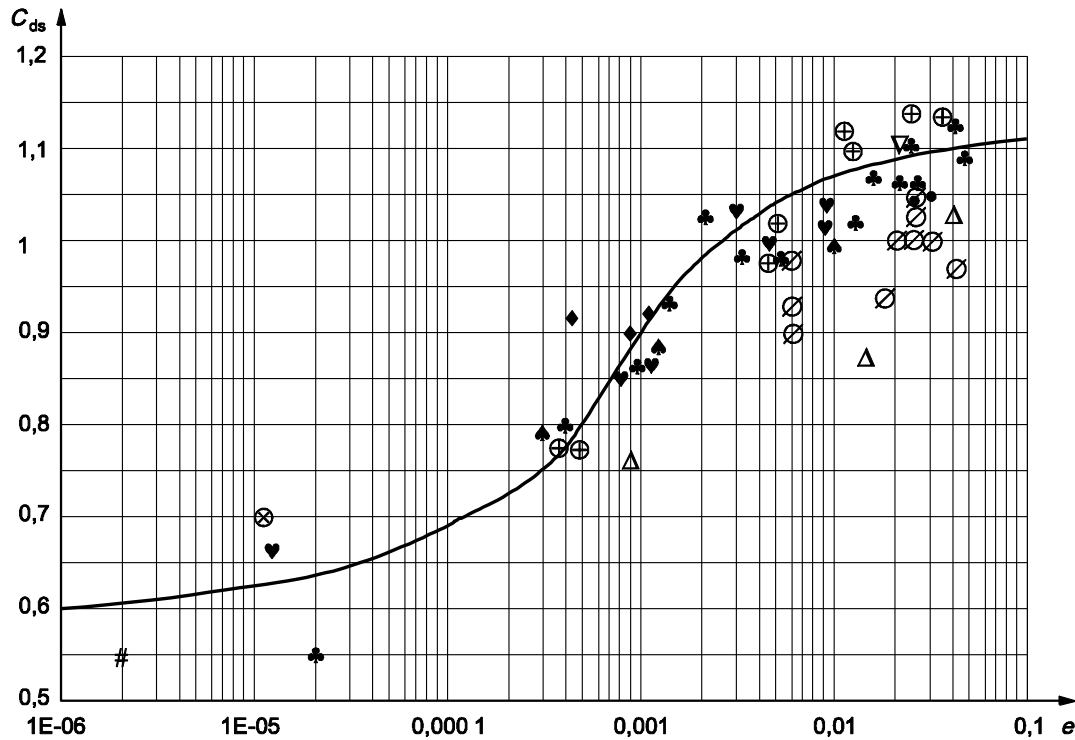
All the data in Figure A.9.5-2 are for cylinders that are densely covered with surface roughness elements. Load measurements (see References [A.9.5-12] and [A.9.5-13]), show that there is little degradation in the effectiveness of surface roughness for surface coverage as sparse as 10 %, but that roughness effects are negligible for surface coverage less than 3 %.

The effect of soft, flexible growth on C_{ds} is poorly understood. Tests reported in Reference [A.9.5-14] indicate that:

- soft, fuzzy growth has little effect, as C_{ds} is predominantly determined by the underlying hard growth, and
- anemones and kelp produce drag coefficients similar to those for hard growth.

For cylindrical elements whose cross-section is not circular, C_{ds} may be assumed to be independent of surface roughness. Suitable values are provided in Reference [A.9.5-15].

Surface roughness also affects the inertia coefficient in oscillatory flow. Generally, as C_d increases with roughness, C_m decreases. More information is provided in subsequent discussions.

**Key**

- e relative roughness
 C_{ds} steady-flow drag coefficient at post-critical Reynolds numbers
 # Jones (1989)
 Δ Blumberg (1961)
 • Wolfram (1985)
 ♣ Miller (1976)
 ♦ Szechenyl (1975)
 ♥ Achenbach (1971, 1981)
 ♠ Wang (1986)
 ⊗ Roshco (1961)
 ⊕ Norton (1983)
 ∅ Nath (1987)
 ∇ Rodenbusch (1983)

Figure A.9.5-2—Dependence of Steady-flow Drag Coefficient on Relative Surface Roughness

A.9.5.2.3.3 Reynolds Number

The hydrodynamic coefficients for elements whose cross-sections have sharp edges are practically independent of the Reynolds number. However, circular cylinders have coefficients that depend on the Reynolds number.

Fortunately, for most offshore structures in the extreme design environment, Reynolds numbers are well into the post-critical flow regime, where C_{ds} for circular cylinders is independent of the Reynolds number. However, in less severe environments, such as considered in fatigue calculations, some platform members can drop into the critical flow regime. Use of the post-critical C_{ds} in these cases would be conservative for static calculations of wave loads/actions but nonconservative for calculating damping of dynamically excited structures.

In laboratory tests of scale models of structures with circular cylindrical members, one should be fully aware of the dependence of C_{ds} on the Reynolds number. In particular, the scale of the model and the surface roughness should be chosen to eliminate or minimize Reynolds number dependence, whereas the difference between model-scale and full-scale C_{ds} should be considered in the application of model test results to full-scale structures. Further guidance on the dependence of circular cylinder C_{ds} on the Reynolds number can be found in References [A.9.5-16], [A.9.5-8], and [A.9.5-5].

A.9.5.2.3.4 Keulegan-Carpenter Number

The Keulegan-Carpenter number, K , is a measure of the unsteadiness of the flow. It is proportional to the distance normal to the member axis travelled by an undisturbed fluid particle in a half wave cycle, normalized by the member diameter. For a typical full-scale space frame structure in design storm conditions, K is generally greater than 40 for members in the “wave zone,” and drag load/action is predominant over inertia load/action. However, for large diameter legs that can be found on self-floating structures, K can be less than 10, and inertia load/action is predominant over drag load/action.

The parameter K is also a measure of the importance of “wake encounter” for nearly vertical (within 15° of vertical) members in waves. As the fluid moves across a member, a wake is created. When oscillatory flow reverses, fluid particles in the wake return sooner and impact the member with greater velocity than undisturbed fluid particles. For larger K , the wake travels farther and decays more before returning to the cylinder and, furthermore, is less likely to strike the cylinder at all if the waves are multidirectional or there is a component of current normal to the principal wave direction. For very large K , wake encounter may be neglected. For smaller K , wake encounter amplifies the drag action for nearly vertical members above its quasi-steady value estimated from undisturbed fluid velocities.

Figure A.9.5-3 shows data for the drag coefficient C_d that are most appropriate for calculating hydrodynamic loads/actions on nearly vertical elements in extreme storm environments. All these data were obtained in the post-critical flow regime, in which C_{ds} is practically independent of the Reynolds number. All data account for wave spreading, i.e. all have two components of motion normal to the element axis. All except the data from “figure 8” orbits implicitly account for random wave motion. The field data also naturally include an axial component of motion and, to some extent, a steady current. For $K > 12$, with K normalized by C_{ds} , the data for smooth and rough cylinders are reasonably well represented by a single curve in Figure A.9.5-3, as suggested by the far-field, quasi-steady wake model described in Reference [A.9.5-17].

Figure A.9.5-4 shows drag coefficient data for $K < 12$, which are more appropriate for calculating hydrodynamic loads/actions on nearly vertical members in less extreme sea states and drag damping in earthquake-excited motion. For $K < 12$, the smooth and rough cylinder data are similar if K is not normalized by C_{ds} . The data in Reference [A.9.5-18] do not agree well with the curves in Figure A.9.5-4, presumably because of the relatively low Reynolds numbers in the tests for the lowest values of K and because of the lack of wave spreading in the tests for the higher values of K .

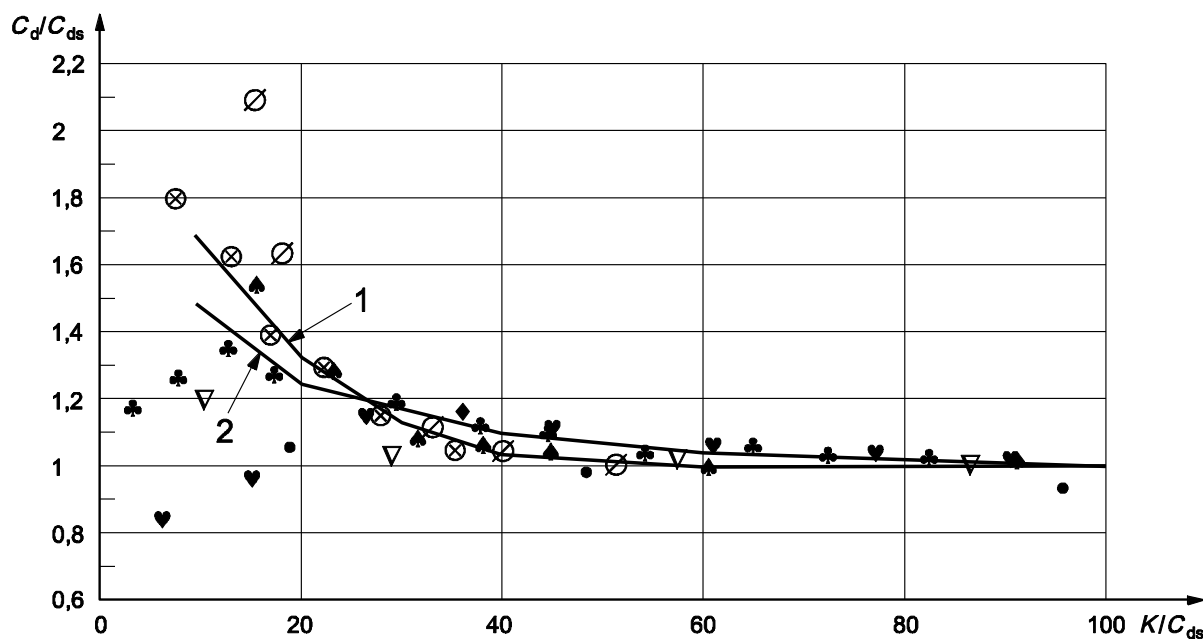
The symbols shown in Figure A.9.5-3 do not represent individual data points. Rather, they represent values from a curve fitted through a scatter of data points. When designing a structure consisting of a single isolated column, the scatter in the C_d data should be considered. The data in Reference [A.9.5-18] for one-dimensional, sinusoidally oscillating motion, which are notably omitted from Figure A.9.5-3, represent a reasonable upper bound. For a structure consisting of many members, the scatter in C_d may be neglected, as the deviations from the mean curve are uncorrelated from member to member (see Reference [A.9.5-19]).

Figures A.9.5-5 and A.9.5-6 show data for the inertia coefficient, C_m , for a nearly vertical circular cylinder. Figure A.9.5-5 shows that C_m for both smooth and rough cylinders approaches the theoretical value of 2.0 for $K \leq 3$. For $K > 3$ with the onset of flow separation, C_m begins to decrease. With the exception of rough cylinder data given in Reference [A.9.5-18], which exhibit a pronounced drop (“inertia crisis”) in C_m at $K \approx 12$, it appears that a single sloping line is adequate for both smooth and rough cylinders up to $K \approx 12$ —beyond which smooth and rough cylinder data begin to diverge. In Figure A.9.5-6, the single line from Figure A.9.5-5 is seen to split into

two lines because K is divided by $C_{ds} = 0.66$ for smooth cylinders and $C_{ds} = 1.1$ for rough cylinders. The value of C_m is taken as 1.6 for smooth cylinders and 1.2 for rough cylinders for $K/C_{ds} \geq 17$.

Although Figures A.9.5-3 to A.9.5-6 are based on circular cylinder data, the figures are also applicable to noncircular cylinders, provided the appropriate value of C_{ds} is used and C_m is multiplied by $C_{m0}/2$, where C_{m0} is the theoretical value of C_m for the noncircular cylinder as $K \rightarrow 0$.

Furthermore, although Figures A.9.5-3 to A.9.5-6 were developed for use with individual, deterministic waves, they can also be used for dynamic analysis (either in the time or in the frequency domain) of fixed structures by using significant wave height and spectral peak period to calculate K .



Key

K Keulegan-Carpenter number

C_d drag coefficient

C_{ds} steady-flow drag coefficient at post-critical Reynolds numbers

⊗ Heideman (1979), $C_{ds} = 1.00$

∅ Heideman (1979), $C_{ds} = 0.68$

♠ Bishop (1985), $C_{ds} = 0.66$

♣ Rodenbusch (1983), $C_{ds} = 1.10$ random directional

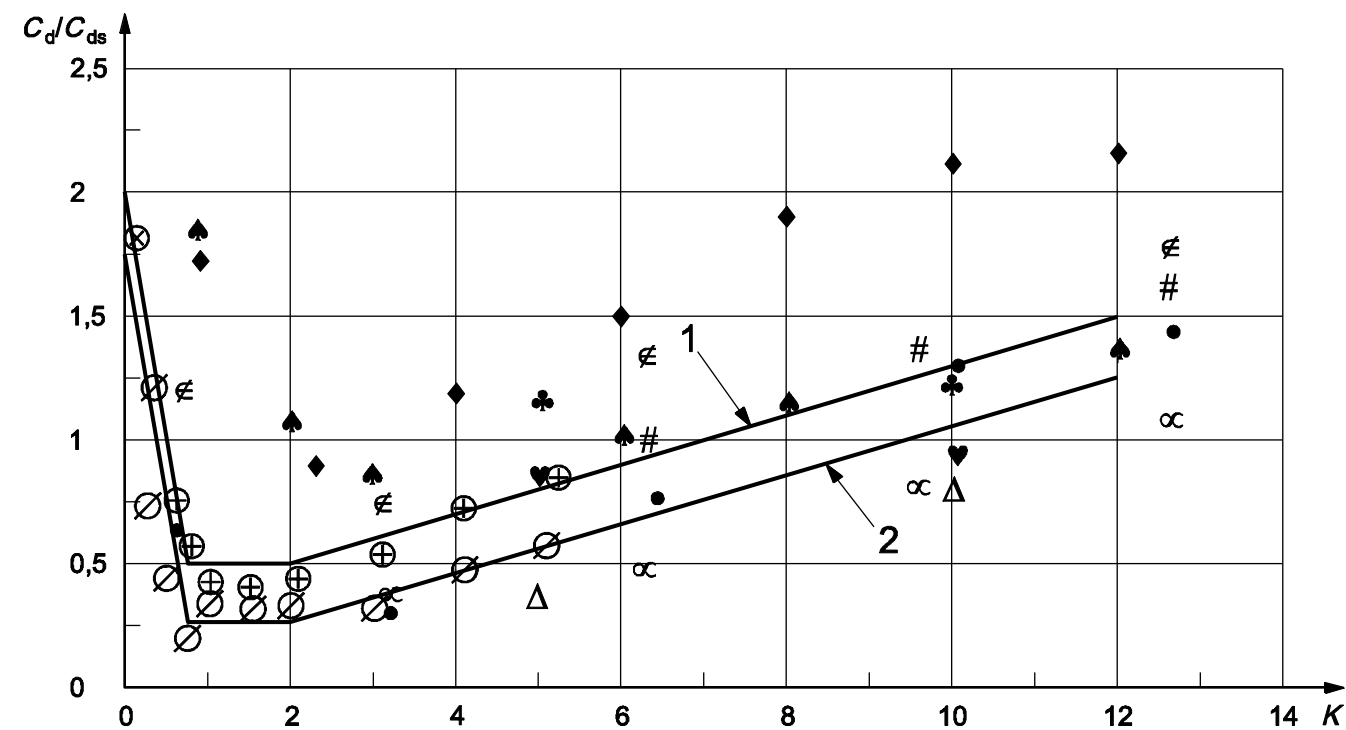
♥ Rodenbusch (1983), $C_{ds} = 0.66$ random directional

▽ Rodenbusch (1983), $C_{ds} = 1.10$ figure 8 orbits

• Rodenbusch (1983), $C_{ds} = 0.66$ figure 8 orbits

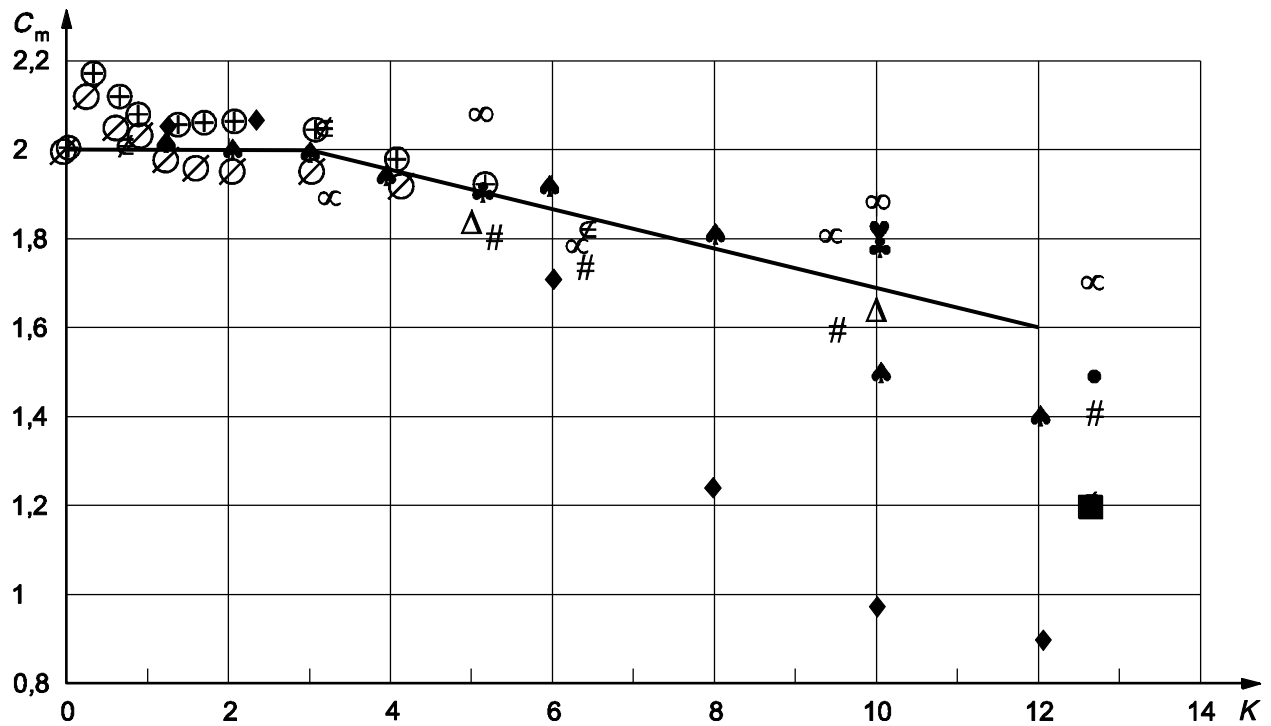
♦ Ohmart and Gratz (1979), $C_{ds} = 0.66$ figure 8 orbits

Figure A.9.5-3—Wake Amplification Factor for Drag Coefficients as Function of K/C_{ds}

**Key**

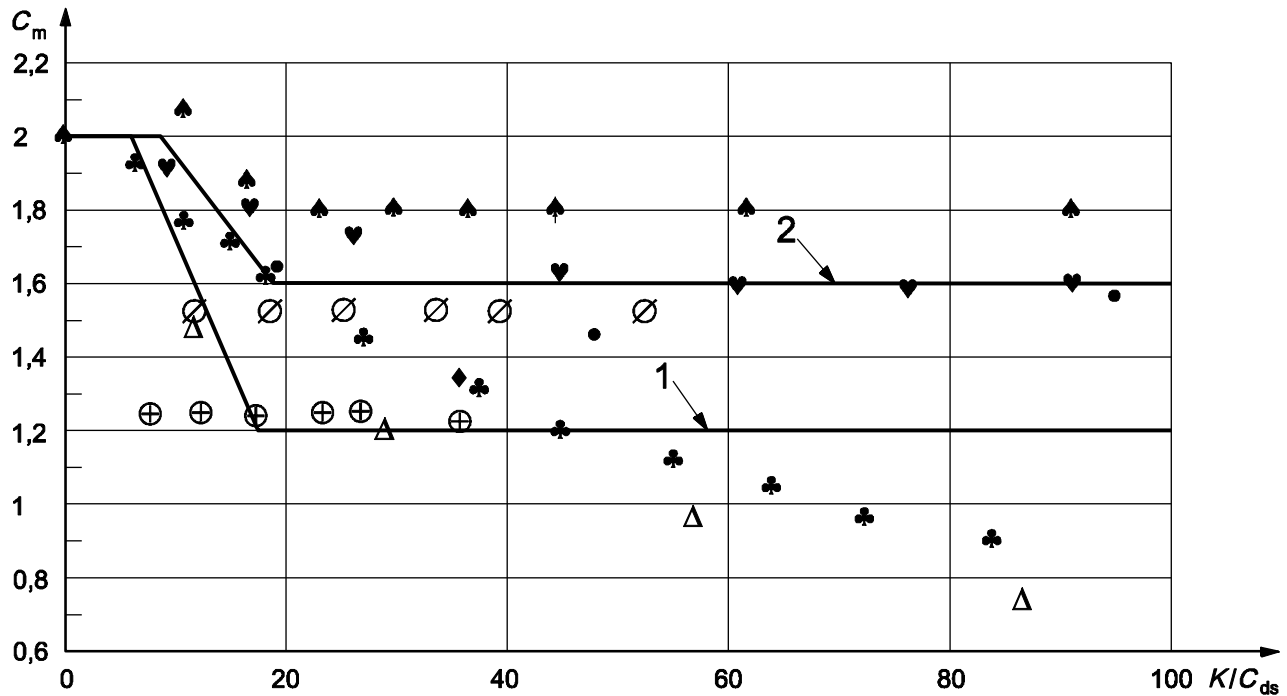
- 1 rough member ($C_{ds} = 1.2$)
- 2 smooth member ($C_{ds} = 0.6$)
- K Keulegan-Carpenter number
- C_d drag coefficient
- C_{ds} steady-flow drag coefficient at post-critical Reynolds numbers
- ♣ Rodenbusch (1983), $C_{ds} = 1.10$ random directional
- ♥ Rodenbusch (1983), $C_{ds} = 0.66$ random directional
- △ Bearman (1985), $C_{ds} = 0.60$
- ≠ Rodenbusch (1983), $C_{ds} = 1.10$ sinusoidal
- Rodenbusch (1983), $C_{ds} = 0.66$ sinusoidal
- ♦ Sarpkaya (1986), $C_{ds} = 1.10$
- ♠ Sarpkaya (1986), $C_{ds} = 0.65$
- ∞ Garrison (1990), $C_{ds} = 0.65$
- ⊕ Marin (1987), $C_{ds} = 1.10$
- # Garrison (1990), $C_{ds} = 1.10$
- ⊗ Marin (1987), $C_{ds} = 0.60$
- ⊗ Iwaki (1991), $C_{ds} = 1.10$

Figure A.9.5-4—Wake Amplification Factor for Drag Coefficients as Function of K


Key

- K Keulegan-Carpenter number
- C_m inertia coefficient
- ♣ Rodenbusch (1983), $C_{ds} = 1.10$ random directional
- ♥ Rodenbusch (1983), $C_{ds} = 0.66$ random directional
- △ Bearman (1985), $C_{ds} = 0.60$
- ⊄ Rodenbusch (1983), $C_{ds} = 1.10$ sinusoidal
- ∞ Bishop (1985), $C_{ds} = 0.66$ random directional
- Rodenbusch (1983), $C_{ds} = 0.66$ sinusoidal
- ♦ Sarpkaya (1986), $C_{ds} = 1.10$
- ♠ Sarpkaya (1986), $C_{ds} = 0.65$
- ∞ Garrison (1990), $C_{ds} = 0.65$
- ⊕ Marin (1987), $C_{ds} = 1.10$
- # Garrison (1990), $C_{ds} = 1.10$
- ⊘ Marin (1987), $C_{ds} = 0.60$
- ⊗ Iwaki (1991), $C_{ds} = 1.10$

Figure A.9.5-5—Inertia Coefficient as Function of K

**Key**

- 1 rough member ($C_{ds} = 1.1$)
- 2 smooth member ($C_{ds} = 0.66$)
- K Keulegan-Carpenter number
- C_{ds} steady-flow drag coefficient at post-critical Reynolds numbers
- C_m inertia coefficient
- ♣ Rodenbusch (1983), $C_{ds} = 1.10$ random directional
- ♥ Rodenbusch (1983), $C_{ds} = 0.66$ random directional
- Δ Rodenbusch (1983), $C_{ds} = 1.10$ sinusoidal
- Rodenbusch (1983), $C_{ds} = 0.66$
- ♦ Ohmart and Gratz (1979), $C_{ds} = 0.60$
- ♠ Bishop (1985), $C_{ds} = 0.66$
- ⊕ Heideman (1979), $C_{ds} = 1.00$
- ∅ Heideman (1979), $C_{ds} = 0.68$

Figure A.9.5-6—Inertia Coefficient as Function of K/C_{ds} **A.9.5.2.3.5 Current/Wave Velocity Ratio**

The effect of a steady in-line current added to oscillatory motion is to push C_d toward C_{ds} , its steady-flow value. Data show that, for practical purposes, $C_d = C_{ds}$ when the current/wave velocity ratio (r) is greater than 0.4. For $r \ll 0.4$, the effect of a steady in-line current can be accommodated by modifying the Keulegan-Carpenter number. A first-order correction would be to multiply K due to wave alone by:

$$1 + 1.6r + 0.8r^2 \quad (\text{for } 0 < r < 0.4) \quad (\text{A.9.5-1})$$

A current component normal to the wave direction also drives C_d toward C_{ds} , since it reduces the impact of wake encounter. Data show that, for practical purposes, $C_d = C_{ds}$ for $V_N T_{app}/C_{ds} D > 8$. On the other hand, wake encounter has nearly its full impact for $V_N T_{app}/C_{ds} D < 1$. Here, V_N is the component of the current velocity normal to the wave direction and T_{app} is the apparent wave period.

A.9.5.2.3.6 Member Orientation

For members that are not nearly vertical, the effect of wake encounter, as characterized by the K dependence shown in Figures A.9.5-3 to A.9.5-6, is small. For horizontal and diagonal members, it is sufficient to use the theoretical value of C_m at $K \rightarrow 0$ and the steady-flow value of $C_d = C_{ds}$ at $K \rightarrow \infty$.

A.9.5.2.4 Current Blockage

Taking account of current blockage can be of benefit to the design of space frame structures with a large number of conductors.

In general, no space frame or lattice type structure is totally transparent to waves and current. These structures cause a global distortion of the incident waves and of the current in and around the structure. Since global hydrodynamic load/action is calculated by summing hydrodynamic loads/actions on individual members, it is important that the local incident flow be used to calculate hydrodynamic load/action on individual members in Morison Equation (9.5-1) to account for global distortion effects.

Space frame structures distort the waves, as well as the current. Some field data indicate that the root-mean-square (RMS) wave orbital velocity very near the structure is slightly reduced from that which is several structure widths up-wave. However, this reduction is not evident in all the data. Until more evidence to the contrary is accumulated, it is appropriate to continue with the assumption that a typical space frame structure does not significantly distort the incident wave kinematics in a global sense.

For currents, however, there now exists a substantial body of evidence that supports a reduction of the current within space frame structures relative to the free stream current. The amount of blockage depends on the type of structure. For dense structures it will be large, whereas for very transparent structures it will be small. Laboratory and field data indicate that the blockage factor can be as low as 0.6 for a dense structure (e.g. the Lena guyed tower [A.9.5-20]), approximately 0.7 for a typical compliant tower [A.9.5-21], and approximately 0.75 to 0.85 for a typical space frame structure [A.9.5-22] (see Table 9.5-2). Figure A.9.5-7 shows the measured current field at 18 m depth around and through the Bullwinkle platform during a Loop Current event in the Gulf of Mexico in 1991 [A.9.5-23]. The average blockage factor within the structure computed from the data is 0.77.

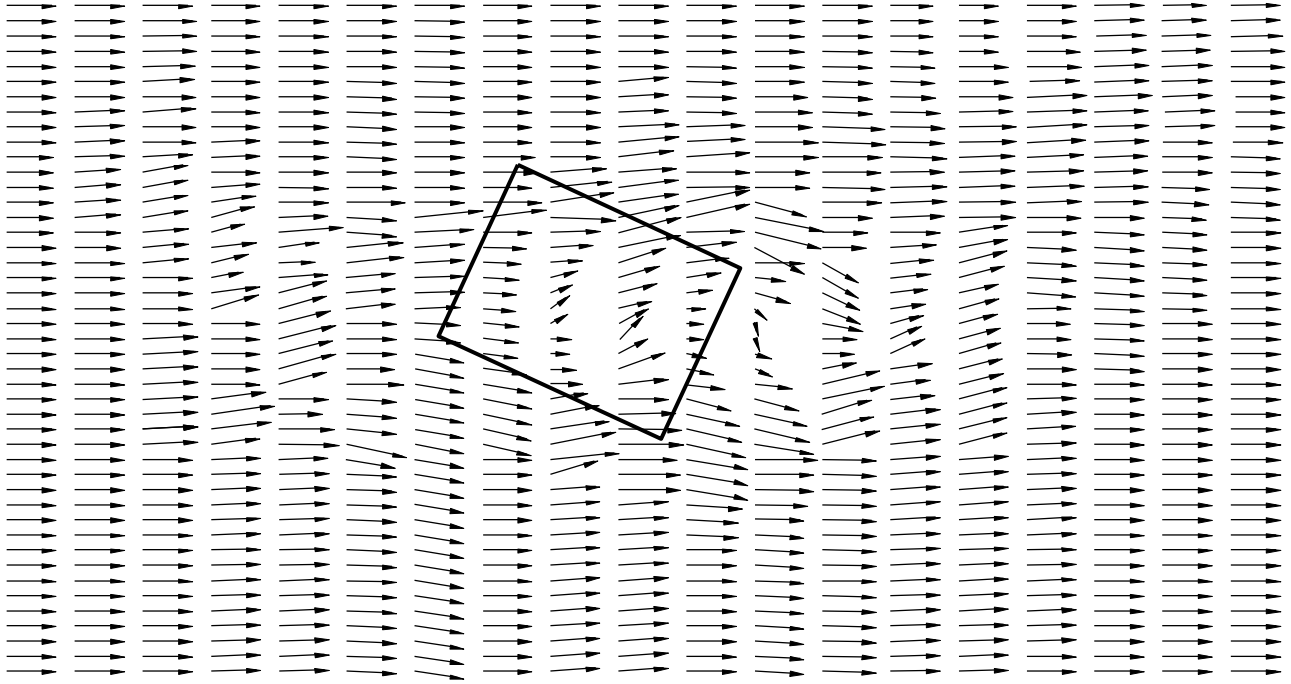


Figure A.9.5-7—Current Blockage Flow—Current Vectors Computed from Doppler Measurements

For structures with other configurations or structures with an atypical number of conductors, a current blockage factor can be calculated using the method described below. Factors calculated to be less than 0.7 should not be used without empirical evidence to support them.

The blockage factor for steady current can be estimated from the “actuator disk” model [A.9.5-22] as:

$$\left[1 + \sum_i (C_d A)_i / 4\bar{A} \right]^{-1} \quad (\text{A.9.5-2})$$

where

$\sum_i (C_d A)_i$ is the summation of the “drag areas” of all members (including horizontal members) in the flow;

$\bar{A}$ is the area within the perimeter area of the structure projected normal to the current.

For structures where geometry changes significantly with depth, the blockage factor can be computed for different depth levels. If the calculated reduction factor is less than 0.7, consideration should be given to modeling the structure as a series of actuator disks rather than a single actuator disk.

An alternative expression for the blockage factor based on a similar approach, but accounting for mixing downstream, is given in Reference [A.9.5-24]. In the case of small values of the ratio $\sum_i (C_d A)_i / \bar{A}$, the alternative expression reduces to Equation (A.9.5-2).

The terms “global blockage,” discussed here, and “shielding,” discussed in 9.5.2.5, are related and are sometimes used to describe the same phenomenon, as is the term “interference.” In this document “shielding” is used only with reference to members in the local wake of neighbouring members (such as conductor arrays), and the shielding factor should be applied to the calculated hydrodynamic load/action due to both waves and currents. “Blockage” is used in relation to the entire structure, and the blockage factor should be applied to the far-field current velocity only. With this distinction, the blockage factor is first used to calculate a reduced current velocity for the entire structure. The reduced current velocity and undisturbed wave kinematics are then used in Morison equation to calculate local hydrodynamic loads/actions on all members. The calculated hydrodynamic load/action on conductors would then be further reduced by the shielding factor.

A.9.5.2.5 Conductor Shielding Factor

The empirical basis for the shielding factor for the hydrodynamic load/action on conductor arrays is shown in Figure A.9.5-8. Data from flow directions perfectly aligned with a row or column of the array are excluded for conservatism.

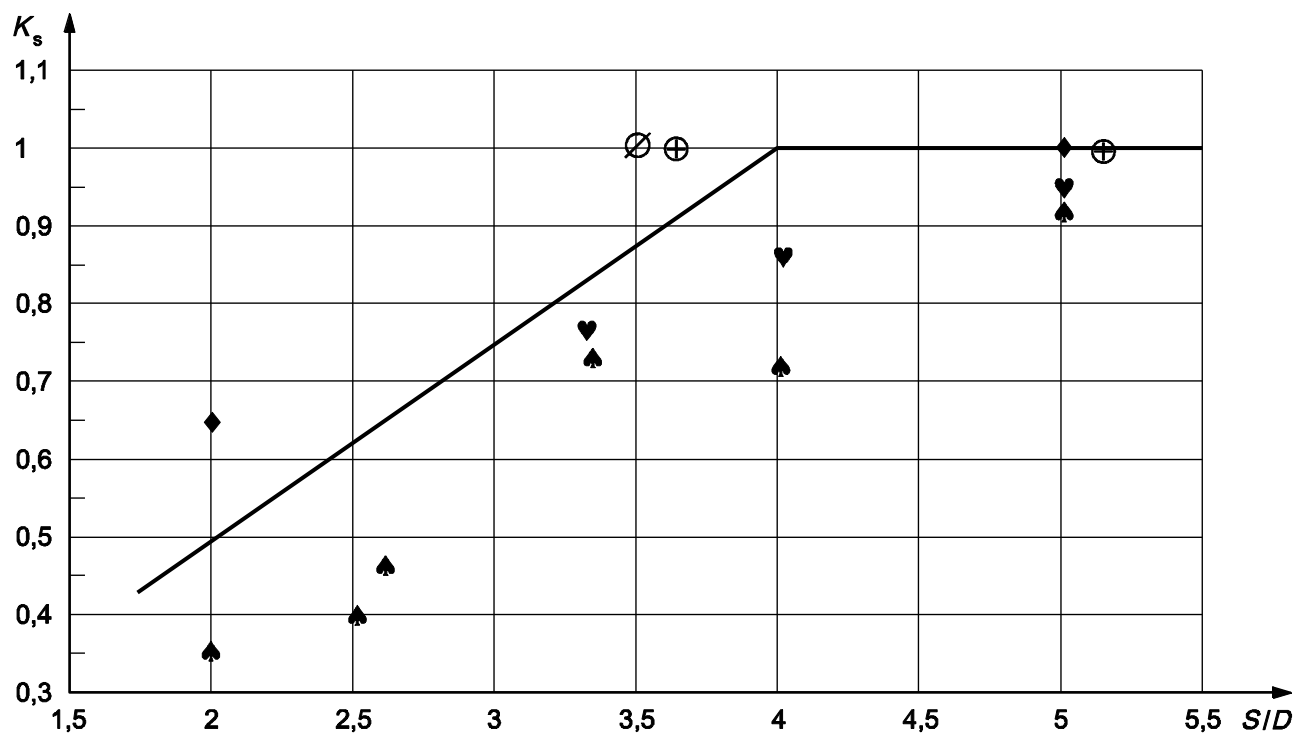
The data in Figure A.9.5-8 are from steady-flow tests and oscillatory-flow tests at very high amplitudes of oscillation. Thus, the factor is strictly applicable only in a steady current with negligible waves or near the mean water level in very large waves. The data in Reference [A.9.5-25] indicate that the factor is applicable if $A/S > 6$, where A is the amplitude of oscillation and S is the center-to-center spacing of the conductors in the wave direction.

The data in Reference [A.9.5-26] indicate that the range of applicability can be expanded to $A/S > 2.5$. For lower values of A/S , there is still some shielding until $A/S < 0.5$ [A.9.5-25]. With $A \approx U_{m0} T_{app}/2\pi$, where U_{m0} is defined in A.9.5.2.3 and T_{app} is the apparent wave period, respectively, the approximate shielding regimes are as follows:

- $A/S > 2.5$, asymptotic shielding, factor from Figure A.9.5-8;
- $0.5 < A/S < 2.5$, partial shielding;
- $A/S < 0.5$, no shielding, factor = 1.0.

In the absence of better information, the shielding factor in the partial shielding regime for any value of S/D may be linearly interpolated as a function of A/S , between the factor from Figure A.9.5-8 for $A/S = 2.5$ and 1.0 for $A/S = 0.5$. Waves considered in fatigue analyses can lie in the partial shielding regime.

NOTE See also A.9.5.2.4 for the relationship of shielding to current blockage.



Key

- S conductor spacing
- D conductor diameter
- K_s shielding factor
- ♥ Reed (1990) waves ($K = 126$)
- ♦ Heideman (1985) waves ($K = 250$) and current
- ♠ Reed (1990) current
- ⊕ Beckman (1979) waves and current
- ⊗ Sterndorff (1990) waves

Figure A.9.5-8—Shielding Factor for Loads/Actions due to Waves on Conductor Arrays as Function of Conductor Spacing

A.9.5.2.6 Apparent Wave Period

Kirby and Chen [A.9.5-27] developed a consistent first-order solution for the apparent wave period of a wave propagating on a current with an arbitrary profile. Their procedure requires the solution of the following three simultaneous equations for T_{app} , λ , and V_l :

$$\frac{\lambda}{T} = \frac{\lambda}{T_{app}} + V_l \quad (A.9.5-3)$$

$$T_{app}^2 = \frac{2\pi\lambda}{g \tanh(2\pi d/\lambda)} \quad (A.9.5-4)$$

$$V_i = \frac{(4\pi/\lambda)}{\sinh(4\pi d/\lambda)} \int_{-d}^0 U_c(z) \cosh\left[\frac{4\pi(z+d)}{\lambda}\right] dz \quad (\text{A.9.5-5})$$

Here λ is wave length, T is the wave period seen by a stationary observer, T_{app} is the wave period seen by an observer moving at the effective in-line current speed V_i , g is the acceleration due to gravity, $U_c(z)$ is the component of the steady current profile at elevation z (positive above storm mean level) in the wave direction, and d is storm water depth. For the special case of a uniform current profile, the solution to these equations is provided in dimensionless form in Figure 9.5-3.

Some other sources consider the wave period seen by a stationary observer to be the apparent period rather than the actual period. Users of wave loading software should determine the assumptions and terminology assumed in developing the software and ensure that the data input is consistent with that approach.

A.9.5.2.7 Two-dimensional Wave Kinematics

There are several wave theories that can be used to predict the kinematics of the two-dimensional, regular waves used for static, deterministic wave load calculations. The different theories all provide approximate solutions to the same differential equation and boundary conditions. All compute a wave that is symmetric about the crest and propagates without changing shape. They differ in their functional formulation and in the degree to which they satisfy the nonlinear kinematic and dynamic boundary conditions at the surface of the wave.

Linear wave theory is applicable only when the linearization of the free surface boundary conditions is reasonable, that is, when the wave amplitude and steepness are infinitesimal. Stokes V (Sarpkaya and Isaacson [A.9.5-5]) is a fifth-order expansion about mean water level and satisfies the free surface boundary conditions with acceptable accuracy over a fairly broad range of applications, as shown in Figure 9.5-4. Chappellear's [A.9.5-28] theory is similar to Stokes V but determines the coefficients in the expansion numerically through a least squares minimization of errors in the free surface boundary conditions, rather than analytically. EXVP-D (Lambrakos [A.9.5-29]) satisfies the dynamic boundary condition exactly and minimizes the errors in the kinematic boundary condition. Stream function theory (Dean and Perlin [A.9.5-30]) satisfies the kinematic boundary condition exactly and minimizes the errors in the dynamic boundary condition.

When Stokes V theory is not applicable, higher-order Chappellear, EXVP-D, or stream function theory may be used. Of these, the most broadly used is stream function. Selection of the appropriate solution order can be based on either the percentage error in the dynamic boundary condition or the percentage change in velocity or acceleration in going to the next higher order. These two methods select comparable solution orders over most of the feasible domain but differ in the extremes of $H > 0.9H_b$ and $d/gT_{\text{app}}^2 < 0.003$. In these extremes, the theory has not been well substantiated with laboratory measurements and should therefore be used with caution. In particular, the curve for breaking wave height H_b shown in Figure 9.5-4 is not universally accepted.

A.9.5.2.8 Wave Kinematics Factor

In wave force computations with regular waves, the kinematics are computed assuming a unidirectional sea (long-crested waves all propagating in the same direction), whereas the real sea surface is comprised of short-crested, directional waves. In fact, the sea surface can be viewed as the superposition of many small individual wavelets, each with its own amplitude, frequency, and direction of propagation. Fortunately, the directional spreading of the waves tends to result in peak forces that are somewhat smaller than those predicted from unidirectional seas. This force reduction due to directional spreading can be accommodated in static, deterministic wave force design procedures by reducing the horizontal velocity and acceleration from a two-dimensional wave theory by a "spreading factor."

There is generally much less directional spreading for wave frequencies near the peak of the wave spectrum than for higher frequencies (Forristall [A.9.5-31], for example). Since the kinematics of the large, well-formed individual waves used in static design are dominated by the most energetic wave frequencies, it is appropriate to use a “spreading factor” corresponding to the spectral peak period. Use of a weighted average spreading factor over all the wave frequencies in the spectrum would be unconservative. The spreading factor can be estimated either from measured or hindcast directional spectral wave data as $\sqrt{(n+1)/(n+2)}$, where n is the exponent in the $\cos^n\theta$ spreading function at the spectral peak frequency. Note that measured directional data from pitch/roll buoys tend to significantly overestimate spreading, whereas directional data from a two-horizontal axis particle velocimeter are thought to provide a good estimate of spreading.

There is some evidence that, even in sea states with very little directional spreading, two-dimensional stream function or Stokes V theory overpredicts the fluid velocities and accelerations (Skjelbreia et al. [A.9.5-32]). This may be attributed to the irregularity of the real wave, that is, its front-to-back asymmetry about the wave crest and its change in shape as it propagates. If an “irregularity factor” less than unity is supported by high-quality wave kinematics data, including measurements in the crest region above mean water level, appropriate for the types of design-level sea states that the platform may experience, then the “spreading factor” can be multiplied by the “irregularity factor” to get an overall reduction factor for horizontal velocity and acceleration.

A.9.5.2.9 Combined Wave/Current Kinematics

Dalrymple and Heideman [A.9.5-33] and Eastwood and Watson [A.9.5-34] showed that waves alternately stretch and compress the current profile under crests and troughs, respectively. Dalrymple and Heideman found that a model that combined Doppler-shifted wave kinematics with a nonlinearly stretched current profile gave the best estimate of global loads on a structure. Nonlinear stretching computes the stretched current for a particle instantaneously at elevation z as the speed $U_c(z')$ evaluated from the specified current profile at elevation z' , the mean elevation of the particle over a full wave cycle. The elevations z and z' are related through linear (Airy) wave theory as follows:

$$z = z' + \eta \frac{\sinh\left[2\pi(z' + d)/\lambda_n\right]}{\sinh(2\pi d/\lambda_n)} \quad (\text{A.9.5-6})$$

Here d is storm water depth, η is the wave surface directly above the water particle, and λ_n is the wave length determined from nonlinear wave theory for a wave of height H and period T_{app} . The elevations z , z' , and η are all positive above storm mean water level.

This equation gives a nonlinear stretching of the current, with the greatest stretching occurring high in the water column, where the particle orbits have the greatest radii. The nonlinearly stretched current profile, coupled with Doppler shifted wave kinematics, produces global platform loads that are within +1 % to –4 % of those produced by the exact solution on a typical drag-dominant structure subjected to representative waves and current profiles.

Another acceptable approximate model for many applications is one that uses a linearly stretched current profile, with:

$$z + d = (z' + d)(d + \eta)/d \quad (\text{A.9.5-7})$$

The stretched current profiles from the two models are compared qualitatively in Figure A.9.5-9 for typical sheared and slab current profiles under a wave crest. The linearly stretched current produces global loads on a typical drag-dominant platform that are nearly as accurate as those produced by the nonlinearly stretched current, being within 0 to –6 % of loads produced by the exact solution. However, it does not simulate the combined wave/current velocity profile from the exact solution as faithfully as nonlinear stretching.

Vertical extrapolation of the input current profile above mean water level produces reasonably accurate estimates of global loads on drag-dominant platforms in most cases. In particular, for a slab profile thicker than about 50 m, like the recommended profiles in API 2MET, vertical extrapolation produces nearly the same result as nonlinear stretching, as illustrated in Figure A.9.5-9. However, if the specified profile $U_c(z')$ has a very high speed at mean water level, sheared to much lower speeds just below mean water level, the global force may be overestimated (by about 8 % in a typical application).

Another approximate model is the linearly stretched model described above, adjusted so that the total momentum in the stretched profile from the seafloor to the wave surface equals that in the specified profile from the seafloor to mean water level. This procedure is not supported by the theoretical analyses of Dalrymple and Heideman [A.9.5-33] or Eastwood and Watson [A.9.5-34].

If the current is not in the same direction as the wave, the methods discussed above may still be used, with one modification. Both the in-line and normal components of current would be stretched, but only the in-line component would be used to estimate T_{app} for the Doppler-shifted wave.

While no exact solution has been developed for irregular waves, the wave/current solution for regular waves can be logically extended. In the first two approximations described above for regular waves, the period and length of the regular wave should be replaced with the period and length corresponding to the spectral peak frequency.

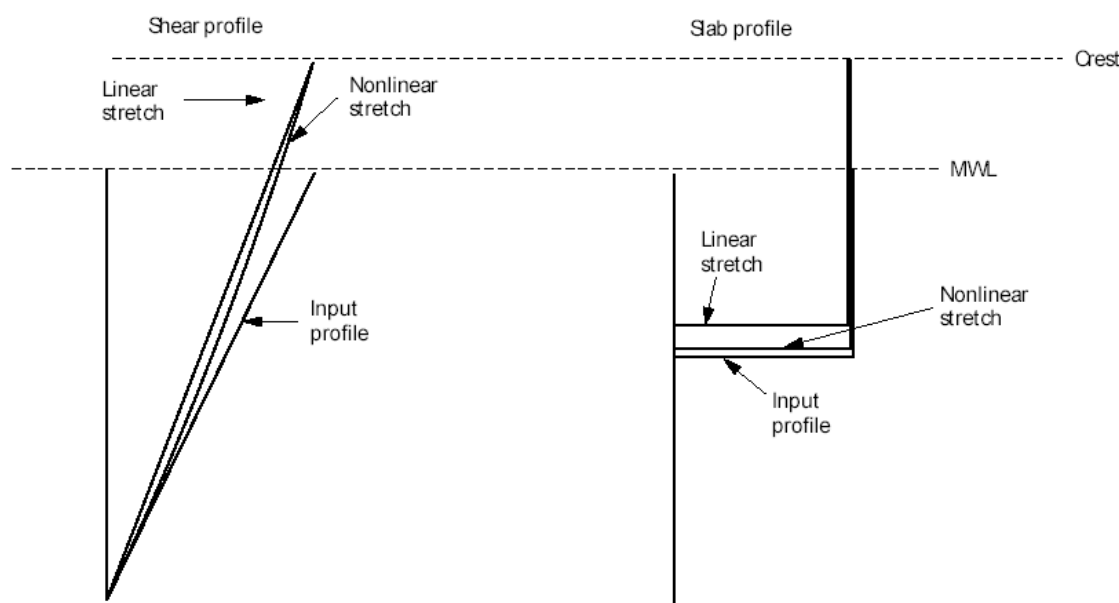


Figure A.9.5-9—Comparison of Linear and Nonlinear Stretching of Current Profiles

A.9.5.3 Hydrodynamic Models for Appurtenances

The hydrodynamic model of a structure is used for the calculation of hydrodynamic loads/actions that represent the global hydrodynamic load/action on the actual structure. The model does not explicitly need to include every component of the structure, provided the dimensions and/or hydrodynamic coefficients of the components included account for the contribution of the omitted components to the global hydrodynamic load/action. The hydrodynamic model should account for the effects of marine growth and for flow interference effects (blockage and shielding) where appropriate.

Appurtenances include components such as boat landings, fenders or bumpers, walkways, stairways, grout lines, and anodes. Although it is beyond the scope of this annex to provide modeling guidance for every appurtenance, some general guidance is provided.

Boat landings generally consist of a large number of closely spaced tubular members. If the components are modeled individually, shielding effects, depending upon the wave direction, may be considered in a similar manner to those for conductor arrays. Alternatively, boat landings may be modeled as either rectangular solids or as one or more plates, with directionally dependent hydrodynamic properties. Guidance for hydrodynamic coefficients for solid shapes and plates can be found in Reference [A.9.5-15].

Conductor guide frames may also be modeled as rectangular solids or, if appropriate, as plates. In both cases, different coefficients are appropriate for vertical and horizontal fluid flow.

Large fenders or boat bumpers and their supports are usually modeled as individual members. They may be treated as nonstructural members, provided that their design has been shown to be adequate for their intended purpose. Walkways, stairways, and grout lines may be modeled using equivalent circular members, although they may be ignored where experience has shown this to be acceptable.

The treatment of anodes depends upon their number and size. Anodes may be modeled as equivalent circular cylinders. Alternatively, hydrodynamic loads/actions on anodes may be approximated by increasing the diameters and/or hydrodynamic coefficients of the components to which they are attached.

A.9.8 Equivalent Quasi-static Load/Action Representing Dynamic Response Caused by Extreme Wave Conditions

A.9.8.1 General

The conditions specified in 9.8 are generally not satisfied in comparatively light sea states. However, while dynamic response in light sea states can be important for checking of fatigue, it will generally not govern structural design for strength. Therefore, the principle of using an equivalent quasi-static load/action may, under the conditions stated, be applied to extreme environmental conditions.

A.9.8.2 Equivalent Quasi-static Load/Action (D_e) Representing the Dynamic Response

The equivalent quasi-static load/action, D_e , is determined at a global level and subsequently distributed over the height of the structure to produce the approximately correct extreme dynamic base shear and overturning moment in the design sea state. Structural design may then proceed as for a quasi-statically responding structure, using design loads/actions that are modified to account for dynamics. This makes design of dynamically responding structures in accordance with the conditions specified in 9.8.1 practical and yet entirely adequate.

The magnitude of D_e should be such that the probability of the total extreme dynamic response due to ($E_{wce} + D_e$) is approximately the same as that of the quasi-static response due to E_{wce} alone. The conditions given in 9.8.1 ensure that phase differences between the dynamic and the quasi-static responses are small enough to assume that this is achieved.

For structures with a sufficiently short fundamental natural period, when compared to the predominant excitation periods present in extreme wave conditions, D_e may be neglected entirely. However, dynamic response depends on several factors and it is difficult to make a universal statement as to when dynamic response is negligibly small. Generally speaking, dynamic response increases as the fundamental natural period of the structure increases and moves closer to the period of the peak of the design wave spectrum, thereby entering a range of periods where the wave spectrum has noticeable energy. Studies, as well as a history of successful performance, indicate that fixed structures of space frame configuration with fundamental natural periods shorter than 2.5 s to 3 s will

experience only a few percent dynamic amplification in the design sea state. For such structures, the dynamic amplification may be neglected. In accordance with the requirements of 9.8.1, this rule of thumb hence assumes peak spectral periods for design sea states of 13 s or longer. If the peak period of the design wave spectrum for such structures is shorter, the possible influence of dynamics should be evaluated. However, these observations should not be generalized to other types of structure or to other design situations.

NOTE See 12.5 for further information and guidance on dynamically responding structures and how dynamic behavior should be treated.

The dynamic response can be influenced by the following, in addition to its dependence on the ratio of the fundamental natural period of the structure to the period of the peak of the design wave spectrum.

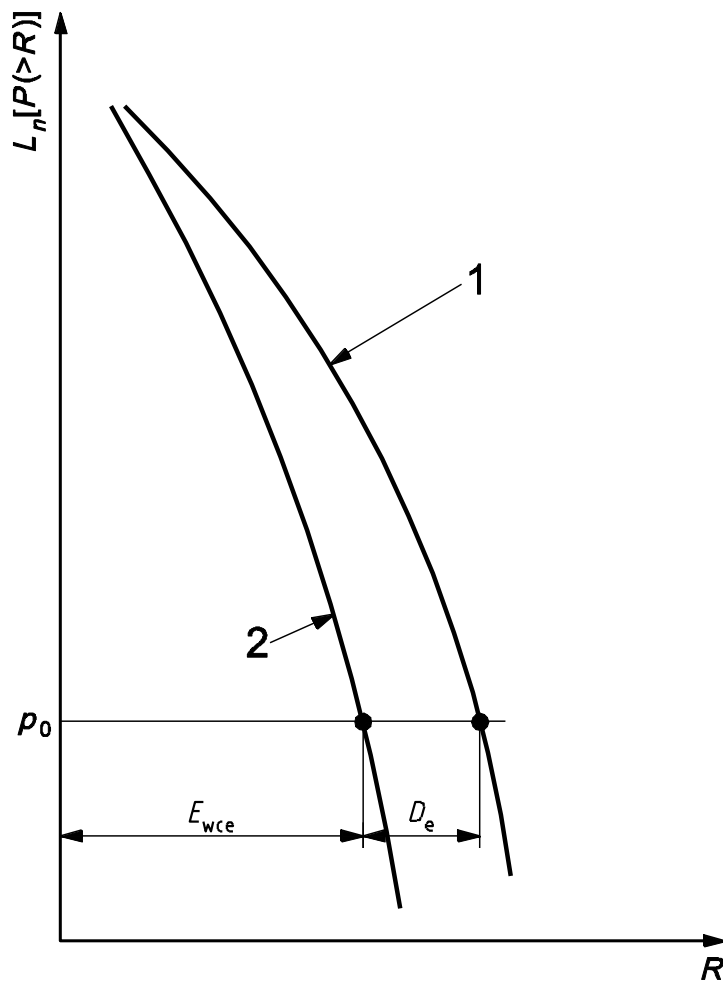
- a) Cancellation or reinforcement of wave load/action due to the spatial separation of members in the space frame, particularly between the legs. Structures that do not benefit from this effect, such as small structures with closely spaced legs or one leg only, are therefore potentially more susceptible to dynamic response from short period waves than large space frame structures. For these cases, the conditions given in 9.8.1 will generally not be satisfied and the influence of dynamics will again have to be assessed more carefully in accordance with Section 12.

NOTE When directional spreading is taken into account, cancellation/reinforcement effects are reduced.

- b) Higher harmonic frequency components of the hydrodynamic load/action originating from nonlinear drag loads/actions and free surface inundation effects. These effects require special evaluation if significant.

The indirect approach presented in 9.8.2 b) for the determination of D_e uses a global dynamic analysis to estimate the DAF, $k_{DAF,BS}$, for base shear, where $k_{DAF,BS}$ is the ratio of the dynamic base shear to the static base shear. In time domain analyses, $k_{DAF,BS}$ can be estimated by taking the average of the values of $k_{DAF,BS}$ for a few dozen short wave segments that include the design wave. A robust method of deducing $k_{DAF,BS}$ from time domain analyses is to plot order statistics of dynamic response and quasi-static response to the same realization of the design sea state. The ratio of the dynamic to quasi-static response, at the level of the exceedance probability of the design wave in the sea state, can be used to obtain a reliable estimate of $k_{DAF,BS}$.

Alternatively, the calculation may be taken into long-term statistics; the curves in Figure A.9.8-1 may be estimated by simulation and then $k_{DAF,BS}$ taken off directly as $(E_{wce} + D_e)/E_{wce}$. For truly linear problems and frequency domain methods, $k_{DAF,BS}$ reduces to the ratio of the RMS dynamic to the RMS static base shear response when there is no current.



Key

- 1 with dynamics
- 2 no dynamics
- R global response

Figure A.9.8-1—Definition of D_e

D_e should next be distributed over the height of the structure to form a set of inertial loads/actions, d_k , that is added to the environmental loads/actions at the elevations, k . This distribution may be done in proportion to the fundamental mode shape of the structure. The procedure is as follows:

$$D_e = \sum_{k=1}^K d_k = \sum_{k=1}^K \alpha_1 \omega_1^2 m_k \phi_{1k} = \alpha_1 \omega_1^2 \sum_{k=1}^K m_k \phi_{1k} \quad (\text{A.9.8-1})$$

$$d_k = \alpha_1 \omega_1^2 m_k \phi_{1k} = \frac{m_k \phi_{1k}}{\sum_{k=1}^K m_k \phi_{1k}} \times D_e \quad (\text{A.9.8-2})$$

where

D_e is the extreme dynamic base shear minus the extreme quasi-static base shear;

d_k is the inertial load/action at elevation k ($k = 1, 2, \dots, K$);

α_1 is a scale factor for the first mode shape (having the dimension of a length);

ω_1 is the first natural frequency of the structure;

m_k is the lumped mass at elevation k ;

ϕ_{1k} is the nondimensional displacement in first mode shape at elevation k .

This results in a set of distributed loads/actions, d_k , that reproduces the dynamic base shear, but generally not the dynamic overturning moment at the same time.

The above procedure for the equivalent quasi-static base shear may also be applied to the equivalent quasi-static overturning moment M_e to determine an analogous distribution. This results in Equations (A.9.8-3) and (A.9.8-4):

$$M_e = \sum_{k=1}^K d_k h_k = \sum_{k=1}^K \beta_1 \omega_1^2 m_k \phi_{1k} h_k = \beta_1 \omega_1^2 \sum_{k=1}^K m_k \phi_{1k} h_k \quad (\text{A.9.8-3})$$

$$d_k = \beta_1 \omega_1^2 m_k \phi_{1k} = \frac{m_k \phi_{1k}}{\sum_{k=1}^K m_k \phi_{1k} h_k} \times M_e \quad (\text{A.9.8-4})$$

where, additionally,

M_e is the extreme dynamic overturning moment minus the extreme quasi-static overturning moment;

h_k is the height of mass point, m_k , above the sea floor;

β_1 is a scale factor for the first mode shape (having the dimension of a length).

As the scale factors, α_1 and β_1 , are usually not the same, the dynamic overturning moment is now correctly reproduced, but the dynamic base shear is not.

Using both the magnitudes of D_e and M_e , a linear combination of the first and the second mode may be used to determine a distribution of inertial loads/actions such that both the dynamic base shear and the dynamic overturning moment are reproduced by the set of inertial loads/actions. This is done in the following manner:

$$D_e = \sum_{k=1}^K d_k = \sum_{k=1}^K \left(\gamma \omega_1^2 m_k \phi_{1k} + \delta \omega_2^2 m_k \phi_{2k} \right) = \gamma \omega_1^2 \sum_{k=1}^K (m_k \phi_{1k}) + \delta \omega_2^2 \sum_{k=1}^K (m_k \phi_{2k}) \quad (\text{A.9.8-5})$$

$$M_e = \sum_{k=1}^K d_k h_k = \sum_{k=1}^K \left(\gamma \omega_1^2 m_k \phi_{1k} h_k + \delta \omega_2^2 m_k \phi_{2k} h_k \right) = \gamma \omega_1^2 \sum_{k=1}^K (m_k \phi_{1k} h_k) + \delta \omega_2^2 \sum_{k=1}^K (m_k \phi_{2k} h_k) \quad (\text{A.9.8-6})$$

$$d_k = \gamma \omega_1^2 m_k \phi_{1k} + \delta \omega_2^2 m_k \phi_{2k} \quad (\text{A.9.8-7})$$

where, additionally,

γ, δ are scale factors for the first and the second mode shape, respectively (having the dimension of a length);

ω_2 is the second natural frequency of the structure;

ϕ_{2k} is the nondimensional displacement in second mode shape at level k .

The two unknown scalars, γ and δ , are determined by solving the two Equations (A.9.8-5) and (A.9.8-6) for D_e and M_e simultaneously, after which the set of inertial loads/actions, d_k , can be determined from Equation (A.9.8-7).

A.9.8.3 Global Dynamic Analysis in Waves

A.9.8.3.3 Design Sea State

Random waves are generally modeled as a linear superposition of a large number of sinusoidal (Airy) components, each with different amplitude, frequency, and phase. For dynamic analyses, random waves can be modeled either in the time domain or in the frequency domain. Time domain simulation may be considered the most general of the methods because nonlinearities can be directly included. Results from such simulations therefore serve as a reference with which other approaches can be compared. Nonlinearities in the environmental loads/actions arise from the squared drag term in Morison equation and the inundation effects as the wave surface moves from trough to crest. Nonlinearities from the structural system come from a nonlinear foundation, and from large displacements and yielding of members and joints. When linearization of these nonlinearities can be justified, frequency domain analysis provides a very efficient and easy-to-interpret means of using random waves for the determination of global k_{DAF} factors.

In the frequency domain, each frequency component is considered separately, one at a time, and the response of the structure to that frequency component is determined. In this manner, a “frequency response function” is established, which is then multiplied by the wave spectrum of the design sea state in order to obtain the response spectrum. For linear problems, this frequency response function is independent of the amplitude chosen for each wave component. The influence of nonlinearities may be included in an approximate manner if each wave component is modeled with a realistic finite steepness. However, for a finite steepness, the linear (Airy) wave kinematics theory that applies below the mean still water level should be modified accordingly. Extrapolation of the theory to elevations above the still water level results in particle velocities that are (much) too high. Adequate models that are commonly employed use vertical extrapolation above water level, Wheeler (linear) stretching [A.9.8-1], and delta stretching [A.9.8-2] (see also API 2MET).

“Frequency response function” is the most correct name for the response of a linear system to harmonic excitation: it is a complex-valued function of frequency having magnitude (the modulus) and phase (the argument). The frequency response function is often also called “transfer function” or “response amplitude operator” (RAO). In general, the transfer function also has magnitude and phase, but in many cases it refers more particularly to the magnitude (the modulus) only. RAO is by definition a scalar function associated with the modulus only.

In the time domain, the response of the structure to the simultaneously applied frequency components is determined in the form of a time series. Each component should be given an appropriate amplitude and phase

before they are combined. Typically, the amplitudes are randomly selected from a Rayleigh distribution in accordance with the discretized design wave spectrum, whereas phases may be randomly selected from a uniform distribution over 0° to 360° . A wave model thus derived is referred to as a simulation model. An alternative method is to determine the amplitude and phase of each component from a Fourier analysis of a measured wave elevation trace that corresponds with the design wave spectrum, if such a measured wave trace is available. Such a wave model is referred to as a “conditional” model. Other methods for selecting amplitudes and phases are the subject of research. Both a simulation and a conditional model require modification of the linear (Airy) kinematics, as with the frequency domain approach. A weakness of the simulation model as it is usually applied is that it produces a Gaussian water surface elevation and does not reproduce the skewness observed in real wave records (higher, sharper crests and shallower, flatter troughs). The conditional model incorporates these effects to some extent but cannot faithfully reproduce simultaneous wave traces at spatially separated points either, because the linear dispersion relationship does not propagate the components correctly. Random wave theories that account for the nonlinear interactions between frequency components and the consequent effects on wave kinematics and wave propagation are under development [A.9.8-3]. Methods that include directional spreading of wave energy are now available.

It should further be recognized that a time domain simulation is a single realization (a sample of finite length) of the full population of an infinite number of possible realizations of infinite length that represent the underlying random process, regardless of whether a simulation or a conditional wave model is used. The interpretation of the results of one (or a few) time domain simulations (samples) and their use to derive statistical properties that provide correct and unbiased results for the underlying random process is far from straightforward. If these difficulties are not carefully managed, the potential error associated with the use of time domain simulations may well be larger than the error involved in linearization to enable frequency domain methods to be applied. When time domain simulation is applied, the use of order statistics as mentioned in A.9.8.2 is strongly recommended.

A.9.8.3.4 Hydrodynamic Load/Action on a Member

For fixed structures, the relative velocity formulation of Morison equation should not be used for regular or random waves. When the displacement of the structure is less than the diameter of a representative member and the motion of the structure is at a high frequency compared to the frequency(ies) of the wave(s), the motion of an individual member interacts with its own wake and does not develop the hydrodynamic loads/actions predicted by the relative velocity formulation of Morison equation (see A.9.5.2.3 and, for example, Reference [A.9.8-4]). Using relative velocity would have the same effect on the dynamic analysis as using too large a value of the damping. In this situation, as the damping force and the environmental load/action do not interact, the use of a formulation with absolute water particle velocities is more realistic. The associated viscous damping coefficient should be estimated in the absence of waves.

A.9.8.3.6 Damping

Estimating damping coefficients is difficult; reliable data for relevant circumstances are extremely scarce if available at all. However, for the conditions to which 9.8 applies, dynamic response is stiffness controlled, thus the influence of damping is moderated and the selection of a damping coefficient is not critical to the results. The value of 2 % to 3 % can be slightly conservative for extreme wave conditions but has little influence on D_e . For the lighter sea states that are relevant for fatigue analyses, the damping coefficient should be reduced to a value of 1 % to 2 % of critical.

A.9.8.3.7 Stiffness

For fixed structures, a linear stiffness model is normally sufficient; second-order geometrical stiffness effects (so-called $P-\Delta$ effects) are only relevant for more flexible structures and do not normally need to be included in the global dynamic analysis. Linearization of the foundation stiffness, if required, should be performed at a suitable response level reflecting the response in the extreme sea state. For fatigue analyses, it is generally appropriate to consider a lower response level and a correspondingly stiffer foundation.

When the objective is to determine a set of inertial loads/actions for use with a nonlinear ultimate strength analysis, nonlinear stiffness properties should be maintained.

A.9.9 Factored Loads/Actions

A.9.9.2 Factored Permanent and Variable Loads/Actions

Partial load/action factors for permanent and variable loads/actions were determined through a calibration process that also included wind, wave, and current loads/actions. No distinction was made between G_1 and G_2 or Q_1 and Q_2 , so the partial load/action factors are the same. Nevertheless, it is useful to establish these subsets, in which the loads/actions have common degrees of uncertainty and duration of occurrence.

The calibration process resulted in a choice of 1.3 for the partial load/action factor on permanent loads/actions and of 1.5 for the partial load/action factor on variable loads/actions. These choices result in a larger component safety index when using load and resistance factor design practice than when using working stress design (WSD) practice for components with both environmental and permanent loads/actions. While the reliability concept of uniform component safety could justify lower factors, the results would be a wide departure from past practice.

The component safety index calibration used the following statistical model: Q_1 and Q_2 have a coefficient of variation (COV) of 14 % and the mean value is the nominal value. G_1 and G_2 have a COV of 8 % and the mean value is the nominal value. For the calibration, the internal force due to permanent and variable loads/actions was assumed to be 25 % of $(G_1 + G_2)$ and 75 % of $(Q_1 + Q_2)$.

A.9.9.3 Factored Extreme Environmental Load/Action

A.9.9.3.1 General

Where no information on partial load/action factors that are specific to the case under consideration (e.g. robustness level ultimate strength analysis required in 6.5.1) is available, these factors may be taken to be $\gamma_{f,E} = 1.35$ and $\gamma_{f,D} = 1.25$ in accordance with the factors that were proposed in Reference [A.9.9-3], based on calibration work done for the Gulf of Mexico. In subsequent calibration work along similar lines, these factors were generally also found to be appropriate for some other areas for L-1 platforms. However, note the reservation expressed in the discussion in A.9.9.3.2. It is important to also consider that the calibration work for the Gulf of Mexico only considered what is now L-2 level extreme environmental loads/actions (L-1 at that point in time before the change in 100-year criteria for the Gulf of Mexico).

A.9.9.3.2 General Discussion of Partial Factors

The numerical values of the partial load/action factors (for example $\gamma_{f,E} = 1.35$ and $\gamma_{f,D} = 1.25$) as well as the resistance factors given in this document have been largely derived on the basis of the original calibration work by Moses [A.9.9-3] to [A.9.9-7]. A calibration like this is suitable for an offshore province, such as the Gulf of Mexico, where the experience base is very extensive and satisfactory. At the same time, there is evidence that these structures are not overdesigned. For other geographical areas where such ample experience and evidence does not exist, calibration to the previous API WSD practice of that area is possible, but it does not achieve harmonization of safety levels worldwide. For the specific case of loads/actions due to extreme storms, it is known that the long-term distribution of environmental loads/actions is a function of the geographic location. For structures with the same geometrical and structural properties, harmonization in safety levels hence requires location dependent partial load/action factors.

Reliability techniques can play a significant role in achieving harmonization, but such techniques need to be used differently from the calibration work referred to above in order to overcome the weaknesses of calibrations relative to the local WSD practice. This discussion lists the properties of more refined reliability models that may be used

to address the above issues and to provide appropriate partial load/action factors for use with joint environmental conditions in different geographical locations. It is necessary that such models achieve results that are close to actuarial results rather than being notional results only.

Appropriate reliability models should include all of the following distinguishing features.

- a) Description of the environment—the models take into account the joint probability of waves, currents, and winds in defining the 100-year extreme as well as the complete long-term distribution of the extreme environmental load/action. For those aspects, both modeling and physical uncertainty will depend on the geographical area.
- b) Wave load models—the models for loads/actions due to waves incorporate a wave load recipe that accounts for the three-dimensional features of real seas and use realistic hydrodynamic coefficients.
- c) Ultimate strength of the structural system rather than of individual components.

A.9.9.3.3 Partial Load/Action Factor, $\gamma_{f,E}$

Reliability models and methods should be tested prior to their application in calculating new partial load/action factors or RSRs. Their predictions of failure rates should be compared with historical values for a geographical area where there is experience of structural survival (and failure) in extreme storms and well-proven metocean data. Calculated values of the partial load/action factor and RSR should be compared with established values for that area. As well as testing the method, the exercise aids the engineering judgment of subsequent results.

Some points should be borne in mind in relation to modeling uncertainty. Different areas can require different treatment of the modeling uncertainty to be associated with metocean data, both in relation to confidence in the design environmental load/action and the statistics of extreme loads/actions. When greater modeling uncertainty is assumed (in any part of the calculation), it will tend to increase the partial factors required to achieve a given target reliability.

A.9.9.3.4 Partial Load/Action Factor, $\gamma_{f,D}$

D_e and E_e are assumed to act in the same direction, implying that the dynamic response is in phase with the applied environmental load/action. This is applicable to stiffness controlled dynamic response where the natural period is shorter than the predominant period of the applied loading, as discussed previously.

The 1.25 factor is based on a reliability analysis by Moses [A.9.9-14]. The calibration for the permanent loads/actions and the extreme quasi-static environmental load/action was repeated with the addition of another random variable, the inertial partial load/action factor ($\gamma_{f,i}$). The random resistance was compared to the random internal force (S); thus S is due to the following F_d :

$$F_d = G + Q + E_e (1 + \gamma_{f,i}) \quad (\text{A.9.9-1})$$

where G , Q , E_e , and $\gamma_{f,i}$ are all random variables.

Note that $(1 + \gamma_{f,i})$ is the ratio of the total dynamic (static and inertial) member internal force to the static member internal force due to only E_e . The variable $\gamma_{f,i}$ was taken to have a COV of 60 %. The mean value of $\gamma_{f,i}$ was taken to have a range of 0 up to 0.8 to represent the fact that $\gamma_{f,i}$ values of individual members can be considerably larger than for a global response, such as base shear. The factor of 1.25 ensures that in any of these cases, the safety index (β) for a member is at least as great as the case when D_e equals 0.

The result of the calibration and the value of $\gamma_{f,D}$ obtained by it depends on the degree of detail and refinement of the dynamic analysis performed. Some experience indicates that $\gamma_{f,D}$ may be taken as 1.0, provided that the requirements of 9.8 and recommendations of A.9.8 are carefully followed.

A.9.10 Design Situations

A.9.10.3 Partial Factor Design Format

A.9.10.3.2 Design Loads/Actions for In-place Situations

A.9.10.3.2.1 Design Loads/Actions for Operating Situations

Where permanent and/or variable loads/actions on the structure during temporary operating conditions are significantly more severe than normal maximum conditions (e.g. during hydro-test), it can be appropriate to define limiting environmental conditions in which the particular mode of operation may proceed. These limiting environmental conditions should then be applied when determining the environmental load/action for that mode of operation.

Typically, a 1-year to 5-year winter storm is used as an operating wind, wave, and current condition in the Gulf of Mexico. In the South China Sea, for example, 1-year return period conditions have traditionally been used. Since these are mild by comparison with the extreme conditions, they do not have an unduly severe impact on the design. In the North Sea, however, 1-year environmental conditions are relatively severe by comparison with 100-year conditions, and their use would have an unduly large impact on design. In this area, the operating condition is most commonly considered in conjunction with specific activities undertaken on the platform, e.g. running drill pipe or crane operations. The operating environment is consequently determined as being the maximum condition in which the associated activities can be undertaken (see API 2TOP [20]).

“Operating environmental conditions” combined with permanent and variable loads/actions (and without a one-third allowable stress increase) has been a load case in API 2A-WSD [A.9.10-1] since the First Edition. This recognizes that some environmental load/action due to wave, wind, and current will be present in combination with the maximum permanent and variable loads/actions. For this reason, and because it is part of the current Gulf of Mexico practice, an operating strength check was maintained in API 2A-LRFD [A.9.10-2].

Unlike the environmental load/action due to extreme wind, wave, and current conditions, the environmental load/action due to operating wind, wave, and current conditions was intentionally not tied to a specific return period event in API 2A-LRFD [A.9.10-2]. The partial factor calibration was performed without making use of the operating wind–wave–current case. The use of Equation (9.10-1) with E_o corresponding to “moderately severe conditions” in the Gulf of Mexico adds a small but uncalibrated increment to member size and safety for members loaded principally by permanent loads/actions. Outside the Gulf of Mexico, if “moderately severe conditions” are defined as the same return period event, Equation (9.10-1) can be conservative beyond the original intention. Equation (9.10-1) can produce smaller design internal forces compared to API 2A-WSD for certain cases, particularly for members that are subjected to mostly (but not 100 %) permanent loads/actions in combination with an environmental load/action due to operating environmental conditions (E_o) that exceed approximately $0.5 E_e$.

In API 2A-LRFD, an overall partial load/action factor of 1.2 was selected to cover the analysis uncertainty that is associated with all load/action factors. Since the operational wind–wave–current check does not check safety for unusual environmental loads/actions but rather primarily for permanent loads/actions, a factor less than 1.35 is appropriate. To generalize the intent and this result in an approximate manner to geographical areas other than the Gulf of Mexico, an additional reduction factor on the environmental loads/actions is introduced, giving $\gamma_{f,Eo} = 0.9 \gamma_{f,E}$.

A.9.10.3.2.2 Design Loads/Actions for Extreme Conditions When the Load/Action Effects Oppose

The partial load/action factors in Table 9.10-1 for extreme conditions are from the reliability calibration performed for the Gulf of Mexico structures [A.9.10-3]. The calibration considered wind, wave, and current loads/actions, permanent loads/actions, and the strengths of various structural components. The extreme environmental conditions exclude Q_2 ($\gamma_{f,Q2} = 0$) due to its short duration and small probability of occurring simultaneously with a storm.

The bottom line of Table 9.10-1 covers the situation of counteracting effects when two representative internal forces are similar in magnitude but opposite in sign. The algebraic sum of the forces is small, which leads to a small required resistance. Due to the greater uncertainties in environmental loads/actions, the safety index (β) can be low (high risk). This phenomenon is intensified when the structure's component strength is different for changes in the sign of the internal force (e.g. piles in tension instead of compression or axial members in compression instead of tension).

In the API LRFD calibration, counteracting effects were considered an important load case for uplift on piles. The reduced partial load/action factors when load/action effects from permanent and variable loads/actions oppose those from environmental loads/actions ensure adequate pile tension capacity. A reduced factor on G_1 implicitly covers cases of reduced topsides loads/actions. The 0.8 on Q_1 is smaller than the 0.9 on G_2 , reflecting its greater likelihood of not being the value assumed. However, reductions are even applied to the very certain and always present G_1 , because it is the uncertainty in wind, wave, and current that this check is intended to guard against. The safety index is thereby raised to a level consistent with the pile compression failure mode.

For brace members in the structure, an analogous case of counteracting effects exists. A calibration structure (Platform “C”) [A.9.10-3] was investigated for this phenomenon by looking at the most critical component in each design group. Although the partial load/action factors in Table 9.10-1 for extreme conditions when the load/action effects are additive serve to raise β compared to WSD for opposing load cases, it was deemed prudent to have the values for when load/action effects oppose apply to all structure components to guard against low β cases.

A.9.11 Local Hydrodynamic Loads/Actions

The Morison equation accounts for local drag and inertia loads/actions but not for the “out-of-plane” (i.e. the plane formed by the velocity vector and the member axis) local loads/actions caused by hydrodynamic lift due to periodic, asymmetric vortex shedding from the downstream side of a member. Hydrodynamic lift can be neglected in the calculation of global loads/actions on the structure. Due to their high frequency, random phasing, and oscillatory (with zero mean) nature, lift loads/actions are not correlated across the entire structure. However, such lift loads/actions should be considered for local member design, particularly for members high in the structure whose internal forces can be dominated by locally generated loads/actions.

The oscillating lift loads/actions can be modeled as a modulated sine function, whose frequency is generally several times the frequency of the wave, and whose amplitude is modulated with U^2 , where U is the time-varying component of fluid velocity normal to the member axis. In the absence of dynamic excitation, the maximum amplitude of the local lift load/action, $F_{L,max}$, per unit length of the member is related to the maximum value of U during the wave cycle (U_{max}), by Equation (A.9.11-1):

$$F_{L,max} = C_{L,max} \times \frac{1}{2} \rho_w D U_{max}^2 \quad (\text{A.9.11-1})$$

The coefficient $C_{L,max}$ has been found empirically [A.9.11-1] to have considerable scatter, with an approximate mean value of $C_{L,max} \approx 0.7 C_d$ for both smooth and rough circular cylinders, and in both steady flow and waves with large Keulegan-Carpenter numbers. Reference [A.9.11-2] focuses on the RMS value of the oscillating lift load, finding that it was less than half $F_{L,max}$.

The frequency of the oscillating lift load/action is $Sr U_{\text{total}}/D$, where Sr is the Strouhal number and U_{total} is the total incident velocity, including the axial component. Laboratory tests [A.9.11-1, A.9.11-3] have shown that $Sr \approx 0.2$ for circular cylinders over a broad range of Reynolds numbers and flow inclination angles in steady flow. If Sr remains constant in waves, the frequency of the oscillating load/action varies as U varies with time during a wave cycle.

In the event that any natural frequency of a member is near the frequency of lift loads/actions, a large amplitude dynamic response called “vortex-induced vibration” (VIV) can occur. When VIV occurs, the motion of the member and the magnitude of the fluid-dynamic loads/actions can increase to unacceptable levels. VIV can occur on long spans due to loads/actions caused by wind in the construction yard and on the tow barge, as well as by waves and currents on the in-place structure.

Horizontal members in the wave zone (between wave crest and trough elevations) of an in-place structure can experience loads/actions due to wave slam. These nearly vertical loads/actions are caused by the local water surface rising and impacting on the underside of the member as a wave passes. Since these loads/actions are nearly vertical, they contribute very little to the base shear and overturning moment of the structure. However, loads/actions due to slamming should be considered in local design of members in the wave zone.

Loads/actions due to slamming can also occur on structural members overhanging a barge while the structure is being towed, or on members that strike the water first during side launching of structures.

In the theoretical case, slamming loads/actions are impulsive. If the slamming load/action is truly impulsive, the member can be dynamically excited. In the real world, the slamming load/action is possibly not impulsive because of the three-dimensional shape of the sea surface, the compressibility of air trapped between the member and the sea surface and the aerated nature of water near the free surface.

Slamming load/action, F_s , per unit length may be calculated using Equation (A.9.11-2):

$$F_s = C_s \times \frac{1}{2} \rho_w D U^2 \quad (\text{A.9.11-2})$$

where U is the component of water particle velocity normal to the member axis at impact.

Reference [A.9.11-2] shows empirically that the coefficient, C_s , lies between 0.5 and 1.7 times its theoretical value of π , depending on the rise time and natural frequency of the elastically mounted cylinder in tests. Reference [A.9.11-4] recommends that if a dynamic response analysis is performed, the theoretical value of $C_s = \pi$ may be used; otherwise, a value of $C_s = 5.5$ should be used.

Additional information on slamming loads/actions and dynamic response is provided in Reference [A.9.11-5].

Axial Froude-Krylov loads/actions have the same form as the inertia load/action in Morison equation, except that C_m is set to unity and the normal component of the local acceleration is replaced by the axial component. Such loads/actions on members that are nearly vertical contribute negligibly to the structure's base shear and overturning moment. Axial Froude-Krylov loads/actions on diagonal and horizontal braces are relatively more important, contributing approximately 10 % as much as the inertia load/action included in Morison equation to base shear and overturning moment, based on computations in Reference [A.9.11-6]. In view of approximations made elsewhere in the computation of global environmental load/action, axial Froude-Krylov loads/actions can generally be neglected.

A.10 Accidental Situations

A.10.1 General

Consideration should in particular be given to accidental events arising from vessel collision and dropped objects. The severity and likelihood of these accidental events depend on many factors, including operating environment, service vessel size, material handling equipment, operator qualifications, and proximity to shipping lanes. The specified accidental design situations and associated design criteria for collisions and dropped objects should therefore depend on the platform location, operational procedures adopted, and safety measures taken.

Design criteria for collisions and dropped objects are difficult to quantify for the following three reasons:

- a) the interaction of human and equipment error that combine to cause an accidental event cannot easily be quantified in probabilistic terms;
- b) satisfying arbitrary or simplistic quantification rules is no guarantee that the required safety for a given accidental event will be achieved;
- c) blindly satisfying simple rules can exclude the option of implementing potentially inexpensive preventive measures derived from a more in-depth understanding of the problem.

Where the consequences of an accidental event are very great, and the probabilities of events occurring are very low, preventive measures can serve to provide the required safety. For example, measures to prevent collision by tanker are acceptable if they are effective. Designing to resist a tanker collision is not a practical approach.

A.10.2 Vessel Collisions

A.10.2.2 Collision Events

The kinetic energy of a vessel may be calculated using Equation (A.10.2-1):

$$E = 1/2 (a \times m) u^2 \quad (\text{A.10.2-1})$$

where

E is the kinetic energy of the vessel;

a is the added mass coefficient, with $a = 1.4$ for a broadside collision, $a = 1.1$ for a bow/stern collision;

m is the vessel mass;

u is the velocity of vessel at impact.

The added mass coefficients given above are typical values for large (5000 t displacement) supply vessels. For smaller vessels, a value slightly higher than 1.4 should be applied, e.g. 1.6 for a typical 2500 t supply vessel.

The representative velocity and size of the vessels used for impact analyses should correspond to those used in the operation and servicing of the platform (e.g. supply boats). By way of example, for the northern North Sea, a vessel mass can be 8000 t, whereas in the southern North Sea a mass of around 2500 t is more normal. For Gulf of Mexico structures in mild environments and reasonably close to their base of supply, a 1000 t vessel represents a typical 55 m to 60 m (180 ft to 200 ft) supply vessel. For deeper and more remote locations in the Gulf of Mexico the vessel mass can be different. The masses of vessels that could collide with the platform when drifting out of control should be specifically considered.

The two energy levels specified for vessel impact analysis represent a frequent condition and a rare condition. The low energy level represents a serviceability condition based on economic considerations, and is intended to ensure that the structure will not require to be shut-in or need major repairs after minor collisions. The high energy level represents a rare condition and an ULS in which progressive collapse should not occur and the safety of personnel should be ensured, although the structure could be substantially damaged.

For low energy impacts, a vessel velocity of 0.5 m/s is commonly used, representing a minor accidental “bump” during normal maneuvering of the vessel while loading or unloading or while standing alongside the platform. For high energy conditions, a vessel velocity of 2 m/s is commonly used, representing a vessel drifting out of control in a sea state with significant wave height of approximately 4 m.

A.10.2.3 Collision Process

The majority of ship impact analyses performed to date have used quasi-static methods based on energy absorption. However, when the duration of the collision is short, dynamic effects can be significant.

The collision duration depends on the size and configuration of both the structure and the vessel, and on the nature of the collision. Dynamic effects can be significant when the duration is of the same order or less than the structure's natural period. In such cases, an assessment of the dynamic behavior during the collision should be considered.

Dynamic collision analyses can be carried out as time domain dynamic calculations in which the collision loads/actions represent both the direct impact and the inertia of the structure. Because the two excitations do not attain their maximum value at the same time, the duration of the time simulation should be long enough to cover all relevant phases of the collision. Typically, the direct impact due to the collision attains its maximum value early during the collision, while the effects of inertia reach their maximum values later during the collision. Energy is absorbed in both the structure and the vessel, but additional absorption sources such as energy imparted to platform vibration and energy dissipating from radiating waves generated as a result of the collision can also be represented.

A.10.4 Fires and Explosions

A number of documents address both hazard management in general (e.g. ISO 17776 [A.10.4-1]) and more specific issues associated with fires and explosions (e.g. API 2FB).

The industry associations have produced their own more detailed guidance applicable to particular types of operation and circumstances and these include API [A.10.4-2], which can be used for Gulf of Mexico type platforms; UKOOA [A.10.4-3, A.10.4-4, A.10.4-5, A.10.4-6], which are suited to larger platforms operated in a safety case regime; and NORSOK [A.10.4-7], which contains explicit analytical requirements.

A.11 Seismic Design Considerations

A.11.3 Seismic Reserve Capacity Factor

In both the simplified and detailed seismic load/action procedures, the ALE return period is determined so that the structure-foundation system is expected to meet the target annual probability of failure. The ELE return period is then determined from the ALE event so that a balance exists between the ELE and ALE requirements, i.e. that the ratio of ALE to ELE spectral accelerations is approximately equal to the inherent static and dynamic reserve strengths of the design. Having this balance, a fixed steel structure that is designed for ELE should have a high likelihood of meeting the ALE design demand so that costly design cycles can be avoided.

A good initial estimate for the seismic reserve capacity factor, C_r , will ensure that a balance exists between the ELE and ALE requirements. If the initial estimate of C_r is too high, the ALE check can lead to major design modifications and costly design cycles. On the other hand, if the initial estimate of C_r is too low, the design would be conservative and the ALE design check could be easily met.

Aside from having a balance between the ELE and ALE requirements, the guidelines also require that the ELE design should lead to an economically viable structure, i.e. the structure should not be severely damaged when subjected to more frequent earthquakes than the ALE. This economic objective is presented in terms of minimum ELE return periods which depend on the platform exposure level (L-1, L-2, or L-3). Because this economic objective is not explicitly checked in the simplified seismic load/action procedure, and to avoid ELE return periods that are too low, the C_r factor is limited to 2.8 for L-1 structures, 2.4 for L-2 structures, and 2.0 for L-3 structures.

A.11.4 Recommendations for Ductile Design

The recommendations given in 11.4 are mainly based on experience of seismic design of offshore California (USA) structures and are consistent with Reference [A.9.1-1]. The industry has shown that these recommendations lead to structural/foundation designs that are ductile and have high seismic reserve capacity factors.

The legs of the structure should be designed against twice the design environmental load/action, E , during the ELE event. This is to ensure that braces fail before the legs and to avoid a premature collapse of the structure. For primary joints, it is desirable that braces achieve their full compressive or tensile strength before the joints fail. In lieu of the recommendation for designing joints in accordance with 14.2.3, the survival of joints against loads/actions calculated in nonlinear ALE time-history analyses may be demonstrated in accordance with 11.6.4. Using this procedure, joints should be checked for internal member forces at each time step of the time-history analysis.

A.11.7 Topsides Appurtenances and Equipment

Due to the structure's dynamic response and amplification of the seismic accelerations, topsides design accelerations may be much larger than those at the sea floor. The response spectra for different points on the topsides can vary significantly depending on the asymmetry of the design and the location of the center of mass of the topsides.

In general, most properly anchored topsides appurtenances are sufficiently stiff that their lateral and vertical responses may be assumed to be the same as the maximum accelerations of their support points on the topsides. Drilling rigs, flare booms, deck cantilevers, tall vessels, large unbaffled tanks, and cranes do not meet this rigid body criterion.

For topsides equipment, the partial load/action factor increase from 0.9 to 1.15 for strength level earthquake loads/actions is intended to provide a margin of safety in lieu of performing an explicit ductility level analysis for those components.

Two methods of analysis for topsides equipment, piping, and appurtenances are allowed: coupled and uncoupled analyses. A coupled analysis can result in more accurate, and often lower, design accelerations than those derived using uncoupled floor response spectra. A coupled analysis may be used when the effects of dynamic interaction are significant. Dynamic interaction becomes significant for more massive topsides components with natural periods that are close to the dominant modal periods of the structure. The variability in natural periods of components can have a significant impact on the reactions between components when performing coupled analyses, and it can be necessary to perform sensitivity analyses to determine the range of the reactions due to modeling uncertainty. A coupled analysis is not necessary when the topsides equipment mass is less than 1 % of the global mass of the whole platform.

Equipment should be restrained by means of welded connections, anchor bolts, clamps, lateral bracing, or other appropriate tie-downs. The ELE design of restraints should include both strength considerations as well as their ability to accommodate imposed displacements.

A.12 Structural Modeling and Analysis

A.12.2 Analysis Principles

A.12.2.2 Calculation Methods

Hand and spreadsheet calculations are generally limited to simple components of the structure (beams, regular

panels, etc.) under simple loads/actions. The methodology used should reflect standard engineering practice with due consideration for the conditions of equilibrium and compatibility. Elastic or plastic design principles may be adopted, dependent on the limit state being checked.

Analysis of the structural system and detailed assessment of its components is normally performed by the finite element method. FEA involves subdivision of the structure or a component into elements of known behavior (e.g. beam, plane stress, plate bending, shell) and subsequent matrix solution on a computer. The method allows the behavior of large and complex structures to be determined in terms of displacements, internal forces, and stresses. Furthermore, once a computer model has been created, different types of analysis can be performed and repeated with ease.

Certain checks need to be undertaken for each analysis, whether by hand calculations or by computer, to ensure that gross errors have not been made and the results are sufficiently accurate. Advice on such checking, and also on the validation and verification of structural analysis software, is given in Reference [A.12.2-1].

A.12.3 Modeling

A.12.3.3 Geometrical Definition for Framed Structures

A.12.3.3.1 General

For steel space frame structures that respond statically, the influence of joint flexibility, offsets, and eccentricities is discussed in A.12.3.3.1 to A.12.3.3.3 in terms of their effects on member forces and on member strengths. For dynamically sensitive structures, these factors can also affect the overall stiffness of the structure and hence the natural period. These aspects are also discussed in A.12.3.3.3.

Changes in dimensions as a result of design changes should be monitored during and after the completion of an analysis. Where these impact on the accuracy of the analysis, the changes should be incorporated and evaluated by reanalysis of the component or the complete structure. Ideally, the final analyses should reflect the as-built structure.

The secondary moments in the chord member of K- and X-joints are proportional to the product of the offset and the component of the axial brace forces that is perpendicular to the chord. The secondary moments may be neglected for joints where the diameter, D , of the chord is substantially greater than the diameters of the braces and where the offset is smaller than $D/4$. However, for joints where all members are of similar diameter, the secondary moments should be assessed. For a K-joint at the end of a chord member, the angle between one of the braces and the chord is approximately 90° (sometimes called an N-joint); in this case, the secondary moment cannot be distributed over the chord member on both sides of the joint, and the secondary moment may only be neglected if the offset is smaller than $D/8$.

A.12.3.3.2 Member Modeling

The contribution of the horizontal plan members to the global lateral stiffness when the structure is in-place depends on the arrangement of the structure framing. The horizontal plan members do, however, contribute to the global torsional stiffness and also add stiffness during temporary phases, such as loadout/lift, transportation, and launch. Furthermore, they often provide lateral support to conductors and other appurtenances. Hence, the main plan bracing members should be regarded as primary structure.

Where secondary members transfer or attract significant loads/actions, they may be represented by equivalent members, with appropriate stiffness and properties, such that the applied loads/actions are appropriately apportioned to the surrounding primary framework. For example, conductor guide frames within the plan framing, which directly support the conductors, may be considered as secondary framework and modeled by equivalent members. The simplification can be used to reduce the complexity of the computer model, the complexity of data input, the amount of output, and the potential for errors. Care should, however, be taken to ensure that load cases

(e.g. the wave crest positions) chosen for analysis are appropriate for the design of all classes of member, whether primary or secondary.

Modeling of structural components such as mudmats, shear plates, and conductor guides can require finite element types other than beam elements (e.g. plane stress, plate bending, shell elements). Deck plating may be simulated in several ways, e.g. by truss (axial) elements to represent in-plane shear stiffness, by plane stress elements to represent both in-plane axial and shear stiffness, or by shell elements to represent membrane, shear, and lateral bending stiffness.

The stiffness of appurtenances, such as launch rails (fitted to legs to provide a smooth launching), mudmats, J-tubes, risers, and skirt pile guides, should be included in the model if they contribute significantly to the overall stiffness of the structure. It should be possible to model appurtenances without oversimplification, although care should be taken to ensure that very short stiff members do not cause numerical ill-conditioning of the global stiffness matrix.

A.12.3.3.3 Joint Modeling

Joint flexibility does not significantly affect the primary axial forces in a simple framework [A.12.3-1]. However, joint flexibility does modify member end bending moments and mid-span moments.

In addition to modifying member bending moments, the inclusion of joint flexibility and member end offsets can reduce the overall stiffness of the framework and thus increase the main periods of vibration by 3 %–6 % [A.12.3-1]. To avoid complex structural analysis, consideration of joint flexibility may be included by appropriate definition of the effective length (K) factors required to evaluate member strength (see 13.5). Similarly, the cross-section characteristics and the member end moments determine the moment reduction factor, C_m , for members subjected to combined axial compression and bending. Both K and C_m may be determined by separate rational analysis or prescribed conservative values in accordance with Table 13.5-1.

In a nonlinear analysis to determine ultimate strength, joint flexibility should be considered and included, particularly if joint failure occurs before member failure and before the ultimate strength is reached. Joint failure will initiate the development of different internal force distributions.

A.12.3.4 Modeling of Material Properties

In linear elastic analysis, the material properties required to determine stiffness are elastic modulus, shear modulus, and Poisson's ratio. To check the strength of members and joints, the SMYS and ultimate tensile strength are required.

In nonlinear analysis, in which internal forces and/or stresses are monitored as the analysis proceeds, the material properties should be defined by the appropriate "true" stress-strain curves incorporating first yield, strain hardening, and tensile fracture.

For frameworks, nonlinear material behavior may be formulated using the plastic hinge theory based on a yield surface, plastic flow normal to the yield surface, and an expanding yield surface (strain hardening). The yield surface for a member is normally defined in terms of the axial yield strength, P_y , and the plastic moment strength about the principal axes of the cross-section, M_{Py} and M_{Pz} . Variables P_y , M_{Py} , and M_{Pz} are all based on the yield strength from the "true" stress-strain curve.

For frameworks, tensile failure (fracture) of members and joints may be simulated by severing members.

A.12.3.6 Appurtenances

In the past, modeling of appurtenances during the design of a structure has often been ignored. Depending on their size and method of attachment to the structure, caissons and risers can provide a load path or can be subjected to load/action effects induced by relative movement between plan elevations arising from overall bending/sway of the structure. Although appurtenances, such as caissons and risers, may be regarded as secondary structures, they are often necessary for production or for safety reasons (e.g. fire water) and should therefore be given proper consideration.

A.12.3.7 Soil-structure Interaction

A.12.3.7.1 General

Nonlinear behavior of a piled foundation arises from nonlinear characteristics of the soil, second-order axial force-displacement ($P-\Delta$) effects and yielding in the pile. There are several ways of modeling single and group loads/actions of piles, based on different implicit assumptions [A.12.3-2]. Their applicability depends on the importance of foundation characteristics in the overall response of the structure, the type of analysis (i.e. static in-place analysis, fatigue analysis, etc.), and the nature of the problem being studied (e.g. use of a detailed model for dealing with local effects).

To allow the soil to be modeled by a set of nonlinear springs, simplifying assumptions are made regarding the stiffness, damping, and added mass. Typically, it is assumed that the loads/actions between the pile and soil at any level are independent of the deflections at any other level, that soil damping is part of the percentage critical damping assumed for the whole structure, and that the mass of soil acting with the piles has a negligible effect on the dynamic behavior of the structure.

For well-spaced piles, interaction effects between piles need not be considered. However, for closely spaced piles, group behavior should be considered (see A.12.3.7.2).

For approximate modeling of foundations and a preliminary estimate of pile behavior, pile fixity may be assumed at a distance of 4 to 10 pile diameters below the sea floor (or depth of scour). Distances of 4 to 5 and 8 to 10 pile diameters are appropriate for stiff clays and very soft silts, respectively. Above this level of fixity, the pile is modeled using its actual cross-section properties and without lateral support (i.e. no soil springs).

The use of a simple foundation model is allowed for determining the behavior of the structure—one such model consists of a beam fully fixed at its lower end and two lateral springs and one torsional spring at the sea floor. The properties of the equivalent system are generally chosen to represent the secant stiffness of the foundation. They are determined by comparing the pile (group) stiffness matrix, with interaction between lateral and rotational movement, and the stiffness matrix for the equivalent system. When used for fatigue analysis, a single linearization for all load cases due to wave loads/actions that are expected to cause the most fatigue damage is usually acceptable, in the knowledge that for some cases it will be over stiff and for other cases over flexible, but that the “average” stiffness will be reasonable.

A full nonlinear soil-structure interaction analysis combines a linear analysis of the structure with a nonlinear analysis of the piled foundation. The nonlinear soil p - y , t - z , and Q - z curves are represented by piecewise linear springs and the piles by a linear or beam-column formulation. The solution involves substructuring of the structure to connection nodes at the pile heads, a nonlinear iterative solution of the foundation and structure to determine the pile head displacements, and back-substitution of these displacements into the reduced equations of the structure to determine the displacements throughout the structure. Both the finite difference and finite element methods have been adopted in offshore structural analysis programs for analyzing piled foundations.

Where an estimate of an existing structure's stiffness is available via natural frequency measurements, this should be used as a basis for the calculation of foundation stiffness; however, due account should be taken of the

difference between the response of the foundation under normal and extreme situations. Any available driving records should also be reviewed for assessment of an existing structure.

A.12.3.7.2 Pile Groups

For closely spaced piles, interaction effects between piles should be considered as their capacity can differ from the sum of the individual piles. Soil failure around the whole group should be prevented.

A pile group may be modeled by simpler means, such as an equivalent single member with equivalent structural and foundation properties. Normally, however, a detailed analysis of the structural components within the pile group is required.

A.12.3.8 Other Support Conditions

In all static analyses, the structure is in equilibrium and should be prevented from rigid body movement. For launch and floating analyses, sufficient restraints should therefore be provided to suppress global motion in each of the six rigid body degrees of freedom (surge, sway, heave, roll, pitch, and yaw) without inhibiting relative displacements within the structure. There will be no reactions at the restraints as long as the applied loads/actions are balanced and the structure has the minimum (usually six) restraints required to avoid rigid body motion. The structure should be similarly restrained for lift situations. It is preferable to provide six translational restraints (two parallel to each axis and arranged to inhibit free rotation as well as free displacement, this requiring three noncollinear nodes that have some form of translational restraint) rather than to suppress all six freedoms (three translations and three rotations) at a single node. This minimizes the effect of any imbalance in the loads/actions, since a minor error in global moment, while insignificant in overall terms, can be significant when applied as a reaction at a single point in the structure.

A.12.3.9 Local Analysis Structural Models

Local analyses are structural analyses performed on a part or component of the overall structural system, their purpose being to provide a more accurate representation of the selected region than that provided by global analysis. They should typically provide more accurate information about the variation of section forces (or stresses) at complex regions of the structure. Local analyses may be nonlinear in nature, where such effects are likely to include material yielding in areas of stress concentration at geometrical discontinuities.

The boundaries of a local analysis, and hence the extent of the local model, should be chosen so that displacements and/or forces from the model of the whole structural system (global analysis) can be readily applied as boundary conditions. The boundaries should be sufficiently far from areas of interest so that the forces and stresses in the areas of interest are not affected by any local effects at the boundaries due to the method of applying the boundary conditions or owing to any inaccuracies in the boundary conditions. For example, in a detailed FEA of a tubular joint the braces should extend approximately two brace diameters to three brace diameters away from the heel of the brace-chord intersection to ensure that the stresses at the intersection are not affected by the boundary conditions at the end of the brace.

Displacements, forces, and stresses at nodes or across a finite element's boundaries may be extracted from the global analysis in order to act as boundary conditions for the local analysis. Force boundary conditions should be used when the force distribution in the global analysis is considered to be correct.

The loads/actions and boundary conditions applied to a local model should represent a self-equilibrating system. If the boundary conditions applied to a local model are extracted from an overall model, then the external loads/actions applied to the idealization of the local model in the global analysis should also be applied to the local model (see A.12.3.10). Force and displacement restraints are both possible. Where it is uncertain which of these types of boundary condition is more appropriate, both should be used and the more onerous results retained.

Any significant loads/actions within the boundaries of a local model should also be applied to the local model. This includes reactions due to support of the structure as well as other external loads/actions. Such locally applied loads/actions should be consistent with the corresponding loads/actions in the global analysis from which the boundary conditions were obtained.

A.12.3.10 Loads/Actions

Crane lift conditions are significant for subassemblies and are often best modeled by simulating the slings as pin-ended ties fixed at the crane hook point or points. Imbalance between sling forces may be incorporated by replacing selected slings with reaction forces calculated to support a percentage of the total lift weight, or by specifying artificial temperature increases in the pin-ended ties representing the slings. In most cases additional fixity is required to restrain the structure against rigid body rotations about the hook point(s). It is usual practice to determine the center of gravity of the structure and align this vertically below the hook point.

During transportation, the structure is supported by external water pressure, either directly applied or through a transportation barge. It is common practice to simulate this condition by applying the external water pressures as loads/actions on the structure equal to and opposing all other loads/actions (self weight, ballast, towing, etc.). The effects of acceleration (roll, pitch, and heave) and tilt of the structure are usually taken into account by applying various combinations of mass, acceleration(s) (varying in magnitude and direction), and loads/actions from self weights to a quasi-static model of the structure.

A.12.3.11 Mass Simulation

In the finite element method, “consistent” element mass matrices are generated if the same interpolation functions are used as in the evaluation of the stiffness matrix. A “consistent” mass matrix has off-diagonal terms, indicating coupling between the various element degrees of freedom.

An approximate mass matrix may be established by “lumping” (distributing) an element’s mass to the nodal points defining its topology. Thus the total mass at a nodal joint will represent the element-contributing volume around the node. The use of lumped (or diagonal) mass matrices can have a considerable computational advantage in the solution of the dynamic equations of equilibrium. For an assessment of the global response of framed structures, the use of lumped masses instead of consistent masses is fully acceptable, as this has only a minor effect on the results of a dynamic analysis. For other structural configurations, or for an assessment of the response of individual components, the influence can, however, be appreciably larger and a consistent mass matrix is recommended.

A.12.3.12 Damping

In a similar way to mass simulation (see A.12.3.11), “consistent” damping matrices can be derived, but in practice it is difficult, if not impossible, to determine the element damping parameters. Experimental results indicate that structural damping is frequency dependent, suggesting that damping is dependent on displacement amplitude. Models that take this into account are a hysteretic model, which is suitable for yielding materials, and a friction damping model, which is suitable for some soils (e.g. sand) and joints where damping arises from friction between sliding surfaces [A.12.3-2].

Although hysteretic and friction damping are more representative of energy dissipation in an offshore structure, viscous damping is assumed for most analyses because it is mathematically more convenient and more important to model the correct energy loss per cycle than the exact mode(s) of damping. For damped vibration, viscous damping is usually expressed in terms of a damping factor, based on the ratio, $\xi = c/c_c$, where c is a viscous damping coefficient and c_c is the critical damping. Critical damping is the amount of damping which separates the two types of response: overdamped decay without oscillation ($c/c_c > 1.0$) and decaying sinusoidal response ($c/c_c < 1.0$).

The choice of damping factor can have a profound effect on the predicted response. Values of 2 % of critical and less have been suggested on the basis of full-scale measurements in low sea states. For extreme sea states, a value of 5 % of critical is commonly used and for extreme seismic events, where inelastic behavior of both structure and foundation is likely, a value of 10 % can be applicable, provided such higher value can be substantiated for the case under consideration. In dynamic analyses for a fatigue assessment, damping factors of 1 % to 2 % are appropriate (see Section 16).

Including structural velocities in the calculation of hydrodynamic drag loads/actions increases the predicted total system damping. For noncompliant structures, this increase is included in the measured values and the damping based on relative velocities should not be used.

A.12.4 Analysis Requirements

A.12.4.3 Other Pre-service and Removal Situations

A structural component (or an appurtenance attached to the structure) can be subjected to large slamming loads/actions if the longitudinal axis of the component is nearly parallel to the water surface as it enters the water. The slamming loads/actions may be determined using the same procedures used to determine slamming loads/actions due to waves.

A.12.4.4 In-place Situations

A.12.4.4.1 General

Combinations of permanent and variable loads/actions, particularly those involving drilling and production equipment that can be added to, removed from, or repositioned on the structure, should be defined carefully so that each structural component will be assessed for the most onerous load case. Modern four-leg, vertical-pile structures, in particular, can experience net tension on the piles, and as the pile tension capacity is lower than its compression capacity, the tension case with minimum permanent and variable loads/actions loading can control the pile design.

The design of legs and primary framing, particularly diagonal bracing in vertical frames, is generally governed by the loads/actions causing maximum base shear or maximum overturning moment. However, primary and secondary structural components in horizontal frames, especially when these are located in the vicinity of the splash zone, are generally not governed by these wave positions (see also A.12.4.4.2).

A.12.4.4.2 Extreme Environmental Conditions

Internal forces in structural components are due to the force distribution associated with global loads/actions as well as local loads/actions on the component itself. It is often not obvious which combination of environmental parameters produces extreme global loads/actions or what constitutes the most severe local situation for a particular structural component. Unless it is clear which wave directions and which crest positions govern the design of all structural components, it will generally be necessary to follow the following procedure.

- a) Analyze eight wave directions corresponding to the main compass directions or the main directions relative to the structure. When there are significant differences in the environmental parameters from different directions, it is usual to align the longer axis of a platform with the prevailing or most onerous direction. Symmetry in structure geometry and in environmental conditions from opposite directions can reduce the analytical requirements, but should be used with care as symmetry requirements are rarely fully satisfied.
- b) Step each wave through the structure using a sufficient number of wave positions (phase angles) relative to the structure, to capture the governing positions, ensuring that governing positions for all structural components are identified.

Wave directions and wave positions should be selected by considering the environment (e.g. wave direction, wave length, wave height) as well as the geometry of the structure (e.g. leg spacing, elevation of plan framing levels).

A.12.4.4.6 Analysis for Reserve Strength

A.12.4.4.6.1 Analytical Software

Three basic types of analytical software are described below and are available for performing reserve strength analyses.

- a) General-purpose nonlinear finite element programs: With these programs, the equations of equilibrium are usually evaluated in the deformed condition and, because the beam element formulations generally do not incorporate geometrical nonlinearity at the element level, multiple beam elements along a member span are required to model global member buckling response accurately.
- b) Nonlinear beam-column space frame programs: These programs have specifically been written for the ultimate strength analysis of space frame structures. The elements embodied in this type of program have been derived to model beam-column behavior, and therefore each member span only requires a single element to model buckling behavior.
- c) Phenomenological space frame programs: In a phenomenological model, the failure of individual elements is prescribed through force-deformation relationships that can be empirically related to the element geometry. Nonlinear analysis is less intensive given these relationships are known prior to the analysis. However, this typically requires some prediction of the type of member behavior prior to the analysis, and can include the following behaviors.
 - 1) Elastic members: These are members that are expected to perform elastically throughout the analysis.
 - 2) Axial force members: These are members that are expected to undergo tensile yielding or axial buckling during ultimate strength analysis.
 - 3) Moment-resisting members: These are members that are expected to yield during an ultimate strength analysis, primarily due to high bending stresses. They should be modeled with beam-column elements that account for the formation of plastic hinges. A plastic hinge occurs when the axial and bending forces acting on a cross-section of the beam lie on the yield surface as defined by an axial force-bending moment (P/P_y , M_y/M_{Py} , M_z/M_{Pz}) interaction surface.

A.12.4.4.6.2 Structural Failure Modes

Care should be taken that the buckling response of beam elements that are used in analyses of type a), b), or c) according to A.12.4.4.6.1 include the possibility of localized failure (local buckling).

The following types of structural failure modes can often be identified. The modeling should be able to describe these failure modes.

- a) Geometrically brittle failure modes, in which only compression members are involved in the failure mechanism (Z- and K-braced structures) and the collapse strength of the framework is achieved at first compression member failure: An accurate model of elasto-plastic buckling will therefore identify the structure's peak ultimate strength, which will occur at relatively small deflections, after which redistribution of internal forces occurs. As the lateral deflections increase, portal frame action occurs. Eventually plastic hinges develop in the legs and a full portal frame failure mechanism occurs in the bay containing the failed compression member. Prior to collapse, the forces in the members are well defined, and therefore the ultimate strength depends on the ability to predict the member's compressive strength, which should be reasonably accurate provided the yield strength is correct and fabrication tolerances for out-of-straightness have been considered.

- b) Ductile failure modes, in which tension members are involved in the failure mechanism: Often, clear portal frame plastic failure mechanisms develop for which, at larger displacements when the internal forces in the compression member have been redistributed, the tensile member is the prime contributor to the collapse strength. In this state, the forces in all members of the failure mechanism are directly defined by equilibrium and the requirement to comply with plasticity limits. Again, it should be possible to predict the collapse strength and behavior accurately.
- c) Other modes of structural failure that are not as easily classified: In these failure modes, a progressive failure of members (and joints) occurs with a corresponding loss of global stiffness. At a certain level of loads/actions, the global stiffness becomes zero and the ultimate strength is achieved. However, the deflections are still relatively small and a clear plastic portal frame failure mechanism has not yet developed. This is often seen in the failure mode for the diagonal wave attack direction of skirt pile structures; a corner leg in the bottom bay will fail in compression and the global stiffness will reduce to zero before a clear plastic failure mechanism develops.

The progressive collapse of a framework structure usually involves compressive and tensile member failures over complete spans. Localized failure (excluding brittle fracture) of a member's cross-section prior to global member failure does not occur often. Although joint failures can be viewed as localized failures, their failure prescribes the forces in and force redistribution from the attached members and hence can be effectively included in a component model of the members.

Material brittle failure modes should be avoided by specifying suitable steel toughness.

A.12.5 Types of Analysis

A.12.5.1 Natural Frequency Analysis

When performing a natural frequency analysis, solution techniques should enable all significant modes of vibration to be determined, i.e. both global and local modes of vibration. Methods involving the definition of a master set of nodes are generally not suitable because local modes are excluded when the stiffness and mass matrices for the complete structural system are condensed to the master nodes.

As predicted and measured natural periods of structures have sometimes been found to vary considerably, it is often prudent to shift the natural periods by as much as $\pm 10\%$ to a more onerous value. This may be accomplished by adjusting mass or stiffness within reasonable limits. Where measured periods are available (e.g. for assessment of existing structures), the model may be modified to reflect the measured values (e.g. by adjusting the elastic modulus or the foundation stiffness).

A.12.5.3 Static and Quasi-static Linear Analysis

The two methods for determining the correction for dynamic effects are applicable if the excitation causing the dynamic effects can be considered to vary approximately harmonically in time. If the variation in time is essentially different from harmonic the influence of dynamic effects should be specially considered.

Guidance on the two methods is given below.

- a) The first method involves the determination of an equivalent set of quasi-static inertial loads/actions that adequately represents the dynamic behavior. When the dynamic response is caused by wave load/action this method is described in 9.8 and A.9.8 at a global analysis level for base shear and/or overturning moment. When the dynamic behavior is due to another cause than wave load/action, the equivalent quasi-static inertial loads/actions corresponding with that cause can be determined in an analogous manner. After adding the equivalent loads/actions to the real loads/actions, both factored by appropriate partial load/action factors if required, one static analysis is performed. The load/action effects resulting from the static analysis are directly used in any further evaluation.

This method is particularly recommended for parts of the structure that are above the application of the dynamic loads/actions. These parts do normally not experience any quasi-static load/action effects, so when multiplied by a DAF [method b)] the quasi-static load/action effects will still be negligible, whereas the properly calculated dynamic load/action effects resulting from inertial loads/actions can be significant.

b) The second method is performed in the following three steps.

- 1) Determine the appropriate range of frequencies for which some influence of dynamic effects is expected to occur. For a number of discrete frequencies of excitation, ω_e , within this range a series of static analyses is performed. At each of these excitation frequencies, the corresponding design loads/actions are applied; note that the loads/actions will in general be different for different frequencies. The calculated load/action effects of interest at each frequency are the static responses $r_{\text{stat}}(\omega_e)$; the static responses will in general also be a function of the frequency of excitation, because the loads/actions are different for different frequencies.
- 2) Determine DAF, k_{DAF} , for an SDOF system for each frequency of excitation, ω_e . The natural frequency of the SDOF system is the relevant natural frequency, ω_n , of the structural system, or structural part, that is responsible for the dynamic effect in the range of frequencies under consideration. The DAF is calculated using Equation (A.12.5-1):

$$k_{\text{DAF}}(\omega_e) = \frac{1}{\sqrt{(1 - \Omega^2)^2 + (2 \times \xi \times \Omega)^2}} \quad (\text{A.12.5-1})$$

where

$$\Omega = \omega_e / \omega_n;$$

ω_e is the frequency of excitation for which a static analysis is performed;

ω_n is the relevant natural frequency of the structural system, or structural part, that is responsible for the dynamic effect;

ξ is the fraction of critical damping associated with the dynamic effects (see A.12.3.12).

k_{DAF} should not be smaller than 1.0, which corresponds to $\Omega \leq \sqrt{2}$.

- 3) Determine the dynamically amplified responses, $r_{\text{dyn}}(\omega_e)$, from Equation (A.12.5-2):

$$r_{\text{dyn}}(\omega_e) = r_{\text{stat}}(\omega_e) k_{\text{DAF}}(\omega_e) \quad (\text{A.12.5-2})$$

The load/action effect that should be used in the further evaluation is the most onerous value of $r_{\text{dyn}}(\omega_e)$ within the range of excitation frequencies considered.

A.12.5.4 Static Ultimate Strength Analysis

The models used for describing tensile, compressive, and bending ultimate strengths of tubular members and their cross-sections may be verified against the ultimate strengths derived from the formulae given in Section 13, with all resistance factors set to 1.0. Differences can be checked to ensure that they fall within the range of expected scatter through statistics provided in A.13. However, values greater than the mean of any distribution should be used with caution.

The post-buckling redistribution of forces, described by the axial force/axial shortening curve, should be verified to be kinematically correct (see References [A.12.5-1] and [A.12.5-2]).

The model for combined compression and bending strength should, if possible, account for any reduction associated with local buckling of the cross-section. However, as local buckling does not usually occur in a well-designed structure until after member failure, local buckling is unlikely to affect the ultimate strength of a structural system. Guidelines are provided in Reference [A.12.2-1], which also provides information on the increased rate of redistribution of forces associated with local buckling of the cross-section.

Models for compressive strength and redistribution of forces should take account of the effects resulting from transverse hydrodynamic member loads/actions. Some nonlinear beam-column and phenomenological space frame analysis programs only consider concentrated loads/actions at the ends of members (often known as joint loads). This is because internal forces and moments in a primary member are predominantly caused by global frame behavior and not by the hydrodynamic load/action imposed on the member itself. In these cases, local bending effects should be separately taken into account.

Hydrostatic pressure can reduce the strength of submerged buoyant members, and this should be taken into account in the modeling.

Phenomenological models for member behavior generally require effective length (K) factors to be defined. The factors are used to establish the strength of members in compression in an ultimate strength assessment. Factors substantially lower than those specified in Section 13 may be applied, if it can be demonstrated that they are both applicable and valid. It is suggested that, prior to performing pushover analyses, detailed elastic analysis be undertaken to determine accurate effective length factors; these factors are subsequently used in the pushover analyses. However, the effective length factors can require revision when joints begin to fail and there is a loss of member end restraint.

When creating structural models for a nonlinear analysis, due consideration should be given to out-of-straightness, out-of-roundness, and locked-in forces arising from the methods of fabrication (see Annex E).

Typical foundation failure modes to be checked include pile punch-through, pile pullout, and pile lateral failure. These are failure mechanisms that are unlikely to be influenced by variations in the local flexibility/stiffness of the framework. It is therefore possible to capture each of these three failure modes using simple strength models, where detailed modeling of the foundation using beam-column elements for the pile and nonlinear springs for the soil by p - y , t - z , and Q - z curves is not required. A description of foundation failure modes is given below.

- a) Pile punch-through or pile pullout: This occurs when the applied loads/actions produce an overturning moment that results in a pile head axial force that exceeds the pile capacity in compression (punch-through) or tension (pullout). The capacity is determined by the skin friction strength given by t - z curves and, for pile compression, by end bearing strength given by Q - z curves (see API 2GEO). The yield strength of the pile usually exceeds the pile capacity, and therefore full axial plasticity is unlikely to form in the piles. As piles pullout or punch through, force redistribution in the piles and/or the structure can occur, depending on whether the soil maintains or loses its strength.

This type of pile failure will occur if the steel framework is stronger than, or of similar strength to, the foundation. If stronger, only the piles will fail; if similar, a combined framework and foundation failure mechanism can occur with the component failures in the framework being influenced by redistribution of the forces in the piles.

- b) Pile lateral failure: This occurs when the applied loads/actions produce a shear force in a pile that cannot be resisted by the lateral p - y resistance of the upper layers of soil. This results in excessive lateral movement of the piles and, therefore, additional bending moments, which combine with the existing pile axial forces to cause yielding and plastic hinges in the piles.

This type of failure will occur if the steel framework is sufficiently stiff and strong against lateral movement and the upper layers of the soil are relatively weak. When the soil fails, a portal frame mechanism develops in the piles below the sea floor with plastic hinges occurring at the sea floor and at some distance below the sea floor. When the portal frame failure mechanism has developed, the structure in the failed area is effectively statically determinate as the forces in the piles are defined by their plastic capacity and equilibrium. Since most of the energy dissipated will be associated with plastic hinge deformation in the piles, it is unlikely that the joint flexibility and/or local variation in the stiffness of the framework will affect the pile lateral failure mechanism.

Variability in structural steel strength and the mechanical behavior of steel structures is relatively well researched and can be rationally quantified (see References [A.12.5-3], [A.12.5-4], and [A.12.5-5]). The variability in behavior of offshore piled foundations is significantly less well researched. Often, when reanalyzing hurricane damaged structures, pushover analyses predict pile failures, although these have not been observed (see References [A.12.5-6] and [A.12.5-7]). Design values for axial and lateral pile performance appear to underpredict actual capacity and performance, and it appears that there is far more modeling and parameter uncertainty in the description of axial pile capacity and lateral pile performance than in the description of the steel structure's strength.

A.12.5.5 Dynamic Linear Analysis

For a specific wave direction, the total load/action on a structure caused by waves depends on the wave frequency and the structure's geometry, e.g. leg spacing. Consequently, for certain wave frequencies, the total wave load/action can approach zero. Should any one of these wave frequencies coincide with a natural frequency of the structure, the dynamic response would be artificially minimized. Care should therefore be taken when selecting waves for dynamic analysis to ensure that the calculation frequencies associated with artificially minimized wave loads/actions do not coincide with the natural frequencies of the structure.

Dynamic analysis should not be limited to one wave direction. A minimum of three directions should be considered, i.e. end-on, broadside, and diagonal wave directions.

A.12.6 Nonlinear Analysis

A.12.6.2 Geometry Modeling

Most finite element formulations are based on polynomial functions that describe the deformation of an element in terms of the displacements and/or rotations at the nodes of the element. The same polynomial functions are often used to describe the element geometry in terms of its nodal coordinates. Increasing the order of the polynomial (linear, quadratic, cubic, etc.) generally improves the accuracy of the element so that fewer elements are required, for example, to model nonlinear geometrical behavior due to elastic buckling. However, sufficient elements should be used to simulate initial imperfections and the various modes of buckling, including buckling due to combined axial forces and bending moments and also lateral torsional buckling of open sections (e.g. I-beams) due to major axis bending. To model lateral torsional buckling, a minimum of two elements are usually required between each laterally restrained section of a member if the mode is essentially sinusoidal.

The requirements for finite element models are generally not applicable to phenomenological models.

The guidance in A.12.4.4.6 also applies to nonlinear analysis.

A.12.6.4 Models for Member Strength

Member failure usually involves yielding of the material. In most finite element formulations, material behavior is monitored at specific locations within an element; for beams, these would be the numerical integration points along the beam and various points over the cross-section, including the extreme fibers, at the integration points. The finite element mesh for critical members should therefore be defined so that there are integration points close to the positions of maximum internal member forces; otherwise, the member strength is likely to be overestimated.

The same mesh requirement is applicable to nonlinear finite element space frame programs based on plastic hinge formulations when the internal forces are monitored at the integration points along a beam.

Special-purpose programs for framework analysis have been developed based on adaptive meshing techniques and also on phenomenological models (see A.12.4.4.6). Their performance is often more reliable than general purpose nonlinear finite element programs. A thorough understanding of the theoretical background and limitations of a nonlinear framework analysis should be gained before use.

A.12.6.5 Models for Joint Strength

Joints that can participate in the failure mechanism should be identified by an initial nonlinear analysis assuming rigid joints and checking the joint strengths using the formulae given in Section 14 with resistance factors set to 1.0. Joint nonlinear behavior may be modeled by uncoupled nonlinear springs, individual springs being used for axial force, IPB, and OPB, as these are the force components used to calculate joint strength. Infinitely stiff springs are usually adopted for shear and torsional joint stiffness.

Empirical formulae have been derived to describe the force-deflection and moment-rotation behavior of joints to ultimate member forces. The formulae cover uncoupled axial force, IPB moments, and OPB moments for simple joints. They do not allow for interaction between force and moment components, e.g. axial force and IPB moments. They should therefore be used with caution, and checks should be undertaken to ensure that joint strengths in Section 14 are not exceeded. Checks should also be carried out to ensure that the behavior is realistic and the interaction equation for joint strength is satisfied. For example, if the axial strength has been reached as defined by the axial force-deflection curve, the magnitude of the other internal forces should be small and not increase unless axial force is redistributed; also, if the chord strength of a joint has been reached, the brace forces should not increase. The effect of redistribution of internal forces on joint behavior should also be carefully examined.

A.12.6.7 Yield Strength of Structural Steel

The steel used in a fixed offshore structure can come from several different sources, e.g. from different steel mills, even if it is purchased from a single vendor. It is also possible for steel made to one specification to be sold as being to a different specification, provided that it meets the chemical and mechanical requirements. It cannot therefore be assumed that all the steel in a fixed offshore structure is from a single statistical population, without examination of the mill certificates and/or analysis of a statistically significant number of coupon tests taken from plates or casts used throughout the structure. Unless sufficient data are available for the particular structure being analyzed, generic data for the steel grades in use should be used. In assessing the strength of particular structural components, it is acceptable for individual measured yield strengths to be used where coupons are taken from the point under consideration, or where it can be demonstrated that the test piece is taken from the same manufactured plate, pipe, or steel cast being assessed.

It should be noted that national standards for tensile tests can specify differing strain rates. Care should be taken to correlate strain rates applied during extreme and accidental situations with test data.

A.12.6.8 Models for Foundation Strength

A.12.5.4 contains descriptions of foundation failure modes, their possible interaction with structural failure modes, and associated modeling features. The most likely modes of failure are pile pullout (insufficient t - z soil strength), pile punch-through (insufficient t - z soil strength and/or Q - z end bearing strength), and pile failure due to excessive bending (weak p - y soil strength). Simplified models for representing foundation failure should include, as a minimum, these three modes of failure.

A.13 Strength of Tubular Members

A.13.1 General

The requirements have been developed specifically for circular tubular shapes that are typical of offshore structure construction. The types of tubulars covered include fabricated roll-bent tubulars with a longitudinal weld seam, hot-finished seamless pipes, and electric resistance welded (ERW) pipes that have undergone some form of PWHT or normalization to relieve residual stresses. Relief of the residual stresses is necessary to remove the “rounded” stress-strain characteristic that frequently arises from the ERW form of manufacture.

The design requirements are tailored to tubulars of dimensions and material yield strengths typical of offshore structure members ($f_y < 500$ MPa and $D/t \leq 120$). Application of the requirements to thin tubulars with high D/t ratios (> 120) and high-strength steels ($f_y > 500$ MPa) can lead to unconservative results. For $D/t > 120$, guidance is, for example, available in References [A.13.1-1] and [A.13.1-2].

The ratio of the specified yield strength to the ultimate tensile strength should not exceed 0.90. The representative value of the yield strength for design should be limited to the minimum of either the tested yield strength or 90 % of the ultimate strength when the actual (tested) yield-strength-to-ultimate-strength ratio exceeds 0.90.

Fabrication imperfections should conform to the following:

- a) global out-of-straightness smaller than length/1000;
- b) global out-of-roundness, i.e. maximum difference between any two diameters measured at one cross-section, smaller than 1 % of the nominal diameter, independent of diameter, and applicable to all D/t ratios;
- c) for $D/t > 60$, local out-of-straightness not exceeding 0.2 % of the nominal diameter when measured using a gauge length of $4\sqrt{R \cdot t}$, where R is the radius.

A.13.2 Tubular Members Subjected to Tension, Compression, Bending, Shear, or Hydrostatic Pressure

A.13.2.1 General

The compression, bending, and hydrostatic pressure requirements contained in this subsection are based on tests carried out since 1975.

The screened test data used in the substantiation comparisons presented in A.13.2.3 to A.13.2.6 were collected from References [A.13.2-1] and [A.13.2-2]. Both of these references presented substantial collections of data, including both acceptable and nonacceptable test results. The main criterion for acceptability was completeness of data. This extended to include yield strength measurements for each specimen as well as individual values of critical geometrical variables such as diameter and thickness. The absence of measurements of elastic modulus alone was not considered reason enough to reject the data. Where elastic modulus was not measured, a value of 205,000 MPa was used when comparing measured and predicted strength. The lack of detailed measurements of critical geometry was usually taken as a reason to ignore data pertaining to columns, beam-columns, and tubulars subjected to external pressure either alone or in combination.

Tubulars with initial geometry outside API 2B tolerances [A.13.2-3] were always rejected except those subjected to external pressure. Correction for out-of-tolerance construction was effected using the expression given in Reference [A.13.2-1] explicitly for this purpose. Column results that exhibited strengths in excess of the Euler buckling strength were rejected along with other specimens in the same series. This excess is only possible if the end fixities do not approach the simply supported end conditions claimed to exist during the test.

Static values of yield strength were used where available, dynamic values were used in their absence. In the form used for normalization of test results, dynamic yield strength values produced conservative points for comparison.

Specimens constructed from steel of thickness of about 1.8 mm and less were rejected because of unease over the potential for being out-of-tolerance and possibly of higher than usual variability in material properties. Most of the specimens in question belonged to early test series when the need and ability to accurately measure circularity was not always appreciated or possible. The possible variability in material properties was judged a consequence of the production processes available at the time. The need to take account of such issues to some extent reflected the inherent variability in the test results in questions.

Figures showing comparisons between test results and predictions using the equations given in 13.2 to 13.4 in support of the requirements of this document are included in the corresponding subsections of this annex. Also included in the figures are the statistics (such as the bias and COV) of the fit.

A.13.2.2 Axial Tension

The equation for checking axial tension is applicable only for yielding of the gross section of circular tubulars. This condition should cover the large majority of tubular members in offshore structures.

The consequences of tensile yielding are less severe than other tubular member failure conditions, such as local buckling and beam-column buckling.

A.13.2.3 Axial Compression

A.13.2.3.1 General

Tubular members subjected to axial compression can fail due to material yield, overall column buckling, local buckling, or combinations thereof. Column buckling is discussed in A.13.2.3.2, while local buckling and material yield are discussed in A.13.2.3.3.

A.13.2.3.2 Column Buckling

The equation for the representative column buckling strength is a function of λ , a normalized form of column slenderness parameter given by $(f_{yc}/f_e)^{0.5}$ where f_{yc} is the local buckling strength of the cross-section (see A.13.2.3.3) and f_e is the Euler buckling strength for a perfect column. The equation is similar in format to the existing API 2A-LRFD equation [A.13.2-4]. Equations given in this subsection are appropriate for tubular column buckling accounting for fabrication imperfections. The buckling strength equates to 90 % of the Euler strength at large slenderness.

The column test database consists of 12 tests on fabricated tubulars, two on seamless pipe and 70 on ERW pipe (see References [A.13.2-5] to [A.13.2-8]). This database is considerably larger than that previously adopted for the purpose of substantiating offshore tubular strength formulations. The increase in the number of tests is primarily due to the inclusion of the relevant results from a large CIDECT test program [A.13.2-8], although in the process of selection based on the criteria cited in A.13.2.1, a number of previously accepted data were rejected.

A comparison between the test data and predictions by the equations is shown in Figure A.13.2-1, together with the statistics of the fit. The representative column strength equation can be seen to approximate a lower bound of the tested strength, although it is not possible to verify this in the range $\lambda > 1.0$ where no relevant data exist. The overall fit achieved is good, with a bias of 1.057 and a COV of 0.041.

For a member composed of two or more different cross-sections along its length, the following steps can be used to determine the representative axial compressive strength, in force terms.

Determine the elastic buckling strength, P_e (in force units), for the complete member, taking into account the end restraints and variable cross-sectional properties. In most cases, the effective length factor for the member needs to be determined. The resulting representative axial compressive strength, $P_{c,r}$ (in force units), is then given by Equations (A.13.2-1) and (A.13.2-2):

$$P_{c,r} = \left(1 - 0.278 \frac{P_{yc,r}}{P_e} \right) P_{yc,r} \quad \text{for } \left(\frac{P_{yc,r}}{P_e} \right)^{0.5} \leq 1.34 \quad (\text{A.13.2-1})$$

$$P_{c,r} = 0.9 P_e \quad \text{for } \left(\frac{P_{yc,r}}{P_e} \right)^{0.5} > 1.34 \quad (\text{A.13.2-2})$$

where

$P_{c,r}$ is the representative axial compressive strength, in force units;

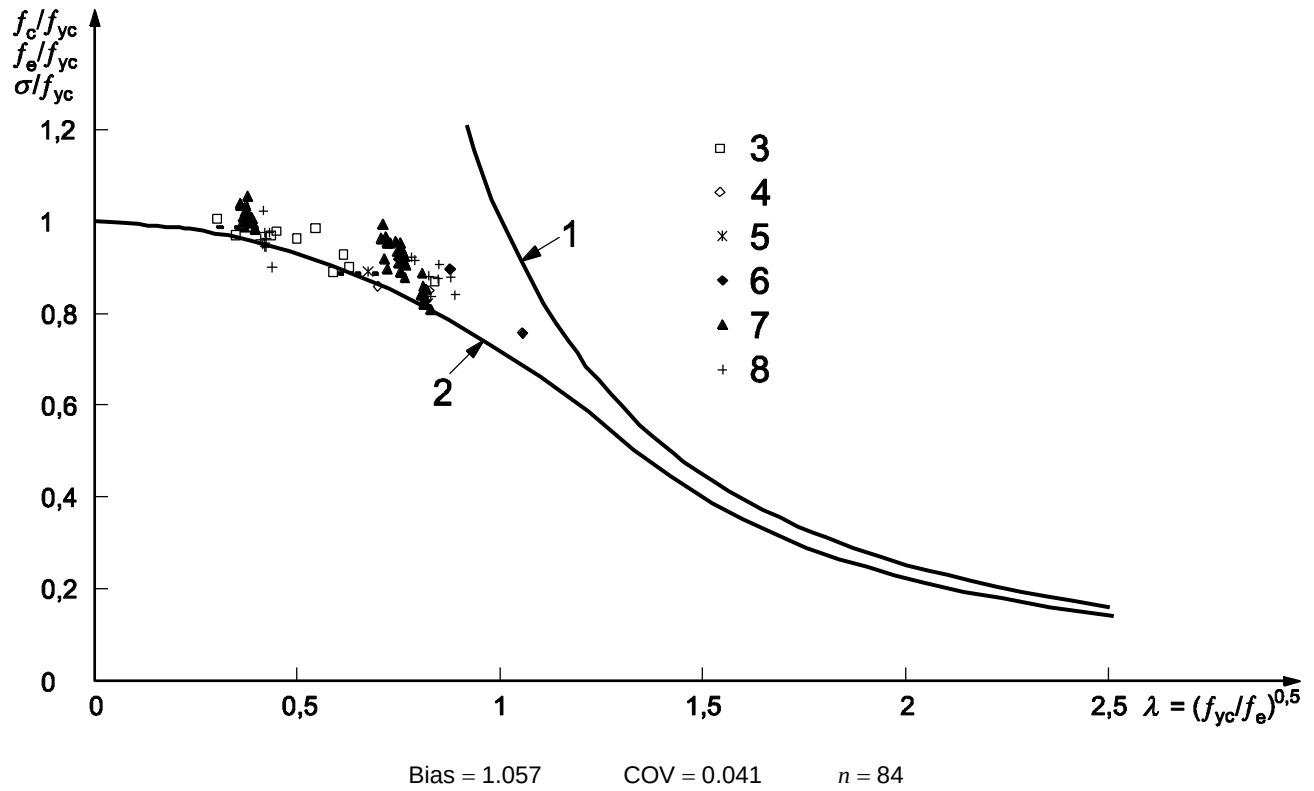
P_e is the elastic buckling strength of the complete member, in force units;

$P_{yc,r}$ is the smallest axial compressive strength of all the cross-sections, in force units, and is given as the minimum of $f_{yc,i} A_i$;

$f_{yc,i}$ is the representative local buckling strength as given by Equations (13.2-8) and (13.2-9) for the i^{th} section;

A_i is the cross-sectional area of the i^{th} section.

The representative axial compressive stress for each section is obtained by dividing $P_{c,r}$ by each of the respective cross-sectional areas, A_i .



Key

- 1 Euler buckling strength, f_e/f_{yc}
- 2 equations in this document, f_c/f_{yc}
- σ compressive stress in test member
- f_c representative axial compressive strength
- f_e Euler buckling strength
- λ column slenderness parameter
- f_{yc} representative local buckling strength
- n frequency (number of occurrences)

Test Data

- 3 Chen and Ross ^a [A.13.2-5]
- 4 Smith, Somerville, and Swan ^a [A.13.2-6]
- 5 Smith, Somerville, and Swan ^b [A.13.2-6]
- 6 Steinmann and Vojta ^c [A.13.2-7]
- 7 Yeomans ^c [A.13.2-8]
- 8 Yeomans ^b [A.13.2-8]

- ^a Fabricated pipe.
- ^b Seamless pipe.
- ^c ERW pipe.

Figure A.13.2-1—Comparison of Test Data with Representative Column Buckling Strength Equations for Fabricated Cylinders Subjected to Axial Compression

A.13.2.3.3 Local Buckling

Short tubular members subjected to axial compression will fail either by material yield or by local buckling depending on the diameter to thickness (D/t) ratio. Tubular members with low D/t ratios are generally not subject to local buckling under axial compression and can be designed on the basis of material failure, i.e. the local buckling stress may be considered to be equal to the yield strength. However, as the D/t ratio increases, the elastic local buckling strength decreases, and the tubular should be checked for local buckling.

Unstiffened thin-wall tubulars subjected to axial compression and bending are prone to sudden failures at stress levels well below the theoretical elastic critical buckling stresses predicted by classical small-deflection shell theory. There is a sudden drop in load-carrying capacity upon buckling. The post-buckling reserve strength of tubular members is small, in contrast to the post-buckling behavior of flat plates in compression, which usually continue to carry substantial load after local buckling. For this reason, there is a need for more conservatism in the definition of the buckling strength for tubulars than for most other structural elements. This is made difficult by the large scatter in test data, and necessitates a relatively conservative design procedure. The large scatter in test data is primarily the result of initial imperfections generated by fabrication tolerances and procedures. In addition to geometrical imperfections, experimental and theoretical evidence has shown that the buckling strength is also affected by boundary conditions and built-in residual stresses (see References [A.13.2-5] and [A.13.2-9]).

In order to achieve a robust design, member geometry should be selected such that local buckling due to axial forces is avoided.

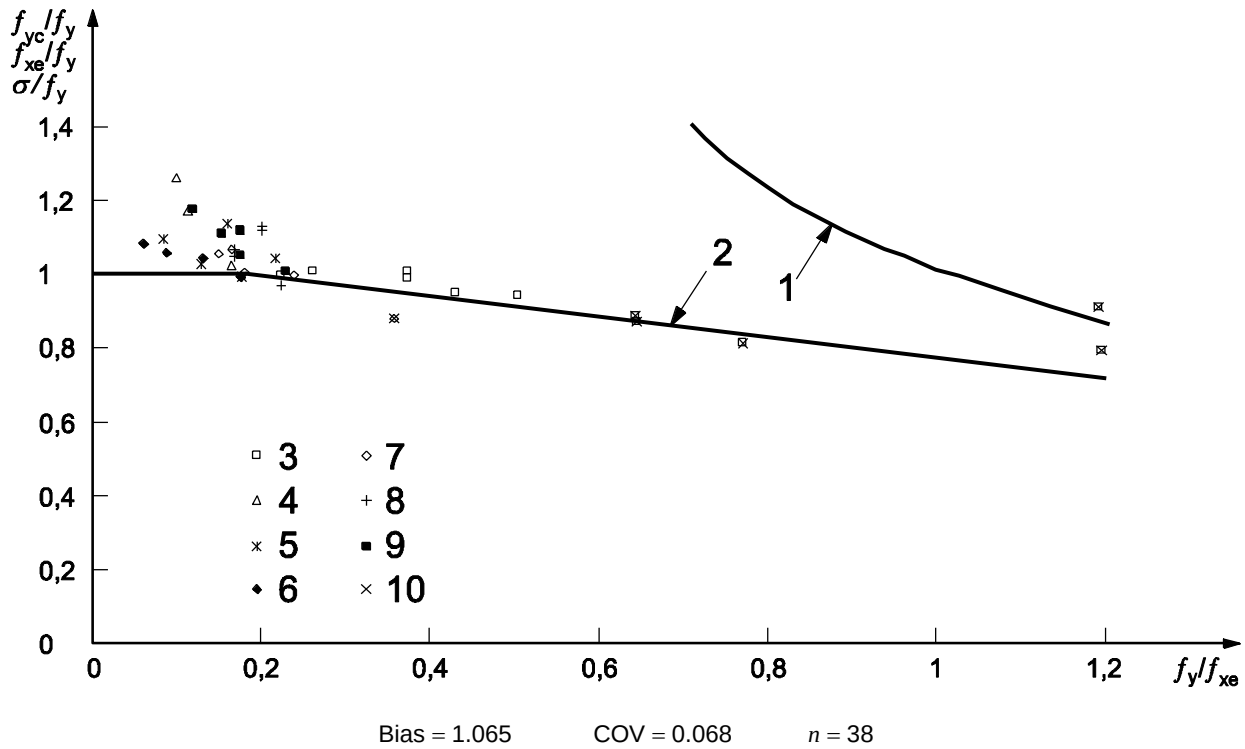
The equations for the representative local buckling strength were developed by screening test data and establishing a strength curve at 95 % success at the 50 % confidence level, which satisfies the following conditions:

- a) it had a plateau equal to the material representative yield strength over the range $0 < f_y/f_{xe} \leq 0.17$;
- b) it had the general form of Equation (13.2-9);
- c) it converged to the elastic critical buckling stress curve with increasing member slenderness ratio;
- d) the difference between the mean minus 1.645 standard deviations of test data and the developed equations was minimal.

The local buckling database consists of 38 acceptable tests performed by several different investigators (see References [A.13.2-5] and [A.13.2-10] to [A.13.2-14]).

A comparison between test data and the representative local buckling strength equations [Equations (13.2-8) and (13.2-9)] is plotted in Figure A.13.2-2. The developed equations have a bias of 1.065, a standard deviation of 0.073, with a COV of 0.068.

The elastic local buckling stress formula of Equation (13.2-10) represents one-half of the theoretical local buckling stress computed using classical small-deflection theory and is created by setting the buckling coefficient C_x to 50 % of its theoretical value, i.e. 0.3 instead of 0.6. This reduction accounts for the prime detrimental effect of geometrical imperfections as well as the secondary effects of residual stresses and end conditions. Based on the test data shown in Figure A.13.2-2, it is considered to be conservative for tubulars with $t \geq 6$ mm and $D/t \leq 120$. Offshore structure members typically fall within these dimensional limits. For thinner tubulars and tubulars with higher D/t ratios, larger imperfection reduction factors can be required. For the design of tubular members beyond these dimensional limits, References [A.13.1-2] and [A.13.2-15] provide guidance.



Key

- 1 elastic local buckling strength, f_{xe}/f_y
- 2 equations in this document, f_{yc}/f_y
- σ compressive stress in test member
- f_y representative yield strength
- f_{yc} representative local buckling strength
- f_{xe} representative elastic local buckling strength
- D/t diameter to thickness ratio
- n frequency (number of occurrences)

Test Data

- 3 Marzullo and Ostapenko [A.13.2-10]
- 4 Chen and Ross [A.13.2-5]
- 5 Eder et al. ^a [A.13.2-13]
- 6 Kiziltug et al. [A.13.2-14]
- 7 Ostapenko and Grimm [A.13.2-11]
- 8 Prion and Birkemoe [A.13.2-12]
- 9 Eder et al. ^b [A.13.2-13]
- 10 $D/t > 170$

^a Fabricated pipe.

^b ERW pipe.

Figure A.13.2-2—Comparison of Test Data and the Representative Local Buckling Strength Equations for Cylinders Subjected to Axial Compression

A.13.2.4 Bending

The representative bending strength of fabricated tubular members, as given in Equations (13.2-13) to (13.2-15), is obtained by dividing the (ultimate) plastic moment strength by the calculated elastic yield moment. A comparison of the representative bending strength with test data is shown in Figure A.13.2-3. The bias and COV of the strength equations are also given. The test data were taken from References [A.13.2-7], [A.13.2-14], and [A.13.2-16] to [A.13.2-19].

The test data show that simply supported beam tests have bending strengths lower than corresponding fixed-end beam tests in which cross-sectional distortion is prevented. However, the end condition has little influence on the rotational capacity of a beam.

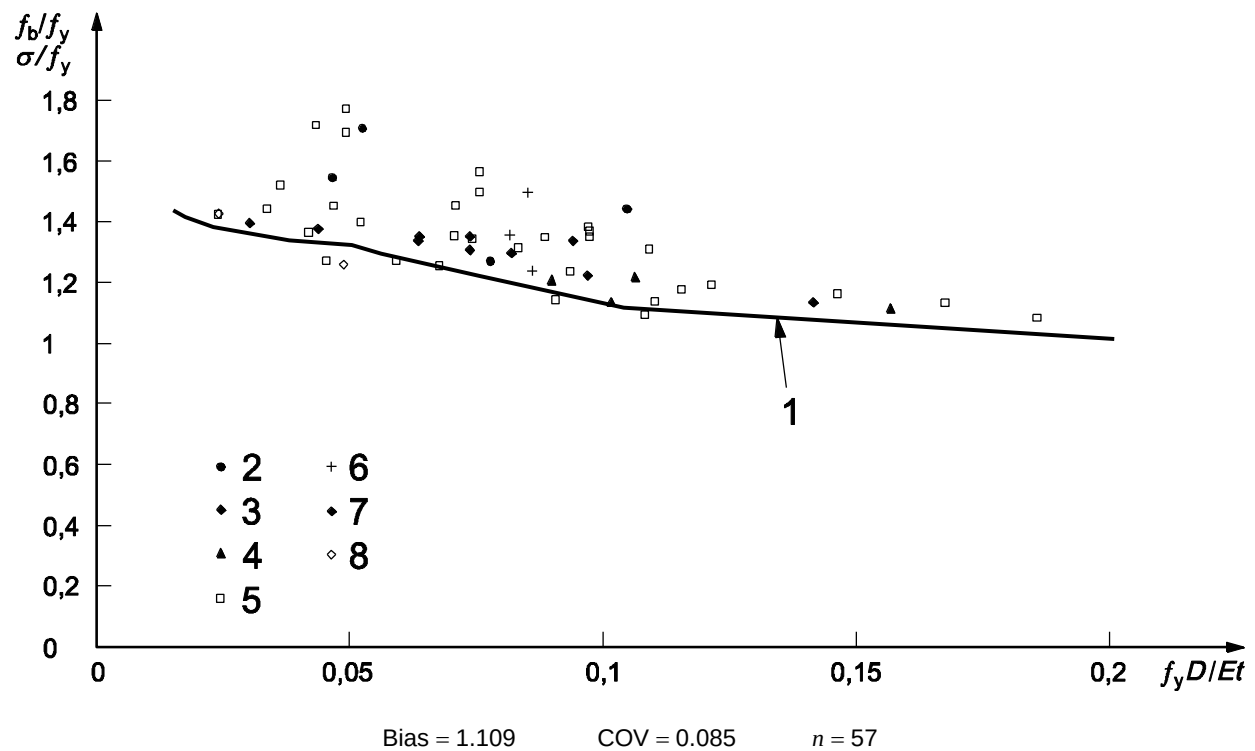
At low $\frac{f_y D}{E t}$, the plastic hinge mechanism forms over a short length of the beam. As the end support rigidity is reduced, the hinge mechanism forms over a longer length with the same rotation.

As $\frac{f_y D}{E t}$ increases, both the bending strength and the rotational capacity decrease.

The behavior of a tubular member under bending is grouped into three regions:

- a) high rotational capacity—ductile failure mode, exhibiting very gradual strength decay [see Equation (13.2-13)];
- b) intermediate rotational capacity—semi-ductile failure mode, exhibiting gradual strength decay [see Equation (13.2-14)];
- c) low rotational capacity—little post-ultimate ductility, exhibiting rapid strength decay and susceptibility to pre-yield local buckling [see Equation (13.2-15)].

The ultimate bending moment strength is the full plastic moment of the member. For members with $f_y = 345$ MPa and $E = 205,000$ MPa, the development of the full plastic moment is realized when $D/t \leq 30$, as reflected in Equation (13.2-13). This strength is linearly reduced to about 10 % in excess of the yield strength when $D/t \approx 60$ as reflected in Equation (13.2-14). Although it is not reflected in Equation (13.2-15), the bending strength of a tubular member with very high D/t should approach the local elastic critical buckling strength for a tubular member subjected to axial compression.



Key

- 1 equations in this document
- σ bending stress in test member
- f_b representative bending strength
- f_y representative yield strength
- n frequency (number of occurrences)

Test Data

- 2 Kiziltug et al. ^a [A.13.2-14]
- 3 Korol and Hudoba ^a [A.13.2-17]
- 4 Sherman (1984) ^a [A.13.2-19]
- 5 Sherman (1987) ^b [A.13.2-16]
- 6 Steinmann and Vojta ^a [A.13.2-7]
- 7 Korol ^a [A.13.2-18]
- 8 Sherman (1984) ^c [A.13.2-19]

- ^a ERW pipe.
- ^b Fabricated pipe.
- ^c Seamless pipe.

Figure A.13.2-3—Comparison of Test Data with Representative Bending Strength Equations for Fabricated Cylinders

A.13.2.6 Hydrostatic Pressure

A.13.2.6.2 Hoop Buckling

Unstiffened tubulars subjected to external hydrostatic pressure are susceptible to elastic or inelastic local buckling of the shell wall between restraints. Once initiated, the collapse will tend to flatten the member from one end to the other. Ring-stiffened tubulars are subject to local buckling of the shell wall between rings, with the rings remaining essentially circular. However, the rings can rotate or warp out of their plane. Ring-stiffened tubular members are also subject to general instability, which occurs when the rings and shell wall buckle simultaneously at the critical stress. It is highly desirable to provide rings with sufficient reserve of strength to prevent general instability.

For tubulars satisfying the maximum out-of-roundness tolerance of 1 %, the representative hoop buckling strength is given by Equations (13.2-25) to (13.2-27). For ring-stiffened tubulars these equations give the hoop buckling strength of the shell wall between rings. To account for the 1 % out-of roundness, the elastic hoop buckling stress is taken as 0.8 times the theoretical value calculated using classical small deflection theory, that is, $C_h = 0.44 t/D$, whereas the theoretical value is $0.55 t/D$. In addition, the other C_h values are lower bound estimates.

For members with an out-of-roundness greater than 1 %, but less than 3 %, a reduced elastic hoop buckling stress, $f_{he,red}$, is applicable [A.13.2-20]. The representative hoop buckling strength (f_h) is then determined using $f_{he,red}$:

$$f_{he,red} = \alpha f_{he}/0.8 \quad (A.13.2-3)$$

where

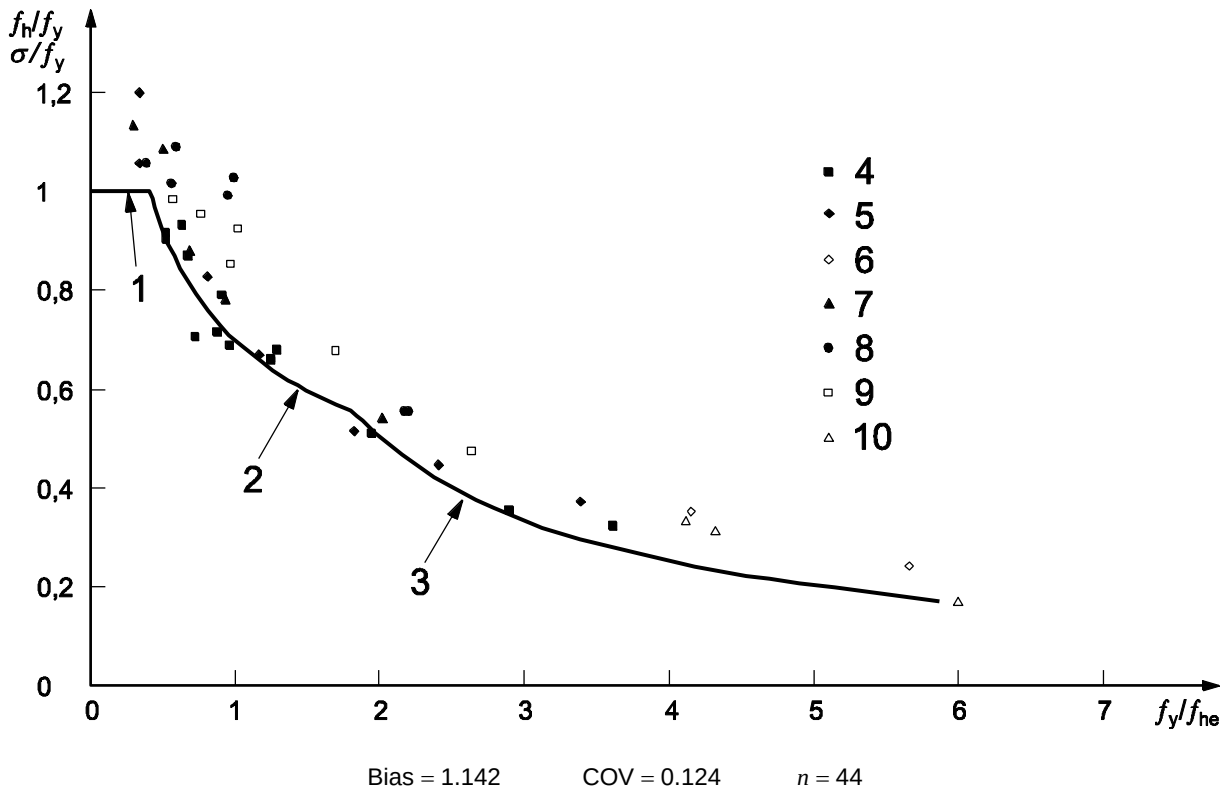
α is a geometrical imperfection factor;

$$\alpha = 1 - 0.2 \sqrt{\frac{D_{max} - D_{min}}{0.01 D_{nom}}}$$

$$\frac{D_{max} - D_{min}}{0.01 D_{nom}} \quad \text{is the out-of-roundness (\%)}$$

where D_{max} and D_{min} are the maximum and minimum values of any measured outside diameter at a cross-section and D_{nom} is the nominal diameter. f_{he} is taken from Equation (13.2-28).

A comparison of the representative hoop buckling strength with test data is shown in Figure A.13.2-4. The bias and COV of the strength equation are also given. The test data were taken from References [A.13.2-7], [A.13.2-13], [A.13.2-14], [A.13.2-20], and [A.13.2-21].



Key

- 1 Equation (13.2-25), f_h/f_y
- 2 Equation (13.2-26), f_h/f_y
- 3 Equation (13.2-27), f_h/f_y
- σ hoop stress in test member
- f_h representative hoop buckling strength
- f_{he} representative elastic critical hoop buckling strength
- f_y representative yield strength
- n frequency (number of occurrences)

Test Data

- 4 Miller and Kinra ^a [A.13.2-21]
- 5 Miller, Kinra, and Marlow ^a [A.13.2-20]
- 6 Miller, Kinra, and Marlow ^b [A.13.2-20]
- 7 Eder et al. ^a [A.13.2-13]
- 8 Eder et al. ^c [A.13.2-13]
- 9 Kiziltug et al. ^d [A.13.2-14]
- 10 Steinmann and Vojta ^d [A.13.2-7]
- ^a Fabricated pipe with ring stiffeners.
- ^b Fabricated pipe unstiffened.
- ^c ERW pipe with ring stiffeners.
- ^d ERW pipe unstiffened.

Figure A.13.2-4—Comparison of Test Data with Representative Hoop Buckling Strength Equations for Fabricated Cylinders Subjected to Hydrostatic Pressure

A.13.2.6.3 Ring Stiffener Design

The formulae for determining the required moment of inertia of the stiffening rings [Equations (13.2-35) and (13.2-36)] provide sufficient strength to resist buckling of the ring and shell even after the shell has buckled between stiffeners. It is assumed that the shell offers no support after buckling and transfers all its force to the effective stiffener section. The stiffener ring is designed as an isolated ring that buckles into two waves ($n = 2$) at a collapse pressure 20 % higher than the strength of the shell.

The dimensional requirements recommended in this subsection are based on an evaluation of References [A.13.2-22] to [A.13.2-24], and [A.13.2-15]. Table A.13.2-1 presents a comparison of the geometry requirements for stiffeners with flanges. The selected values are the same as those for Eurocode 3 [A.13.2-23], but the proposed expressions are presented nondimensionally (they are also the same as those used for Class I in NS 3472 [A.13.2-24], as required for the plastic design of beams subjected to compression).

Table A.13.2-1—Comparison of Geometry Requirements in Different Codes for Stiffeners with Flanges When Supported Along Two Edges

Reference	b/t_f	h/t_w
AISC [A.13.2-22]	16.6	32
Eurocode 3 [A.13.2-23]	13	25
NPD [A.13.2-15]	10	20
This document	13.4	24.5
The data in the table apply for $E = 210,000$ MPa and a yield strength of 420 MPa.		

A.13.3 Tubular Members Subjected to Combined Forces without Hydrostatic Pressure

A.13.3.1 General

This subsection describes the background of the design requirements given in 13.3, which cover tubular members subjected to combined axial and bending forces without hydrostatic pressure.

In this subsection, as for 13.4, the designer should include second-order frame moments, i.e. $P-\Delta$ effects in the bending stress determination, when they are significant. The $P-\Delta$ effects could be significant in the design of unbraced deck legs, piles, and laterally flexible structures.

A.13.3.3 Axial Compression and Bending

This subsection provides an overall beam-column stability check, Equation (13.3-3), and a strength check, Equation (13.3-4), for components subjected to combined axial compression and bending.

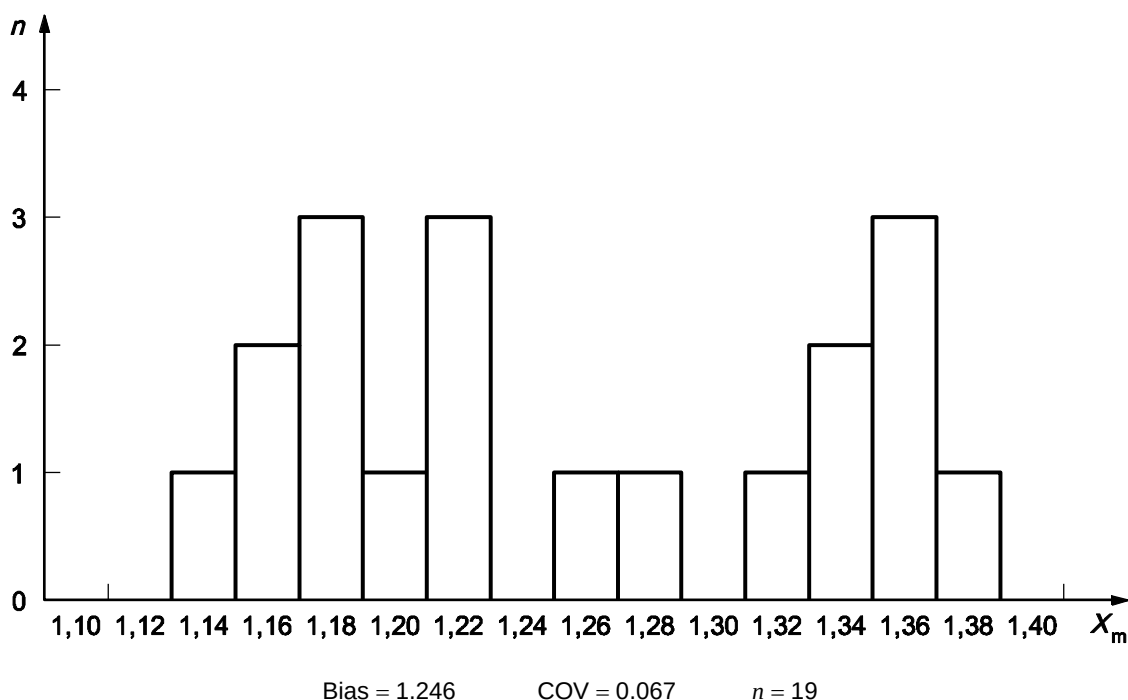
Equation (13.3-3) maintains the AISC-ASD beam-column stability interaction equation [A.13.2-22]. This equation is conservative with respect to small scale tubular member tests and analyses of larger members performed with assumed imperfections and residual stresses. As data accumulate on this subject, a reliability calibration of the form of this interaction will be appropriate.

Equation (13.3-4), the interaction equation for the strength check, is of the “linear form” which is simpler and slightly conservative for $D/t < 25$ compared with that of the “cosine form.” Thus, comparisons for members with dominant bending can show higher interaction ratios. The influence of D/t on the strength interaction equation is contained in both f_{yc} in Equations (13.2-8) to (13.2-10), and in f_b in Equations (13.2-13) to (13.2-15). For low D/t

ratios (< 25) there is some justification for using a “cosine” interaction equation. However, the test evidence for this is limited and much of it relates to short specimens ($L/D = 3$), so there does not appear to be strong support for its use at this time. To use the cosine form of interaction equation for low D/t ratios (< 25), the appropriateness needs to be demonstrated on each occasion.

NOTE The ISO equation for axial compression and bending is identical to that used in Reference [A.13.3-1].

The test data for tubular members subjected to combined axial compression and bending were taken from References [A.13.2-9], [A.13.2-12], [A.13.2-14] and [A.13.3-2] to [A.13.3-4]. The comparisons between the test data and the predicted strength of tubulars subjected to combined axial compression and bending are plotted in Figures A.13.3-1 and A.13.3-2, for short and long columns, respectively. The statistics of the fits are also included in the figures.



Key

n frequency (number of occurrences)

X_m measured strength/predicted strength

Figure A.13.3-1—Comparison of Measured and Predicted Strengths of Short Tubulars Subjected to Combined Compression (Local Buckling) and Bending

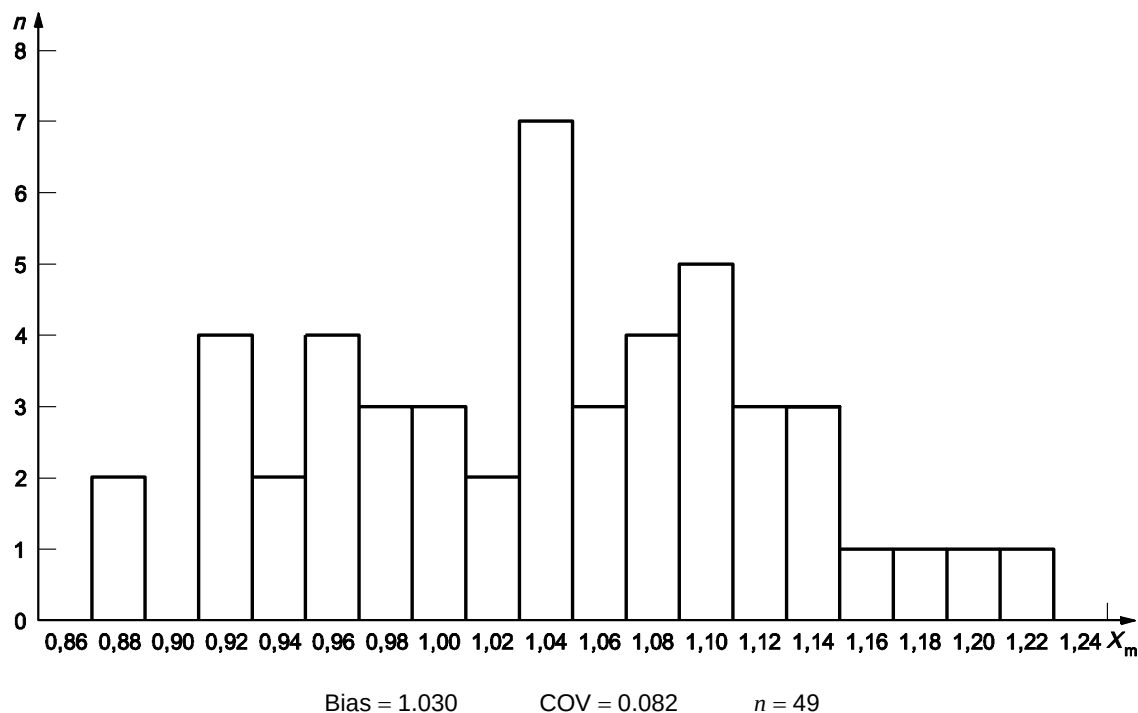
**Key** n frequency (number of occurrences) X_m measured strength/predicted strength

Figure A.13.3-2—Comparison of Measured and Predicted Strengths of Long Tubulars Subjected to Combined Compression (Column Buckling) and Bending

A.13.4 Tubular Members Subjected to Combined Forces with Hydrostatic Pressure

A.13.4.1 General

When hydrostatic pressure is considered in code checks, the code check equations need to be carefully chosen according to the adopted analysis method [marine buoyancy analysis method vs. rational buoyancy analysis method (see Figure A.13.4-1)].

For analyses using factored loads/actions that do not include the capped-end loads/actions (i.e. only frame loads/actions considered in analyses; namely, marine buoyancy analysis method), the hydrostatic capped-end stress is approximated as half of the hoop stress in the code check equations ($0.5 \sigma_h$) (Table A.13.4-1). The marine buoyancy analysis method assumes a Poisson's ratio of 0.5, resulting in a conservative estimate of hydrostatic capped-end stress.

For analyses using factored loads/actions that include capped-end loads/actions (i.e. frame actions and hydrostatic effects considered in analyses; namely, rational buoyancy analysis method), the effects of hydrostatic pressure are included in the analyses. Hence, the analysis outcomes include hydrostatic pressure effects, such that no additional values need to be further included in code check equations (Table A.13.4-1). The rational buoyancy analysis method, however, explicitly includes the local hydrostatic stress in the analysis, resulting in more accurate results.

Analyses using marine or rational buoyancy will have different member stresses, which in turn, affect member code checks with hydrostatic pressure and need to be carefully considered. Table A.13.4-1 differentiates the uses of the code check equations in current industry design codes when different analysis methods are used (i.e. marine buoyancy analysis method vs. rational buoyancy analysis method).

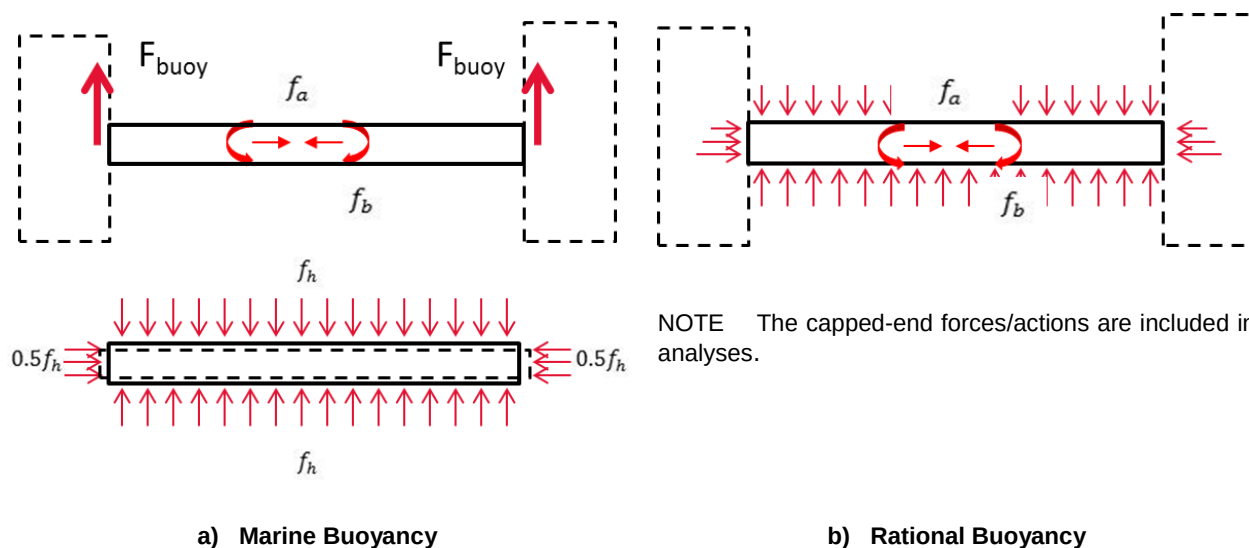


Figure A.13.4-1—Marine Buoyancy Analysis Method vs. Rational Buoyancy Analysis Method

Table A.13.4-1—Code Check Equation Differences between Marine Buoyancy Analysis Method and Rational Buoyancy Analysis Method

Maine Buoyancy	Rational Buoyancy
Axial Tension and Hydrostatic Pressure $U_m = A^2 + B^2 + 2\sqrt{A B}$ $A = \frac{\gamma_{R,t}\sigma_{t,c} + \gamma_{R,b}\sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_y}$ $B = \frac{\gamma_{R,h}\sigma_h}{f_h}$ where $\sigma_{t,c} = \sigma_t - \sigma_q \quad \sigma_q \cong 0.5 \sigma_h$	Axial Tension and Hydrostatic Pressure $U_m = A^2 + B^2 + 2\sqrt{A B}$ $A = \frac{\gamma_{R,t}\sigma_{t,c} + \gamma_{R,b}\sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_y}$ $B = \frac{\gamma_{R,h}\sigma_h}{f_h}$ where $\sigma_{t,c} = \text{analysis outcome}$
Axial Compression and Hydrostatic Pressure $U_m = \frac{\gamma_{R,c}\sigma_{c,c}}{f_{yc}} + \frac{\gamma_{R,b}\sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_y}$ where $\sigma_{c,c} = \sigma_q - \sigma_t \quad \text{if } \sigma_t < \sigma_q$ $\sigma_{c,c} = \sigma_c + \sigma_q \quad \sigma_q \cong 0.5 \sigma_h$	Axial Compression and Hydrostatic Pressure $U_m = \frac{\gamma_{R,c}\sigma_{c,c}}{f_{yc}} + \frac{\gamma_{R,b}\sqrt{\sigma_{b,y}^2 + \sigma_{b,z}^2}}{f_y}$ where $\sigma_{c,c} = \text{analysis outcome}$

Further discussion of the development of the equations in this subsection can be found in References [A.13.4-1] through [A.13.4-4].

A.13.4.2 Axial Tension, Bending, and Hydrostatic Pressure

The interaction formula recommended to check axial tension and hydrostatic pressure interaction is based on the Beltrami and Haigh maximum total energy theory, with the critical hydrostatic buckling stress substituted for the yield stress and Poisson's ratio set equal to 0.3. The Beltrami and Haigh failure theory reduces to the Hencky–von Mises distortion energy theory with Poisson's ratio equal to 0.5. A comparison with available test data, shown in Figure A.13.4-2, confirms that the recommended interaction formula is appropriate for D/t values typically used for offshore platform members. The test data that fall inside the ellipse correspond to stretch failures and tests with very low D/t values.

A.13.4.3 Axial Compression, Bending, and Hydrostatic Pressure

The combination of hydrostatic pressure and axial load may produce a different critical buckling stress than either of these load systems acting independently. Figure A.13.4-6 illustrates the recommended interaction equations for various possible stress conditions. These interaction equations imply that no interaction occurs if the axial stress is less than one-half the allowable hoop stress.

The recommended interaction equations have been checked against the results of available tests and found to give conservative results, as shown in Figure A.13.4-3, Figure A.13.4-4, and Figure A.13.4-5. Figure A.13.4-3 shows the results of elastic buckling tests on Mylar, plexiglass, and fabricated steel cylinders, whereas Figure A.13.4-4 shows the results of fabricated steel cylinders alone. In Figure A.13.4-3, the test results are compared with the recommended equation for elastic interaction, Equation (13.4-10), using F_{xe} and F_{he} values determined from the tests. This comparison validates the form of Equation (13.4-10). In Figure A.13.4-4, the fabricated steel cylinder test results are compared with Equation (13.4-10), using F_{xe} and F_{he} values computed from the design equations in 13.2. This confirms that Equation (13.4-10) is conservative. In Figure A.13.4-5, the recommended interaction equations are compared with the results of test data for unstiffened steel pipe with an elastic hydrostatic buckling stress and an inelastic axial buckling stress. This comparison demonstrates the validity of the recommended interaction equations for combined elastic and inelastic behavior.

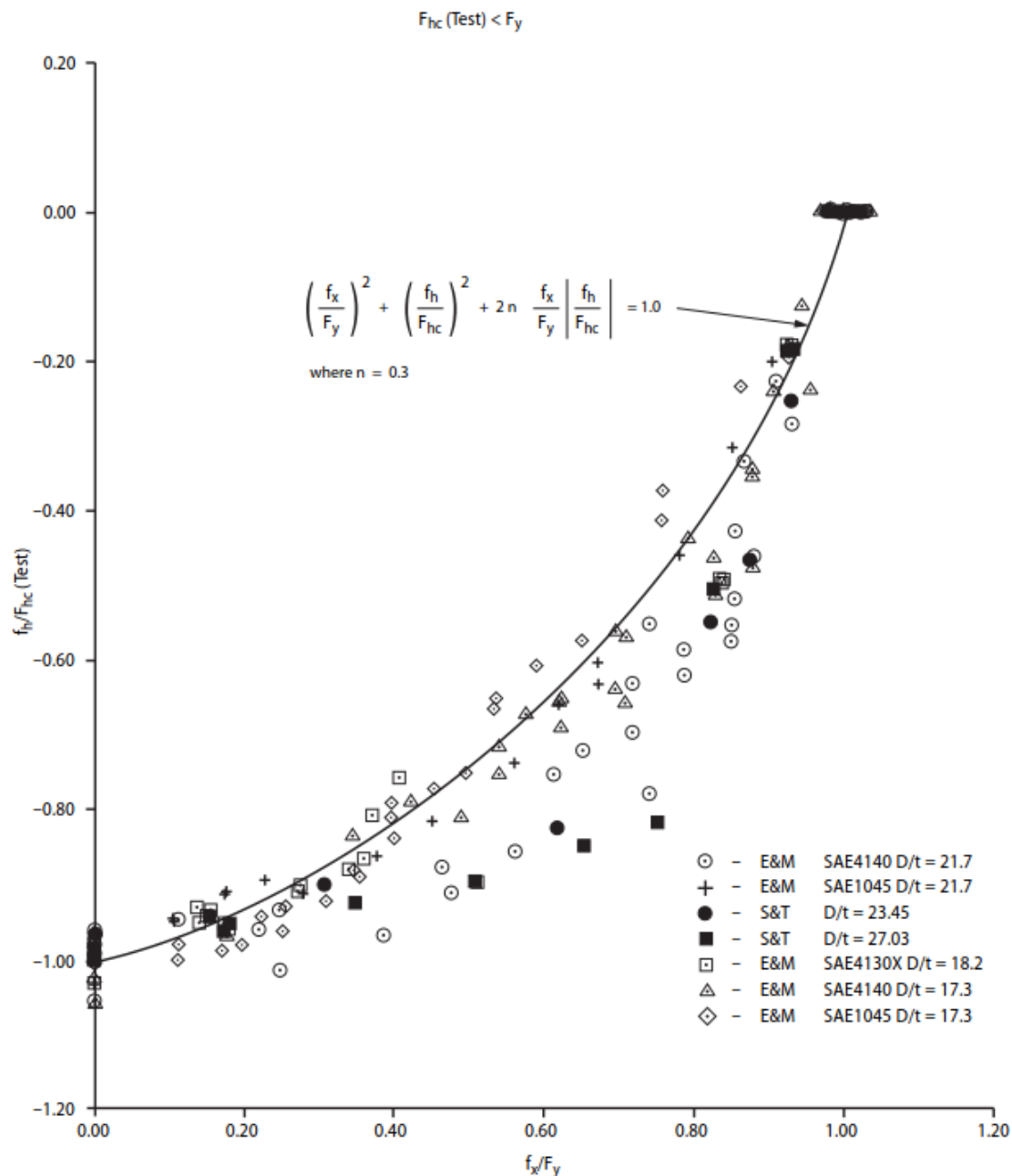


Figure A.13.4-2—Comparison of Test Data with Interaction Equation for Cylinders under Combined Axial Tension and Hydrostatic Pressure (F_{hc} Determined from Tests)

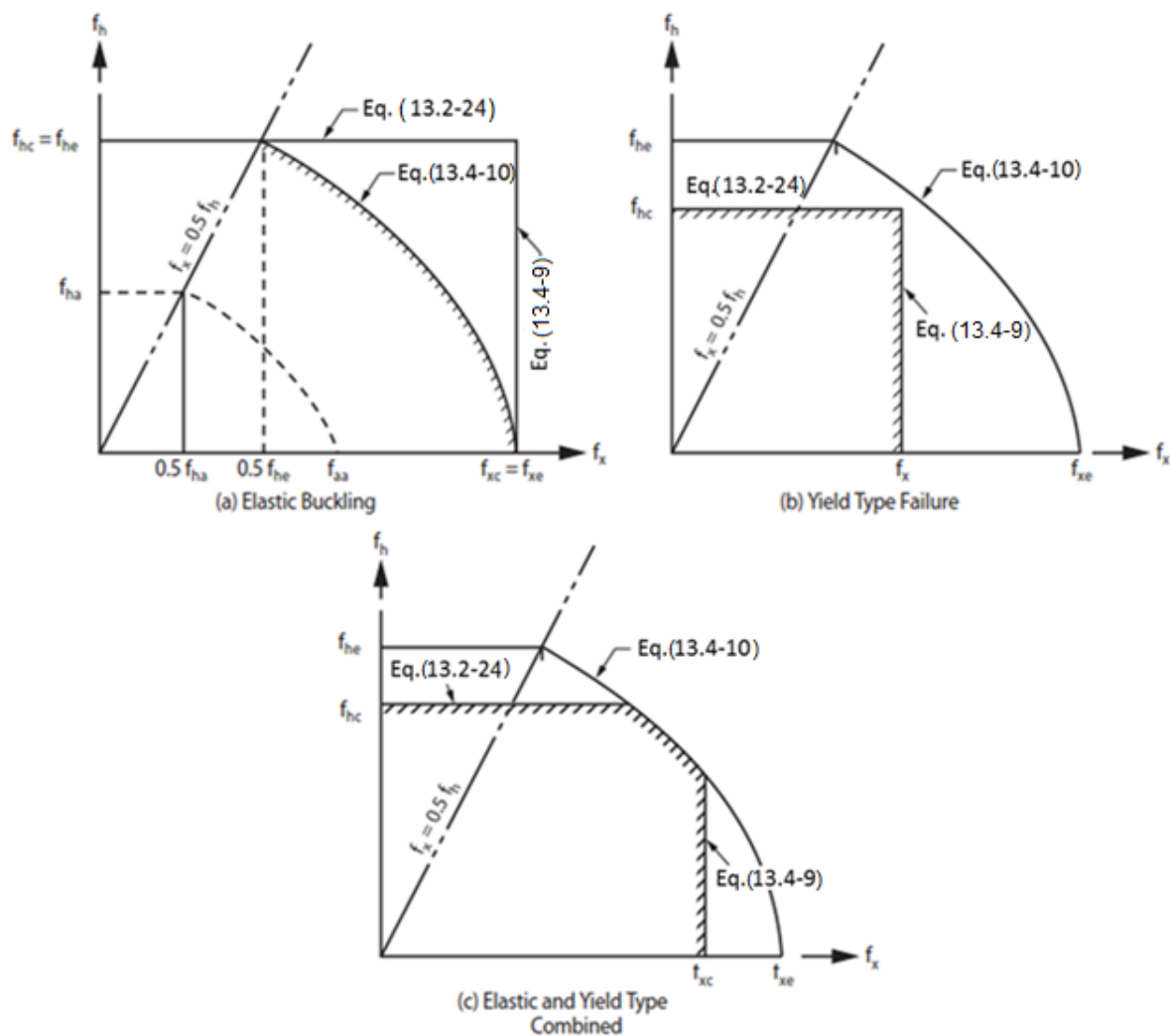
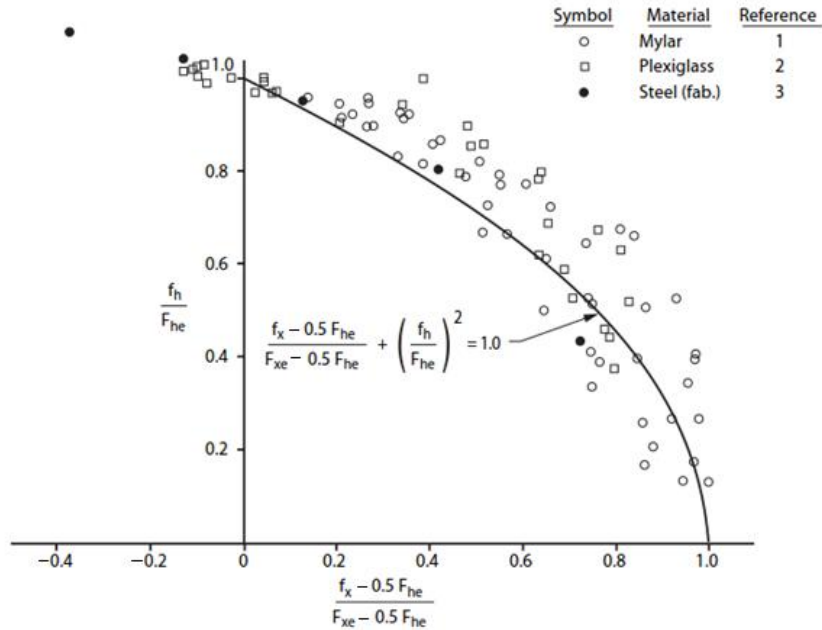
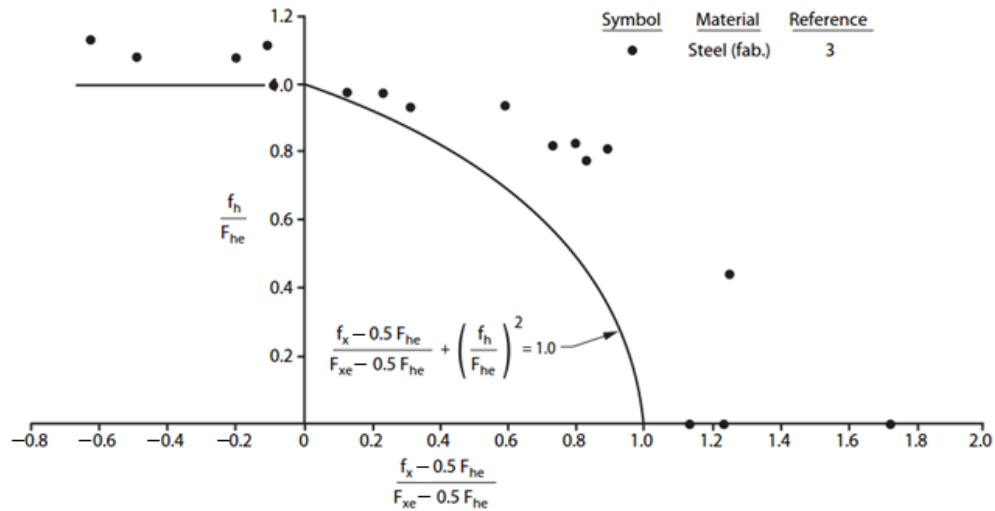


Figure A.13.4-3—Comparison of Interaction Equations for Various Stress Conditions for Cylinders under Combined Axial Compressive Load and Hydrostatic Pressure



NOTE F_{xe} and F_{he} are determined from tests.

Figure A.13.4-4—Comparison of Test Data with Elastic Interaction Curve for Cylinders under Combined Axial Compressive Load and Hydrostatic Pressure



NOTE F_{xe} and F_{he} are determined from recommended design equations.

Figure A.13.4-5—Comparison of Test Data on Fabricated Cylinders with Elastic Interaction Curve for Cylinders under Combined Axial Load and Hydrostatic Pressure

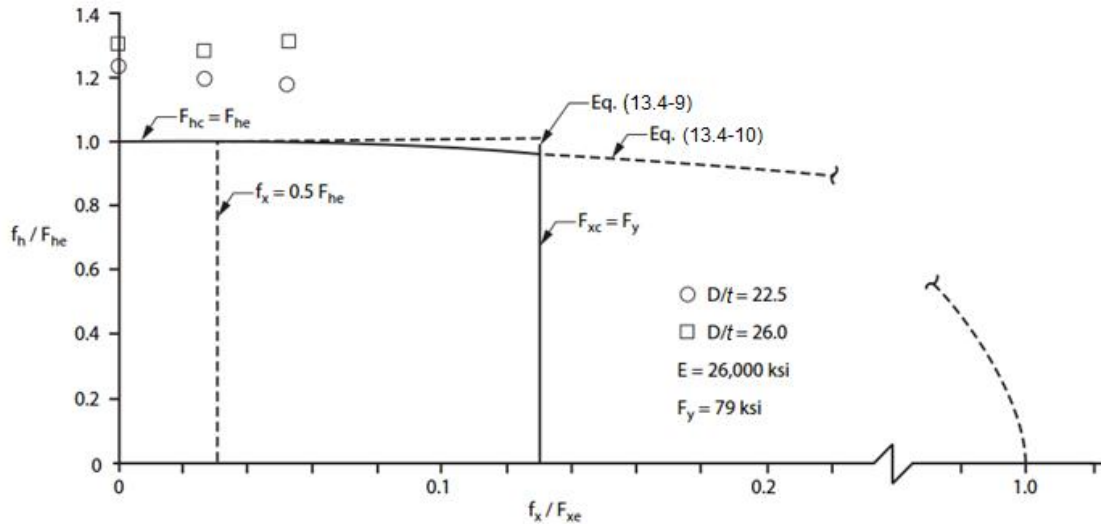


Figure A.13.4-6—Comparison of Test Data with Interaction Equations for Cylinders under Combined Axial Compressive Load and Hydrostatic Pressure (Combined Elastic and Yield Type Failures)

A.13.5 Effective Lengths and Moment Reduction Factors

The effective length used for a bracing member in framed structures is important for determining its axial compressive strength.

Based on experience with the use of nonlinear pushover analyses, it was concluded in Reference [A.13.5-1] and others that the accuracy of linear design predictions is significantly improved by using buckling lengths determined from refined analysis. Studies indicate that buckling lengths for real, three-dimensional structures demonstrate little variation when loads/actions are increased up to collapse. Thus, buckling lengths calculated for the initial elastic loading regime are, in fact, representative throughout the complete loading history.

The effective lengths for initial conditions can be calculated for an X-frame through the use of the differential equations given in Reference [A.13.5-2] or for a general structural component through the use of ϕ functions derived from differential equations (see Reference [A.13.5-3]). Use of ϕ functions implies that exact solutions are obtained for the considered buckling problem. General effective buckling lengths have been derived using ϕ functions incorporating end flexibility of members. The results for an X-brace with four equal-length members are shown in Figure A.13.5-1 as a function of the force distribution in the system, expressed by the ratio Q/P (where P is the maximum compressive force in the X-brace structure), and of the nondimensional end rotational stiffness parameter, ρ , which is defined by:

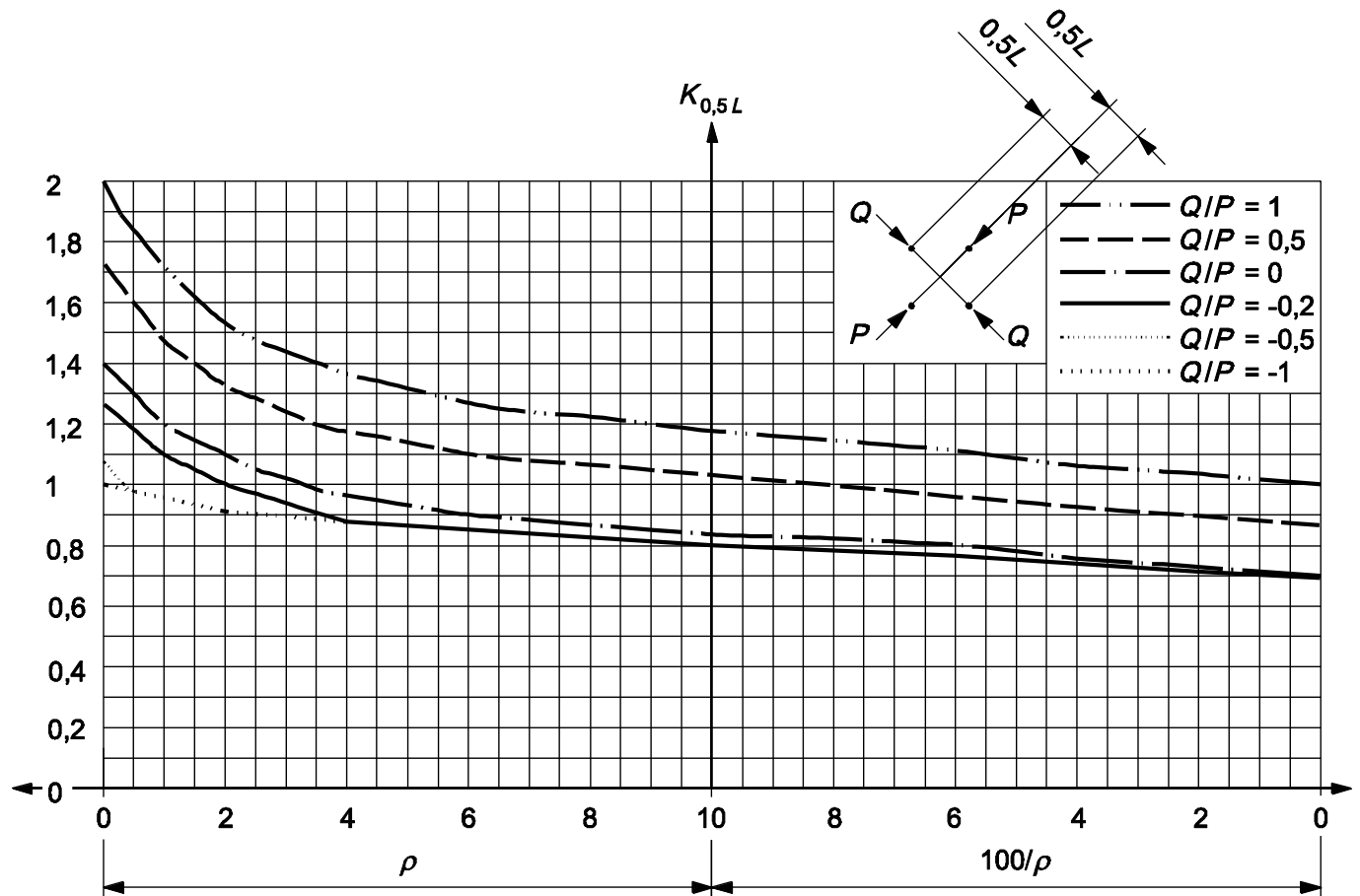
$$\rho = \frac{CL}{EI} \quad (\text{A.13.5-1})$$

In this equation, C is the rotational stiffness at a node and I is the moment of inertia of the tubular member. C should take into account the local joint flexibility [A.13.5-4] and the flexural and torsional stiffnesses of the incoming members. L refers to centerline-to-centerline distances between end nodes, i.e. excluding the intermediate X-joint node.

Figures A.13.5-1 to A.13.5-3 provide effective length factors as a function of the rotational stiffness of the member ends and the ratio of the axial forces in the braces for three X-brace configurations:

- equal segment lengths;
- a segment length ratio of 0.6 to 0.4;
- a segment length ratio of 0.7 to 0.3.

In developing Figures A.13.5-1 to A.13.5-3, the local bending stiffness at the X-joint is assumed to be infinite. The torsional stiffness of incoming members at the X-joint has been conservatively neglected.



Key

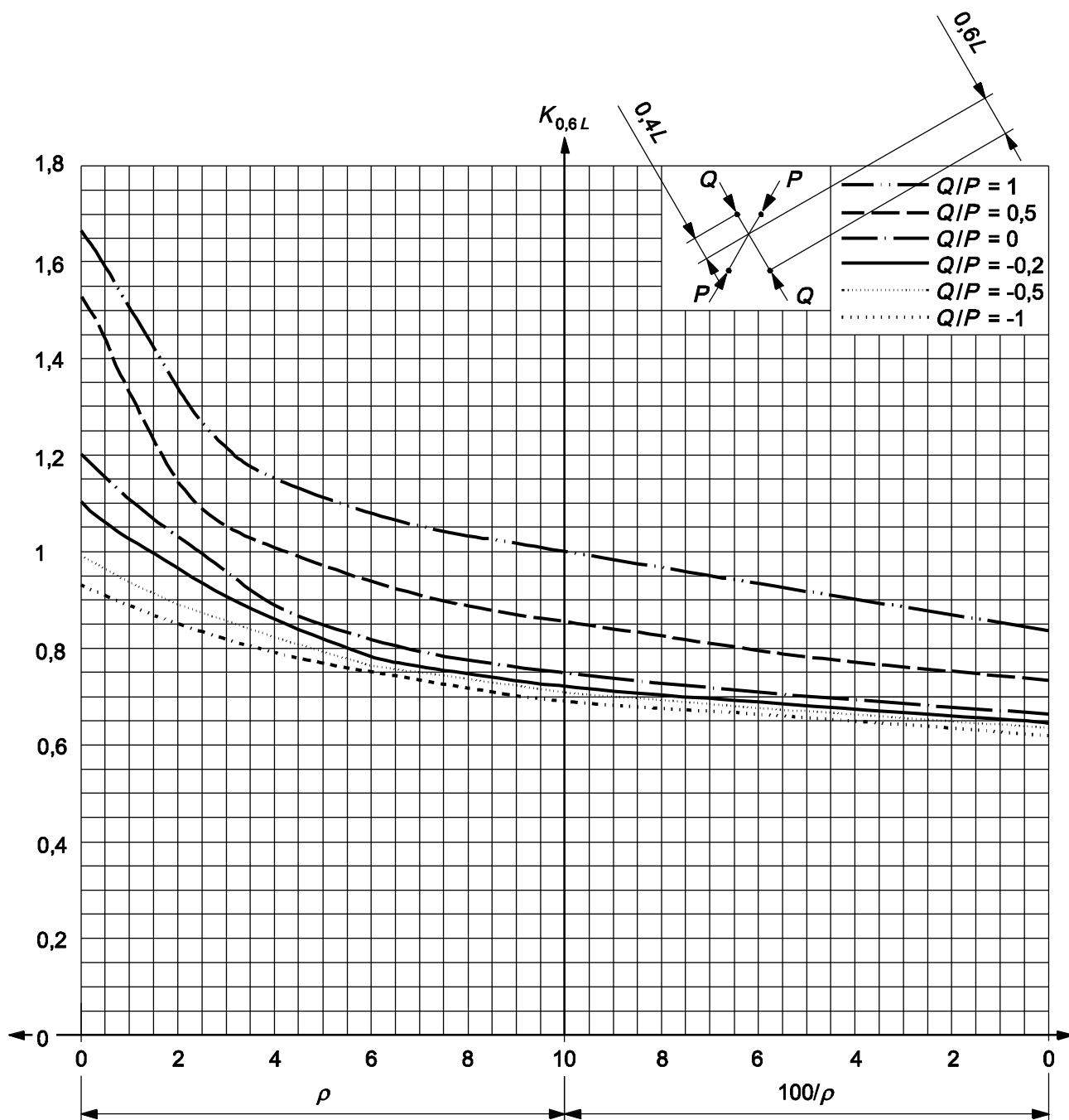
$K_{0.5L}$ effective length factor relative to 50 % of member length

ρ rotational stiffness parameter from Equation (A.13.5-1)

P compressive force in member under consideration

Q compressive force in other member of X-frame

Figure A.13.5-1—Effective Length Factors for Symmetric X-braces as Function of Force Distribution and Rotational End Stiffnesses for Equal Lengths (Torsional Stiffness of Crossing Member Included)

**Key**

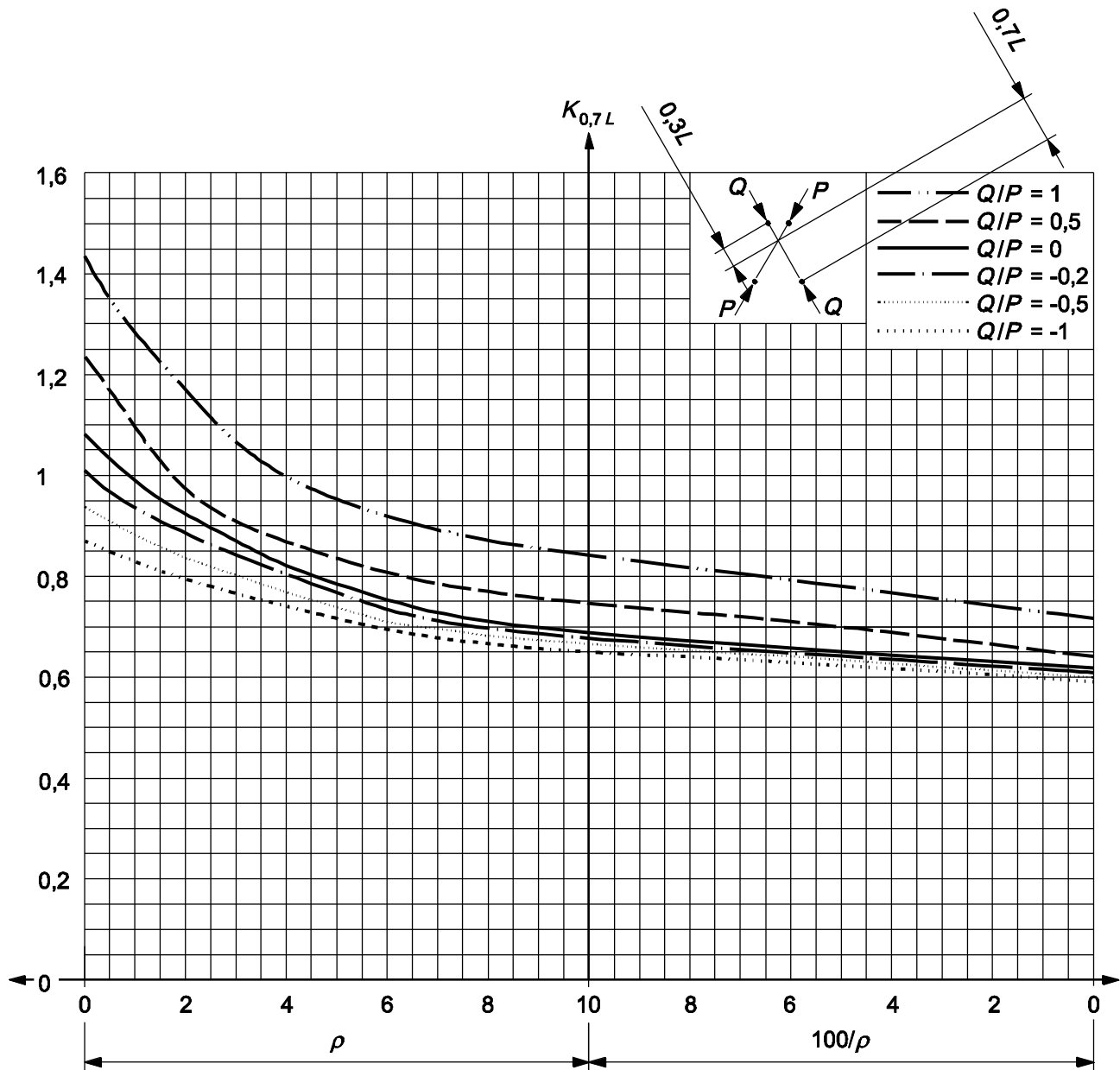
$K_{0.6L}$ effective length factor relative to 60 % of member length

ρ rotational stiffness parameter from Equation (A.13.5-1)

P compressive force in member under consideration

Q compressive force in other member of X-frame

Figure A.13.5-2—Effective Length Factors for Nonsymmetric X-braces as Function of Force Distribution and Rotational End Stiffnesses (Longer Segment = 0.6 Times the Brace Length)



Key

$K_{0.7L}$ effective length factor relative to 70 % of member length

ρ rotational stiffness parameter from Equation (A.13.5-1)

P compressive force in member under consideration

Q compressive force in other member of X-frame

Figure A.13.5-3—Effective Length Factors for Nonsymmetric X-braces as Function of Force Distribution and Rotational End Stiffnesses (Longer Segment = 0.7 Times the Brace Length)

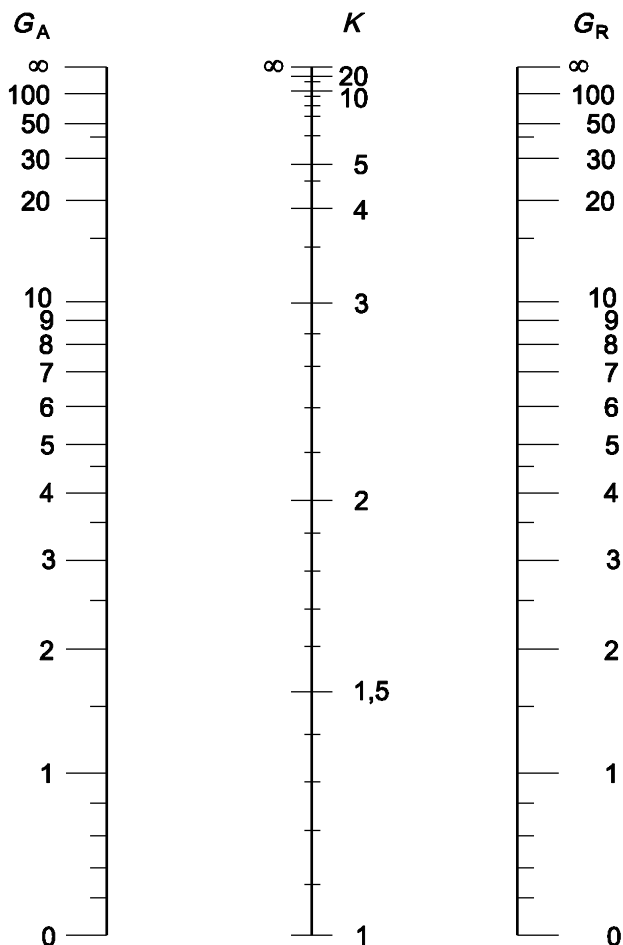
Figure A.13.5-1 indicates that for realistic end conditions ($\rho > 3$) and for single braces, i.e. when $Q/P = 1$ (if $Q/P = 1.0$, the member with Q load has no effect on the P loaded member), a K factor with respect to the full

X-brace length of less than 0.8 (i.e. less than 1.6 of $0.5 L$) is acceptable provided end joint stiffnesses do not diminish as the braces become fully loaded. Experimental results relating to frames seem to confirm 0.7 with respect to the full brace length is acceptable, which is confirmed by the results in Figure A.13.5-1 and recommended in Table 13.5-1.

In Table 13.5-1, a K factor of 0.8 with respect to the longer segment length is recommended for X-braces in which $Q/P \leq 0$. This value is well supported by the results in Figure A.13.5-3 for unequal X-braces with realistic end conditions ($\rho = 3$ to 5). For equal X-braces, however, an unusually high rotational end stiffness is required ($100/\rho < 7$) to obtain a K factor of 0.8 in Figure A.13.5-1—notwithstanding that the experimental results studied in Reference [A.13.5-5] support a K factor value of 0.8 for equal X-braces too. This may be attributed to the fact that significant rotational restraint is provided at the X-joint and at member ends.

For cases in which both X-braces are in compression, for realistic end conditions a K factor of 0.7 with respect to the full member length may be used. Implicitly, the X-joint is not braced.

To estimate the effective length of an unbraced column, such as topsides legs, the use of the alignment chart in Figure A.13.5-4 provides a fairly rapid method for determining adequate K values. The alignment chart may be modified to allow for conditions different from those assumed in developing the chart.



The subscripts A and R refer to the joints at the two ends of the column section being considered. G is defined as:

$$G = \frac{\frac{I_c}{L_c}}{\sum \frac{I_0}{L_0}}$$

where Σ indicates a summation of all members rigidly connected to that joint and lying in the plane in which buckling of the column is being considered, I_c is the moment of inertia, L_c the unsupported length of the column section, I_0 is the moment of inertia, and L_0 the unsupported length of a girder or other restraining member. I_c and I_0 are taken about axes perpendicular to the plane of buckling being considered.

For column ends supported by, but not rigidly connected to, a footing or foundation, G is theoretically infinity; but, unless actually designed as a true friction free pin, it may be taken as 10 for practical designs. If the column end is rigidly attached to a properly designed footing, G may be taken as 1.0. Smaller values may be used if justified by analysis.

Figure A.13.5-4—Alignment Chart for Determining the Effective Length of Unbraced Columns

For cantilever tubular members, an independent rational analysis is required for the determination of appropriate effective length factors. Such an analysis should take full account of all large deflection ($P-\Delta$) effects. For a cantilever tubular member, $C_m = 1.0$, where C_m is the moment reduction factor.

The purpose of using of C_m in the interaction Equation (13.3-3) is to obtain an equivalent moment for the moment pattern to which the beam-column is subjected that more accurately reflects the effect of the moment pattern on beam-column buckling of the member. Further discussion can be found in Reference [A.13.5-6], where more accurate expressions are provided—particularly, for relatively low levels of axial compression. The C_m values recommended in Table 13.5-1 are similar to those recommended in Reference [A.13.5-7] and its commentary.

A.13.7 Dented Tubular Members

A.13.7.1 General

The design recommendations are tailored to dented tubulars with dimensions and material yield strengths typical of offshore structure members ($f_y < 500$ MPa and $D/t \leq 120$). Because of the lack of test data for larger dent depths, the applicability of the proposed equations for dented members should be limited to $h/D \leq 0.3$ and $h/t \leq 10$.

The most significant parameter affecting the residual strength of dented members is dent depth. Experimental and analytical data (see References [A.13.7-1] to [A.13.7-4]) have shown that dent length and shape all have relatively negligible effects on the strength of a dented member. Furthermore, provided the dent is located length-wise in the middle half of the member, the loss of strength in a dented member seems insensitive to dent location (see Reference [A.13.7-2]).

The equations in this subsection are based largely on Reference [A.13.7-5], although the force format adapted therein has been converted to a “stress” format for inclusion in this document.

A.13.7.2 Dented Tubular Members Subjected to Tension, Compression, Bending, or Shear

A.13.7.2.1 General

The tension, compression, and bending recommendations contained in this subsection are based on a program of testing carried out since the late 1970s. Hydrostatic pressure recommendations have not been included in 13.7 due to limited data. Applicable tests up to 1995 have been documented in the references cited. Figures showing test results supporting the recommendations of 13.7 are included below in the relevant subsections.

A.13.7.2.2 Axial Tension

The strength equation for axial tension relies on yielding of the steel of the dented cross-section whose cross-sectional area is assumed unaltered by the damage.

A.13.7.2.3 Axial Compression

Dented tubular members subjected to axial compression fail primarily due to overall column buckling, with or without local buckling. The capacity of the dented member has been reduced by ξ_c with respect to the undamaged cross-section to account for the reduced effective cross-sectional area of the dented section.

The parameter ξ_c , defined in Equation (13.7-14), is discussed below.

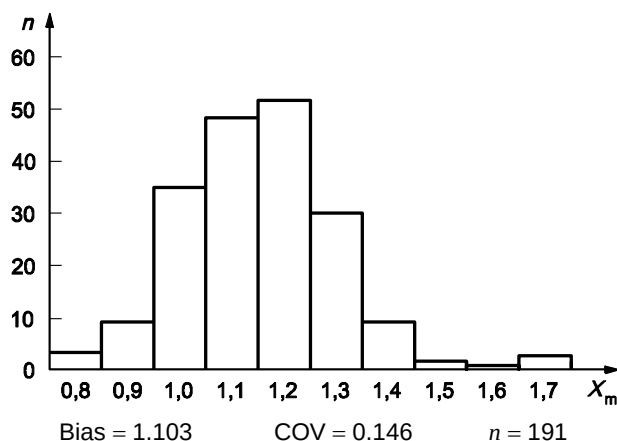
- a) Column buckling: The representative axial compressive strength for members with an out-of-straightness, Δy , which is not necessarily a consequence of the damage, is determined from Equations (13.7-5) and (13.7-6), which account for the additional bending stress on the dented cross-section due to the member out-of-straightness. Equation (13.7-5) deals with an out-of-straightness for dented tubulars that is still within the corresponding fabrication tolerance for undamaged members, i.e. 0.001 of the length. As such, only the effect

of the reduction in effective cross-sectional properties at the dent needs to be taken into account. For an out-of-straightness greater than the fabrication tolerance, only the out-of-straightness in excess of the tolerance needs to be considered [Equation (13.7-6)] because the column buckling Equations (13.2-5) and (13.2-6) already account for Δy values up to the level of the tolerance.

The column buckling equations are similar to the undamaged member equations [Equations (13.2-5) and (13.2-6)] except that λ has been replaced by λ_d , the column slenderness parameter that accounts for the effects of denting on cross-sectional properties. The test database was obtained from several different investigators (see References [A.13.2-6], [A.13.2-9], [A.13.7-1] to [A.13.7-4], and [A.13.7-6] to [A.13.7-13]). A comparison between the test data and the predictions using Equations (13.7-5) and (13.7-6) is given in Figure A.13.7-1.

- b) Local buckling: The local buckling equation for dented members is similar to that for undamaged tubular members except that the parameter ξ_c has been introduced [Equations (13.7-7) and (13.7-8)] to account for the reduced strength of the dented cross-section. Discussion of the dented member parameters ξ_c and ξ_m (see A.13.7.2.4) can be found in Reference [A.13.7-14].

The results of column tests on relatively short damaged tubulars ($\lambda_d < 0.252$) are plotted normalized with respect to their undamaged strength [Equations (13.2-8) and (13.2-9)] against the dent damage ratio h/t in Figure A.13.7-2. Also plotted is the cross-sectional area dented member parameter, ξ_c [Equation (13.7-14)]. The appropriateness of ξ_c to represent the effects of dent damage on the local buckling strength is clearly seen.

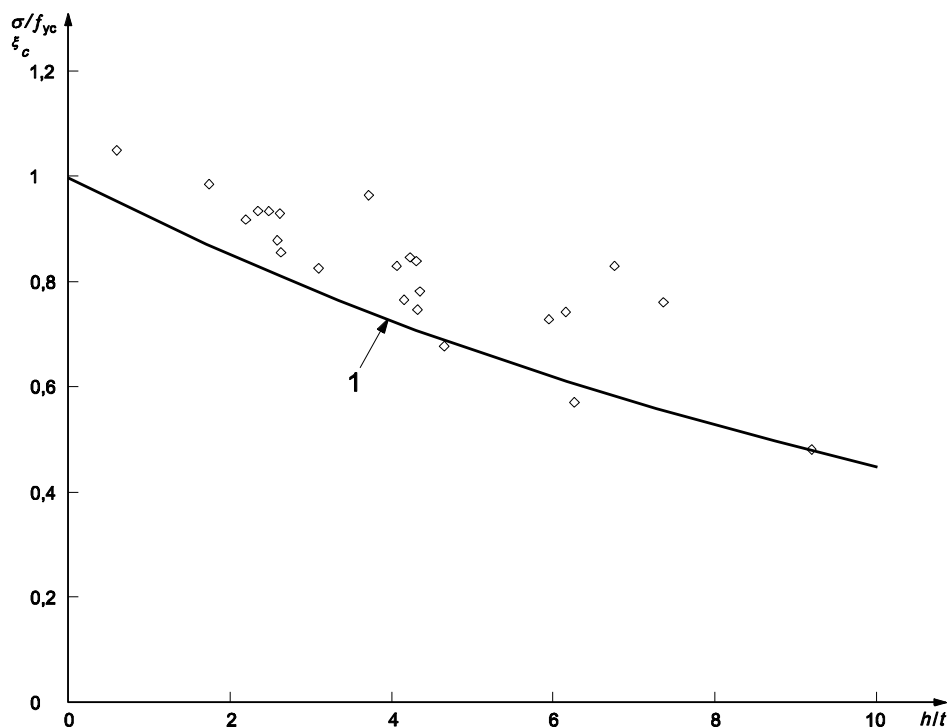


Key

n frequency (number of occurrences)

X_m measured strength/predicted strength

Figure A.13.7-1—Comparison of Measured and Predicted Strengths for Dented Tubulars Subjected to Axial Compression



Key

- 1 $e^{-0.08 h/t}$
 h dent depth
 t wall thickness
 σ stress in test member
 f_{yc} representative local buckling strength (undamaged)
 ξ_c compression strength knock-down factor

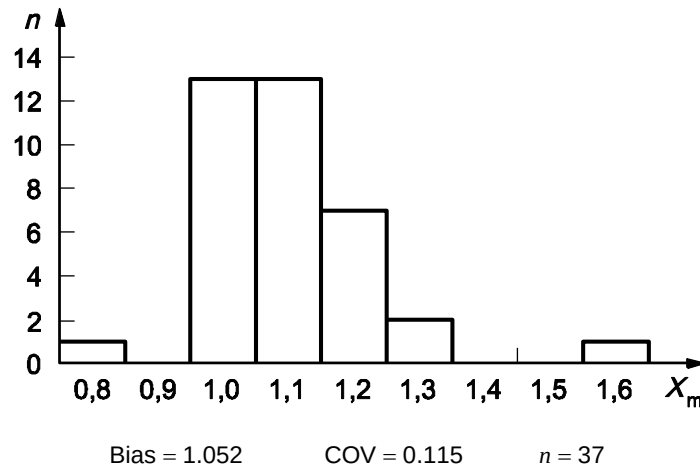
Figure A.13.7-2—Comparison of Normalized Test Data of Dented Tubulars with Compression Strength Knock-down Factor for $\lambda_d < 0.252$

A.13.7.2.4 Bending

When dented tubulars are subjected to bending moments resulting in tensile (positive) or zero (neutral) stresses in the center region of the dent, the bending strength of the dented tubular is not appreciably lower than that of an undamaged member. The representative bending strength equations for members under positive [Equation (13.7-16)] or neutral [Equation (13.7-20)] bending are thus similar to that for undamaged members [Equation (13.2-11)].

For members subjected to bending moments resulting in compressive (negative) stress in the center region of the dent, the bending strength of the dented member is reduced compared with that of the undamaged member. Test data from References [A.13.7-9] and [A.13.7-15] indicate that the reduction in strength can be approximated by the application of Equation (13.7-15) to Equation (13.2-11).

A comparison between test data and the prediction of the representative bending strength equation, $\xi_m \times f_b$, is plotted in Figure A.13.7-3. The statistics of the fit are indicated in the figure.

**Key**

n frequency (number of occurrences)

X_m measured strength/predicted strength

Figure A.13.7-3—Comparison of Measured and Predicted Strengths for Dented Tubulars Subjected to Bending (Dent in Compression)

A.13.7.3 Dented Tubular Members Subjected to Combined Forces

A.13.7.3.1 Axial Tension and Bending

The tension-bending interaction equations for dented tubulars, Equations (13.7-24) and (13.7-26), are similar in form to the corresponding equation for undamaged members, Equation (13.3-1), although, for Equation (13.7-26), a less compact format is used because of the need to cater to the differences between positive and negative bending effects. The α term in Equation (13.7-26) reflects the change in the interaction between the two orthogonal moments when a dent is present (see Reference [A.13.7-14]).

Recommendations for the interaction of axial tension, bending, and hydrostatic loads have not been included due to limited test data.

A.13.7.3.2 Axial Compression and Bending

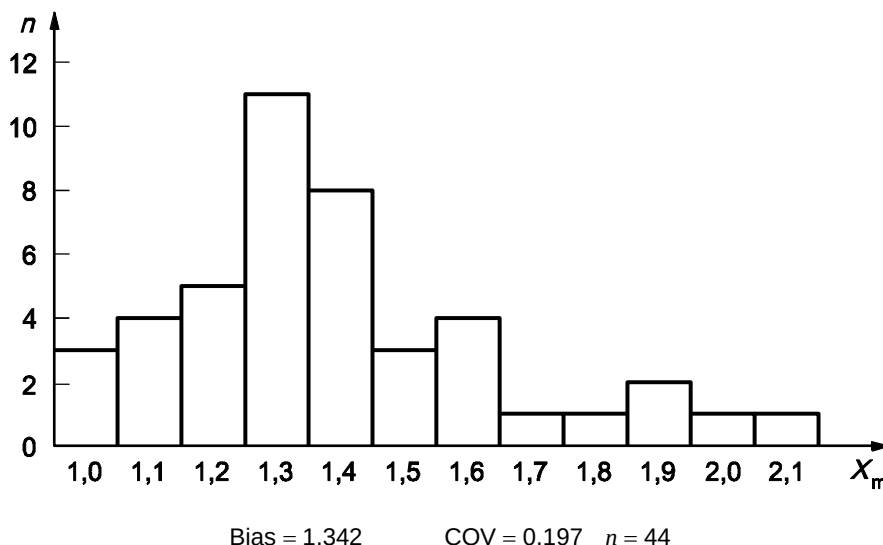
Both overall beam-column stability checks [Equations (13.7-29) and (13.7-34)] and strength checks [Equations (13.7-30) and (13.7-35)] for dented tubular members subjected to combined axial compression and bending are provided, again in a format similar to that accepted for undamaged tubulars in 13.3.3.

For members with different effective length factors in the negative and neutral bending directions, the Euler buckling strengths, f_e and $f_{e,d}$, will be different and should be taken into account in Equation (13.7-34).

The value of C_m used in the stability equations for dented tubulars has been conservatively set to 1.0, although an upper limit of 0.85 is normally recommended for end-restrained undamaged members in laterally braced frames typical of offshore steel structures as described in Table 13.5-1. For dented members, values of C_m smaller than 1.0 may be justified, but these should be considered on a case-by-case basis.

A comparison between test data and the combined axial compression and bending strength equations [Equations (13.7-29), (13.7-30), (13.7-34), and (13.7-35)] is plotted in Figure A.13.7-4. The statistics of the fit are indicated in the figure.

Recommendations for the interaction of axial compression, bending, and hydrostatic loads have not been included due to limited test data.



Key

n frequency (number of occurrences)
 X_m measured strength/predicted strength

Figure A.13.7-4—Comparison of Measured and Predicted Strengths for Dented Tubulars Subjected to Combined Compression and Bending (Dent in Tension, Compression, or on Neutral Axis)

A.13.8 Corroded Tubular Members

One approach to estimate the strength of an approximately uniformly corroded member is to assume a reduced thickness for the entire member. The reduced thickness should be consistent with the average material loss due to corrosion. The member with the reduced thickness can then be evaluated as an undamaged member. This reduced thickness approach is generally conservative, as discussed in Reference [A.13.8-1]. However, caution should be exercised with respect to fatigue.

Another common case of corroded members is the presence of severe localized corrosion in the form of patches. This form of corrosion cannot be approximated as uniform. References [A.13.7-13], [A.13.8-2], and [A.13.8-3] provide equations to estimate the local buckling strength of these corroded members as a function of the thickness loss. No information is available on the effect of localized corrosion on overall member strength. In this case, FEA or a numerical segment approach based on empirical or analytical moment-axial force-curvature ($M-P-\phi$) relations can be used. For fatigue sensitive conditions, a fatigue evaluation of the corroded member should also be considered.

A.13.9 Grouted Tubular Members

A.13.9.1 General

Grout filling offers benefits to undamaged members, but offers greater benefits to members suffering limited damage following impacts from vessels or dropped objects. For undamaged members, greater member strength and stiffness can be achieved at the expense of weight, but without any corresponding increase in environmental loads/actions. For damaged members, growth in dent damage is inhibited, enabling the full strength of the damaged cross-section to be achieved.

Most of the available test data relates to tubulars in which grouting has been introduced following damage, usually in the form of local dents but frequently accompanied by overall bowing. The reason for this is that the use of grouting in offshore tubulars has largely been driven by the need to strengthen/stiffen damaged members.

The formulations presented in 13.9 are only applicable to fully grouted tubular members. Based on actual experience, it can be difficult to achieve a fully grouted condition in offshore repair or strengthening operations. Consequently, some reduction in the calculated residual strength will be necessary to account for the potential incomplete grouting at the end of the grouted member. Because only limited data are available for partially grouted tubular members (especially undamaged ones) and because defining the extent of the grout filling of tested specimens is not precise, no design equations for partially grouted tubular members can be given at this time. However, the strength equations for such partially filled tubulars can be formulated by assuming that the grout does not contribute to either the axial compressive strength or to the moment capacity. This is achieved by setting the terms for the fully grouted members that relate specifically to the grout to zero (see References [A.13.9-1] and [A.13.9-2]).

A.13.9.2 Grouted Tubular Members Subjected to Tension, Compression, or Bending

A.13.9.2.1 General

The tension, compression, and bending requirements contained in this subsection are based largely on the studies reported in References [A.13.9-1] and [A.13.9-2]. Shear and hydrostatic pressure requirements are not included because of lack of data. Applicable tests up to 1995 have been documented in the above references. Figures showing test results supporting the requirements of 13.9 are included in the appropriate subsections.

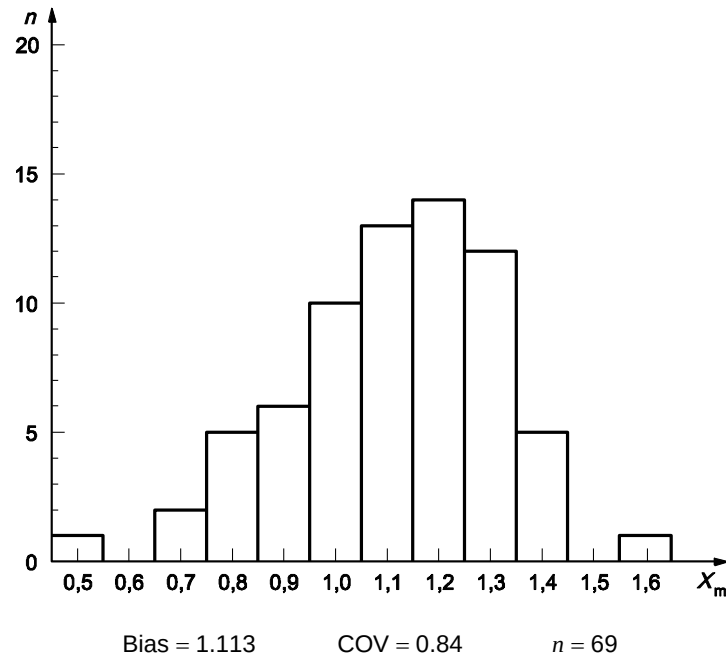
A.13.9.2.2 Axial Tension

The strength equation for axial tension reflects only yielding of the steel cross-section. The contribution of the grout to the tensile capacity of the fully grouted member has been conservatively ignored. When complete grouting can be demonstrated, advantage can be taken of the benefits of the grouting.

A.13.9.2.3 Axial Compression

The axial compressive strength of fully grouted tubular members is determined from Equations (13.9-8) and (13.9-9). These column buckling equations are similar to the undamaged and nongrouted member strength equations [Equations (13.2-5) and (13.2-6)], except that the slenderness parameter λ has been replaced by λ_g and the representative local buckling strength f_{yc} has been replaced by the axial squash strength f_{ug} of the fully grouted member.

The test data for fully grouted undamaged tubulars are taken from References [A.13.9-1] and [A.13.9-3] to [A.13.9-5]. Similar data for fully grouted damaged tubulars are taken from References [A.13.9-3] to [A.13.9-8]. A comparison between the test data and the predictions using the strength equations is presented in Figure A.13.9-1.

**Key**

n frequency (number of occurrences)
 X_m measured strength/predicted strength

Figure A.13.9-1—Comparison of Measured and Predicted Strengths Using the Representative Axial Compressive Strength Equations for Fully Grouted Damaged and Undamaged Tubular Members

A.13.9.2.4 Bending

The bending strength of fully grouted tubular members may be calculated from Equation (13.9-19). The bending strength of the dented section is found from a second-order fit to the exact equation, simply expressed in terms of dent depth to diameter ratio. The effect of the presence of the grout is based on an approximation to the semi-graphical representation of this feature in BS 5400-5 [A.13.9-9].

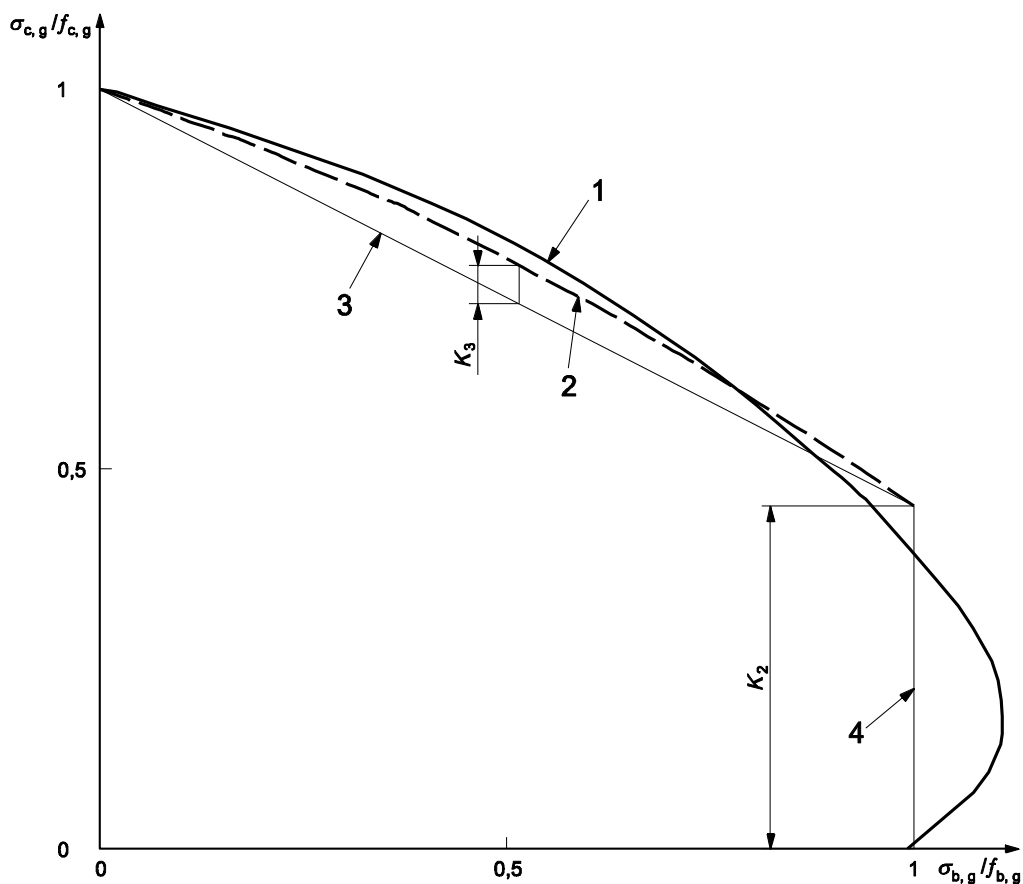
A.13.9.3 Grouted Tubular Members Subjected to Combined Forces**A.13.9.3.1 Axial Tension and Bending**

For a fully grouted, undamaged tubular subjected to combined axial tension and bending forces, there are several ways to estimate its strength. The member may be assessed using Equation (13.3-1) by neglecting the effect of the grout or using Equation (13.9-19) if the maximum stress due to tension is small in comparison to that due to bending. For a fully grouted, dented tubular member, Equations (13.7-24) and (13.7-26) may be used by replacing f_b with $f_{b,g}$.

A.13.9.3.2 Axial Compression and Bending

The design requirements are based on those given in Reference [A.13.9-9] for concrete-filled tubular members subjected to combined axial compression and bending. The background to the design requirements is documented in References [A.13.9-10] and [A.13.9-11]. The interaction equations for fully concrete-filled tubular members are judged to be applicable to grout-filled tubular members because their behavior and the strength contribution of the grout are very similar to those of concrete filled tubulars.

The parameter, K_1 , as given in Equations (13.9-28) and (13.9-29), is the nondimensional axial compressive strength of the member. A slightly different K_1 equation is given in Reference [A.13.9-9]. The significance of the parameters K_2 and K_3 is illustrated in Figure A.13.9-2. The influence of the parameter K_2 is low for tubular members having a small outer diameter and a high slenderness ratio, regardless of the D/t ratio. Low values of K_2 generally also mean a low value of K_3 , which can be negative for very small diameters and a high slenderness ratio. In general, the advantageous effect of grouting resulting in increased strength diminishes as the diameter becomes smaller.



Key

- 1 interaction curve from analytical solution
- 2 approximating parabolic curve in Equation (13.9-24)
- 3 Equation (13.9-38) for fully utilized and fully grouted member with $K_3 = 0$
- 4 Equation (13.9-39) for fully utilized and fully grouted member
- $\sigma_{c,g}$ axial compressive stress in fully grouted member
- $\sigma_{b,g}$ bending stress in fully grouted member
- $f_{c,g}$ representative axial compressive strength of grouted member
- $f_{b,g}$ representative bending strength of grouted member
- K_2 variable from Equation (13.9-30)
- K_3 variable from Equation (13.9-31)

Figure A.13.9-2—Illustration of Relevance of Parameters K_2 and K_3

A.14 Strength of Tubular Joints

A.14.1 General

The requirements of Section 14 are intended to provide guidance on the design of tubular joints covering the wide range of joint configurations, geometries, and load/action combinations that can be found in practice. The joint equations are consistent with API 2A-WSD, Twenty-Second Edition, which incorporates a more up-to-date joint test and analysis database. Additional data on the background and modifications of joint design can be found in B.7 of API 2A-WSD, Twenty-Second Edition.

The informative that follows is basically the same as contained in ISO 19902:2007.

Despite continuing industry-wide efforts, combinations of test data and analytical techniques continue to be used as a basis for design. Guidance on using tests and analyses is contained in 7.7, whereas Reference [A.14.1-1] provides methods for selecting an appropriate reduction factor for joint strength to account for a small number of data.

A.14.2 Design Considerations

A.14.2.1 Materials

All of the empirical strength equations have been based on measured yield strengths. Very few test results have indicated unexpected low strength due to substandard material properties. However, it is recognized that some limits are implied by the database.

Over the period 1994 to 1996, MSL Engineering (UK) ² undertook an update of the tubular joint database and guidance through a joint industry project (JIP) (see References [A.14.2-1] and [A.14.2-2]). One important change resulting from the MSL work concerns steels with high yield-to-ultimate strength ratios. Many existing codes do not allow the designer to assume more than a ratio of about 0.7 or $\frac{2}{3}$. If the ratio exceeds this limit, the chord yield strength is downgraded to the specified ratio of the tensile strength. The MSL work found that the database justified a limit on the ratio of 0.8 for joints with a chord yield strength of up to at least 500 MPa.

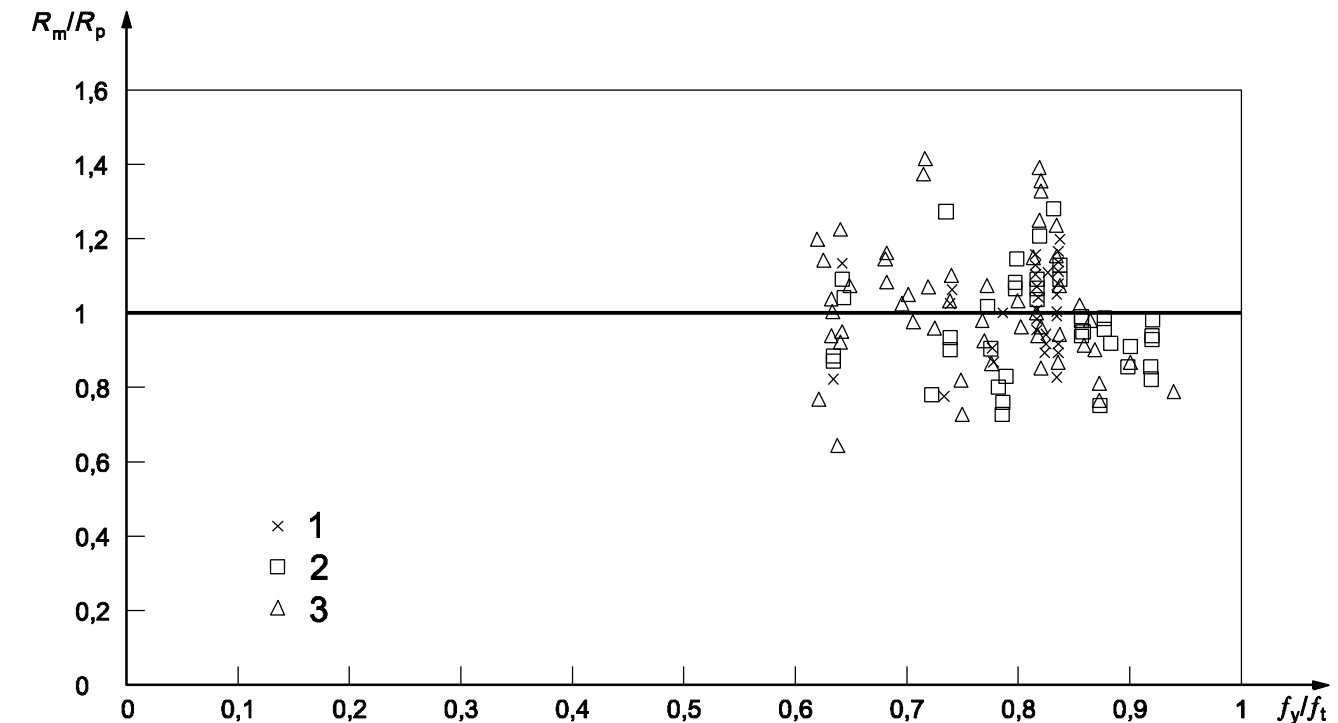
Figure A.14.2-1 shows a plot of the ratio of measured to predicted K-joint strength against the chord material yield-to-ultimate strength ratio. This is based on the updated MSL database [A.14.2-1] with the predicted representative joint strength taken from Section 14 using actual chord yield strength values and the predicted representative strength suitably upgraded to determine the mean strength of the joint. It can be seen from Figure A.14.2-1 that no effect of yield strength occurs for yield-to-tensile strength ratios up to 0.8. Similar observations have been made with respect to other joint types and load cases. Some steels exhibit adequate toughness even when the ratio exceeds 0.8. However, use of this limit is still considered appropriate because strength equations generally imply some degree of strain hardening. Lack of strain hardening can lead to large local strains and premature cracking. Use of the limiting yield-to-tensile strength ratio of 0.8 ensures that joints exhibit strength and ductility behavior representative of those in the test database.

The material property range is limited to $f_y \leq 500$ MPa. There has, historically, been a concern that the strength of joints with chord yield strength in excess of 500 MPa does not increase in proportion to the yield strength. The concern relates to the potential that the higher yield strength is obtained at the expense of lower ductility and lower strain hardening capacity, thereby compromising the post-yield reserve strength on which the strength equations rely. This matter is discussed in Reference [A.14.2-3]. A re-evaluation of the test results reported in Reference [A.14.2-3] has been carried out. This re-evaluation has revealed that use of the limiting yield-to-tensile strength ratio of 0.8 is generally adequate to permit the strength equations to be used for joints with

² This information is given for the convenience of users of this document and does not constitute an endorsement.

$500 \text{ MPa} < f_y \leq 800 \text{ MPa}$, provided adequate ductility can be demonstrated in both the HAZ and in the parent material. It should, however, be noted that the test data reported in Reference [A.14.2-3] is limited to a small number of joint types and loading modes with just 11 joints tested in total.

A later JIP [A.14.2-4] investigated the static strength of high-strength steel X-joints. The JIP involved the testing of four compression joints (two with a representative yield strength of 355 MPa and one each with 500 MPa and 700 MPa) and three tension joints (one each with a representative yield strength of 355 MPa, 500 MPa, and 700 MPa). The findings presented in Reference [A.14.2-4] appear to indicate that all the joints performed satisfactorily in the tests in terms of strength and ductility, confirming the practicality of using higher-strength steels. The results from this limited data identified that a yield-to-tensile strength ratio of 0.8 could be used to estimate the ultimate compression and tension strengths of the joints. However, no detailed assessments were presented for the joints subjected to tension forces in which cracking was observed prior to reaching the ultimate strength.



Key

- 1 near zero gap/overlap K-joints
- 2 overlapped K-joints
- 3 gapped K-joints
- f_y yield strength
- f_t tensile strength
- R_m measured joint strength
- R_p predicted joint strength

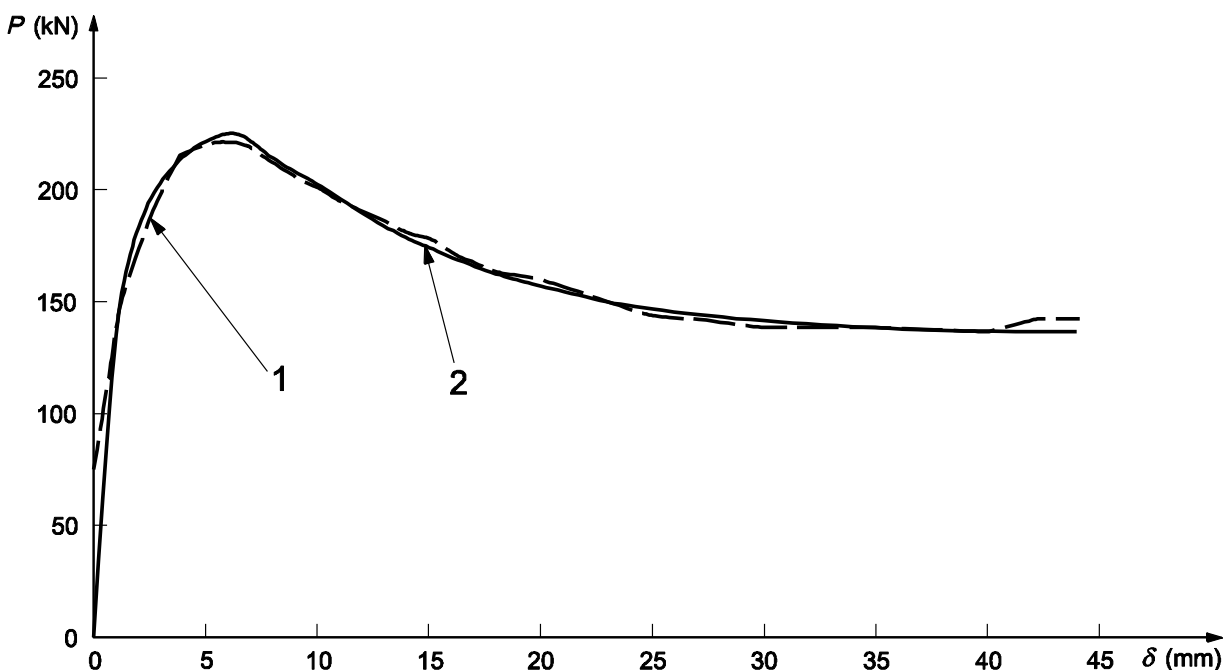
Figure A.14.2-1—Effect of f_y/f_t for K-joints on Joint Strength under Balanced Axial Forces

A.14.2.2 Design Forces and Joint Flexibility

Historically, designers of fixed steel offshore structures have usually assumed that the joints are rigid and that the frame can be modeled with members extending to working points at chord centerlines. It has long been recognized, however, that tubular joints can show significant linear elastic flexibility at locations such as skirt pile bracing. Joint flexibility has sometimes been included in calculating fatigue life estimates. Fatigue life estimates for conductor framing connections can sometimes be substantially greater when linear elastic flexibilities are included in the analyses. Full nonlinear force-deformation curves for joints can be required for pushover analyses to determine a system's ultimate strength, especially where joint failures participate in the sequence leading to system collapse. Such pushover analyses are common in studies for maintenance and life extension of structures.

In 1993, Buitrago et al. [A.14.2-5] published a robust set of equations for the linear elastic flexibility and stiffness of simple tubular joints. Although a number of other sets of formulations are available in the literature, a comparative exercise shows that the Buitrago formulations are wide-ranging, have physical meaning, compare well with hard data, and are simple to use both manually and computationally.

As part of the MSL JIP [A.14.2-1], the understanding of linear elastic flexibility was extended, through analyses of the updated database, and a range of closed-form expressions was established which permit the creation of nonlinear force-deformation (P - δ or M - θ) curves, in both loading and unloading regimes, for simple joints across the practical range of load cases and geometries. Details are presented in Reference [A.14.2-6]. Figure A.14.2-2 shows a fitted curve superimposed on the experimental curve.



Key

- 1 experimental curve
- 2 fitted curve
- P force
- δ deformation

Figure A.14.2-2—Comparison of Fitted Curve to Experimental Data for Force-deformation Behavior of a Simple Tubular Joint

Reference [A.14.2-7] reports on a pilot study to assess the effect of hydrostatic pressure on tubular joint strength. Double T-joints [DT-joints (see 14.2.4)] were studied, and the results indicate that strength can be reduced by up to 30 %, depending on geometry, brace force or moment, and hydrostatic pressure. Apart from Reference [A.14.2-7], no other studies on this subject have been identified. Hydrostatic pressure effects can be significant, especially for deep water structures.

A.14.2.3 Minimum Strength

Historically, API had a minimum strength requirement that equated to 50 % of the strength of the brace. For earthquake loads/actions, the requirement was essentially 100 % of the brace strength. Aside from earthquake regions, the 50 % strength requirement generally dominated secondary joint design [A.14.2-8] and often took precedence for primary joints. Although it was not intended that this 50 % rule should govern the design of joints it was an inadequate requirement both for accidental situations and for joints critical to the reserve/residual strength of the structure. In general, joint failure prior to member failure is undesirable, due to uncertainty in failure behavior and the effect on surrounding braces.

The intent in this document is to simplify the requirement for critical load paths. In order to ensure that joints do not fail before members, joint utilization is limited to $1/\gamma_{zj}$ of the member utilization.

A.14.2.4 Joint Classification

Some special cases of the basic planar joint types have their own names. A special case of a Y-joint with a brace angle of approximately 90° is called a T-joint. Another special case of a Y-joint is a double T- or DT-joint, which has braces on opposite sides of the chord, each with a brace angle of around 90°. By its configuration, a DT-joint appears to be a special case of an X-joint with braces under 90° and the through member serving as the chord. However, the axial forces in both braces of a DT-joint are reacted by beam shear in the chord, whereas in an X-joint the axial brace force is transferred through the chord to the opposite side without a beam shear reaction of the chord. A joint should be classified as a DT-joint if the brace on one side of the chord is in tension and the brace on the other side of the chord is in compression. Both braces should then be treated separately as for a T-joint.

A K-joint where one of the two braces has a brace angle of approximately 90° [see, for example, case a) in Figure 14.2-2] is sometimes referred to as an N-joint.

Joint classification is based on axial force pattern, as well as on joint configuration. The favorable strength factor Q_u (see 14.3.3) for K-joints reflects their reduced joint can ovalization as compared to Y-joints. The Q_u values for X-joints are typically worse than for Y-joints, because X-joints have an even more severe ovalization tendency.

If the axial force in a given brace corresponds within ± 10 % to one of the standard joint types, it is permissible to classify the brace end as totally of that joint type: no interpolation is required. Many real joints have braces that are not clearly of one given type. In other words, the distribution of forces is complex, in the sense that an individual brace axial force should be divided into portions that are each treated as a Y-, K-, or X-joint. Figure 14.2-2 c) and h) contain braces with mixed classifications.

Several schemes for automating the classification process have evolved over the years. None is unique. In all of them, braces belonging to a particular joint type are identified and the geometrical information is catalogued. Braces lying in a common plane and on the same side of the node are identified and the gap between them is computed. Each brace is evaluated based on its axial force for each load case. Classification may change from load case to load case and is often different for each brace at a given joint. Mixed classifications generally occur.

In a typical classification scheme, braces whose axial force component perpendicular to the chord is essentially balanced by axial forces in other braces on the same side of the chord are treated as K-joints. Figure 14.2-2 a), d), e), and g) are such cases, as are the lower braces shown in Figure 14.2-2 c) and h). Braces whose perpendicular force components are reacted by beam shear in the chord are treated as Y-joints, as in Figure 14.2-2 b), even

though its geometrical appearance suggests a K-joint. Finally, braces whose perpendicular forces follow neither K-joint nor Y-joint classification are treated as X-joints. In other classification schemes, the hierarchy is K- followed by X-, with T/Y- being the default.

The classification scheme does not quantitatively address multiplanar connections, even though fixed steel offshore structures are generally space frames, not planar trusses. Furthermore, the scheme does not recognize that several braces in a given plane can simultaneously contribute to ovalization of the chord, for example, in launch frames and the other examples illustrated in Figure A.14.2-3. Such cases can produce a more adverse force distribution than is recognized in the classification scheme.

An alternative approach to joint classification is to use the ovalization parameter, α , from AWS D1.1/D1.1M [A.14.2-9]. The attraction of the α -based classification in AWS D1.1/D1.1M is that it does not require any decisions concerning classification. In a general sense, it encompasses the simple joint classification scheme recommended in this document and provides a promising methodology for multiplanar joints. However, deficiencies exist with the AWS approach where:

- a) α significantly exceeds baseline X-joint classification,
- b) the diameter ratio β is above 0.9, or
- c) a low value of α is not the result of classical K-joint load/action.

AWS D1.1/D1.1M does not properly incorporate the effect of the gap or address tension failures in the same manner as this document.

Another JIP [A.14.2-10] has generated a considerable database of finite element data for multiplanar joints having no overlapped braces and being subjected to axial brace forces. Refined expressions are given for the ovalization parameter, α , which are used within the AWS D1.1/D1.1M approach.

In cases such as those illustrated in Figure A.14.2-3, it is conservative to first find the sum of the force components perpendicular to the chord that are passed through the chord section and then assume that the strength is the minimum of any one of the brace-chord intersections when acting as an X-joint. General collapse calculations or FEA can reduce the conservatism.

Additional guidance specific to multiplanar T/Y-, K-, and X-joints with axial forces can be found in Reference [A.14.2-11]. More contemporary information on multiplanar T/Y- and K-joints is available in References [A.14.2-12] to [A.14.2-14]; however, the guidance provided in these is not especially robust, and there are general restrictions as to force pattern, as well as to joint configuration.

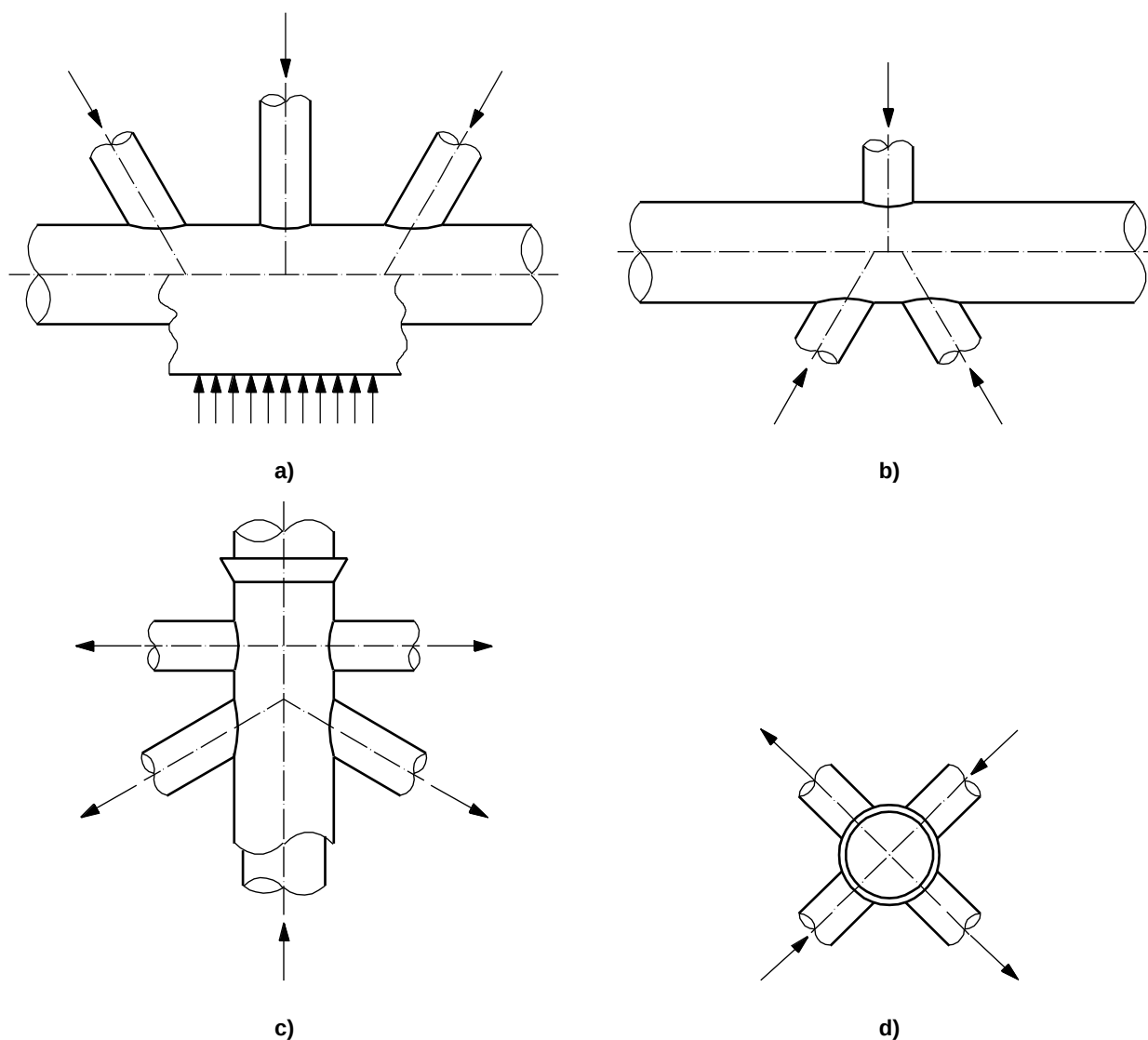


Figure A.14.2-3—Force Patterns Causing Chord Ovalization Greater Than Considered in Strength Equations

A.14.2.5 Detailing Practice

The API guidelines [A.9.1-1] have been changed in this document in several important ways. The can and stub length dimensions are unchanged, but length measurement does not include thickness tapers.

The guidance on overlap dimension has been changed to simplify analysis and make measurement easier during fabrication. The minimum overlap dimension is a recommendation rather than an absolute requirement. There are many practical instances where only minor overlap occurs. These cases are fully amenable to contemporary analysis for both strength and fatigue. Fabrication of small overlaps has not proved particularly difficult; however, any overlap in a joint reduces the extent to which a joint can be inspected.

In many instances, using the minimum chord can lengths provides significantly less strength than can be achieved by lengthening the can. Equation (14.3-11) can be used to optimize the strength of the joint by providing sufficient can length.

A.14.3 Simple Tubular Joints

A.14.3.1 General

A.14.3.1.1 Historical Background

The requirements and recommendations contained in this document are based on interpretation of available data, both experimental and numerical. Additional data on the background and modifications of joint design can be found in B.7 of API 2A-WSD, Twenty-Second Edition. As with all empirically based practice, the recommendations have specific validity ranges, although for this document the qualification has little significance since the ranges cover the wide spectrum of geometries and materials currently used in practice.

Apart from the yield strength limitations given in A.14.2, the recommendations can still be used for joints with geometries that lie outside the validity ranges, by taking the usable strength as the lesser of the strengths calculated on the basis of the following:

- actual geometrical parameters, and
- imposing the limiting value parameters for the validity range, where these limits are infringed.

API requirements [A.9.1-1] included a limit on (punching) shear stress. This limit is unnecessary for joints within the given parametric ranges that have empirical equations assigned to them. However, a limit on shear stress can still be appropriate for cases where the parametric ranges are exceeded or where there are no empirical equations founded on a suitable database. Both experimental and calibrated finite element results can constitute an acceptable database.

Most of the detailed requirements and recommendations provided in the First Edition of API 2A-LRFD relate to simple joints consisting of circular hollow sections. Other offshore design documents are based on an early 1980s experimental database. Many data have since been added, often as a consequence of testing a simple joint as a reference in the course of examining a complex configuration. The database for IPB has doubled [A.14.3-1], while, in addition to experimental data, FEA studies have improved the understanding of tubular joint behavior [A.14.3-1]. Hence, a distinct need existed for a complete review and update of simple joint practice.

The MSL JIP [A.14.2-1, A.14.2-2], over the period 1994–1996, significantly influenced practice for simple and overlapping joints. The general approach adopted in the MSL JIP was the following:

- a) collation of comprehensive databases of worldwide experimental and pertinent FEA results,
- b) validation and screening of the databases,
- c) fitting curves to the data,
- d) comparison of databases and of derived strength formulations with previous practice, and
- e) selection of appropriate formulations.

Baseline screening criteria adopted in the MSL JIP were as follows:

- chord diameter, and
- gap dimension for K-joints.

A.14.3.1.2 Data Range for Chord Diameter

A lower limit of 100 mm was imposed in collecting data for the validated database. The argument for a cut-off limit stems from a concern that small scale specimens are not representative of fabricated joints found offshore from a material, weld, or tolerance standpoint. It is recognized however, that failure modes for tubular joints subjected to predominantly static forces are generally dominated by gross, plastic ovalization of the chord cross-section and, therefore, size effects are expected to be less dominant than for joints subjected to repetitive loads/actions simulating fatigue. In the MSL JIP, no distinct influence of scale down to 100 mm was observed.

A.14.3.1.3 Data Range for Gap Dimension for K-joints

The validated database for K- and T/Y-joints contains both nonoverlapped and overlapped geometries. A filter based on the gap dimension was required. One possible criterion was to consider all joints with a gap greater than or equal to zero as nonoverlapped. However, this can lead to ambiguity, as some references only record the eccentricity with calculated gaps falling into a narrow range around zero, e.g. ± 1 mm. It was therefore not clear that the zero gap criterion should be strictly applied. From a practical viewpoint, the feasibility of achieving a zero gap is questionable. To overcome these difficulties, it was decided to select nonoverlapped joints as those having a finite gap with a small lower limit. Similarly, overlapped joints were defined as those having a finite overlap of at least a specific small amount. Nominally zero gap specimens, g_0 were defined by the following criterion:

$$-g_0 \leq g \leq g_0 \quad (\text{A.14.3-1})$$

where

g_0 = 7.5 mm or $2 T$, whichever is the greater;

T is the thickness of the chord.

Use of the above criterion excluded only a relatively small number of data from either the gapped or overlapped databases.

A.14.3.1.4 Data Range for Other Dimensions

No other limitations on dimensions were imposed. However, the derived nondimensional quantities were screened as follows.

- a) Length/diameter ratio: Specimens having very low ratios of chord length to radius were removed, as they would be unduly affected by short chord effects and by end restraints, and because unusual chord stress flow patterns would exist at the joint. A limit on the ratio of 3 was initially selected, although this limit was tightened following data analyses.
- b) Thickness ratio: The vast majority of the data have joints with nominally the same or greater chord wall thickness than that of the brace. This is in accordance with general offshore practice. Some data, however, have relatively thick braces (no doubt an attempt by the investigator to induce joint failure rather than brace failure) with thickness ratios τ as high as 5.5. For nominally the same chord and brace thickness and allowing for a $\pm 5\%$ variation of both, an upper limit of $\tau = 1.1$ was selected. Similarly, some investigators have attempted to force joint failure rather than brace failure by reinforcing the brace with stiffeners or by using concrete within the brace. These specimens were also rejected from the database.

A.14.3.1.5 Data Range for Material Properties

All specimens where the representative chord yield strength was not recorded have been excluded from the screened database.

A.14.3.1.6 Utilization of Chord and Brace

The data were segregated into two groups:

- a) joints where nominal stresses in both brace and chord were below yield;
- b) joints where nominal stresses in brace or chord have exceeded yield.

The first group was treated as true joint strength data with no reservation. The second group was treated as lower bounds to joint failure.

A total of 1066 simple joint specimens with D greater than 100 mm were screened. Of these, 653 specimens were validated and provided the database. The significance of establishing a suitably screened database cannot be overemphasized. The differences in the various previous/other code requirements on joint strength are partly due to differences in databases. The screened data have been used to assist in the creation of suitable expressions for joint strength, using regression analysis based on minimizing the percentage differences and using statistical calculations, which are characterized by a 95 % survivability level and a 50 % confidence level.

A.14.3.2 Basic Joint Strength

The basic API format for strength equations has been retained. A change to API practice is that where interpolation of Q_u was previously adequate for braces with mixed classification subjected to axial forces, interpolation is now based on a weighted average of P_{uj} , since Q_f also varies with axial force classification. Taking Figure 14.2-2 h) as an example, the diagonal brace has a 50 % K- and a 50 % X-joint classification. In this case, P_{uj} is calculated separately for K-joint classification and for X-joint classification. In the calculation for X-joint classification, any effects of short can lengths are included in accordance with 14.3.5. The representative joint axial strength can thereafter be calculated as follows:

$$P_{uj} = 0.50 P_{uj,K} + 0.50 P_{uj,X} \quad (\text{A.14.3-2})$$

where

P_{uj} is the representative joint axial strength, in force units;

$P_{uj,K}$ is the representative joint axial strength for K-joint classification, in force units;

$P_{uj,X}$ is the representative joint axial strength for X-joint classification, in force units.

A.14.3.3 Strength Factor, Q_u

The various Q_u factors have been derived from appraisals of screened steel model test data, supplemented by FEA data, where available, for each joint and force type. In deriving the factors, the formulations in other codes were examined and the best formulations for capturing the effects of the joint parameters (e.g. β and γ) were selected and the coefficients were adjusted to give representative strength values. In some cases, new formulations provided significant improvements in the COV or have a wider range of applicability. In particular, the former applies to the formulation for overlapped K-joints with axial forces, while the latter applies to the OPB formulation. Full details of the assessments appear in Reference [A.14.2-1].

In the derivation of design formulations for the static strength of tubular joints, it has been traditional practice to fit Q_u functions to empirical data, without accounting for the effects of chord force (now taken into account via the Q_f factor).

Further improvements to the strength factor can be found in References [A.14.3-2] and [A.14.3-3].

A.14.3.4 Chord Force Factor, Q_f

Compared to previous codes, a substantial change to the chord force factor (Q_f) is made in this document—at least in respect of T/Y- and K-joints subjected to axial forces. This change follows an exhaustive appraisal of the effect of chord forces on joint strength [A.14.2-1].

Inspection of the Q_f factor shows that no dependency on γ is given. Previous codes included a dependency that was based on forcing DT-joints of one specific γ and K-joints of another specific γ to align. The new appraisals have indicated that any γ dependency in K-joints is small, and its influence is therefore not included in the requirements in this document.

Comparison of the Q_f formulation in this document with those of existing guidance such as API shows the following differences.

The dependency of Q_f on γ has been removed.

The factor $C_1 = 14$ for K-joints in Table 14.3-2 has been obtained from an examination of data from both tests for which chord axial forces are only in equilibrium with brace forces and tests in which additional chord forces were applied.

For T-joints with axial brace forces, the effect of chord IPB has been relaxed. Substantial IPB moments can be generated by axial brace forces in T-joints. These chord moments can be expected to affect joint strength, as confirmed in FEAs by van de Valk [A.14.3-4]. However, due to chord length and end plate effects and inherent experimental scatter, doubts over true chord boundary conditions exist. The extraction of a chord stress effect from the FEA data proved most difficult. There is a strong correlation between the chord IPB and joint failure with β , with the effect of high chord IPB being captured by the β term in the Q_u function. Therefore, the Q_f effect is hidden in the Q_u function. Recognizing that chord IPB moments in structures can be higher than those attained in tests, an allowance for Q_f was made. This was achieved by assuming that Q_f reaches a value of 0.8 when chord IPB reaches the yield moment.

It was recognized in the MSL JIP [A.14.2-1] that chord forces are one of the least understood parameters affecting joint strength, there being few data to properly define Q_f . Recent work by EWI/API [A.14.3-5] has continued to develop alternative formulations. To date, the findings from a comprehensive study for DT- and X-joints subjected to brace compression or brace IPB on the effects of chord forces, including axial, IPB, and OPB, have been published [A.14.3-6]. In light of the EWI/API work, the coefficients for X-joints given in Table 14.3-2 were adjusted to bring Q_f more in line with the EWI findings.

Further improvements to the chord force factor can be found in References [A.14.3-2] and [A.14.3-3].

A.14.3.5 Y- and X-joints with Chord Cans

The reduced strength for simple Y- and X-joints with axial forces and short can lengths is supported by numerical and experimental data (see Reference [A.14.2-1]). No reduction in strength is required for K-joints with axial forces. The lack of data has precluded an assessment of strength reduction (if any) for joints subjected to moments or for complex joints.

The API provisions for force transfer across chords have been extended to cover T-joints with axial forces. X-joints with axial forces and with $\beta > 0.9$ increasingly transfer force across the chord through membrane action, and this beneficial mechanism is recognized. The requirements of this document are also intended for application to other cases where force transfer through chords occurs, e.g. launch frame joints.

A.14.3.6 Strength Check

The interaction of axial force and bending moments on joint performance is evaluated using an interaction equation that represents a change from the trigonometric equations that have historically existed in API and other codes. However, Equation (14.3-12) is identical to that already in use in the United Kingdom [A.14.3-7] and is supported by experimental studies carried out at the University of Texas in mid-1980s (see References [A.14.3-8] and [A.14.3-9]). Although there is little improvement in reliability between the present Equation (14.3-12) over that of API when the utilization is less than 1.0, the present equation is more sensible when the utilization exceeds 1.0, as can be found in assessments. An improved understanding is beneficial when deciding on any remedial measures for overstressed existing structures.

Although the discussion in A.14.3 describes the changes that have been made to offshore practice for simple joints, in most instances the overall impact is limited. MSL Engineering has made a brief study of the amalgamated impact of the new guidance on a four leg structure in the central North Sea [A.14.2-1]. Only the changes in simple joint design guidance were considered. The effect of the new guidance on the design was modest, although the general perception is that overall reliability has improved significantly.

A.14.4 Overlapping Circular Tubular Joints

Guidance on the strength of overlapping joints has existed in API [A.9.1-1], HSE [A.14.3-7], CIDECT [A.14.2-11], and other practices for more than a decade. However, previous guidance has never addressed the effects from moments or from out-of-plane overlap. Recent work [A.14.4-1] to [A.14.4-6] has shown that the previous guidance for the axial force strength of joints overlapping in-plane is inadequate. A relatively complete summary of the problems with the previous guidance and the background database can be found in Reference [A.14.4-4].

The guidance in this document has been based on MSL's JIP results [A.14.2-1]. Pertinent test evidence of calibrated FEA results may be applied to specific cases following the principles in 7.7.

In several respects, the guidance in this document is simplified compared with the API and HSE guidance. For example, there is no longer a need to routinely calculate weld lengths. However, in more precise analyses such lengths can still be necessary. Reference [A.14.4-5] reproduces equations for these calculations.

The guidance expands the MSL JIP recommendations into a set of considerations that should avoid the need for FEA in all but the most unusual or failure-critical cases. There are simple, but conservative suggestions for addressing both in-plane and out-of-plane force situations, as well as out-of-plane overlap situations, which are not uncommon offshore. The hope is that ongoing research using FEA (see References [A.14.4-5] and [A.14.4-6]) will eventually help future requirements to be more definitive.

A.14.5 Grouted Circular Tubular Joints

Grouted joints are becoming more common in new steel frame structures and joint grouting is generally a cost-effective means of strengthening older structures. It is now possible to provide guidance [A.14.5-1] based on engineering approximations and some experimental evidence (see References [A.14.5-2] to [A.14.5-6]). The experimental evidence is primarily for double-skin joints subjected to axial brace forces. However, a JIP by MSL and TNO [A.14.5-7] provides additional data for fully grouted joints, especially those subjected to brace bending moments.

Special joint strength investigation can be warranted when braces are grouted, whether or not they frame into grouted chords. Although joint strength is heavily dependent on chord parameters, grouting a brace can reduce its effective diameter which, in turn, reduces β . For joints subjected to moments, the strength in many instances is expected to be limited by the local buckling strength of the brace, even if the brace is compact. Unless evidence to the contrary exists, strength for grouted joints subjected to bending moments should be limited to the plastic moment design strength of the brace.

For double-skin joints, a further limiting strength has been introduced to cater for the potential of failure due to chord ovalization. In such cases, the design strength is the lesser of the following:

- a) brace strength,
- b) strength calculated on the basis of effective thickness, and
- c) strength calculated on the basis of Q_u values for grouted joints.

Consideration of the effects of grouted joints should include review and perhaps revision of the structural model used to determine the sectional forces for the joint. Presence of grout clearly stiffens the joint so that the most appropriate model is likely to be one with a rigid offset from the chord centerline to the chord wall at each incoming brace. If the structure has been modeled with rigid joints located at the chord centerline, the conservatism, or otherwise, of the internal forces should be assessed. If joint flexibility has been introduced at the chord surface, while using a rigid offset to that point, only the flexibility needs to be altered. It is generally conservative to assume grouted joints to have no local flexibility, i.e. that they are assumed to be rigid up to failure.

A.14.6 Ring Stiffened Circular Tubular Joints

Some studies on the strength of ring-stiffened joints are given in References [A.14.6-1] to [A.14.6-3]. There are no published ultimate strength results of such joints, apart from those generated in MSL's JIP [A.14.2-1] and in recent studies at EWI [A.14.6-4]. Future data from EWI could assist in providing further guidance in the design of ring-stiffened joints.

Since robust, codified design practices are not yet available, ring-stiffened joints require more engineering attention than many of the simpler joint types. Consequently such joint designs are often more conservative than would be required using test results or calibrated FEA results.

At least three approaches exist for determining the required number and sizes of stiffeners. In all three, the first step is to assume ring dimensions, while being careful to avoid the possibility of local buckling. Then the required number of rings is evaluated. If the number is excessive, the stiffening ring geometry is modified, possibly including the addition of an inner edge flange, and the number of required stiffeners is recalculated. The three approaches are described below.

- a) The joint forces are assumed to be fully resisted by the rings assuming elastic behavior. The ring cross-sectional properties are calculated using an effective flange width from the chord can. The effective flange width and the elastic analysis of the ring are based on Roark's formulae [A.14.6-5]. Usually, a partial resistance factor on yield strength is applied.
- b) The joint forces are assumed to be fully resisted by the rings assuming plastic behavior. An effective flange width is assumed, and this value is often the same as in approach a). Based on a simple interaction expression for axial force, shear, and moment in the ring, a ring ultimate strength is derived. This strength is downgraded by a partial resistance factor that is normally assumed to be the same as for simple joints.
- c) The joint forces are assumed to be resisted by a summation of simple joint strength and ultimate strength behavior of the rings. The residual ultimate ring strength may simply be calculated as being the shear strength of two cross-sections of the ring. Partial resistance factors are applied to both the simple joint and ring strengths.

Several questions can arise with all of the above methods. It is not always clear how to address brace moments. The usual approach is to break them into couples and take the absolute sum of the forces from axial and moment couple as the applied load/action. Rings that are outside any brace footprint are also not addressed by any of the approaches listed above.

Although outside rings offer little advantage with respect to fatigue (by reducing SCFs), they can be much more effective than internal rings where ultimate strength is of concern. Often, external rings can be assumed to be fully effective if the clear distance from the edge of a given brace does not exceed $D/2$, although the shear transfer strength of the chord wall between the brace and outer ring should still be examined. The effectiveness of rings under a given footprint is normally assumed to be limited to the particular brace involved. The $D/2$ limit is only of concern for rings at the end of the chord can.

A more general procedure for the design of ring-stiffened joints is to consider sections or planes through the joint and ensure that the strength of all parts severed by the plane is sufficient to resist the applied forces. This approach is quite general, although difficult to automate. Its advantages are that it can address even the most complex conditions, and it often provides a better physical feel for load paths. This approach is recommended as a manual check of expected behavior whenever possible. However, this approach does not implicitly address potential local buckling or premature cracking.

As with grouted joints, use of ring-stiffened joints warrants a review of the structural model used to determine the sectional forces for the joint. Rings often increase the joint stiffness substantially, such that rigid offsets to the chord surface are appropriate.

A.14.7 Other Circular Joint Types

A general approach based on material strength principles can be suggested. The potential for local buckling or premature cracking should be investigated. Further information on circular joints with doubler or collar plates can be found in References [A.14.7-1] to [A.14.7-4].

A.14.8 Damaged Joints

Reduction in axial or moment strengths may be estimated by taking into account the reduced area or section modulus due to the presence of cracks. References [A.14.8-1] to [A.14.8-3] cover some of the research carried out in this area.

A.14.9 Noncircular Joints

Detailed design practices and guidance for some connections with noncircular chords and braces are available in References [A.14.9-1] and [A.14.9-2]. These were developed for on-shore use. Care should be taken with the use of these documents to ensure consistency with the design approach and philosophy in this document.

A.15 Strength and Fatigue Resistance of Other Structural Components

A.15.1 Grouted Concentric-tubular Connections

A.15.1.4 Computation of Allowable Axial Force

A.15.1.4.2 Plain Pipe Connections

Tests indicate that the strength of a grouted pile to structure connection using plain pipe is due to the bond and confinement friction between the steel and grout. Failure of test specimens normally occurs by slippage between the grout and steel.

Figure A.15.1-1 shows a plot of available test data for plain pipe grouted connections. Ordinates are failing values of the ultimate load transfer stress, f_{bu} , which were computed by dividing the failing value of axial load by the contact area between the grout and pipe at the surface of failure. Abscissas are corresponding values of unconfined grout compressive strength, f_{cu} . Only tests in which $f_{cu} \geq 17$ MPa (2500 psi) are included (see 15.1.4.4). A comparison between the recommended factored axial load transfer stress f_{ba} and the available test results and load data indicates a safety index exceeding 4.1.

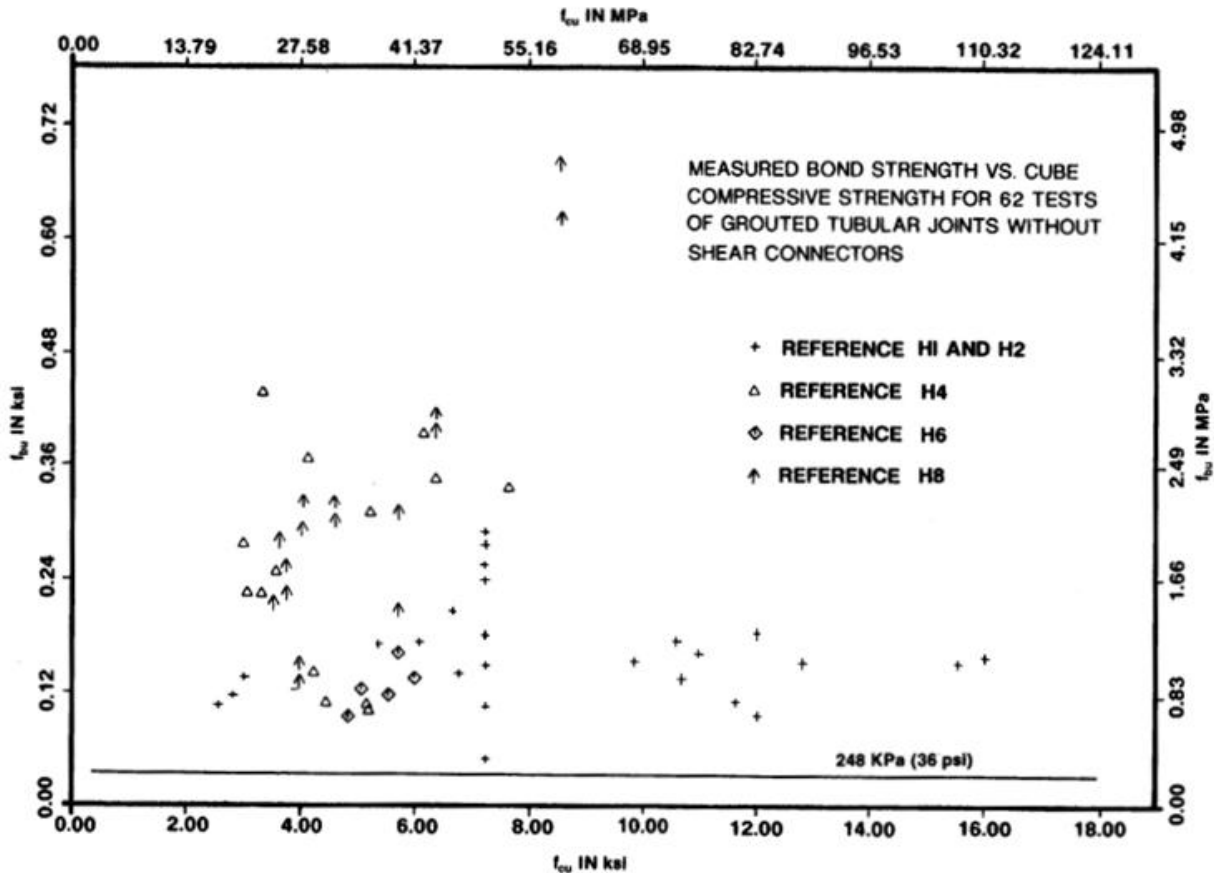


Figure A.15.1-1—Measured Bond Strength vs. Cube Compressive Strength

A.15.1.4.3 Shear Key Connections

Tests of grouted pile to structure connections using shear keys indicate that two separate sources of strength contribute to the ultimate strength of the connection: first, the contribution of bond and confinement friction between the steel and grout, and second, the contribution of bearing of the shear keys against the grout. At failure, two separate mechanisms occur: first, a slippage between the steel and grout, and second, a crushing of the grout against the shear keys. These specimens normally fail in a ductile manner, with both mechanisms acting, so that the ultimate strength of the connection is the sum of the two separate sources of strength. At some time prior to final failure, diagonal cracks tend to open across the grout, generally between diagonally opposite shear keys, or from one shear key to the opposite pipe.

The basic equation for allowable load transfer stress [see Equation (15.1-1)] is based on an ultimate strength formulation of the mechanisms of failure described above, with the application of a safety factor (see Reference [A.15.1-1]). Figure A.15.1-2 shows a plot of available test data for shear key grouted connections. Ordinates are failing values of load transfer stress, f_{bu} , which were computed by dividing the failing value of axial load by the contact area (π times diameter times length) between the grout and the pipe at the surface of failure. Abscissas are corresponding values of $f_{cu} \times h/s$. Only tests in which $f_{cu} \geq 17$ MPa (2500 psi) are included (see 15.1.4.4). A comparison between the recommended axial load transfer stress f_{ba} and the 85 available test results and load data gives safety index exceeding 3.75.

The provision for the design of shear key cross-section and weld (see Figure A.15.1-2) is intended to provide a shear key whose failing capacity is greater than the failing capacity of the grout crushing against the shear key.

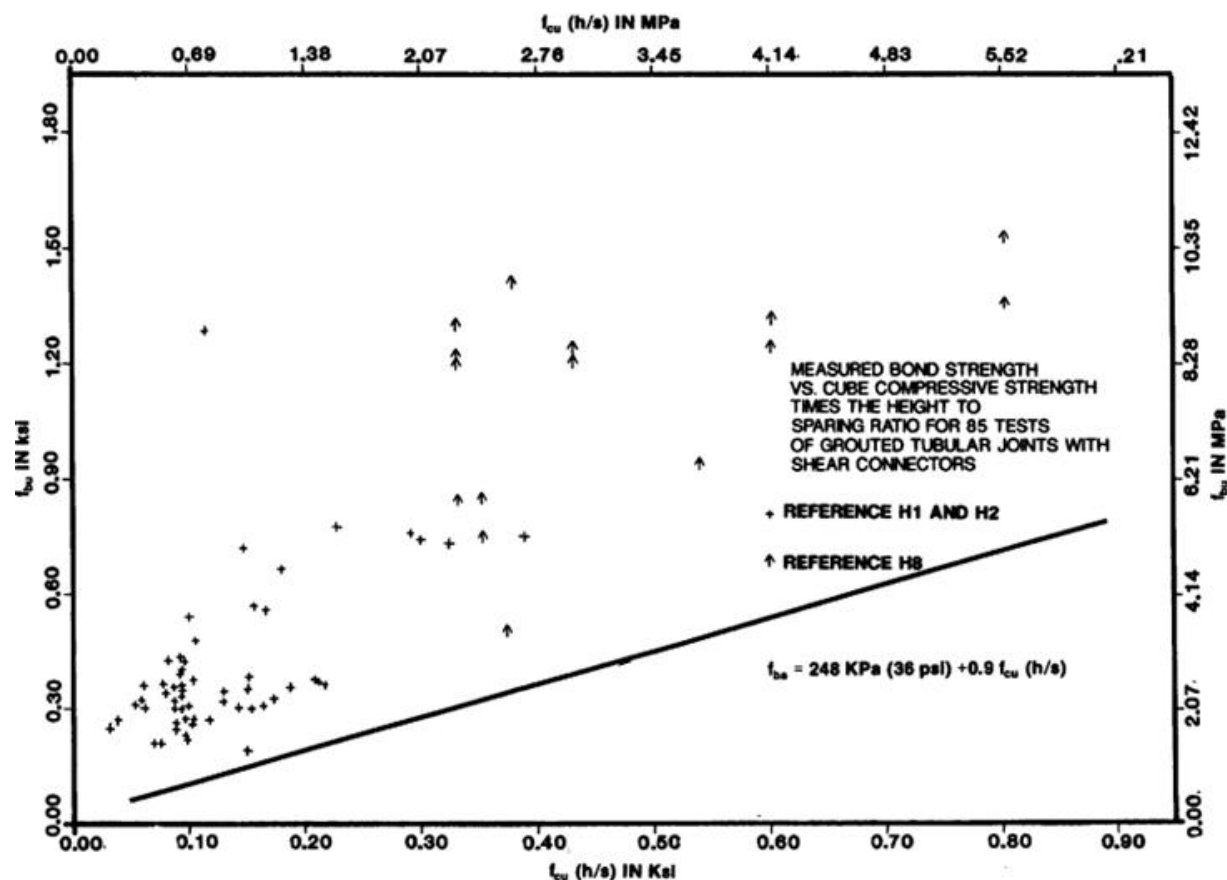


Figure A.15.1-2—Measured Bond Strength vs. Cube Compressive Strength Times the Height-to-Spacing Ratio

A.15.1.4.4 Limitations

The maximum values of important variables that are specified in this paragraph correspond closely to the maximum values of those variables in the tests on which the allowable stress equations are based. Use of values outside of these limits should be based on additional testing.

A.15.1.4.5 Other Design Methods

In recent years, the design method included in the U.K. Department of Energy (DOE) Code has received considerable use in the design of connections using shear keys (see References [A.15.1-2], [A.15.1-3], and [A.15.1-4]). The allowable load transfer stress, f_{ba} , by the DOE Code is calculated from the following equation, using SI units:

$$f_{ba} = \frac{1}{6} K C_L \left(9 C_s + 1100 \frac{h}{s} \right) (f_{cu})^{1/2} \text{ MPa} \quad (\text{A.15.1-1})$$

where, in this case, f_{cu} is the characteristic grout compressive strength as defined in the DOE Code in units of megapascals.

K = a stiffness factor defined as follows:

$$K = \frac{1}{m} \left(\frac{D}{t} \right)_g^{-1} + \left[\left(\frac{D}{t} \right)_p + \left(\frac{D}{t} \right)_s \right]^{-1} \text{ dimensionless} \quad (\text{A.15.1-2})$$

where

- C_L is a length coefficient as specified in the DOE Code;
- C_S is a surface coefficient as specified in the DOE Code;
- h is the minimum shear connector outstand, in millimeters;
- s is the nominal shear connector spacing, in millimeters;
- m is the modular ratio of steel to grout;
- D is the outside diameter;
- t is the wall thickness;
- suffixes g , p , and s refer to grout, pile, and sleeve, respectively.

The safety factor of 6 in Equation (A.15.1-1) is specified for normal loading conditions on a connection in which the grout displaced water, and the safety factor is adjusted for other conditions. The stiffness factor, K , which is defined in Equation (A.15.1-2) and is used in Equation (A.15.1-1), is intended to introduce into the equation the effect of the hoop flexibility of the pile, sleeve, and grout on the connection strength. The DOE equations are based on extensive testing performed at the Wimpey Laboratories near London (see References [A.15.1-2], [A.15.1-3], and [A.15.1-4]). Detailed instructions for the use of these equations and limitations on their use are set out explicitly in the DOE Code (see Reference [A.15.1-2]), to which the designer is hereby referred.

A.15.1.5 Fatigue Assessment

Fatigue testing of grouted connections has demonstrated the following fatigue characteristics:

- a) plain grouted tubular connections (without shear keys) do not suffer fatigue damage unless the applied variable force amplitude approaches the ultimate strength of the connection;
- b) the fatigue performance of shear key connections increases rapidly as the R ratio of minimum to maximum variable force amplitude moves from -1 (reverse direction cycling of equal magnitude) to 0 (single direction cycling);
- c) fatigue damage is characterized by a shouldered force-displacement hysteresis loop, where the shoulder, i.e. the point of initiation of rapid slip, has been shown to correspond to the plain grouted tubular strength of the connection.

The combination of these three observed characteristics indicates that fatigue damage can be averted by ensuring that the peak reverse force cycle is less than the plain grouted tubular strength. Under these circumstances, fatigue damage will not accumulate as the rapid slip mechanism, producing the impact damage which ultimately fails grouted connections in reverse cycling, is averted. Therefore, if the connection's effective grouted length is such that the plain grouted tubular strength exceeds the force amplitude in reverse cycling, fatigue damage will not accumulate by this observed mechanism. If the plain grouted tubular strength is not sufficient, the connection length should be increased. An upper limit on the increase in the connection's effective grouted length by a factor of 1.5 is suggested when used with the partial load/action and resistance factors given in this document, as there is evidence of an endurance limit in the data with $R = -1$ at an σ_p/f_g ratio of 0.18.

If it is not possible to increase the connection length to provide the necessary plain grouted tubular strength, a detailed fatigue evaluation should be carried out. Reference [A.15.1-5] provides further information on this subject.

It has been observed that the fatigue performance of grouted connections is reduced by movements during grout setting as the h/s ratio increases (see A.15.1.4.3). Therefore, it is advisable to minimize, as far as is practicable, both the h/s ratio and the early age movement. For most practical designs, a structurally efficient pile/pin-to-sleeve connection, with an L_e/D_p of approximately 5 to 6, can be achieved with an h/s of less than 0.04, using a grout with a 28-day strength of 50 MPa. Such an approach is recommended.

A.15.2 Mechanical Connections

A.15.2.2 Design Requirements

A.15.2.2.2 Static Strength Requirements

If connectors are designed to be stronger than the weakest of the connected tubular members, strain hardening effects may be relied upon to ensure that the connector has sufficient strength to mobilize the ductile deformation capacity of the tubular members. The factor by which the connector strength should exceed that of the weakest tubular member depends on the uncertainty of the analysis procedure, the connector material strength, and the consequence of connector failure.

The connector, if not stronger than the weakest of the connected tubular members, should have sufficient ductility to sustain all loads/actions due to deformations, as well as accidental loads/actions, without impairing the safety of the structure.

A.15.2.3 Loads/Actions and Forces on the Connector

When the connector can be considered to be axi-symmetric and the connected tubular is subjected to both an axial force (P) and a bending moment (M), the moment can be converted into an equivalent axial force (P_{eq}). The concept of the equivalent axial force is simple and slightly conservative for relatively thin tubulars ($D/t > 20$). The design axial force (P_d) can be expressed as in Equation (A.15.2-1):

$$P_d = P + P_{eq} \quad (A.15.2-1)$$

with

$$P_{eq} = \frac{M(D-t)}{2I} \times A = M \times \frac{32t(D-t)^2}{D^4 - (D-2t)^4} \quad (A.15.2-2)$$

where

P_{eq} is the equivalent axial force representing the bending moment;

M is the bending moment in the tubular member;

I is the line moment of inertia of the cross-section of the tubular member;

A is the cross-sectional area of the tubular member;

D is the outside diameter of the tubular member;

t is the wall thickness of the tubular member.

A.15.2.4 Resistance of the Connector

Nonlinear FEA can be used to determine the global force-deformation behavior of the connector. In this case, the connector's ultimate strength can be established by progressing the analysis until the connector can no longer sustain the force, or significant plastic deformation takes place that impairs the functional requirements of the connector, or precipitates its disengagement. In conducting these analyses, due consideration should be given to the connector material properties in terms of yield, strain hardening, ductility, and ultimate strength. However, both the connector modeling and the solution methodology should be calibrated to experimental data (see 7.7).

In developing the model for the FEA, the modeling of each adjoining tubular member should extend at least four diameters. For nonlinear analysis, a force-displacement diagram should be developed from which the connector resistance can be obtained. The force should correspond to the equivalent axial force on the nominal tubular member, whereas the displacement is the elongation of the connector measured between the two circumferential welds joining the connector to the tubular members. To achieve this, the connector portion of the model should include the nonlinear modeling aspects, while the portion of the model corresponding to the nominal tubular, extending beyond each circumferential weld, is linear elastic.

A.15.2.5 Strength Criteria

A.15.2.5.1 General

FEA may be used to calculate the stresses induced in the connector for the purpose of strength checks. Strength checks may use linear analysis to determine stress component distributions across particular cross-sections. The stresses induced in connectors by the factored loads/actions can be classified as primary or secondary.

Primary stresses are normal and shear stress distributions across a connector section that result from the application of external loads/actions and are required to satisfy equilibrium. Primary stresses are not self-limiting and, if not controlled, will induce failure. Stresses arising from axial force and bending moments acting in the tubular members are primary stresses.

Secondary stresses are normal and shear stress distributions across a connector section that are self-limiting and self-equilibrating. Stresses resulting from assembly of connectors can be primary or secondary stresses.

Other stresses are pure shear stresses and bearing stresses.

A pure shear stress is an average stress over a connector cross-section that is parallel to the direction of the acting force. An example of this stress is the average stress along the thread roots due to an axial force on the connector.

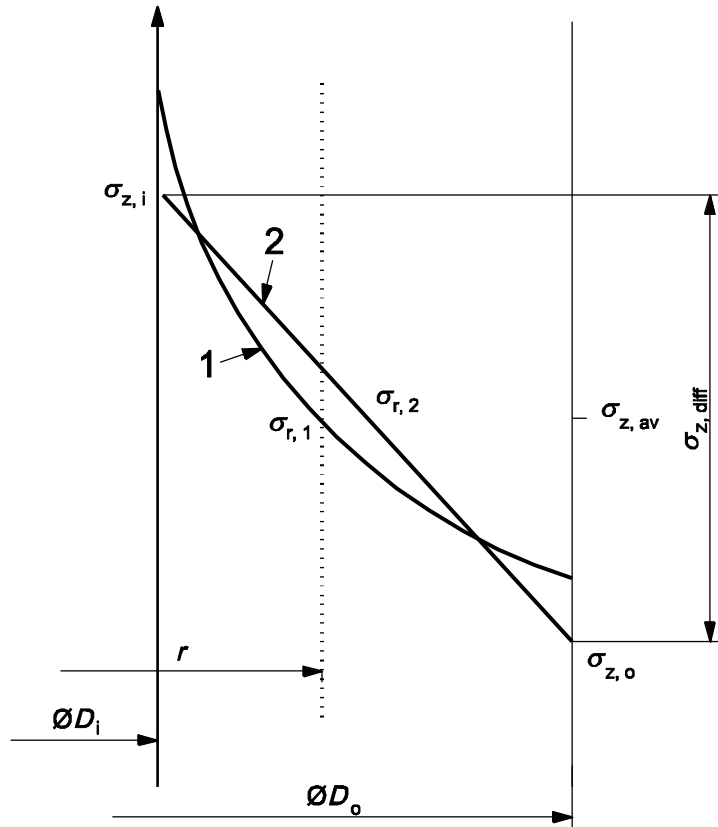
A bearing stress is an average stress normal to contact surfaces between connector parts mutually or between a connector part and the connector body, such as those between engaged threads, at preload shoulders, or between washers, dogs, etc.

Connector design using stresses resulting from linear FEA is based on criteria from ASME [A.15.2-1]. In order to take advantage of ductile steels and eliminate stress limits based on ultimate stress, material for connectors should:

- a) have at least 20 % elongation,
- b) have ratios of yield-to-ultimate stress no greater than 0.75, and
- c) be sufficiently tough at the LASTs.

A.15.2.5.2 Linearization and Von Mises Stresses

In order to use the stress criteria given in A.15.2.5.3 to A.15.2.5.6, each of the stress component distributions across the section of interest should first be linearized, from which an average and a differential stress can be obtained, as shown in Figure A.15.2-1.



$\sigma_{z,i}$ and $\sigma_{z,o}$ are determined by satisfying

$$\sum_{r=D_i/2}^{D_o/2} \sigma_{r,1} dr = \sum_{r=D_i/2}^{D_o/2} \sigma_{r,2} dr$$

and

$$\sum_{r=D_i/2}^{D_o/2} (\sigma_{r,1} \times r^2) dr = \sum_{r=D_i/2}^{D_o/2} (\sigma_{r,2} \times r^2) dr$$

Average and differential stresses are

$$\sigma_{z,av} = 0.5 (\sigma_{z,i} + \sigma_{z,o})$$

$$\sigma_{z,diff} = 0.5 (\sigma_{z,i} - \sigma_{z,o})$$

Key

- 1 elastic stress distribution
- 2 linearized stress distribution
- $\sigma_{z,i}$ axial stress at inner face
- $\sigma_{z,o}$ axial stress at outer face
- r radius to point being considered
- D_i inner diameter
- D_o outer diameter
- $\sigma_{r,1}$ stress at radius r for elastic stress distribution
- $\sigma_{r,2}$ stress at radius r for linearized stress distribution
- $\sigma_{z,av}$ average linearized axial stress
- $\sigma_{z,diff}$ difference between average and maximum linearized axial stresses

Figure A.15.2-1—Linearization of Axial Stress Components

The Von Mises effective stress can be calculated using Equation (A.15.2-3):

$$\sigma_e = 0.707 \left[(\sigma_r - \sigma_\theta)^2 + (\sigma_\theta - \sigma_z)^2 + (\sigma_z - \sigma_r)^2 + 6(\tau_{r\theta}^2 + \tau_{\theta z}^2 + \tau_{zr}^2) \right]^{1/2} \quad (\text{A.15.2-3})$$

where

σ_e is the Von Mises effective stress;

σ_r is the normal radial stress;

σ_θ is the normal hoop stress;

σ_z is the normal axial stress;

$\tau_{r\theta}$, $\tau_{\theta z}$, τ_{zr} are the shear stress components.

The average Von Mises stresses ($\sigma_{e,av}$) can be calculated from the average values of the linearized stresses and the maximum Von Mises stresses ($\sigma_{e,max}$) can be calculated from the maximum values of the linearized stresses, taken as the average $\pm$ the differential stress.

If detailed radial and hoop stresses are not available, then the radial stress at the internal wall, $\sigma_{r,i}$, may be assumed to equal the internal pressure; $\sigma_{r,o}$ may be assumed to equal the external pressure; and the hoop stresses may be assumed to equal half the radial pressures.

The intent of linearization is to:

- a) eliminate the effect of the peak stresses arising from elastic FEA at the location of stress raisers, and
- b) account for the global section behavior.

Linearization involves finding, for each stress component distribution across the thickness of the section, a linear distribution whose equivalent force and moment are the same as those of the actual distribution (see Figure A.15.2-1). The average and differential components of the normal axial stress distribution physically correspond to the membrane and bending components acting on that section in accordance with P/A and $M(D - t)/2I$.

Linearization of the shear stress is not recommended. Instead, an average value across the section suffices for the calculation of the effective Von Mises stresses.

In cases where torsion is a significant external load/action, the average shear stress ($\tau_{r\theta}$) at the cross-section of interest should be approximated by the shear stress calculated by $T r/J$, where T is the applied torque, r is the mean radius of the section, and J is the polar moment of the section.

Strength checks should be made at both connector cross-sections and through other sections. For instance, in flanges cylindrical sections parallel to the flange axis and containing the bolt circle should be checked. For arbitrary sections, the stress components are rotated from the global coordinates to the local coordinates parallel and perpendicular to the section in question. The hoop component, however, remains the same for axi-symmetric models.

A.15.2.5.3 Primary Stress Criteria

For primary stress checks, the portion of the stress due to secondary forces, such as preload due to assembly, may be subtracted from the total stress calculated due to loads/actions and forces following preloading.

The primary average Von Mises effective stress, $\sigma_{e,av}$, and the maximum Von Mises effective stress, $\sigma_{e,max}$, resulting from factored loads/actions should satisfy both the following conditions:

$$\sigma_{e,av} \leq \frac{f_y}{\gamma_{R,t}} \quad (\text{A.15.2-4})$$

$$\sigma_{e,max} \leq \frac{f_y}{\gamma_{R,b}} \quad (\text{A.15.2-5})$$

where, in addition to the definitions in A.15.2.5.2,

f_y is the representative yield strength of the connector material;

$\gamma_{R,t}$ is the partial resistance factor for tension; $\gamma_{R,t} = 1.35$ for cases that are dominated by environmental load/action, $\gamma_{R,t} = 1.6$ for cases where environmental loads/actions are not included;

$\gamma_{R,b}$ is the partial resistance factor for bending; $\gamma_{R,b} = 0.9$ for cases that are dominated by environmental load/action, $\gamma_{R,b} = 1.1$ for cases where environmental loads/actions are not included.

Strains developed under primary stress conditions should also be limited so that the connector's sealability, its ability to disassemble, and/or the function of the tubular string are not adversely affected.

A.15.2.5.4 Primary Plus Secondary Stress Criteria

The average plus differential Von Mises effective stress, $\sigma_{e,av + diff}$, as calculated by linear analysis under preload and factored loads/actions, should additionally satisfy the following condition:

$$\sigma_{e,max} \leq \frac{f_y}{\gamma_{R,b}} \quad (\text{A.15.2-6})$$

where $\gamma_{R,b}$ is the partial resistance factor for bending; $\gamma_{R,b} = 0.5$ for cases that are dominated by environmental load/action, $\gamma_{R,b} = 0.6$ for cases where environmental loads/actions are not included.

A.15.2.5.5 Shear Stress Criteria

The average shear stress on a gross section, τ_{av} , resulting from factored loads/actions should satisfy the following condition:

$$\tau_{av} \leq \frac{f_y}{\gamma_{R,v}\sqrt{3}} \quad (\text{A.15.2-7})$$

where, further to previous definitions, $\gamma_{R,v}$, the partial resistance factor for shear stresses, = 1.25.

A.15.2.5.6 Bearing Stress Criteria

The average bearing stress, σ_{br} , from factored loads/actions should satisfy the following condition:

$$\sigma_{br} \leq \frac{f_y}{\gamma_{R,br}} \quad (\text{A.15.2-8})$$

where, further to previous definitions,

σ_{br} is the bearing stress;

$\gamma_{R,br}$ is the partial resistance factor for bearing stresses = 1.0.

A.15.2.6 Fatigue Criteria

A.15.2.6.1 General

In order to be able to predict the fatigue life at a particular location of a structural component, both the stress history at the point under consideration and the resistance of the material to time-varying stresses need to be determined.

For the $S-N$ curve method, the stress history during the lifetime may be given in statistical form as the long-term stress range distribution(s) at the most highly stressed location(s) of the connector. The stress ranges at the location(s) of interest may be determined by linear FEA of a suitable structural model, using all forces and moments occurring in the tubular member as input. The resistance of the connector material should be given by an appropriate $S-N$ curve. If such an $S-N$ curve is not known, it should be determined experimentally. The damage due to each stress range is the number of times, n , that the stress range, S , occurs in the long-term stress range distribution, divided by the allowable number of cycles to failure, N , from the $S-N$ curve. Total damage is obtained by the linear accumulation of the damage done by each stress range, in accordance with Palmgren-Miner rule. The method is described in detail in Section 16. In the calculations for mechanical connections, the effect of the mean stress due to make-up and preload on the connector may be taken into account.

In the initiation life method, the fatigue resistance of the connector is compared to the lowest fatigue resistance of the circumferential welds in the tubular members connected by the connector. The lowest circumferential weld resistance is characterized by the usual $S-N$ curve. The fatigue resistance of the connector is determined in an equivalent form by determining the local stress ranges S' at notches in the connector, due to the same forces that cause S at the relevant circumferential weld in the tubular member. Pairs of S' versus the allowable number of cycles N' to crack initiation can be calculated using the local strain approach of Reference [A.15.2-2]. The effect of the mean stress at the notch due to make-up and preload on the connector should be included.

If the $S'-N'$ curve characterizing the fatigue resistance of the connector is higher than the $S-N$ curve characterizing the lowest fatigue resistance of the circumferential welds, the connector does not control the fatigue life of the connection. However, if the curves cross, the relative fatigue performance of the connector and the circumferential weld depends on the stress history. In this case, both fatigue lives should be calculated, using the same stress history as input for the fatigue life calculation of the connector and the fatigue life calculation of the circumferential weld.

A.15.2.6.2 Initiation Life Method

The calculation of initiation lives at notches in the connector requires the following:

- detailed FEA of the connector to generate local stresses at the notches, and
- material data for both the monotonic (static) and the cyclic properties of the material in question.

Reference [A.15.2-2] contains one such set of data typical for connector material.

The crack initiation life or number of cycles to failure, N_f , at a generic notch of a mechanical connector may be evaluated by solving for the number of reversals to failure, $2N_f$, in the nonlinear Equation (A.15.2-9):

$$\sigma_{re} \varepsilon_{re} E = \sigma'_f{}^2 (2 N_f)^{2b} + \sigma'_f \varepsilon'_f E (2 N_f)^{b+c} \quad (\text{solve for } N_f) \quad (\text{A.15.2-9})$$

where

σ_{re} is the equivalent fully reversible elasto-plastic stress amplitude at the notch (not known);

ε_{re} is the equivalent fully reversible elasto-plastic strain amplitude at the notch (not known);

E is the modulus of elasticity (material constant);

σ'_f is the fatigue strength coefficient (material constant);

ε'_f is the fatigue ductility coefficient (material constant);

b is the fatigue strength exponent (material constant);

c is the fatigue ductility exponent (material constant).

The coefficients and exponents defined above are generated experimentally for the material in question, and ε_{re} may be obtained using the cyclic stress-strain relation for the material using Equation (A.15.2-10).

$$\varepsilon_{re} = \frac{\sigma_{re}}{E} + \left(\frac{\sigma_{re}}{K'} \right)^{\frac{1}{n'}} \quad (\text{A.15.2-10})$$

where

K', n' are the cyclic strain hardening coefficient and exponent, respectively, which are known as material constants and are determined experimentally;

σ_{re} is the equivalent fully reversible elasto-plastic stress amplitude at the notch (not known at this point).

The equivalent fully reversible stress amplitude at the notch, σ_{re} , includes the effect of the mean stress at the notch and may be calculated using Equation (A.15.2-11):

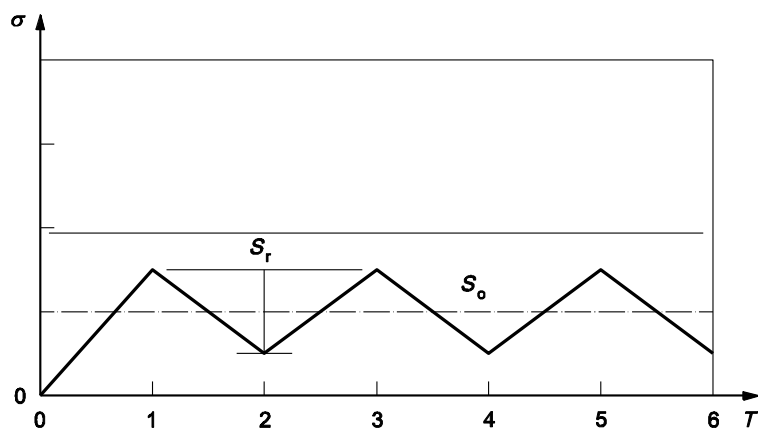
$$\sigma_{re} = \left[\frac{\sigma_a}{1 - \sigma_0 / \sigma'_f} \sqrt{(\sigma_0 + \sigma_a) \times \sigma_a} \right]^{0.5} \quad (\text{solve for } \sigma_{re}) \quad (\text{A.15.2-11})$$

where

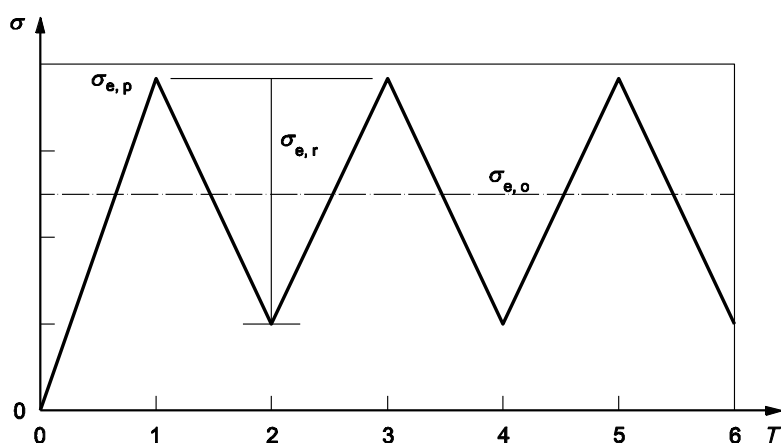
σ_0 is the elasto-plastic mean stress at the notch [see Figure A.15.2-2 c)] (not known at this point);

σ_a is the elasto-plastic stress amplitude at the notch, given by $\sigma_a = \frac{1}{2} \sigma_r$;

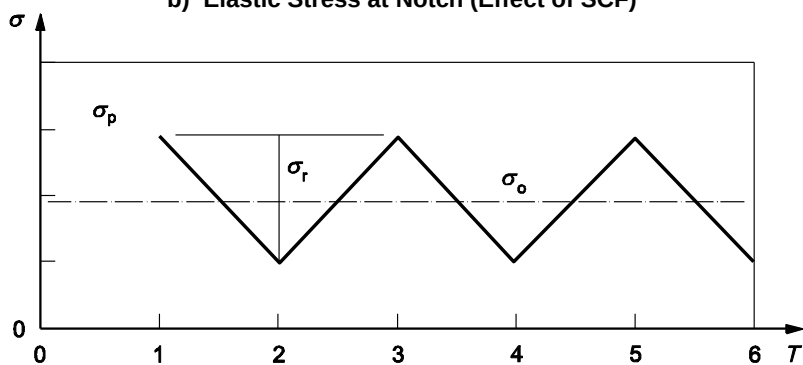
σ_r is the elasto-plastic stress range at the notch [see Figure A.15.2-2 c)] (not known at this point).



a) Nominal Pipe Stress Variation



b) Elastic Stress at Notch (Effect of SCF)



c) Transformed Notch Stress (Effect of Elasto-plastic Behavior)

Key

σ	stress	$\sigma_{e,r}$	elastic stress range at notch
T	time step	$\sigma_{e,o}$	mean elastic stress at notch
S_r	nominal pipe stress range	σ_p	peak elasto-plastic stress at notch
S_o	mean nominal pipe stress	σ_r	elasto-plastic stress range at notch
$\sigma_{e,p}$	peak elastic stress at notch	σ_o	mean elasto-plastic stress at notch

Figure A.15.2-2—Definition of Pipe and Notch Stresses

The local elasto-plastic notch stresses, mean and range, may be obtained by transforming the elastic notch response obtained from FEA. This transformation accounts for the local plasticity that takes place at the notch. The elastic notch response may be given in terms of the stresses directly calculated at the notch, the nominal pipe tubular stresses and a geometrical SCF, which relates the pipe tubular stress, S , to the notch stress, σ (preferred), or a combination thereof. The elastic notch stresses may be transformed to elasto-plastic stresses by one of two methods: the strain energy density or Neuber's.

- a) Strain energy density method: The energy method simply equates the elastic strain energy at the notch to the elasto-plastic energy given by the actual cyclic stress-strain behavior obtained experimentally for the material in question (see Figure A.15.2-3). This transformation is general and accurate. Knowing the elastic notch stress range and peak stress [see Figure A.15.2-2 b)], Equations (A.15.2-12) to (A.15.2-14) apply:

$$\frac{\sigma_{e,r}^2}{2E} = \frac{\sigma_r^2}{2E} + \frac{\sigma_r}{n'+1} \left(\frac{\sigma_r}{K'} \right)^{1/n'} \quad (\text{solve for } \sigma_r) \quad (\text{A.15.2-12})$$

and

$$\frac{\sigma_{e,p}^2}{2E} = \frac{\sigma_p^2}{2E} + \frac{\sigma_p}{n'+1} \left(\frac{\sigma_p}{K'} \right)^{1/n'} \quad (\text{solve for } \sigma_p) \quad (\text{A.15.2-13})$$

where

$\sigma_{e,r}$ is the elastic stress range at the notch from FEA;

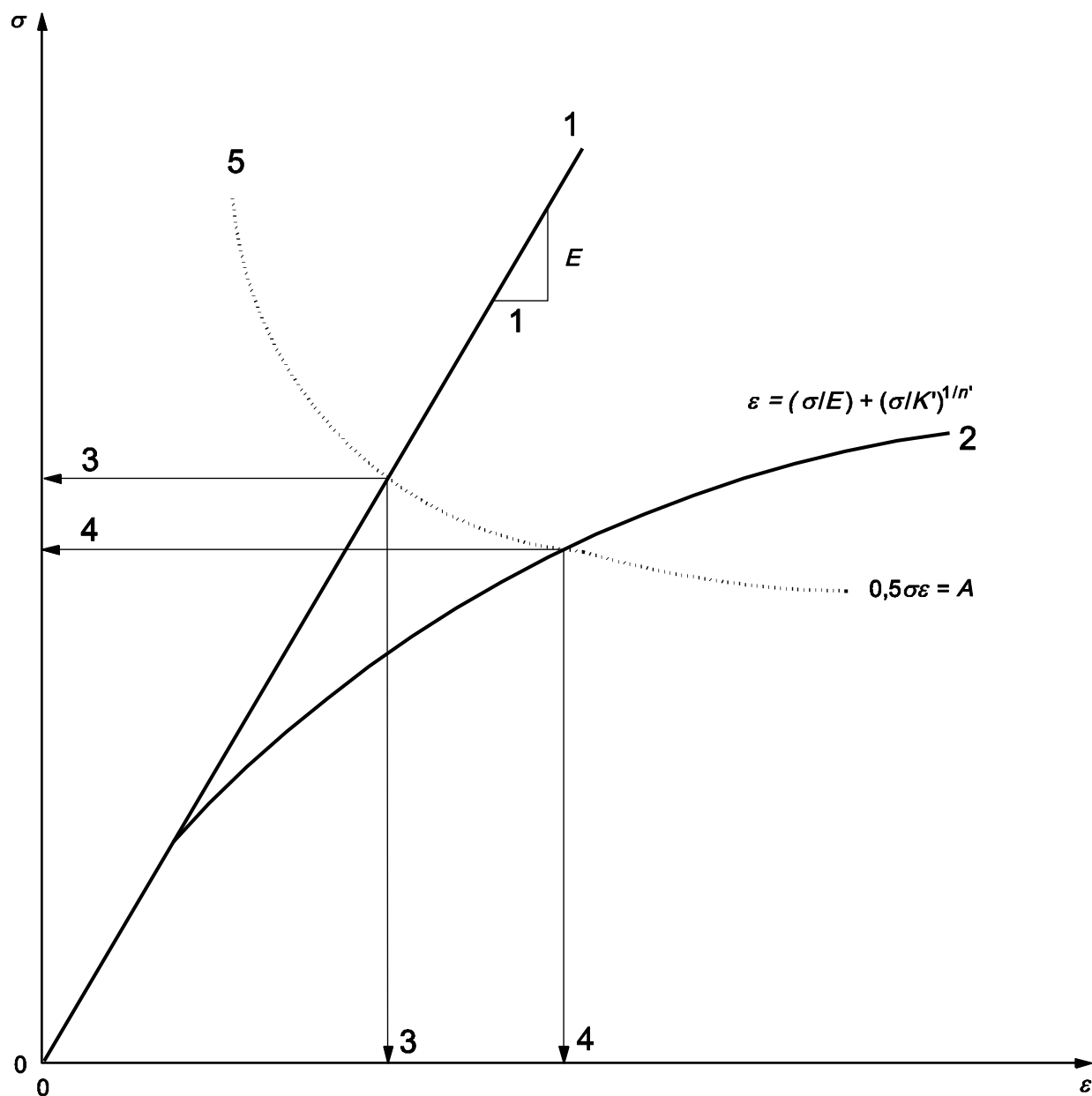
σ_r is the elasto-plastic stress range (transformed) at the notch;

$\sigma_{e,p}$ is the elastic peak stress at the notch from FEA;

σ_p is the elasto-plastic peak stress (transformed) at the notch.

Then, the elasto-plastic mean stress, σ_o , at the notch is:

$$\sigma_o = \sigma_p - \frac{\sigma_r}{2} \quad (\text{A.15.2-14})$$

**Key**

- 1 elastic behavior
- 2 elasto-plastic behavior
- 3 intersect with elastic behavior
- 4 intersect with elasto-plastic behavior
- 5 line of equal strain energy (area below stress-strain curve)
- σ notch stress
- ϵ notch strain

Figure A.15.2-3—Transformation of Elastic Stress via Strain Energy

- b) Neuber's method: Neuber's method represents an engineering approximation to the same transformation and is based on Neuber's rule (the geometrical mean of the strain and SCFs remains constant with load) and Peterson's empirical fatigue strength reduction factor, K_f (see Reference [A.15.2-3]). Referring to Figure A.15.2-4, the notch stress is linearly related to the pipe tubular stress via an SCF and there is no initial prestress at the notch. Two approaches can be followed to obtain the notch peak stress and stress range to calculate the notch mean stress.

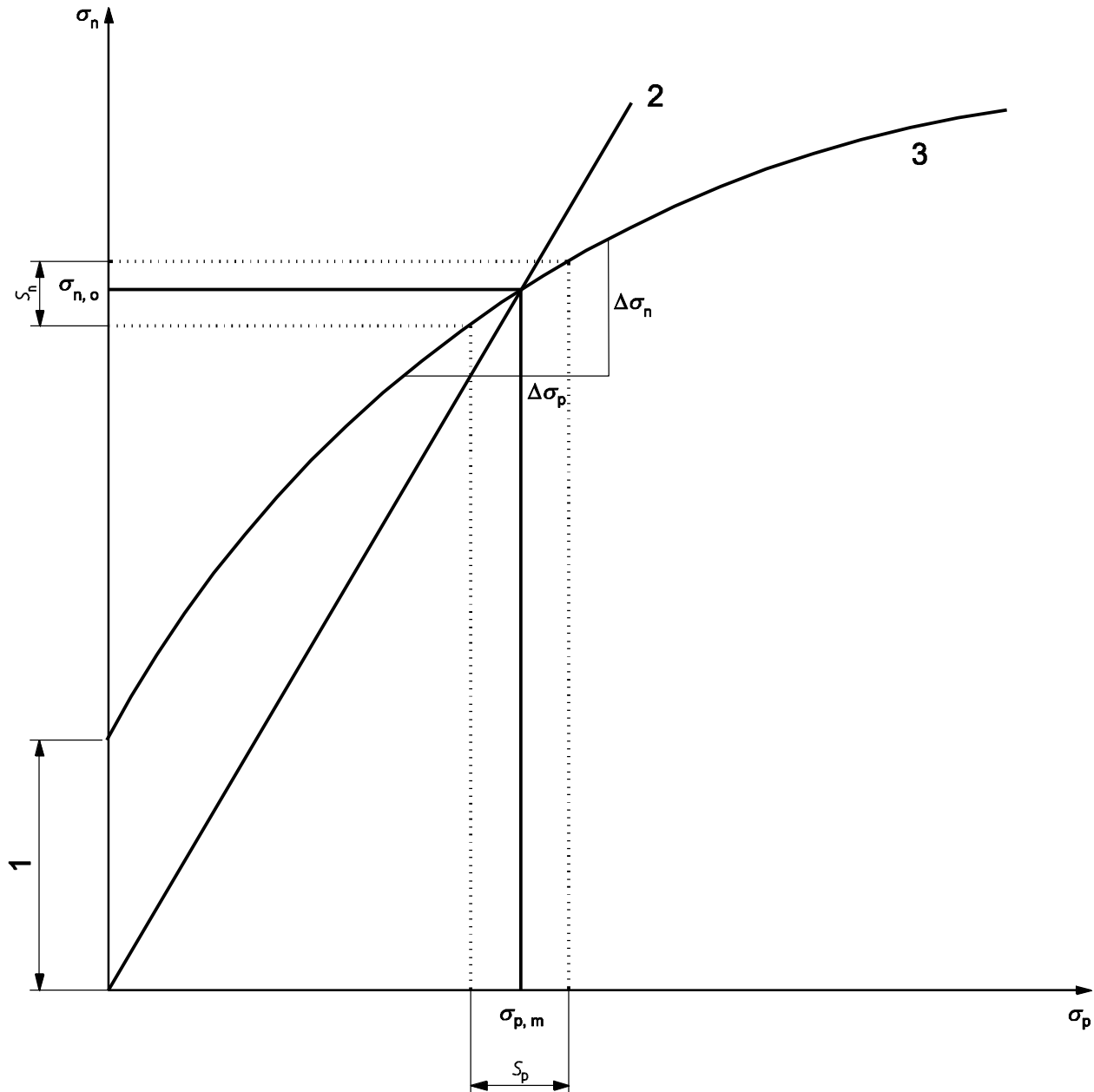


Figure A.15.2-4—Relation between Notch and Pipe Stresses

Knowing the nominal pipe tubular stress range (S_r), peak stress (S_o), the elastic SCF, the transformed elasto-plastic notch stress range, σ_r , and the peak notch stress, σ_p , can be obtained directly from the nominal pipe tubular stresses [see Figure A.15.2-2 a)] according to the following two relations:

$$\sigma_r \left[\frac{\sigma_r}{E} + \left(\frac{\sigma_r}{K'} \right)^{1/n'} \right] = \frac{1}{E} (K_f \times S_r)^2 \quad (\text{solve for } \sigma_r) \quad (\text{A.15.2-15})$$

and

$$\sigma_p \left[\frac{\sigma_p}{E} + \left(\frac{\sigma_p}{K'} \right)^{1/n'} \right] = \frac{1}{E} K_f^2 \left(S_o + \frac{S_r}{2} \right)^2 \quad (\text{solve for } \sigma_p) \quad (\text{A.15.2-16})$$

where

S_r is the nominal stress range in the pipe tubular (known);

S_o is the mean nominal stress in the pipe tubular (known);

K_f is the fatigue strength reduction factor, or effective fatigue SCF given by:

$$K_f = 1 + \frac{\text{SCF} - 1}{1 + A/r}$$

where

r is the Notch radius, in millimeters;

$$\text{SCF} = \frac{\Delta \sigma}{\Delta S} \quad \text{known from FEA;}$$

$$A = 0.821 \left(\frac{300}{\sigma_u} \right)^{1.8};$$

σ_u is the ultimate stress of the material in megapascals.

Then, the elasto-plastic mean stress, σ_o , at the notch is as follows:

$$\sigma_o = \sigma_p - \frac{\sigma_r}{2} \quad (\text{A.15.2-17})$$

The accuracy of the transformation by Neuber's method depends on the relative contribution of the connector preload to the notch stress and on the degree of nonlinearity between the notch stress and the pipe tubular stress. If linearity between the notch and pipe tubular stresses exists and there is no preload, the notch stresses can be calculated using the pipe tubular stresses and an SCF, and the energy method can alternatively be used.

If the local notch stress is not linearly related to the pipe tubular stress and the connector is preloaded (Figure A.15.2-4), then:

- 1) the SCF varies with the pipe tubular stress, and
- 2) there is an initial stress at the notch that is not related to the pipe tubular stress but to dimensional interferences in the connector.

In this case, the instantaneous SCF may be taken as the derivative of the notch stress with respect to the pipe tubular stress to relate the pipe tubular stress range to the notch stress range. However, the pipe tubular mean stress can no longer be related to the notch stress via an SCF. To avoid this situation, the elastic mean stress directly calculated at the notch via FEA is used.

Knowing the elastic mean notch stress (σ_o) and the pipe tubular stress range (S_r) and the instantaneous SCF, the transformed notch stress range, σ_r , and the notch peak stress, σ_p , can be obtained using Neuber's method by:

$$\sigma_r \left[\frac{\sigma_r}{E} + \left(\frac{\sigma_r}{K'} \right)^{1/n'} \right] = \frac{1}{E} (K_f \times S_r)^2 \quad (\text{solve for } \sigma_r) \quad (\text{A.15.2-18})$$

and

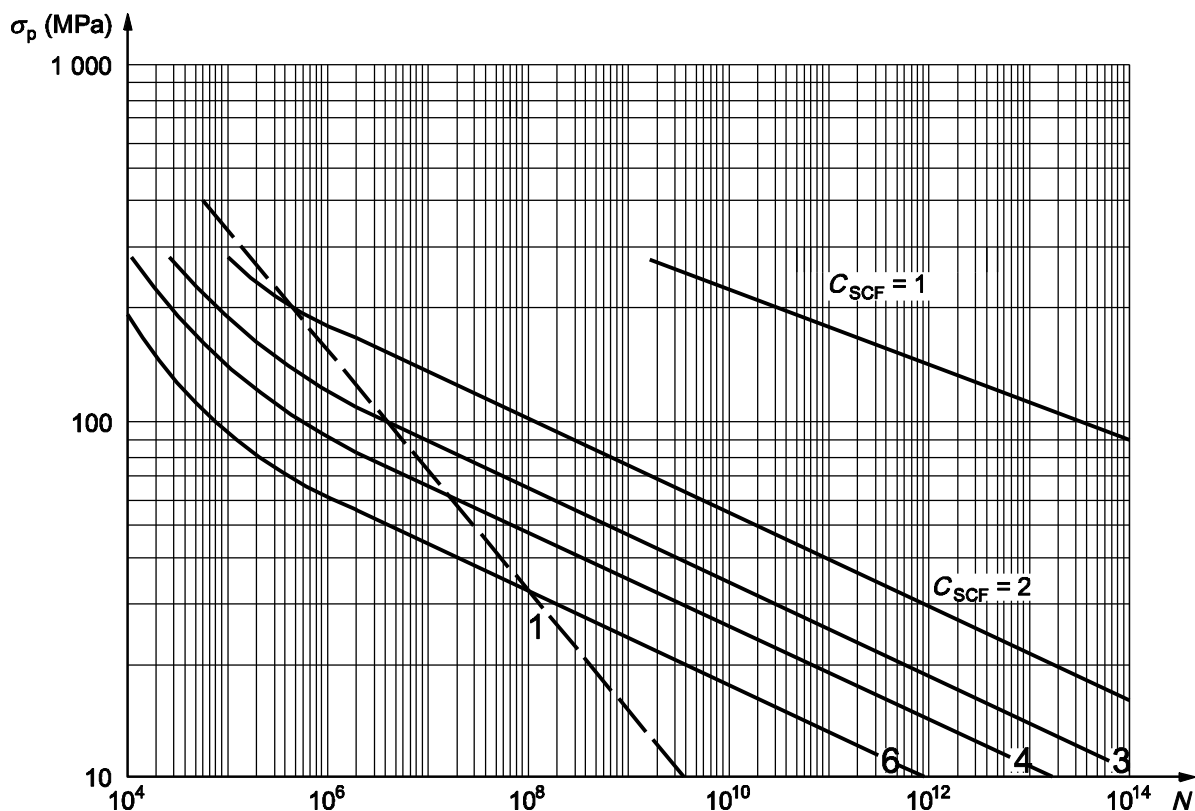
$$\sigma_p \left[\frac{\sigma_p}{E} + \left(\frac{\sigma_p}{K'} \right)^{1/n'} \right] = \frac{1}{E} \left[\sigma_{e,o} + K_f \left(\frac{S_r}{2} \right) \right]^2 \quad (\text{solve for } \sigma_p) \quad (\text{A.15.2-19})$$

Again, the elasto-plastic mean stress, σ_o , at the notch is:

$$\sigma_o = \sigma_p - \frac{\sigma_r}{2} \quad (\text{A.15.2-20})$$

Using σ_r and σ_o in the equation for the equivalent, fully reversible, σ_{re} , the corresponding strain, ε_{re} , can be calculated. With σ_{re} and ε_{re} , it is possible to solve for N_f in the first equation.

An example of S - N curves developed for an actual connector is presented in Figure A.15.2-5, compared to a ground flush circumferential weld estimated as the D curve from the UK HSE guidance, shifted by a factor of two on life. The connector in question develops preload upon makeup, the notch stress is not linear with the pipe tubular stress, and the nominal pipe tubular is maintained at a mean tension of 690 MPa. For this example, the last set of equations presented above was used to transform the stresses. The material used is HY-80 from Reference [A.15.2-2].



Key

- 1 S-N curve for ground circumferential weld
- solid lines S-N curves for varying SCF
- σ_p pipe stress
- N number of cycles to failure

HY-80 material property data

- $n = 0.147$
- $K' = 1184 \text{ MPa}$
- $\sigma'_f = 1249 \text{ MPa}$
- $\epsilon'_f = 1.456$
- $a = -0.101$
- $b = -0.693$

Figure A.15.2-5—Example of Calculated S - N Curves for Various SCFs and Constant Pipe Mean Stress for HY-80 Material Compared with the Circumferential Weld between Connector and Pipe

A.15.2.8 Threaded Fasteners

A.15.2.8.2 Threaded Fastener Materials and Manufacturing

Threaded fasteners should comply with the requirements of BS 4882 [A.15.2-5] or similar.

Rolled threads, as opposed to cut threads, tend to improve fatigue performance, provided the rolling process is undertaken after the heat treatment. However, the $S-N$ curve provided in A.15.2.8.7 does not take advantage of an improvement associated with rolled threads, if any. Furthermore, fatigue test results in air show only a marginal fatigue performance improvement due to rolling of the threads [A.15.2-6].

If threaded fasteners or studs are partially threaded, the location and geometry of thread run-out should be such that stress concentrations are minimized.

A.15.2.8.3 Threaded Fastener Installation

Threaded fasteners may be preloaded by controlling the applied load, the stretch or the torque. The load control method to preload threaded fasteners is considered the most reliable. In this case, threaded fastener preload can be achieved by applying an axial load to the threaded fastener and running the nut to a hand-tight position before releasing the load. Typically, hydraulic bolt tensioners are used, which are modified hydraulic jacks. When allowed to react against a rigid, even support and calibrated to a standard traceable to a recognized national standard, axial hydraulic bolt tensioners can provide a very precise prestress.

When the load is released from the tensioner and transferred to the nut, the resulting deformations as the nut engages the threaded fastener lead to what is known as a transfer loss. This loss should be taken into account during tensioning. The percentage of transfer losses for standard metric and UNS (unified screw thread, standard series) threads bearing on a rigid steel plate may be taken as:

$$\Delta P_{\text{ave}} = \left(0.9 \frac{d}{l} \right) \times 100 \% \quad (\text{A.15.2-21})$$

$$\Delta P_{\text{max}} = \left(1.0 \frac{d}{l} \right) \times 100 \% \quad (\text{A.15.2-22})$$

where

ΔP_{ave} is the average percentage transfer loss;

ΔP_{max} is the maximum percentage transfer loss;

d is the threaded fastener diameter;

l is the stressed length of the threaded fastener, which can be taken as the distance between the first and the last engaged threads.

Where the surface supporting fasteners is flexible, reduced transfer losses will occur. The reduction in losses may be taken into account by calculating an equivalent effective stressed length.

When using hydraulic tensioners, care should be taken to ensure that threaded fasteners are not tensioned beyond yield, since this may lead to excessive deformation. The maximum tensioner load applied to the threaded fastener should not exceed the yield load less a safety margin equal to two standard deviations, as obtained from the tensioning calibration test.

In the absence of specific data, the threaded fastener preload should also include a material relaxation loss of approximately 2 % for steel threaded fasteners with a minimum yield strength above 600 MPa at room temperature. Steel threaded fasteners with a minimum yield strength less than 600 MPa are not recommended for use where a controlled preload is required.

Other methods for preloading threaded fasteners are direct torquing and the turn of the nut, a stretch controlled method. The latter method differs from the former in that a specified torque is initially applied, followed by a measured rotation between the nut and the threaded fastener. Although more practical than direct preload, these methods are inherently less reliable due to the friction on the threads and between the nut and the bearing surface. Use of the direct torque method is not recommended unless carefully calibrated tests, traceable to a

national standard, are carried out in order to consistently account for friction. Control of lubrication and avoidance of galling is important to achieve consistent results.

For threaded fastener connections intended for multiple assembly and disassembly operations, in which the threaded fasteners are to be reused, the calibration procedure should include multiple make-up and break-out operations.

A.15.2.8.4 Threaded Fastener Inspection

The level of cathodic protection of individual bolts should be checked. The cathodic potential of any individual bolt should be minus 900 mV $\pm$ 50 mV compared with an Ag-AgCl cell.

When inspecting the bolts for tension, the bolts should be re-tensioned to the same level to which they were intended to be preloaded during installation, using the original installation procedure and calibrated tools. If any bolt is found to fall below the minimum design value, the bolt should be re-tensioned to its originally required value.

A.15.2.8.5 Threaded Fastener Strength Criteria

The extreme fiber stress in the connector (axial plus bending), induced by preload and factored loads/actions on the connected tubular members, should satisfy the following criteria:

$$\sigma_{bt} \leq \frac{f_y}{\gamma_{R,bt}} \quad (\text{A.15.2-23})$$

where

σ_{bt} is the extreme fiber stress (axial plus bending), acting on the minimum section;

f_y is the representative yield strength of the material;

$\gamma_{R,bt}$ is the resistance factor for axial tensile strength = 1.0.

In addition, the size and number of threaded tension fasteners used should be such that the total factored tension force, P_t , due to factored loads/actions, acting on the connected tubular members, does not exceed:

$$P_t \leq \frac{f_y A_b N_b}{\gamma_{R,ba}} \quad (\text{A.15.2-24})$$

where, additionally,

A_b is the area of the shank of the threaded element;

N_b is the total number of threaded elements;

$\gamma_{R,ba}$ is the resistance factor for threaded fastener area = 1.15.

Allowance should be made for time-dependent relaxation losses in threaded fasteners, the magnitudes of which depend on the material and service temperature. In determining the pretensioning force for a threaded fastener, allowance should also be made for losses associated with the pretensioning procedure, such as transfer losses, as well as for the precision of the procedure itself.

A.15.2.8.6 Threaded Fastener Fatigue Criteria

To minimize the stress fluctuation response in threaded fasteners caused by fluctuating loads/actions, the fastener stressed length and cross-sectional area should be chosen to give maximum flexibility of the fastener within the constraints of the design (i.e. maximize the fastener preload extension).

FEA may be used to evaluate threaded fastener stress variations due to fluctuating loads/actions. The FEA model for this case may be developed with the intent of only simulating stiffness, requiring only a coarse mesh. Ignoring the fastener preload in evaluating the fastener force variations is always conservative.

To calculate fatigue damage, Palmgren-Miner sum may be used, in which the damages induced in the fastener by each of the stress variations are linearly superimposed. The damage due to a given variable force may be expressed in closed form when a spectral fatigue analysis of the structure has been performed. Alternatively, the damage may be calculated as the discrete sum of the ratios of the acting number of cycles of a given stress range in the fastener, as defined by the long-term distribution of annual occurrence versus stress range, to the allowed number of cycles for the same stress range, as given by an S – N curve.

In the absence of specific fatigue data for threaded fasteners, the following S – N curve may be used; however, the user should be aware that the derivation of Equation (A.15.2-25) does not necessarily account for the effects of preload:

$$\log N = 11.55 - 3 \log S \quad (\text{A.15.2-25})$$

where N is the number of cycles to failure and S is the stress range in the fastener.

The S – N curve provided above is lower bound to fatigue tests in air given in Reference [A.15.2-6] augmented by tests conducted at TWI in the United Kingdom on behalf of HSE [A.15.2-7]. The data include tests in which the applied load is transferred to the fastener from the loading plate via a single or double nut arrangement. In the latter arrangement, the load is transferred from a perforated plate, through which the fastener passes. The fastener is connected to the plate by means of two nuts torqued against each other, one on each side of the plate. The above S – N curve provides for reasonably conservative designs for fasteners made of ASTM A193 [A.15.2-8], Grade B7 and L7 steels, as well as of ISO Grades 8.8 and 12.2 to BS 3692 [A.15.2-9].

Unless demonstrated by pertinent tests under relevant conditions, no advantage may be taken of an endurance limit.

Although fatigue data show that rolling the threads after heat treatment tends to improve the fatigue performance of fasteners, the above given S – N curve conservatively ignores this effect. Heat treating fasteners after rolling the threads essentially results in a fastener with a fatigue performance similar to that of fasteners with cut threads.

Tests conducted at TWI in the United Kingdom showed that the fatigue performance of Grades 8.8, B7 and L7 bolts, tested under cathodically protected conditions, experienced reductions on fatigue life relative to in-air performance. Therefore, a reduction factor of 3.0 on the life obtained by the S – N curve given above may be used to account for the effect of using bolts underwater with adequate cathodic protection in the range –850 mV to –1000 mV. However, the underwater fatigue performance of fasteners made of higher grade steels should be specifically addressed via a validated test program, as should the effect of hydrogen embrittlement.

Tests on threaded fasteners varying from 19 mm to 44 mm in diameter made of Grades 8.8, B7 and L7 support a correction factor on stress, relative to the 20 mm size, given by Equation (A.15.2-26):

$$k_{\text{corr}} = \left(\frac{d}{20} \right)^{0.3} \quad (\text{A.15.2-26})$$

where d is the fastener diameter, in millimeters.

The factor, k_{corr} , is applied to the stress range prior to entering the S – N curve given above.

A.15.3 Clamps for Strengthening and Repair

A.15.3.1 General

The general clamp design process is presented in Figure 15.3-1.

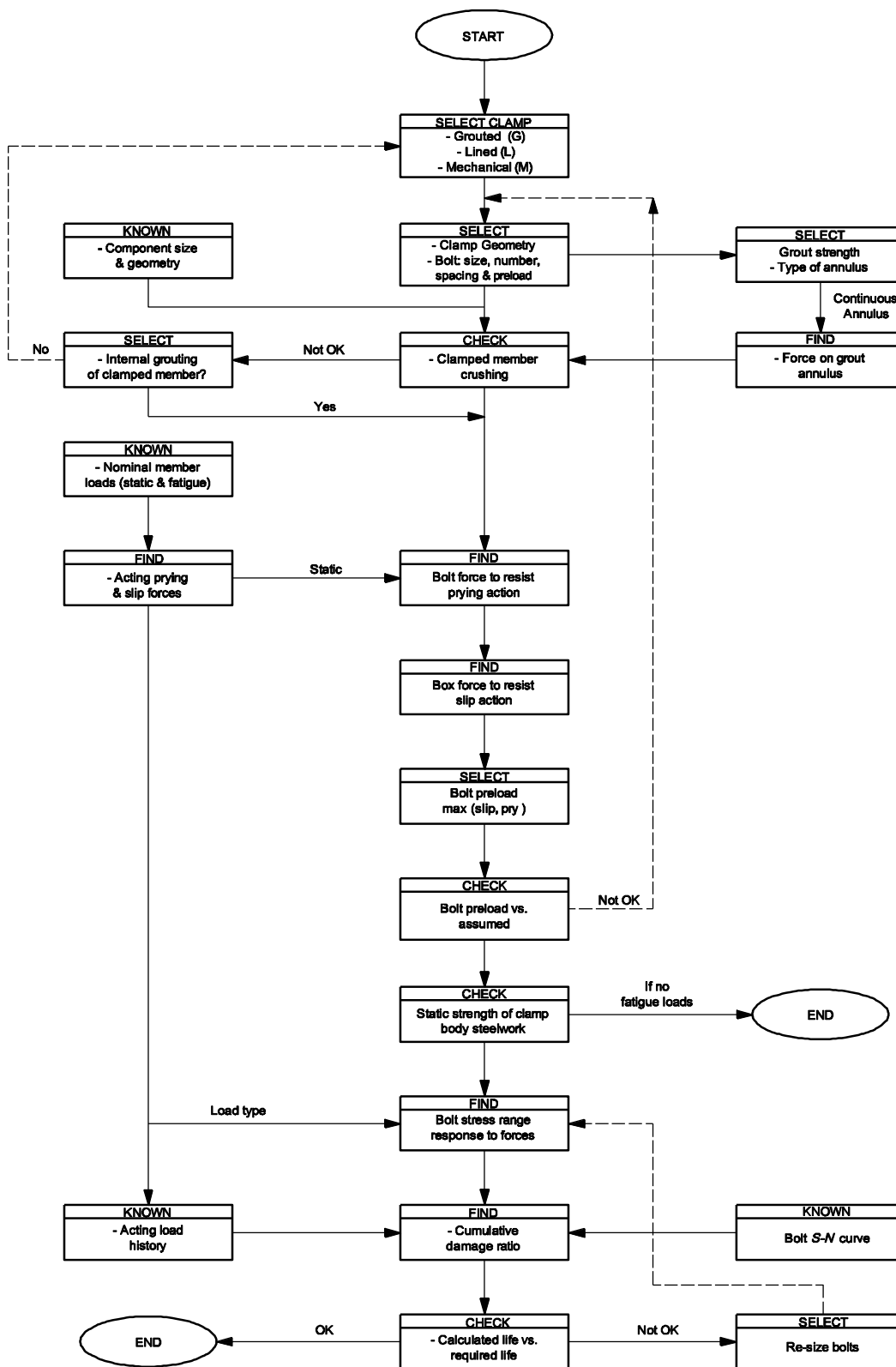


Figure A.15.3-1—General Design Procedure for Stressed Clamps

A.15.3.4 Forces on Clamps

A.15.3.4.3 Bolt Forces

A.15.3.4.3.1 General

Figures A.15.3-2, A.15.3-3, and A.15.3-4 illustrate typical applications of clamps for strengthening or repair of structures for, respectively:

- a) end-to-end connections;
- b) addition of members;
- c) tubular joints.

The capability of prestressed clamps to transfer forces is determined by the bolt prestress. Bolts should be prestressed so that the clamp does not open or slip due to the forces acting on the clamp. Shear forces and bending moments from the substrate member tend to pry or separate the clamp, whereas axial force and torsion tend to cause the clamp to slip.

The prying load/action on the clamp arising from the shear forces and bending moments is directly resisted by extension in the bolts. However, the forces that tend to pull out or slip the substrate member from the clamp are resisted by the friction developed at the clamp-member interface. This frictional resistance is induced by the contact pressure developed as the bolts are tightened or prestressed. Because the prying load/action tends to relieve the contact pressure and, thus, decrease the resistance against slippage, the bolt prestress should correspond to the greater of the following two bolt forces:

- bolt forces required to resist the prying (without separation);
- bolt forces required to prevent slippage.

When calculating the bolt forces in clamps intended to strengthen or repair joints between tubulars, advantage may be taken of the sharing of forces between the clamp and the substrate member, provided that such load sharing of forces is demonstrated by analysis or tests. Otherwise, it is conservative to assume that no sharing of forces takes place and that the clamp fully transfers the substrate forces across the joint.

Methods for calculating the required prestressing forces in the bolts to prevent prying and slipping at the clamp-member interface for each of the three clamp applications are presented in A.15.3.4.3.2 to A.15.3.4.3.4, ignoring sharing of forces between the clamp and the substrate member.

A.15.3.4.3.2 End-to-end Connection Clamps

Figure A.15.3-2 shows a generic clamp over an end-to-end connection and a typical set of substrate member end forces. The prying forces from the substrate member are transferred to the clamp and, thus, to the bolts over the contact length L_c . To calculate the axial force sustained by the bolts due to the prying forces, in the absence of test data, it may conservatively be assumed that the vectorial resultants from the acting prying moments and shear forces superimpose on one another to separate the clamp (Figure A.15.3-3, section A-A).

where

M is the resultant bending moment on the substrate member assumed acting as if in-plane;

M_{ipb} is the IPB moment on the substrate member;

M_{opb} is the OPB moment on the substrate member;

V is the resultant shear force on the substrate member assumed acting as if in-plane;

V_{ip} is the in-plane shear force on the substrate member;

V_{op} is the out-of-plane shear force on the substrate member.

Tests of an end-to-end joint have shown that in the case of grouted clamps with the bolts oriented as shown in Figure A.15.3-2, only a fraction of the applied OPB moment (M_{opb}) results in bolt force variation so long as no separation of the clamp halves occurs. In absence of pertinent data, one half of the acting M_{opb} can be substituted in the above equation when calculating the total bending moment, M . However, for neoprene-lined clamps, tests show that the full M_{opb} should be used.

The overall separation load/action on the clamp results in a contact pressure distribution at the clamp-member interface that is maximum at the free end of the clamp and decreases toward the center of the clamp. The distribution of this pressure and the length over which it acts is not well understood. In the absence of a detailed analytical or experimental evaluation of such pressure, the pressure distribution arising from the resulting bending moment (M) may be assumed to decrease over a distance, $2a$, from the free end, where it is maximum, as indicated in Figure A.15.3-2. The distance $2a$ can be taken as equal to the contact length, L_c . The pressure arising from the resulting shear force (V) may be assumed to be uniform over the same length ($2a$) over which the bending moment was assumed to have dissipated.

In general, the force resisted by the end bolts due to prying load/action should include the effect of the resultant bending moment and resultant shear force and, if applicable, any other components of forces acting directly on the clamp (e.g. add-on members) that are parallel to the bolt axes. Thus, the bolt force, $P_{B,b}$, assuming two rows of bolts parallel with the axis of the substrate members (one on either side) as shown in Figure A.15.3-2, can in general be given by:

$$P_{B,b} = \frac{1}{2} (P_{B,M} + P_{B,V} + P_{B,P}) \quad (A.15.3-3)$$

where

$P_{B,b}$ is the total force on the end bolt (assuming two parallel rows of bolts);

$P_{B,M}$ is the force on the end bolt induced by the resultant bending moment, M ;

$P_{B,V}$ is the force on the end bolt induced by the resultant shear force, V ;

$P_{B,P}$ is the bolt force induced by other forces parallel to the bolt axes and acting on the clamp, if present.

For clamps installed over end-to-end connections $P_{B,P} = 0$, therefore, only the bending moments and shear forces on the substrate member contribute to the bolt force through $P_{B,M}$ and $P_{B,V}$. As indicated in Figure A.15.3-2 by the shaded area, these forces can generally be obtained from the corresponding pressure distributions brought about by the bending moment and shear force resultants acting over a distance equal to one half of the bolt spacing on each side of the bolt.

The force on the end bolt induced by the resultant shear force is equal to the average shear force over the contact length ($V/2a$) times the distance, b :

$$P_{B,V} = \frac{V}{2a} \times b \quad (\text{A.15.3-4})$$

where, further to previous definitions,

$$a = L_c/2;$$

b is the larger of the two distances s and $(s/2 + e)$;

s is the bolt spacing;

e is the distance between the end bolt and the end of the clamp (see Figure A.15.3-2).

In addition, the substrate member tends to slip relative to the clamp. The total slip force may be taken as the vectorial sum of the slip forces in the longitudinal and circumferential directions. The slip force associated with the longitudinal shear force induced by the bending moment (M) in the substrate member may be ignored. Hence, the total acting slip force is according to Equation (A.15.3-5):

$$P_{Sl} = \sqrt{P_{ax}^2 + \left(\frac{2T}{D}\right)^2} \quad (\text{A.15.3-5})$$

where

P_{Sl} is the slip force between the substrate member and the clamp;

P_{ax} is the axial force on the substrate member;

T is the torsional moment on the substrate member;

D is the outside diameter of the substrate member (see Figure A.15.3-2).

The interface transfer stress σ_p acting on the clamp is according to Equation (A.15.3-6):

$$\sigma_p = \frac{P_{Sl}}{\pi D L_s} \quad (\text{A.15.3-6})$$

where L_s is the length of the clamp over which the slip force, P_{Sl} , is assumed to transfer.

For clamps used in end-to-end joints, L_s should be taken as the shortest length from the joint to either end of the clamp, i.e. the shortest contact length L_c , if the clamp is not placed symmetrically relative to the joint. Both halves of the clamp are assumed to contribute to the slip resistance.

A.15.3.4.3.3 Addition-of-member Clamps

Figure A.15.3-3 illustrates the use of clamps to add a member. In this instance, the substrate member only supports the clamp and the forces acting on the add-on member tend to pry and slide the clamp relative to the substrate member.

The prying moment, M , may be taken as:

$$M = \sqrt{(M_{opb} \cos \theta + T \sin \theta)^2 + (M_{ipb})^2} \quad (\text{A.15.3-7})$$

where

- M is the total prying moment on the clamp;
- M_{ipb} is the IPB moment on the added member;
- M_{opb} is the OPB moment on the added member;
- T is the torsional moment on the added member;
- θ is the included angle between the added member and the substrate member (see Figure A.15.3-3).

Both M_{ipb} and M_{opb} are calculated about the intersection of the add-on member axis and the axis of the substrate member (the work point).

The bolt force, $P_{B,b}$, is analogous to Equation (A.15.3-3), without a contribution from shear forces in the substrate member:

$$P_{B,b} = \frac{1}{2}(P_{B,M} + P_{B,P}) \quad (\text{A.15.3-8})$$

where

- $P_{B,b}$ is the total force on the end bolt (assuming two parallel rows of bolts);
- $P_{B,M}$ is the force on the end bolt pair induced by the total prying moment, M ; $P_{B,M}$ is calculated as for end-to-end clamps (see A.15.3.4.3.2);
- $P_{B,P}$ is the force on the end bolt pair parallel to the bolt axes, induced by the forces in the added member:

$$P_{B,P} = \frac{P \sin \theta - V \cos \theta}{N_{bp}} \quad (\text{A.15.3-9})$$

where

- P is the axial force on the added brace;
- V is the resultant shear force on the added brace;
- N_{bp} is the number of bolt pairs.

The total acting slip force can be taken as:

$$P_{Sl} = \sqrt{\left(P \cos \theta + V_{ip} \sin \theta\right)^2 + \left(\frac{2(T \cos \theta + M_{opb} \sin \theta)}{D} + V_{op}\right)^2} \quad (\text{A.15.3-10})$$

where, further to previous definitions,

- P_{Sl} is the slip force between the substrate member and the clamp;
- V_{ip} is the in-plane shear force on the added member;
- V_{op} is the out-of-plane shear force on the added member;
- D is the outside diameter of the substrate member.

The interface transfer stress is assumed resisted by that half of the clamp to which the add-on member is attached, and can be determined as follows:

$$\sigma_p = 2 \times \frac{S}{\pi D L_s} f_p = 2 \times \frac{S}{\pi D L_s} \quad (\text{A.15.3-11})$$

where, furthermore,

σ_p is the interface transfer stress;

L_s is the length of the clamp over which the slip force is assumed to transfer.

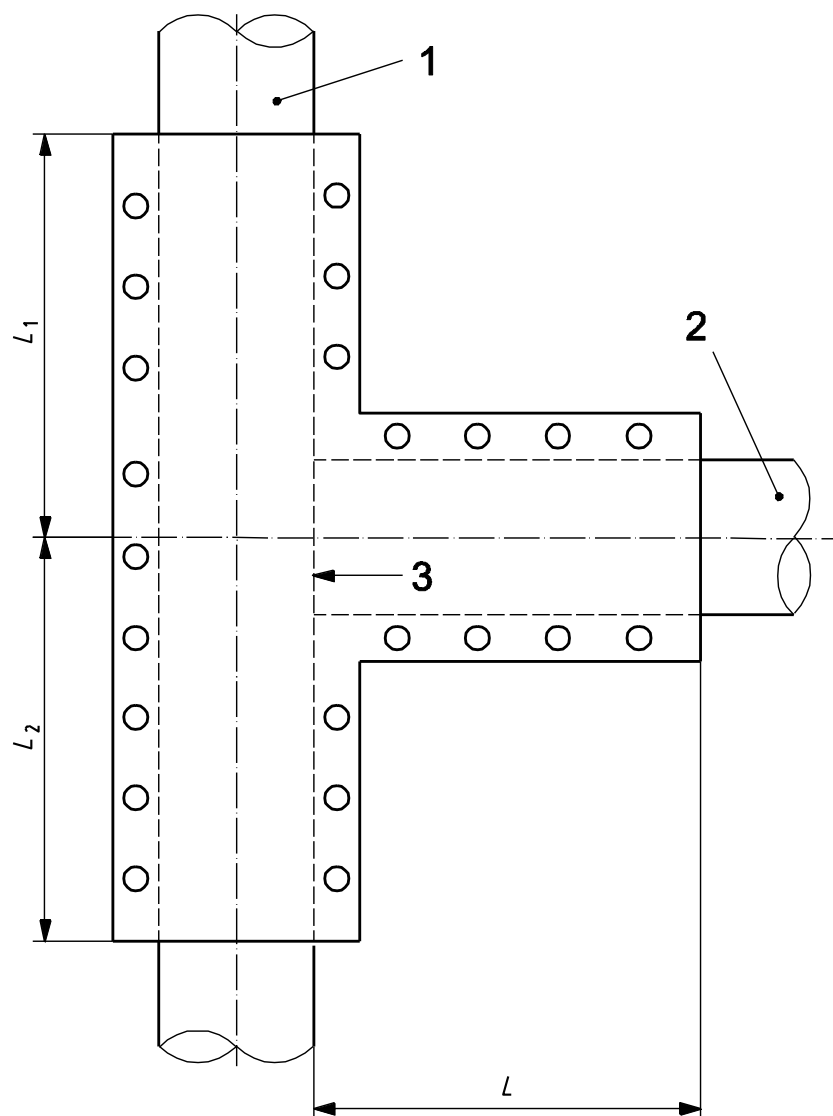
The above slip force is assumed to act only on the clamp's half to which the add-on member attaches. The other half of the clamp is not active due to the inability of the bolts to transfer shear. However, both halves of the clamp can be made to act simultaneously by providing shear transfer between the two clamp halves using rigidly engaged longitudinal hinges, placed on each side of the clamp.

A.15.3.4.3.4 Clamps on Tubular Joints

Referring to Figure A.15.3-4, the prying forces on clamps placed on tubular joints may conservatively be determined by treating each of the tubular joint member ends as the substrate member of a clamp on an end-to-end connection. The substrate member in Figure A.15.3-4 can be the brace or the chord parts of the tubular joint. Assuming that the clamp is to fully transfer the forces, the lengths over which force transfer can be assumed to take place are as follows:

- a) L for the brace side, and
- b) L_1 and L_2 for the chord sides.

Likewise, the calculation of the slip force may conservatively be treated as a case of an end-to-end clamp for each of the free ends of the joint.

**Key**

- 1 chord substrate member
- 2 brace substrate member
- 3 break

Figure A.15.3-4—Typical Application of a Clamp on a Tubular Joint

A.15.3.5 Clamp Design

A.15.3.5.4 Fatigue Design of Bolts

FEA can be used to evaluate bolt force fluctuations due to the time-varying loads/actions. The FEA meshes in analyses used for evaluating force fluctuations may be coarse and the geometry may be defined by mid-surface planes. The bolts may be modeled with single beam elements, and the clamp and the substrate members may be modeled with plate and thin shell elements. Alternatives for modeling the bolt preload and clamp-member interface include:

- a) explicitly modeling a bolt preload that is sufficient to maintain contact between the clamp and the substrate member, or
- b) ignoring the bolt preload and modeling the interface between the clamp and the substrate member with contact elements that allow gaps to open.

In FEA, the bolts should be prestressed to the intended level prior to applying any member forces to the clamp. Then, a linear elastic analysis should be conducted for each of the member forces relevant to the clamp in question and the corresponding variation of the axial stresses in the bolt determined. In this way, transfer functions between the forces on the clamp and the bolt stresses can be established. Hence, knowing the clamp force histories and the corresponding transfer functions, the actual stress history in the bolt can be obtained.

A.15.3.6 General Requirements for Bolted Clamps

A.15.3.6.1 Mechanical Clamps

The structural performance of mechanical clamps is sensitive to the fit-up achieved during installation, which in turn depends on the tolerances between the clamp and the structure. If a fit is too loose, the clamp can have substantially reduced strength. If a fit is too tight, the clamp can be installed incorrectly. Also, it is important that the contact between the clamp and the structure be as uniform as possible.

To achieve proper installation and effective performance of mechanical clamps, it is important to clean the surfaces to bare metal and to grind welds, seam or circumferential welds, flush with the substrate member outer diameter. Also, the clamp design should be based on an accurate dimensional survey of the actual structure.

Tests have shown [A.15.3-1] that corrosion products, formed immediately after blasting the substrate member(s) underwater, can affect performance. Therefore, mechanical clamps for underwater applications should be installed within 24 hr of cleaning, ensuring that the surfaces are wiped clean of any corrosion products immediately before installation.

Generic flat-plate friction tests given in Reference [A.15.3-1] show that, for grit blasted steel surfaces exposed to salt air and water, the coefficient of friction varies between 0.4 and 0.57, with a mean value of 0.51.

A.15.3.6.3 Lined Clamps

Elastomeric materials, such as polychloroprene (commonly known as neoprene), have been used for lining the inside of friction repair clamps. The lining can be formed or notched to accommodate local irregularities (e.g. doubler plates) on existing members and can be bonded to the clamp saddle at room temperature.

Extreme care should be taken in the design and use of elastomer lined clamps. The design should consider the long-term effectiveness of the elastomer with regard to creep and seawater resistance. For example, soft and thick linings will have a higher initial deformation under force, and hence higher long-term creep, than hard and thin linings. Ribbed linings (linings with longitudinal grooves) should not be used for applications of primary force transfer, due to creep and their added flexibility.

A.16 Fatigue

A.16.1 General

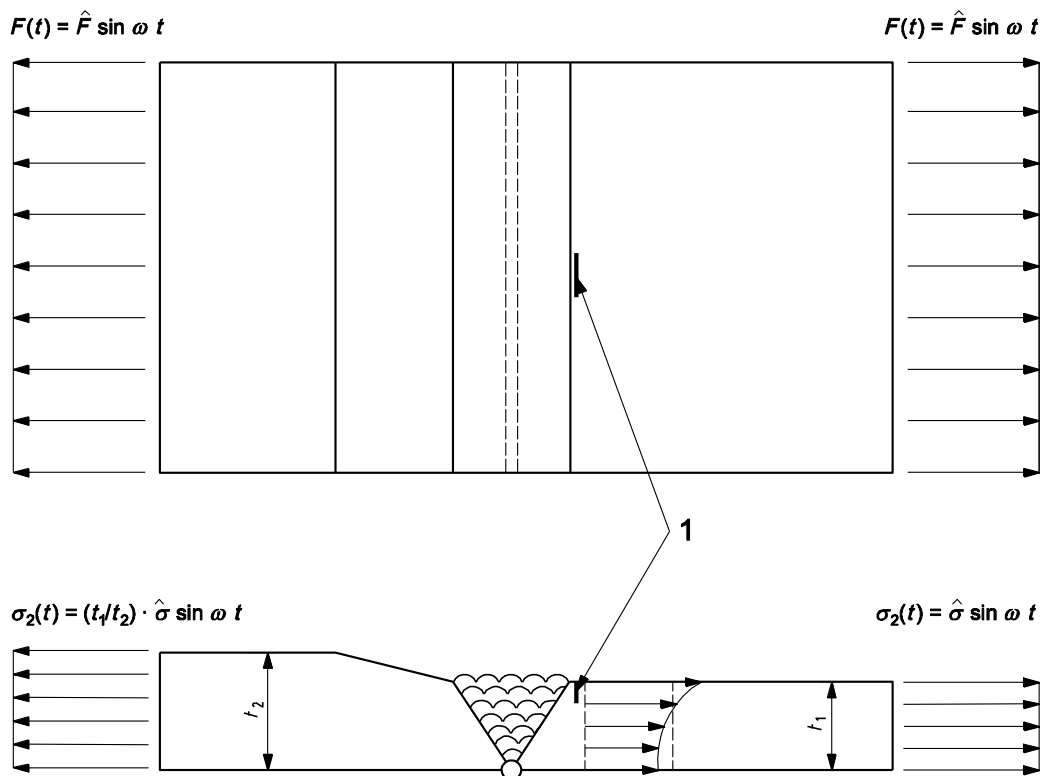
A.16.1.1 Applicability

Fatigue has long been recognized as an important consideration for the design of offshore structures. Tubular joints in a fixed steel structure are especially subjected to large forces and are sensitive to fatigue damage; intensive co-operative industry research on tubular joints occupied the full decades from the 1960s to the 1980s.

A.16.1.2 The Fatigue Process

Figure A.16.1-1 gives an idealized overview of the mechanics of the fatigue process. Its purpose is to assist basic understanding of the overall process and of the many separate factors involved. It further serves as a general reference for features and explanations of fatigue issues that are discussed in Section 16. The basic description given here refers specifically to fatigue cracks emanating from pre-existing defects in welded connections, but the general concepts are equally applicable to fatigue in general.

Consider two plates of unequal thickness, connected by a full penetration butt weld perpendicular to the longitudinal direction of the plates (see Figure A.16.1-1). The plates are subjected to a variable load/action, $F(t) = \hat{F} \sin \omega t$, of constant amplitude and frequency. At sufficient distance away from the weld this load/action results in nominal stress distributions of constant amplitude, which are uniform over the cross-section of the plates. In the thinner plate the stress is $\sigma_1(t) = \hat{\sigma} \sin \omega t$, whereas in the thicker plate the stress is $\sigma_2(t) = (t_1 / t_2) \times \hat{\sigma} \sin \omega t$.



Key

- 1 small defect at weld toe

Figure A.16.1-1—Fatigue Illustration: Two Plates of Unequal Thickness Connected by a Full Penetration Butt Weld

At various locations in the body of the weld, or the HAZ in the plate material adjacent to the weld, small defects will inevitably exist. These defects will be very small, because large defects are avoided by proper design, workmanship and control of the weld process, or are captured by inspection during fabrication and repaired before the weld is accepted. The defects have in common that they are generally too small for detection, but otherwise they are due to different causes (e.g. lack of fusion, slag inclusions, shrinkage) and have different shapes, sizes, and orientations. However, their most likely orientation will be more or less parallel to the weld. This orientation is in any case the most relevant and potentially the most onerous with a view to fatigue damage.

The defects represent crack-like features of microscopic size that are continuously subjected to variable stresses perpendicular to their most likely orientation. By way of example, a small defect at the toe of the weld is shown in Figure A.16.1-1. Under the influence of variable tensile stresses the defect will begin to grow in depth across the thickness of the plate, as well as in length across the width of the plate. The question is how many cycles of a particular magnitude the plate can withstand before it fails. To answer this question the following questions need first to be addressed.

- a) What is the magnitude and number of cycles of the nominal stress variations in the plate as a result of the applied load/action?

The predominant source of time-varying loads/actions on offshore structures is wave load/action. For flare towers and parts of topsides structures, wind or mechanically induced vibrations can dominate. However, the present discussion will concentrate on wave-induced stress variations; hence, wave load/action is the main driver of fatigue damage. The nominal stress in a component depends on the load/action effects (the internal forces) resulting from the applied loads/actions (which vary with wave height, wave period, and wave direction) and the structural response to the applied loads/actions (which varies with type, location, and orientation of the component; its boundary conditions; and other features). Wave conditions vary widely during the design service life of a structure; all these conditions should be taken into account. For the same wave condition the nominal stresses in different components will vary widely; all components should thus be checked for adequate resistance to fatigue. A fatigue assessment hence requires consideration of a large number of different load cases and a large number of different locations in the structure, and involves a very substantial computational effort.

- b) What is the local stress variation at the actual location of the initial defect?

At the actual location of the initial defect the stress distribution will generally deviate from the uniform nominal stress distribution due to local variations of geometry (see Figure A.16.1-1). However, the stress-raising effect caused by such changes of geometry at the location of the defect can be related to the nominal stress by the introduction of a so-called SCF. The SCF expresses the ratio of the local stress variation to the nominal stress variation and can both be larger and smaller than 1.0.

- c) Which stress parameter controls fatigue damage (e.g. the tensile stress amplitude, the ratio of minimum to maximum stress, the stress range between the maximum and the minimum stress in a cycle)?

The loads/actions on an offshore structure generally consist of a combination of permanent and time-varying loads/actions. In welded connections the weld process and any constraints during the heat cycle additionally introduce residual stresses. The magnitude of these residual stresses is not known, but it is commonly accepted that tensile stresses of yield strength magnitude will exist around the weld. From this, it logically follows that all, or at least the major part of, the full variable stress range will be in the tensile region. Therefore, the stress range S is used as the controlling stress parameter.

- d) How can the resistance of the material (the number of stress cycles to failure) be determined?

Crack growth due to time-varying (tensile) stresses can theoretically be described by the Paris-Erdogan law of fracture mechanics. However, the number of cycles that a material or a structural detail can withstand can generally only be empirically determined by testing the particular detail. The results of such tests are presented in terms of a pair of values of the applied stress range, S , and the number of cycles to failure, N . By

repeating the tests for different values of S and counting the corresponding number of cycles N , a so-called S - N curve is determined, which defines the resistance of the material (or of the structural detail) to constant amplitude variable stresses of different magnitudes. The relationship between S and N has been found to be exponential and the S - N curve plots as a straight line on double logarithmic scales. Duplicating tests under the same circumstances (for so far as test conditions can be precisely controlled) demonstrates that the experimental outcome for a given value of S is not one unique value of N , but that significant scatter is observed. Therefore, the experimental results are presented as $\log S$ versus the mean of $\log N$. To account for the uncertainty associated with this scatter, an S - N curve of $\log S$ versus the mean minus two standard deviations of $\log N$ is used for design purposes.

e) What is the definition of failure?

In view of the sensitivity of the test results it is of great importance to clearly define which definition of stress and which definition of failure is adopted. Definitions for both these parameters have been agreed upon internationally. The governing stress parameter is either the nominal stress range, $S_{\text{nom}} = \sigma_{\text{max}} - \sigma_{\text{min}}$ (which, in the present example, is equal to $S = 2\hat{\sigma}$), or a specially defined local stress range derived from $\sigma(t)$ as experienced at an identified location. The number of cycles to failure is defined as the number of cycles, N , for the crack to grow through the thickness of the test sample. In using and interpreting S - N curves for a fatigue assessment, these definitions should clearly be borne in mind. From the definition of N , it further follows that S - N curves can be dependent on the thickness of the plate.

For geometries other than the simple example used here (for the purpose of illustration only), the definition of the stress parameter becomes more difficult. This is due to the more complex geometry of the structural components connected, as well as to the shape of the weld connecting them. In some cases the stress-raising effect resulting from the geometry of the structural detail in combination with the type of local stresses can be rather well determined; in such cases, preference is generally given to a local stress range. This is, for example, often the case for tubular joints, where the nominal stress range, $S_{\text{nom}} = \sigma_{\text{max}} - \sigma_{\text{min}}$, is multiplied by a geometrical SCF (C) so that $S = C \times S_{\text{nom}} = C(\sigma_{\text{max}} - \sigma_{\text{min}})$. The SCF differs with the type of forces to which the members that are connected at the joint are subjected (axial forces, IPB, or OPB), with the structural details of the tubular joint, and with the location around the circumference of the connecting weld under consideration. However, in other cases the geometrical stress-raising effect is not easily established, or it is much too cumbersome to determine for each and every different type of detail and load case; in such situations, it is more convenient to use the nominal stress range $S_{\text{nom}} = 2\hat{\sigma}$. This latter case implies that a different S - N curve applies to each and every detail and load case.

It is, in any event, the stress-raising effect due to the geometry of the structural detail only that is included in the SCF. Additional stress-raising effects at the location of the actual defect, which are due to the profile and internal structure of the weld (the so-called notch effect), are incorporated in the empirical S - N curve.

f) How can the damage done be determined when the stresses are not variable stresses of constant amplitude?

When the stresses in the structural component are not of the constant amplitude type, the number of cycles, n_i , of stress range class, S_i , that occur during a time period, T , are counted. The partial damage done by stress range class, S_i , is equal to the number of cycles, n_i , actually experienced divided by the corresponding number of cycles to failure, N_i . The total damage done is then the sum of the partial damages over all stress

range classes; i.e. the total damage done is assumed to be equal to the nondimensional number $D = \sum_i \frac{n_i}{N_i}$.

Failure is assumed to occur when $D = 1.0$. This hypothesis of the linear accumulation of damage is known as the Palmgren-Miner rule. The calculated fatigue life, L , is the inverse of the total fatigue damage done multiplied by the time period T , i.e. $L = D^{-1} \times T = T/D$.

A.16.3 Description of the Long-term Wave Environment

A.16.3.2 General

Ocean waves are irregular in shape; vary in height, length, and speed of propagation; and can approach a structure from one or more directions simultaneously. During a sea state, these features of a real sea can be described by means of a random wave model. The linear random wave model views the sea as the superimposition of many small individual frequency components, each of which is a periodic wave with its own amplitude, frequency, and direction of propagation, and which have random phase relationships with respect to each other. A unidirectional random sea is a special case of this in which all frequency components propagate in the same direction. The surface of a unidirectional sea is hence long-crested, whereas the surface of a real or directional sea is short-crested. In the linear random wave model, a unidirectional sea is completely described by the two-parameter frequency spectrum, $S(\omega)$ (see API 2MET).

A.16.3.3 Wave Scatter Diagram

For the representative period, T_R , of a sea state, several choices are in common use. T_R may be chosen to be the peak spectral period, T_p , or the average zero-crossing period, T_z , or an alternative mean period, T_1 (see API 2MET). These periods are distinctly different, and it is important to ensure that the period by which the data are given and the period that is used in the analysis are consistent.

Wave scatter diagrams usually refer to a (typical) year and are normally given in a normalized form of number of occurrences per thousand observations. Generally available data usually refer to two-dimensional wave scatter diagrams supplemented with a distribution of mean wave directions, which is often assumed to be equal to that of the mean wind directions. An example of a two-dimensional wave scatter diagram is shown in Figure A.16.3-1.

See API 2MET for guidance on the determination and adequacy of the metocean database.

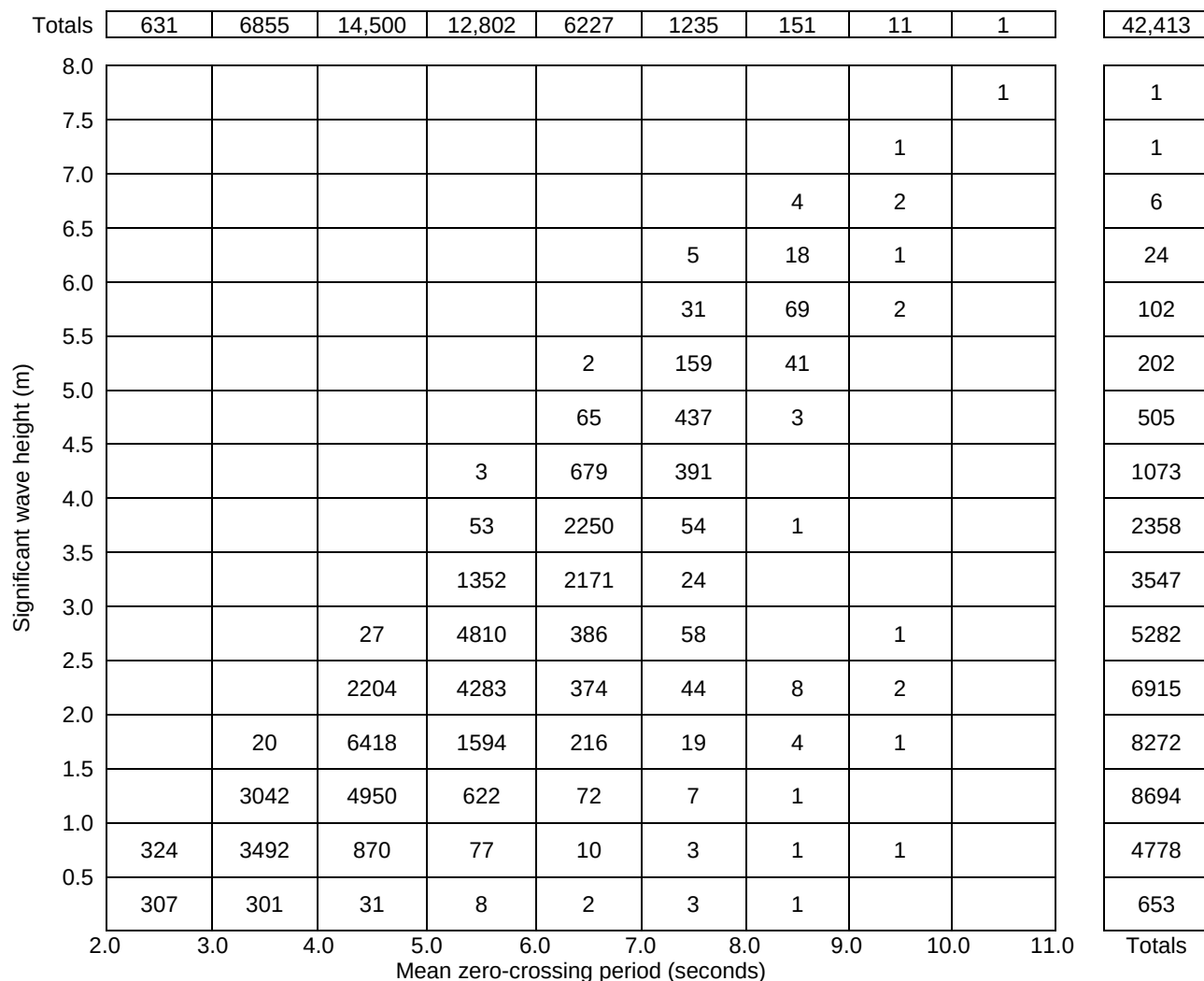


Figure A.16.3-1—Example of Two-dimensional Wave Scatter Diagram

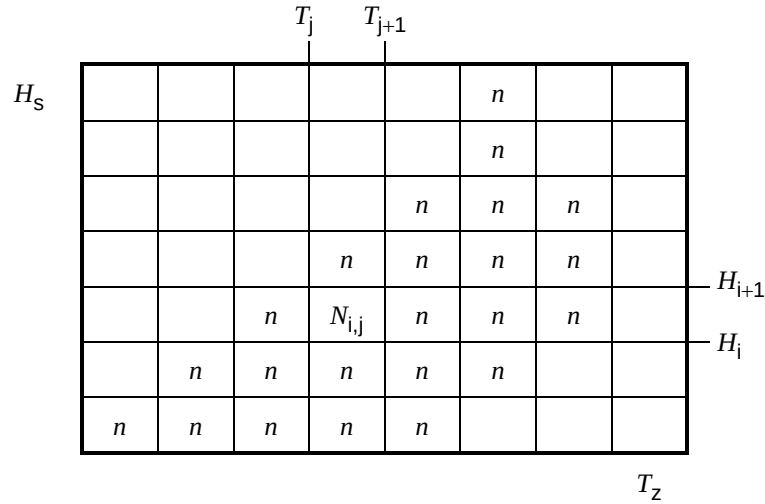
A.16.3.6 Wave Directional Spreading Function

Experience has shown that in most cases the influence of wave directional spreading around the mean is of secondary importance and can be neglected. However, if it is considered that the influence of wave directional spreading on the stress variations is of some significance and it is desired that this effect is included, this can be done by choosing an appropriate wave spreading function in accordance with API 2MET.

A.16.3.8 Long-term Distribution of Individual Wave Heights

Figure A.16.3-2 is a schematic of a wave scatter diagram in which the sea states are defined by the significant wave height, H_s , and the average zero-crossing period, T_z . If the representative period, T_R , of the data is not the average zero-crossing period, T_R should first be converted into T_z . The selected sea state indicated in the Figure occurs $N_{i,j}$ times out of a total of N_T observations, where i is the subscript for significant wave height and j the subscript for representative period and

$$N_T = \sum_j \left(\sum_i N_{i,j} \right) = \sum_i \left(\sum_j N_{i,j} \right) \text{ is the total number of observations in the diagram.}$$



Key

H_s significant wave height

T_z mean zero-crossing period

$N_{i,j}$ number of occurrences of sea states with $H_i < H_s < H_{i+1}$ and $T_j < T_z < T_{j+1}$

n number of occurrences of sea states falling within indicated range of H_s and T_z

Figure A.16.3-2—Example of Wave Scatter Diagram

The probability that the individual wave height, H , in a sea state with significant wave height, H_s , has a particular value, H^* , is given by $p(H^*)dH$, where $p(H^*)$ is the Rayleigh probability density function:

$$P(H^*) = \frac{4H^*}{H_s^2} \times \exp \left[\frac{-2(H^*)^2}{H_s^2} \right] \quad (\text{A.16.3-1})$$

The cumulative probability, $p(H > H^*)$, that the individual wave height, H , exceeds value H^* in the sea state with significant wave height H_s is thus:

$$P(H > H^*) = \int_{H^*}^{\infty} p(H)dH = \int_{H^*}^{\infty} \frac{4H}{H_s^2} \times \exp \left(\frac{-2H^2}{H_s^2} \right) \times dH = \exp \left[\frac{-2(H^*)^2}{H_s^2} \right] \quad (\text{A.16.3-2})$$

The number of times in any one year that individual wave height H^* is exceeded in the selected sea state, i, j , is indicated by the symbol $(N_{H^*})_{i,j}$ and given by:

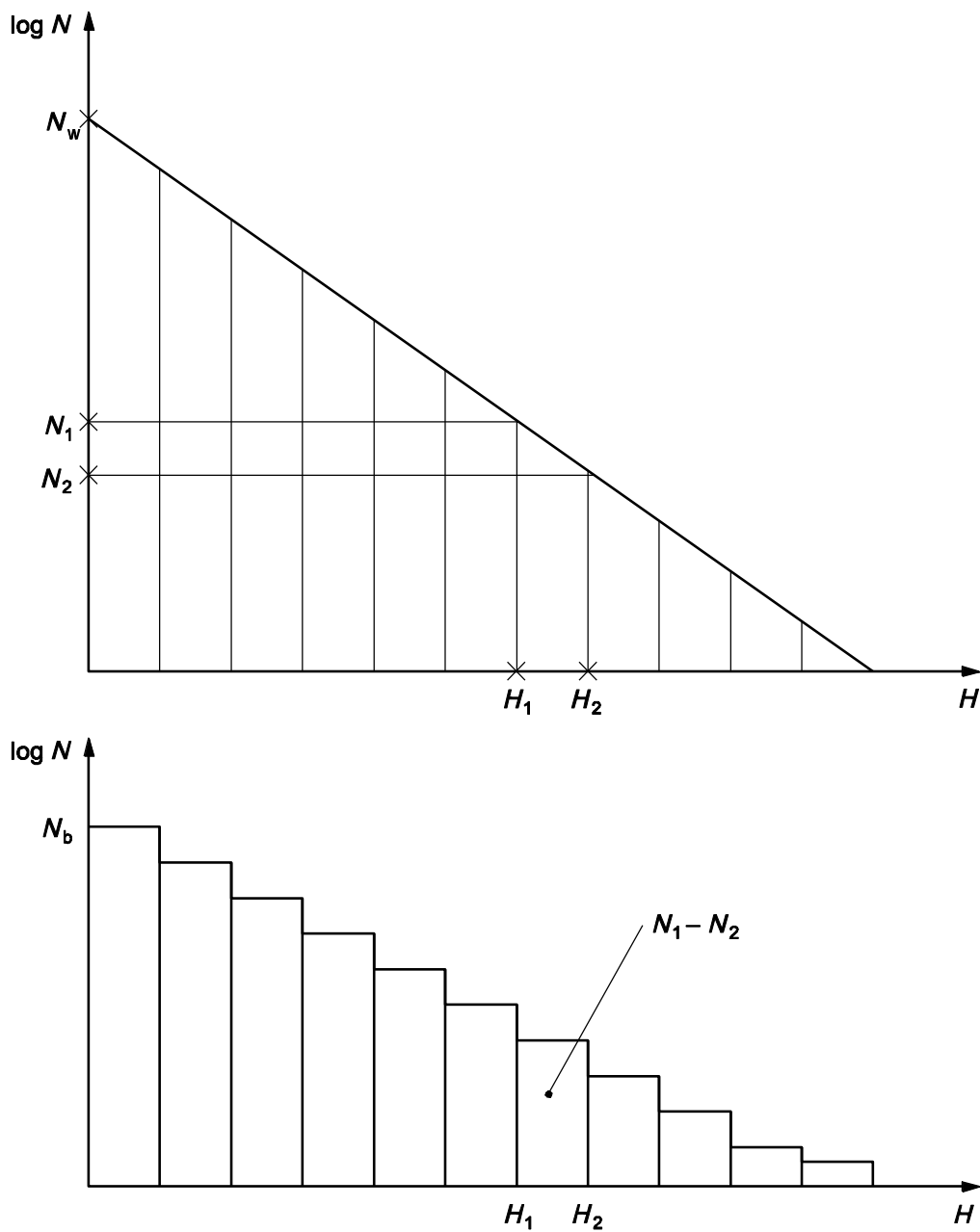
$$(N_{H^*})_{i,j} = \left(\frac{N_{i,j}}{N_T} \right) \times \left(\frac{(365)(24)(3600)}{(0.5)(T_{z,j} + T_{z,j+1})} \right) \times P(H > H^*) \quad (\text{A.16.3-3})$$

The total number of exceedances per year, N_{H^*} , of the individual wave height, H^* , is found by repeating this calculation for all sea states in the scatter diagram and summing the results:

$$N_{H^*} = \frac{(365)(24)(3600)}{N_T} \times \sum_i \left\langle \exp \left\{ \frac{-2(H^*)^2}{[(0.5)(H_{s,i} + H_{s,i+1})]^2} \right\} \sum_j \frac{N_{i,j}}{(0.5)(T_{z,j} + T_{z,j+1})} \right\rangle \quad (\text{A.16.3-4})$$

The long-term cumulative distribution of individual wave heights H is obtained by repeating the calculation of Equation (A.16.3-4) for all values of H^* . The total number of waves per year N_W is the sum of N_{H^*} for all values of H^* for which the calculations are made.

Figure A.16.3-3 shows a typical example of the long-term cumulative distribution of individual wave heights and its discretization into wave height blocks. It is usually close to an exponential distribution that plots approximately as a straight line when plotted on log-linear scales.

**Key**

- H individual wave height (on linear scale)
- N_w total number of individual waves (on logarithmic scale)
- N_b number of waves in a block
- H_1 wave height 1
- H_2 wave height 2
- N_1 number of waves exceeding H_1
- N_2 number of waves exceeding H_2

Figure A.16.3-3—Typical Long-term Cumulative Distribution of Individual Wave Heights

A.16.3.9 Current

In design practice for strength design, it is common to use an in-line design current in combination with a design wave. This is not appropriate for a fatigue analysis. Each combination of H_S and T_R in the wave scatter diagram represents a large number of occurrences of different sea states. All these occurrences have the same statistical parameters of H_S and T_R , but are in other respects likely to be (very) different. Currents usually vary widely over time in both speed and direction, and no specific current information can normally be associated with each individual sea state.

Furthermore, for a fatigue assessment it is not the absolute value of the stress that matters but the stress range. Therefore, it is also the range rather than the absolute magnitude of the applied loads/actions that should be considered.

The presence of a current in conjunction with waves affects the drag component of the hydrodynamic loads/actions on the structure from the Morison equation in two ways; i.e. it will produce a mean value of the load/action and it will modify the variable part of the wave and current load/action, compared to the variable part of the load/action in waves alone. The mean value is not relevant for fatigue. For an in-line current, the maximum load/action under the wave crest and the minimum load/action under the wave trough are both increased. Therefore, the range of the applied load/action (the maximum minus the minimum load/action) is much less affected than the peak load/action in an absolute sense. For current directions that are not in-line with the wave, the influence is even smaller.

Sensitivity analyses have also shown that the influence of currents on the range of the calculated stress variations and hence on the fatigue damage accumulation in the structure is normally rather small.

In view of the above, it is not only impractical, but normally fully justified, to neglect current in calculating the applied wave loads/actions for fatigue analyses. The remaining uncertainty is considered to be included in the additional fatigue damage design factor and/or the field experience factor (see 16.2.8 and 16.12).

A.16.3.10 Water Depth

In contrast with analyses for strength design in extreme environmental conditions, mean sea level should be used for a fatigue analysis, accounting for any settlement of the structure where appropriate. If, additionally, sea floor subsidence is expected to occur, the water level used in the analysis should reflect the average subsidence during the design service life; if the expected subsidence is substantial, several mean sea levels should be considered.

A.16.3.11 Marine Growth

Marine growth development is usually most pronounced during the first few years after installation, after which a more or less stable situation arises. The design marine growth thickness is the marine growth that is expected to accumulate on the structure without being removed. It therefore represents a situation that can exist during the entire design service life and which should be taken into account in a fatigue assessment.

For the fatigue assessment of an existing structure, the actual marine growth distribution on the structure can be used, provided this can reliably be established and shown to be stable.

A.16.4 Performing Global Stress Analyses

A.16.4.2 Loads/Actions Caused by Waves

To determine the stress range during the passage of a wave, a full periodic wave cycle should be stepped past the structure. The number of wave positions should be sufficient to accurately determine the maximum and the minimum stress during the passage of the wave. The definition of a full stresses cycle requires, in general, a substantial number of points. For 12 equally spaced points in a perfectly sinusoidal variation, the peak can still be missed by 15° ; as $\cos 15^\circ = 0.966$, the error in the true stress range can amount to 3.4 %. As fatigue damage is governed by the stress range to the power of the slope of the $S-N$ curve, this can easily introduce an error of some 10 %, i.e. $(1.034)^3 \approx 1.10$.

Each calculated point in the stress cycle corresponds with a wave position, thus requiring the calculation of distributed wave loads/actions over the full structure and a subsequent structural analysis for each wave position. Some computer programs therefore perform interpolation of the stress cycle between calculated points in a post-processing operation, by which the number of individual wave positions and associated structural analyses can be reduced. The interpolation procedure used for this purpose should at least be of second order.

If the marine growth thickness is such that certain assemblies of components (e.g. conductor guide frames, fenders, or boat landings) are completely blocked, then the associated wave load/action will be substantially increased. If this is likely to occur, its effect should be properly incorporated in modeling the hydrodynamic loads/actions on these components.

As the periodic waves used for fatigue analyses are low and short compared to design waves for a strength analysis, the associated Keulegan-Carpenter numbers, $K = U_m T/D$ (see A.9.5.2.3), are also much lower. They are largest at the free water surface. Assuming deep water waves, the maximum attainable value is $K = \pi H/D$, while the value decreases exponentially through the water column with depth below the surface. The corresponding value of the inertia coefficient is therefore larger than that recommended for the design wave analysis. For many combinations of waves and typical member sizes, the inertia coefficient will be approximately equal to 2.0. The theoretical value of the inertia coefficient for negligible wake effects around members is also 2.0. For these reasons, $C_m = 2.0$ is adopted for fatigue analyses.

A.16.4.3 Quasi-static Analyses

The dynamic amplification of quasi-static stress responses is governed by the magnitude and the frequency range of the excitation of the structure in relation to the platform natural frequencies. Wave periods that coincide with structural natural periods of 2.5 s and below correspond with wave lengths of approximately 10 m and less. While these can still be effective in producing local wave loads/actions on typical member sizes of space frame structures in accordance with Morison equation, global load/action on the whole structure rapidly diminishes with increasing frequency due to partial cancellation resulting from phase differences between the local loads/actions on spatially distributed members.

For monotowers, or configurations with highly correlated members, such partial cancellations do not occur. Therefore, the decrease in global wave load/action on such structures can only be due to wave diffraction effects for wave length to diameter ratios of less than approximately 5. For such configurations, the neglect of dynamic effects should be judged on the basis of the magnitude of the excitation at the platform natural frequencies.

A.16.4.4 Dynamic Analyses

A.16.4.4.1 General

For calculating global structural responses, modal analysis techniques can offer an efficient and reliable method in some circumstances. However, for local responses, pure modal techniques are usually unreliable. Unless a

very large number of modes are included, a modal solution will not accurately represent the local (quasi-static) member deformations, which are important for a fatigue analysis. Choosing a very large number of modes is unlikely to improve the situation with confidence due to numerical inaccuracies associated with the high frequency modes. To overcome both of these aspects, the so-called method of modal analysis plus static back substitution has been developed, which superimposes the dynamic responses due to a limited number of modes onto a full static solution. The method is also known by the name "mode acceleration method." This method offers the benefits of a combination of accuracy and computational efficiency. When using this procedure, it is still necessary to include a sufficient number of modes to accurately represent all significant dynamic responses.

However, a direct solution technique is generally preferred over a modal type of solution, provided this solution method is available in the structural analysis software used. When using a direct solution technique, a modal analysis should also be performed to obtain natural frequencies and mode shapes for the principal modes of the structure. This serves generally to verify that the dynamic behavior of the model is realistic, and provides the values of the natural frequencies that are used in the selection of the frequency grid as described in 16.7.2.2. Further particulars of the above alternative solution techniques are described in Reference [A.16.4-1].

If a natural frequency falls in a valley in the transfer function of the applied wave load/action it should be shifted to a more conservative location. A realistic range of natural frequency values should be considered, based on lower and upper bound values for the platform mass and stiffness as discussed in A.16.4.4.2 and A.16.4.4.3. The choice of which parameter to adjust for achieving such a shift is structure specific and depends upon deck mass, soil conditions, and structural configuration. It should be recognized that adjusting the foundation stiffness can unintentionally alter the member forces in the base of the structure, which can be fatigue sensitive.

A.16.4.4.2 Mass

For slender members, the added mass should be based on the displaced volume for motion transverse to the longitudinal axis, with an appropriate added mass coefficient to account for the member shape. Guidance on added mass coefficients can be found in the literature. For complex assemblies, the added mass should be determined for each motion direction and should account for the actual geometry. Account should further be taken of the increase in volume due to marine growth as well as possible blocking effects associated with marine growth.

The equipment and consumable masses in the model should include all items supported by the structure during any given operation on the platform. If the mass of such items is predicted to vary significantly for different operational phases during the structure life, it is appropriate to perform separate analyses for each phase and combine the associated fatigue damage.

The masses that are used in the model should be reconciled with the topsides and structure weights; centers of gravity should be established with the weight-estimating procedure used. When assessing the range of natural period values that can occur, upper and lower bound estimates of the masses should be considered. The lower bound mass should be determined with marine growth omitted.

A.16.4.4.3 Stiffness

The structural model of a space frame structure consists primarily of beam elements and the modeling detail should be adequate to provide accurate stresses at all relevant connections. The member intersections should be modeled such that the resulting nominal member end stresses are consistent with their subsequent use in the fatigue analysis. For typical space frame members, nominal brace stresses are normally computed at the intersections of the brace and chord centerlines. For large diameter chords or short braces, local joint geometry can require more refined modeling.

For stiffness modeling, when determining the wall thickness of members that include a corrosion allowance, it should normally be assumed that half of the allowance has been consumed, to represent the average condition during the life of the member.

The stiffness of appurtenances (conductors, caissons, risers, J-tubes, boat landings, fenders, launch rails, skirt piles, pile-to-sleeve connections, mudmat framing, etc.) should be included if their connection details contribute to the overall stiffness and transfer of forces in the structure. The transfer of forces at support points should be realistically modeled. Detailed stiffness modeling of appurtenances is also required if a fatigue analysis of the appurtenances themselves is to be performed.

Complex assemblies such as pile-to-sleeve connections, mudmats, conductor guides, etc., can require finite element types other than beam elements (e.g. shell, plate, or solid elements). Detailed FEAs of such assemblies may be performed separately from the global analysis. These may be used to determine equivalent stiffness properties for simplified beam element representations for inclusion in the global stiffness model and can also assist in fatigue checking of the assembly details. Alternatively, some computer programs allow the use of super elements within the global stiffness model for representing such complex assemblies.

A.16.4.4.4 Damping

A relative velocity formulation of the Morison equation inseparably links wave excitation and hydrodynamic drag damping. It reduces applied wave loads/actions and causes implicit hydrodynamic damping associated with drag loads/actions, which is additional to any damping from a viscous damping coefficient introduced in accordance with 16.4.4.4. For small structural displacements, as are typical for fatigue excitation, this additional damping is not observed in measurements and is inappropriate. Consequently, the relative velocity formulation should not be used for a fatigue analysis.

A.16.5 Characterization of the Stress Range Data Governing Fatigue

In this document, the terms “geometrical stress” (GS) and “geometrical stress range” (GSR) are used, replacing the terms hot spot stress and hot spot stress range, which were previously commonly used. These new terms have been introduced to emphasize that stress concentrations occur to a varying degree all along the welds and not only at specific locations.

The nominal axial stress at the member end is denoted by $\sigma_{ax}(t)$; the nominal IPB stress is denoted by $\sigma_{ipb}(t)$ and is assumed to be due to the moment, M_y , about the local member y-axis; the nominal OPB stress is denoted by $\sigma_{opb}(t)$ and is assumed to be due to the moment about the local member z-axis. The geometrical stress, $\sigma_{GS,A}(t)$, at a specific location, A, around the tubular connection is then determined using Equation (A.16.5-1) (see Figure A.16.5-1):

$$\sigma_{GS,A}(t) = C_{ax} \sigma_{ax}(t) \pm C_{ipb} \sigma_{ipb}(t) \sin \theta \pm C_{opb} \sigma_{opb}(t) \cos \theta \quad (\text{A.16.5-1})$$

where the signs in the formula are determined by the sign conventions employed in the analysis. The range of $\sigma_{GS,A}(t)$ is the GSR (S). Due to phase differences between the various stress components, the GSR is normally not equal to the algebraic sum of the stress components.

For location A, at least the four main points around the circumference for $\theta = 0^\circ, 90^\circ, 180^\circ$, and 270° (the four main clock positions) should be considered. Further details of GSRs for tubular and other joints are discussed in A.16.10.

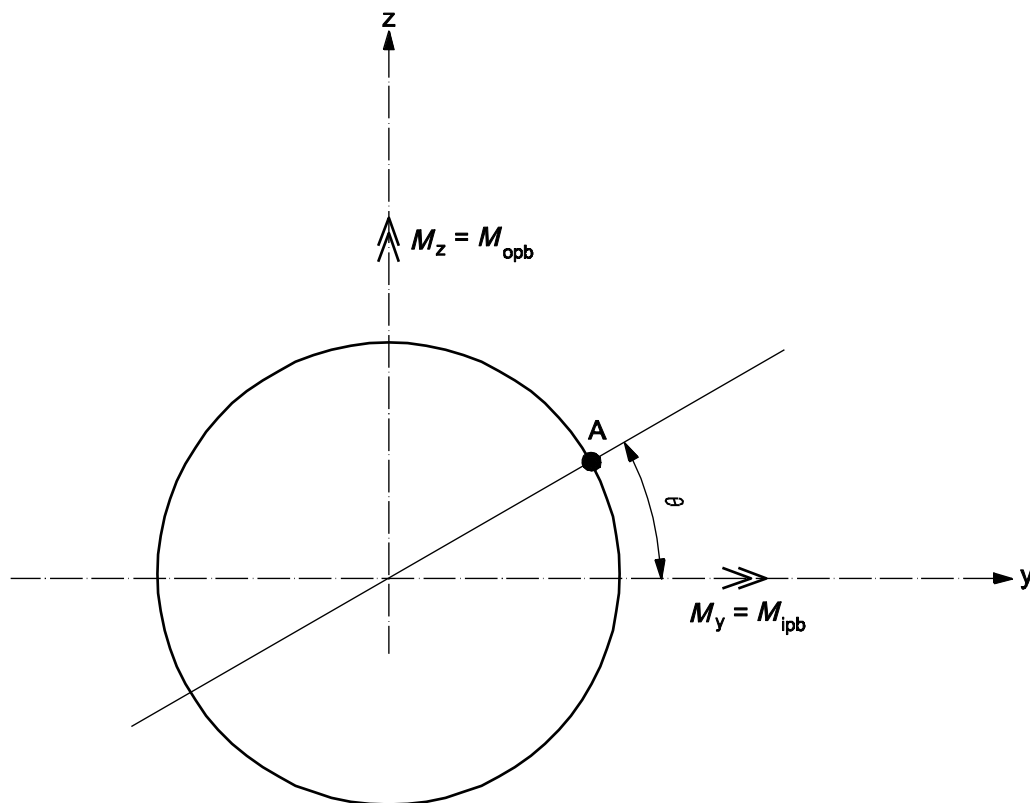


Figure A.16.5-1—Determination of Geometrical Stress at Point A around Tubular Connection

A.16.7 Determining Long-term Stress Range Distribution by Spectral Analysis

A.16.7.1 General

Spectral analysis is an established technique for the prediction of the responses of a linear system to random excitation (see, for example, References [A.16.7-1] and [A.16.7-2]). Its application to fatigue assessments in the offshore environment is described in, for example, References [A.16.7-3] and [A.16.7-4]. The method has been applied to a wide range of offshore structures of various types in various environments.

A.16.7.2 Stress Transfer Functions

A.16.7.2.1 General

A flowchart for the spectral analysis procedure including the calculation of the stress transfer function is shown in Figure A.16.7-1.

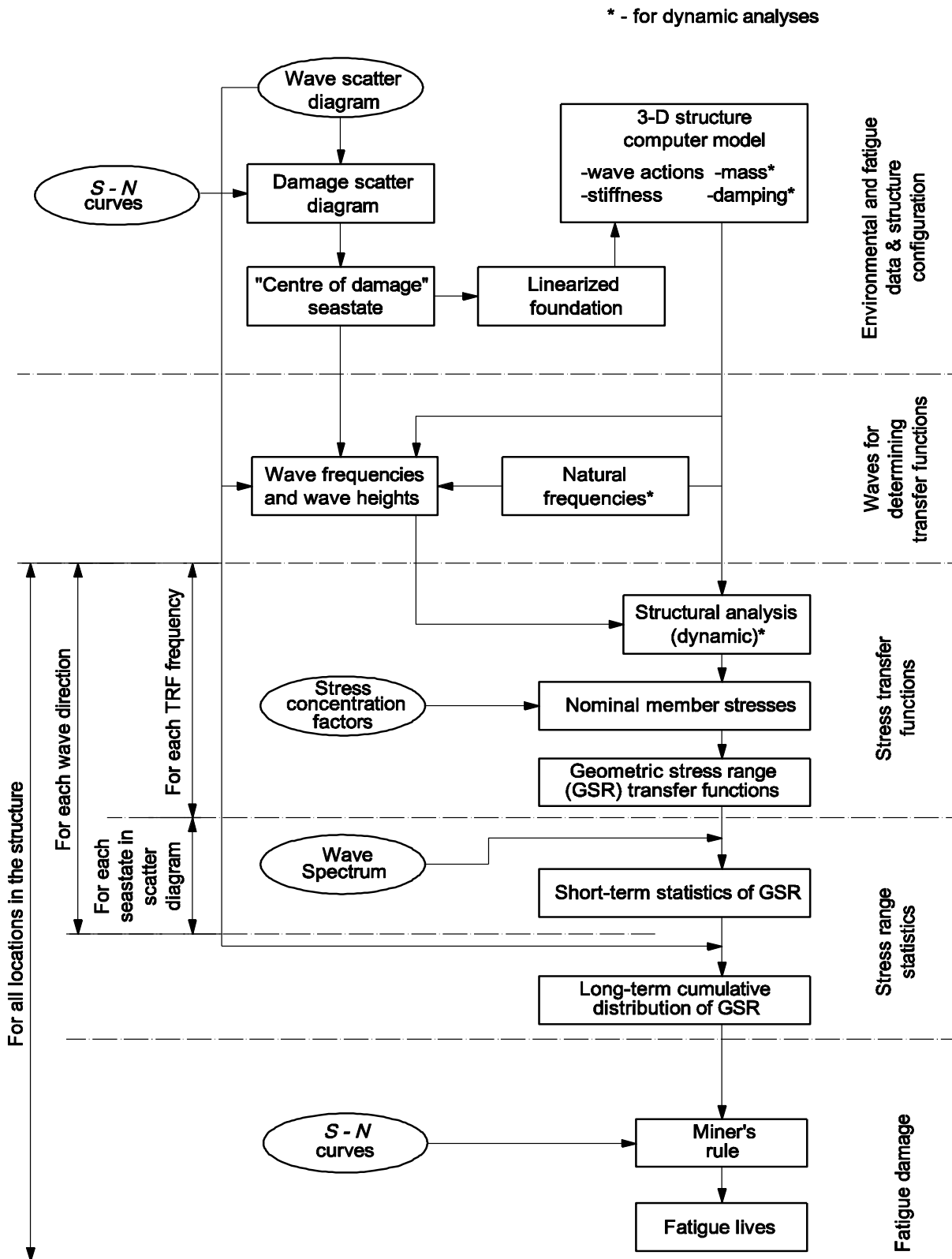


Figure A.16.7-1—Flowchart of Spectral Analysis Procedure

For a spectral analysis, the GSR transfer functions are required for all locations in the structure that should be checked for fatigue. These functions define the GSR per unit wave height over a range of wave frequencies for each wave approach direction. A GSR transfer function is determined as follows.

For a given wave approach direction and a given wave frequency, a global structural analysis is performed using a periodic (regular) wave. If the analysis is performed in the frequency domain, a linear wave theory should be selected. The wave height is not relevant and usually a unit wave amplitude (or wave height) is chosen. If the analysis is performed in the time domain, a nonlinear wave theory may be chosen; in this case, the wave height is selected in accordance with 16.7.2.3. Further guidance on the global structural analysis is given in 16.4.

At each location in the structure and for each relevant stress component, the nominal stress and the associated phase with respect to the wave is determined. Using these stress components, the geometrical stress $\sigma_{GS,A}(t)$ at point A is determined as described in 16.5. The range of $\sigma_{GS,A}(t)$ is the GSR at point A. Dividing the GSR by the wave height, one point of the GSR transfer function for point A is defined; from this moment onwards, the phase angle information can be dropped in the further processing. This procedure is repeated for all points A around the connection and all locations in the structure.

The above calculations are next repeated for all wave frequencies in order to fully define the GSR transfer functions for one particular wave approach direction; subsequent to that, they are repeated for all wave approach directions.

A.16.7.2.2 Selection of Wave Frequencies

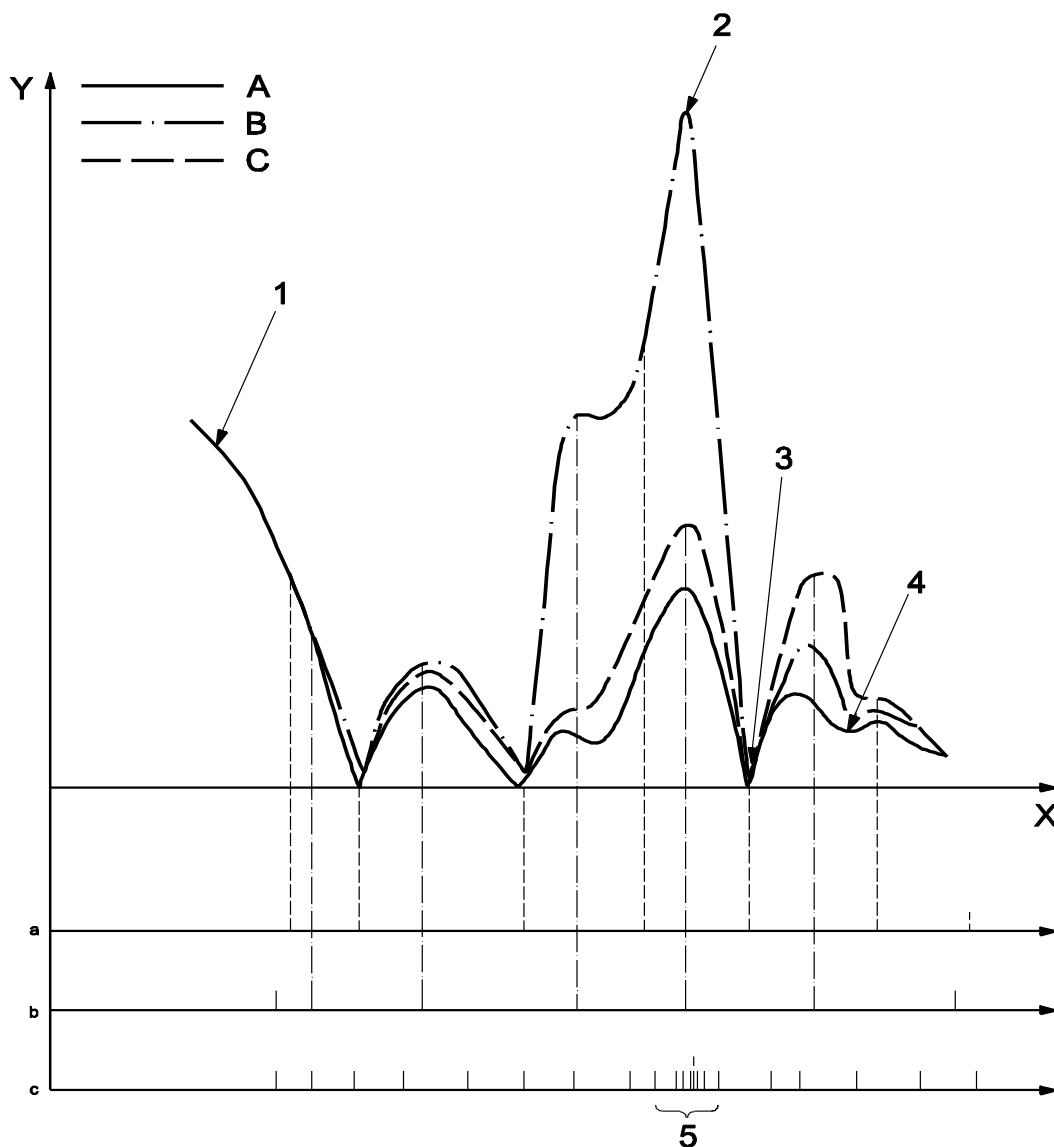
Figure A.16.7-2 shows an example of the shapes of a transfer function of static and dynamic loads/actions on the structure due to waves and depicts several typical features as a function of frequency.

The following guidance is offered for the selection of wave frequencies in order to take the characteristics of the environment as well as the characteristics of the structure duly into account.

- a) Lowest and highest frequency: The lowest and highest frequency used should be based on the range of frequencies over which there is significant energy in the sea states in the wave scatter diagram. For dynamic analyses, special attention should be paid to the high frequency cut-off.
- b) Cancellation and addition frequencies: Wave frequencies in the vicinity of peaks and troughs in the transfer functions of applied wave load/action (curve A in Figure A.16.7-2) should be included to ensure that the structural response at these frequencies is properly defined. For simple structures, the frequencies at which addition and cancellation of wave loads/actions on structural components occurs can be reasonably predicted, as they correspond with wave lengths that are specific multiples or fractions of key dimensions of the structure, such as the main leg spacing. For complicated structures, these features are often not so obvious from the structural drawings, and it can be necessary to determine the corresponding frequencies by inspection of the calculated wave loads/actions and/or the wave-induced responses.
- c) Intermediate frequencies: A sufficient number of intermediate frequencies should be included, in order to ensure a complete definition of the shape of the transfer function. To reduce the number of stress analyses required, some computer programs perform interpolation to derive transfer function ordinates at intermediate frequencies. However, care should be exercised with such interpolation routines due to the often complex shape of the transfer functions. Interpolation is more reliable when this is performed on the real and imaginary parts (the in-phase and out-of-phase components) of the transfer function separately, because these are normally smooth functions.

Selection of intermediate frequencies corresponding with peaks of the wave spectra in the scatter diagram is further necessary in order to provide accurate definition of wave input energy.

- d) Natural frequencies: For dynamic analyses, the natural frequencies of the structure should also be included (curves B and C in Figure A.16.7-2). In addition, a number of closely spaced frequencies on both sides of a natural frequency should be included so as to accurately define the sharp dynamic response peak of a lightly damped system over a frequency range of $(1 \pm 4\beta) \omega_n$ around resonance, where ω_n is the natural frequency in radians per second and β is the fraction of critical damping. The spacing of these frequencies should preferably be $0.5 \beta \omega_n$, and definitely not greater than $\beta \omega_n$.

**Key**

- | | | | |
|---|---|---|--|
| A | quasi-static transfer function | X | wave frequency, ω |
| B | dynamic transfer function—resonance at “peak” | Y | normalized applied wave load/action at wave frequency, E_w |
| C | dynamic transfer function—resonance at “valley” | | |
| 1 | low frequency, long waves, no cancellation effects | | |
| 2 | structure natural frequency at peak in static transfer function | | |
| 3 | structure natural frequency at valley in static transfer function | | |
| 4 | peaks and valleys due to interaction between wave length and structure geometry | | |
| 5 | fine grid around natural frequency | | |

Frequency Selection

- a Poor: underpredicts response.
- b Poor: overpredicts response.
- c Good: adequately represents transfer function.

Figure A.16.7-2—Typical Transfer Functions of Total Applied Loads/Actions due to Waves

A.16.7.2.3 Selection of Wave Heights

A.16.7.2.3.1 Determination of Wave Height

The linearization for an analysis in the time domain is performed by the rational selection of wave heights used in the regular wave analyses for the determination of the stress transfer functions. Alternative approaches can be used for the selection; one well- tested method is described below.

Calibration of the wave steepness should be based on the matching of global response parameters, which are representative of the predominant wave load/action to which the structure is subjected under fatigue conditions. For typical space frame structures, the total applied wave load/action (quasi-static base shear) is normally used for this purpose. The matching should be performed for the sea state at the center of the fatigue damage scatter diagram; the derivation of this sea state is described in A.16.7.2.3.2.

The calibration determines a wave steepness value, which matches the spectrally calculated range of applied wave load/action to the deterministically calculated range of applied wave load/action for the structure under consideration and the sea state at the center of the fatigue damage scatter diagram. The calibration should be performed for various wave directions. Typically, a broadside, an end-on and a diagonal wave direction are considered as a minimum. The calibration process consists of the following five steps.

- a) Determine the transfer function of the total applied wave load/action (quasi-static base shear) for each wave direction and for a range of wave steepness values. Note, however, that use of a constant wave steepness gives unrealistically large wave heights at low wave frequencies (large wave lengths). Therefore, the wave heights at the low frequency side should be capped. A wave height equal to the wave height with a 1-year return period should normally be used as a maximum.
- b) Calculate the RMS value of the applied wave load/action for each wave direction and the sea state at the center of the fatigue damage scatter diagram (see A.16.7.2.3.2), using standard spectral analysis techniques. This provides RMS values of the applied wave load/action for each wave direction and the initially chosen wave steepness values.

The linearized spectral response is a Gaussian process and, assuming a standard duration of some 3 hr with approximately 1000 response peaks, the most probable maximum (MPM) values of the applied loads/actions due to waves are 3.7 times the RMS values. In this manner, the MPM values of the applied wave load/action from the spectral calculations are determined for each wave direction and wave steepness. The applied ranges of wave load/action that should be used in the calibration are twice the MPM values of the applied wave load/action.

- c) Next, a deterministically calculated range of applied wave load/action is also determined by stepping a deterministic wave through the structure for each wave direction. This deterministic wave should have a height that is equal to the MPM wave height in the sea state at the center of the fatigue damage scatter diagram, and a period that corresponds with the peak period of the wave spectrum for this sea state. The deterministic range of applied wave load/action is the maximum minus the minimum total applied wave load/action during a full wave cycle.
- d) For each wave direction, the calibrated value of the wave steepness is the steepness that matches the spectrally calculated MPM value of the range of applied wave load/action from b) to the deterministically calculated range of applied wave load/action from c).
- e) From the set of calibrated wave steepness values for all wave approach directions obtained from d) a single representative wave steepness value should next be chosen.

The wave height for each wave frequency is now determined using the calibrated wave steepness value. The wave heights at the low frequency side should again be limited to the maximum wave height used in the calibration process. The resulting wave heights should further be compared to the elevations of plan framing levels near the water surface, to assess if any unreasonable discontinuities in the applied wave load/action are likely to occur. Where appropriate, some final adjustment should be made.

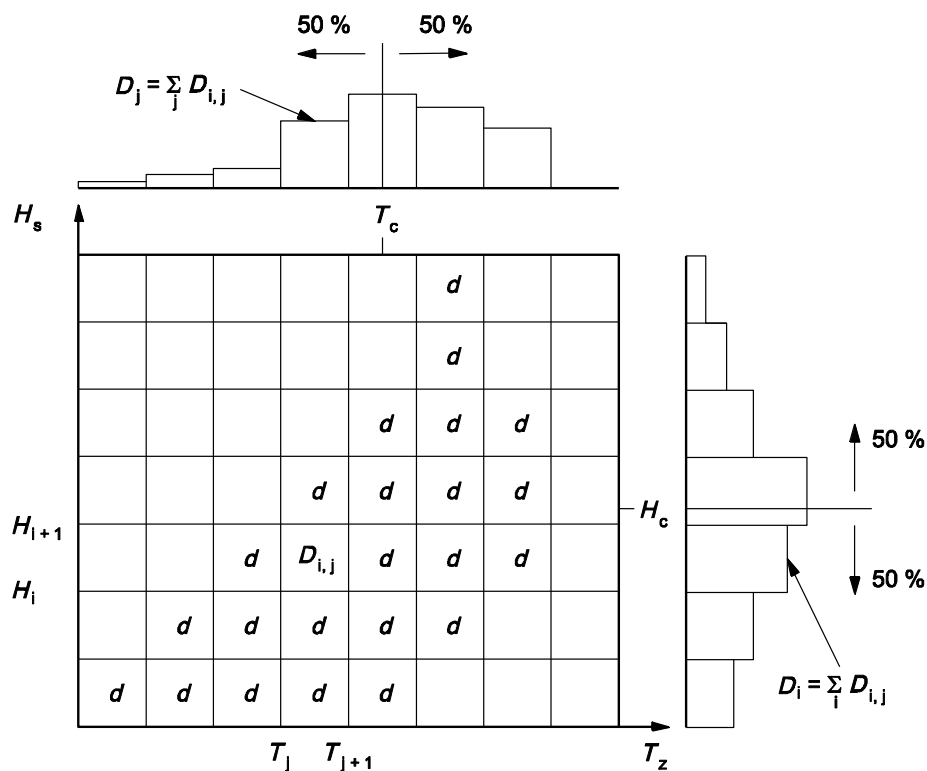
A.16.7.2.3.2 Derivation of the Sea State at the Center of the Fatigue Damage Scatter Diagram

Stress is assumed to be a linear function of wave height, whereas the number of stress cycles can initially be assumed to be equal to the number of wave cycles in a sea state. Fatigue damage can further be assumed to be proportional to the stress range raised to the power of the S – N curve slope.

The partial damage, $D_{i,j}$, caused by a particular sea state is hence proportional to the number of occurrences of the sea state, $N_{i,j}$, and the significant wave height, H_s , raised to the power (m) of the slope of the S – N curve. Proportionality to the number of stress cycles in the sea state translates into an inversely proportional relationship to the mean zero crossing period, T_z . Consequently:

$$D_{i,j} \propto \frac{N_{i,j} [0.5(H_i + H_{i+1})]^m}{0.5(T_j + T_{j+1})} \quad (\text{A.16.7-1})$$

The above calculation is repeated for each sea state in the wave scatter diagram to produce a damage scatter diagram with relative damages in the sea state bins as shown in Figure A.16.7-3.



Key

- T_z mean zero-crossing period
- H_s significant wave height
- T_c central value of the mean zero crossing period
- H_c central value of the significant wave height
- $D_{i,j}$ fatigue damage from sea states with $H_i < H_s < H_{i+1}$ and $T_j < T_z < T_{j+1}$
- D_i fatigue damage from sea states with $H_i < H_s < H_{i+1}$
- D_j fatigue damage from sea states with $T_j < T_z < T_{j+1}$
- d fatigue damage from sea states falling within indicated range of H_s and T_z

Figure A.16.7-3—Typical Fatigue Damage Scatter Diagram

The sea state at the center of the damage scatter diagram is defined as having:

- a) a significant wave height, H_C , such that 50 % of the damage occurs in sea states with a lower H_S and 50 % with a higher H_S , and
- b) a mean zero crossing period, T_C , such that 50 % of the damage occurs in sea states having a lower T_Z and 50 % with a higher T_Z .

The MPM wave height, H_{ref} , and the associated wave period, T_{ref} , for this center of damage sea state are calculated as follows:

$$\begin{aligned} H_{ref} &= (1.86)H_C \\ T_{ref} &= T_C / (0.81) \end{aligned} \quad (A.16.7-2)$$

For linear $S-N$ curves on a log-log plot, these calculations can be performed analytically and independent of the magnitude of the stress response. However, for bilinear $S-N$ curves, the calculations can be done only numerically using actual stress responses. In lieu of this more rigorous procedure, an approximation using the slope of the first branch of the $S-N$ curve (usually with $m = 3$) can be used, because most damage usually accumulates from this part.

A.16.7.3 Short-term Stress Range Statistics

The GSR transfer function for a specific location in the structure and one wave direction is combined with the wave spectrum describing a sea state in the wave scatter diagram to calculate the GSR response spectrum for the sea state. The RMS value and mean period of the GSR response are then computed by integrating the response spectrum and determining its moments. These parameters define the probability distribution function for GSR values and establish the short-term GSR statistics for the sea state. The Rayleigh distribution is an appropriate distribution function for narrow-banded processes and is normally applied. The Rice probability distribution function is valid for Gaussian random processes of any width and may be used as an alternative. The Rayleigh distribution returns stress peaks (or stress ranges) that are all positive. However, the Rice distribution includes a small fraction of the total number of stress peaks (or stress ranges) that are negative; for fatigue analysis purposes, these negative stress ranges are removed from the distribution and ignored. See Figure A.16.7-1 for an outline of the overall procedure.

A.16.8 Determining Long-term Stress Range Distribution by Deterministic Analysis

A.16.8.1 General

A flowchart for the procedure is shown in Figure A.16.8-1.

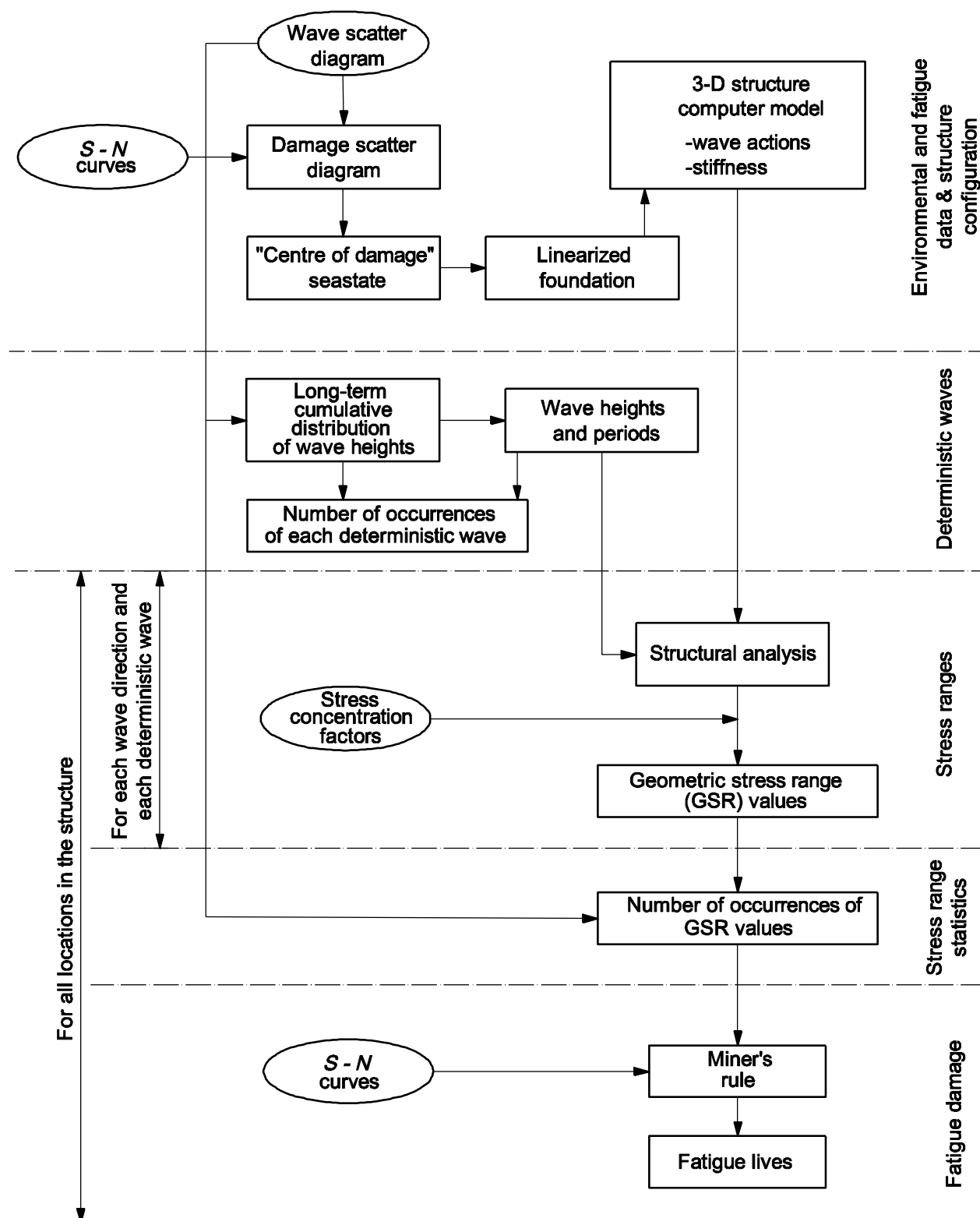


Figure A.16.8-1—Flowchart of Deterministic Analysis Procedure

As for a spectral analysis, for a deterministic analysis the GSR values are required for all locations in the structure that should be checked for fatigue. These are now calculated using a series of periodic (regular) waves.

For a given wave approach direction, a global structural analysis is performed using a periodic (regular) wave of suitable height and period. The analysis is performed in the time domain stepping the wave through the structure over a full cycle.

At each location in the structure and for each relevant stress component, the nominal stress and the associated phase with respect to the wave is determined. Using these stress components, the geometrical stress $\sigma_{GS,A}$ at point A is determined as described in 16.5. The range of $\sigma_{GS,A}$ is the GSR at point A. This procedure is repeated for all points A and all locations in the structure.

The above calculations are next repeated for all wave periods for one particular wave approach direction, and subsequent to that for all wave approach directions.

The selection of the height and period of the regular waves is specific to the deterministic analysis and is described in 16.8.2 and 16.8.3.

A.16.8.2 Wave Height Selection

The long-term distribution of individual wave heights is calculated from the wave scatter diagram as described in 16.3.7. The number of waves occurring in each discrete wave height block is determined as shown in Figure A.16.3-3. The upper wave height of each block is normally used as the wave height of each regular wave for the global structural analysis.

A.16.8.3 Wave Period Selection

As the expected joint distribution of individual wave heights and periods for the offshore location under consideration is not normally available as routine information, the wave period selection should be established in consultation with an oceanographer.

The selected periods should also be verified against cancellation and addition frequencies in the shape of the transfer function of the total applied wave load/action (see 16.7.2.2). If any wave periods fall close to a relative minimum, the wave period should be adjusted to ensure a proper representation of the total wave energy. The lowest and highest, as well as intermediate, wave frequencies are less influential in the deterministic than in the spectral method, but should nonetheless be considered carefully. Considerations with respect to the natural frequencies of the structure are not relevant, as the deterministic method should not be used for dynamically responding structures.

A.16.9 Determining Long-term Stress Range Distribution by Approximate Methods

A.16.9.1 The Weibull Distribution of Long-term Stress Ranges

The general three-parameter Weibull distribution of stress ranges is given by Equation (A.16.9-1):

$$P(S_i > S_i^*) = \exp \left[- \left(\frac{S_i^* - A}{B} \right)^C \right] \quad (\text{A.16.9-1})$$

where

- S_i is the local GSR;
- $P(S_i > S_i^*)$ is the probability that the stress range S_i exceeds the particular value S_i^* ;
- A is the location parameter of the distribution ($S_i^* - A \geq 0$);
- B is the scale parameter ($B > 0$);
- C is the shape parameter ($C > 0$).

A Weibull distribution with the parameters $(A, B, 1)$ is an exponential distribution, whereas a Weibull distribution with the parameters $(0, 1, 2)$ is the normalized Rayleigh distribution.

For GSRs (S_i) , the location parameter is zero $(A = 0)$, whereas the probability $P(S_i > S_i^*)$ can further be expressed by Equation (A.16.9-2):

$$P(S_i > S_i^*) = \frac{N_{Si^*}}{N_T} \quad (\text{A.16.9-2})$$

where

N_{Si^*} is the number of times that S_i^* is exceeded during a period, T ;

N_T is the total number of stress ranges occurring during period, T ;

T is the duration considered, which for the in-place situation should be at least a year.

Substituting $A = 0$ and Equation (A.16.9-2) into Equation (A.16.9-1) and taking the natural logarithm results in:

$$\ln N_{Si^*} - \ln N_T = -\left(\frac{S_i^*}{B}\right)^C = -(B)^{-C} (S_i^*)^C \quad (\text{A.16.9-3})$$

All stress ranges exceed $S_i^* = 0$, so that $N_{Si^*=0} = N_T$, while there is only one maximum stress range, $S_{i,\max}^*$, giving $N_{Si^*,\max} = 1$. The scale parameter B can be determined by introducing $N_{Si^*,\max} = 1$ into Equation (A.16.9-3):

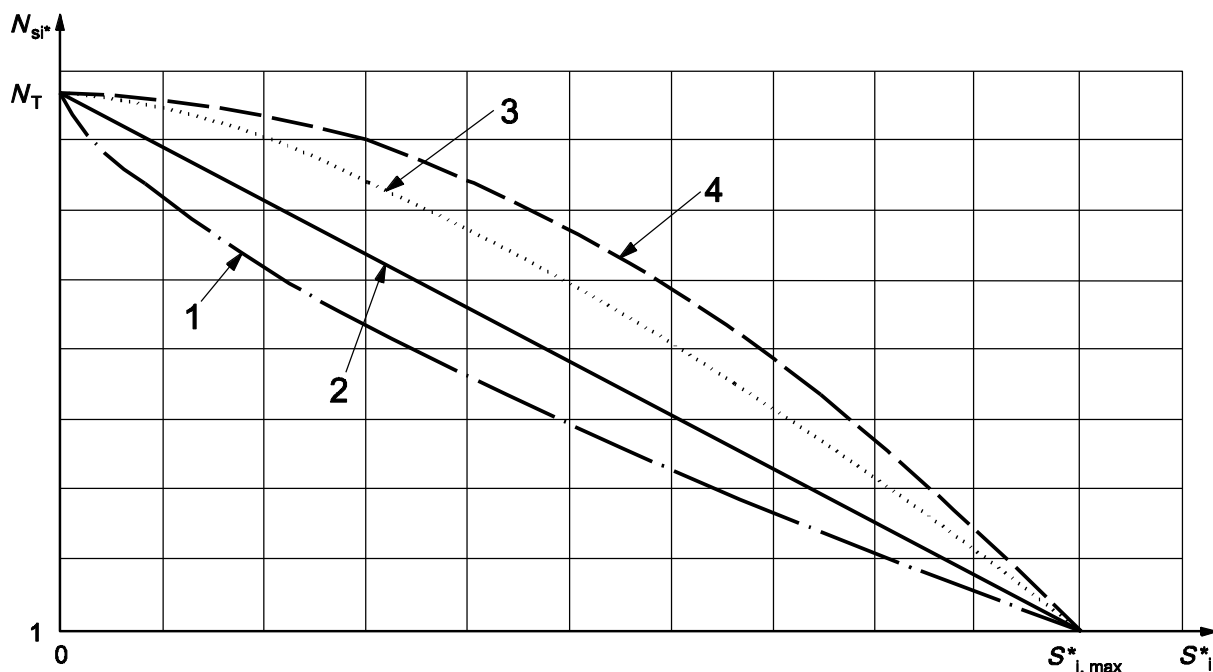
$$(B)^{-C} = \frac{\ln N_T}{(S_{i,\max}^*)^C} \quad (\text{A.16.9-4})$$

Substituting this into Equation (A.16.9-3) provides the final equation of the Weibull distribution of local GSR as:

$$\ln N_{Si^*} = \ln N_T - \frac{\ln N_T}{(S_{i,\max}^*)^C} (S_i^*)^C \quad (\text{A.16.9-5})$$

If desired, the natural logarithm can be replaced by the logarithm on base 10 without any modification.

For a shape parameter, $C = 1$, this distribution plots as a straight line on a semi-logarithmic graph, with $\ln N_{Si^*}$ (or $\log_{10} N_{Si^*}$) on a logarithmic scale and S_i^* on a linear scale. For shape parameters, $C < 1$, the line is concave, whereas for shape parameters, $C > 1$, the line is convex. All lines pass through the two points $(0, \ln N_T)$ and $(S_{i,\max}^*, 1)$ (see Figure A.16.9-1).



Key

1 Weibull distribution for $C = 0.7$

2 Weibull distribution for $C = 1.0$

3 Weibull distribution for $C = 1.5$

4 Weibull distribution for $C = 2.0$

S_i^* local GSR (on linear scale)

$S_{i,max}^*$ maximum value of local GSR

N_{Si^*} number of times that S_i^* is exceeded during period T (on logarithmic scale)

N_T total number of stress ranges occurring during period T

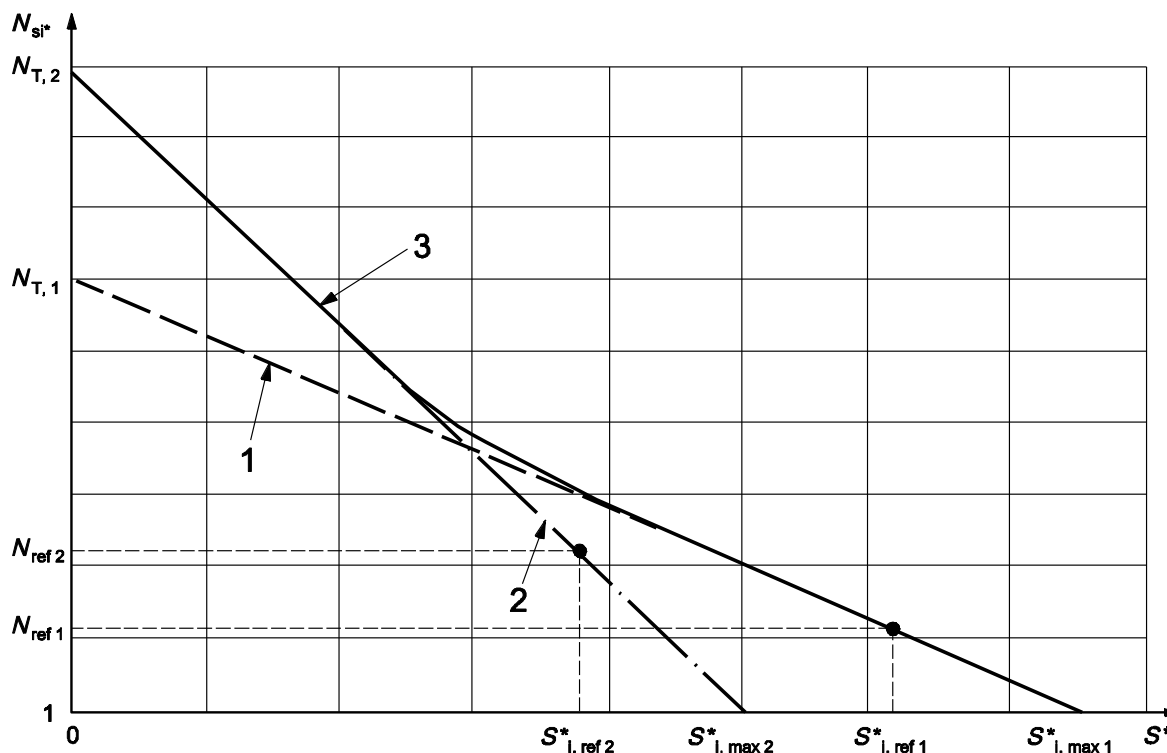
Figure A.16.9-1—Weibull Distributions

A.16.9.2 Determination of the Distribution Parameters

A.16.9.2.1 General

The Weibull distribution parameters of the transformed Equation (A.16.9-5) are N_T , $S_{i,max}^*$, and C . In general, these parameters can only be determined empirically by fitting Weibull distributions to measured or spectrally calculated long-term stress range distributions. For shape parameters, $C \neq 1$, this is the only possibility; there is no sound theoretical basis for estimating them otherwise.

However, experience has shown that in many cases a practical and reasonably accurate approximation is possible with a shape parameter, $C = 1$. For these cases, estimates of two points fully determine the resulting straight line graph on semi-logarithmic paper. One simple and convenient point that is usually chosen is $(0, \ln N_T)$, whereas the other point may be generally depicted as $(S_{i,ref}^*, N_{ref})$ (see Figure A.16.9-2). If the long-term distribution of local GSRs is approximated by combining two Weibull distributions, the parameters of each distribution should be estimated separately (see Figure A.16.9-2).



Key

- 1 Weibull distribution for distribution 1
- 2 Weibull distribution for distribution 2
- 3 combined Weibull distributions
- S_i^* local GSR (on linear scale)
- $S_{i, \max}^*$ maximum value of local GSR
- N_{Si^*} number of times that S_i^* is exceeded during period T (on logarithmic scale)
- N_T is the total number of stress ranges occurring during period T

Figure A.16.9-2—Combination of Weibull Distributions

A.16.9.2.2 Determination of the Total Number of Stress Range Cycles

The total number of stress ranges, N_T , occurring during the period, T , may be derived from the total number of wave cycles occurring during T .

Where one exponential distribution adequately describes the long-term stress range distribution, T is normally chosen to be 1 year. The total number of waves per year, N_W , can be determined from the wave scatter diagram as discussed in A.16.3.7. N_T may next be related to N_W by a multiplication factor, R :

$$N_T = R \times N_W \quad (\text{A.16.9-6})$$

Multiplication factor R reflects that shorter waves (high frequency waves, which form a large percentage of N_W) act over smaller depths below the surface than longer waves (low frequency waves, which form a relatively low percentage of N_W). In lieu of more pertinent information, the following values of R may be used as a guide:

- $R = 1.5$ at and above 15 m below the still water surface for the fatigue analysis;
- $R = 0.5$ at 30 m below the still water surface;
- $R = 0.2$ at 60 m below the still water surface.

Between 30 m and 60 m below the still water surface, R may be determined by linear interpolation.

Where two exponential distributions should be combined to describe the long-term stress range distribution, T should normally be chosen to be the design service life (usually some 20 years). The first distribution then describes the stress ranges during the design service life that are due to wave conditions arising from normal atmospheric conditions, whereas the second distribution describes those that are due to the extraordinary wave conditions arising from cyclones. Reference [A.16.9-1] contains a typical example of such applications from the Gulf of Mexico, with $N_{T,1} = 10^9$ and $N_{T,2} = 10^6$.

A.16.9.2.3 Determination of Point $S_{i,ref}^*$, N_{ref}

The local GSR values at all locations of interest are estimated using a single periodic (regular) wave in the same way as for the deterministic analysis procedure (see 16.8). A reference wave with height, H_{ref} , and period, T_{ref} , needs to be selected and is used as a fatigue design wave in the procedure. For cases governed by one exponential distribution, two options readily present themselves:

- a) the annual wave height occurring once a year with an associated wave period;
- b) the MPM wave at the center of damage sea state [see A.16.7.2.3 and Equation (A.16.7-2)].

Other options reflecting and calibrated to local experience can be taken.

A stress analysis (see 16.4) is performed by stepping the reference wave through the structure, after which the GSR is determined (see 16.5) at all locations of interest in the structure. To ensure that no undue effects of partial cancellation occur due to phase differences between local wave loads/actions on spatially distributed members, the wave length of the reference wave should be checked against the principal horizontal dimensions of the structure. The wave length should normally be more than four times the maximum dimension between the outer legs of the structure. The stress analysis is repeated for all wave directions relative to the structure. At each location of interest, the largest GSR calculated for any of the wave directions considered is $S_{i,ref}^*$. The corresponding N_{ref} is determined as T/T_{ref} .

Where two exponential distributions are involved, these steps should be executed for each of the two distributions separately.

The procedure for the Gulf of Mexico covered in Reference [A.16.9-1] is not based on estimating a second point of the distribution. Instead, it defines long-term wave height distributions, one for regular wave conditions and one for hurricane conditions, then assumes a relationship between stress ranges and wave heights (regardless of wave period) and develops a closed-form formulation for fatigue damage.

A.16.9.2.4 Fatigue Assessment

From the long-term distribution of local GSR values, the number of occurrences in discrete GSR blocks are determined as in Figure 16.7-1. These numbers are subsequently used in the fatigue damage calculation described in 16.12, as for the spectral and deterministic analysis procedures.

A.16.10 Geometrical Stress Ranges

A.16.10.2 Stress Concentration Factors for Tubular Joints

A.16.10.2.1 General Requirements for the Determination of the Stress Concentration Factor

A.16.10.2.1.1 Definition of Stress Concentration Factor

The GSR concept has evolved as the most practical basis for fatigue design of tubular joints. This concept places many different structural geometries on a common basis, enabling them to be treated using a single $S-N$ curve. The basis of this concept is to capture a stress (or strain) in the proximity of the weld toes, which characterizes the fatigue life of the joint, but excludes the very local microscopic effects such as the sharp notch, undercut and crack-like defects at the weld toe. These local weld notch effects are included in the $S-N$ curve.

In this document, the terms “geometrical stress” (GS) and “geometrical stress range” (GSR) replace the previously used terms “hot spot stress” and “hot spot stress range,” except in the description of past (existing) practice. The SCF can be defined as follows:

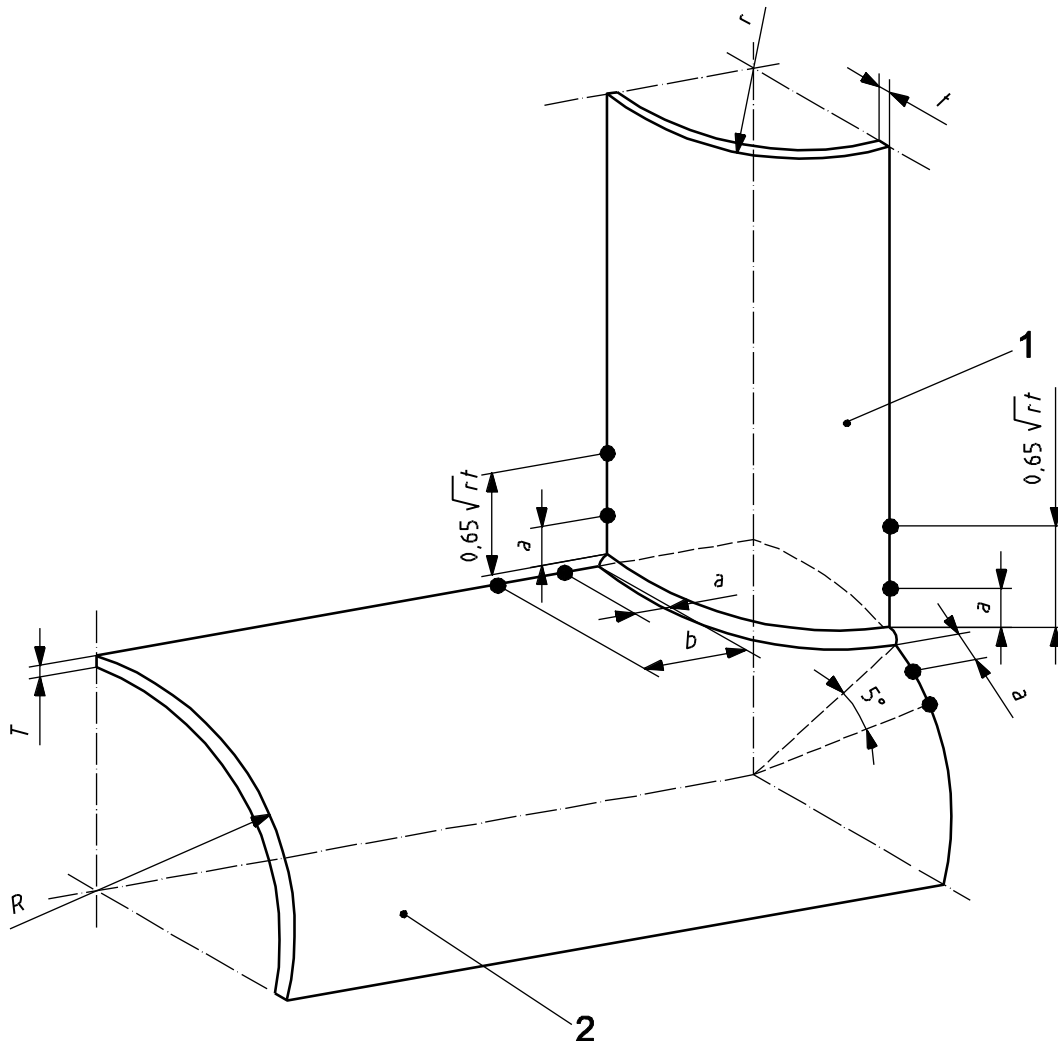
$$\text{SCF} = \frac{\text{the range of the GS at a particular location of the intersection weld (excluding notch effect)}}{\text{the range of the nominal brace stress}}$$

Consistency with the $S-N$ curve is established by using the same method for estimating the GSR during the fatigue test as used when obtaining SCFs.

In US practice that is codified in API 2A-LRFD [A.16.10-1] and AWS D1.1/D1.1M [A.16.10-2], the geometrical stress or strain is defined as the total range that would be measured by a strain gauge placed adjacent to the toe of the weld and oriented perpendicular to the weld so as to reflect the stress which will be amplified by the weld toe discontinuities [A.16.10-3]. Typical geometrical strain gauges are centered within 6 mm to $0.1\sqrt{rt}$ from the weld toes with a gauge length of 3 mm. Here r and t refer to the outside radius and thickness of the member instrumented, whether chord or brace.

Within Europe, the geometrical stress concept has been retained but applied differently. The local weld notch effect is excluded by using stress values just outside the weld notch region and extrapolating these (linearly) to the weld toe. The locations of the stress points used for extrapolation have been agreed upon by researchers in Europe (see Reference [A.16.10-4]) and are defined in Figure A.16.10-1. The European definition is based on maximum principal stress, i.e. the stress components are extrapolated to the weld toes and then used in Mohr's Circle to establish the maximum principal stress at the toe. The stress normal to the weld toe, used in the US definition, is somewhat lower than this, but for the all-important saddle location the two are virtually identical.

The recommended $S-N$ curves and SCF equations used in this document are based on the European definition and are consistent.

**Key**

- 1 brace $a = 0.2\sqrt{rt}$, but $a \geq 4$ mm
 2 chord $b = 0.4\sqrt[4]{rtRT}$

Figure A.16.10-1—Distances for Extrapolation to Weld Toes

A.16.10.2.1.2 Deriving SCFs

SCFs may be derived from FEA, model tests, or empirical equations based on such methods. When deriving SCFs using FEA, solid (volume) elements should be used to represent the weld region (as opposed to thin shell elements). In models using volume elements, the SCFs can be derived by extrapolating stress components to the relevant weld toes and combining these to obtain the maximum principal stress and hence the SCF. The extrapolation direction should be normal to the weld toes.

If thin shell elements are used, the results should be interpreted carefully since no single method is guaranteed to provide consistently accurate stresses [A.16.10-5]. Extrapolation to the mid-surface intersection generally, but not consistently, overpredicts SCFs, whereas extrapolation to the notional weld toes would generally underpredict SCFs. In place of extrapolation, it is possible to directly use the nodal average stresses at the mid-surface

intersection. This will generally overpredict stresses, especially on the brace side. This last method is expected to be more sensitive to the local mesh density than the extrapolation methods.

When deriving SCFs from model tests, care should be taken to cover all potential hot spot locations with strain gauges. Furthermore, it should be recognized that the strain concentration factor is not identical to the SCF but is related to it via the transverse strains and Poisson's ratio. If the chord length in the joint tested is less than about 6 diameters (chord length parameter $\alpha < 12$), the SCFs should possibly be corrected using the Efthymiou short-chord correction factors [A.16.10-6]. The same correction can be needed in FEA if $\alpha < 12$.

A.16.10.2.1.3 Joint Classification

For the purpose of SCF evaluation, tubular joints are usually classified into joint types of T/Y-, X-, K-, or KT-joints, as defined in Figure A.16.10-2. The geometrical parameters needed to define each joint type, e.g. chord diameter (D), chord thickness (T), brace diameter (d), brace thickness (t), etc. are shown in the figure. Appropriate nondimensional, geometrical parameters, β , γ , τ , α , θ , and ζ , are also defined in Figure A.16.10-2 for each joint type.

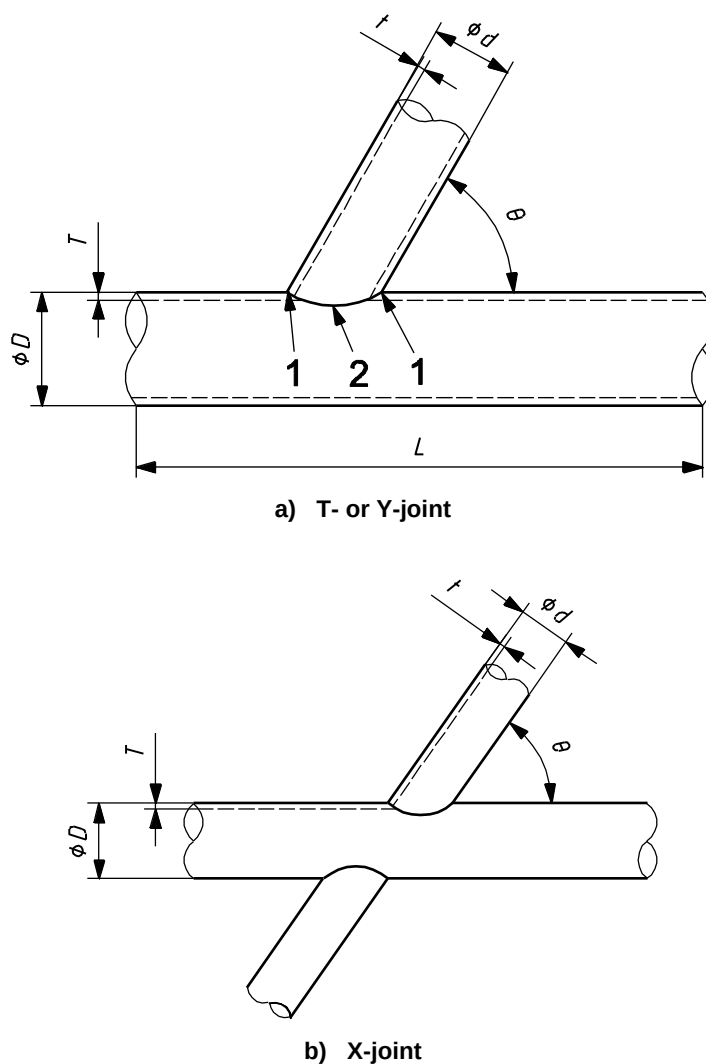
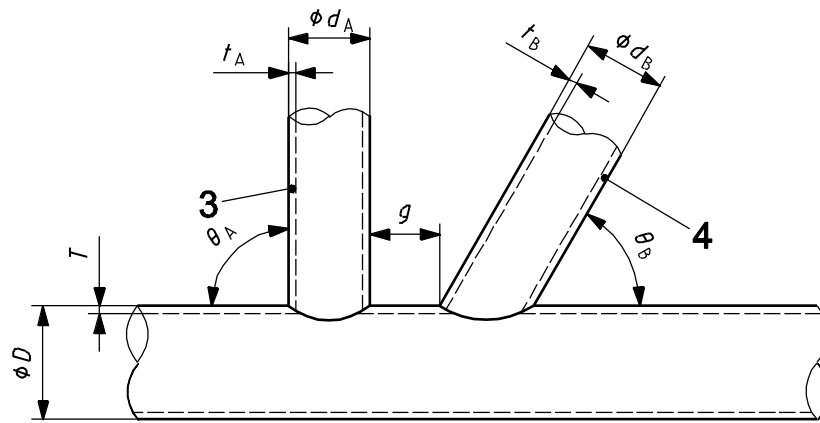
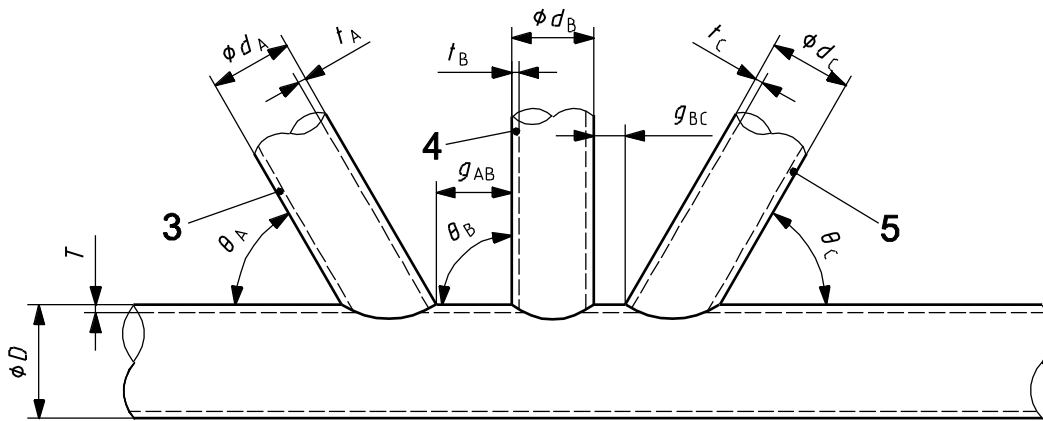


Figure A.16.10-2—Joint Classification and Geometrical Parameters for SCFs



c) K-joint



d) KT-joint

Key

1	crown	$\beta = d/D$	$\beta_A = d_A/D$	$\beta_B = d_B/D$	$\beta_C = d_C/D$
2	saddle	$\tau = t/T$	$\tau_A = t_A/T$	$\tau_B = t_B/T$	$\tau_C = t_C/T$
3	brace A	$\zeta = g/D$	$\zeta_{AB} = g_{AB}/D$	$\zeta_{BC} = g_{BC}/D$	
4	brace B	$\gamma = D/2 \ T$			
5	brace C	$\alpha = 2L/D$			

Figure A.16.10-2—Joint Classification and Geometrical Parameters for SCFs *(continued)*

For many joints, it is sufficiently accurate to use a classification based on geometry only, following the rules given in Table A.16.10-1, or the assumptions on forces in brace members recommended below. Alternatively, a classification methodology can be used, including the more rational influence function approach that handles classification automatically on the basis of the actual load paths. This is discussed further in A.16.10.2.2.

Table A.16.10-1—Classification Table for a Brace End

N_{near}	N_{far}	Classification ^a
1	0	T/Y
1	1	X
1	2	T/Y
1	3	T/Y
2	Any	K
3	Any	KT
N_{near} number of braces on same side of chord as reference brace (including reference brace)		
N_{far} number of braces on far side of chord with respect to reference brace		
^a Nonplanar braces are ignored.		

In joint types where more than one brace is involved, i.e. in K-, KT-, and X-joints, the relative magnitude and direction of the nominal brace forces (and moments) has a significant influence on SCFs. For instance, for a typical K-joint under OPB, the SCF increases from 2 when the brace moments are exactly balanced, to approximately 6 when they are unbalanced.

The following recommendations are given with respect to categorization of tubular K-, KT-, and X-joints based on force patterns in braces for fixed offshore structures (see also 14.2.4).

a) K- and KT-joints:

- 1) normal components of the axial brace forces are assumed to be balanced;
- 2) IPB moments on braces are assumed to be unbalanced;
- 3) OPB moments on braces are assumed to be unbalanced.

b) X-joints:

- 1) axial forces on braces are assumed to be balanced;
- 2) SCFs in X-joints are not sensitive to the sign of the IPB moment;
- 3) OPB moments on braces are assumed to be balanced.

These recommendations are based on the assumption that, in a well-braced structure, the response of primary joints is usually governed by axial forces; bending and shear due to frame action are of secondary importance. Hence the (normal components of the) axial forces in the braces at a joint (X-, K-, or KT-joints) should be approximately balanced so that shear remains small. IPB and OPB moments can be significant when they are caused by direct wave load/action (as opposed to frame action) and for secondary braces. Hence, IPB and OPB moments in K- and KT-joints tend to be unbalanced, whereas OPB moments in X-joints tend to be balanced for the same reason.

A.16.10.2.1.4 Evaluation of GSRs

The key GSR locations at the tubular joint intersection are termed saddle and crown (see Figure A.16.10-2). A minimum of eight stress range locations need to be considered around each chord/brace intersection weld in order to adequately cover all relevant locations. These are: the chord sides at two crown positions, the brace sides at two crown positions, the chord sides at two saddle positions, and the brace sides at two saddle positions.

The GSRs for the chord and the brace side of the weld are determined from the geometrical stresses given by Equation (A.16.10-1):

$$\begin{aligned}\sigma_{GS,s}(t) &= C_{ax,s} \sigma_{ax}(t) \pm C_{opb,s} \sigma_{opb}(t) \\ \sigma_{GS,c}(t) &= C_{ax,c} \sigma_{ax}(t) \pm C_{ipb,c} \sigma_{ipb}(t) + \sigma_{C,c}(t)\end{aligned}\quad (A.16.10-1)$$

where

- σ_{GS} is the geometrical stress on the chord or the brace side of the weld between chord and brace;
- σ_{ax} is the nominal axial stress in the brace (or stub);
- σ_{ipb} is the nominal IPB stress in the brace (or stub);
- σ_{opb} is the nominal OPB stress in the brace (or stub);
- $\sigma_{C,c}$ is the nominal stress in the chord (or chord can) at the crown position (see A.16.10.2.1.5);
- C_{ax} is the SCF for axial brace stress;
- C_{ipb} is the SCF for IPB stresses in the brace;
- C_{opb} is the SCF for OPB stresses in the brace;
- t is time;
- s is the subscript denoting the saddle position;
- c is subscript denoting the crown position.

The GSR, denoted by S , is the maximum minus the minimum stress during one full stress cycle. Equations (A.16.10-1) apply to both the brace side and the chord side of the weld at the intersection of brace and chord. Note that the nominal chord stress ($\sigma_{C,c}$) only affects the chord side of the weld at the crown position and should be deleted from Equation (A.16.10-1) when the geometrical stress for the brace is determined.

Since the nominal brace stresses, σ_{ax} , σ_{opb} , and σ_{ipb} , are functions of wave position (i.e. time t), it follows that in combining the contributions from the various stresses, the phase differences between them are automatically taken into account.

For some joints and certain individual brace force patterns, the point of highest stress can lie at a location between the saddle and the crown. Examples include balanced axial brace forces in K-joints where the hot spot generally lies between the saddle and the crown toe. For IPB, the hot spot might not be precisely at the crown, but could lie within a sector of $\pm 30^\circ$ from the crown depending on the γ - and β -values. The recommended SCF equations capture these higher SCFs even though they are referred to as occurring notionally at the crown or the saddle for simplicity.

For combined axial forces and bending moments, it is possible for the maximum GSR to occur at a location between the saddle and crown, even when the individual hot spots occur at the saddle or crown. These cases occur when OPB and IPB contributions are comparable in terms of GSR and are in-phase, and when, additionally, the axial contributions are small or relatively constant around the intersection. For such cases, Equation (A.16.10-1) can underpredict the maximum stress range. To overcome this, the GSR around the entire joint intersection can be estimated using location dependent SCFs and a generalized form of Equation (A.16.5-1):

$$\begin{aligned}S(\chi) &= C_{ax,C}(\chi) \times \sigma_{ax}(t) \pm C_{ipb,C}(\chi) \times \sigma_{ipb}(t) \pm C_{opb,C}(\chi) \times \sigma_{opb}(t) \\ S(\chi) &= C_{ax,B}(\chi) \times \sigma_{ax}(t) \pm C_{ipb,B}(\chi) \times \sigma_{ipb}(t) \pm C_{opb,B}(\chi) \times \sigma_{opb}(t)\end{aligned}\quad (A.16.10-2)$$

where, additionally,

$S(\chi)$	is the GSR at the location around the chord to brace intersection defined by angle χ ;
$C_{ax,C}(\chi)$	describes the variation of the SCF due to axial brace force on the chord side around the chord to brace intersection defined by angle χ ;
$C_{ipb,C}(\chi)$ and $C_{opb,C}(\chi)$	similarly describe the variation of the SCFs due to IPB and OPB around the chord to brace intersection defined by angle χ , respectively.

SCFs with subscript B instead of C describe the same variables for the brace side of the chord to brace intersection. The SCF distribution functions around the circumference can be obtained from parametric expressions given in Reference [A.16.10-7].

A.16.10.2.1.5 Effect of Nominal Chord Stress, $\sigma_{C,c}(t)$

Nominal variable stresses in the chord member also contribute to stress ranges for fatigue damage accumulation. Their contribution is usually small because (unlike brace forces) chord forces do not cause any significant local distortion of the chord walls. Hence any stress-raising effects are minimal. The effect of nominal variable stresses in the chord member can be covered by including the stress ($\sigma_{ax,C}$) due to axial force in the chord can member, combined with an axial SCF of 1.25, i.e. $\sigma_{C,c}(t) = 1.25 \sigma_{ax,C}(t)$, while accounting for sign and phase differences with other effects due to brace force patterns in Equation (A.16.10-1). The contribution, $\sigma_{C,c}(t)$, is applied at the chord crown location only; contributions at other locations, i.e. at the chord saddle and at the brace side, are considerably smaller and may be neglected.

A.16.10.2.2 Unstiffened Tubular Joints

A.16.10.2.2.1 Review of SCF Equations

Several sets of parametric equations have been derived for estimating SCFs in tubular joints (see References [A.16.10-5], [A.16.10-6], and [A.16.10-8] to [A.16.10-11]).

The performance of the various sets of SCF equations in terms of accuracy, degree of conservatism and range of applicability has been assessed in a number of studies, notably in a study by EWI funded by API (see Reference [A.16.10-12]), and a study by Lloyd's Register funded by HSE (see Reference [A.16.10-7]).

The main conclusion from the EWI study was that the Efthymiou equations [A.16.10-6] and the Lloyd's design equations [A.16.10-7] have considerable advantages in consistency and coverage in comparison with other available equations. When discussing the Lloyd's SCF equations, it is important to clarify that two modern sets of Lloyd's SCF equations exist, namely:

- mean SCF equations created from the database of acrylic test results that were available in 1988;
- design SCF equations defined as mean-plus-one standard deviation from the same database.

When assessed by EWI against the latest SCF database, the Lloyd's mean SCF equations are found to generally underpredict SCFs and fail the HSE assessment criteria. The mean SCF equations are not recommended for design.

A second conclusion from the EWI study was that the option of mixing-and-matching equations from different sets would lead to inconsistencies and is not recommended. An inconsistency of this type is already present in the Lloyd's design SCF equations, due to the variable partial factor used, but it is not very significant.

For the α -Kellogg equations, which are given in Reference [A.16.10-13], the EWI study concluded that they are generally more conservative than both the Efthymiou and the Lloyd's design SCF equations. Perhaps the most significant weakness of the α -Kellogg equations is that the predicted SCFs for all joint types are independent of β . This is clearly not the case as evidenced from test data and FEA results. Furthermore, the equations imply that chord SCFs are proportional to $\sqrt{\gamma}$ as opposed to observations that indicate that they increase linearly with γ . One advantage of the α -Kellogg equations is their simplicity.

In the comparison studies by Lloyd's Register, the Efthymiou SCF equations were found to provide a good fit to the screened SCF database, with a bias of about 10 % to 25 % on the conservative side [A.16.10-7]. They generally pass the HSE criteria for goodness-of-fit and conservatism. For the important case of K-joints under balanced axial forces, the Efthymiou equations did not pass the HSE criteria. A closer examination of this specific case revealed that these equations are satisfactory for both the chord and the brace side. For the chord side in particular, the Efthymiou equations provide the best fit to the database (COV = 19 %) and have a bias of 19 % on the conservative side. The second best equation (Lloyd's) has a COV of 21 % and a bias of 41 % on the conservative side. The HSE criteria were deliberately designed to favor those equations that overpredict SCFs and to penalize underpredictions. This is the key reason why the Efthymiou equations for K-joints marginally failed the HSE criteria, even though they provide a good fit and also err on the conservative side.

The Lloyd's design SCF equations generally pass the HSE criteria, except for T/Y-joints under axial force and IPB on the brace crown side. The reason for this is that there seems to be a systematic difference between the acrylic results and steel results for T/Y-joints at the brace crown.

Use of the Efthymiou SCF equations is recommended because this set of equations is considered to offer either the best option or a very good option for all joint types and types of brace forces and is the only set that covers overlapped K- and KT-joints.

Mix-and-match between different sets of equations is not recommended. The Efthymiou equations are also recommended in the Twenty-Second Edition of API 2A-WSD. The Efthymiou equations are given in Tables A.16.10-2 to A.16.10-5 and briefly discussed in A.16.10.2.2.2.

A.16.10.2.2.2 The Efthymiou Equations

- a) Overview: The Efthymiou equations cover SCFs in unstiffened T/Y-, X-, K-, and KT-joints under all relevant brace force conditions. Overlapped K- and KT-joints are also covered. These expressions are based on extensive FEA using the PMBSHELL program. The program uses thick shell elements for modeling the chord and braces and 3-D brick elements for the welds. The weld profiles are in accordance with AWS D1.1/D1.1M. This modeling enables direct extrapolation of stresses to the weld toes. The modeling and extrapolation removes the need for corrections, such as those attributed to Marshall [A.16.10-3], which were aimed at the brace side SCFs derived from thin shell FEA. Inclusion of the weld profiles with appropriate cut-back for high diameter ratios ensures realistic behavior when modeling $\beta = 1.0$ joints.
- b) T/Y-joints: For T/Y-joints under axial brace forces (see Table A.16.10-2), the SCFs are significantly influenced by the chord length and the fixity conditions at the ends of the chord. Beam bending of the chord influences primarily the crown SCFs, whereas for short chords ($\alpha < 12$), the shell distortion at the chord to brace intersection and hence the SCFs are affected by the fixity at the ends.

For beam bending the chord end fixity is defined by a parameter C , which is analogous to the effective length factor for buckling and has the range of 0.5–1.0. When $C = 0.5$, the ends are fully fixed and the equations degenerate to those of the fixed case. When $C = 1.0$, the chord ends are pinned. There are instances where beam bending of the chord is limited (see below). In such cases the chord length should be taken to be small (e.g. $\alpha = 12$ or less).

For joints with short chords ($\alpha < 12$), the correction factors, F_1 or F_2 , in the Efthymiou equations should be applied when the ends of the chord are radially restrained. Factor F_1 is applicable when the ovalization at the chord ends is completely suppressed, for instance by a diaphragm, or if the ends are welded onto another stiff member.

- c) X-joints: In X-joints under axial brace forces (see Table A.16.10-3), the SCFs at the saddles tend to dominate for all values of β , including the common case of $\beta = 1.0$. In joints with short chords, the correction factors, F_1 or F_2 , in the Efthymiou equations may be used, provided that ovalization of the ends is restrained to some extent. If the ends are completely free, the saddle SCFs should be increased. An approximate way of achieving this is to increase them by the ratio $1/F_2$.

The chord crown SCF (C_{X2}) under axial forces is derived from the corresponding SCF (C_{T2}) for T/Y-joints by suppressing beam bending of the chord, i.e. setting α to zero. For the brace crown SCF (C_{X4}), a better fit to the PMBSHELL database was obtained not by setting α to zero but by deleting completely the second term in the corresponding SCF (C_{T4}) for T/Y-joints.

- d) K-joints: K-joints with a gap greater than one chord diameter ($\zeta > 1$) under balanced axial brace forces (see Table A.16.10-4) should be classified as Y-joints for the purpose of SCF evaluation. The chord length parameter, α , should be set to 12 to reflect the fact that beam bending of the chord is limited. For OPB when $\zeta > 1$, the SCF equations degenerate to the Y-joint equations, and hence it is not necessary to re-classify these joints.
- e) KT-joints: For KT-joints under balanced axial brace forces (see Table A.16.10-5), the SCFs on a diagonal brace are evaluated by considering the axial brace force to be balanced by the other diagonal brace, i.e. ignoring the central brace and hence degenerating the joint to a K-joint. For the central brace, it is generally sufficient to consider that its axial force is balanced by one of the diagonal braces. If the diagonal braces are identical but the gaps differ, the maximum of the two gaps should be used, to be conservative. If the diagonal braces are not identical, then the central brace should be successively paired with each of the diagonal braces and the maximum resulting SCFs selected.

- f) Validity ranges: The validity ranges for the Efthymiou equations are as follows:

$$\begin{array}{rclcl}
 0.2 & \leq & \beta & \leq & 1.0 \\
 0.2 & \leq & \tau & \leq & 1.0 \\
 8 & \leq & \gamma & \leq & 32 \\
 4 & \leq & \alpha & \leq & 40 \\
 20^\circ & \leq & \theta & \leq & 90^\circ \\
 \frac{-0.6\beta}{\sin\theta} & \leq & \zeta & \leq & 1.0
 \end{array} \tag{A.16.10-3}$$

For cases where one or more parameters fall outside this range, the following procedure should be adopted:

- 1) evaluate SCFs using the actual values of geometrical parameters;
- 2) evaluate SCFs using the limit values of geometrical parameters;
- 3) use the maximum of 1) or 2) above in the fatigue analysis.

Table A.16.10-2—Equations for SCFs in T/Y-joints

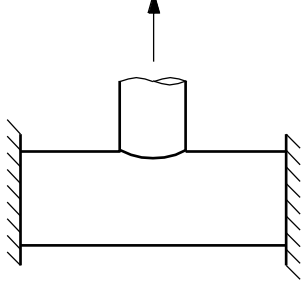
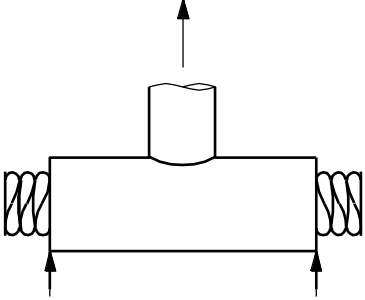
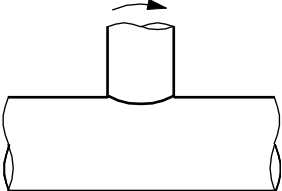
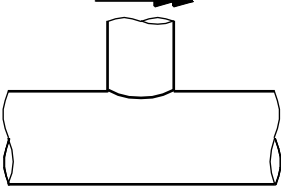
Type of Brace Force and Fixity Conditions	SCF Equation
Axial brace force, chord ends fixed 	Chord saddle: $C_{CS} = F_1 C_{T1} \quad C_{T1} = \gamma \tau^{1.1} \left[1.11 - 3(\beta - 0.52)^2 \right] \sin^{1.6} \theta$ Chord crown: $C_{CC} = C_{T2} \quad C_{T2} = \gamma^{0.2} \tau \left[2.65 + 5(\beta - 0.65)^2 \right] + \tau \beta (0.25\alpha - 3) \sin \theta$ Brace saddle: $C_{BS} = F_1 C_{T3} \quad C_{T3} = 1.3 + \gamma \tau^{0.52} \alpha^{0.1} \left[0.187 - 1.25\beta^{1.1} (\beta - 0.96) \right] \sin^{(2.7-0.01\alpha)} \theta$ Brace crown: $C_{BC} = C_{T4} \quad C_{T4} = 3 + \gamma^{1.2} \left[0.12 \exp(-4\beta) + 0.011\beta^2 - 0.045 \right] + \beta \tau (0.1\alpha - 1.2)$
Axial brace force, general chord fixity 	Chord saddle: $C_{CS} = F_2 C_{T5} \quad C_{T5} = C_{T1} + C_1 (0.8\alpha - 6) \tau \beta^2 (1 - \beta^2)^{0.5} \sin^2 2\theta$ Chord crown: $C_{CC} = C_{T6} \quad C_{T6} = \gamma^{0.2} \tau \left[2.65 + 5(\beta - 0.65)^2 \right] + \tau \beta (C_2 \alpha - 3) \sin \theta$ Brace saddle: $C_{BS} = F_2 C_{T3} \quad \text{See above}$ Brace crown: $C_{BC} = C_{T7} \quad C_{T7} = 3 + \gamma^{1.2} \left[0.12 \exp(-4\beta) + 0.011\beta^2 - 0.045 \right] + \beta \tau (C_3 \alpha - 1.2)$
In-plane bending (IPB) 	Chord crown: $C_{CC} = C_{T8} \quad C_{T8} = 1.45 \beta \tau^{0.85} \gamma^{(1-0.68\beta)} \sin^{0.7} \theta$ Brace crown: $C_{BC} = C_{T9} \quad C_{T9} = 1 + 0.65 \beta \tau^{0.4} \gamma^{(1.09-0.77\beta)} \sin^{(0.06\gamma-1.16)} \theta$
Out-of-plane bending (OPB) 	Chord saddle: $C_{CS} = F_3 C_{T10} \quad C_{T10} = \gamma \tau \beta (1.7 - 1.05\beta^3) \sin^{1.6} \theta$ Brace saddle: $C_{BS} = F_3 C_{T11} \quad C_{T11} = \tau^{-0.54} \gamma^{-0.05} (0.99 - 0.47\beta + 0.08\beta^4) \times C_{T10}$

Table A.16.10-2—Equations for SCFs in T/Y-joints (continued)

	Short chord correction factors ($\alpha < 12$): $F_1 = 1 - (0.83\beta - 0.56\beta^2 - 0.02)\gamma^{0.23}\exp(-0.21\gamma^{-1.16}\alpha^{2.5})$ $F_2 = 1 - (1.43\beta - 0.97\beta^2 - 0.03)\gamma^{0.04}\exp(-0.71\gamma^{-1.38}\alpha^{2.5})$ $F_3 = 1 - 0.55\beta^{1.8}\gamma^{0.16}\exp(-0.49\gamma^{-0.89}\alpha^{1.8})$ where $\exp(x) = e^x$
	Chord-end fixity parameter, C: $0.5 \leq C \leq 1.0 \quad (\text{Typically } C = 0.7)$ $C_1 = 2(C - 0.5)$ $C_2 = C/2$ $C_3 = C/5$

Table A.16.10-3—Equations for SCFs in X-joints

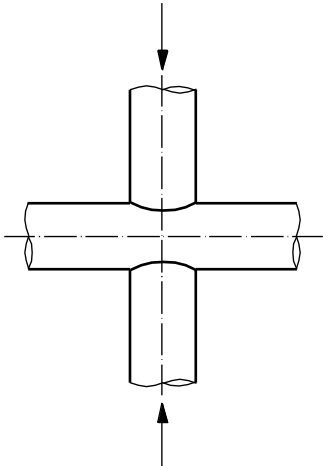
Type of Brace Force	SCF Equation
Axial force (balanced) 	Chord saddle: $C_{CS} = C_{X1} \quad C_{X1} = 3.87\gamma\tau\beta(1.10 - \beta^{1.8})\sin^{1.7}\theta$ Chord crown: $C_{CC} = C_{X2} \quad C_{X2} = \gamma^{0.2}\tau\left[2.65 + 5(\beta - 0.65)^2\right] - 3\tau\beta\sin\theta$ Brace saddle: $C_{BS} = C_{X3} \quad C_{X3} = 1 + 1.9\gamma\tau^{0.5}\beta^{0.9}(1.09 - \beta^{1.7})\sin^{2.5}\theta$ Brace crown: $C_{BC} = C_{X4} \quad C_{X4} = 3 + \gamma^{1.2}\left[0.12\exp(-4\beta) + 0.011\beta^2 - 0.045\right]$
NOTE 1 In joints with short chords, $\alpha < 12$, which have stiffened ends, both the chord saddle and the brace saddle SCF may be reduced by multiplying them by the short chord factor, F_1 or F_2. Factor F_1 can be used for stiff end reinforcements preventing ovalization as well as rotation of the chord wall, while factor F_2 can be used for end reinforcements partially preventing ovalization only. NOTE 2 If the chord ends are completely free, both the chord saddle and the brace saddle SCF can increase significantly. An approximation can be obtained by increasing them by the ratio $1.0/F_2$ (see A.16.10.2.2.2), but FEA is recommended. NOTE 3 F_1 and F_2 are given in Table A.16.10-2.	

Table A.16.10-3—Equations for SCFs in X-joints (continued)

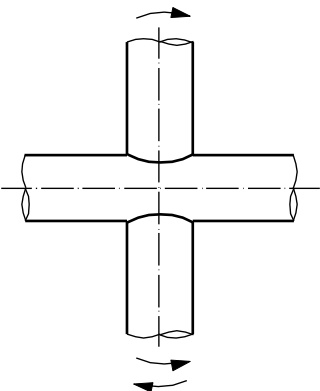
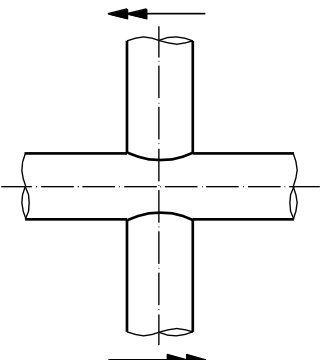
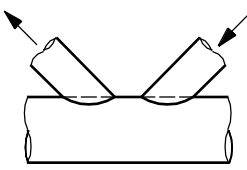
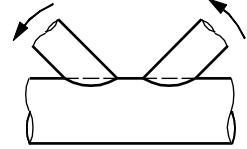
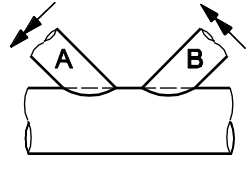
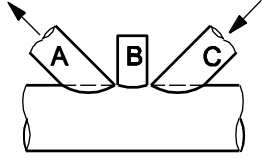
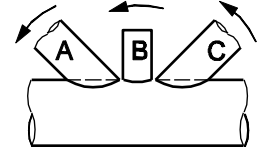
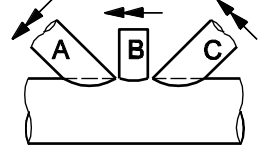
Type of Brace Force	SCF Equation
In-plane bending (IPB) (balanced or unbalanced) 	Chord crown: $C_{CC} = C_{T8}$ see Table A.16.10-2 Brace crown: $C_{BC} = C_{T9}$ see Table A.16.10-2
Out-of-plane bending (OPB) (balanced) 	Chord saddle: $C_{CS} = C_{X5}$ $C_{X5} = \gamma \tau \beta (1.56 - 1.34\beta^4) \sin^{1.6} \theta$ Brace saddle: $C_{BS} = C_{X6}$ $C_{X6} = \tau^{-0.54} \gamma^{-0.05} (0.99 - 0.47\beta + 0.08\beta^4) \times C_{X5}$
NOTE 1 In joints with short chords, $\alpha < 12$, which have ends stiffened with a diaphragm or ring stiffener, both the chord saddle and the brace saddle SCF may be reduced by multiplying them by the short chord factor F_3 (see Table A.16.10-2). NOTE 2 If the chord ends are completely free, saddle SCFs can increase significantly. An approximation can be obtained by increasing them by the ratio $1.0/F_3$ (see A.16.10.2.2.2), but FEA is recommended.	

Table A.16.10-4—Equations for SCFs in Gap/Overlap K-joints

Type of Brace Force	SCF Equation
Axial forces (balanced) 	Chord: $C_C = C_{K1} \quad C_{K1} = \left[\tau^{0.9} \gamma^{0.5} (0.67 - \beta^2 + 1.16\beta) \sin \theta \right] \left(\frac{\sin \theta_{\max}}{\sin \theta_{\min}} \right)^{0.30} \times \left(\frac{\beta_{\max}}{\beta_{\min}} \right)^{0.30} \times \left[1.64 + 0.29\beta^{-0.38} \arctan(8\zeta) \right]$ Brace: $C_B = C_{K2} \quad C_{K2} = 1 + C_{K1} (1.97 - 1.57\beta^{0.25}) (\tau^{-0.14} \sin^{0.7} \theta) + \left[K \beta^{1.5} \gamma^{0.5} \tau^{-1.22} \sin^{1.8} (\theta_{\max} + \theta_{\min}) \right] \times \left[0.131 - 0.084 \arctan(14\zeta + 4.2\beta) \right]$ where $K = 0$ for gap joints; $K = 1$ for the through brace; $K = 0.5$ for the overlapping brace; the arctangents are evaluated in radians; τ, β, θ and the nominal stress relate to the brace being considered.
In-plane bending (IPB) (unbalanced) 	Chord crown, nonoverlapping joint or overlap ≤ 30 % of contact length: $C_{CC} = C_{T8}$ see Table A.16.10-2 Chord crown, overlap > 30 % of contact length: $C_{CC} = 1.2 C_{T8}$ see Table A.16.10-2 Brace crown, nonoverlapping joint: $C_{BC} = C_{T9}$ see Table A.16.10-2 Brace crown, overlapping joint: $C_{BC} = C_{T9} \times (0.9 + 0.4\beta)$
Out-of-plane bending (OPB) (unbalanced)^a 	Chord saddle adjacent to brace A: $C_{CS} = F_4 C_{K4} \quad C_{K4} = C_{T10,A} \left[1 - 0.08 (\beta_B \gamma)^{0.5} \exp(-0.8x) \right] + C_{T10,B} \left[1 - 0.08 (\beta_A \gamma)^{0.5} \exp(-0.8x) \right] \times \left[2.05 \beta_{\max}^{0.5} \exp(-1.3x) \right]$ where $x = 1 + \frac{\zeta \sin \theta_A}{\beta_A}$ $C_{T10,A}$ and $C_{T10,B}$ (see Table A.16.10-2) are calculated with the parameters for braces A and B, respectively. Brace saddle adjacent to brace A: $C_{BS} = F_4 C_{K5} \quad C_{K5} = \tau^{-0.54} \gamma^{-0.05} (0.99 - 0.47\beta + 0.08\beta^4) \times C_{K4}$
	Short chord correction factor ($\alpha < 12$): $F_4 = 1 - 1.07\beta^{1.88} \exp \left[-0.16\gamma^{-1.06} \alpha^{2.4} \right]$ where $\exp(x) = e^x$.

^a The designation of braces A and B is not geometry dependent. It is nominated by the user.

Table A.16.10-5—Equations for SCFs in KT-joints

Type of Brace Force	SCF Equation
Axial force (balanced) 	Chord: $C_C = C_{K1} \quad \text{see Table A.16.10-4}$ Brace: $C_B = C_{K2} \quad \text{see Table A.16.10-4}$ where $\zeta = \zeta_{AB} + \zeta_{BC} + \beta_B \quad \text{for the diagonal braces A and C;}$ $\zeta = \text{maximum of } \zeta_{AB} \text{ and } \zeta_{BC} \quad \text{for the central brace B.}$
In-plane bending (IPB) 	Chord crown: $C_{CC} = C_{T8} \quad \text{see Table A.16.10-2}$ Brace crown: $C_{BC} = C_{T9} \quad \text{see Table A.16.10-2}$
Out-of-plane bending (OPB) (unbalanced) 	Chord saddle at diagonal brace A: $C_{CS} = C_{KT1} \quad C_{KT1} = C_{T10,A} \left[1 - 0.08(\beta_B \gamma)^{0.5} \exp(-0.8 x_{AB}) \right] \times \left[1 - 0.08(\beta_C \gamma)^{0.5} \exp(-0.8 x_{AC}) \right]$ $+ C_{T10,B} \left[1 - 0.08(\beta_A \gamma)^{0.5} \exp(-0.8 x_{AB}) \right] \times \left[2.05 \beta_{\max}^{0.5} \exp(-1.3 x_{AB}) \right]$ $+ C_{T10,C} \left[1 - 0.08(\beta_A \gamma)^{0.5} \exp(-0.8 x_{AC}) \right] \times \left[2.05 \beta_{\max}^{0.5} \exp(-1.3 x_{AC}) \right]$ where $x_{AB} = 1 + \frac{\zeta_{AB} \sin \theta_A}{\beta_A};$ $x_{AC} = 1 + \frac{(\zeta_{AB} + \zeta_{BC} + \beta_B) \sin \theta_A}{\beta_A}.$ Chord saddle at central brace B: $C_{CS} = C_{KT2} \quad C_{KT2} = C_{T10,B} \left[1 - 0.08(\beta_A \gamma)^{0.5} \exp(-0.8 x_{AB}) \right]^{(\beta_A/\beta_B)^2}$ $\times \left[1 - 0.08(\beta_C \gamma)^{0.5} \exp(-0.8 x_{BC}) \right]^{(\beta_C/\beta_B)^2}$ $+ C_{T10,A} \left[1 - 0.08(\beta_B \gamma)^{0.5} \exp(-0.8 x_{AB}) \right] \times \left[2.05 \beta_{\max}^{0.5} \exp(-1.3 x_{AB}) \right]$ $+ C_{T10,C} \left[1 - 0.08(\beta_B \gamma)^{0.5} \exp(-0.8 x_{BC}) \right] \times \left[2.05 \beta_{\max}^{0.5} \exp(-1.3 x_{BC}) \right]$ where $x_{AB} = 1 + \frac{\zeta_{AB} \sin \theta_B}{\beta_B};$ $x_{BC} = 1 + \frac{\zeta_{BC} \sin \theta_B}{\beta_B}.$ Brace saddle: $C_{BS} = C_{KTB} \quad C_{KTB} = \tau^{-0.54} \gamma^{-0.05} (0.99 - 0.47 \beta + 0.08 \beta^4) \times C_{CS}$

A.16.10.2.2.3 Guidance on Reducing the SCF

Increasing the chord wall thickness is an effective way of reducing stress concentrations. For T/Y- and X-joints, a doubling of the chord wall thickness reduces the saddle SCFs by a factor of 4; crown SCFs are also reduced considerably.

Increasing the brace wall thickness (by introducing brace stubs) is NOT an effective way of reducing stress concentrations. For many joint configurations, the introduction of thicker brace stubs has little or no effect on the GSR on the chord side. It actually leads to an increase in the SCF, which is counteracted by a corresponding decrease in the nominal brace stub stress, and hence there is no significant change in the fatigue damage calculation. The use of thicker brace stubs tends to reduce the GSR on the brace side, but this reduction is small and usually not worthwhile in terms of the controlling stress value.

It is also relevant to note that the introduction of thicker brace stubs has little or no effect on the static strength of joints. It does, of course, increase the strength of the brace end and hence can be an attractive design detail for a brace that carries substantial transverse loads/actions (e.g. direct wave load/action or loads/actions due to accidental ship impact on braces in the splash zone). In general, however, the use of thicker brace stubs is an inefficient use of steel and should be avoided.

The use of thicker brace stubs further introduces additional stress concentrations at the brace to stub closure welds in nodal constructions because of the thickness transition. This can create additional problems, and the use of thicker brace stubs should be avoided.

A.16.10.2.2.4 Influence Functions

The concept of influence functions as a generalization of the SCF method of evaluating GSRs is described in References [A.16.10-6], [A.16.10-11], and [A.16.10-15]. This method is more accurate than the SCF approach because it can handle generalized forces and moments on the braces forming the joint, as opposed to an approach that is based on individual planes and joint classification.

An influence function is an expression for the GSR at a certain location around the chord-to-brace intersection arising from a nominal stress of unit magnitude acting on any brace of the joint. The GSR at a given location can be obtained by multiplying each influence function with its respective nominal stress and superimposing the contributions from all braces of the joint, including the brace under consideration.

The influence function algorithm is consistent with the SCF approach in the sense that it will lead to identical results for a joint that is subjected to forces and classed in the manner that is assumed by the SCF approach. For instance, it leads to identical results for a K-joint when the axial forces are exactly balanced and the OPB moments are exactly unbalanced.

In addition to being more accurate than the SCF approach, the influence function concept obviates the need to classify joints and hence is more convenient to use. An additional advantage is that it has been extended [A.16.10-6, A.16.10-11] to handle multiplanar joints for the important case of axial brace forces.

A disadvantage of the influence function algorithm is that it is less transparent than the direct SCF approach.

For complex joints of particular interest, specific influence coefficients and geometrical stresses can be accurately established by developing a detailed local FEA model of the joint and incorporating this model into the overall fatigue analysis (frame) model of the structure [A.16.10-16]. The advantage of this approach is that it captures all brace and chord forces and moments and their phase differences and all geometrical stress concentration effects, including multiplanar effects. However, introducing local joint flexibility can also have an effect on global stresses; therefore, when taking this approach, all joints should in principle have flexibility modeled.

A.16.10.2.2.5 Tubular Joints Welded from One Side

Single-sided welding is used as the principal method for connecting braces to chords in tubular joints for offshore structures in many areas of the world. Single-sided welding presupposes that a fatigue crack always initiates at the weld toe. However, if the SCF at the internal weld root of a tubular joint is relatively high compared to that at the external weld toe (e.g. the internal SCF is 70 % of the external SCF), then the crack can initiate at the internal weld root, due to more onerous $S-N$ curves that are relevant for the root detail than for the external weld toe. This is particularly important when weld improvement techniques are employed.

The appropriate $S-N$ curve from the other joints (OJ) group (see Table 16.11-1) should be used for fatigue analysis of the weld root. For further information, see References [A.16.10-17] and [A.16.10-18].

A.16.10.2.2.6 Tubular Thickness Transitions

Thickness transitions introduce additional stress concentrations at the connection weld and hence potential fatigue problems. The highest stresses occur on that side of the transition that is not flush. Possible fatigue problems are therefore made worse if the transition is made internally because the highest stresses then occur at the weld root, which has the highest potential for welding defects. Furthermore, this location cannot easily be inspected. Where thickness transitions are necessary, they should therefore be external (i.e. internally flush connections).

A.16.10.2.3 Internally Ring-stiffened Tubular Joints

The Lloyd's equations for ring-stiffened joints are given in Reference [A.16.10-19]. The following points should be noted regarding the equations:

- a) the derived SCF ratios for the brace to chord intersection and the SCFs for the ring edge are mean values, although the degree of scatter and proposed design factors are given;
- b) short chord effects should be taken into account where relevant;
- c) for joints with diameter ratios $\beta \geq 0.8$, the effect of stiffening is uncertain; it may even increase the SCF;
- d) the maximum of the saddle and crown values should be applied around the whole brace to chord intersection;
- e) the minimum SCF for the brace side under axial and OPB moments should be taken as 2.0, whereas a minimum value of 1.5 is recommended for all other locations.

The following additional observations can be made about the use of ring stiffeners in general:

- a) thin shell FEA should be avoided for calculating the SCF if the maximum stress is expected to be near the brace to ring crossing point, and special consideration should be given to this crossing point in the fatigue analysis;
- b) ring stiffeners have a marked effect on the circumferential stress in the chord, but have little or no effect on the longitudinal stress;
- c) ring stiffeners outside the brace footprint have little effect on the SCF but can be beneficial for static strength.

Failures in the ring inner edge or brace to ring interface occur internally and will probably only be detected after through-thickness cracking, at which time the majority of the fatigue life will have been expended. These areas should therefore be considered as noninspectable, unless sophisticated inspection methods are used.

A.16.10.2.4 Grouted Tubular Joints

Grouted joints have either the chord completely filled with grout (single skin grouted joints) or the annulus between the chord and an inner member filled with grout (double skin grouted joints). The SCF of a grouted joint depends on the stress history. The SCF is smaller when the bond between the chord and the grout is unbroken. Due to the grout, the tensile and compressive SCFs can be different. Only the highest value should be used in the fatigue analysis.

Grouted joints may be treated as simple joints, except that for the brace and the chord saddle points the chord thickness in the γ -term for saddle SCF calculations may be substituted with an equivalent chord wall thickness given by Equation (A.16.10-4):

$$T_e = (5 D + 134 T)/144 \quad (\text{A.16.10-4})$$

where

D is the chord outside diameter;

T is the chord wall thickness.

The formulation in Equation (A.16.10-4) has been derived on the basis of engineering mechanics.

Joints with high β - or low γ -ratios gain little benefit from grouting. Although fully substantiated evidence is not available, the benefits of grouting should be neglected for joints with $\beta > 0.9$ or $\gamma \leq 12.0$, unless documented otherwise. A minimum SCF value of 1.5 is recommended for all locations.

A.16.10.2.5 Cast Joints

It is recommended that FEA be used to determine the magnitude and location of the maximum stress range in castings sensitive to fatigue. The FEA model should use volume elements at the critical areas and properly model the shape of the joint. Consideration should be given to stresses at the inside of the castings. The brace to casting circumferential butt weld, which is designed to the appropriate $S-N$ curve from the OJ group (see Table 16.11-1), can be the most critical location for fatigue.

In the absence of fatigue data for castings tested in seawater under normal cathodic protection conditions, a minimum SCF of at least two should be applied.

A.16.10.3 Geometrical Stress Ranges for Other Fatigue-sensitive Locations

In any welded connection, there are several locations at which a fatigue crack can develop, e.g. in the parent metal at the weld toe, at the weld ends, and at the weld root (in the case of the weld throat of load-carrying fillet or partial penetration butt welds). Each location should be classified separately.

Constructional details for nonwelded material and welded connections with their corresponding joint classifications are given in, firstly, Table A.16.10-6, which affords an overview, and in Tables A.16.10-7 to A.16.10-12.

Table A.16.10-6—Joint Classification Tabulation

Table	Type	Constructional Detail
A.16.10-7	Type 1	Material free from welding
A.16.10-8	Type 2	Continuous welds essentially parallel to the direction of applied stress
A.16.10-9	Type 3	Transverse groove welds in plates and tubulars (essentially perpendicular to the direction of applied stress)
A.16.10-10	Type 4	Welded attachments on the surface or edge of a stressed member
A.16.10-11	Type 5	Load-carrying fillet and T-butt welds
A.16.10-12	Type 6	Details in welded girders and tubulars

The types and associated joint classification have been established on the basis of stresses generally along (within 45°) the direction indicated in the tables by the arrow for the potential crack on the surface of the parent metal or, in the case of weld throat cracking, on the shear stress calculated in the weld throat. The classification is based on a minimum quality level for assembly of joint and weld (see, for example, Reference [A.16.10-2]).

For details that are not expressly classified, the following minimum classification class should be used, unless a higher class can be justified from published experimental work, or by specific tests:

- a) W_1 for load-carrying fillet or partial penetration weld metal;
- b) F_2 for other cases.

The appropriate stress range indicated in Tables A.16.10-7 to A.16.10-11 and used as the GSR is the maximum principal stress range adjacent to the detail under consideration, except for the throat of load-carrying fillet or partial penetration welds, for which it is the shear stress range calculated on the minimum throat area. The stress range should be amplified to reflect the effect of any stress concentration in the vicinity of the joint that is not characteristic of the detail itself. For example, effects of holes, cut-outs, re-entrant corners, joint eccentricity and misalignment during fabrication that is greater than allowable should be taken into account, either in the structural model used to calculate stress range or by appropriate SCFs. SCFs for these typical details may be obtained from References [A.16.10-20] and [A.16.10-21], and other published literature.

Table A.16.10-7—Type 1: Material Free from Welding

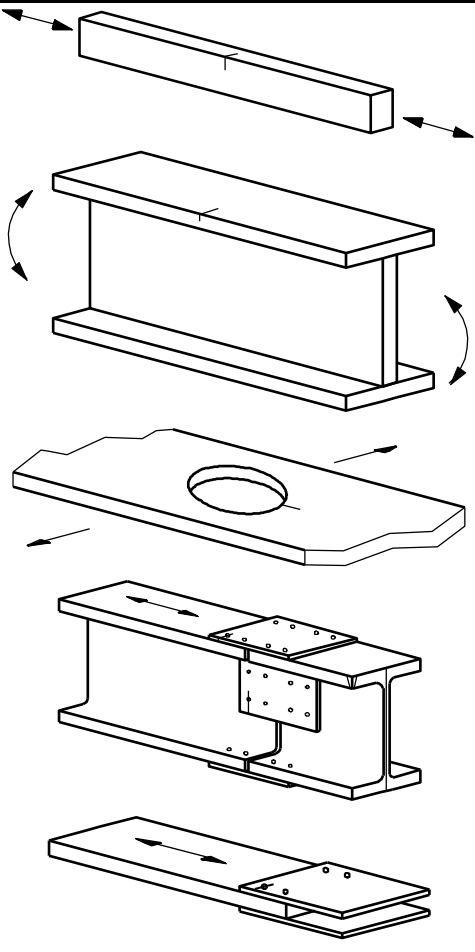
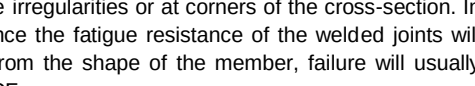
Type Number and Description, Explanatory Comments	Class	Example (Including Failure Mode)
1.1 Plain steel a) In the as-rolled condition or ground smooth or machined after cutting. b) With edges flame cut. No repair by weld refill.	B C C	
1.2 Bolted connection Unsupported one-sided connections should be avoided or the effect of eccentricities taken into account in the calculation of stresses. Stresses to be calculated in the gross section for friction grip connections or in the net section for all other connections.	C	
NOTE In plain steel, fatigue cracks initiate at the surface, usually either at surface irregularities or at corners of the cross-section. In welded construction, fatigue failure will rarely occur in a region of plain material since the fatigue resistance of the welded joints will usually be much lower. In steel with holes or other stress concentrations arising from the shape of the member, failure will usually initiate at the stress concentration. In this case, the stress range should include an SCF.		

Table A.16.10-8—Type 2: Continuous Welds Essentially Parallel to the Direction of Applied Stress

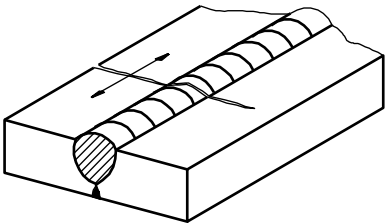
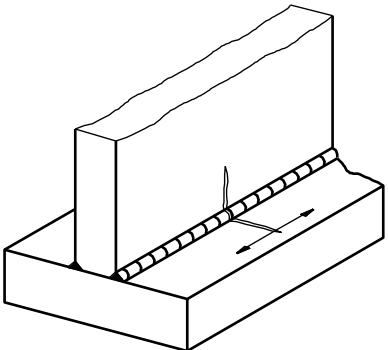
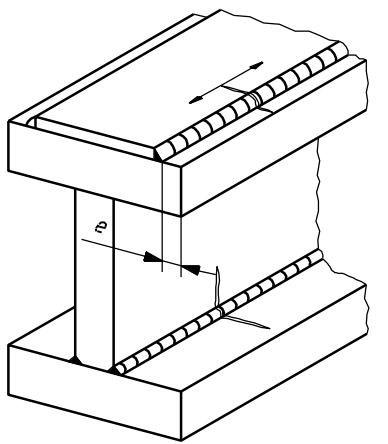
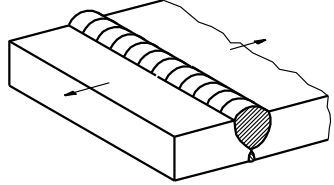
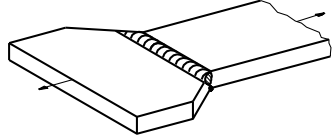
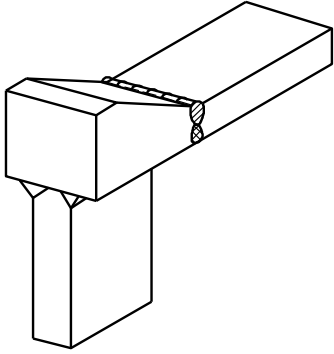
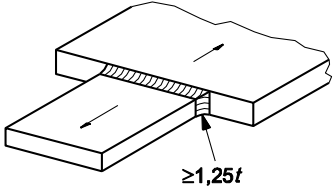
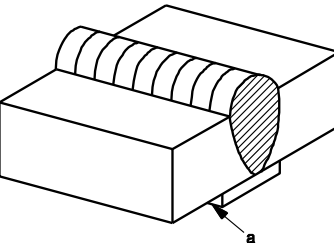
Type Number and Description, Explanatory Comments	Class	Example (Including Failure Mode)
2.1 Complete or partial joint penetration groove weld. Parent or weld metal in members, without attachments, built up of plates or sections and joined by continuous welds. a) Complete joint penetration groove welds with the weld overfill dressed flush with the surface and machine-finished in the direction of stress, and with the weld proved free from significant defects by NDT. b) Groove or fillet welds with the welds made from both sides by an automatic submerged or open arc process and with no stop-start positions within the length. If an accidental stop-start occurs remedial action should be taken so that the finished weld has a similar surface and root profile to that intended. c) As b) weld from one side. d) As b) but with the weld containing stop-start positions within the length. For the ends of flange cover plates, see joint Type 6.4. e) As b) but with manual welding.	B B C D D D	   e is the distance from weld toe to edge of flange, $e > 10$ mm
2.2 Discontinuous weld: refer to Type 6.5.		
NOTE 1 (Remarks on Potential Modes of Failure) With the excess weld metal dressed flush, fatigue cracks would be expected to initiate at weld defect locations. In the as-welded condition, cracks might initiate at stop-start positions or, if these are not present, at weld surface ripples. NOTE 2 (General Comments) i) Backing strips: if backing strips are used in making these joints: a) they should be continuous, and b), if they are attached by welding, those welds should also comply with the relevant type requirements (note particularly that tack welds, unless subsequently ground out or covered by a continuous weld, would reduce the joint to joint Type 6.5). ii) Edge distance: an edge distance criterion exists to limit the possibility of local stress concentrations occurring at unwelded edges as a result, for example, of undercut, weld spatter, or accidental over weave in manual fillet welding (see also notes on joint Type 4). Although an edge distance can be specified only for the "width" direction of an element, it is equally important to ensure that no accidental undercutting occurs on the unwelded corners of, for example, cover plates or box girder flanges. If it does occur, it should subsequently be ground smooth. iii) Weld ends: weld ends are classified as Type 4 or 5.		

Table A.16.10-9—Type 3: Transverse Groove Welds in Plates and Tubulars (i.e. Essentially Perpendicular to the Direction of Applied Stress)

Type Number and Description, Explanatory Comments	Class	Example
3.1 Parent and weld metal at complete penetration butt joints welded from both sides. Changes in thickness should be machined to a smooth transition not steeper than 1 in 4. Changes of less than 1.15 times the thinner member thickness may be accommodated in the weld profile without machining. a) With the weld cap ground flush with the surface and the weld proved to be free from significant defects by NDT. b) With the weld made, either manually or by an automatic process other than submerged arc, provided all runs, including repairs, are made in the down-hand position. c) In general, welds made by the submerged arc process, or in positions other than down-hand, tend to have a poor reinforcement shape, from the point of view of fatigue resistance. Hence, such welds are downgraded from c) to d). d) Welds made other than in a), b), or c). In both c) and d) the corners of the cross-section of the stressed component at the weld toes should be dressed to a smooth profile. e) Weld between plates of unequal width, with the weld ends ground to a radius not less than 1.25 times the thickness, t. Step changes in width can often be avoided by the use of shaped transition plates. For this detail, the stress concentration has been taken into account in the joint classification.	C D E F₂	     a No tack weld.
3.2 Parent and weld metal at complete joint penetration butt joints made from one side on a permanent backing strip. If the backing strip is fillet welded or tack welded to the member, the joint should be assessed using joint Type 4 c).	F	
3.3 Parent and weld metal at full penetration weld made from one side without permanent backing strip. All circumferential welds in tubulars of less than 650 mm diameter should be considered single-sided.	F ₂	

NOTE 1 (Remarks on Potential Modes of Failure)

With the weld ends machined flush with the plate edges, fatigue cracks in the as-welded condition normally initiate at the weld toe, so that the fatigue resistance depends largely upon the shape of the weld overfill. If this is dressed flush, the stress concentration caused by it is removed and failure is then associated with weld defects. In welds made on a permanent backing strip, fatigue cracks initiate at the weld metal to strip junction, and in partial joint penetration groove welds at the weld root.

Welds made entirely from one side, without a permanent backing, require care to be taken in the making of the root bead in order to ensure a satisfactory profile.

NOTE 2 (General Comments)

The stress range considered should be that at the actual crack location. It should include SCFs arising from any thickness change, or due to the overall form of the joint (e.g. butt joint at the end of a conical transition piece).

Table A.16.10-10—Type 4: Welded Attachments on the Surface or Edge of a Stressed Member

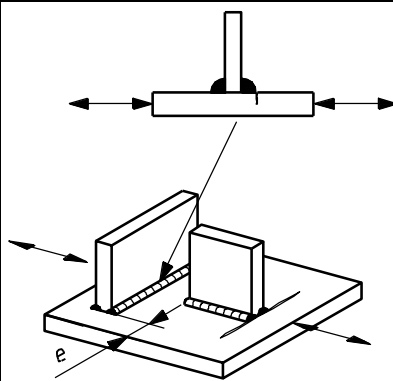
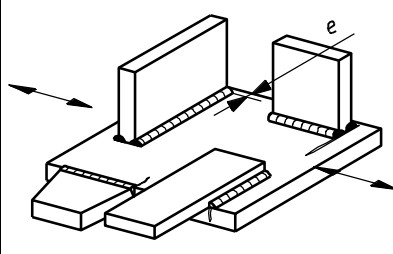
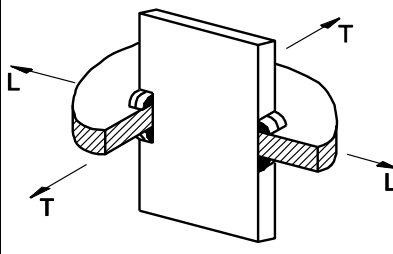
Type Number and Description, Explanatory Comments	Class	Examples (Including Failure Mode)
4. Welded attachments Parent metal of the stressed member adjacent to toes or ends of groove or fillet welded attachments (both stressed and unstressed attachments), regardless of the orientation of the weld to the direction of applied stress, and whether or not the welds are continuous around the attachment. Groove welds should have an additional reinforcing fillet so as to provide a similar toe profile to that which would exist in a fillet welded joint. Parent metal of the stressed member at the toe of a full penetration weld connecting the stressed member to another member slotted through it.		 e = edge distance
a) With attachment length (parallel to the direction of the applied stress) ≥ 150 mm and with edge distance ≥ 10 mm. This classification includes ring stiffeners on tubulars away from nodes.	F	 e = edge distance
b) With attachment length (parallel to the direction of the applied stress) > 150 mm and with edge distance ≥ 10 mm. The decrease in fatigue resistance with increasing attachment length is caused by more force being transferred into the longer gusset, giving an increase in stress concentration.	F ₂	
c) Weld within 10 mm of the edges or corners of a stressed member. The classification applies to all sizes of attachment. It would therefore include, for example, the junction of two flanges at right angles. In such situations, a low fatigue classification can often be avoided by the use of a transition plate [see also joint Type 3.1 e)].	G	
NOTE 1 (Remarks on Potential Modes of Failure) When the weld is parallel to the direction of the applied stress, fatigue cracks normally initiate at the weld ends, but when it is transverse to the direction of stressing, they usually initiate at the weld toe; for attachments involving a single, as contrasted to a double, weld, cracks can also initiate at the weld root. The cracks then propagate into the stressed member. When the welds are on or adjacent to the edge of the stressed member, the stress concentration is increased and the fatigue resistance is reduced; this is the reason for specifying an "edge distance" in some of these joints (see also remark on edge distance in joint Type 2). NOTE 2 (General Comments) i) The stress range considered should be the nominal stress range in the member on which the attachment is welded. ii) When the attachment is subjected to significant forces, affecting the stress flow in the member, another joint type has to be used, e.g. Type 6.6 for gusseted connections.		

Table A.16.10-11—Type 5: Load-carrying Fillet and T-butt Welds

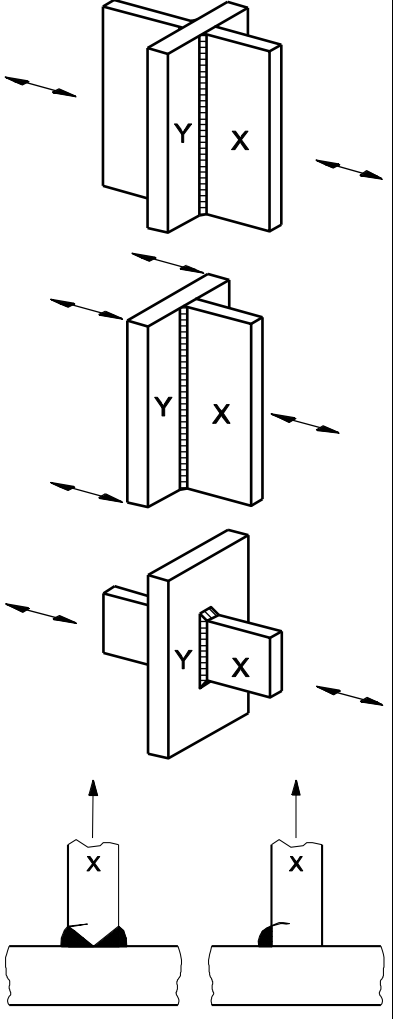
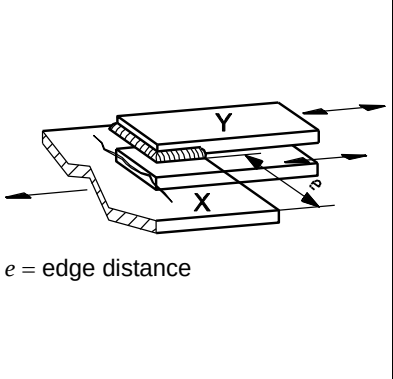
Type Number and Description, Explanatory Comments	Class	Example (Including Failure Mode)
5.1 Parent metal adjacent to cruciform joints or T-joints (member marked X in sketches). a) Joint made with complete joint penetration groove welds and with any undercutting at the corners of the member dressed out by local grinding. b) Joint made with partial penetration or fillet welds with any undercutting at the corners of the member dressed out by local grinding. In this type of joint, failure is likely to occur in the weld throat unless the weld is made sufficiently large (to be checked with joint Type 5.4).	F F₂	
5.2 Parent metal adjacent to the toe of load-carrying fillet welds that are essentially transverse to the direction of applied stress (member X in sketch). a) Edge distance ≥ 10 mm. b) Edge distance < 10 mm. The relevant stress in member X should be calculated on the assumption that its effective width is the same as the width of member Y. These classifications also apply to joints with longitudinal welds only.	F₂ G	 $e = \text{edge distance}$

Table A.16.10-11—Type 5: Load-carrying Fillet and T-butt Welds (*continued*)

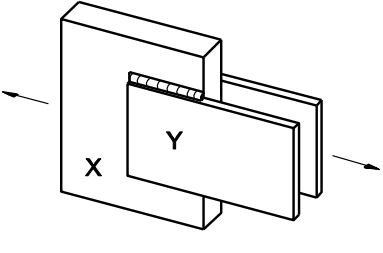
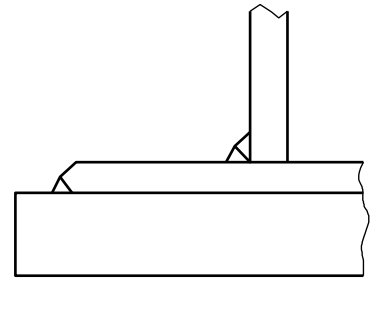
Type Number and Description, Explanatory Comments	Class	Example (Including Failure Mode)
5.3 Parent metal at the ends of load-carrying fillet welds that are essentially parallel to the direction of applied stress, with the weld end on plate edge (member Y in sketch).	G	
5.4 Weld metal in load-carrying joints made with fillet or partial joint penetration groove welds, with the welds either transverse or parallel to the direction of applied stress (based on nominal shear stress on the minimum weld throat area). This includes welds in doubler plates, anodes, pipe supports. In these cases, an assessment of the stress range in the weld throat should be made. In each case, the thickness correction should be based on the weld throat minimum thickness.	W_1	
NOTE (Remarks on Potential Modes of Failure) Failure in cruciform or T-joints with complete joint penetration groove welds will normally initiate at the weld toe, but in joints made with load-carrying fillet or partial penetration groove welds, cracking may initiate either at the weld toe and propagate into the plate or at the weld root and propagate through the weld. In welds parallel to the direction of the applied stress, however, weld failure is uncommon; cracks normally initiate at the weld end and propagate into the plate perpendicular to the direction of applied stress. The stress concentration is increased and the fatigue resistance is therefore reduced, if the weld end is located on or adjacent to the edge of a stressed member rather than on its surface.		

Table A.16.10-12—Type 6: Details in Welded Girders and Tubulars

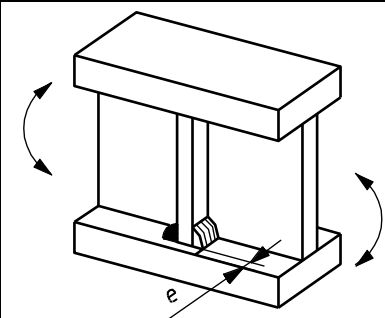
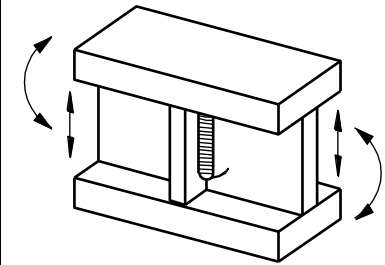
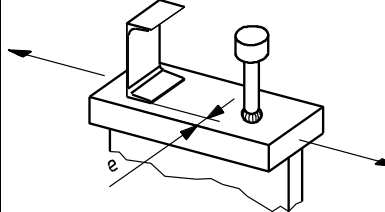
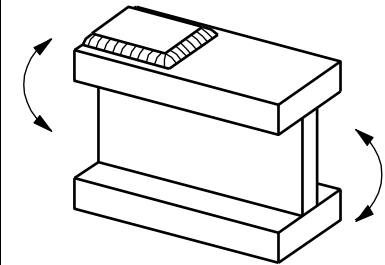
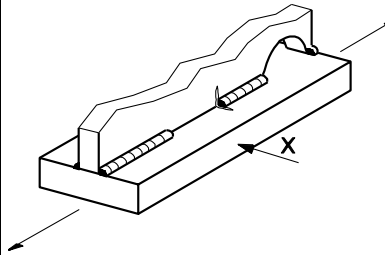
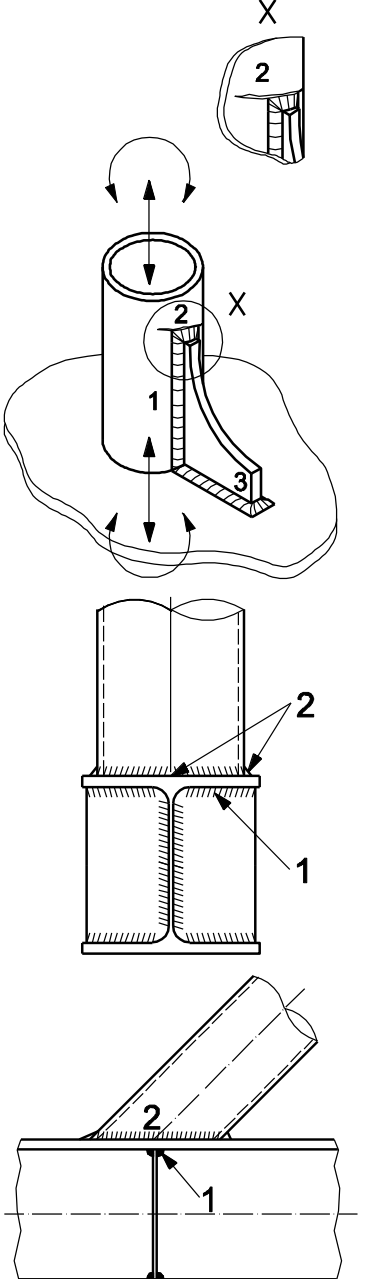
Type Number and Description, Explanatory Comments	Class	Example (Including Failure Mode)
6.1 Parent metal at the toe of a weld connecting a stiffener, diaphragm, etc. to a girder flange. a) Edge distance ≥ 10 mm. b) Edge distance < 10 mm. Edge distance refers to distance from a free, i.e. unwelded, edge. In this example, therefore, it is not relevant, as far as the (welded) edge of the web plate is concerned.	F G	 e = edge distance
6.2 Parent metal at the end of a weld connecting a stiffener, diaphragm, etc. to a girder web in a region of combined bending and shear. This classification includes all attachments to girder webs.	E	 e = edge distance
6.3 Parent metal adjacent to welded shear connectors. a) Edge distance ≥ 10 mm. This classification includes pile-to-sleeve shear connectors. b) Edge distance < 10 mm [see Type 4 c)].	F G	 e = edge distance
6.4 Parent metal at the end of a partial length welded cover plate, regardless of whether the plate has square or tapered ends and whether or not there are welds across the ends. The class includes cover plates that are wider than the flange. However, such a detail is not recommended because it will almost inevitably result in undercutting of the flange edge where the transverse weld crosses it, as well as involving a longitudinal weld terminating on the flange edge and causing a high stress concentration.	G	 e = edge distance
6.5 Parent metal adjacent to the ends of discontinuous welds, e.g. intermittent web/flange welds, tack welds unless subsequently buried in continuous runs. The relevant stress is the nominal stress in the member marked X in the sketch. If the weld is load-carrying, it should be checked with joint Type 5.4. The same, adjacent to cope holes. The existence of the hole is allowed for in the joint classification; it should not be regarded as an additional stress concentration.	F	 X

Table A.16.10-12—Type 6: Details in Welded Girders and Tubulars (*continued*)

Type Number and Description, Explanatory Comments	Class	Example (Including Failure Mode)
6.6 Gusseted connections When the connection is subjected to significant forces, the stress range should be that in the tubular member, the gusset or the weld, including an SCF that arises from the overall form of the joint. To reduce the SCF, it is recommended in example 1 to slot the tubular and place the gusset through. a) Parent metal of gusset plate adjacent to a member or weld in full penetration welds connecting a gusset plate to a member—Location 1. b) Parent metal of a member adjacent to fillet, full or partial penetration welded gusseted connection—Location 2. Full penetration welds are normally required in such joints. c) Weld metal in fillet or partial penetration welds attaching a gusset plate to a member—Location 3.	F F W_1	
NOTE 1 (Remarks on Potential Modes of Failure) Fatigue cracks generally initiate at weld toes and are especially associated with local stress concentrations at weld ends, short lengths of return welds, and changes of direction. Concentrations are enhanced when these features occur at or near an edge of a part (see remarks on Table A.16.10-10). NOTE 2 (General Comments) Most of the joints in this table are also shown, in a more general form, in Table A.16.10-10; they are included here for convenience as being the joints that occur most frequently in girder and tubular joints.		

A.16.11 Fatigue Resistance of the Material

A.16.11.1 Basic $S-N$ Curves

The fatigue assessment of welded joints is based on the assumption that the connection has full-penetration single- or double-sided welding, unless otherwise stated.

Offshore structures are subjected to fatigue due to variable amplitude stresses. However, the prediction of fatigue damage under variable amplitude stresses is a complex subject and the most commonly adopted approach for the assessment of offshore structures is the use of the Palmgren-Miner summation law in combination with fatigue resistance in the form of an $S-N$ curve for constant amplitude stresses. A limited number of variable amplitude fatigue tests on tubular joints have been undertaken and the results compared with constant amplitude $S-N$ curves using an equivalent stress range, which has been defined as the cube root of the average value of (stress)³. This indicates that Palmgren-Miner sum for the mean $S-N$ curve falls essentially within the range of 0.5 to 2.0, with an average value of 1.8. This is comparable with the results from constant amplitude tests. A significantly larger number of test results are available for plate joints, which give an average Palmgren-Miner's sum of 1.1.

The $S-N$ curves for tubular joints and castings are based on a comprehensive review of fatigue data for both tubular and plated joints. The background information is presented in References [A.16.11-1] and [A.16.11-2]. The $S-N$ curves apply to crack growth associated with through-thickness cracking.

The tubular joint $S-N$ curves have been derived from an analysis of data on tubular joints manufactured using welds conforming to a standard flat profile as given in AWS D1.1/D1.1M [A.16.10-2]. Therefore, the fatigue recommendations apply to joints, which conform to this AWS standard flat profile.

US investigations in this field have been carried out by EWI [A.16.11-2] on behalf of API. Both the HSE [A.16.11-1] and EWI [A.16.11-2] investigations concur on the general form of the $S-N$ curves. The basic $S-N$ curves relate to in-air conditions. Separate curves are presented for joints in seawater with adequate corrosion protection (i.e. from -850 mV to -1100 mV). Fatigue data for tubular joints indicate that, in general, there is a reduction in the fatigue performance in seawater under cathodic protection for stress range magnitudes corresponding with resistances less than 10^6 cycles for in-air conditions. For smaller stress range magnitudes and corresponding resistances greater than 10^6 cycles in-air, the fatigue lives in seawater are restored to that of in-air conditions. References [A.16.11-3] and [A.16.11-4] present the results from fracture mechanics evaluations and illustrate the detrimental effect of seawater relative to air for joints with and without adequate cathodic protection. In normal design of offshore structures, the most significant damaging stress range cycles correspond with resistances greater than $N = 10^6$ cycles, such that the detrimental reduction in fatigue life for cathodically protected joints is not apparent. The basic $S-N$ curves given in Table 16.11-1 are therefore applicable for these conditions. In instances where significant damage occurs for stress range cycles corresponding with resistances less than $N = 10^6$ cycles, a penalty factor of 2 on N is recommended. For joints in freely corroding conditions, or for joints with corrosion protection levels more negative than -1100 mV, a factor of 3 on N for all resistances, without a change of slope at 1×10^7 cycles, is recommended.

A number of tubular joints used in deriving the basic TJ $S-N$ curve had chord and braces with equal diameters (i.e. $\beta = 1.0$). Some of these joints showed extensive weld inter-run cracking in preference to weld toe cracking. This can be significant in relation to the application of weld toe grinding improvement techniques, since clearly toe grinding of the chord or brace weld toes alone does not improve the fatigue performance of these joints. Improvement would only be achieved if the weld face is also toe ground to remove all of the inter-run toes. However, an assessment of the $\beta = 1.0$ joints, using the recommended SCF equations in A.16.10.2.2, indicated that the predicted fatigue lives are significantly above the TJ curve.

A.16.11.2 High-strength Steels

A limited amount of test data for plate joints with representative yield strengths up to 540 MPa [A.16.11-1] and tubular joints manufactured from some modern high-strength steels [A.16.11-3] have suggested that the fatigue performance in seawater under cathodic protection and under free corrosion is similar to that for medium strength structural steels, thus allowing the $S-N$ curves in Table 16.11-1 also to be used for these steels. Alternatively, test data or fracture mechanics analysis should be used to determine appropriate $S-N$ curves.

High-strength steels are increasingly being used in the fabrication of offshore structures, particularly for jack-up legs, which are made from steels with typical yield strengths of 700 MPa to 800 MPa. The effect of seawater on the fatigue performance of these materials is thought to be more detrimental than for medium strength structural steels because of their greater susceptibility to cracking from hydrogen embrittlement under variable stresses in seawater [A.16.11-3]. The susceptibility to hydrogen embrittlement increases with increasing yield strength and increasingly negative cathodic protection potential. A number of studies have identified excessively negative cathodic protection potential as a cause of cracking due to the generation of hydrogen which enhances crack growth rates at the crack tip [A.16.11-3]. Evidence of hydrogen cracking found in jack-ups during routine surveys has been reported in Reference [A.16.11-5]. It is therefore important that the fatigue performance of selected high-strength steels is understood and that appropriate levels of cathodic protection are applied.

A.16.11.3 Cast Joints

The $S-N$ curve for cast joints has been derived from tests in air on large-scale cast joints with thicknesses in the range of 18 mm to 40 mm, tested principally at $R = -1$ (i.e. stresses varying from a minimum in compression to an equal maximum in tension), and cruciform specimens with thicknesses in the range of 38 mm to 125 mm tested at $R = 0$. Similar mean curves are obtained from the two sets of data using an inverse slope of 4 [A.16.11-1]. Since cast joints are stress relieved, the R -ratio has an influence on the fatigue behavior. The $S-N$ curve for the test data could therefore overestimate the fatigue performance of cast joints tested at $R > -1$. Hence, allowance has been made for the influence of mean stresses by applying a 20 % reduction to the maximum experimental stress range used to determine the CJ curve.

There is insufficient experimental evidence to justify a change in slope in the low stress/high cycle region, the highest experimental resistance being 5×10^6 cycles. A conservative approach of using a constant slope of $m = 4$ for all numbers of cycles is therefore recommended.

Fracture mechanics analysis shows that casting defects can have a significant effect on the fatigue resistance, and the design curve corresponds to four standard deviations in N below the mean curve to allow for the possibility of undetected defects [A.16.11-6]. The curve is applicable to castings, which satisfy defect acceptance criteria compatible with current offshore practice; Reference [A.16.11-1] contains further information.

In order to determine whether weld repairs could be detrimental to the fatigue performance of cast joints, fatigue tests on cruciform specimens in both air and seawater were undertaken [A.16.11-7]. These tests showed that, provided weld repaired surfaces are ground flush to the as-cast profile and free from weld toe defects, the CJ $S-N$ curve may be used for cast joints with weld repairs (see also Reference [A.16.11-1]).

The limited data for castings in seawater (under conditions of cathodic protection and free corrosion) suggest that the environmental reduction factors on resistance from A.16.11.1 should be used.

The fatigue assessment of cast joints requires that FEA is performed to determine the location of the maximum stress range in the casting. For cast tubular nodal connections, the brace to casting circumferential butt weld can be the most critical location.

A.16.11.4 Thickness Effect

An assessment by HSE [A.16.11-1] and EWI [A.16.11-2] of a wide range of data for various combinations of member forces and/or moments has shown that the fatigue performance is dependent on the member thickness, the performance decreasing with increasing thickness for the same stress range.

The TJ and OJ representative curves are based on a material thickness of 16 mm. EWI derived a thickness exponent of 0.29, compared with 0.3 in the HSE investigations. API uses a value of 0.25, whereas in NORSOK the range is from 0.15 to 0.25 and work by DNV has indicated that 0.25 is generally applicable unless there are high SCFs; consequently a value of 0.25 has been selected for Equation (16.11-2).

The basic material thickness for the CJ representative curve is 38 mm. Fracture mechanics predictions [A.16.10-2] show that the thickness effect in castings is smaller than in welded joints and an exponent of 0.15 is specified.

A.16.12 Fatigue Assessment

A.16.12.1 Cumulative Damage and Fatigue Life

The Palmgren-Miner rule is not precise. Its accuracy depends on issues such as stress range history, type of constructional detail, and failure definition. For tubular joints in offshore structures, the Palmgren-Miner rule can be somewhat unconservative. However, this unconservatism is generally more than compensated by bias (and scatter) associated with the GSR and the $S-N$ curve. Hence, adjustment of the Palmgren-Miner rule is not considered necessary at this time.

Fatigue life estimates should be viewed with scepticism. The $S-N$ curves used in design are representative curves (e.g. 95 % confidence of 97.5 % survival), as is typical of land based structures like bridges. This survival level implies that there is little chance of failure at the predicted life. An estimate of mean fatigue life using the mean $S-N$ curve is generally several times the calculated life using the design $S-N$ curve. In addition to uncertainties associated with wave load/action, the primary uncertainties and probable conservatisms in fatigue life estimates stem from:

- a) the GSR history, and
- b) the $S-N$ curve assumption in the low- S –high- N (the very long-life) regime, for which there are few supporting data.

The reliability of fatigue damage estimates is not yet established because full calibration of the analysis techniques, including both failures and nonfailures, has not broadly occurred.

Fatigue life estimates of very short duration are even more doubtful when they are based on long-term wave histories, taken over a period of years. Short-term conditions can easily deviate from average conditions and can be better or worse than average.

Fatigue life estimates can be evaluated in a probabilistic framework. For more information, the user should refer to References [A.16.12-1] and [A.16.12-2]. However, broad ranges of probabilities are possible. The uncertainties mentioned above affect probabilistic as well as deterministic fatigue life estimates. Probability estimates are most useful when used in relative terms.

For structures that need their life to be extended or are being re-used or converted to a new application, prior damage [D_1 in Equation (16.12-3)] should be estimated via inspection findings. An absence of crack discoveries should *not* be assumed to mean no prior damage has occurred. For example, if MPI reveals no evidence of defects, it may be assumed that the prior damage in terms of the Palmgren-Miner sum is limited to 0.3 for a welded tubular joint and to 0.5 for a welded plate detail. Assuming a defect-free inspection, lower values of assumed damage can be justified by the designer based on analysis, if the prior history of the structure can be established with confidence. However, a value of zero is normally only used for those details that will be modified so as to eliminate prior damage.

A.16.12.2 Fatigue Damage Design Factors

Fatigue damage design factors for fatigue of steel components primarily depend on failure consequence and in-service inspectability. Failure criticality is normally established on the basis of redundancy analyses. A structure with redundancy, capability for in-service inspection, and the possibility for repair/strengthening is preferred, especially for the design of a new structural concept or a conventional structure for new environmental conditions. Joints in the splash zone are considered as “not inspectable.”

Both the factor of 5 and the factor of 10 in Table 16.12-1 imply that a significant change in fatigue reliability only occurs when there is a significant change in the predicted life (or Palmgren-Miner damage sum) for the design service life of the structure. As the slopes of the $S-N$ curves vary between 3 and 5, values $1 < \gamma_{FD} \leq 10$ are equivalent to much smaller changes in the calculated stresses of from 1.38 S to 1.71 S for a factor of 5 and from 1.58 S to 2.15 S for a factor of 10.

The fatigue damage design factors do not differentiate between fatigue analysis procedures. At present, there is little certainty in how the various procedures compare in terms of reliability, so the same set of explicit fatigue damage design factors is generally applied to all of them. The fatigue damage design factors also do not differentiate such aspects as risk to assets and difficulties associated with repairs, or lost production due to repairs. It is the owner's responsibility to assess how these sorts of risk should be addressed in the design phase.

There are instances where the cited fatigue damage design factors may be reduced. An example could be a structural component above water, for which inspection can be either easier or more frequent. A reduction in fatigue damage design factor can also be appropriate if loss of the component does not jeopardize personnel safety or the environment. Finally, smaller fatigue damage design factors can be justified if the fatigue analysis algorithm has been calibrated to the structural type and design situations being considered. However, use of reduced factors should generally require some form of documented justification.

In selecting fatigue damage design factors, inspectability and inspection technique need careful consideration. In general, the in-service inspection for fatigue damage should be more thorough than the general level II survey described in API 2SIM. Locations below 200 m under the water surface should generally be assumed to be uninspectable. For some complex joints, such as internally stiffened ones, cracking can originate from the inside (hidden) surfaces. Hence, the need and possibility for inspection prior to crack penetration through the thickness should be considered at the design stage. A trade-off exists between, for example, introducing a fatigue damage design factor of 5 (for a component that is not failure critical) and inspecting in-service with a more complex technique such as ultrasonics. As ultrasonic inspection normally requires the use of diving technicians, the risk of diving should be included in the considerations.

Even though a given component is considered readily inspectable from exposed surfaces, inspection frequency should still be balanced with the fatigue damage design factor. In some structural components, such as those with low SCFs, the rate of crack growth can surpass normal circumstances and inspection intervals.

Despite the need to address inspectability during the design phase, there is no implied requirement to perform a regular, detailed inspection of each and every joint for which a fatigue damage design factor from the inspectable category is adopted. The scope and frequency associated with the inspection plan involve considerations that extend well beyond the issue of the fatigue analysis recipe alone. However, if no inspection is clearly intended from the start for a particular class of joint, then the fatigue damage design factor should be selected from the not inspectable category.

A.16.12.3 Local Experience Factor

The default value of the local experience factor, k_{LE} , is 1.0. (See the regional information in API 2MET for some cases where available data support the use of a factor that is different from 1.0.) In other cases, a factor different from 1.0 should be substantiated by reliable evidence and should be fully documented.

A local experience factor may be used for the fatigue life estimation of a new structure design, for the extension of the fatigue life of an existing structure, and for an assessment of the fatigue life of a structure that is re-used or converted to a new application. If such a structure has accumulated prior fatigue damage, any local experience may only be used for an assessment of the remaining fatigue life.

As the fatigue process is complex and fatigue damage is sensitive to many different factors, local experience with existing structures can only be considered relevant for the fatigue assessment of a structure under consideration if the following conditions are satisfied:

- a) there is a database with fatigue data from existing structures in-place that have not suffered from noticeable fatigue damage during time periods exceeding the predicted fatigue lives for these structures;
- b) the environment to which the in-place structures are subjected is essentially the same as the environment that the structure under consideration will be subjected to; this applies in particular, but not exclusively, to the wave environment;
- c) the structure under consideration is of the same general type, is of similar configuration, and has comparable dynamic characteristics as existing structures;
- d) the structure under consideration contains similar constructional details as existing structures;
- e) the structure under consideration is of similar material and fabricated in a comparable manner as existing structures.

A.16.13 Other Causes of Fatigue Damage Than Wave Load/Action

A.16.13.3 Wind-induced Vibrations

Where necessary, measures should be taken to avoid wind-induced vibrations of members during fabrication. If this is not possible, or the measures are not successful, the resulting fatigue damage during the fabrication period for the members involved should be estimated and included in the overall fatigue damage assessment.

A.16.13.4 Transportation

Guidance on fatigue assessment during transportation is given in Reference [A.16.13-1]. The damage assessment during transportation should use reliable statistical information on the wave conditions along the tow route in the expected tow season. Normally, these should be provided in the form of wave scatter diagrams.

Spectral analysis techniques should be used to account for the random nature of the wave environment and for compatibility with standard vessel motion prediction techniques. The motions of the structure during the tow should be evaluated using either model tests or a suitable vessel motions analysis program. The possibility of different wave directions relative to the tow occurring should be taken into account. Structural analyses should be performed to evaluate the stresses due to motions and deformations during tow conditions. For structures transported on barges, the stiffnesses of the barge, structure, and sea-fastening should be taken into account.

As the actual weather conditions experienced during the tow can deviate significantly from the statistical information for the tow season, no matter how reliable this information is, it is recommended that sensitivity analyses be performed using a modified scatter diagram or diagrams. The modified scatter diagram(s) could, for example, include one occurrence of the most severe sea state that can occur along the tow route during the tow period, increased probabilities of occurrence of lesser sea states, and/or increased probabilities of occurrence of unfavorable wave directions relative to the tow. If the sensitivity analyses indicate that fatigue damages during transportation are always acceptable, no further consideration is necessary. However, if the sensitivity analyses suggest that unacceptable fatigue damage can occur, further consideration of the transportation phase is warranted.

A.16.13.5 Installation

Driving fatigue should be considered for piles and driven caissons/monotowers.

Design measures to minimize potential fatigue damage during pile driving include:

- relocating attachments to avoid the sensitive areas,
- using substantial connections for those attachments that cannot be relocated,
- using doubler plates for connections to main members, and
- using welded rather than screwed or bolted fittings for piping.

A.16.14 Further Design Considerations

A.16.14.4 Miscellaneous Load-carrying Attachments

Where attachments are located close to supporting structural members, the effect of flow enhancement of the free stream flow should be taken into account.

Where doubler plates are used, these should be designed such that the connection of the attachment to the doubler plate fails before the connection of the doubler plate to the primary member.

A.16.14.6 Members in the Splash Zone

Plan framing levels immediately below or above the water surface are immersed and emerged during the passage of waves exceeding a certain threshold height. Normal wave load/action on these members is highly nonlinear and is most likely not reliably represented in the spectral or other analysis methods. The manner in which the load/action on these members is included will differ from program to program and should be carefully reviewed. A separate evaluation based on deterministic analysis principles using estimated loads/actions and numbers of cycles for those waves that exceed the threshold should be performed to account for these effects.

Variations in buoyancy due to the passage of waves can also cause cyclic forces in members in the splash zone. If these buoyancy variations are not explicitly represented in the calculation algorithms of the computer program used, the effect of variable buoyancy loads/actions on member stresses should be assessed and included, if significant, before fatigue damage calculations are performed.

Members located near the water surface can further be subjected to wave slamming loads/actions in addition to the normal wave loads/actions that are included in the fatigue analysis. The slamming loads/actions cause vibrations of the member and are largest for horizontal members lying parallel to the wave crest. Guidance on estimating slamming loads/actions can be found in the general literature. Where relevant, the fatigue damage associated with vibrations due to slamming should be included in the assessment.

A.16.14.8 Inspection Strategy

During the design of new structures, it is common to make initially conservative assumptions (e.g. with respect to SCF or $S-N$ curve selection) when performing the fatigue analysis to screen the structure with a view to identifying potential problem areas. These areas are then examined in more detail by refining the analysis assumptions and eliminating conservatism where possible and justified. Consequently, the ranking of calculated fatigue lives does not necessarily reflect the true ranking if all structural details are not refined to a common level. This observation should be considered if fatigue lives are used for prioritizing areas of the structure for inspection.

Similarly, the approximations in the analysis procedure should be reviewed if low calculated fatigue lives based on approximate methods of analysis are used as the basis for an extensive inspection effort. In such cases, consideration should be given to performing a spectral analysis to refine the calculated lives where possible in order to try and reduce the inspection effort.

Generally, an initial inspection interval of one quarter of the calculated fatigue life is recommended for flat plate and welded tubular joints. Consequently, joints with calculated fatigue lives of up to four times the structure design service life would normally require inspection for fatigue once during the design service life.

A.16.15 Fracture Mechanics Methods

A.16.15.1 General

The benefits of defect assessment procedures (see, for example, Reference [A.16.15-1]) for the fitness-for-service assessment of offshore structures are widely recognized, and defect assessment is increasingly being used in design and during fabrication and in-service inspection. However, existing procedures are based on general principles, and their application to tubular joints—which is a particularly complex problem due to the nature of the loads/actions and the geometry of the joint—requires special care. Detailed information is given in References [A.16.15-2], [A.16.15-3], and [A.16.15-4].

A.16.15.2 Fracture Assessment

Static failure can occur by fracture or by plastic collapse, if a crack propagates to a critical size. Failure assessment procedures combining considerations of failure by fracture and by plastic collapse are given in Reference [A.16.15-1]. The fracture assessment should be performed for each increment of crack growth.

The possibility of a fracture failure is normally greater in the brace than in the chord, since there is a greater likelihood of the crack propagating into material with lower toughness in the weld region. Further details can be found in Reference [A.16.15-1].

A.16.15.3 Fatigue Crack Growth Law

Recommended values for the fatigue crack growth rate parameters, C and m , are presented in Reference [A.16.15-5].

The complexity of the normalized stress intensity factor (Y) usually requires the use of numerical integration methods. It is important that a sensitivity analysis is performed to establish that convergence is achieved using the selected increment of crack growth.

Knowledge of the initial crack (defect) size is essential in any fracture mechanics analysis. The predicted number of cycles to failure is sensitive to this parameter. It can be necessary to perform a sensitivity analysis to determine the significance of the assumed initial defect size.

For assessments performed during design, it is normal to assume that the initial crack (defect) is located at the position of the maximum GSR. The assumed initial crack (defect) size, a_i , should be based on the accuracy of the NDT method used during fabrication. Inspection data can be highly variable, and there is an increasing tendency toward the use of probability of detection (POD) curves. Where there is sufficient information, the minimum detectable defect size may be based on the 90 % POD, 95 % confidence level. However, this depends on the inspection method used and the selection of the minimum detectable defect size requires specialist advice.

The final crack size, a_f , is limited to the critical crack size in view of either fracture toughness or plastic collapse (see A.16.15.2).

The assessment of a crack that is detected during inspection requires information on the crack location, depth from the surface, and length along the surface of the material.

A.16.15.4 Stress Intensity Factors

The principal sources of normalized stress intensity factors, Y , are as follows:

- a) standard solutions for semi-elliptical cracks in plates (see, for example, Reference [A.16.15-5]) used in conjunction with magnification factors and the moment release method);
- b) numerical methods by FEA or boundary element analysis;
- c) analytical methods (e.g. using weight functions);
- d) empirical methods, involving the assessment of fatigue crack growth rate data.

The use of plate solutions is generally conservative, particularly for deep cracks, but provides stress intensity factors for a very wide range of parameters. The use of the moment release method in conjunction with plate solutions reduces the excessive conservatism, and this approach is recommended for general use. The moment release method assumes that the net force acting across the cross-section of a cracked tubular member in an offshore structure, and hence the stress intensity factor, is reduced by the redistribution of forces. The linear moment release method involves the reduction of the bending stress component using the following simple expression:

$$\sigma_{b,c} = \sigma_b (1 - a/T) \quad (\text{A.16.15-1})$$

where $\sigma_{b,c}$ and σ_b are the bending stress components in the cracked and uncracked joint, respectively, and T is the wall thickness.

The availability of stress intensity factor solutions for welded joints is limited. In Reference [A.16.15-1], the stress distribution through the wall thickness is approximated to a combined bending and membrane stress field; further known stress intensity factor solutions for cracks in plates (see, for example, Reference [A.16.15-5]) are used in conjunction with a magnification factor to account for the influence of the stress concentration at the weld toe.

Magnification factors have been derived from two-dimensional FEA of butt and fillet welded joints, and this approach is generally conservative. Alternatively, stress intensity factors can be evaluated by analysis or by approximate analytical methods, e.g. weight function analysis.

Models that take account of the variation of the GSR around the brace/chord weld periphery have also been developed [A.16.15-6], but validation is limited.

Below a certain threshold value of the stress intensity factor range (ΔK), no crack growth will occur. Recommended threshold values (ΔK_{th}) for carbon and carbon manganese steels in air as well as in seawater with cathodic protection are $\Delta K_{th} = 170 \text{ Nmm}^{-3/2}$ at $R = 0$ (tensile stress ranges between zero and a maximum stress) and $\Delta K_{th} = 63 \text{ Nmm}^{-3/2}$ at $R = 0.5$ (tensile stress ranges between half the maximum and the full maximum stress). For seawater with free corrosion, a threshold stress intensity factor range $\Delta K_{th} = 0$ is recommended.

A.16.15.5 Fatigue Stress Ranges

For a fatigue crack growth analysis, the full stress range should always be used, even when the calculated stress range cycle is partially compressive, since the stress range can become effectively tensile due to the effects of mean and residual stresses.

A.16.15.6 Castings

The through-thickness stress distribution should be determined using FEA. The fatigue crack growth rate data for typical structural steels are generally found to be also applicable to cast material. However, where significant differences in chemical composition or mechanical properties exist, it can be necessary to obtain specific fatigue crack growth rate data.

A.16.16 Fatigue Performance Improvement Techniques

A.16.16.1 General

Post-weld improvement techniques may be used to improve fatigue performance. These techniques, discussed below, improve fatigue performance by improving the local geometry at the weld toe, by reducing the stress concentrations and/or by modifying the residual stresses. All these effects are included in the improvement factors on fatigue performance for each technique. Specific requirements for the various techniques are noted or referenced below.

The GSR to be used for an assessment of the improved performance should be obtained from equivalent joints before the improvement technique is applied, from FEA or from SCF equations. They should not be obtained from measurements on improved joints.

The designer should be wary when applying weld improvement techniques, especially a powerful one like peening. If later fatigue cracking occurs, it should *not* be expected to originate at the treated location. However, if cracking does originate at a treated weld toe, the life associated with subsequent propagation is likely to be much shorter than is normal for untreated details.

Except as noted below, multiple improvement factors should not be considered for a single joint location. If more than one technique is applied, only the one giving the highest improvement factor should be considered.

A.16.16.2 Post-weld Heat Treatment

As-welded joints contain significant tensile residual stresses induced by the welding process, which combine with the operating stresses to promote fatigue failure. This is due to the increase of the effective mean stress and, for situations where the stress range consists of a compressive component, of the effective stress range. It follows that the reduction of tensile residual stresses can increase the fatigue performance.

A comparison of the fatigue behavior of as-welded and post-weld heat treated joints has confirmed that PWHT can have a beneficial effect on the fatigue behavior of welded joints. However, the effect of PWHT diminishes with increasing R ratio and is negligible at $R > 0$ (all tensile stress ranges). Thus, the fatigue performance of post-weld heat treated and as-welded joints at R ratios greater than zero are very similar and the same $S-N$ curves apply.

A significant drawback of accepting PWHT as a fatigue improvement measure in design is that knowledge of the residual stress distribution, including the contribution of long-range fit-up stresses, is required. This information is not usually available.

Nevertheless, prefabricated welded nodes with fully ground profiles and PWHT may be treated as the equivalent of cast nodes with weld repair, provided the local stress intensification of the fillet radius is accounted for in design.

A.16.16.3 Weld Profiling

Limited investigations of the influence of weld profile on the fatigue resistance of tubular joints have shown there is no clear evidence that weld profiling leads to improved fatigue performance [A.16.16-1]. The basic TJ $S-N$ curve has been derived from an analysis of data on tubular joints manufactured using welds conforming to a standard flat profile given in AWS [A.16.10-2]. Therefore, the fatigue recommendations apply to joints, which conform to this AWS standard flat profile.

Improvements through any form of profiling can be justified using information from an appropriate test program for tubular joints for the condition being considered.

A.16.16.4 Weld Toe Grinding of Tubular Joint Welds

For welded joints in-air and for joints in seawater with adequate cathodic protection, the fatigue performance can be increased by controlled local machining or grinding to produce a smooth concave profile at the weld toe. This is especially beneficial at low stress ranges. Experimental data indicate that this technique can lead to an increase in the fatigue performance by a factor of approximately 2. It should be noted that the beneficial effect of weld toe grinding can be reduced by free corrosion, though it tends to be restored by cathodic protection (see References [A.16.16-2] and [A.16.16-3]).

A limited number of tests have demonstrated the importance of QC of the grinding procedure. The grinding procedure should ensure that all defects in the weld toe region have been removed by grinding to a depth not less than 0.5 mm below the bottom of any visible undercut or defect. The maximum depth of local grinding should not exceed 2 mm or 5 % of the plate thickness, whichever is less. NDT of the joint is required after grinding to verify that no significant defects remain, and, for fillet welded connections, it is important that the required throat size is maintained. Further QC aspects apply, particularly on the shape of grinding burrs and their orientation (see Reference [A.16.16-2]).

A.16.16.5 Grinding of Butt Welds

For butt welded connections, additional benefit can be gained by grinding the weld cap flush. The effect of this is to improve the joint classification (see Table A.16.10-9). For welded tubular nodes, full grinding of the surface profile to a radius of not less than $0.5r$ qualifies for both the life improvement factor of 2 on curve for welded joints, and the 0.15 size effect exponent applicable to geometrically similar notch-free scale-ups.

A.16.16.6 Hammer Peening

By hammer peening the toes of welded connections, surface defects can be eliminated, the transition between the parent and weld material is smoothed out, and beneficial compressive residual stresses are induced at the surface, all of which contribute to the enhancement of the fatigue performance of the treated weld. The net effect is to delay crack development and retard or eliminate growth of cracks already present.

The objective in hammer peening is to obtain a smooth groove at the weld toe. The grooved depth should be at least 0.3 mm, but should not exceed 0.5 mm [A.16.16-4, A.16.16-5]. The equipment and procedure required to attain this groove configuration should be established via trials on detail mock-ups. Note that the number of passes required is determined by the equipment and procedure. Heavy duty pneumatic hammers are preferred. The bit tip radius should be about 3 mm, so as to expedite the process and facilitate treatment right at the weld toe. Extensive use of peening has ergonomic implications. Consideration should be given to limiting the consecutive hours spent peening by one individual and vibration dampening gloves should be used. Peening can result in metal rollovers along the sides of the groove. Tests have shown that such rollovers can be deleterious to fatigue performance and should be removed [16.16-6]. Removal eliminates difficulty with interpretation of later inspection findings. Peened weld toes should be inspected by MPI directly after peening and any burr grinding.

The recommended fatigue performance improvement factor is 4. This value is significantly less than that found in many test programs and varies with stress range magnitude and other variables. The reduced value takes into account uncertainties in:

- a) mean stress,
- b) dominant stress range magnitude, and
- c) the effects of overloads.

The fatigue performance improvement factor may be applied to both tubular and nontubular weld details. At nodes constructed by nodal fabrication, the effectiveness of peening is improved if it is performed after fit-up of the associated brace.

The benefits of hammer peening on fatigue performance can only be realized through adoption of adequate QC procedures. References [A.16.16-4] and [A.16.16-5] contain the state-of-the-art in this field and should be consulted in the preparation of adequate QC procedures prior to taking benefit for fatigue performance enhancement.

A.17 Foundation Design

A.17.1 General

A.17.1.3 Exposure Levels

The reasons for the distinction between new and existing structures include the following.

- a) The reliability of the foundation design depends on the type of soil, its variability, the amount of data collected, and the suitability of the design method to the specific conditions. Experience indicates that there is a tendency for less data to be available at the design stage for exposure level L-2 and exposure level L-3 structures than for exposure level L-1 structures. For existing structures, installation records, e.g. pile driving records, generally provide further relevant data for exposure level L-2 and L-3 structures.
- b) If a foundation component such as a pile fails, it will probably cause the structure to become unserviceable. Repair is likely to be impossible, and at best will be difficult and expensive. Thus the balance between reduced cost and higher reliability normally favors reliability. In particular, exposure level L-2 and L-3 structures frequently have nonredundant foundations and the failure of one pile can lead to collapse. However, for existing structures the cost and uncertainty of intervening in the foundation systems puts the balance in favor of reduced reliability for exposure level L-2 and L-3 structures.

A.17.2 Pile Foundations

A.17.2.3 Drilled and Grouted Piles

There are two types of drilled and grouted piles.

- a) Single-stage piles: For the single-stage drilled and grouted pile, an oversized hole is drilled to the required penetration, a pile is lowered into the hole and the annulus between the pile and the soil is grouted. This type of pile can be installed only in soils that will hold an open hole to the surface. As an alternative method, the pile with expendable cutting tools attached to the tip can be used as part of the drill string to avoid the time required to remove the drill bit and insert a pile.
- b) Two-stage piles: The two-stage drilled and grouted pile consists of two concentrically placed piles grouted to become a composite section. A pile is driven to a penetration that has been determined to be achievable with the available equipment and below which an open hole can be maintained. This outer pile becomes the casing for the next operation, which is to drill through it to the required penetration for the inner or “insert” pile. The insert pile is then lowered into the drilled hole, and the annuli between the insert pile and the soil and between the two piles are grouted. The diameter of the drilled hole should be at least 150 mm (6 in.) larger than the insert pile diameter.

A.17.3 General Requirements for Pile Design

A.17.3.4 Foundation Capacity

- c) Pile overturning resistance: Experience has shown that due to the prolonged large deflection of caissons and other more flexible structures, the soil at and near the surface is subject to substantial degradation and frequently loses contact with the caisson for a short distance below the surface. This loss of soil strength due

to remolding and the effective increase in unsupported length of the caisson should be considered in determining dynamic effects and the resulting bending stresses, as well as in estimating tilt.

After severe storms in the Gulf of Mexico, caissons have been observed to be leaning with no visible overstress or damage to the caisson. This may have been caused by inadequate penetration that resulted in the ultimate lateral resistance of the soil being exceeded. Caissons and similarly founded structures should be designed for lateral loading in accordance with 17.8 with sufficient penetration to ensure that the analysis is valid. Analysis procedures using “fixity” at an assumed penetration should be limited to preliminary designs only. The partial resistance factors are given to provide a foundation overturning capacity equivalent to that provided with coupling the axial resistance of multiple piles.

The conventional beam-column model assumed in the development of the original basis for the requirements of 17.3.4 c) should not be expected to be suitable for short, large diameter caissons, e.g. monopiles, and suction piles, even though the penetration satisfies the criteria in 6.1.1. Annex E of API 2SK ^[A.17.3-1], Third Edition, contains some guidance on the limitations of the conventional beam-column model for short piles, as well as some alternative models. It should be recognized, however, that specific design recommendations in that document are intended for the design of anchoring components that have loading and service conditions different from structures included in the scope of this document. Those recommendations should be used for fixed offshore structures only to the extent that the designer can determine that they are relevant.

The Second Edition of API 2A-LRFD allows, but does not require, inclusion of the contribution of skin friction to the representative foundation overturning capacity of 17.3.4 c). This document provides no guidance on methods for determining the contribution. The guidance should be obtained from an experienced geotechnical specialist familiar with the considerations to be evaluated when including the skin friction contribution. Caution should be exercised so that inclusion of the skin friction contribution does not lead to an inadequate penetration condition similar to those that resulted in exceeding the ultimate lateral resistance of the soil as previously observed in the Gulf of Mexico.

A.17.10 Pile Wall Thickness

A.17.10.4 Check for Load Case due to Weight of Hammer during Hammer Placement

Designers should use the K factor that is appropriate for the cantilevered end-span of a continuous beam, which differs from the theoretical or commonly recommended K value for a fixed-end cantilever. The minimum effective length factor, K , of 2.1 is suggested for pile and conductor add-ons for typical installations with a minimum moment reduction factor, C_m , of 1.0.

In the beam column resistance check, for either nearly vertical or inclined add-on sections, the recommended method of calculating the secondary $P-\Delta$ moments due to first-order lateral deflections is more accurate ^[A.17.10-1] than the familiar procedure of modifying the primary support moment (M) by means of the moment reduction factor C_m as discussed in Section 13.

A.17.10.8 Driving Shoe

The designer should be aware that pile buckling and pile refusal incidents in very dense sands have been associated with the use of external chamfers at the pile tip. Although factors other than the shape of the pile tip contribute to buckling, the use of an external chamfer can increase the potential for buckling and/or refusal.

A.19 Materials

A.19.2 Design Philosophy

A.19.2.1 Material Characterization

Key characteristics of structural steel are as follows.

- a) **Strength:** Strength is the primary parameter for structural steel and the yield strength is used throughout Sections 13 to 16 to determine the resistance of structural components. The strength is measured by tensile testing of specimens. Fatigue resistance of as-welded structures (Section 16) tends to fall in the same scatter band, regardless of strength, so increasing design stresses for high-strength steel tends to shorten fatigue lives.
- b) **Toughness:** Toughness becomes more important as the magnitudes of varying loads/actions increase and as service temperatures decrease. Whereas fracture control is based on preventing the initiation of cracks, notch toughness of the steel affects both terminal fatigue condition and early fracture resistance. Toughness can be measured or controlled by the following:
 - 1) controlling the chemistry and processing of the steel,
 - 2) CVN testing, or
 - 3) CTOD testing.

Both Charpy testing and CTOD testing are undertaken at temperatures related to the lowest temperature to which a structure is expected to be exposed.

A.19.2.2 Material Selection Criteria

A.19.2.2.3 Component Criticality

Connections whose failure would endanger the entire structure should receive detailed consideration, particularly if they are subject to factors known to enhance the risk of fracture. Such factors include conditions of high restraint (e.g. adverse geometry, multi-axial stresses, thick sections), high residual stresses arising from fabrication, through-thickness shrinkage strains after welding, material properties, and the possibility of hydrogen pickup.

Practice will vary from one region to another, but for more severe environments such as the North Sea consideration should be given to the notch toughness and its variation with temperature. In some areas, the structural steel used for construction of a fixed steel structure will generally be the same above and below water and will be selected with reference to the most severe anticipated conditions. Applications such as subsea templates and other structures where the steel is completely and permanently submerged may use less stringent temperature requirements for notch toughness. Where high-strength steels have been utilized in fatigue conditions, the level of cathodic protection provided should be taken into account.

Particularly critical joints can benefit from the use of tougher class CV2 steels.

Those structures continuously above the water-line, or only intermittently submerged in the splash zone, usually experience lower temperatures in winter and can additionally suffer shock loading from causes such as wave slam and boat impact. For these structures and any other critical structures or connections where it is recognized that a significant risk of brittle fracture exists, the tougher steels of class CV2 should be considered. Special attention should be given to welding procedures for higher-strength steels of group IV and above [above 460 MPa (67 ksi)].

For critical joints liable to the load/action of any of these factors, active consideration should be given to the use of CV2Z and CV2ZX quality steel having improved through-thickness (Z direction) properties as well as high toughness, including the possible specification of crack arrest and CTOD testing. The use of PWHT for these joints may be considered as another alternative. Although the sensitivity of structural steels to the factors noted above generally increases with yield strength, this tendency does not necessarily preclude the use of high-strength steels, as the processing route can have considerable consequences for the mechanical behavior.

Particular consideration should be given to through-thickness behavior under constrained loadings. Although the brace ends at tubular connections are also subject to stress concentration, the conditions of service are not quite as severe as for node cans. For critical braces, for which brittle fracture would be catastrophic, consideration should be given to the use of stub-ends in the braces having the same class as the node can, or one class lower. This recommendation need not apply to the body of braces (between joints).

A.19.2.2.4 Lowest Anticipated Service Temperature

The bulk of an offshore structure is submerged in seawater. Except for arctic structures, temperatures are above freezing, with the minimum temperature in the range of -2°C to 10°C (28°F to 50°F), depending on geographical location.

The values of the LAST presented in Table A.19.2-1 have been suggested for various offshore operating areas. For the Gulf of Mexico, a somewhat higher temperature of 21°C (70°F) prevails for both air and water during the hurricane season, which corresponds to the occurrence of significant loads/actions approaching design level. At depths below 300 m (1000 ft), the Gulf of Mexico water temperature is 4°C (40°F), like deep oceans worldwide.

Table A.19.2-1—Recommended Lowest Anticipated Service Temperatures

Location	LAST in Air	LAST in Water
Gulf of Mexico	-10°C ($+14^{\circ}\text{F}$)	$+10^{\circ}\text{C}$ ($+50^{\circ}\text{F}$)
Southern California	0°C ($+32^{\circ}\text{F}$)	$+4^{\circ}\text{C}$ ($+40^{\circ}\text{F}$)
Cook Inlet, Alaska	-29°C (-20°F)	-2°C ($+28^{\circ}\text{F}$)
North Sea, south of Latitude 62°	-10°C ($+14^{\circ}\text{F}$)	$+4^{\circ}\text{C}$ ($+40^{\circ}\text{F}$)
North Sea, north of Latitude 62°	Site-specific data should be used	
Mediterranean Sea, north of Latitude 38°	-5°C ($+23^{\circ}\text{F}$)	$+5^{\circ}\text{C}$ ($+41^{\circ}\text{F}$)
Mediterranean Sea, south of Latitude 38°	0°C ($+32^{\circ}\text{F}$)	$+10^{\circ}\text{C}$ ($+50^{\circ}\text{F}$)

A.19.2.2.5 Other Considerations

Resistance to lamellar tearing is an important property for steels strained normal to the plane of rolling (i.e. parallel to the thickness). The formation of planar discontinuities during the steel manufacturing process creates potential sites for steel separation if tensile loads/actions are applied in the through-thickness direction.

Weldability is a measure of the steel's resistance to cracking due to hydrogen and other embrittlement mechanisms in the HAZ immediately adjacent to a weld, using normal welding techniques. The effects of welding can cause modifications to the microstructure within the steel adjacent to the weld; consequently, achieving the toughness requirements in the HAZ can be affected by the steel selection as well as by the welding processes and parameters.

A.19.2.4 Material Category Approach

The MC method has evolved from practices in the Gulf of Mexico and other worldwide applications where ASTM, API, and AWS standards are used. Each structure to be designed and built is assigned to a particular category. MC3 includes small structures in water depths less than 130 m (400 ft) that are unmanned during the design environmental event. MC2 refers to larger structures in deeper water where failure consequences and fatigue loading become more significant. MC1 is reserved for structures with very high failure consequences in terms of monetary loss, wasted natural resources, pollution, and/or loss of human life. These definitions are closely aligned with the exposure levels L-3, L-2, L-1 discussed in 6.6.

A.19.3 Strength Groups

A.19.3.1 General

The steel strength groupings also provide for distinct categories of CVN impact toughness. The impact energy and test temperature required for each class varies depending on the steel strength, details of the application, the steel specification, and the location of the structure (affecting the LAST). Care should be taken to ensure that the impact toughness level is appropriate for each intended application. The impact toughness should be in accordance with the specific steel specification.

A.19.3.2 Group I Steels

Group I designates mild steels with the SMYS of 275 MPa (40 ksi) or less. Routine weldability can generally be expected as long as the value of the carbon equivalent (P_{CE}), computed from Equation (A.19.3-1), is less than or equal to 0.45:

$$P_{CE} = P_C + P_{Mn}/6 + (P_{Cr} + P_{Mo} + P_V)/5 + (P_{Ni} + P_{Cu})/15 \quad (A.19.3-1)$$

where

P_C is the weight percentage of carbon;

P_{Mn} is the weight percentage of manganese;

P_{Cr} is the weight percentage of chromium;

P_{Mo} is the weight percentage of molybdenum;

P_V is the weight percentage of vanadium;

P_{Ni} is the weight percentage of nickel;

P_{Cu} is the weight percentage of copper.

A.19.3.3 Group II Steels

Group II designates intermediate strength steels with the SMYS of 280 MPa (41 ksi) to 395 MPa (57 ksi). Their weldability should be ensured by requiring approved weldability data from the supplier, normally by testing in accordance with a recognized standard, or by limiting the carbon equivalent of the steel depending on toughness class.

P_{CE} calculated as in Equation (A.19.3-1) should not exceed 0.43 for all toughness categories (except for NT steels, where P_{CE} may be 0.45).

P_{CM} (carbon equivalent parameter for cracks, modified), computed from Equation (A.19.3-2), should not exceed 0.23 for all toughness categories. This requirement does not apply to NT steels.

$$P_{CM} = P_C + P_{Si}/30 + (P_{Mn} + P_{Cu} + P_{Cr})/20 + P_{Ni}/60 + P_{Mo}/15 + P_V/10 + 5 P_B \quad (\text{A.19.3-2})$$

where, in addition to the symbols given in A.19.3.2,

P_{Si} is the weight percentage of silicon;

P_B is the weight percentage of boron.

A.19.3.4 Group III Steels

Group III designates high-strength steels with the SMYS in the range of 400 MPa to 455 MPa (58 ksi to 66 ksi). These steels, when supplied in plate form, are usually manufactured by either Q&T or TMCP methods. They share the weldability characteristics of group IV.

For group III and above, the following should be considered:

- weldability—some modern, low P_{CE} (low P_{CM}) plates have lean compositions and may be more readily weldable than a lower grade/lower strength plate that has been made to a higher carbon composition and normalized;
- fatigue problems, including corrosion fatigue, which can result from the use of higher working stresses—consequently, the higher tensile strengths of these steels is sometimes not exploited in such situations;
- notch toughness in relation to other elements of fracture control, such as fabrication practices, inspection procedures, and the service environment, including corrosion and hydrogen embrittlement, whether from cathodic protection or microbiological origins, the effects of temperature, and the effects of high rates of loading.

A.19.3.5 Group IV Steels

Strength groups IV and V have been introduced to accommodate higher-strength structural steels (beyond current usage) when available. Specific strength levels for these should be designated by the designer. The designer should take into account any difficulties of matching the yield strength of the weld metal to those of the steels, and factors such as thickness and shrinkage stresses should be considered when specifying welding processes.

Group IV designates high-strength steels with the SMYS between 460 MPa (67 ksi) and 500 MPa (72 ksi), which can be supplied in the Q&T condition or produced by TMCP. Such steels are usually characterized by a relatively lean chemistry and good weldability and possess high fracture toughness in addition to high strength. Normalized steel at this strength generally cannot provide the combination of tensile properties and fracture toughness in conjunction with good weldability, and so the normalized condition is not specified for class CV2.

Most of the experience to date has been accumulated for the Q&T condition. The user should take steps to ensure that the SMYS is clearly defined for both Q&T and TMCP steels at all thicknesses. The user is reminded of the tendency for steels to be more susceptible to hydrogen embrittlement at higher yield strength. Hydrogen can arise from a number of sources, including excessive cathodic protection of submerged structures and the action of sulphate-reducing bacteria under anaerobic conditions. Some TMCP steels with SMYS in the range 450 MPa to 480 MPa have been resistant to hydrogen attack in pipeline service. TMCP steel with 420 MPa SMYS is being used in fracture-critical bridge members. Conventional high-strength steels that can be hardened by use of higher alloy content are likely to be more susceptible to hydrogen damage.

A.19.5 Applicable Steels

Listed steels in Annex C generally have an acceptable track record of weldability. When specifying or allowing unlisted steels, weldability and suitability should be assessed.

Offshore grade steel castings are most readily obtained in strength levels equivalent to Grade 355 (50 ksi), but higher-strength castings of 540 MPa (78 ksi) and 690 MPa (100 ksi) are also obtainable, albeit in more limited size ranges. Examples of the use of castings include the following:

- a) cast nodes for fatigue sensitive area of fixed steel structures,
- b) J-tube trumpet sections,
- c) complex nodes for topsides modules or integrated decks,
- d) padeye or trunnion (padear) assemblies, and
- e) spreader bar ends.

Castings can offer solutions to specific problem areas, such as the avoidance of massive welding in the fabrication of trunnions and improved transitions between heavy and light sections or between tubular members and open section steelwork. Reference [A.19.5-1] contains information on steel castings for offshore structures.

A.19.6 Cement Grout for Pile/Pin-to-sleeve Connections and Grouted Repairs

A.19.6.2 Onshore Grout Trial

75 mm (3 in.) cubes have been used for a large majority of the grout compressive strength determination of laboratory test specimens used to develop the connection capacity formulation. Therefore, the specified grout compressive strength, f_{cu} , used in the capacity equations given in 15.1 should be based on, or corrected to, a 75 mm cube.

Tests reported [A.19.6-1] show that cube conversion factors are dependent on the grout age at testing as well as the size. This is probably due to the different rates of increase in strength and elastic moduli properties observed in grouts. Therefore, care is required in obtaining appropriate conversion factors.

The volume of grout mixed and the number of samples tested at each age should be sufficiently large to be representative of the production grouting.

A.19.6.3 Offshore Grout Trial

The offshore grouting process is unusual in that only one chance is available to achieve a satisfactory connection. Therefore, it is imperative that all measures possible be taken to ensure that the grout and grouting equipment used in the process are satisfactory prior to pumping the grout. The offshore trial provides a functional check on the equipment, including the calibration of gauges and material compliance. The volume of grout mixed and the number of samples tested at 30 hr should be sufficiently large to be representative of the production grouting.

A.19.6.4 Offshore Quality Control

The measurements of the specific gravity of the grout slurry prior to grouting and on the return through the annulus, are the only quantitative QC methods available at the time of grouting. Therefore, it is important that these measurements be taken frequently and accurately to ensure, as far as is reasonably practical, that the in-place grout achieves its strength compliance.

In cement and water grouts, the specific gravity is an accurate measure of the mix proportions. However, when fillers and admixtures are included in the mix, additional care is required to control batching and dosing. If prebatched bulk cement-fillers-admixtures are used, care should be taken to ensure that segregation of the constituent materials does not occur during transportation, handling, and storage.

ASTM [A.19.6-2] is a suitable standard for casting and testing grout cubes.

A.20 Welding, Fabrication, and Weld Inspection

A.20.2 Welding

A.20.2.1 Selected Generic Welding and Fabrication Standards

The standard(s) selected by the owner should be an international standard, a de facto international standard, or a national standard in use by the steel fabrication industry. To meet the requirements of a specific project application, the owner may use several standards. In addition, the contractual arrangements should identify clearly the responsibilities of all parties involved in the project. To this end, appropriate supplemental documents can be provided to correlate the special needs of the project with the relevant sections of this standard and those of the selected standard(s).

Examples of standards that satisfy the requirements of 20.2.1 are as follows:

- a) EEMUA 158 [A.20.2-1] or other generic standards providing the guidance listed in 20.1;
- b) AWS D1.1/D1.1M [A.20.2-2], generally compatible with the MC method and Annex D;
- c) NORSOK M-001 [A.20.2-3], generally compatible with the DC method and ISO 19902;
- d) CSA W59 [A.20.2-4].

When WPQ by test is required (i.e. when the procedure is not prequalified, when comparable impact performance has not been previously demonstrated, or when the welding consumables are to be employed outside the range of essential variables covered by prior testing), qualification shall include CVN testing of the as-deposited weld metal, impact tested in accordance with the selected standard.

A.20.2.2 Weld Metal and HAZ Properties

A.20.2.2.1 General

In addition to weld metal toughness, consideration should be given to controlling the properties of the HAZ. Although the heat cycle of welding sometimes improves base metals of low toughness, this region will more often have degraded properties. A number of early failures in welded tubular joints involved fractures that either initiated in, or propagated through, the HAZ, often before significant fatigue loading.

The average HAZ toughness values in Annex D and ISO 19902 have been found by experience to be reasonably attainable. Due to the large scatter usually observed, single specimen (one out of three) energy values of 7 J (5 ft-lb) or lower are allowed without requiring retest. As criticality of the component performance increases, lower testing temperatures are usually required. More extensive sampling increases the likelihood of finding local brittle zones (LBZ) with low toughness values.

Since HAZ toughness is as much dependent on the steel as on the welding parameters, a preferable alternative for addressing this issue is through weldability prequalification of the steel. For example, API 2Z [A.20.2-5] spells

out such a prequalification procedure, using CTOD as well as Charpy testing. This prequalification testing is presently being applied as a supplementary requirement for high-performance steels such as API 2W [A.20.2-6] and API 2Y [A.20.2-7] and is accepted as a requirement by several steel producers.

Conservative requirements in MC1 and DC structures and in API 2Z [A.20.2-5] on the subject of LBZ give little credit for ways and means of having managed to avoid LBZ induced failures in the recent past. These design countermeasures include:

- a) the use of steels with moderate crack-arrest capabilities, as demonstrated by no-break in the NRL drop-weight test [A.20.2-8] (small flaw);
- b) overmatch and strain hardening in conventional normalized 290 MPa to 345 MPa (42 ksi to 50 ksi) carbon-manganese steels, in which the weld metal and HAZ are harder than adjacent base metal, forcing plastic strains to go elsewhere;
- c) the tendency for fatigue cracks in welded tubular joints to grow out of the HAZ before they reach appreciable size (assuming unfavorable osculation of joint can weld seam with the brace footprint is avoided [see Figure 20.2-1]);
- d) traditional limits on layer thickness or heat input in welding procedures, which promote grain refinement in the HAZ and avoid extensive LBZ.

Many standards permit testing one 345 MPa (50 ksi) steel to qualify all other grades of 345 MPa (50 ksi) and below. Consequently, selection of a premium grade of 345 MPa steel [such as API 2H, 50 Z with very low sulphur, or 270 J (200 ft-lb) upper shelf Charpy] [A.20.2-9], for qualification test plates will virtually ensure the satisfying of an HAZ impact requirement of 34 J (25 ft-lb), even when welded with high heat inputs and high interpass temperatures. There is no reasonable way to extrapolate this test to ordinary 345 MPa (50 ksi) steel (such as ASTM A572 Gr 50) with the expectation of similar HAZ impact energies. Thus separate Charpy testing of each steel class is appropriate, if HAZ toughness is being addressed via weld procedure qualification testing.

A.20.2.2.4 Charpy V-notch Toughness

Charpy impact testing is a method for qualitative assessment of material toughness. Although lacking the technical precision of CTOD testing, the method has been, and continues to be, a reasonable measure of fracture safety, when employed with a definitive program of nondestructive examination to eliminate weld area imperfections.

The requirements specified in 20.2.2.4 are based on practices that have generally provided satisfactory fracture experience in structures located in moderate temperature environments [e.g. 4 °C (40 °F) seawater and -10 °C (14 °F) air exposure].

For other environments, impact testing temperatures should be reconsidered, based on local temperature exposures. Since most welding procedure requirements are concerned primarily with tensile strength and soundness (with minor emphasis on fracture toughness), it is appropriate to consider additional essential variables that have an influence on fracture toughness (e.g. combinations of specific brands of wire and flux).

A.20.2.2.5 CTOD Toughness

For critical welded connections, the CTOD testing is technically more exact than Charpy testing. CTOD tests are run at realistic temperatures and strain rates, representing those of the engineering application, using specimens having the full prototype thickness. This yields quantitative information useful for engineering fracture mechanics analysis and defect assessment in which the required CTOD is related to anticipated stress levels (including residual stress) and flaw sizes. When CTOD data for the HAZ are not provided by the steel manufacturer, the

contractor can use weld procedure specification tests to CTOD qualify the welds necessary for the specific project. This is a shortened form of the API 2Z [A.20.2-5] prequalification testing usually performed by the steel manufacturer. These sections cover the testing requirements for the HAZ zone only.

Previous CTOD testing data performed by the steel manufacturer in accordance with a recognized standard, e.g. API 2Z [A.20.2-5] or EN 10225 [A.20.2-10], is an acceptable alternative for the HAZ requirements, provided the contractors' WPS is within the range of heat inputs, interpass temperature, thickness, and other essential variables qualified.

A.20.2.3 Tubular T-, Y-, and K-joints

A.20.2.3.1 General

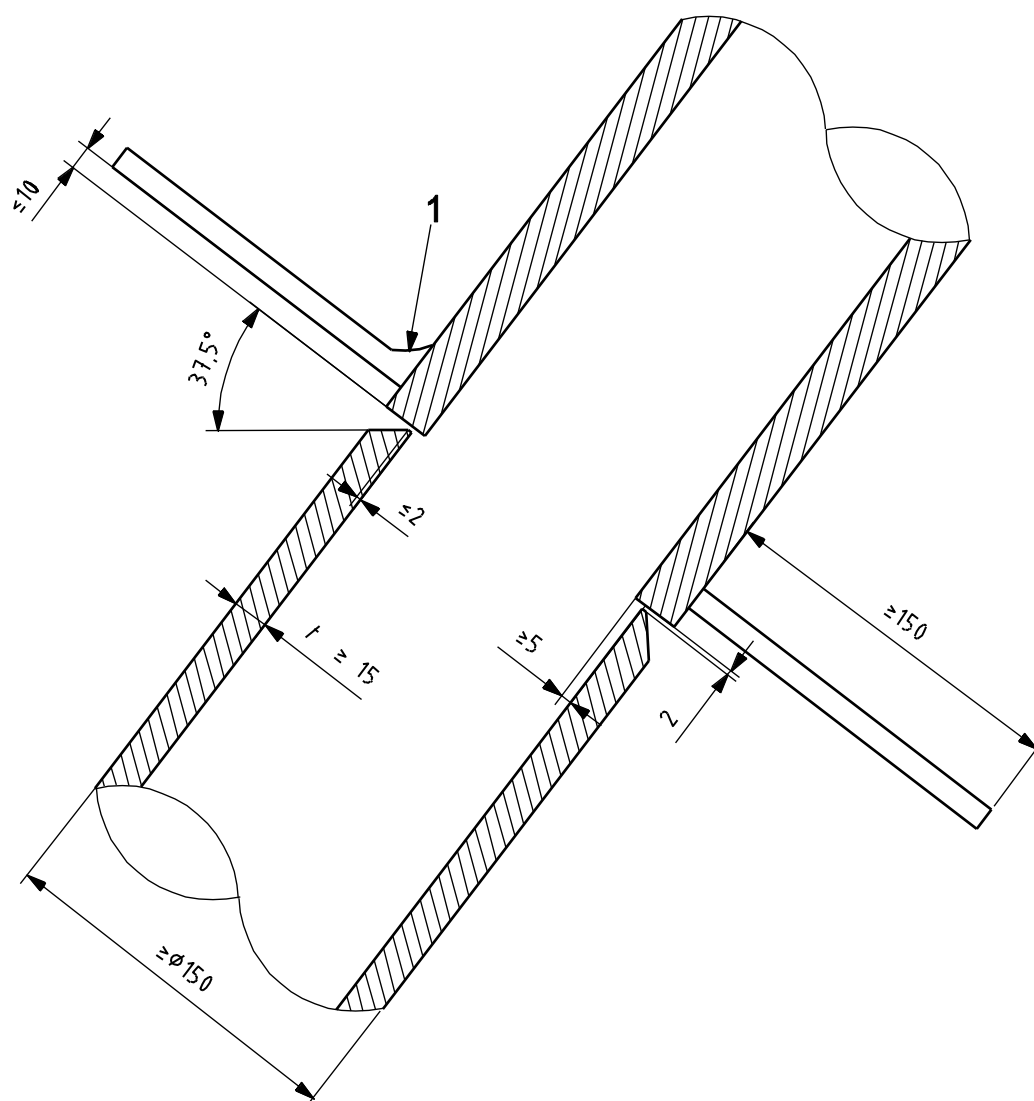
Specialized considerations for single-sided T-, Y-, and K-joints are described comprehensively in AWS D1.1/D1.1M [A.20.2-2].

As examples, some of these considerations are briefly, though not comprehensively, discussed in A.20.2.3.2 and A.20.2.3.3.

A.20.2.3.2 Welder Qualification

All welders making full penetration welds from one side only in tubular joints should qualify on a test that represents all welding positions and incorporates a shoulder at the open root, as well as restricted access from the chord side—for example, the 6GR test specimen as shown in Figure A.20.2-1. Typically, four prismatic bend tests are taken from the 45° locations. Macrosections adjacent to the bend specimens should conform to the applicable visual and profile requirements.

Dimensions in millimeters

**Key**

1 disk tack weld

Figure A.20.2-1—Welder Test Piece for Single-sided Welds**A.20.2.3.3 Production Weld Joint Details**

Examples of weld details for tubular T-, Y-, and K-joints welded by shielded metal arc (manual stick) welding are shown in Figure A.20.2-2 and presented in Tables A.20.2-1 and A.20.2-2. Other processes are covered in References [A.20.2-1], [A.20.2-2], and [A.20.2-3].

Table A.20.2-1—Root Gap Requirements for Tubular Joints—Shield Metal Arc Welding

Groove Angle, b	Root Opening, g	
	mm	in.
Over 90°	0 to 4.8	0 to $\frac{3}{16}$
45° to 90°	1.6 to 4.8	$\frac{1}{16}$ to $\frac{3}{16}$
Under 45°	3.2 to 6.4	$\frac{1}{8}$ to $\frac{1}{4}$

Table A.20.2-2—Minimum Weld Thicknesses for Tubular Joints—Shield Metal Arc Welding

Weld Angle, a	Weld Thickness, t_w
Over 135°	To continue line of brace/stub, except that t_w need not exceed $1.75 t$ Smooth profile required between brace/weld/chord
50° to 135°	$1.25 t$
35° to 50°	$1.50 t$
Under 35°	$1.75 t$

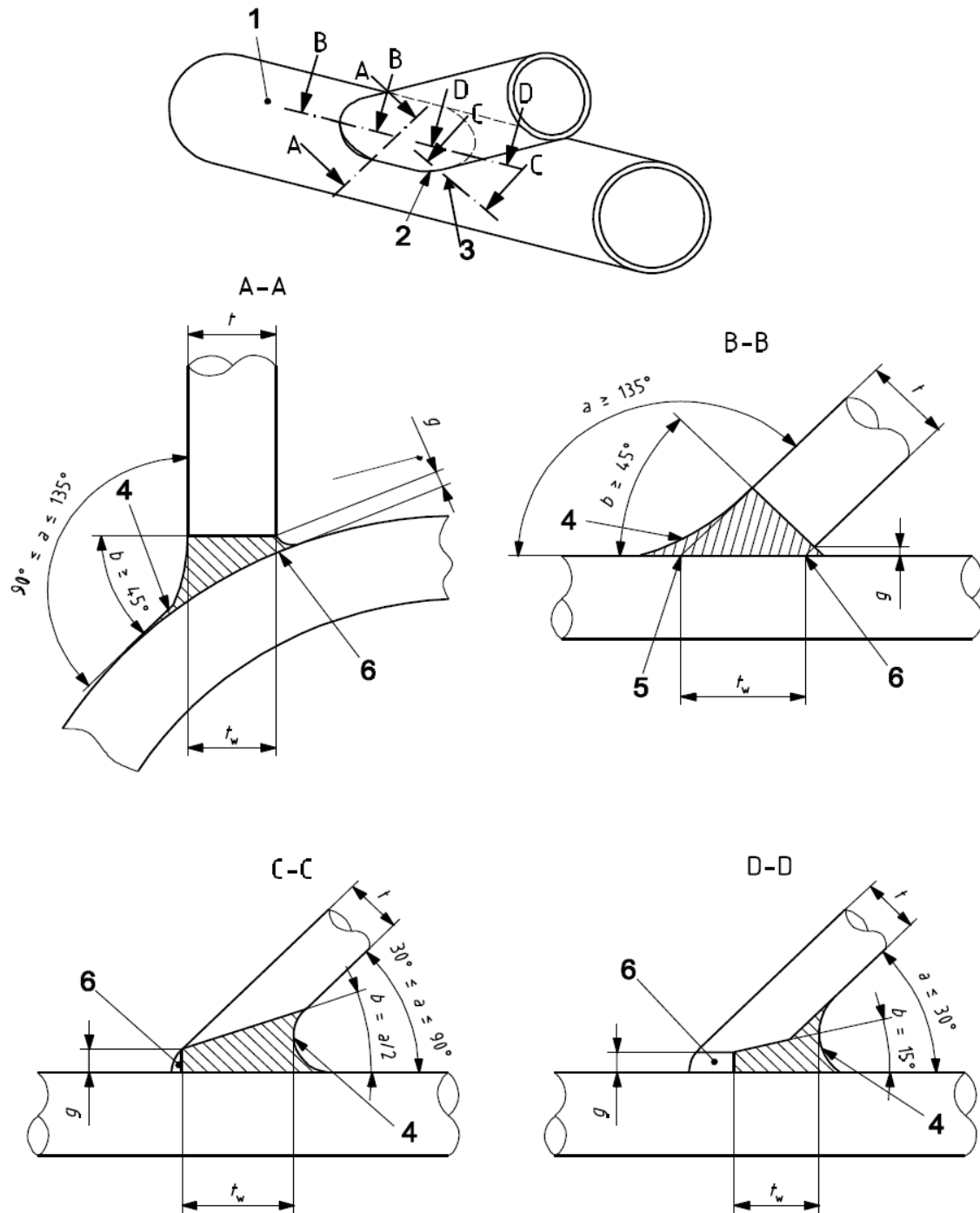


Figure A.20.2-2—Welded Tubular Joints—Shield Metal Arc (Stick) Welding

Key

- 1 chord
- 2 joint intersection and weld line
- 3 smooth transition between details C-C and D-D and external fillet of $1.75 t$
- 4 smooth profile between brace/weld/chord
- 5 build out to fit thickness, but t_w need not exceed $1.75 t$
- 6 weld root not subject to inspection beyond required t_w
- a local dihedral angle formed by exterior surfaces of brace and chord at point being considered
- b groove angle formed by brace weld preparation and chord at point being considered
- g root gap (see Table A.20.2-1)
- t brace or stub thickness
- t_w weld thickness (see Table A.20.2-2)

Figure A.20.2-2—Welded Tubular Joints—Shield Metal Arc (Stick) Welding *(continued)***A.20.3 Inspection**

In some cases, it is not necessary to undertake 100 % inspection with all methods. The requirements given in Annex D and ISO 19902 allow for inspection of a proportion of the length of the welds of certain types and for certain categories or classes of structure. The reduced extent of inspection is dependent on the welds being of good quality; consequently, if the defect identification is too high, the extent of inspection should be increased as described below.

The defect rate is calculated from:

$$\text{defect rate} = \frac{\text{sum of length of defects identified by method}}{\text{total weld length inspected by method}} \quad (\text{A.20.3-1})$$

The extent of NDT should be increased as follows:

- a) defect rate exceeds 5 %—proportion of welds inspected should be doubled; spot inspection proportion should increase to 20 %;
- b) defect rate exceeds 5 % for increased proportional inspection—proportion of welds inspected should increase to 100 %;
- c) defect rate exceeds 10 %—proportion of welds inspected should increase to 100 %.

The required level of increased extent should be maintained until a defect rate below 5 % is re-established and documented.

A.21 Quality Assurance, Quality Control, and Documentation**A.21.1 General**

Exposure level L-1 structures have the most severe failure consequences and hence the most stringent inspection and documentation requirements.

In selecting the inspection category, welds in joints more than 150 m below the water surface should be assumed inaccessible for in-service inspection by divers. Accessibility for inspection at greater depths may be considered if the use of ROV methods (e.g. visual, wet MPI, or flooded member detection) is planned.

When only partial testing is required for welds in an area, the weld testing should be distributed such that the most essential members and nodes are included in the inspection and such that areas of welds most susceptible to weld defects are examined.

A.21.3 Quality Control Plan

A.21.3.1 General

Proactive QC by the contractor is different from after-the-fact QA inspection, testing, and documentation, which is normally conducted by the owner or by a third party.

The use of “negative reporting” systems for QC inspection, whereby a formal report is only raised when a defect is found, may be considered, subject to the contractor furnishing evidence of appropriate procedures controlling such systems to the owner’s satisfaction. Available logs, markings, or other records for both positive and negative QC inspection outcomes should be retained—at least temporarily, in any case.

Further considerations for the QC plan are presented in the following subsections.

A.21.3.2 Inspector Qualifications

Inspectors should be knowledgeable in the general areas of welding technology, inspection and testing procedures, as well as in construction methods for those areas of their responsibility during fabrication. They should know how and where to look for problems and situations which lead to problems, as well as the practical limitations on making repairs.

Suitable standards for the qualification of inspectors include the following:

- AWS QC-1 [A.21.3-1],
- CSA W178.2 [A.21.3-2],
- CSWIP [A.21.3-3],
- BGAS-CSWIP [A.21.3-4], and
- PCN [A.21.3-5].

Other standards may be agreed upon between owner, contractors, and regulators.

A.21.3.3 Nondestructive Testing Personnel Qualifications

All NDT personnel should have demonstrated ability and experience, and should preferably be qualified to an internationally recognized offshore code or guidance. Continuing qualification should be based on satisfactory performance on the job.

Internationally recognized requirements for NDT personnel are contained in the following:

- CSWIP [A.21.3-3],
- BGAS-CSWIP [A.21.3-4],
- PCN [A.21.3-5],
- ACCP (ASNT) [A.21.3-6],
- SNT-TC-1A (ASNT) [A.21.3-7], and
- ISO 9712 [A.21.3-8].

Additionally, API 2X [A.21.3-9] provides specific offshore guidance.

Personnel who perform other types of inspection during the fabrication of an offshore structure should be required to demonstrate ability and experience, or be qualified to an appropriate code for the required inspections.

A.21.3.4 Inspection of Materials

Receipt of the correct material should be verified by cross-checking with appropriate original mill certificates and heat stamps, or with other appropriate documentation for nonstructural material and structural materials other than steel.

A.21.3.5 Inspection of Fabrication

In general, the inspection of fabrication should confirm that each component incorporated into the structure is of the correct material, orientation, size, and dimension, etc., and is fitted, aligned, and permanently fastened according to the specified requirements. Legs and pile sleeves through which piles will be field installed should be carefully checked for internal clearance and straightness to ensure that required tolerances have been met, and, if possible, a template of the pile of nominal length should be passed through the leg or sleeve to demonstrate adequate clearance. Particular attention should be given to field joints (such as the tops of structure legs) that should be checked to ensure that all dimensions are within tolerance.

A.21.3.6 Inspection of Welding

Inspection and testing of welding should be performed during all phases of fabrication, with an aim to preventing introduction of defects into the weld. To the maximum extent possible, inspection and testing should be performed as construction progresses and be scheduled so as not to delay the progress of the job. All weld fit-ups (joint preparation prior to welding) should be visually inspected to ensure acceptable tolerances before welding.

It should be verified that the welder or welding operator is currently qualified for the procedure being used and that the appropriate qualified procedure is being followed. In addition, inspection should ensure that appropriate consumables are being used and that the consumables are being stored, handled, and used in accordance with the manufacturer's recommendations and other appropriate requirements.

A.21.7 Documentation

A.21.7.1 General

In addition to document preservation, document control should have an information flow from contractors to owners, and vice versa, and be able to track and distribute revisions and drawing issuances in a timely manner.

Document retention for the life of the structure is generally the responsibility of the owner. Permanent records should not be put into a “destroy after X years” system.

The following considerations apply only where indicated in Table 21.7-1.

- a) Weight and center of gravity reports: The contractor should prepare a weight and center of gravity report for each structure. Weights should be based on design, shop, or as-built drawings as the job progresses. Guidance on weight control is given in ISO 19901-5 [5].
- b) Dimensional control documentation: Prior to and during fabrication, documentation should be prepared on the dimensions and compliance of components and of assemblies.
- c) Materials and fabrication inspection: During the fabrication phase, material inspection documentation covering the mill certificates and material identification records (as described in Section 19), as well as any additional materials, testing, or special inspections that were conducted and that reflect on the final condition of the structure, should be prepared and compiled. This need not include documentation for inspection related to the assembly of the structure.
- d) Weld inspection: A set of structural drawings should be marked with an appropriate identification system detailing the location of each weld to be examined and referenced as an integral part of the inspection record. All welds should be uniquely identified and be traceable during fabrication to the individual welder or welding operator. A report should be prepared for each nondestructive test performed (other than visual), the details of which shall be documented sufficiently to permit repetition of the examination at a later date. Sketches and drawings incorporating the weld identification system should be used to augment descriptions of the part and locations of all discontinuities required to be reported. Forms should be provided to show the required details of documentation, and sketches of typical weld configurations should also be provided to clarify the written description. Discontinuities required to be reported should be identified on sketches by the appropriate weld number and position.
- e) Other inspection: Inspection of all nonstructural systems and tests should be documented to confirm details of the inspection and results. Any deviations from the specified requirements shall be properly recorded, including sketches if necessary.

A.21.8 Drawings and Specifications

A.21.8.1 General

Drawings should be prepared using consistent units.

For use in connection with fixed offshore structures and related facilities, the drawings and specifications are defined as described in the following subsections.

A.21.8.2 Conceptual Drawings

Conceptual drawings are intended to supply a general idea of the facility under consideration. These drawings should include preliminary layouts and elevation views of the overall facility showing the number, type of construction, and approximate size of each platform, as well as the more important auxiliary features, such as heliports and boat landings.

Simplified process or mechanical flow diagrams and electrical one-line diagrams should be included for all production or utility systems. A generalized equipment layout drawing should be included, which also indicates buildings, storage of supplies, and so forth.

All information that contributes to clarifying the overall intent of the facility should be shown. Specifications are not generally required. However, if included, they should be of a general descriptive nature, so as to supplement the drawings in adequately describing the facility.

A.21.8.3 Bid Drawings and Specifications

Bid drawings are intended to show the total facility with its configuration and dimensions in sufficient detail to accurately define the scope of the project. With supplemental specifications, bid drawings are suitable for generally defining the scope of the proposal, or suitable to be furnished by the owner to a contractor when requesting a quotation in which the design is to be part of the contractor's bid. In the latter case, all essential information needed by the designer should be included.

Bid structural drawings should show major overall dimensions, deck arrangements, operational loading requirements, and any preferred type of construction and materials. Structural details and member sizes are not necessarily furnished, since these are considered "design" drawings. All auxiliary items that are to be included in the bid, such as boat landings, barge bumpers, stairs, walkways, fence, and handrail, should be shown on these drawings. Typical preferred construction details of these items also should be included.

Equipment layout drawings should be included for all decks. Sufficiently detailed process, mechanical and utility flow diagrams, and electrical one-line diagrams should be included for all systems that are covered by the bid.

Specifications for equipment, machinery, and other engineered components should include an itemized list and description of all items not shown in the drawings, but which are to be included in the bid—even such items as lighting and cathodic protection. Specifications for materials and fabrication should include all types of material allowed for use and any particular requirements for dimensional tolerances, inspection, testing, and welding.

A.21.8.4 Design Drawings and Specifications

Design drawings give descriptive information about the major components of the facility. Emphasis in these drawings is placed on overall layouts and definition of critical items, supplemented by essential details. They should indicate all appurtenances and should include all dimensions where strict adherence is required.

Design drawings should include a layout of the location and orientation of the structure or structures in the field, as well as the location of equipment on the decks of each structure. Structural drawings showing member sizes of all major structural members and all controlling dimensions should be included. General locations and preliminary or typical details of miscellaneous structural items, such as joints, cover plates, and web plate stiffeners, should be indicated. In addition, any other typical structural details that are not normally standard to this type of construction should be included.

Design drawings should also include all items necessary for installation purposes, such as lifting eyes and launch frames, which are critical to the design of the structure.

Mechanical and utility flow diagrams showing sizes of all equipment, piping, and valves, and electrical line diagrams showing rating and sizes of feeders and controls, should be included. Equipment layout drawings of all equipment shown on the flow diagrams or one-line diagrams, manifolds, and major instrumentation items, such as large control valves, meter runs, control valve stations, and control panels, should be shown. Piping plan and elevation drawings should show major piping only and indicate adequate space reserved for minor piping and for conduit and cable runs.

Design drawings should be supplemented by all specifications necessary to convey the intent of the design. Standard specifications for material and fabrication that are referred to in this document can be properly referenced on appropriate drawings. However, any deviations from these specifications should be detailed. Specifications should be included for equipment, machinery, and other engineered items.

Design drawings and specifications are often used as part of the solicitation package, or as part of the contract document. As such, they need to be sufficiently detailed and suitable for furnishing by the owner to the contractor to be used for making accurate material take-offs for bidding purposes when no design is required on the part of the contractor, or suitable for submittal by the contractor to the owner to completely define the proposal. When design drawings are used for bid or contract purposes, all auxiliary items such as stairs, boat landings, and walkways should be shown in sufficient detail for estimating purposes.

A.21.8.5 Fabrication Drawings and Specifications

Fabrication drawings are intended to supply sufficient information that fabrication can be performed directly from these drawings. They should contain all design data fully detailed and dimensioned. At the fabricator's option, they may be supplemented by shop drawings.

A set of fabrication drawings includes completely detailed design drawings with descriptions, exact locations, sizes, thicknesses, and dimensions of all structural members and stiffeners. This information should also be shown for all structural items, such as brackets, stiffeners, and cover plates, and for all auxiliary items, such as stairs, walkways, fence, and handrail. Connections and joints should be completely detailed, including welding symbols, unless standard procedures apply. Methods of attaching timber, grating, and plate should be included.

In addition to complete piping plan and elevation drawings, a set of fabrication drawings should include piping isometric drawings and details for all pipe supports, if required by the complexity of the facility. Instrumentation location plans and supports, electrical location diagrams showing general routing, and wire and cable tie-ins to electrical equipment should be included.

Fabrication drawings should clearly indicate the components or "packages" scheduled for assembly as units in the fabrication yard. Welds and connections to be performed in the field should be indicated.

Detailed specifications should be included for all work to be done by the fabricator, such as welding, fabrication, and testing, and for all materials, equipment, or machinery to be purchased by the fabricator. However, for standard specifications covered under the recommendations of this document, no copies need be furnished, provided reference is made on key drawings. Specifications for equipment and other engineered items not purchased by the fabricator may also be included with fabrication drawings, for general information.

A.21.8.6 Shop Drawings

Shop drawings or sketches are prepared by or for the fabricator, at their option, to facilitate the fabrication of parts and/or components of the structure. They are intended to provide all information and instructions needed for that purpose. Due to differences in methods and procedures of various fabricators, shop drawings can vary in appearance.

Shop drawings may include typical shop details to supplement details and dimensions shown on either fabrication drawings or patterns for coping the ends of members, detailed piece marked drawings for each member, and pipe spool drawings.

Shop drawings are the responsibility of the fabricator. Approval or review of shop drawings by the designer or owner should not relieve the fabricator of their responsibility to complete the work in accordance with the contract or fabrication drawings and specifications.

A.21.8.7 Installation Drawings and Specifications

Installation drawings furnish all pertinent information necessary for the construction of the total facility on location at sea. They contain relevant information not included on fabrication drawings.

If special procedures are required, a set of installation drawings may include installation sequence drawings. Details of all installation aids such as lifting eyes, launch rails or frames, jacking brackets and stabbing points should be included, if not shown on fabrication drawings. For structures installed by flotation or launching, drawings showing launching, upending, and flotation procedures should be provided. Details should also be provided for piping, valving, and controls of the flotation system, closure plates, etc.

Erection of temporary struts or supports should be indicated. All rigging, cables, hoses, etc. that are to be installed prior to loadout should be detailed. Barge arrangement, loadout, and tie-down details should be provided.

Installation drawings are intended to be used in connection with fabrication drawings. They should be supplemented by detailed installation specifications, installation procedures, or special instructions, as required, to provide all information required to complete the field installation.

A.21.8.8 As-built Drawings and Specifications

As-built drawings show in detail the manner in which the structure was actually constructed. These drawings are usually made by revising the original fabrication drawings, supplemented by additional drawings if necessary. As-built drawings are intended to reflect all changes, additions, corrections, or revisions made during the course of construction. They are prepared for use by the owner to provide information related to the operation, servicing, maintenance, and future expansion of the facility.

When the preparation of as-built drawings has been authorized by the owner, it is the responsibility of the fabricator and the field erector to furnish to the owner or to the designer adequate information regarding all variations between the drawings and the facility as actually constructed. This is usually furnished as corrections from the yard, the shop, and the field, marked on prints of the original drawings or by supplementary sketches, if required. This information should be sufficiently complete that the owner or the designer can correct and revise the original drawings without additional data or field measurements. Since the fabricator and erector are responsible for the accuracy of the corrections, a review and/or approval of the corrected drawings should be made by both the fabricator and the erector.

Minor deviations from the original drawings are generally numerous. Differences between the actual dimensions and those shown on the drawings need not be reported if they are within the specified allowable tolerances.

Specifications should also be corrected to reflect any changes made during the purchase of material, equipment, or machinery.

A.22 Loadout, Transportation, and Installation

A.22.5 Pile Installation

A.22.5.7 Driven Pile Refusal

Two examples of driven refusal criteria are given below.

- a) In soft soils, pile driving refusal for a properly operating hammer is defined as the point where pile driving resistance exceeds either 1000 blows/m (300 blows/ft) for a consecutive 1.50 m (5 ft) of penetration, or 800 blows for 300 mm (1 ft) of penetration. This definition applies when the weight of the pile does not exceed four times the weight of the hammer ram. If the pile weight exceeds this, the above blow counts are increased proportionally, but in no case should they exceed 800 blows for 150 mm (6 in.) of penetration.
- b) In hard clays and dense sands, pile driving refusal can be defined as the point where driving resistance exceeds one of the following criteria:

- 1) in continuous driving—a minimum of 125 blows/250 mm over six consecutive intervals of 250 mm, or a minimum of 200 blows/250 mm over two consecutive intervals of 250 mm;
- 2) in the last interval of 250 mm at the end of driving—325 blows/250 mm;
- 3) at restart of driving after a stoppage for 1 hr or longer—325 blows/250 mm over two consecutive intervals of 250 mm.

In soils where hard driving conditions are anticipated, such as in the presence of boulders or of strong cemented layers, the definition of pile refusal criteria cannot be based solely on a blow count value, and the potentially high local driving stresses induced in the pile should also be taken into account. The stress level in the pile steel can be calculated from wave equation analyses and can be estimated from the stress measurements from pile instrumentation. An example of refusal criteria for pile driving in strongly cemented carbonate soils is given in References [A.22.5-1] and [A.22.5-2].

The potential consequences of hard driving conditions in strong cemented layers (i.e. damage of the pile, hammer or structure) are highly dependent on the hammer type and size, on the pile wall thickness (D/t ratio, presence of a driving shoe), and on possible defects and irregularities in the pile shape, as well as on the soil conditions (in particular, strength and thickness of the rock layer, and soil type below the rock formation). Moreover, the reflected stress level (ratio of the maximum reflected stress to the initial peak stress), as measured from pile instrumentation at the pile head, only gives an estimate of the average stress in the pile wall; more severe stresses can be experienced locally at the pile tip during driving. Therefore, the definition of driven pile refusal criteria in cemented soils should preferably be based on local piling experience at the site. Correlation charts, similar to the one proposed in Reference [A.22.5-3], can be developed as an aid in deciding whether pile driving through a cemented layer can be attempted, or if drilling of the rock below the pile tip is necessary.

A.23 Structure Reuse

A.23.4 Inspection of Structures to Be Reused

A.23.4.3 Extent of Weld Inspection

As the costs and safety implications of inspection are very much lower when the structure is out of the water, it is recommended that inspection out of the water is undertaken for structures to be reused.

The following describes the minimum recommended inspection extent, but this should be modified in light of the structural assessment for the reuse condition and the previous in-service inspection history.

When only partial testing is required for welds in an area, the weld testing should be distributed such that the most critical components are included in the inspection, and such that areas of welds most susceptible to weld defects or to damage from previous service are examined.

- a) Primary tubular members: The extent of inspection of primary structure should be determined by comparing the design loads/actions and stresses (including removal and reinstallation loads/actions and stresses) for the new site with those to which the welds previously have been designed for and/or exposed to.

Where new design loads/actions and stresses are less than or equal to initial design or actual loads/actions, then the extent of inspection, if any, should be determined based on NDT documentation or the results of the initial spot survey.

Where new design loads/actions and stresses are significantly greater than initial design or actual loads/actions and stresses, or when comparison based on initial design or actual loads/actions (stresses) is not possible, a minimum of one bracing member and one leg spanning between each level should be inspected.

Additional inspection should be performed where in-service damage is known or suspected.

Tubular member inspection should be done using UT or MPI.

- b) Primary tubular joints: The brace-to-chord and brace-to-stub welds of all tubular joints with high static or fatigue utilizations or prone to accidental damage should be 100 % inspected. Where NDT inspection of these connections reveals significant defects, additional inspection of other connections should also be performed.

A minimum of one brace to chord connection at each level and one X-brace connection between levels, as applicable, should be 100 % inspected.

Tubular joint inspection should be done using UT or MPI. For tubular connections not having Class CV2Z or better steel in the heavy wall joint-cans both UT and MPI should be performed.

- c) Piles: A minimum of 10 % of each individual longitudinal and circumferential seam on piles to be reused should be inspected using UT or MPI.

100 % of each individual field splice in piles should be inspected using UT or radiographic testing (RT).

- d) Nonredundant bracing and subassemblies: Nonredundant bracing and subassemblies (e.g. padeyes, lifting bracing, single-level conductor guide framing level above sea floor) should be 100 % inspected using UT or MPI.

Attachment welds connecting nonredundant bracing/subassemblies to main members should be 100 % inspected using UT or MPI.

- e) Redundant bracing and subassemblies: A minimum of 10 % of each individual weld on redundant bracing and subassemblies (e.g. multilevel conductor guide framing, secondary splash zone and sea floor bracing, boat landings) should be visually inspected.

A minimum of 10 % of each individual attachment weld connecting redundant bracing/subassemblies to main members should be visually inspected.

- f) Deck members and connections: The requirements for reuse of topsides structures are given in API 2TOP [20].

A.23.5 Removal and Reinstallation

Some particular considerations for reuse that do not apply to new structures include the following.

- a) Removal: Structures that cannot be lifted onto barges can be removed by controlled de-ballasting and skidding the structure back onto a properly configured launch barge. Such operations require precise control of barge ballasting, positioning, and alignment between structure and barge. Environmental conditions for such operations can be more restrictive than usual for installation.
- b) Buoyancy and re-floating: When removal from a prior site requires re-floating of platform components, such as the structure, additional buoyancy can be required in excess of that provided when the structure was originally installed to compensate for loss of buoyancy and for additional weights not present during the original installation, e.g. grouted piling.
- c) Marine growth removal: When removing structures for reuse, appropriate equipment for marine growth removal from sea-fastening locations should be provided. If the structure is to be skidded back onto a launch barge, marine growth should be removed from launch rails to ensure reasonable predictions of the coefficient of friction and the sling forces on padeyes and winches. Water blasting or sand blasting to remove marine growth has been found effective.

Annex B

(informative)

CTOD Testing Procedures

B.1 Testing Procedure Requirements

When required by 20.2.2.5, CTOD testing shall be conducted in accordance with the CTOD-related sections of ISO 12135. Annex B provides supplementary requirements, based on a specimen geometry as shown in Figure B.1.

NOTE Additional alternative guidance can be found in API 2W, API 2Y, and API 2Z or BS 7448 [B.1] or ASTM E1290-02 [B.2].

B.2 Test-assembly Welding

The test assembly shall be prepared as shown in Figure B.1. The square bevel shall be machined as square as is possible. The following requirements also apply.

- a) Back-gouging shall be carefully controlled, gouging into the square bevel shall be avoided. Careful grinding shall be used to achieve the final groove profile.
- b) Surfaces shall be machined to obtain as large a thickness of the test specimen, B , as is practical for the test weld.
- c) Mechanized welds in the 1G position may be made with the test plate assembly rotated 15° to 20° about the weld axis so that the angle of attack between the electrode and the edge preparation, the degree of weld bead overlap, and the clearance between the electrode tip and the edge preparation simulate the production welding arrangement. Temporary flux dams may be attached for submerged arc welding (SAW). For welding the cap passes, the test plate assembly may be restored to its flat position.

B.3 Number and Location of CTOD Specimens

CTOD specimens shall be located within the weld metal (and the HAZ if required) such that the notch positions are in accordance with Figure B.2.

Three valid weld metal specimens from the highest heat input/interpass temperature and three valid specimens from the lowest heat input/interpass temperature to be qualified shall be obtained for each welding procedure.

When the preproduction steel qualification requirements of 20.2.2.5.2 have not been met, the HAZ specimens shall be obtained from the highest heat input/interpass temperature and from the lowest heat input/interpass temperature. Twelve specimens shall be prepared from cross-section B in Figure B.2, and five each from cross-sections C and D in Figure B.2.

The three “cross-section B” specimens with the lowest CTOD values from each of the high and low heat input sets shall be sectioned as shown in Figure B.3, to characterize the fracture initiation site.

B.4 Specimen Preparation

The specimen shall be prepared as follows:

- for plate thickness $T < 75$ mm (3 in.), the specimen width shall be $W = 2B$; for $T > 75$ mm (3 in.), $W = B$, where T , W , and B have the meaning shown in Figure B.1;
- the machined groove, saw cut, or electron discharge machine (EDM) notch, and fatigue cracking shall comply with the selected CTOD test procedure.

The procedure for determining notch locations is shown in Figure B.2 and is as follows:

- a) locate the thickest pass that is both in the central $\frac{2}{3}$ of the specimen thickness and immediately adjacent to the nonbevelled side of the weld;
- b) locate line A (centerline) that lies through the center of this pass;
- c) locate the reference line R, perpendicular to the specimen surface and dividing the fusion line such that over the middle 80 % of the specimen width, equal portions of the fusion line are to each side of the reference line;
- d) locate lines B (fusion line + 0.4 mm), C (HAZ + 2 mm), and D (HAZ + 5 mm), at 0.4 mm, 2 mm, and 5 mm from the reference line.

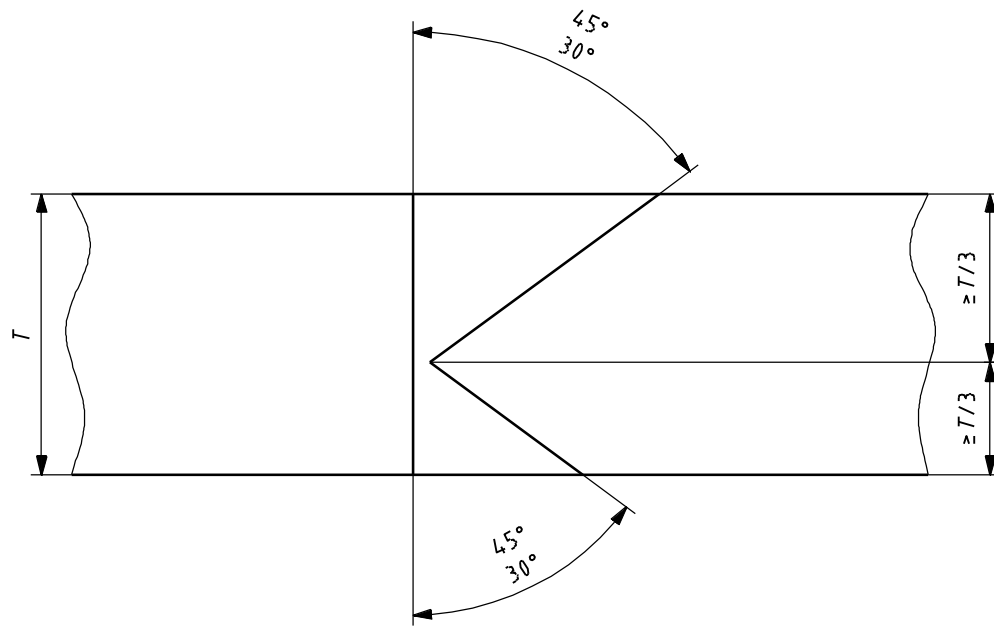
Surface-notched subsidiary-geometry specimens with $B = W$ shall be used for repair weld tests unless the repair weld consumable has been qualified by CTOD testing of a full penetration butt weld made using the same welding parameters in the same position.

B.5 Pre-compression

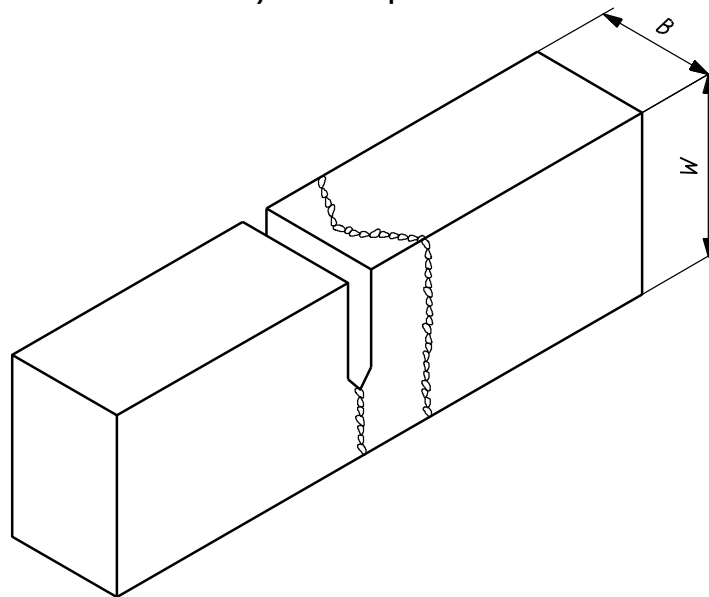
For specimens in the as-welded condition, a pre-compression dimple shall be applied to both sides of the specimen at the tip of the machined notch. The depth of the dimple shall not exceed 0.5 % strain per side (i.e. 0.01 B total). After the first series of specimens have been tested, this amount may be reduced in order to achieve a more uniform fatigue pre-crack front. The dimple shall be made with hardened indenters with a minimum diameter of 19 mm ($\frac{3}{4}$ in.). Multiple overlapping indentations may be used. The location of the dimple may overlap the machined notch. However, it shall be at least 6.4 mm ($\frac{1}{4}$ in.) from the side of the specimen opposite from the machined notch (to prevent distorting the back surface).

B.6 Sectioning

Specimen sectioning and examination shall be in accordance with Figure B.3.



a) Weld Preparation



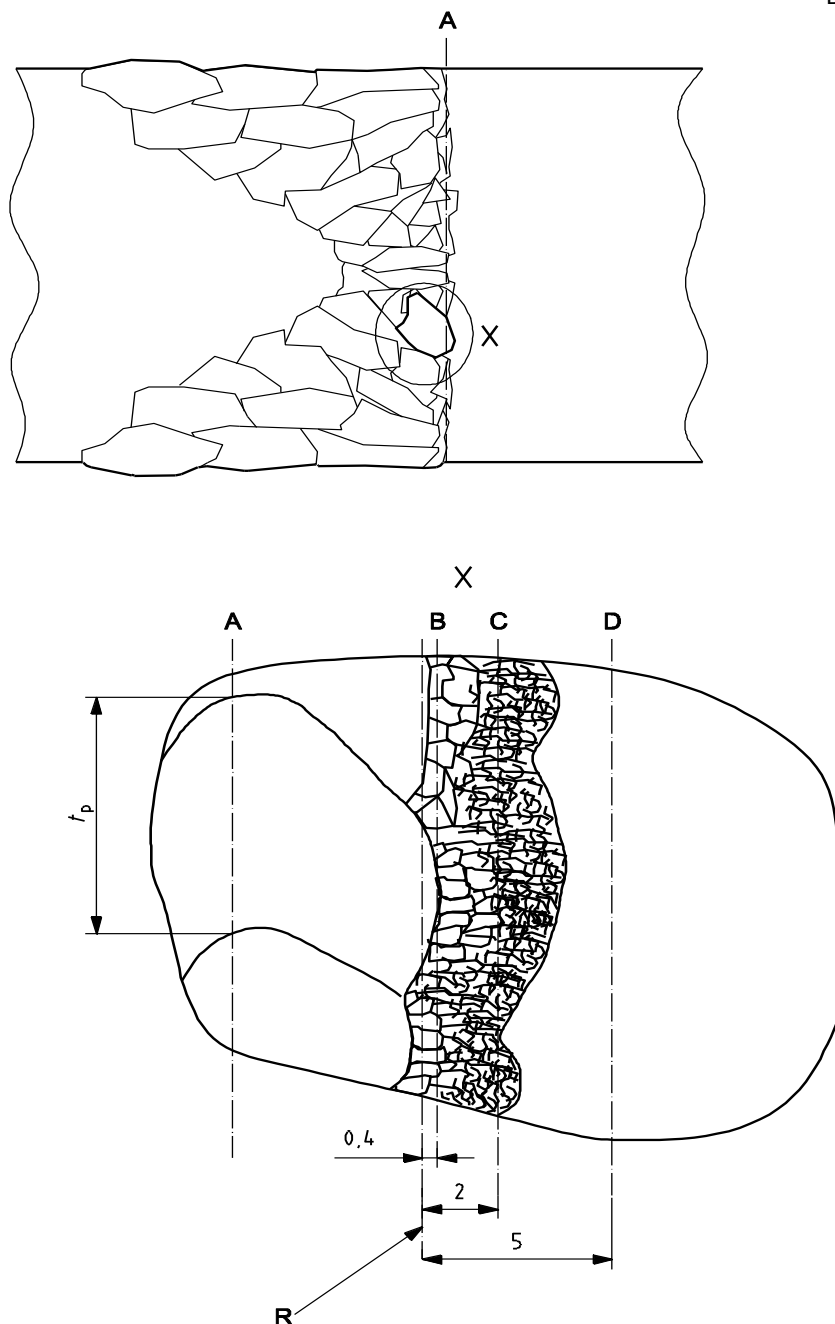
b) CTOD Specimen

Key B specimen thickness T plate thickness W specimen width

NOTE See B.4.

Figure B.1—CTOD Specimen Preparation

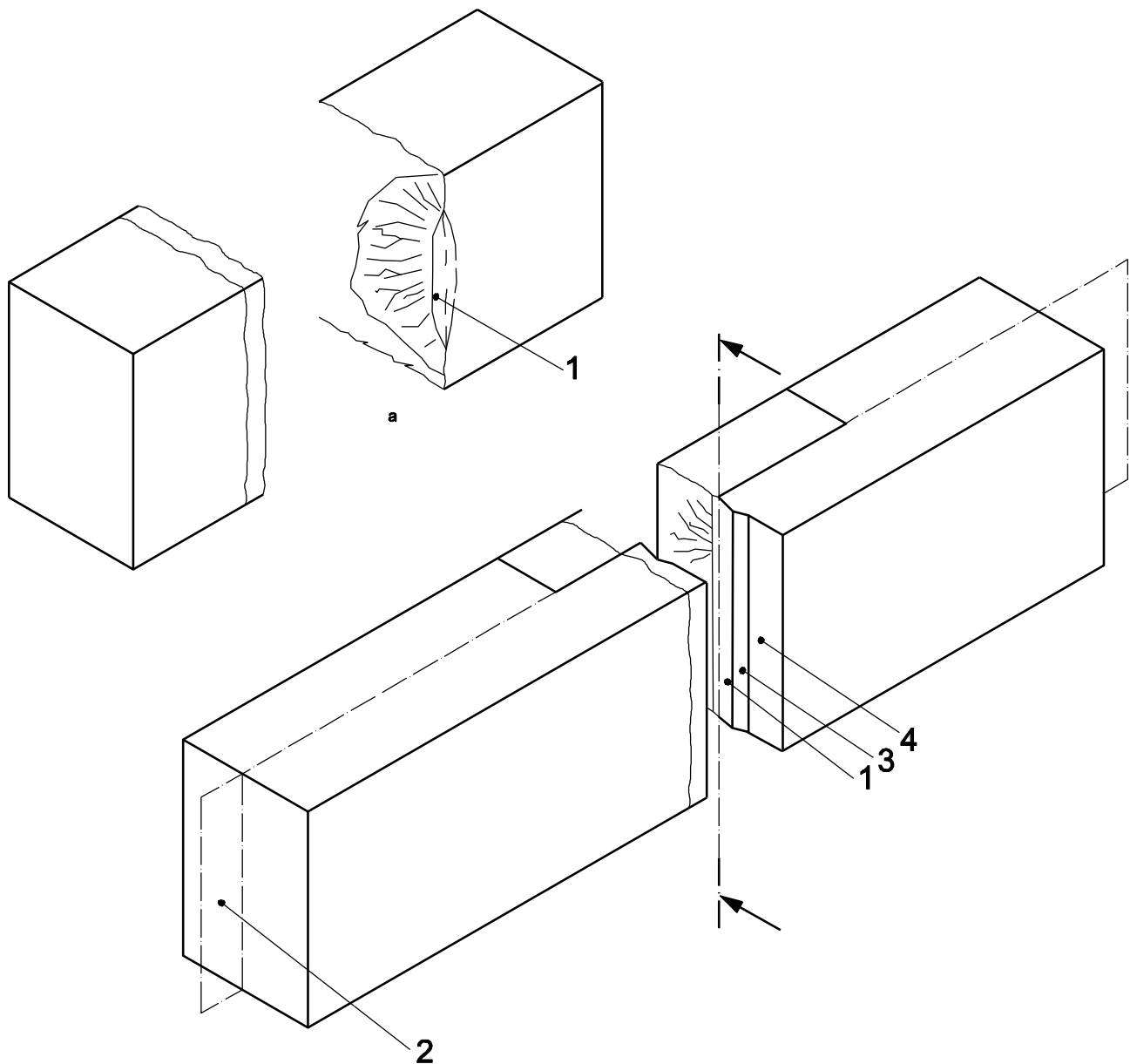
Dimensions in millimeters

**Key**

- A centerline of thickest weld pass
- B fusion line + 0.4 mm—coarse grain HAZ
- C HAZ + 2 mm—fine grain HAZ
- D HAZ + 5 mm—subcritically reheated base metal
- t_p thickness of thickest weld pass in middle $\frac{2}{3}$ of weldment
- R reference line [see B.4 c)]

NOTE See B.4.

Figure B.2—Notch Locations for CTOD Specimens



Remove a full-thickness section from each side of the fracture. Polish and etch on the plane of examination. Examine fracture features in relation to the weld beads and other artefacts.

Key

- 1 fatigue crack front
- 2 plane through specimen
- 3 saw cut
- 4 machined notch
- a Plane of examination.

Figure B.3—CTOD Specimen Sectioning and Examination

Annex C
(informative)

Material Category Approach

C.1 Selection of Material Category

The responsibility for selecting an appropriate MC for the structure belongs to the owner. This decision can be based on the guidelines below.

MC1—This category is generally used for exposure level L-1 structures. It was modeled on advanced fracture control methods used for world-record fixed platforms and compliant towers.

MC2—This category is generally used for exposure level L-2 structures. It was modeled on common practices for major structures in moderate water depths.

MC3—This category is generally used for exposure level L-3 structures. It was modeled on common practices for minor (but structurally redundant) platforms in shallow water.

C.2 Selection of Toughness Class

The required toughness class is determined from Table C.1, based on the following:

- a) type of component,
- b) MC of the structure, and
- c) steel group, in turn based on SMYS, where the required specific SMYS has previously been determined, along with thickness, by design analysis and calculation.

C.3 Specific Steel Selection

Specific material selection shall be shown in design drawings and specifications.

The strength groups and toughness classes are used to reference welding requirements, e.g. preheat and electrode selection, where these tend to follow the steel. Plates, sections, and rolled tubular sections specified by the designer should conform to a recognized specification. Supplementary specifications can be required for steel used to fabricate items intended for particularly arduous duty or where recognized steel grades are used in a thickness that is above the stated limits.

Table C.2 shows a correlation of steel group and toughness class for steel plates to US specifications. Tables C.3 and C.4 do the same for wide flange and tubular members. Steels that are not included in the lists given in C.1 and C.2 above may be used, as long as they satisfy the provisions of Sections 19 and 20, with particular attention to weldability (see also A.19.2.2.5).

Table C.1—Material Category: Material Selection for the Structure

Component Location in the Structure		SMYS Group ^e	Toughness Class		
			MC1	MC2	MC3
Joint cans ^a	Up to 50 mm (2 in.) thick	II	CV2Z	CV2Z	CV2 ^b
	Up to 50 mm (2 in.) thick	III	CV2Z	CV2Z	—
	Greater than 50 mm (2 in.) thick	II	CV2ZX	CV2Z ^c	CV2 ^b
	Greater than 50 mm (2 in.) thick	III	CV2ZX	CV2ZX	—
Bracing	Primary ^d	I	NT	NT	NT
		II	CV2 or CV1	CV1	—
		III	CV2	—	—
	Secondary	I	NT	NT	NT
	Brace end stubs at nodes	I	CV2	—	—
		II	CV2	—	—
Legs	Launch legs	I	—	—	NT
		II	CV2	CV2 or CV1	—
		III	CV2	—	—
	Launch rail	I	—	—	NT
		II	CV2	CV1	—
		III	CV2	—	—
	Legs elsewhere	I	CV1	CV1 or NT	NT
		II	CV2 or CV1	CV1	—
		III	CV2	—	—
Stiffeners	At nodes	II	CV2	CV2	CV2
	Elsewhere	I	NT	NT	NT
		II	CV1	NT	—
Piling	Heavy wall at sea floor	II	CV2	CV2 or CV1	CV1
	Elsewhere	I	NT	NT	NT
		II	CV1	—	—
	Compliant tower	III	CV2	N/A	N/A
Padeyes	Major lifts, padeye greater than 50 mm (2 in.) thick	II	CV2ZX ^c	CV2Z ^c	CV2Z
	Major lifts, padeye greater than 50 mm (2 in.) thick	III	CV2ZX	CV2ZX	—
	Major lifts, padeye 50 mm (2 in.) thick or less	II	CV2Z	CV2Z	CV2Z
	Major lifts, padeye 50 mm (2 in.) thick or less	III	CV2Z	CV2Z	—
Miscellaneous	Conductor panels	I	NT	NT	NT
	Boat landings, walkways	I	NT	NT	NT

— indicates that this combination is not normally used.

N/A not applicable.

^a Includes joint cans in legs and in primary bracing.

^b Specify steel with a low sulphur content, below 0.006 %.

^c Consider CV2ZX for nodes greater than 75 mm (3 in.) thick.

^d A higher toughness class is recommended for thicker primary bracing, especially in areas subject to collisions.

^e SMYS groups IV and V have had limited historical use, and are therefore omitted from this table. Selection of toughness class is left to the designer.

Table C.2—Correlation of Steel Group and Toughness Class for Steel Plates to US Specifications

Steel Group	Toughness Class	Specification	Grade	Thickness mm (in.)	SMYS MPa (ksi)	Comments and Recommended Supplements a, b, c, d, e
I	NT	ASTM A36		≤ 50 (≤ 2)	250 (36)	b
		ASTM A131	A	≤ 12 (≤ 1/2)	235 (34)	
	CV1	ASTM A131	B, D		235 (34)	
		ASTM A516	65	≤ 37 (≤ 1.5)	240 (35)	b, c
		ASTM A573	65		240 (35)	b, c
		ASTM A709	36T Zone 2		250 (36)	b ($P_{CE} \leq 0.40$), c (in accordance with A709 supplement S83)
	CV2	ASTM A131	E		235 (34)	c (in accordance with A131 supplement S86), d
II	NT	ASTM A572	42	≤ 50 (≤ 2)	290 (42)	a, b ($P_{CE} \leq 0.43$), $P_V \leq 0.10$
		ASTM A572	50	≤ 50 (≤ 2)	345 (50)	a, b, $P_V \leq 0.10$
	CV1	API 2MT1			345 (50)	
		ASTM A709	50T Zone 2		345 (50)	Gulf of Mexico: b, c (in accordance with A709 supplement S83)
		ASTM A709	50F Zone 3		345 (50)	Cold, nonarctic: b, c (in accordance with A709 supplement S84)
		ASTM A131	AH32		315 (46)	Option A131 supplement S87 [normalize T > 19 mm (0.75 in.)]
		ASTM A131	AH36		355 (51)	Optional A131 supplement S87 [normalize T > 19 mm (0.75 in.)]
	CV2	API 2H	42		290 (42)	d
		API 2H	50	≤ 65 (≤ 2 1/2)	345 (50)	d
		API 2H	50	> 65 (> 2 1/2)	325 (47)	d
		API 2W	50		345 (50)	
		API 2Y	50		345 (50)	
		ASTM A131	DH36 EH36		355 (51)	c (in accordance with A131 supplement S86), d
		ASTM A537	Class 1	≤ 65 (≤ 2 1/2)	345 (50)	b, c [Charpy at −40 °C (−40 °F)], d
		ASTM A633	A		290 (42)	b, c [Charpy at −40 °C (−40 °F)], d
		ASTM A633	C, D		345 (50)	b, c [Charpy at −40 °C (−40 °F)], d
		ASTM A678	A		345 (50)	b, c [Charpy at −40 °C (−40 °F)], d
III	CV2	API Spec 2W	60		415 (60)	
		API Spec 2Y	60		415 (60)	
		ASTM A537	Class 2	≤ 65 (≤ 2 1/2)	415 (60)	b, c [Charpy at −40 °C (−40 °F)], d
		ASTM A678	B		415 (60)	b, c [Charpy at −40 °C (−40 °F)], d
		ASTM A710	Grade A Class 3	≤ 50 (≤ 2)	515 (75)	b, c [Charpy at −40 °C (−40 °F)], d
		ASTM A710	Grade A Class 3	50 to 100 (2 to 4)	450 (65)	b, c [Charpy at −40 °C (−40 °F)], d
		ASTM A710	Grade A Class 3	> 100 (> 4)	415 (60)	b, c [Charpy at −40 °C (−40 °F)], d
IV	CV1	ASTM A709	HPS70WF	≤ 65 (≤ 2 1/2)	485 (70)	b, c (in accordance with A709 supplement S84)

^a With $t > 13\text{mm}$, ASTM A6 S29 is required.

^b $P_{CE} \leq 0.45$ unless noted otherwise; for pressure vessel steels, see ASTM A20 supplement S20.

^c CVN testing in accordance with ASTM A6 or A20 supplement S5, frequency H. For plate testing, use transverse Charpy specimens. Unless noted otherwise in steel specification, absorbed energy requirement should be as stated in Table 19.4-1.

^d For joint cans, add ASTM A6 supplement S8 (or A20 supplement S12), ultrasonic examination to ASTM A578 level B; and also limit $P_S \leq 0.006\%$.

^e ASTM supplement references are from 2013 book of standards. A6 is for structural steel; A20 for pressure vessels.

Table C.3—Correlation of Steel Group and Toughness Class for Structural Steel Shapes to US Specifications

Steel Group	Toughness Class	Specification	Grade	Thickness mm (in.)	SMYS MPa (ksi)	Comments and Recommended Supplements a, b, c, d
I	NT	ASTM A36		≤ 50 (≤ 2)	250 (36)	Limited availability
		ASTM A131	A	≤ 12 (≤ 1/2)	235 (34)	Limited availability
	CV1	ASTM A709	36T Zone 2		250 (36)	b ($P_{CE} \leq 0.40$), c (in accordance with A709 supplement S83)
II	NT	API 2MT2	Class C		345 (50)	
		ASTM A572	42	≤ 50 (≤ 2)	290 (42)	a, b ($P_{CE} \leq 0.43$), $P_V \leq 0.10$
		ASTM A572	50	≤ 50 (≤ 2)	345 (50)	a, b, $P_V \leq 0.10$
		ASTM A992			345 (50)	
	CV1	ASTM A709	50T Zone 2		345 (50)	Gulf of Mexico: b, c (in accordance with A709 supplement S83)
		ASTM A709	50F Zone 3		345 (50)	Cold, nonarctic: b, c (in accordance with A709 supplement S84)
		ASTM A131	AH32		315 (45)	
		ASTM A131	AH36		350 (50)	
		API 2MT2	Class B		345 (50)	Gulf of Mexico
	CV2	ASTM A913	50		345 (50)	
		API 2MT2	Class A		345 (50)	Cold, nonarctic

^a With $t > 13\text{mm}$, ASTM A6 S29 is required.

^b ASTM A6 supplement S74 with $P_{CE} \leq 0.45$, unless noted otherwise; for pressure vessel steels see ASTM A20 supplement S20.

^c CVN testing in accordance with ASTM A6 or A20 supplement S5, frequency H; test temperatures shall be as stated or in accordance with specifications cited; unless noted otherwise in steel specification, absorbed energy requirement should be as stated in Table 19.4-1.

^d ASTM supplement references are from 2013 book of standards. A6 is for structural steel; A20 for pressure vessels.

Table C.4—Correlation of Steel Group and Toughness Class for Structural Steel Pipe to US Specifications

Steel Group	Toughness Class	Specification	Grade	SMYS MPa (ksi)	Comments
I	NT	API 5L	B	240 (35)	PSL 2 pipe recommended
		ASTM A53	B	240 (35)	
		ASTM A106	B	240 (35)	
		ASTM A135	B	240 (35)	
		ASTM A139	B	240 (35)	
		ASTM A500 (round)	A	230 (33)	
		ASTM A500 (shaped)	A	270 (39)	
		ASTM A501	A	250 (36)	
		ASTM A501	A	250 (36)	
	CV1	ASTM A106	B	240 (35)	Normalized; Charpy at 5 °C (40 °F); recheck yield strength and tensile strength after normalizing
II	NT	ASTM A333	6	240 (35)	
		ASTM A334	6	240 (35)	
		API 5L	L290 (X42)	290 (42)	PSL 2 pipe highly recommended
		API 5L	L360 (X52)	360 (52)	
		ASTM A500 (round)	B	290 (42)	
		ASTM A500 (shaped)	B	315 (46)	
		ASTM A1085	B	345 (50)	
	CV1	API 5L	L360NO (X52NO)	360 (52)	Charpy requirement 40 J (30 ft-lb) at −18 °C (0 °F); may be normalized (N), quenched and tempered (Q&T), or thermomechanical formed (M).

NOTE In addition to the above, API 2B structural pipe made from any of the plates listed in Table C.2 may be used.

Annex D
(informative)

Welding and Weld Inspection Requirements—Material Category Approach

D.1 General

This annex contains the weld toughness and NDT requirements for the fabrication, welding, and inspection of fixed steel offshore structures that have been designed using the MC methodology.

It is anticipated that the material category (MC number), steel group (strength level), and toughness class will have been determined by specific steel selections given in the design drawings and specifications.

The weld toughness requirements follow a hierarchy that parallels the toughness class of the steels to be joined, and that has been previously defined (see Section 19) in consideration of criticality, strength level, thickness, and other aspects of the corresponding structural component.

The NDT inspection requirements use a hierarchy that follows the structural component type directly.

D.2 Weld Toughness

D.2.1 Weld Metal Toughness

Table D.1 gives the minimum weld metal toughness requirements. These shall be demonstrated by testing during WPQ. An exception is provided for low-criticality applications, where as-classified welding consumable toughness may be relied upon. Steel group V is not included in Table D.1; the toughness shall be specified by the user.

D.2.2 HAZ Toughness

Table D.2 gives the minimum HAZ toughness requirements. For critical applications as indicated, these shall be demonstrated by tests during WPQ. CTOD testing of the HAZ is required only for CV2X and CV2ZX steels, only in MC1 structures, only for thicknesses over 50 mm, and only where not covered by a manufacturer’s steel prequalification. CTOD testing of the HAZ shall not be required if the steel has been preproduction tested in the HAZ and the production WPS heat input range is within the preproduction test range.

Table D.1—Material Category: Minimum Weld Metal Toughness Requirements

Steel Group	Toughness Class	MC1		MC2		MC3	
		Energy J (ft-lb)	Temperature below LAST °C (°F)	Energy J (ft-lb)	Temperature below LAST °C (°F)	Energy J (ft-lb)	Temperature below LAST °C (°F)
I	NT	27 (20)	10 (18) ^a	27 (20)	10 (18) ^a	27 (20)	10 (18) ^a
	CV1	27 (20)	10 (18) ^a	27 (20)	10 (18) ^a	N/A	N/A
	CV2	27 (20)	20 (36)	27 (20)	20 (36)	N/A	N/A
II	NT	27 (20)	20 (36) ^a	27 (20)	10 (18) ^a	27 (20)	10 (18) ^a
	CV1	27 (20)	10 (18)	27 (20)	10 (18)	27 (20)	10 (18)
	CV2	34 (25)	20 (36)	34 (25)	20 (36)	34 (25)	20 (36)
	CV2Z	34 (25)	20 (36)	34 (25)	30 (54)	34 (25)	30 (54)
	CV2ZX	34 (25)	30 (54)	34 (25)	30 (54)	N/A	N/A
III	CV2	40 (30)	20 (36)	40 (30)	20 (36)	N/A	N/A
	CV2Z	40 (30)	30 (54)	40 (30)	30 (54)	N/A	N/A
	CV2Z	40 (30)	30 (54)	N/A	N/A	N/A	N/A
	CV2ZX	40 (30)	30 (54)	40 (30)	30 (54)	N/A	N/A
IV	CV2ZX	40 (30)	10 (18)	N/A	N/A	N/A	N/A
NOTE N/A indicates that this category is not normally applicable. Where such steels are present, use the nearest toughness requirements in the same row of the table.							
^a CVN test shall not be required if the electrode specification has a minimum CVN value that is greater than or equal to 27 J (20 ft-lb) satisfying the test temperature or lower.							

Table D.2—Material Category: Minimum Heat Affected Zone Toughness Requirements

Steel Group	Toughness Class	MC1		MC2		MC3	
		Energy J (ft-lb)	Temperature below LAST °C (°F)	Energy J (ft-lb)	Temperature below LAST °C (°F)	Energy J (ft-lb)	Temperature below LAST °C (°F)
I	NT	N/A	N/A	N/A	N/A	N/A	N/A
	CV1	20 (15)	0 (0)	20 (15)	0 (0)	N/A	N/A
	CV2	20 (15)	10 (18)	N/A	N/A	N/A	N/A
II	NT	N/A	N/A	N/A	N/A	N/A	N/A
	CV1	27 (20)	0 (0)	27 (20)	0 (0)	20 (15)	0 (0)
	CV2	34 (25)	10 (18)	34 (25)	10 (18)	34 (25)	0 (0)
	CV2Z	34 (25)	10 (18)	34 (25)	10 (18)	34 (25)	0 (0)
	CV2ZX	34 (25)	20 (36)	34 (25)	20 (36)	N/A	N/A
III	CV2	40 (30)	10 (18)	40 (30)	10 (18)	N/A	N/A
	CV2Z	40 (30)	10 (18)	40 (30)	10 (18)	N/A	N/A
	CV2ZX	40 (30)	20 (36)	40 (30)	20 (36)	N/A	N/A
IV	CV1	40 (30)	0 (0)	N/A	N/A	N/A	N/A
NOTE N/A indicates that this category is not required.							

D.3 Inspection

100 %VT applies to all components for all material categories. See Table D.3 for additional extent of NDT requirements.

Proportional inspection rates are further described in A.20.3.

Personnel requirements and reject criteria for NDT, other required inspections (e.g. visual and dimensional checks), and document requirements are described in:

- a) the selected generic standard for welding, fabrication, and inspection;
- b) Annex E for fabrication tolerances;
- c) the project QC plan;
- d) the owner’s stipulated QA requirements; and/or
- e) the contractor’s accredited QMS (see 21.2).

Annex E (normative)

Fabrication Tolerances

E.1 Measurements

Tolerances shall be based on theoretical setting out points and centerlines of the structure referenced to permanent approved datum points (e.g. coordinated survey stations) and corrected to a temperature of +20 °C (68 °F). Compensation shall be made for significant deflection differences between temporary support and final conditions.

Fabrication and yard assembly supports shall be set to within ± 5 mm ($^1/4$ in.) of the appropriate position shown on the approved cutting out drawings. Where no such drawings exist, fabrication shall be carried out from a level plane to within ± 5 mm ($^1/4$ in.).

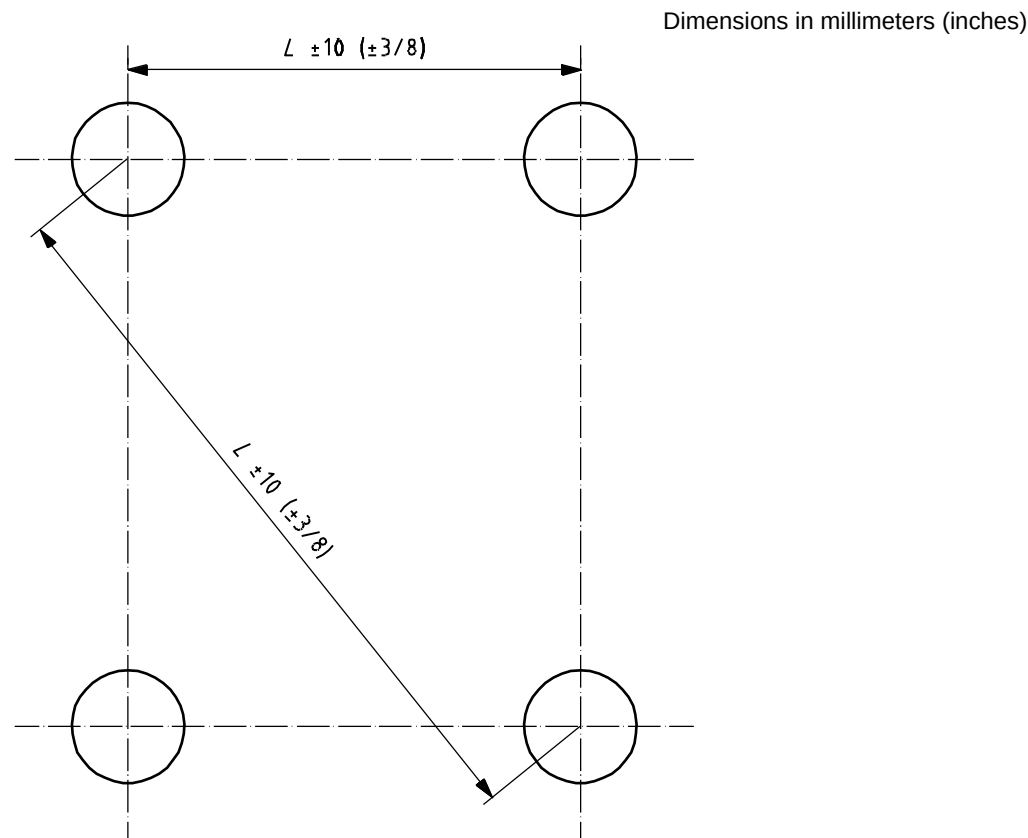
E.2 Launch Rails

The dimensional tolerance of launch rail centerlines shall be within ± 20 mm ($^3/4$ in.) of the theoretical position and shall also be within ± 6 mm ($^1/4$ in.) of its reference elevation. The variation in elevation between any two points on a launch rail shall not exceed 3 mm ($^1/8$ in.) within any 3 m (10 ft).

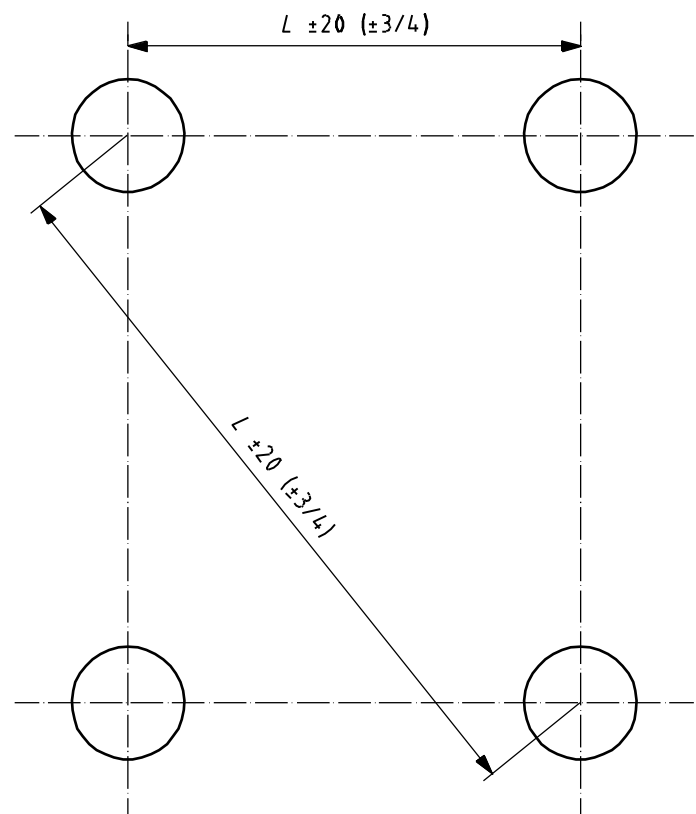
E.3 Global Horizontal Tolerances

The global tolerances for leg spacing at plan bracing levels are as shown in Figure E.1 and detailed below:

- a) the horizontal center-to-center distance between adjacent legs at the top of a structure where a deck or other structure is to be placed (stab-in nodes) shall be within 10 mm ($^3/8$ in.) of the design values;
- b) the horizontal center-to-center distance between legs at other locations shall be within 20 mm ($^3/4$ in.) of the design values;
- c) the horizontal center-to-center diagonal distances between legs at the top of a structure where a deck or other structure is to be placed (stab-in nodes) shall be within 10 mm ($^3/8$ in.) of the design values;
- d) the horizontal center-to-center diagonal distance between legs at other locations shall be within 20 mm ($^3/4$ in.) of the design values.



a) Top Elevation for Deck Stab-in



b) Other Elevations

Figure E.1—Global Horizontal Tolerances

E.4 Global Vertical Tolerances

The global tolerances for vertical levels of plan bracing are as shown in Figure E.2 and are detailed as follows:

- a) the elevation of plan bracing levels shall be within 13 mm (¹/₂ in.) of the design values;
- b) the vertical level of braces within a horizontal plane shall be within 13 mm (¹/₂ in.) of the design values;
- c) the vertical distance between plan bracing elevations shall be within 13 mm (¹/₂ in.) of the design values.

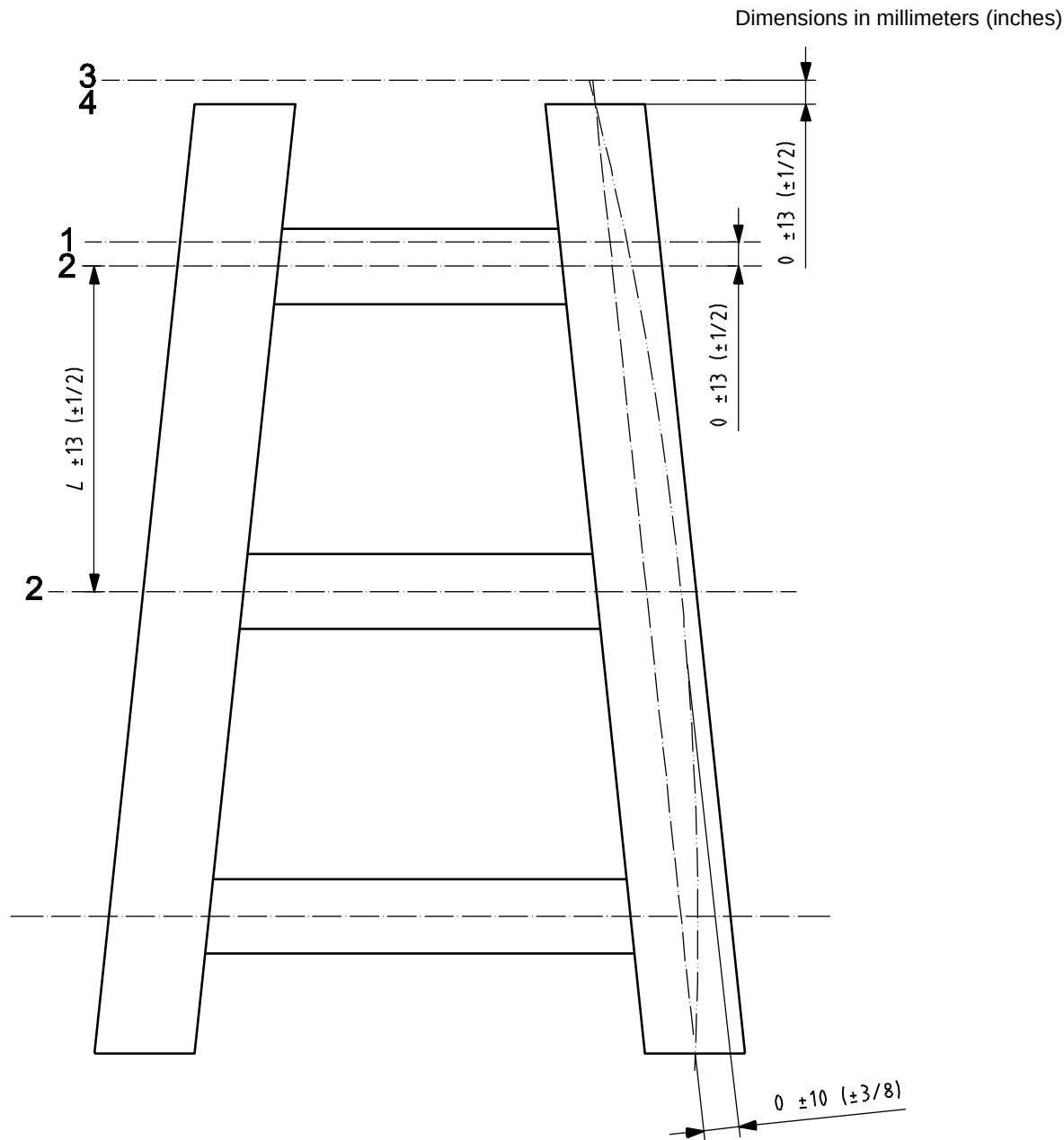


Figure E.2—Global Vertical Tolerances

E.5 Roundness of Tubular Members

For tubular members with thicknesses of 50 mm (2 in.) or less, the difference between the major and the minor outside diameters (the out-of-roundness) at any point of the tubular member shall not exceed the smaller of either 1 % of the diameter or 6 mm (¹/₄ in.).

For tubular members with thicknesses of more than 50 mm (2 in.), the difference between the major and the minor outside diameters at any point of the tubular member shall not exceed 12.5 % of the wall thickness.

For tubular members with nominal outside diameters greater than or equal to 1200 mm (50 in.) and with wall thicknesses less than or equal to 100 mm (4 in.), the maximum difference between the major and minor actual outside diameters at any section of the tubular shall be 13 mm (¹/₂ in.), provided the actual circumference is within 6 mm (¹/₄ in.) of the nominal circumference.

E.6 Circumference of Tubular Members

The difference between the actual and nominal outside circumferences at any point of a tubular member shall not exceed the lesser of either 1 % of the nominal circumference or 13 mm (¹/₂ in.).

E.7 Straightness and Circumferential Weld Locations of Tubular Members

E.7.1 The tolerances for the straightness of tubular members are as shown in Figures E.3 and E.4, and as follows:

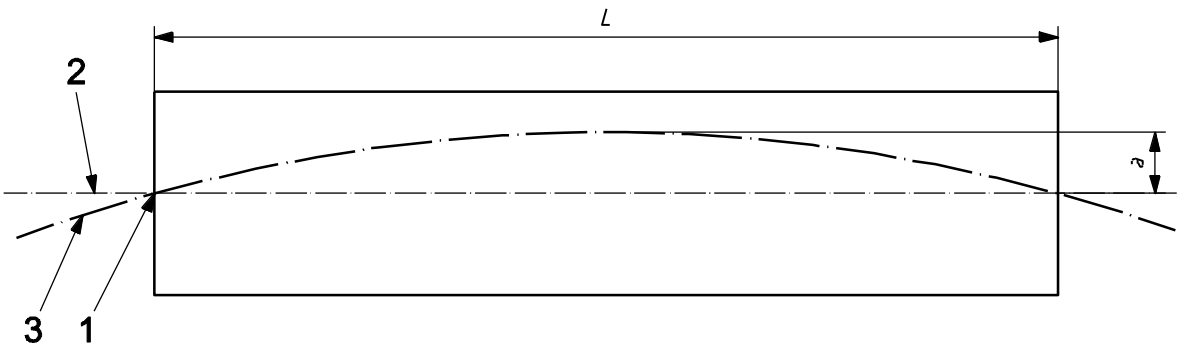
- a) lengths equal to or less than 12 m shall be straight to within 10 mm (³/₈ in.);
- b) lengths exceeding 12 m shall be straight to within 13 mm (¹/₂ in.).

E.7.2 Additionally, the tolerances for the straightness of tubular members when assembled into the structure are as follows:

- a) brace members shall be straight to within 0.12 % of the nominal length measured between work points;
- b) chord members shall be straight to within 0.10 % of the nominal length measured between the work points of major framing members.

E.7.3 The tolerances for the location of cans with a different wall thickness to the member, and other wall thickness transitions within a tubular member, are as shown in Figure E.5 and as follows:

- a) for joint cans, within 25 mm (1 in.) of the design value;
- b) for other changes of wall thickness, within 50 mm (2 in.) of the design value.

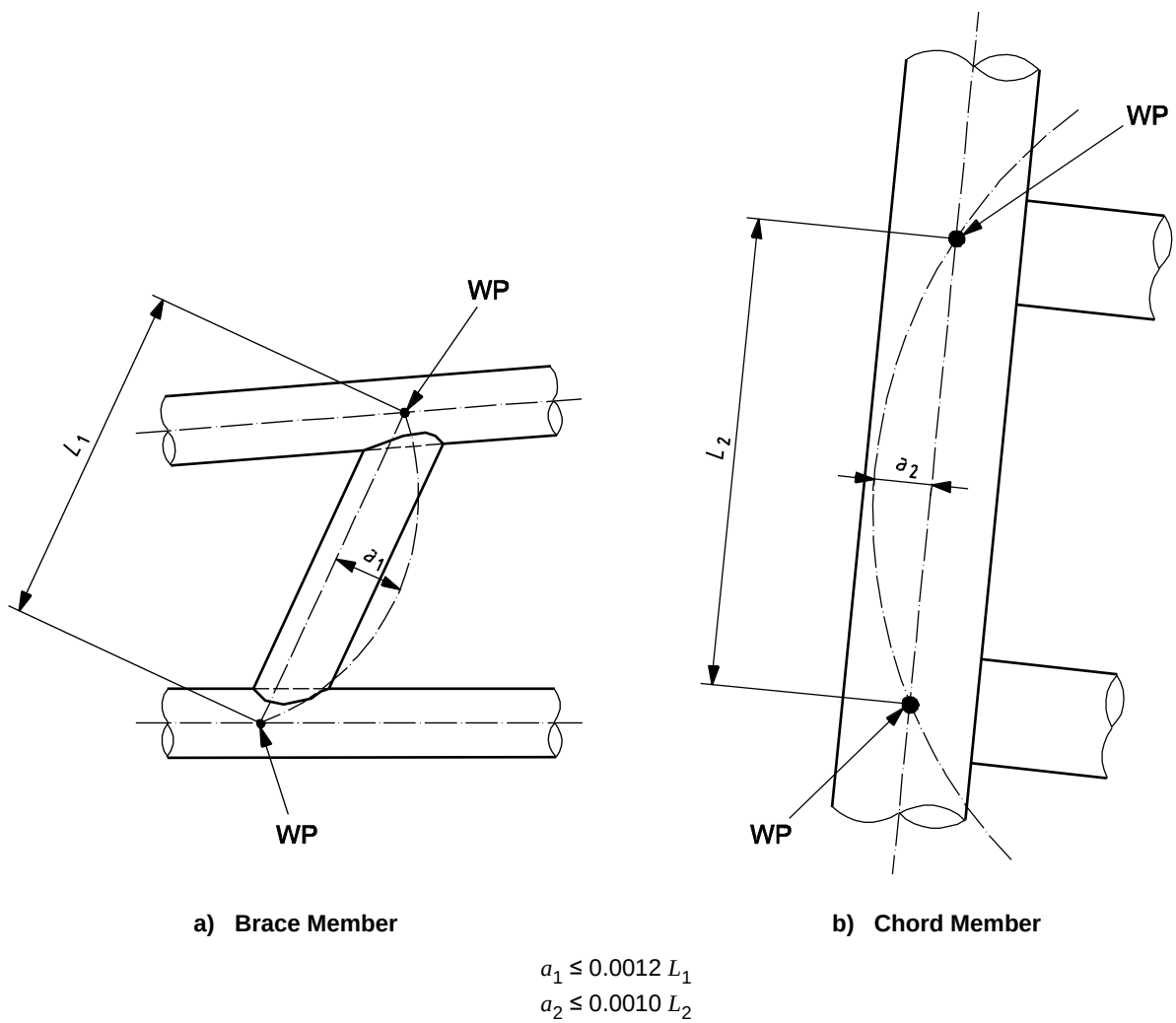


For $L \leq 12 \text{ m (40 ft)}$, $a \leq 10 \text{ mm (0.375 in.)}$
For $L > 12 \text{ m (40 ft)}$, $a \leq 13 \text{ mm (0.5 in.)}$

Key

- 1 datum: centerline at member end
- 2 theoretical centerline
- 3 actual centerline
- L total length of fabricated tubular prior to structural assembly; L may consist of one or more tubular sections welded together
- a maximum actual centerline deviation from the theoretical centerline, measured normal to the theoretical centerline and anywhere within L in any plane

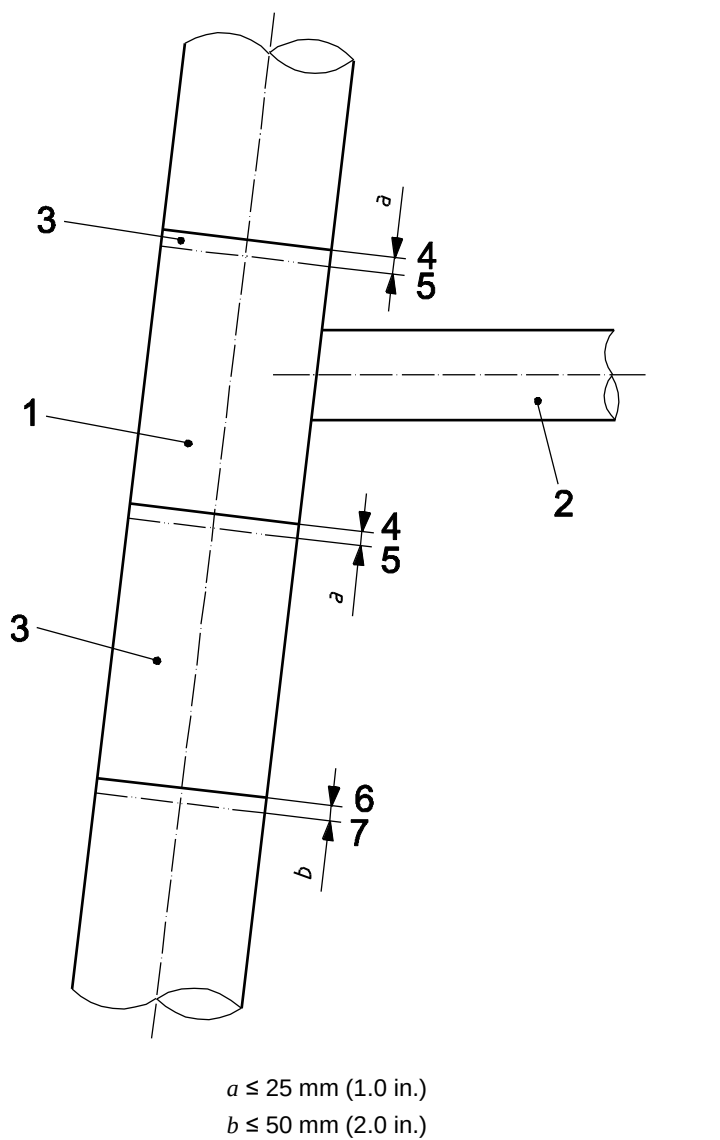
Figure E.3—Tolerance on Straightness of Tubular Members



Key

- WP work point
- L_1 nominal length of brace member between work points
- L_2 nominal length of chord member between work points of major framing members
- a_1 maximum deviation at brace centerline from theoretical centerline
- a_2 maximum deviation of chord centerline from theoretical centerline established between nearest framing member working points

Figure E.4—Tolerances on Straightness of Tubular Members Assembled into Structure



Key

- 1 joint
- 2 brace
- 3 member
- 4 theoretical location of thickness transition between joint and member
- 5 actual location of thickness transition between joint and member
- 6 theoretical location of thickness transition within member
- 7 actual location of thickness transition within member
- a distance between design and as-built location thickness change between joint can and member
- b distance between design and as-built location thickness change within member

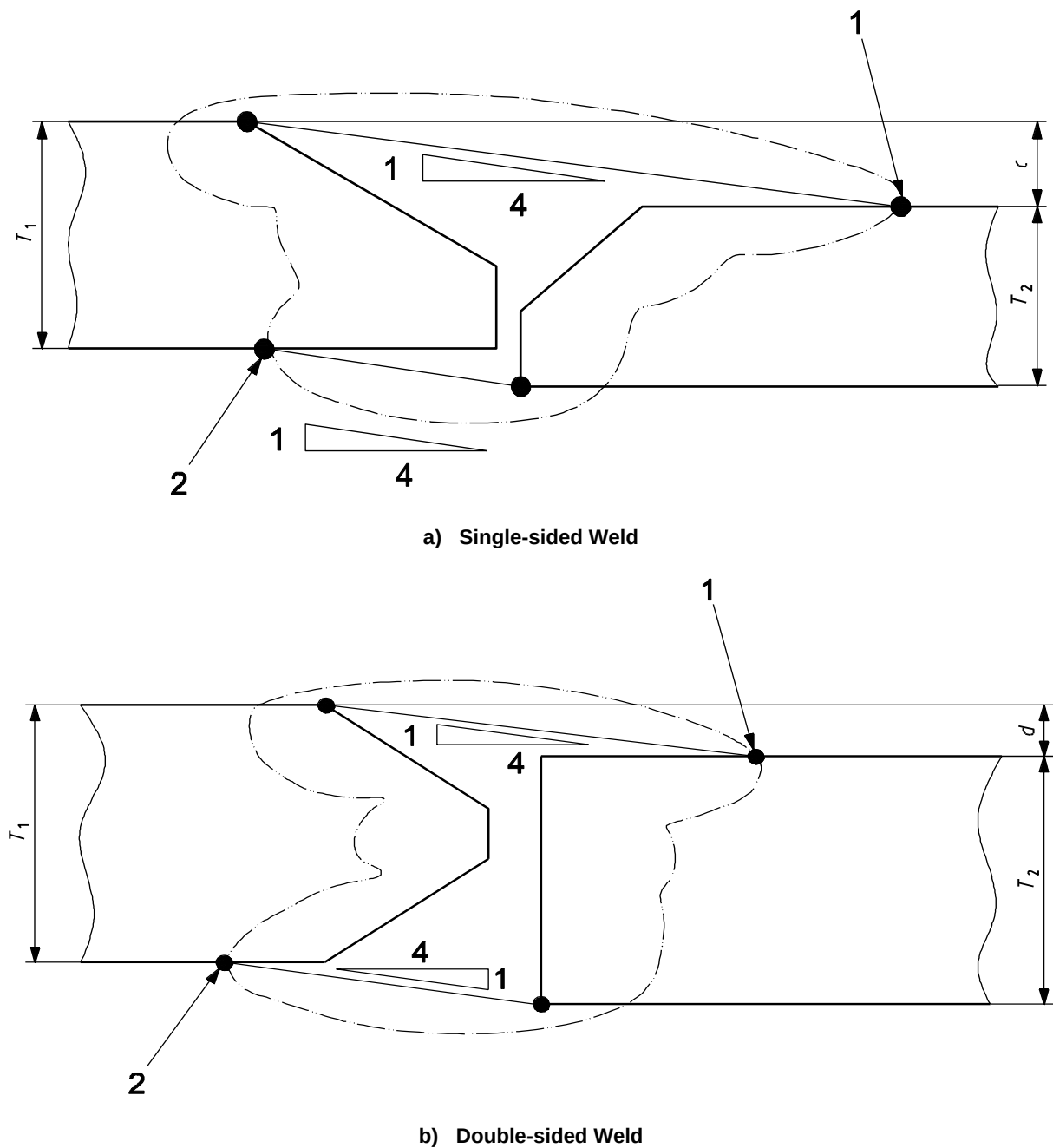
Figure E.5—Tolerances on Thickness Change Locations

E.8 Joint Mismatch for Tubular Members

The tolerances for the mismatch at circumferential welds and longitudinal seams in tubular members are as shown in Figure E.6 and detailed below:

- a) for double-sided welds, the lesser of 10 % of the thicker member thickness, or 6 mm ($\frac{1}{4}$ in.);
- b) for single-sided welds, the lesser of 10 % of the thicker member thickness, or 3 mm ($\frac{1}{8}$ in.).

Where these tolerances are exceeded, the transitions shall be made by weld profiling to a maximum slope of 1:4 (see Figure E.6).



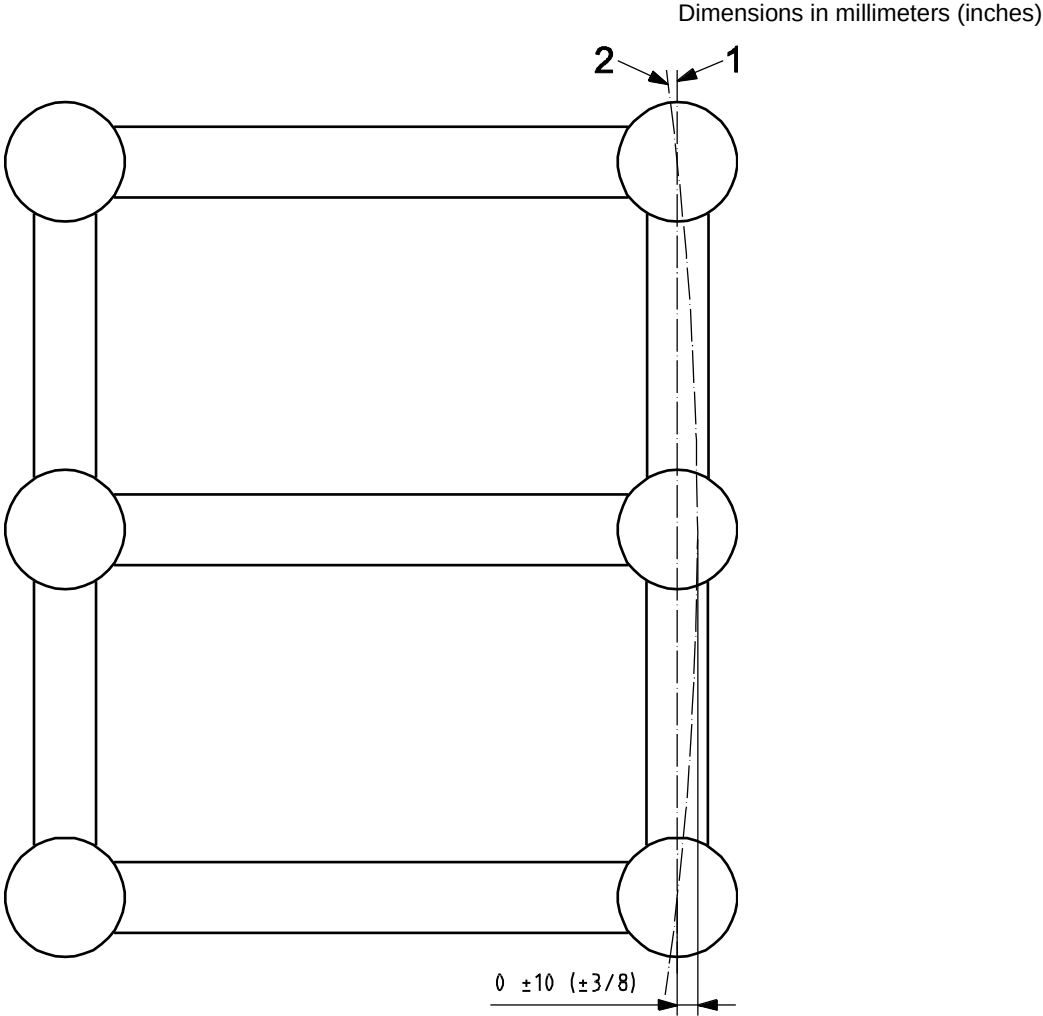
- Key**
- 1, 2 points established by the transition weld toe that can be connected by a theoretical line to the prepared groove high point (on-top surface) and low point (on-bottom surface)
 - c less than the smaller value of 1) $0.10 T_1$, 2) $0.10 T_2$, 3) 3 mm ($\frac{1}{8}$ in.)
 - d less than the smaller value of 1) $0.10 T_1$, 2) $0.10 T_2$, 3) 6 mm ($\frac{1}{4}$ in.)
- NOTE Grooves shown are for illustration only.

Figure E.6—Joint Mismatch Tolerances

E.9 Leg Alignment and Straightness Tolerances

In addition to the tolerances for individual members (see E.7), the tolerances for the alignment and straightness of legs shall be as shown in Figure E.7 and as follows:

- a) at each plan bracing level the legs shall be aligned horizontally within 10 mm ($\frac{3}{8}$ in.) of the design location;
- b) each leg shall be straight to within 10 mm ($\frac{3}{8}$ in.) over the entire length (see Figure E.2).



Key

- 1 theoretical centerline
- 2 actual centerline

Figure E.7—Leg Alignment Tolerances at Plan Bracing Level (Structures with More Than Four Legs)

E.10 Tubular Joint Tolerances

The alignment and tolerances for tubular joints (braces, brace stubs, and chords) are shown in Figures E.8 and E.9 and detailed below.

The as-built work point of a brace stub on a chord centerline shall be within 13 mm (¹/₂ in.) of the design work point location.

The as-built work point of a brace on a chord centerline built with point-to-point construction shall be within 20 mm (³/₄ in.) of the design work point location.

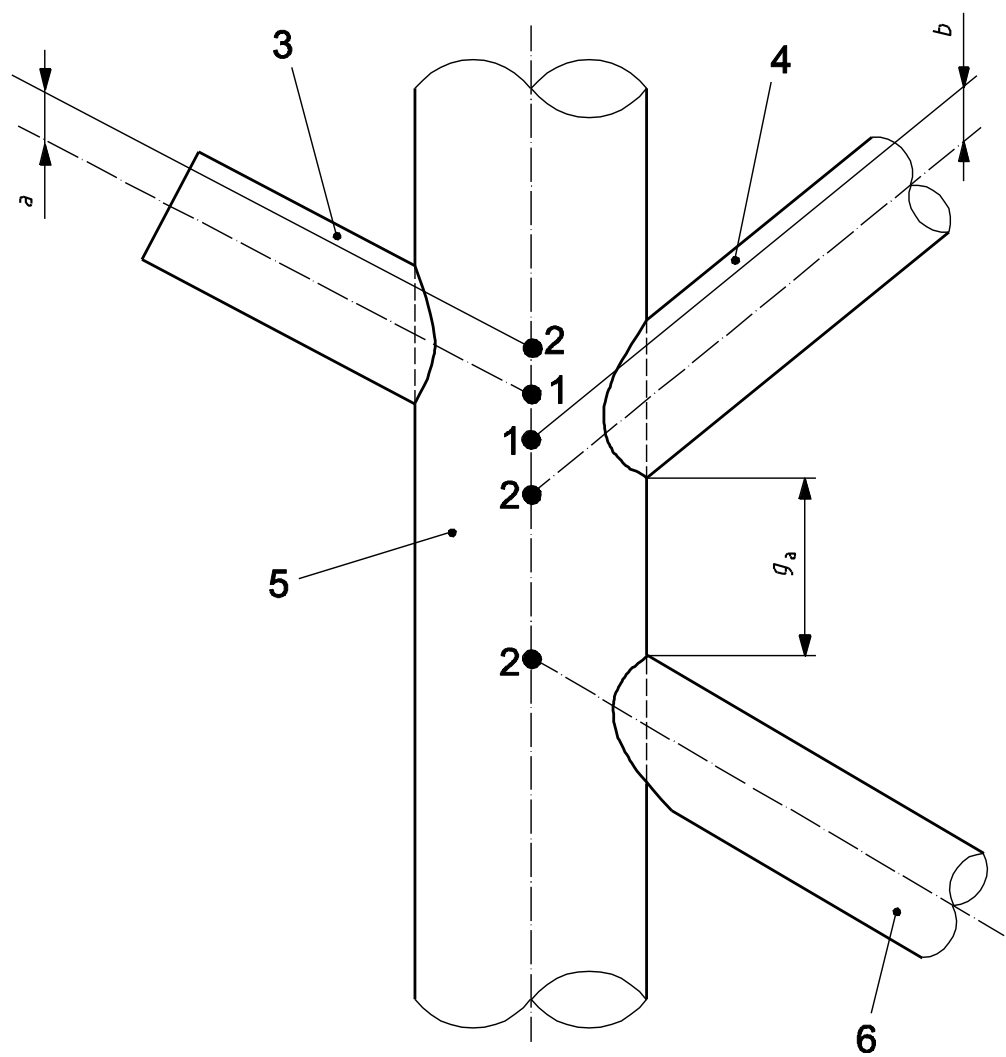
For braces built on the basis of a design minimum gap, *g*, between brace toes on the chord diameter, the as-built distance between brace toes shall not be less than *g* – 3 mm (¹/₂ in.) or greater than *g* + 25 mm (1 in.).

If a joint can or a brace stub contains intermediate circumferential welds, the individual lengths within the joint can or brace stub shall be within the range of design length + 10 mm (³/₈ in.) to design length – 5 mm (¹/₄ in.) and the location of the circumferential welds shall be within 5 mm (¹/₄ in.) of the design location.

The overall lengths of the brace stub of a tubular joint shall be within the range of design length + 50 mm (2 in.) to design length – 5 mm (¹/₄ in.).

The tolerances on the positioning and alignment of stubs on tubular joints are as follows:

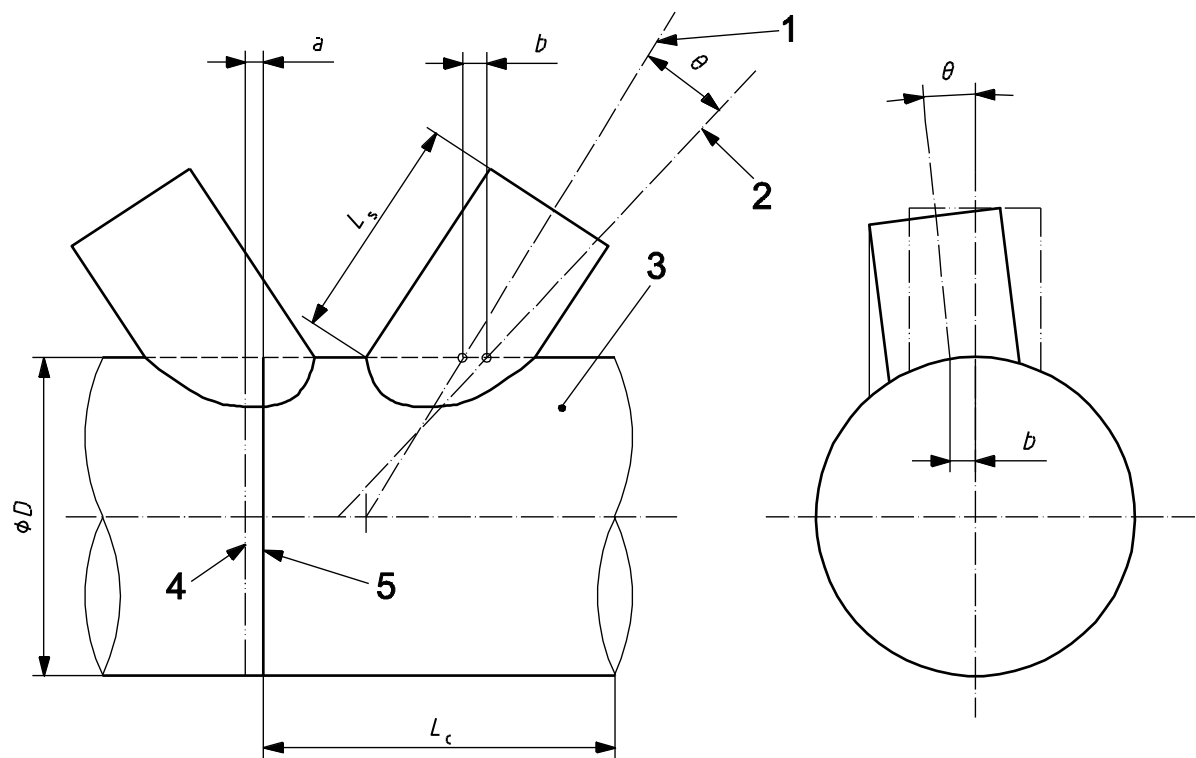
- a) the centerline of the brace at the intersection with the chord wall shall be within 5 mm (¹/₄ in.) of the design position where the chord diameter is less than 3.5 m (11.5 ft);
- b) the centerline of the brace at the intersection with the chord wall shall be within 10 mm of the design position where the chord diameter is greater than or equal to 3.5 m (11.5 ft);
- c) the angular orientation of the brace stub shall be within 10 min (0.17°) of the design orientation;
- d) the position of the centerline of the brace end of the stub shall be within 5 mm (¹/₄ in.) of the design position.



Key

- 1 design work point
- 2 as-built work point
- 3 brace stub
- 4 brace built point-to-point
- 5 chord
- 6 brace built to minimum gap clearance
- a distance between design and as-built location of brace stub work points on chord centerline $a \leq 13 \text{ mm (0.5 in.)}$
- b distance between design and as-built location of brace work points on chord centerline $b \leq 20 \text{ mm (0.75 in.)}$
- g_a as-built gap between brace toes
- ^a $g - 12 \text{ mm (0.5 in.)} \leq g_a \leq g + 25 \text{ mm (1.0 in.)}$, where g is the design gap between brace toes.

Figure E.8—Tolerances on Alignment of Brace Stubs at Tubular Joints



for $D < 3.5 \text{ m (11.5 ft)}$, $b < 5 \text{ mm (}^1\text{/}_4 \text{ in.)}$
for $D \geq 3.5 \text{ m (11.5 ft)}$, $b < 10 \text{ mm (}^3\text{/}_8 \text{ in.)}$

Key

- 1 theoretical brace stub centerline
- 2 actual brace stub centerline
- 3 node can
- 4 theoretical circumferential weld
- 5 actual circumferential weld
- a offset between actual and theoretical node can position, $a < 5 \text{ mm (}^1\text{/}_4 \text{ in.)}$
- D outside diameter of chord
- b offset between actual and theoretical brace centerline intersection at chord outer diameter
- L_s theoretical brace stub length
- L_c theoretical node can length
- θ angle between actual and theoretical centerline of brace, $\theta < 0.17^\circ$
- a $L_s - 5 \text{ mm (0.25 in.)} \leq L_{s,\text{actual}} \leq L_s + 50 \text{ mm (2.0 in.)}$
- b $L_c - 5 \text{ mm (0.25 in.)} \leq L_{c,\text{actual}} \leq L_c + 10 \text{ mm (}^3\text{/}_8 \text{ in.)}$

Figure E.9—Tolerances for Tubular Joints

E.11 Cruciform Joints

Where plates carrying in-plane forces are arranged to form a cruciform joint, the misalignment shall not exceed the lesser of 50 % of the thickness of the thinnest noncontinuous member or 10 mm (³/₈ in.), whichever is smaller.

E.12 Stiffener Tolerances

E.12.1 Stiffener Location

Stiffeners shall be positioned to the following tolerances:

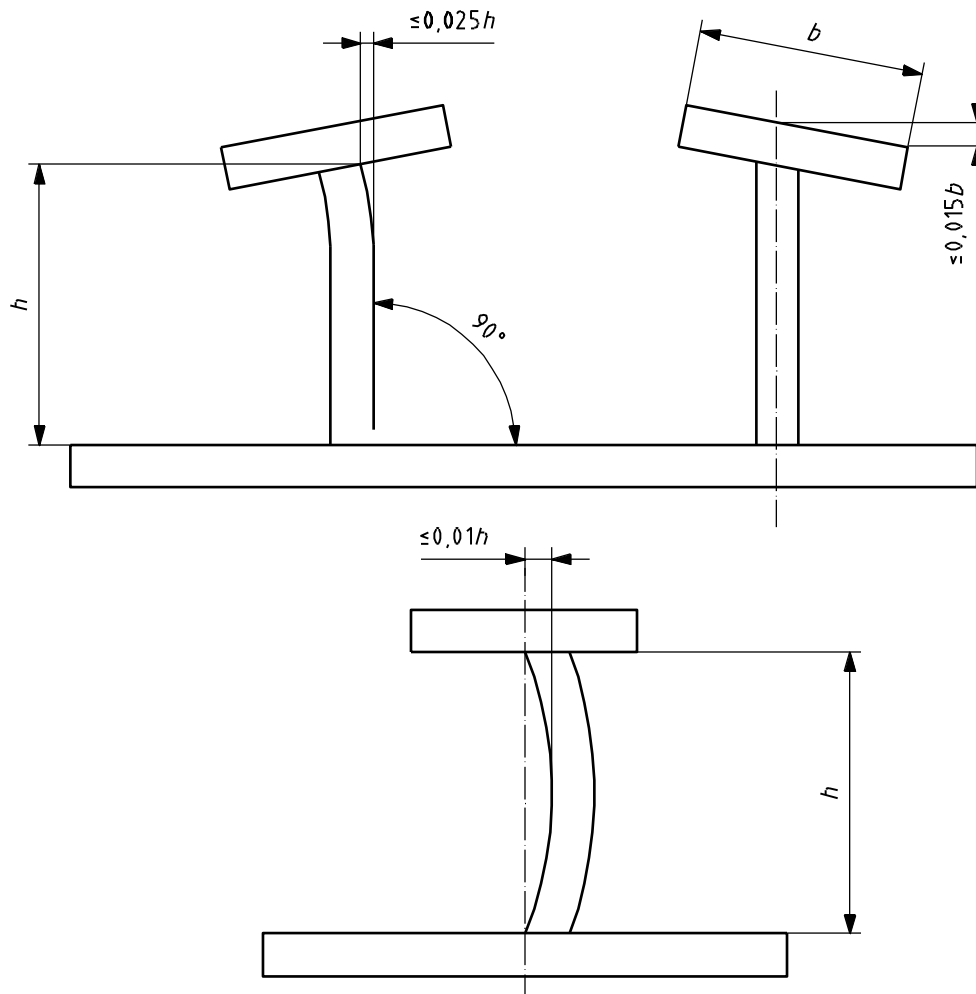
- a) when within 150 mm (6 in.) of a conical transition, accurate to within 3 mm (¹/₈ in.) of the design position;
- b) in launch legs, to within 3 mm (¹/₈ in.) of the design position;
- c) in tubular joints other than at conical transitions of launch leg nodes, to within 5 mm (¹/₄ in.) of the design position;
- d) at all other locations, to within 10 mm (³/₈ in.) of the design position.

E.12.2 Stiffener Cross-section

The tolerances on ring stiffener cross-sections are as shown in Figure E.10 and as follows:

- a) the web of the stiffener shall be perpendicular to the centerline of the tubular to within 2.5 % of the web height;
- b) the flange of the stiffener shall be parallel to the centerline of the tubular to within 1.5 % of the flange width;
- c) the web of the stiffener shall be flat over its height to within 1.0 % of the web height.

Out-of-straightness for longitudinal or diaphragm stiffeners in tubular members shall be limited to 0.15 % L or 3 mm, whichever is larger (see Figure E.11).



Key

h dent depth

Figure E.10—Ring Stiffener Cross-section Tolerances

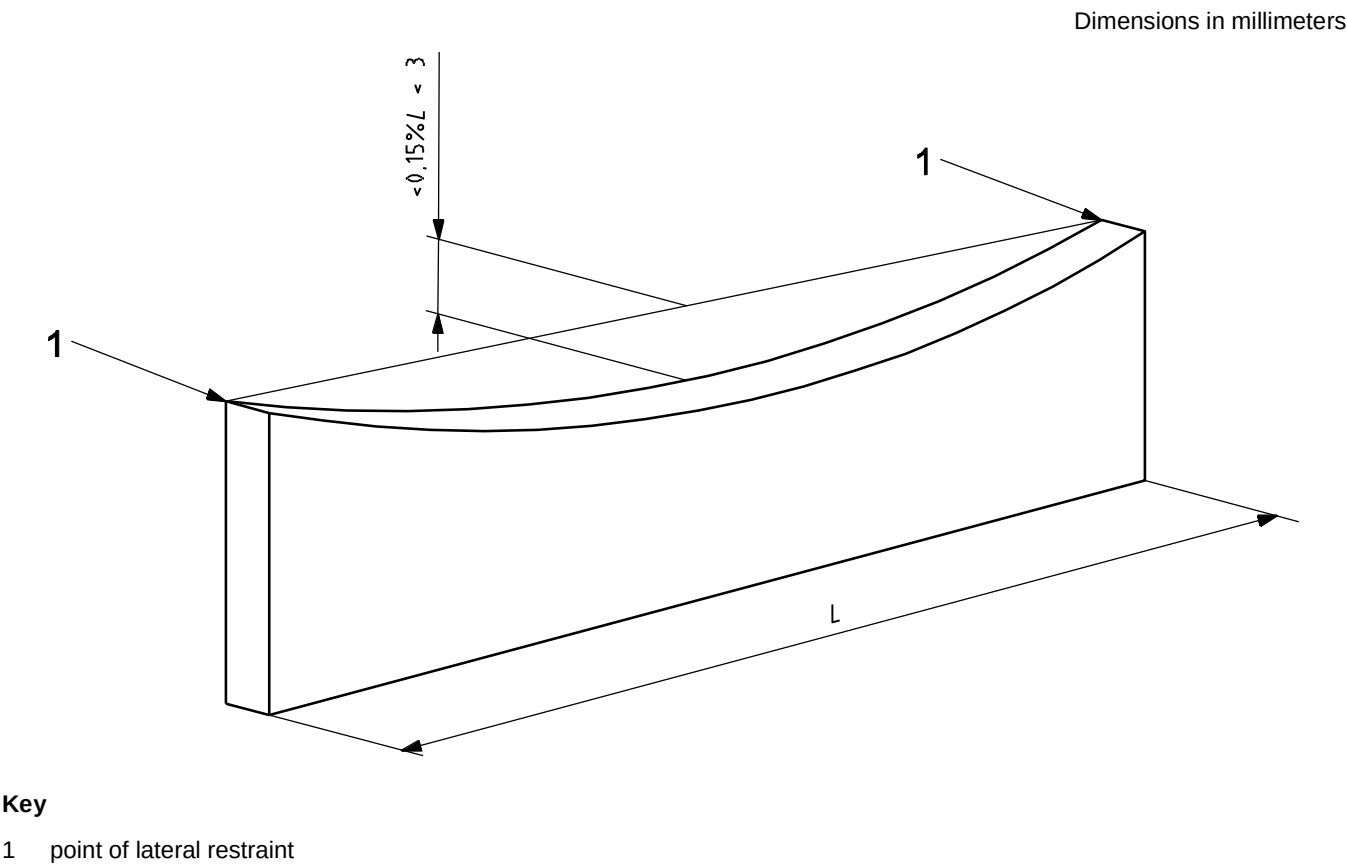
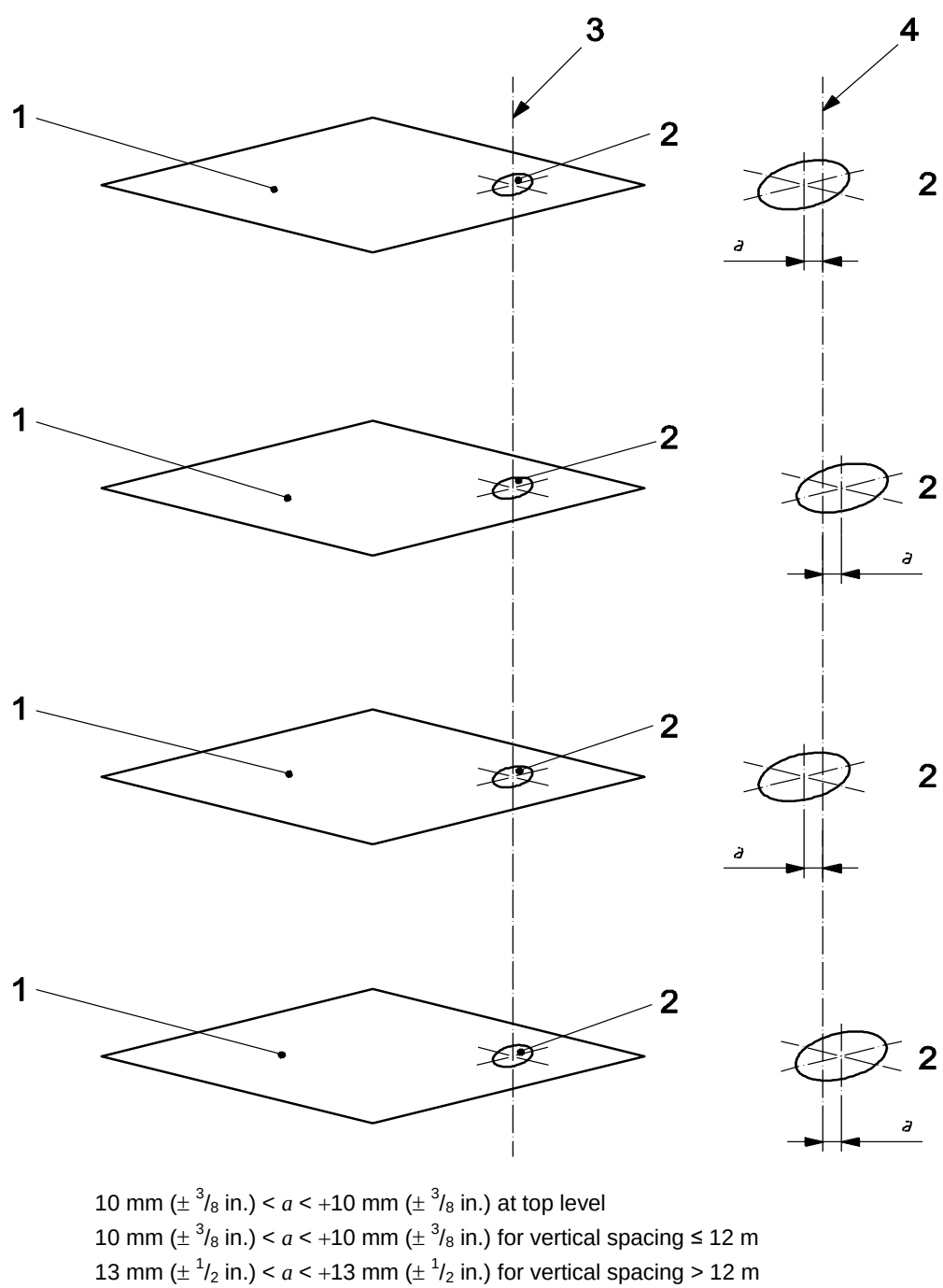


Figure E.11—Ring Stiffener Straightness Tolerance

E.13 Conductor, Pile Guide, Pile Sleeve, and Appurtenance Support Tolerances

The misalignment on conductor and pile guides, and sleeves and appurtenance supports shall be related to a best-fit line through the centers of the guides, sleeves, and supports (see Figure E.12). The tolerance of the center of each guide/sleeve and the best-fit line shall be less than 10 mm (³/₈ in.) at the top guide and where the vertical spacing between guides is less than 12 m (40 ft). The tolerance of the center of each guide/sleeve and the best-fit line shall be less than 13 mm (¹/₂ in.) in other cases. For pile sleeves, the misalignment shall be checked at mid-height of each set of centralizers.

The tolerance of the center of each appurtenance support and the best-fit line shall be less than 10 mm (³/₈ in.) at the top and less than 25 mm (1 in.) elsewhere.



Key

- 1 plan bracing levels
- 2 centers of conductor guides at each level
- 3 theoretical centerline
- 4 best fit centerline
- a tolerance of alignment of guides

NOTE Also applies to pile guides/sleeves where provided.

Figure E.12—Conductor Guide Alignment Tolerances

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